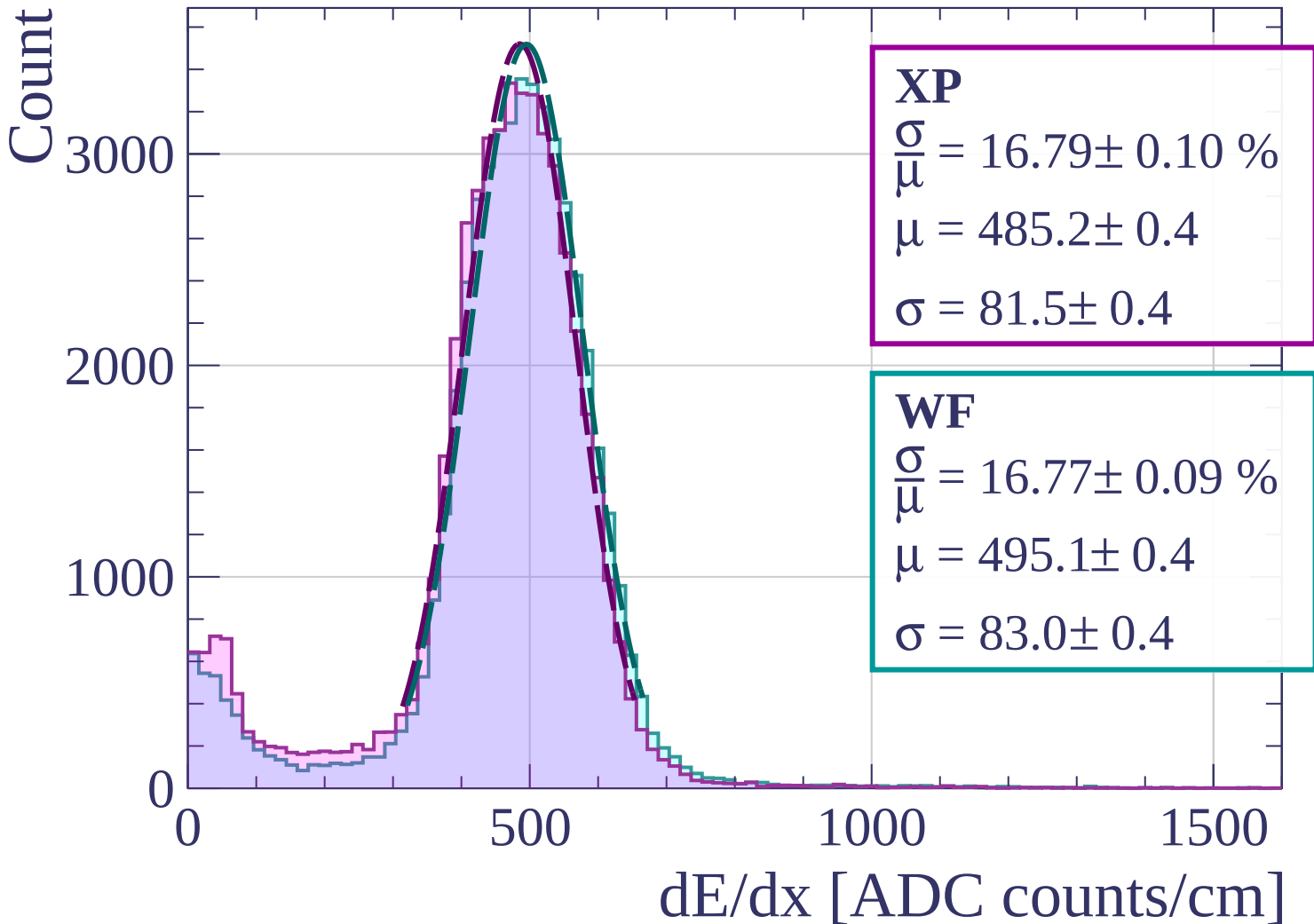
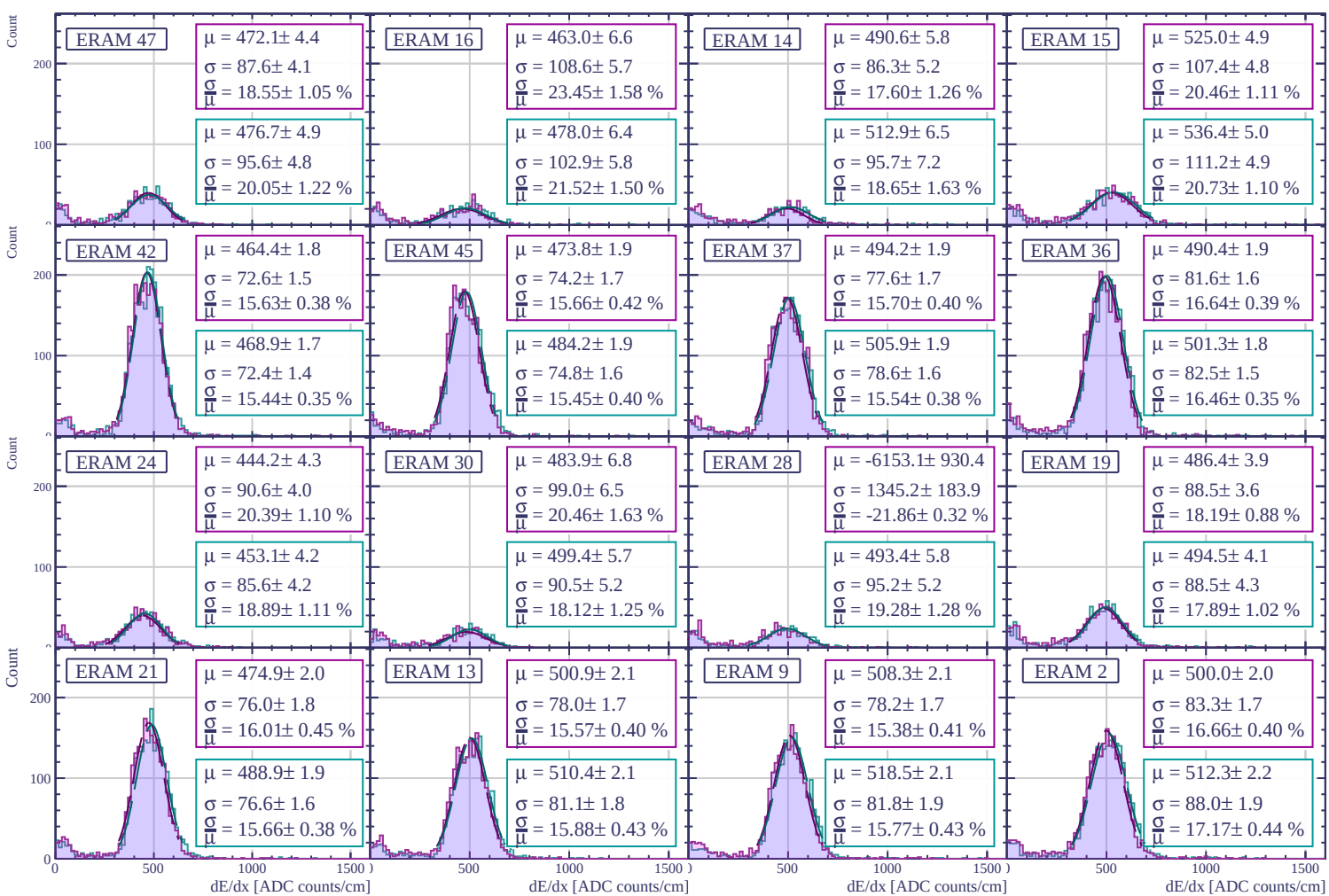
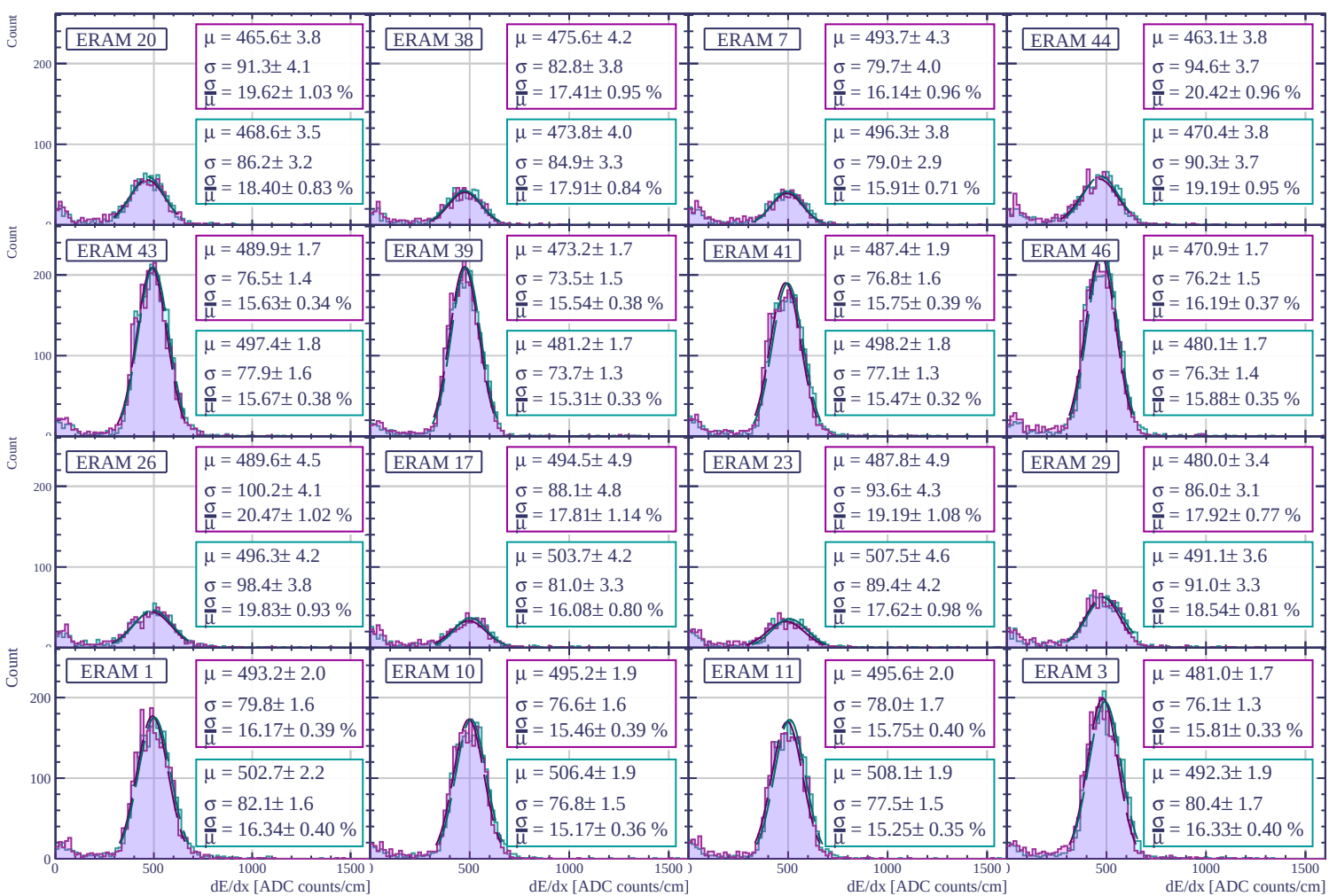
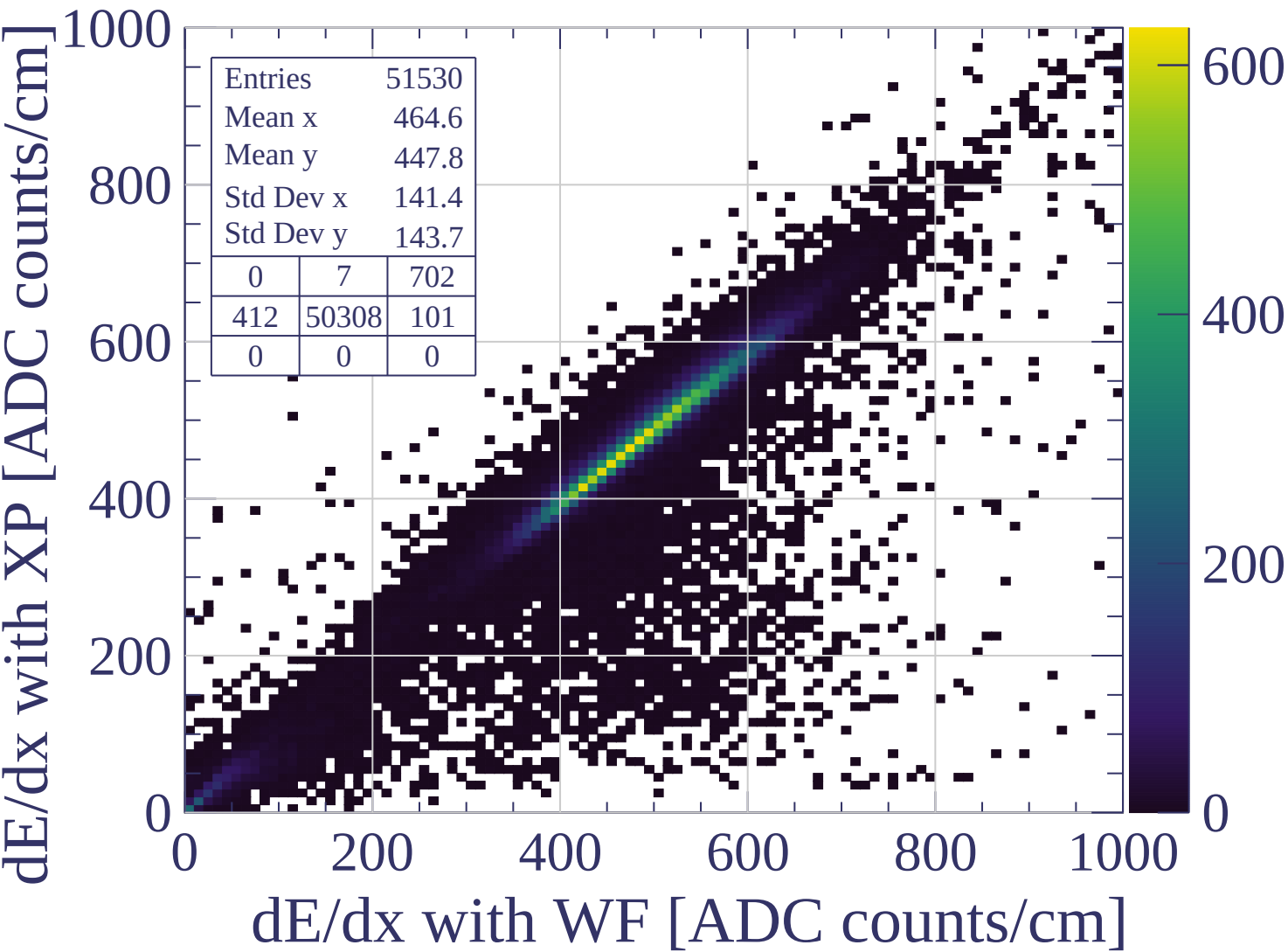
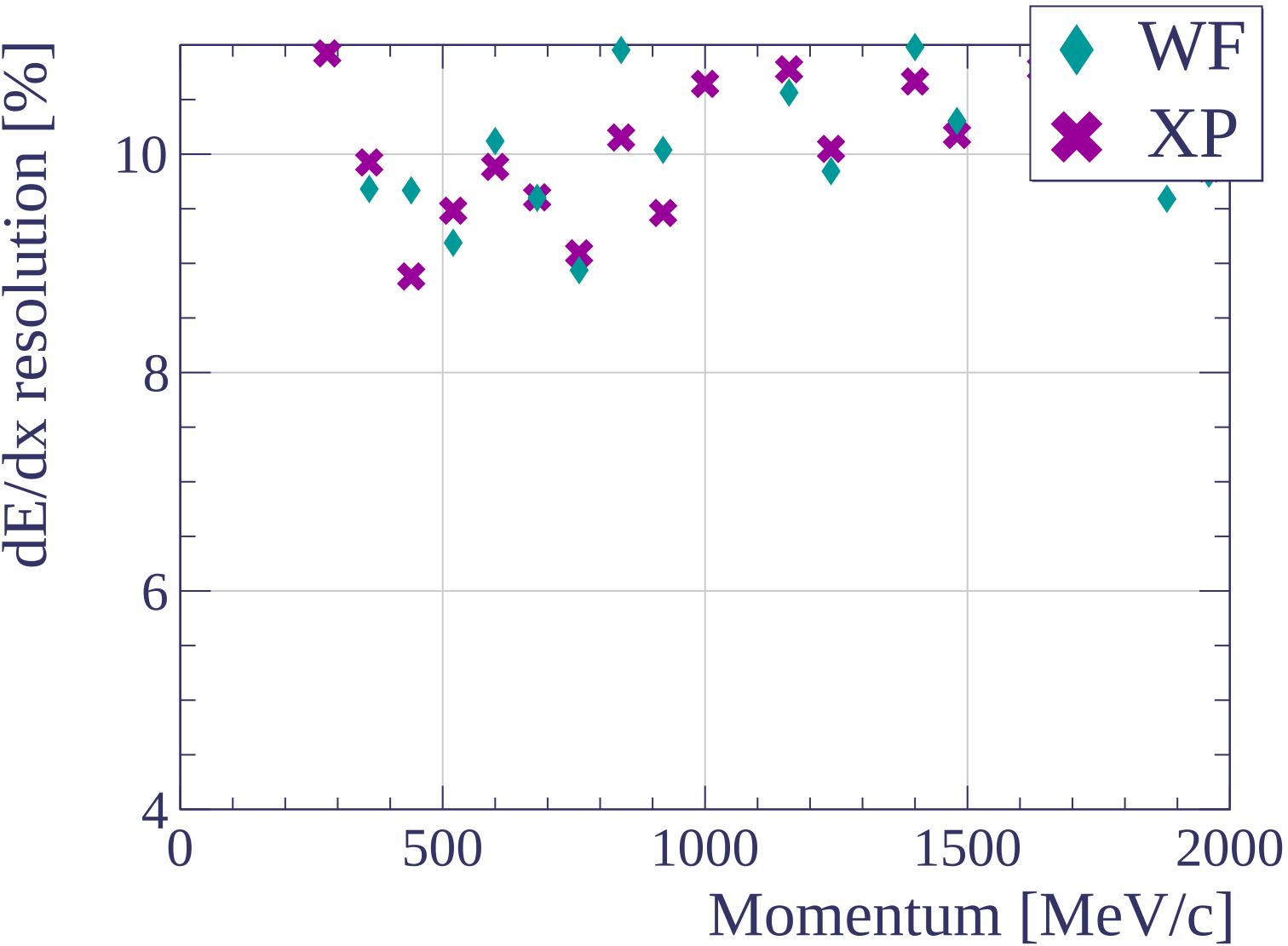


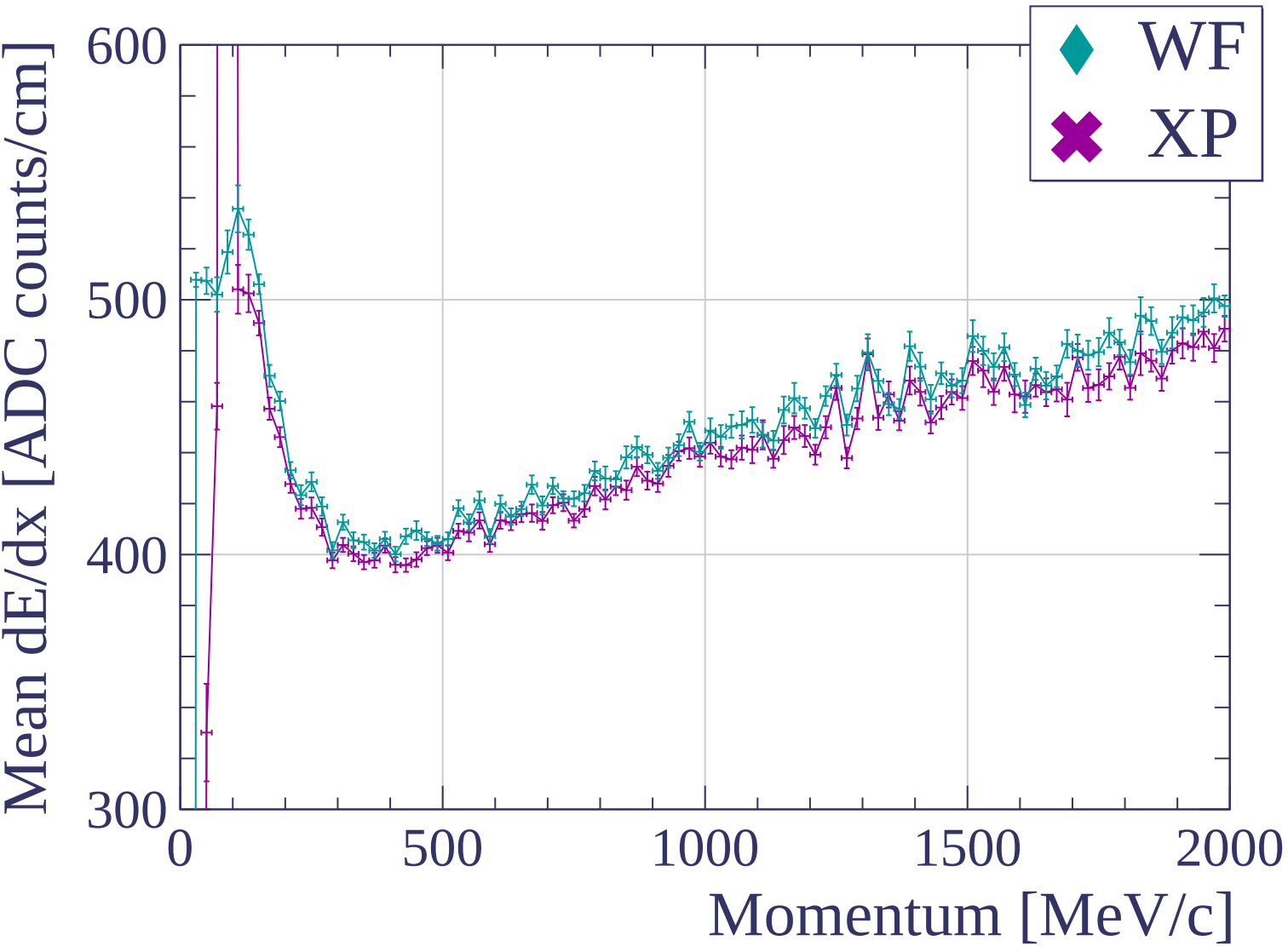


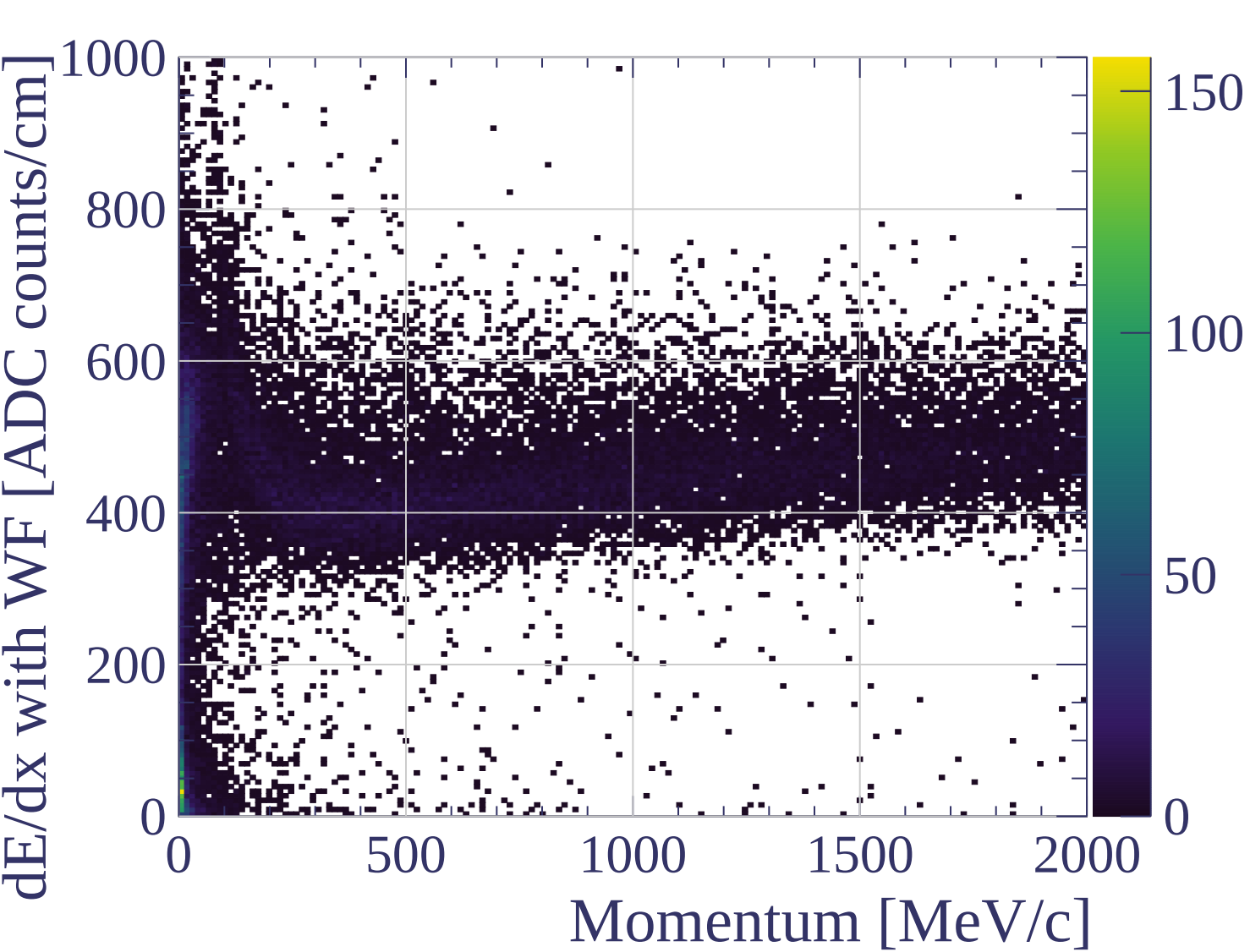


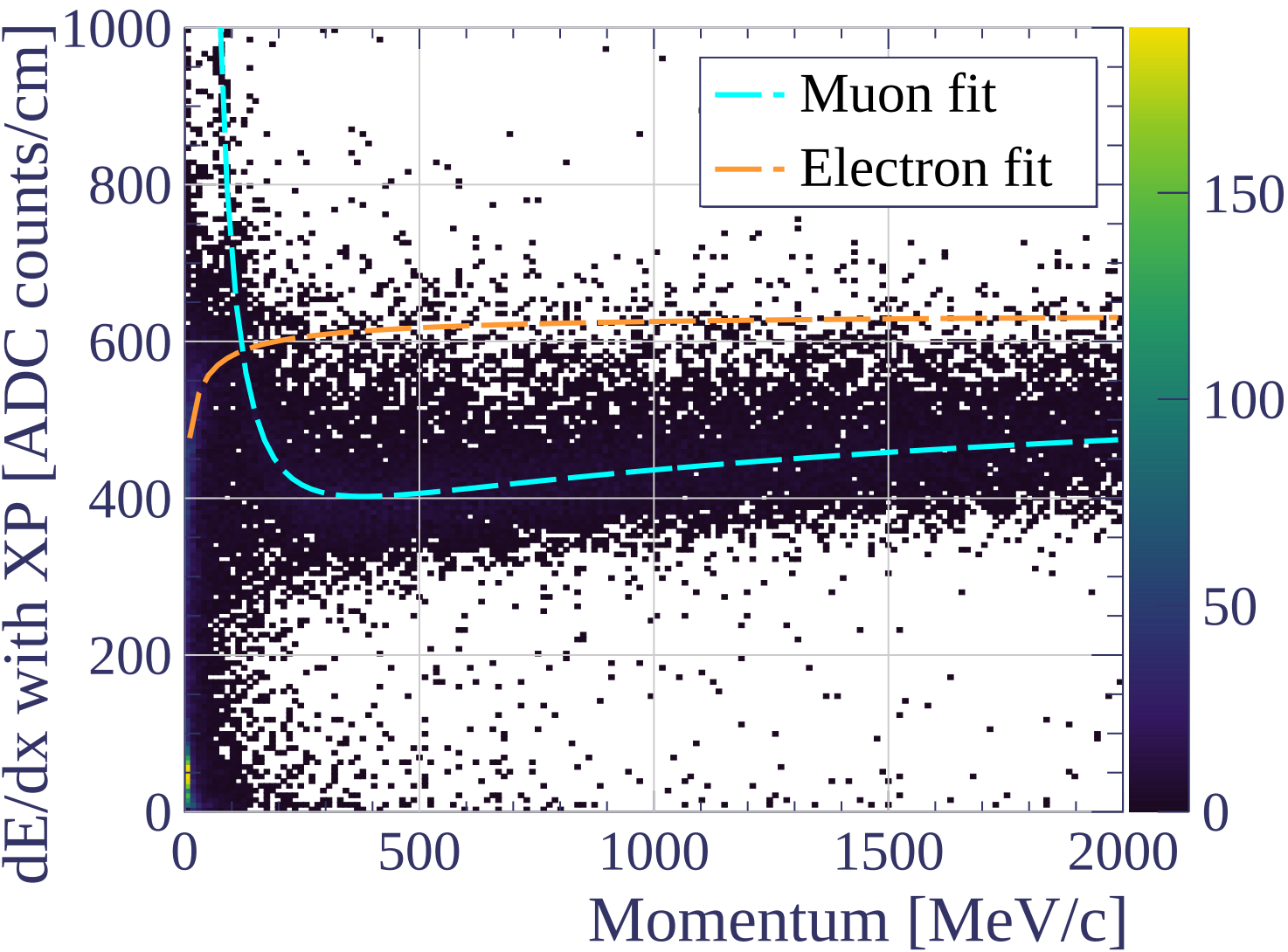




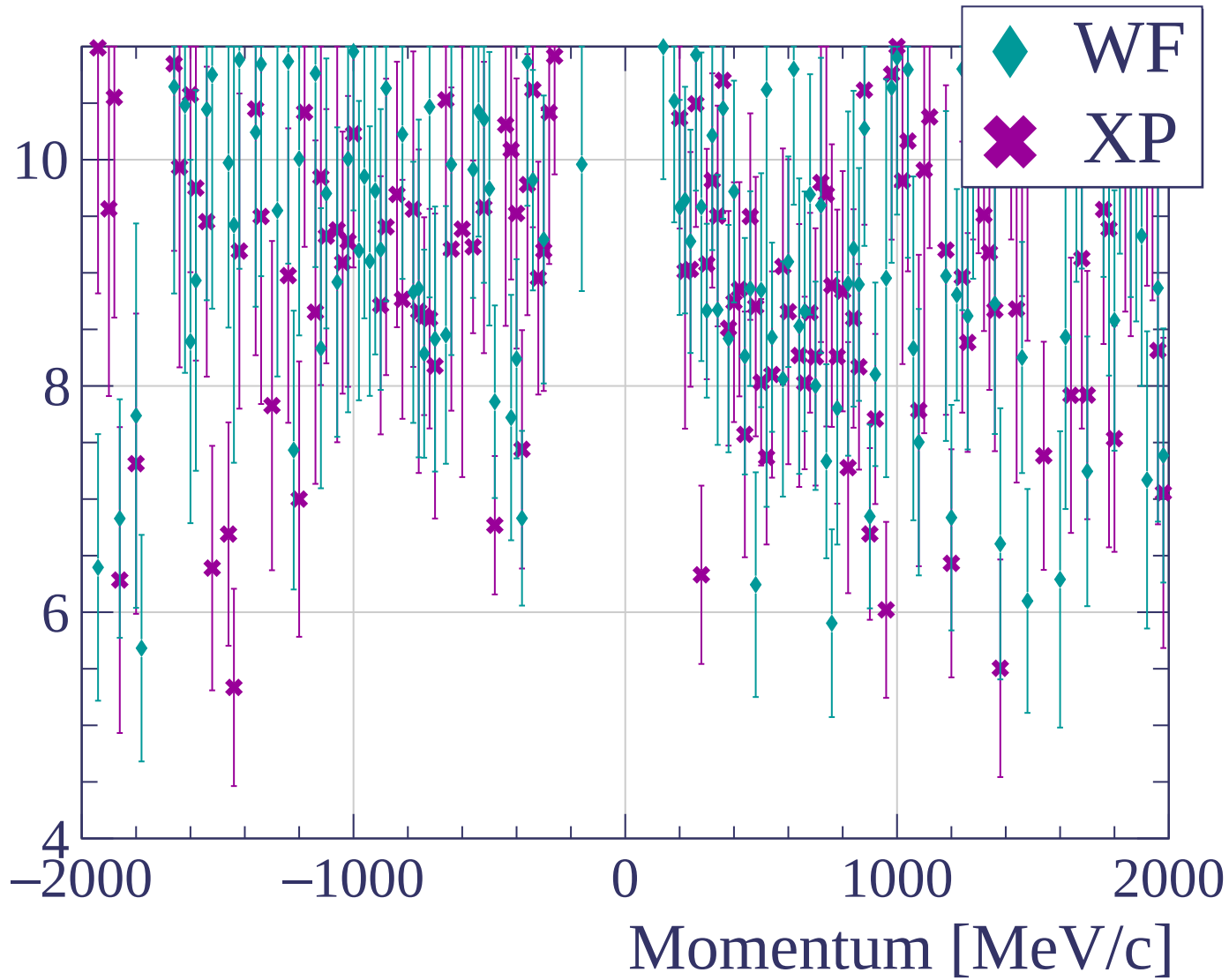


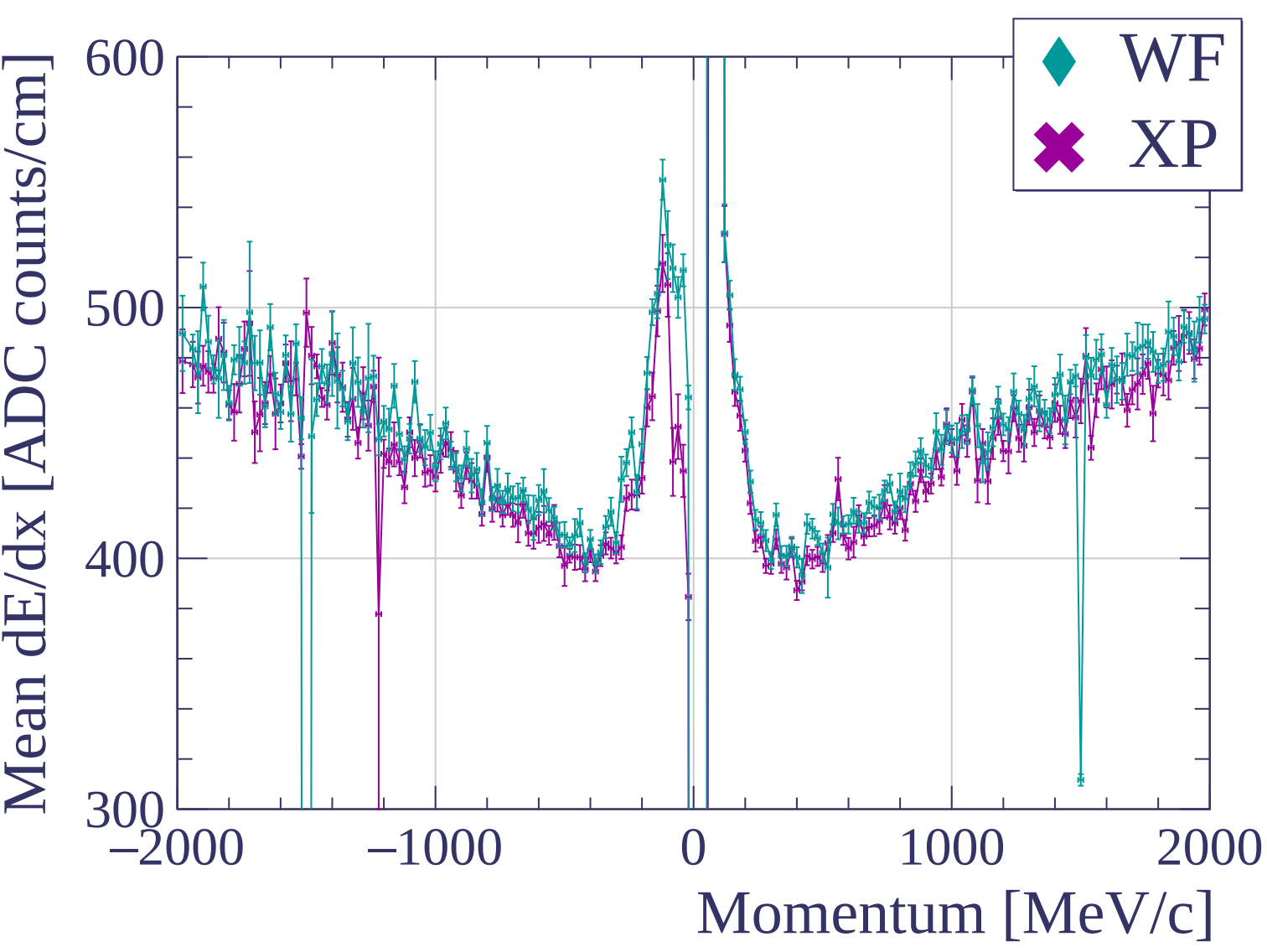


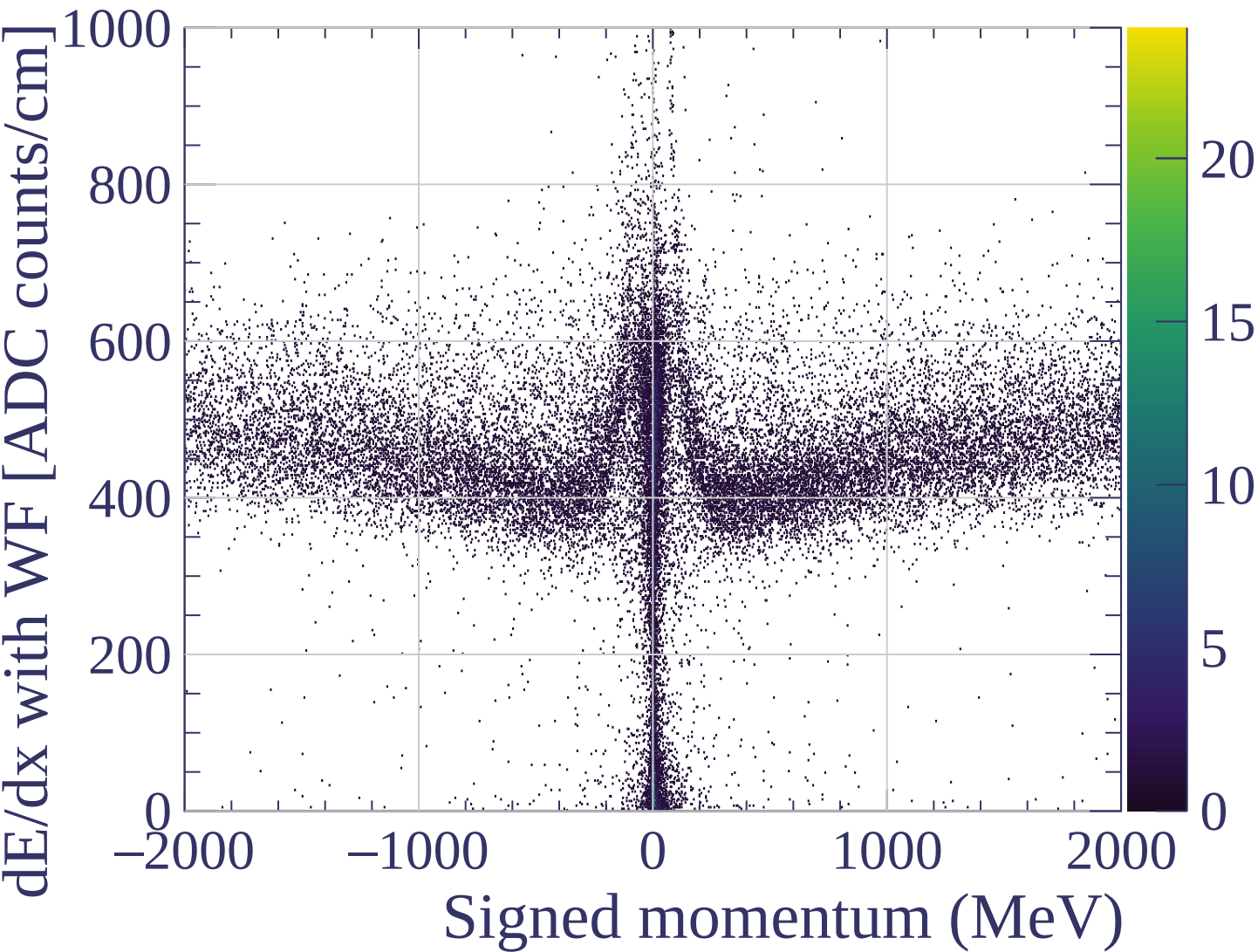


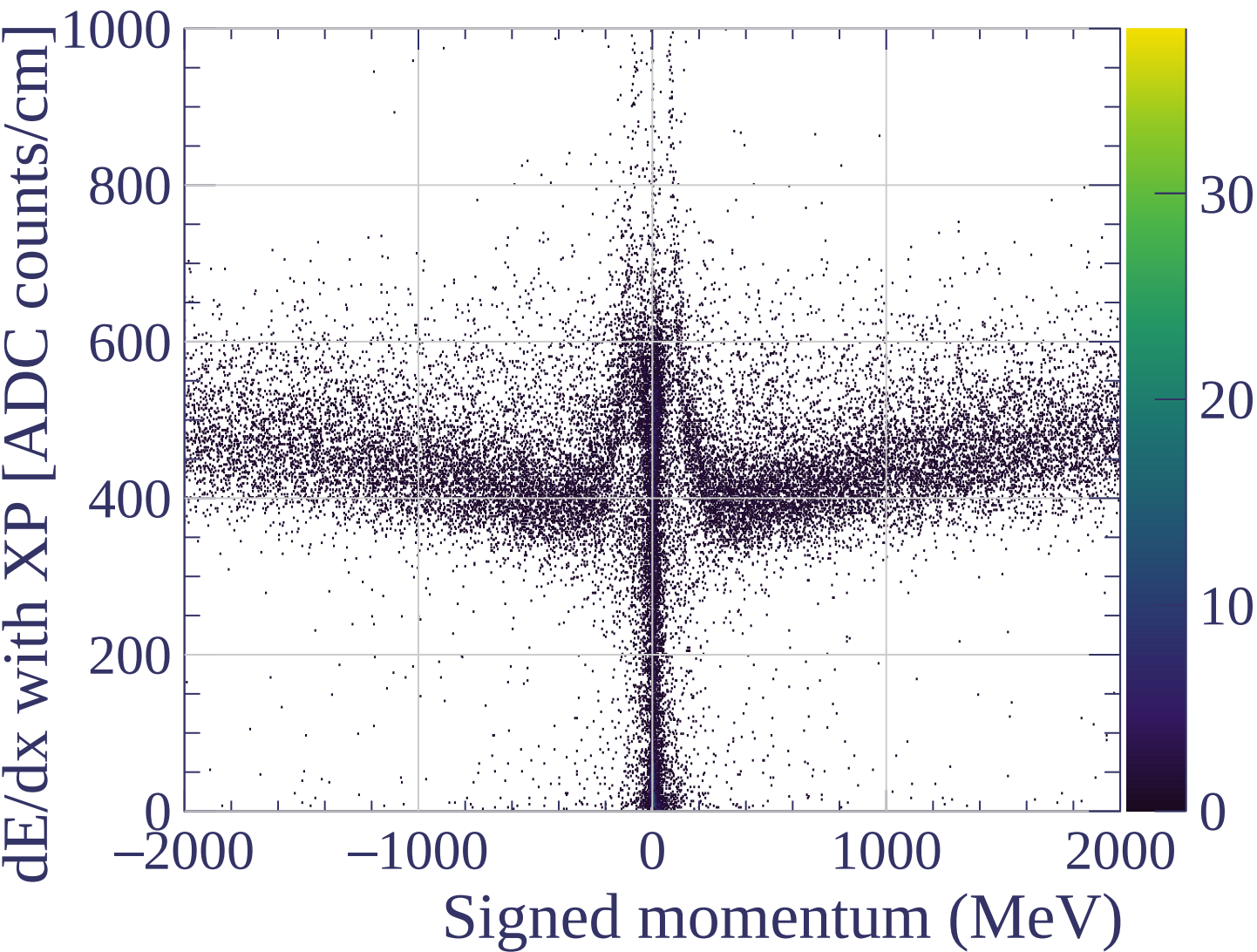


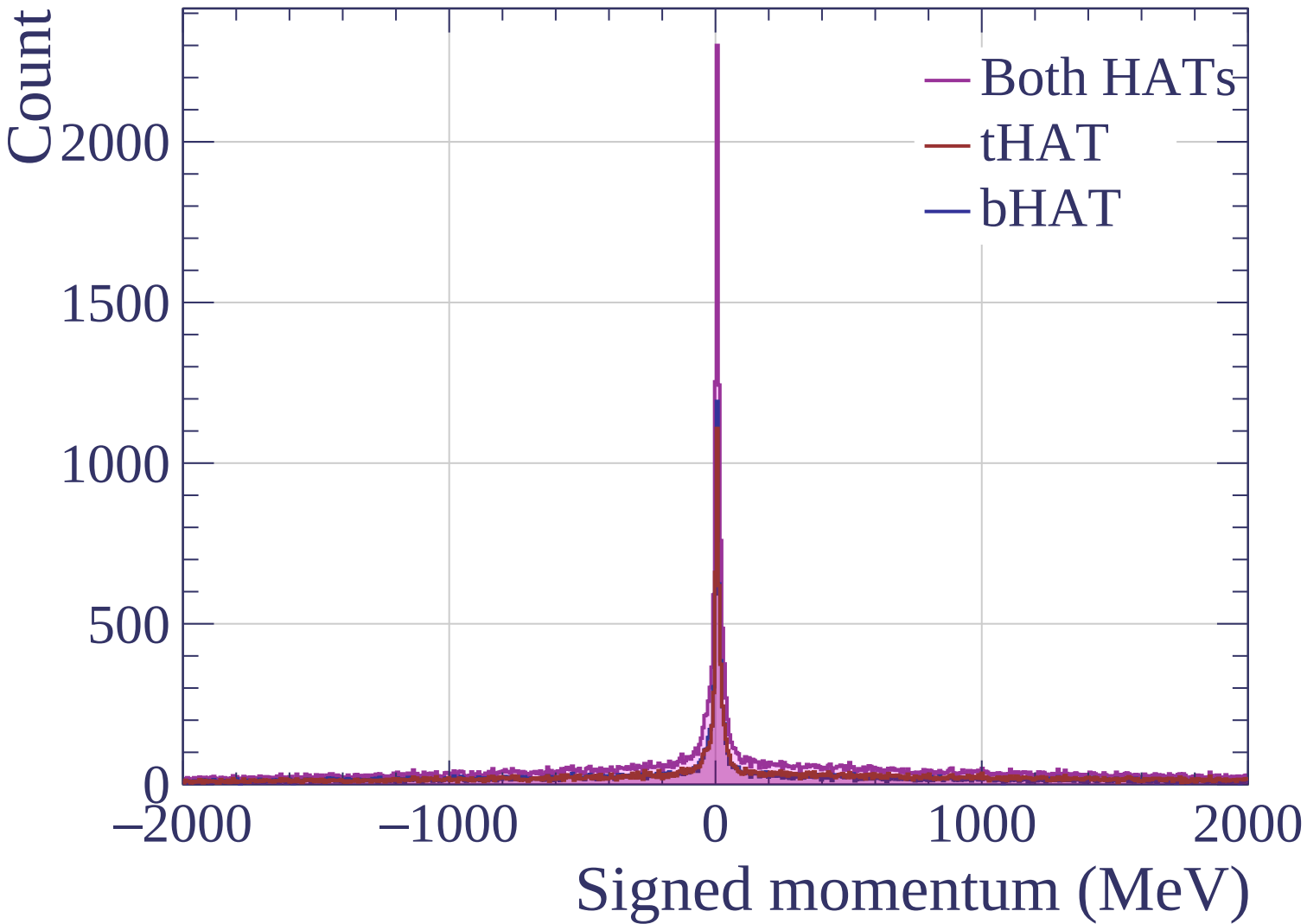
dE/dx resolution [%]

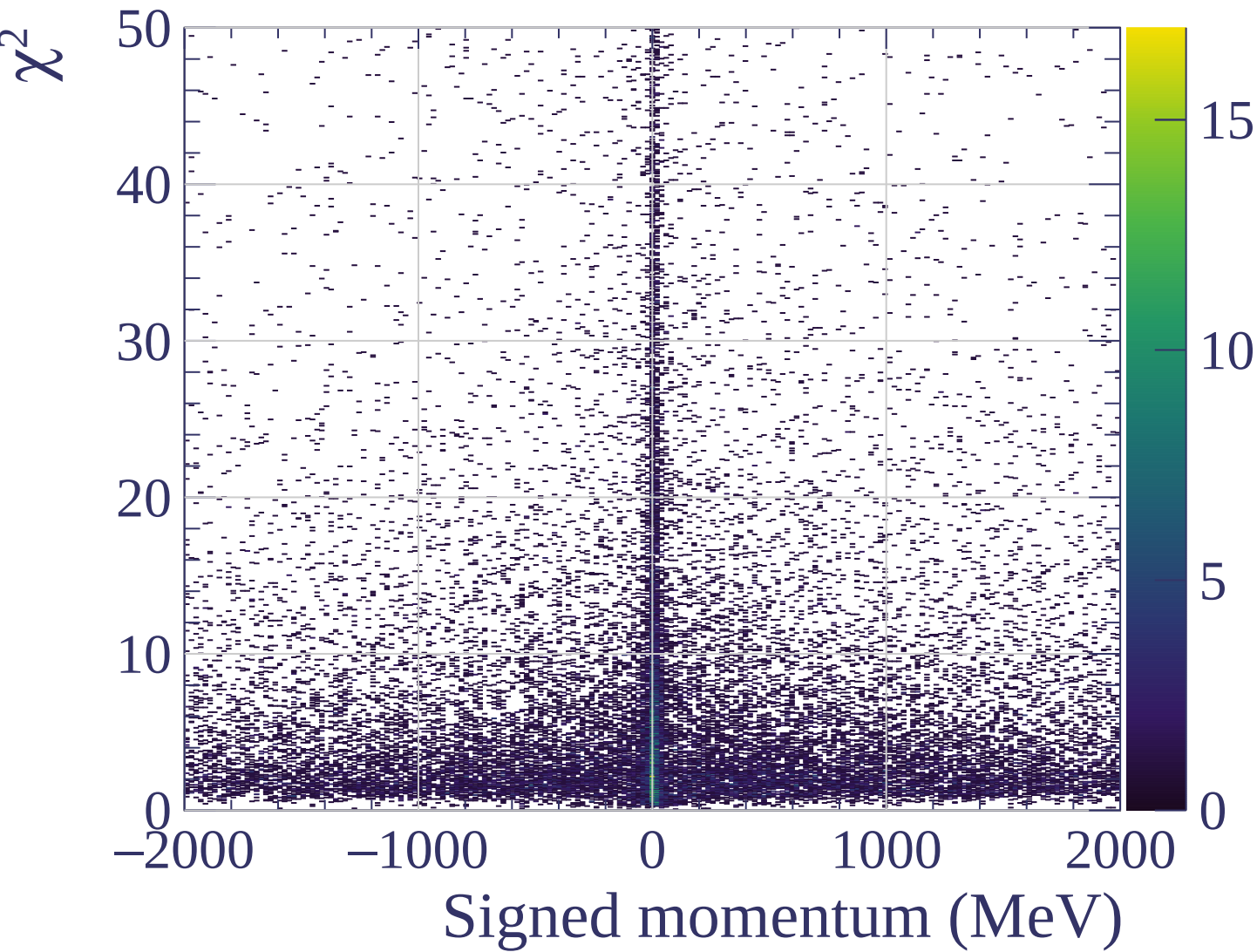


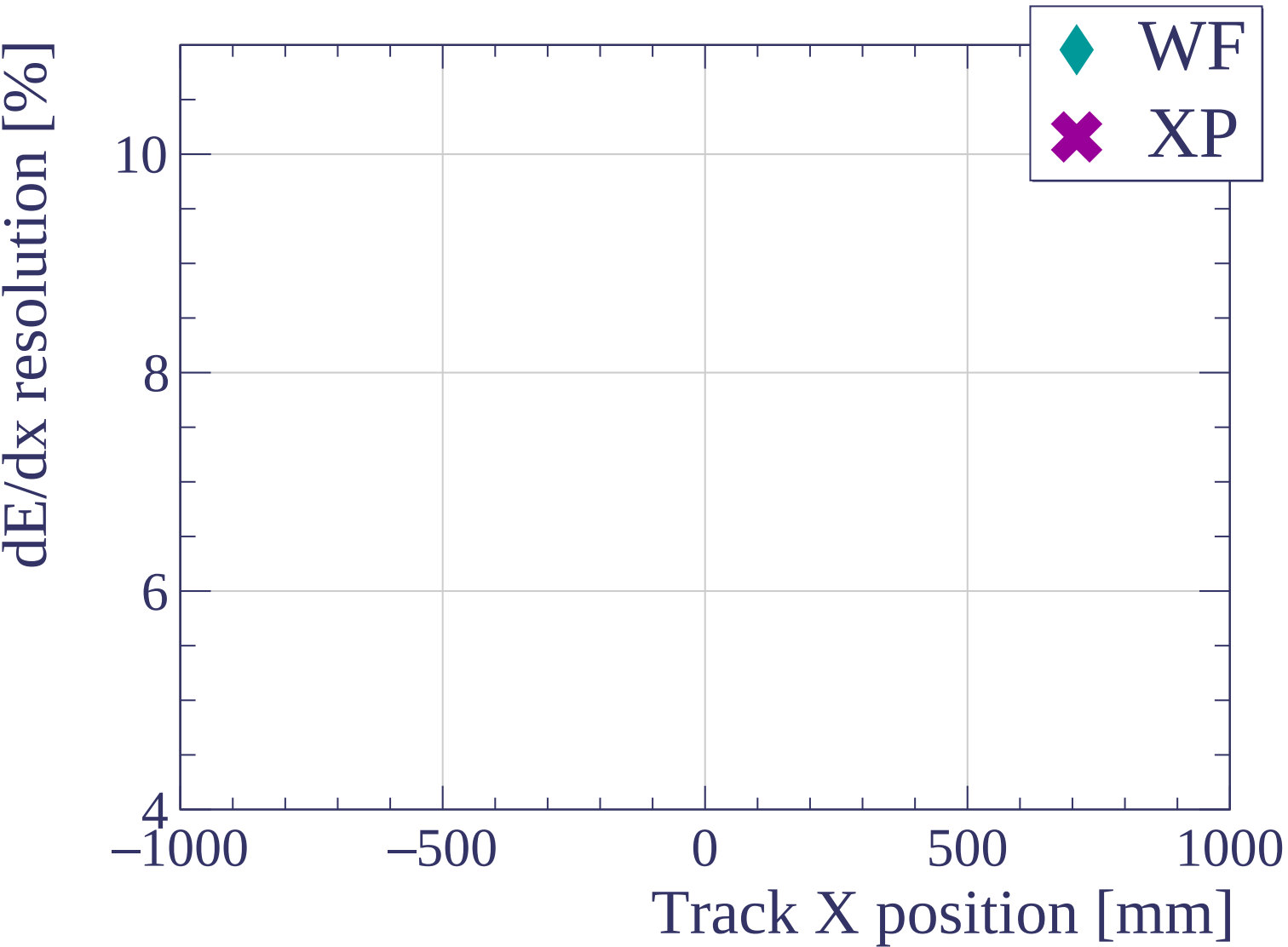


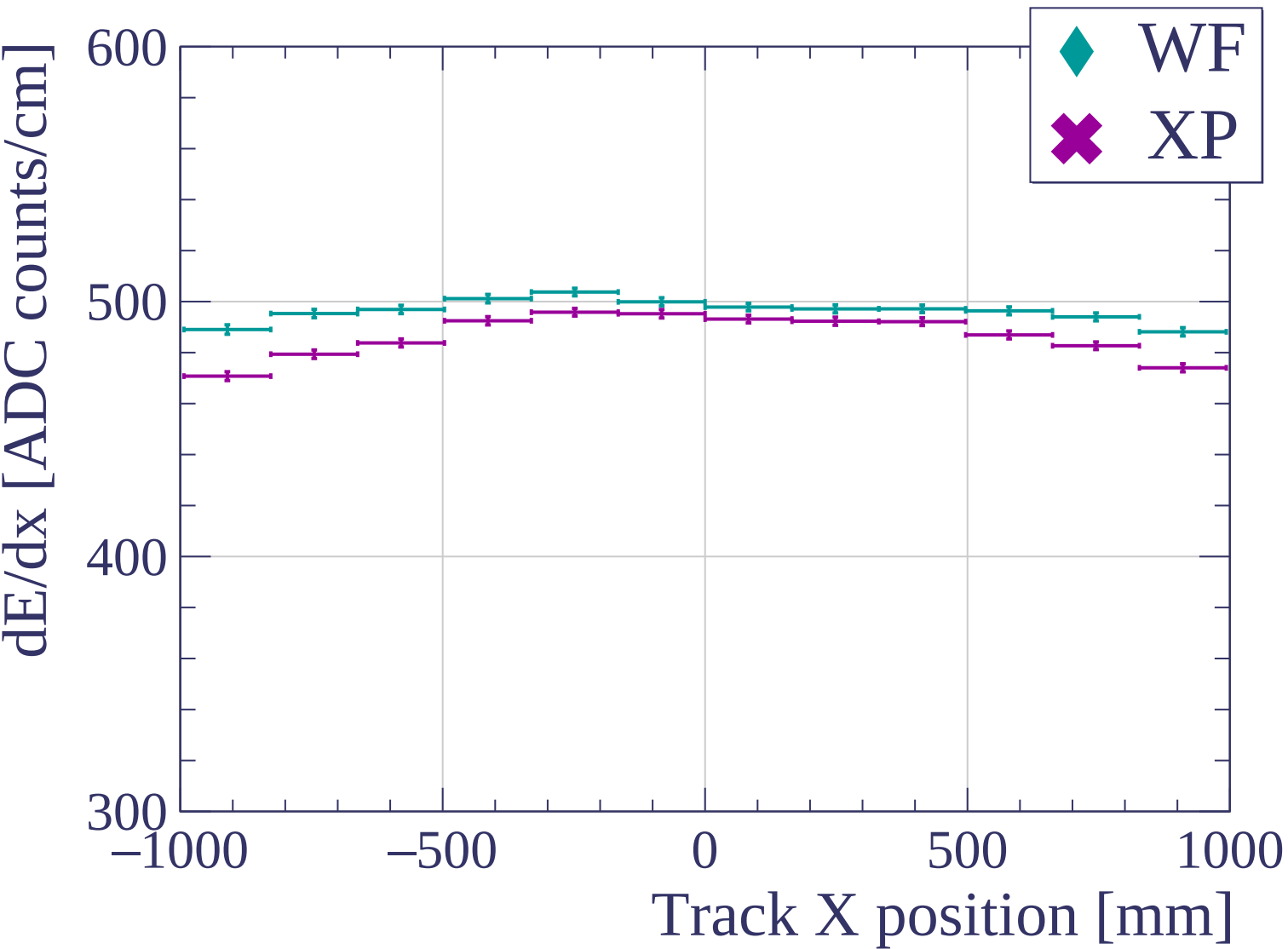


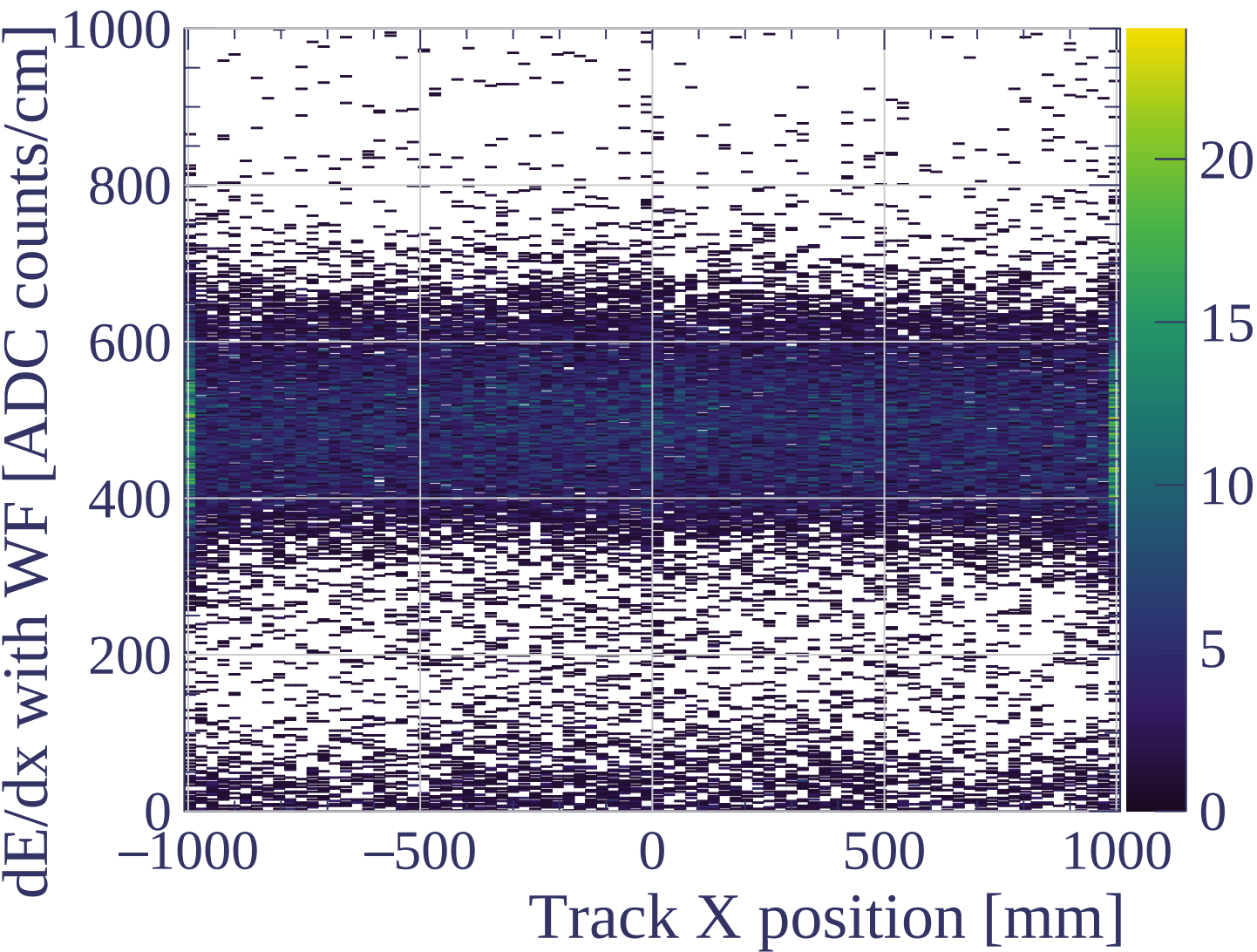


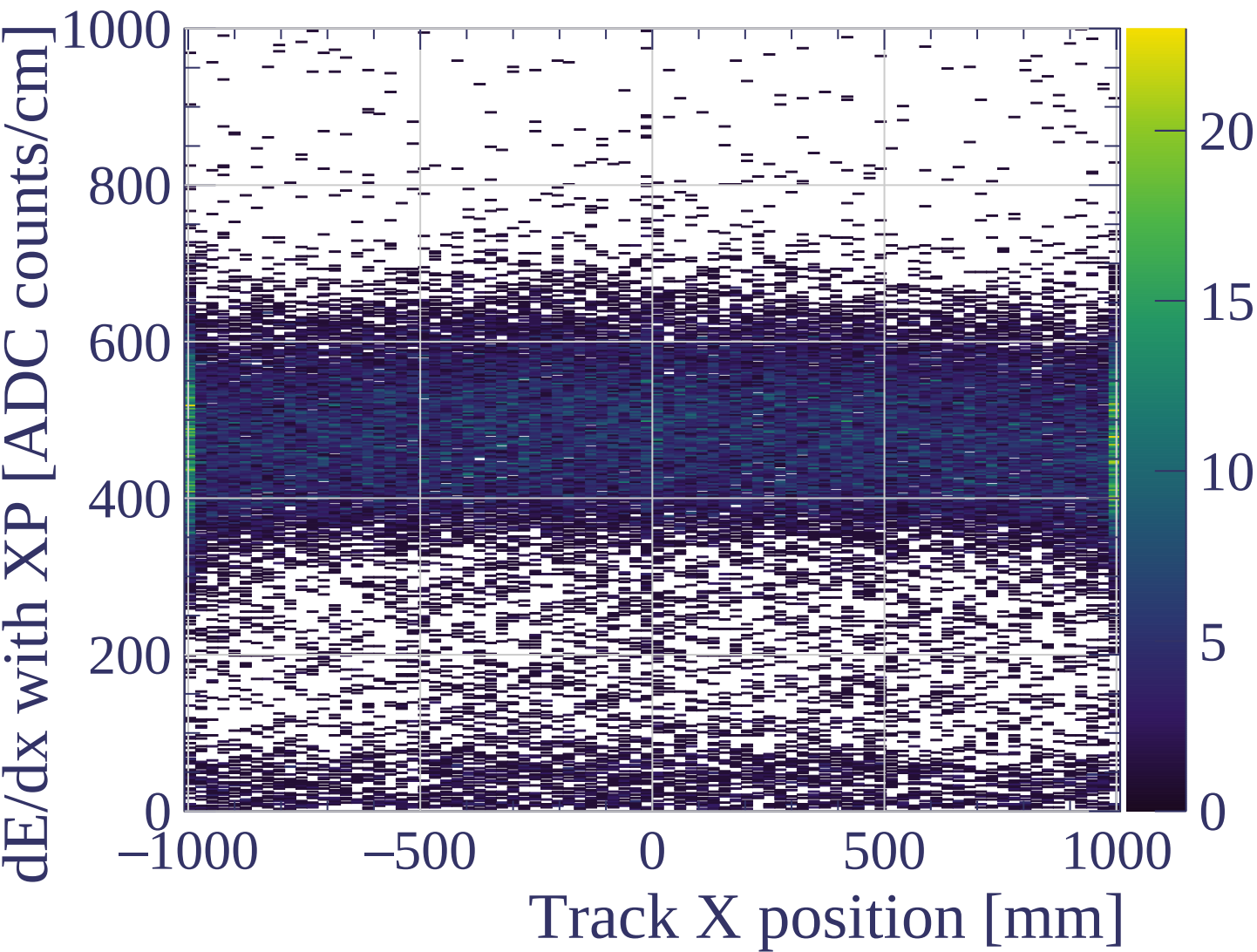


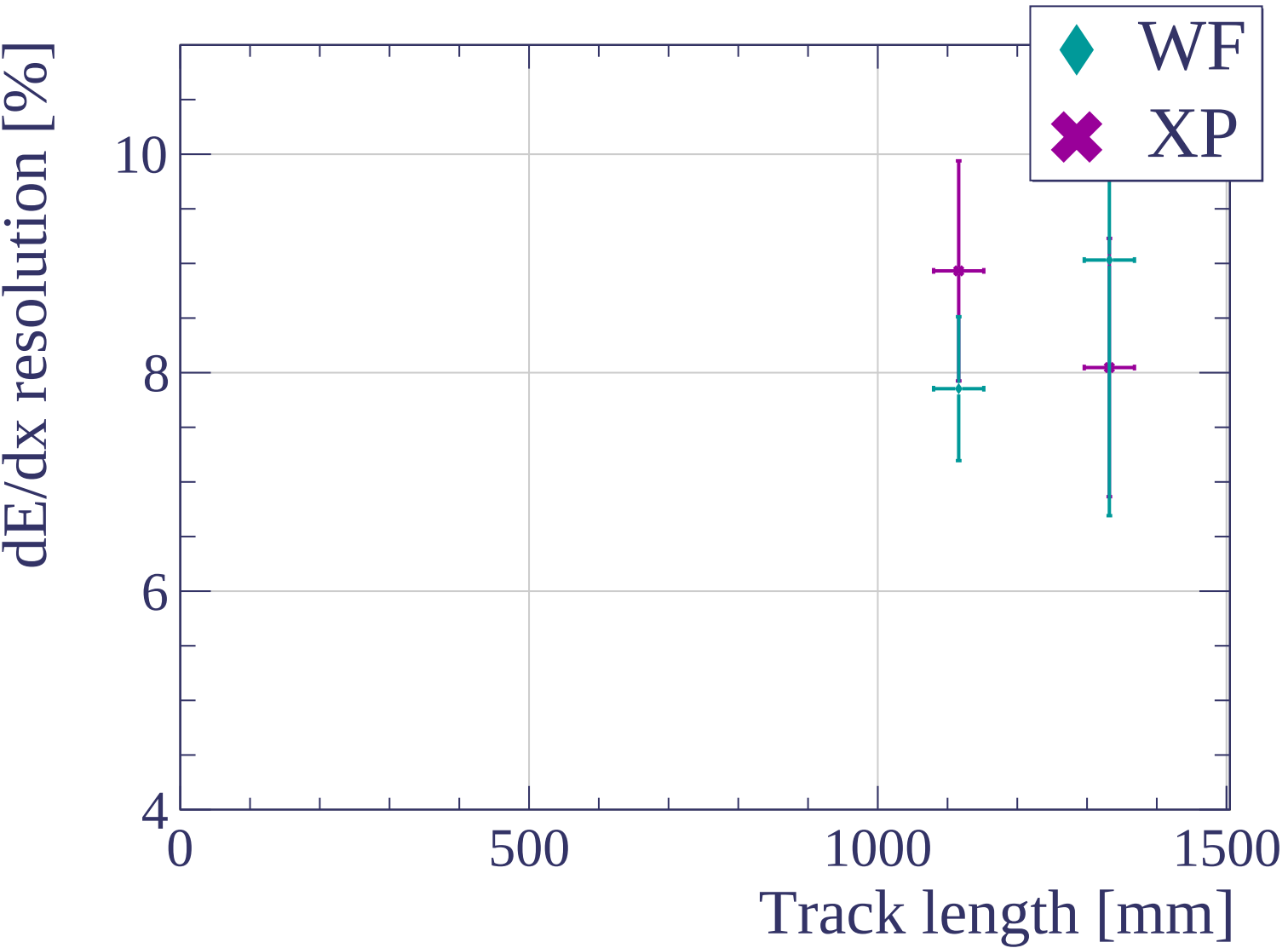


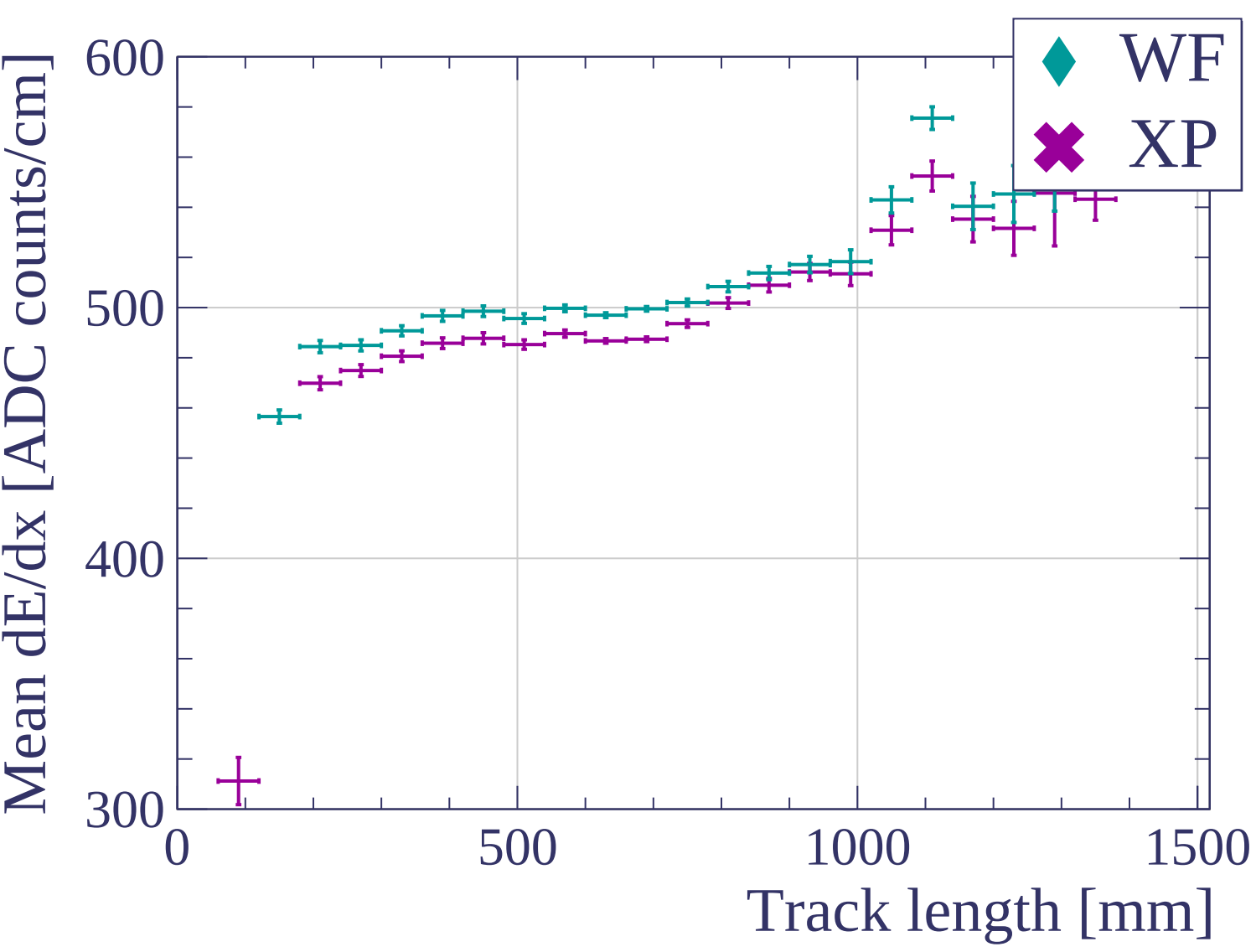


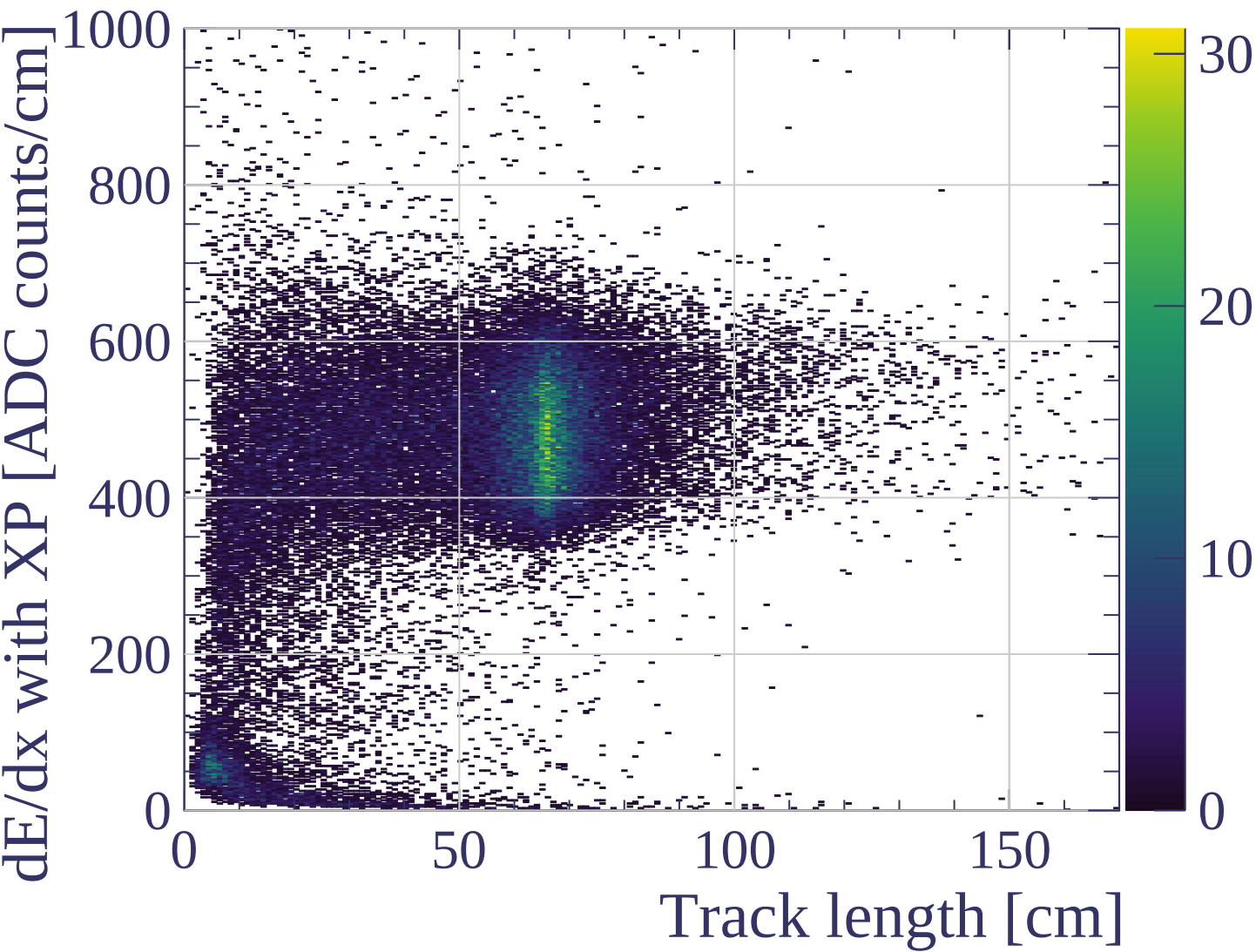


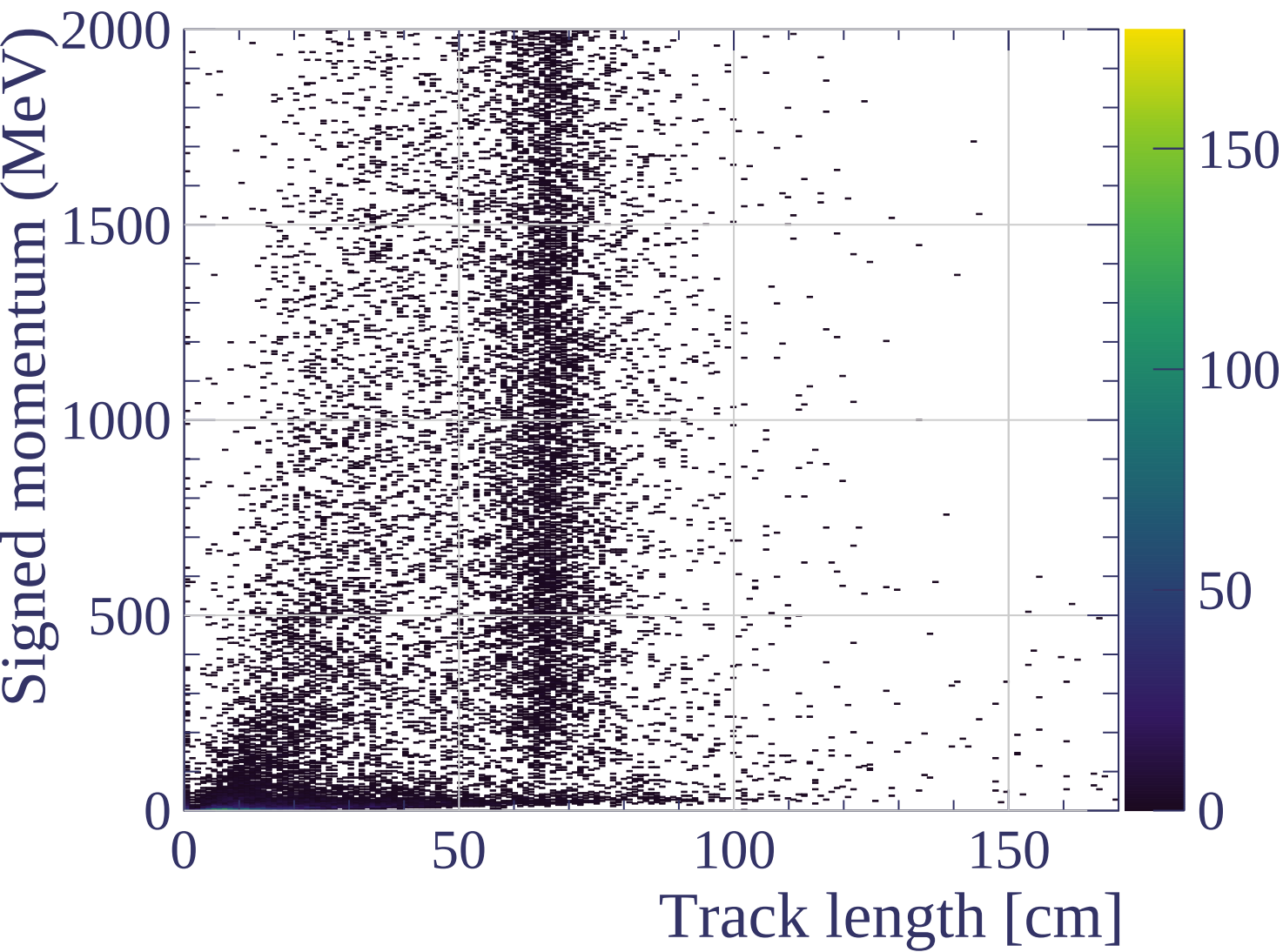


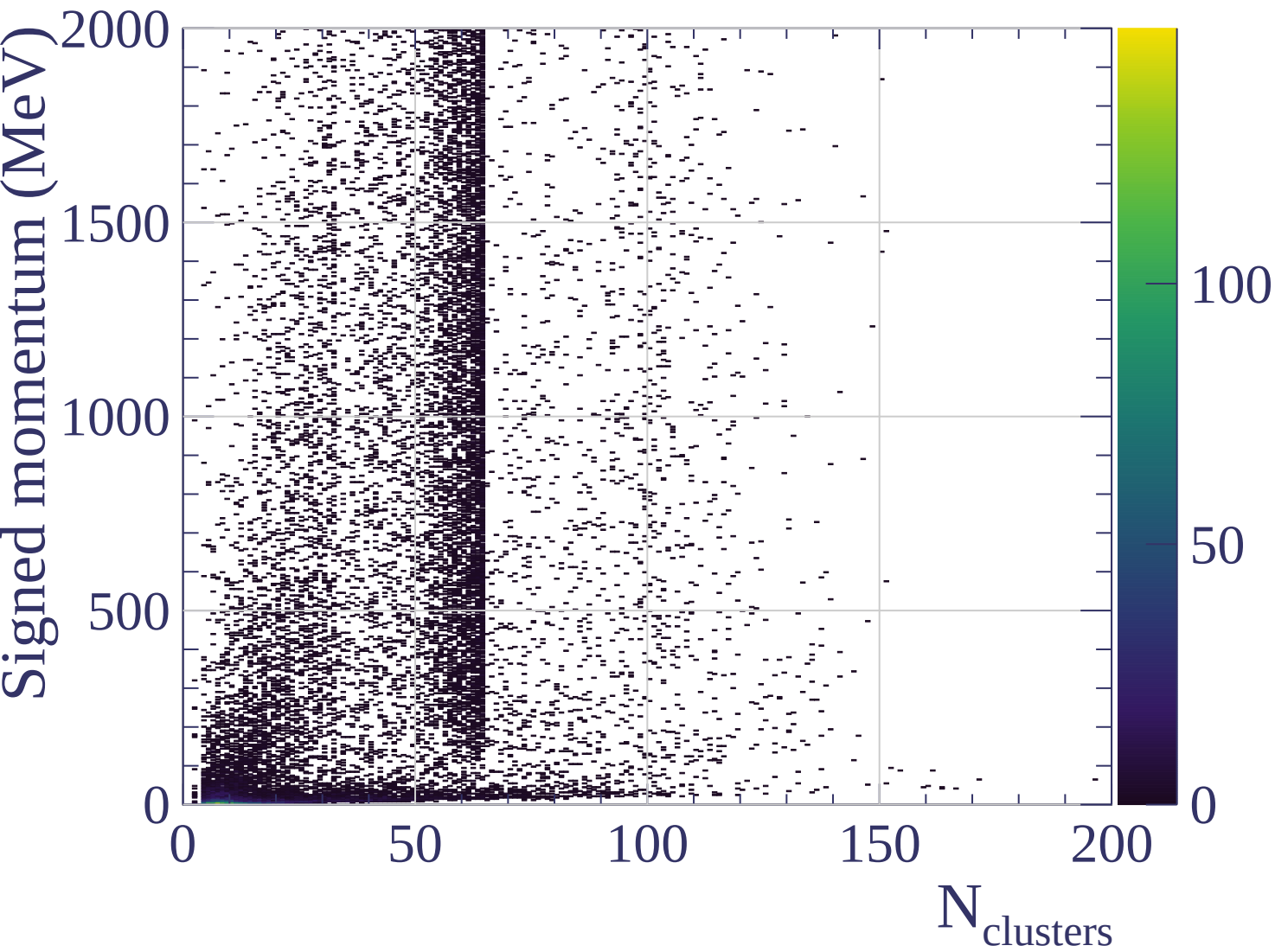


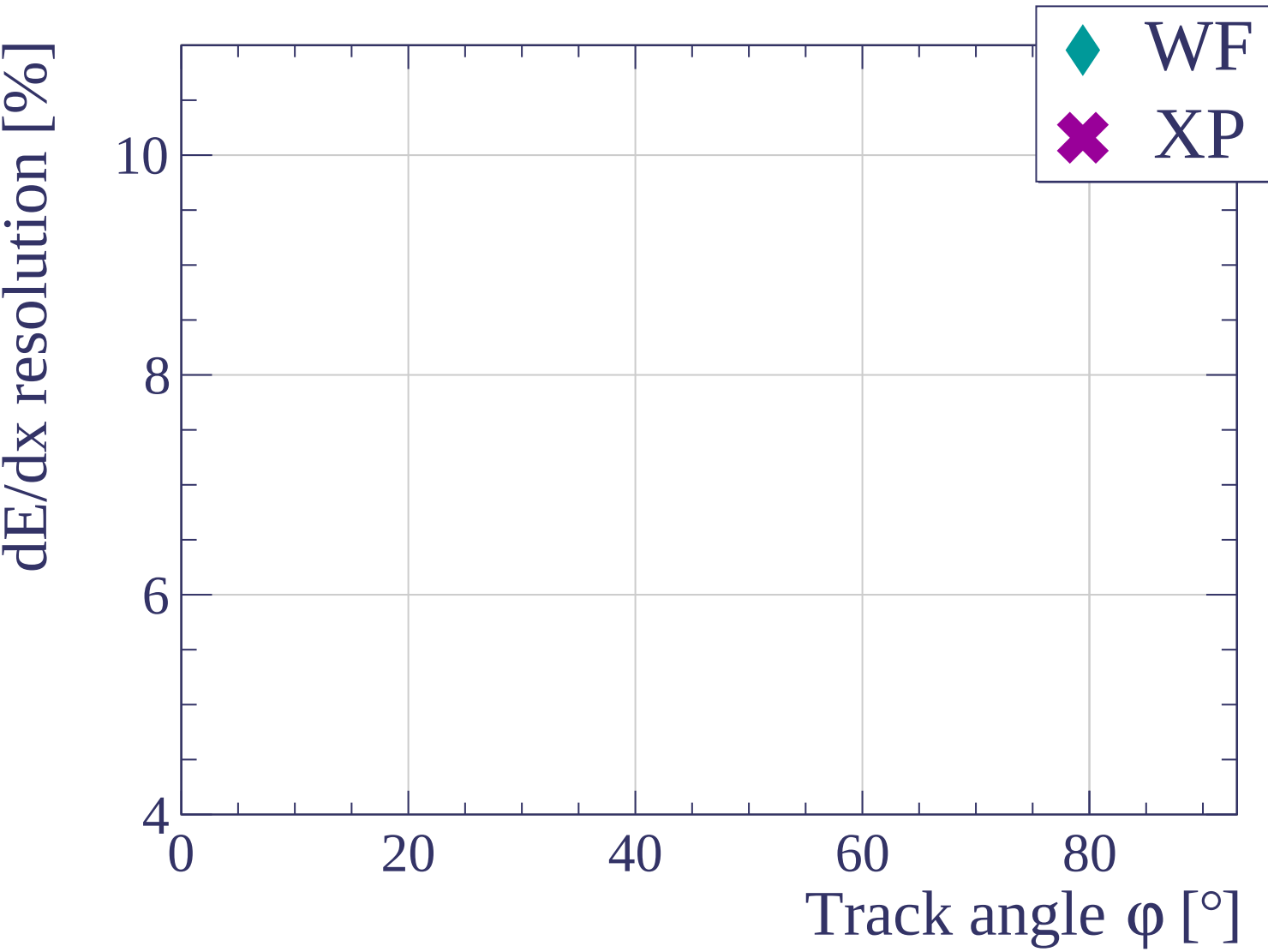


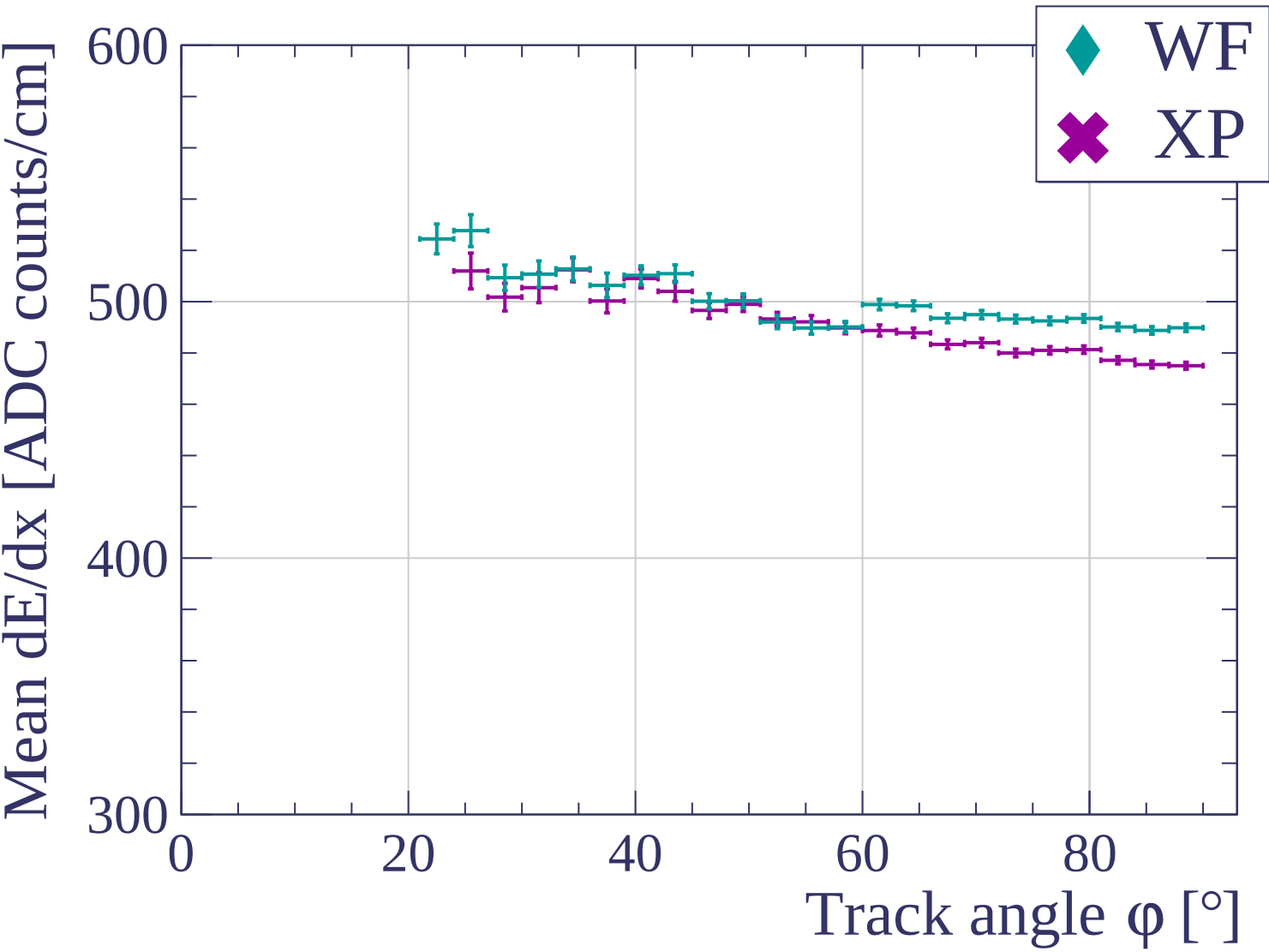


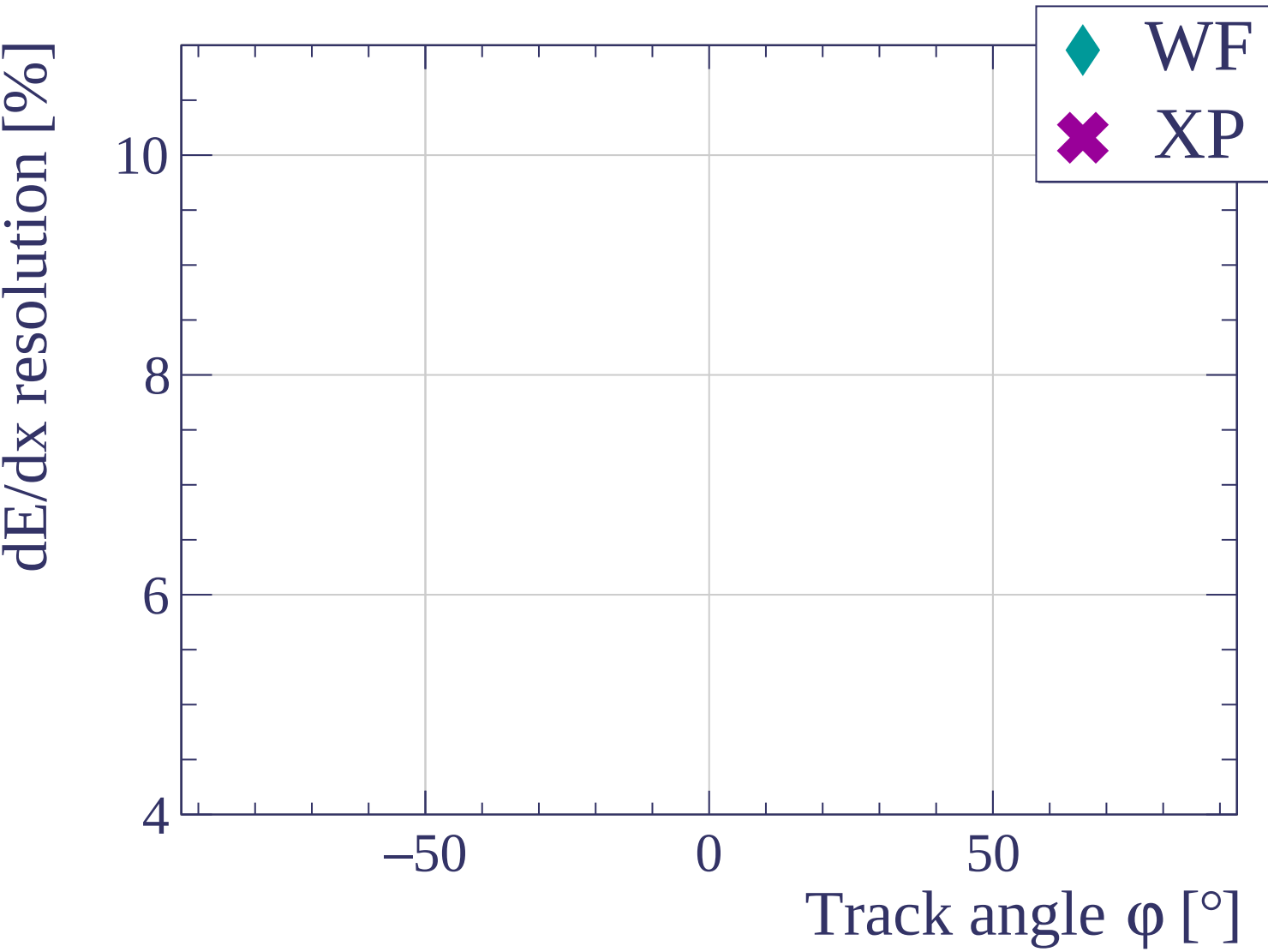


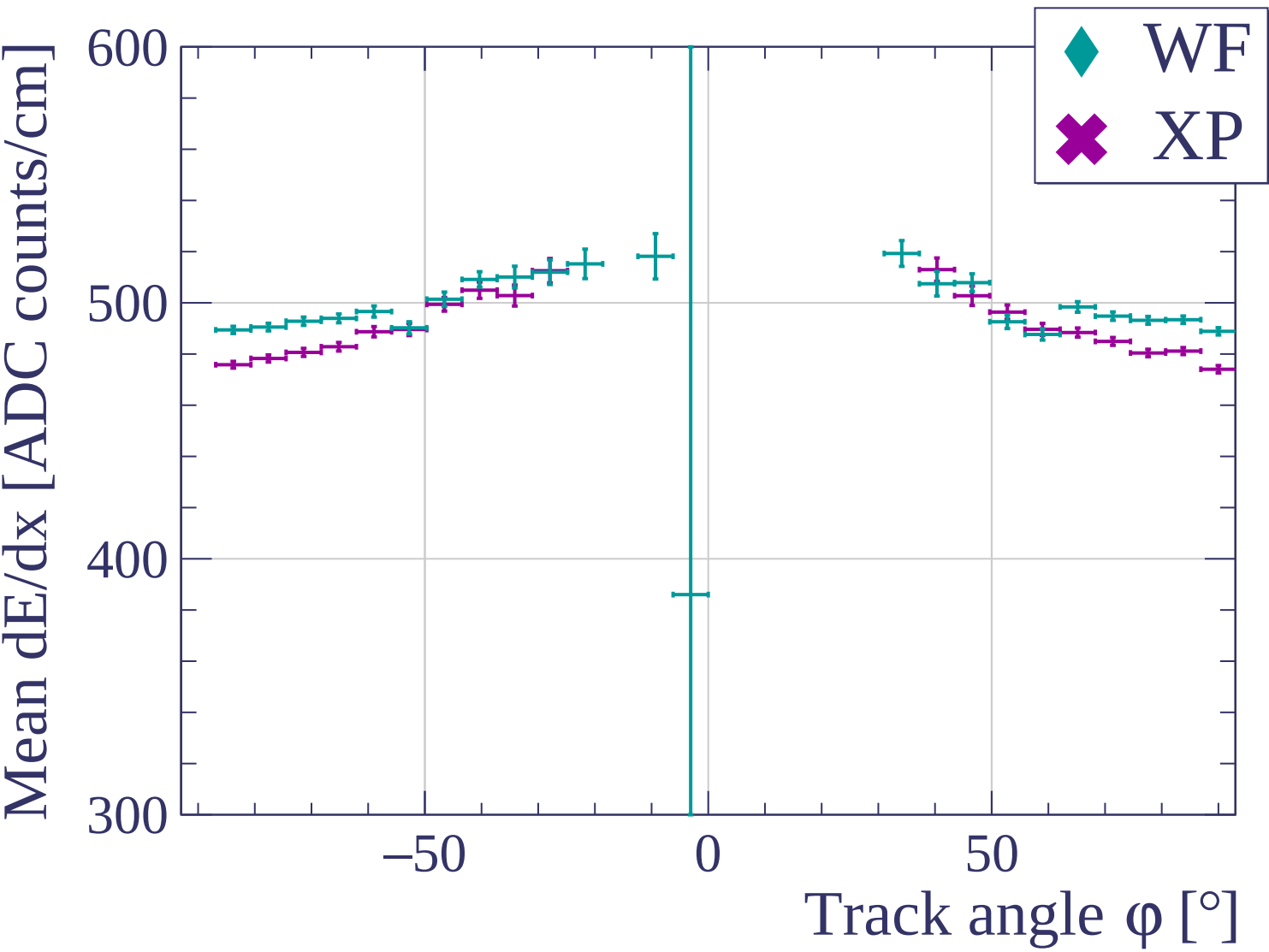


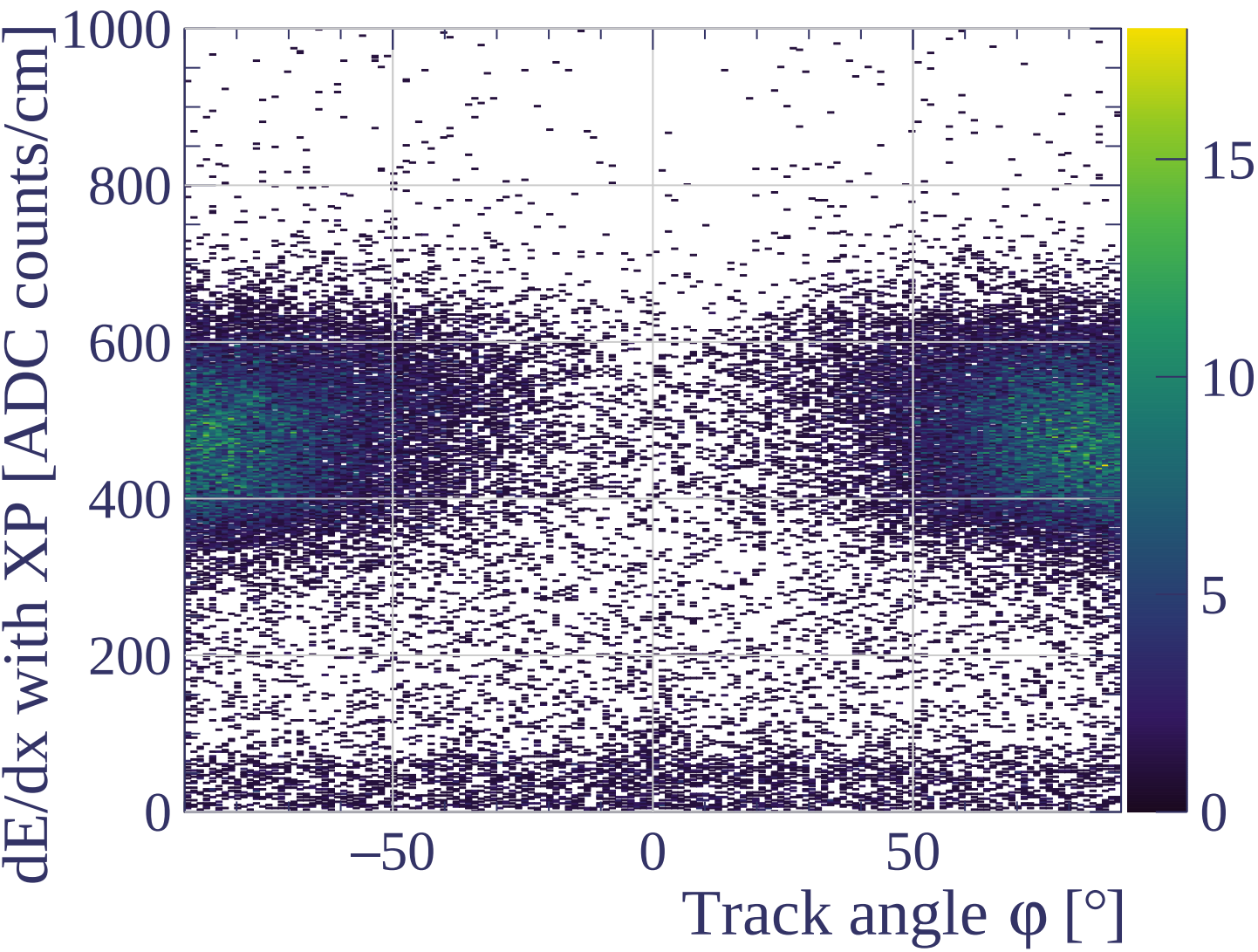


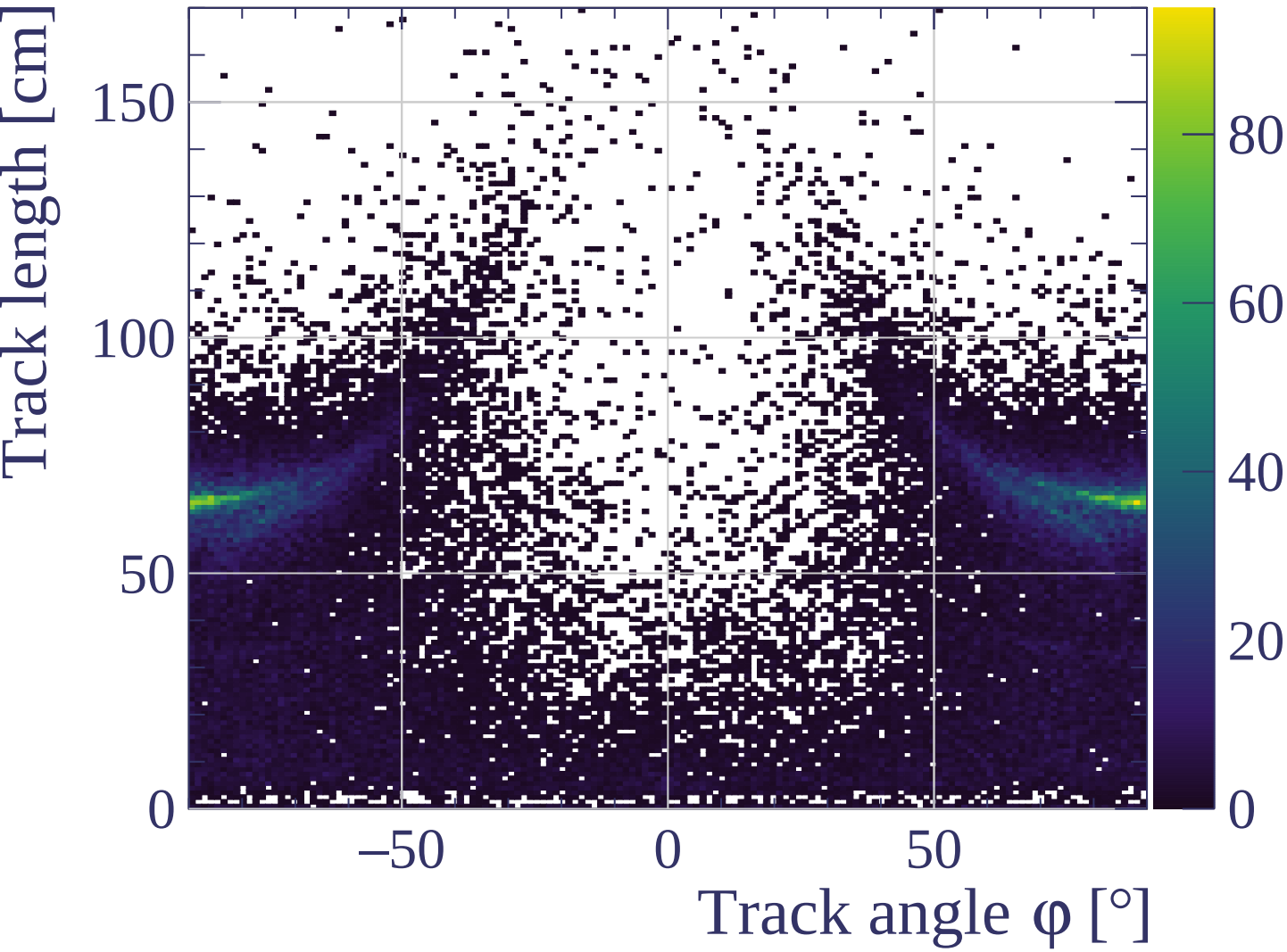


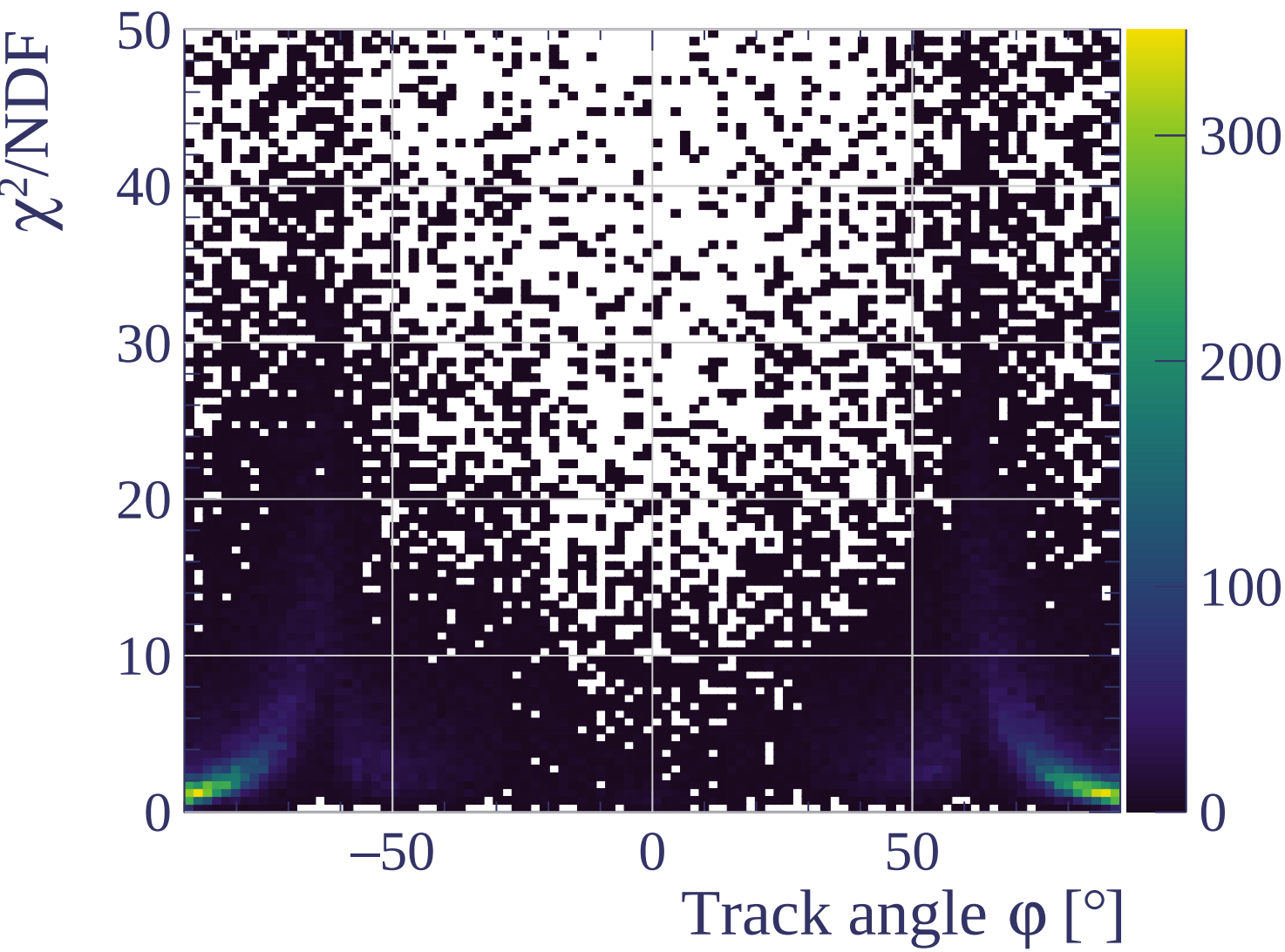


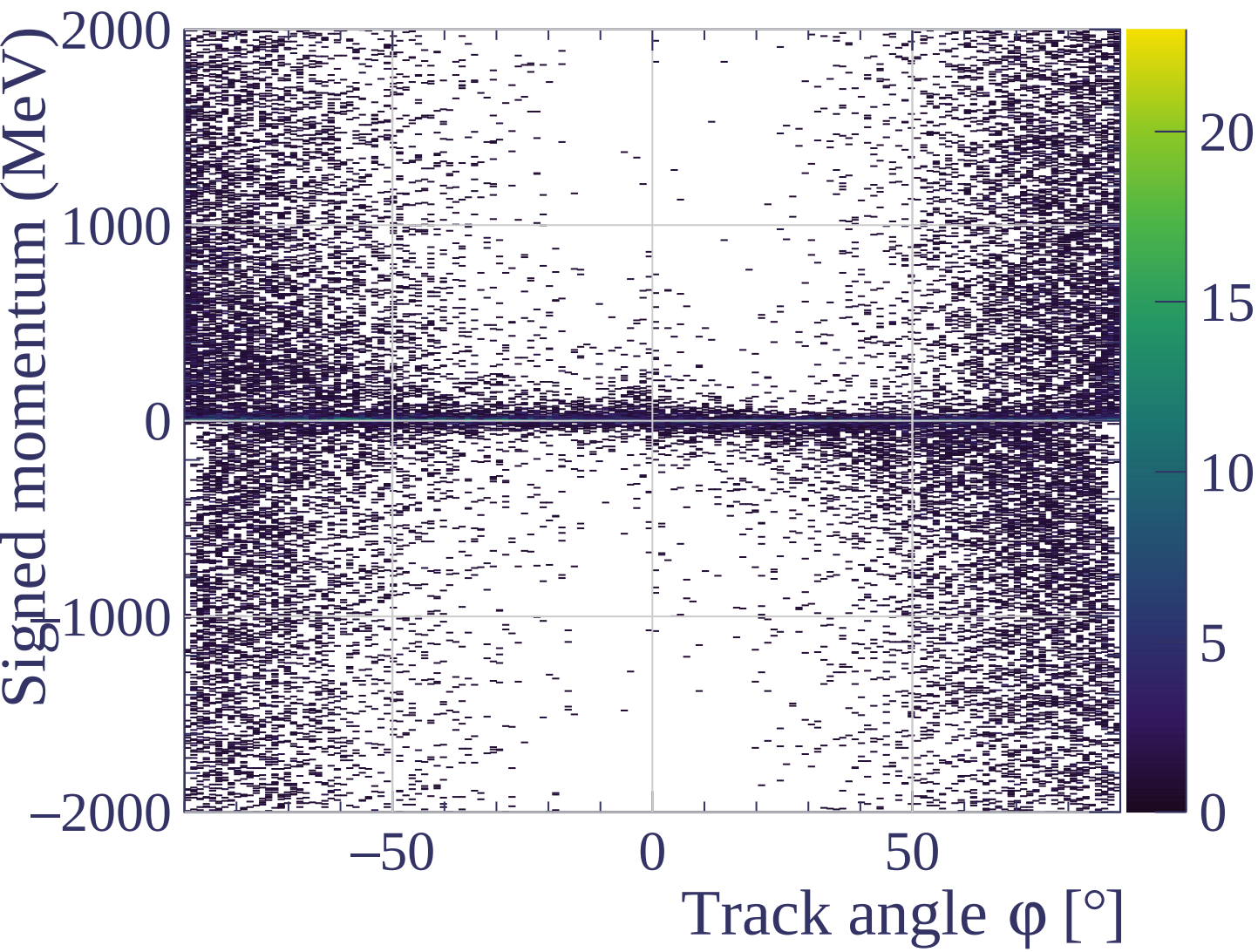




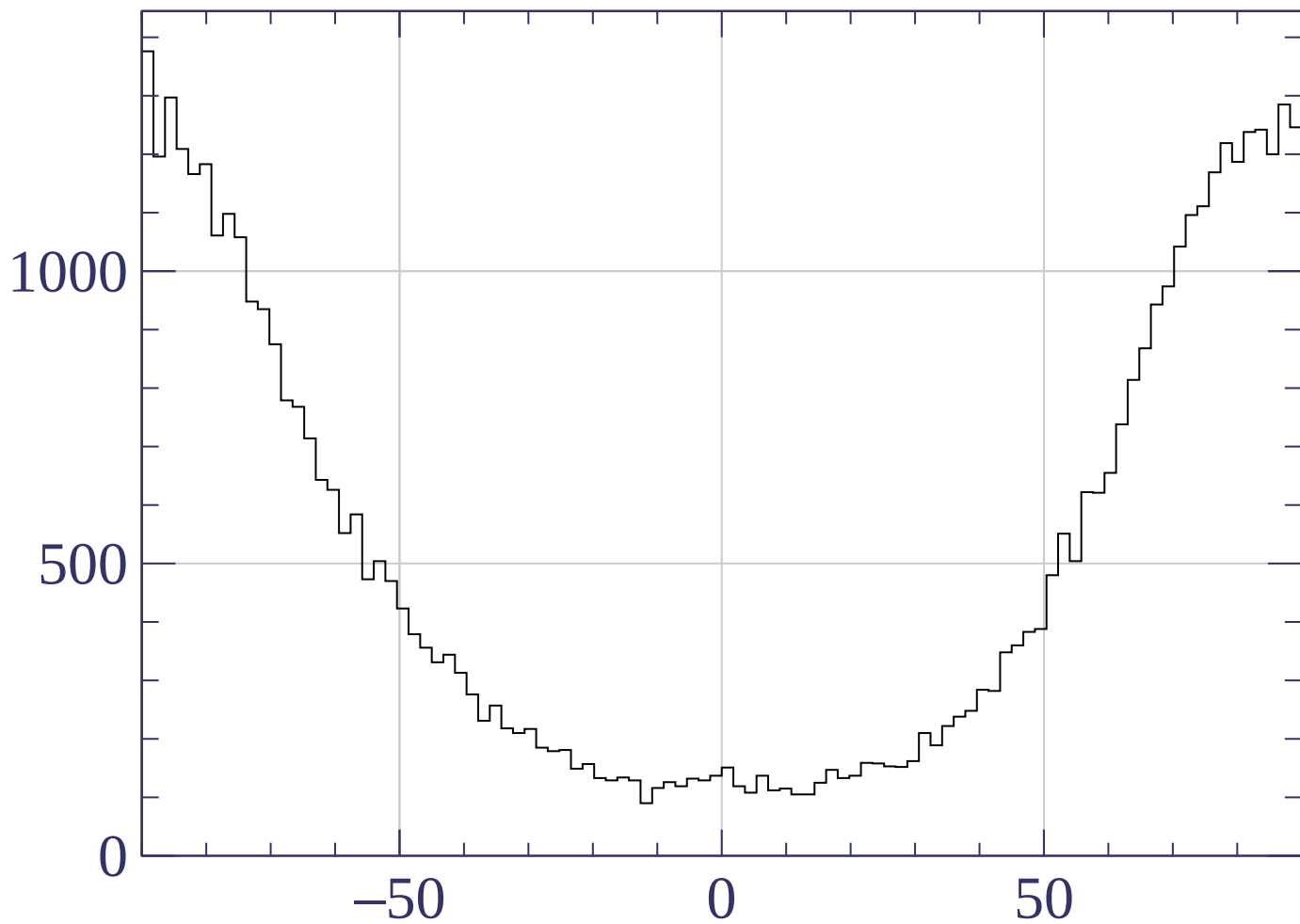




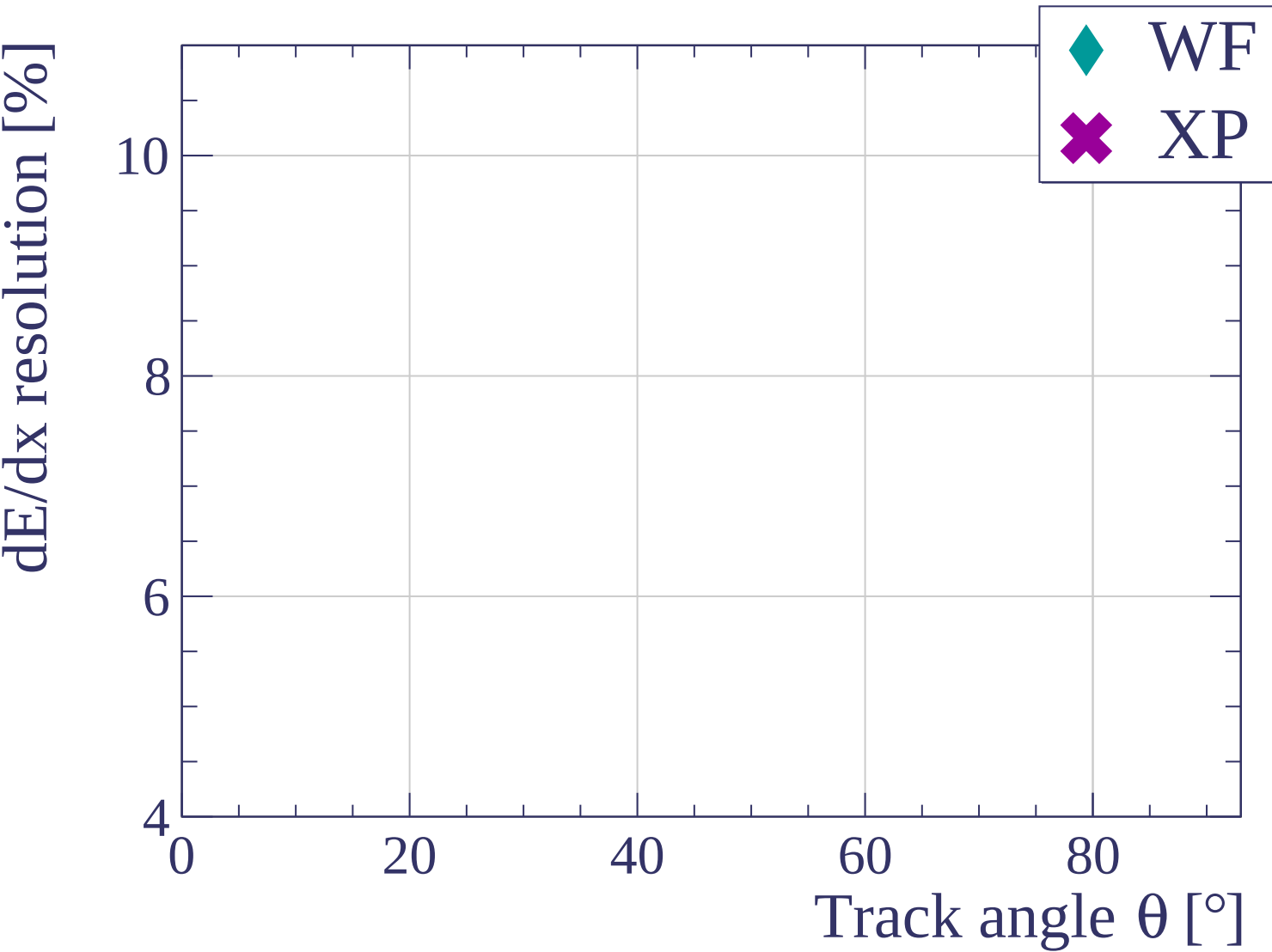


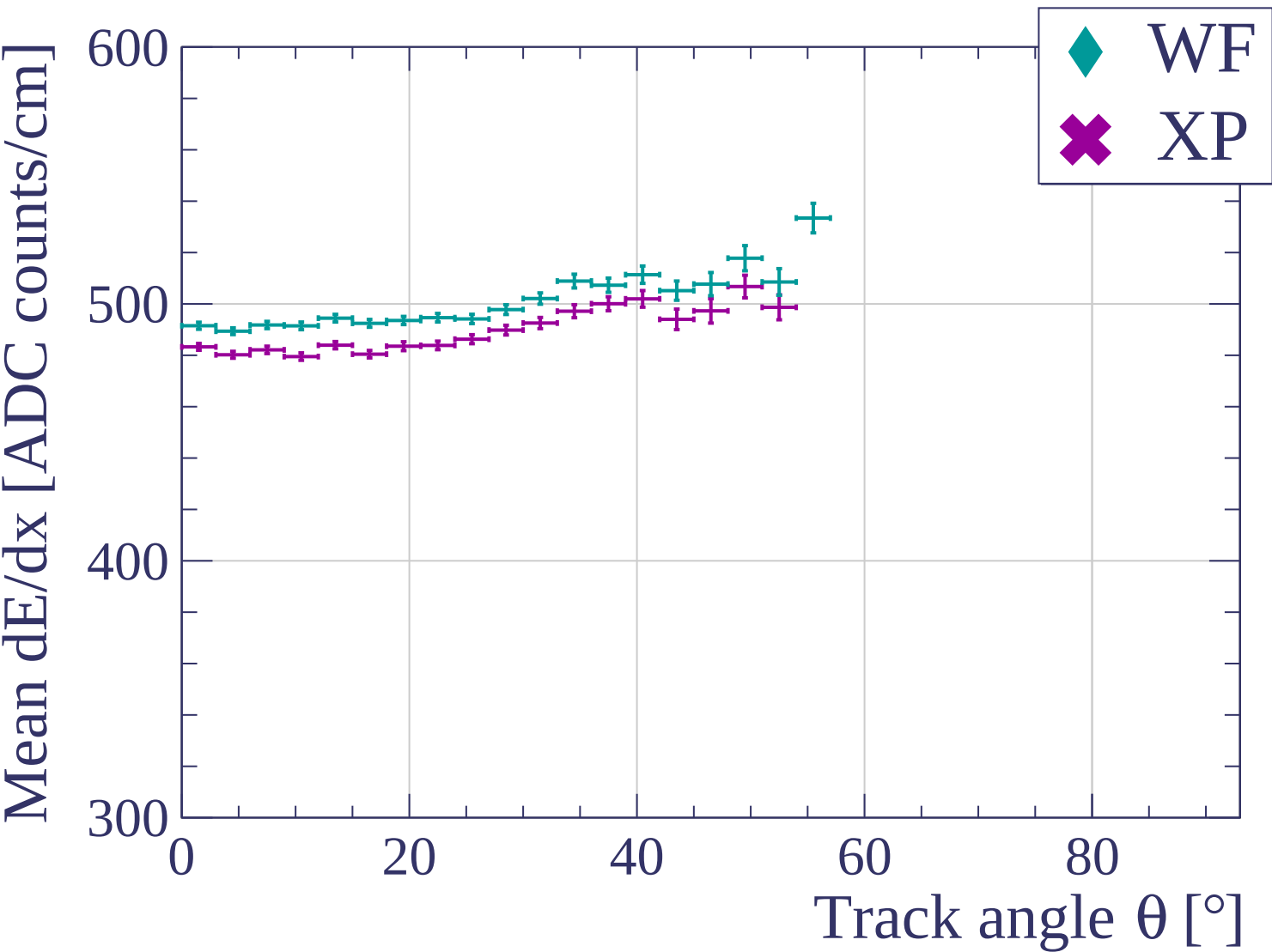


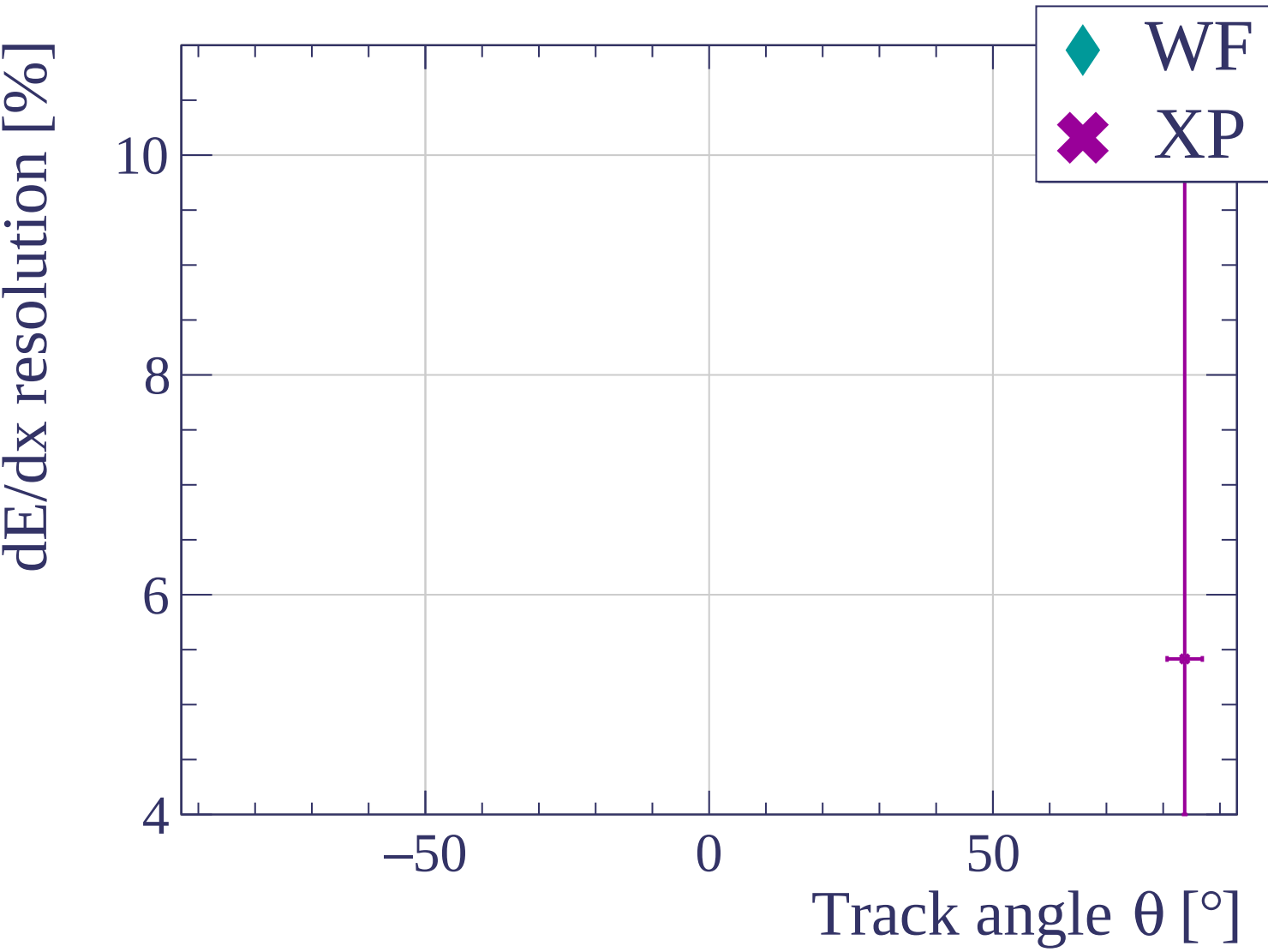
Count

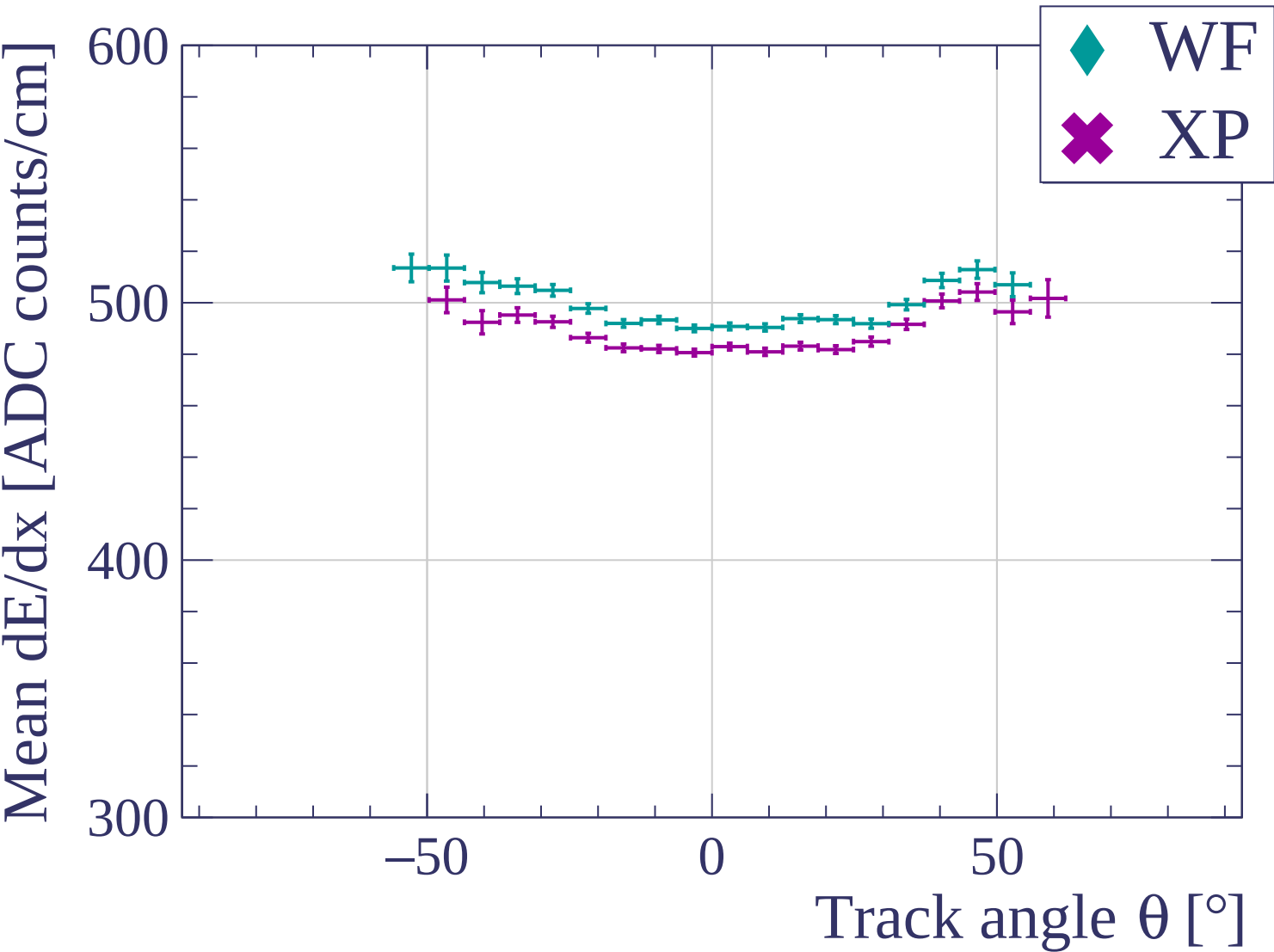


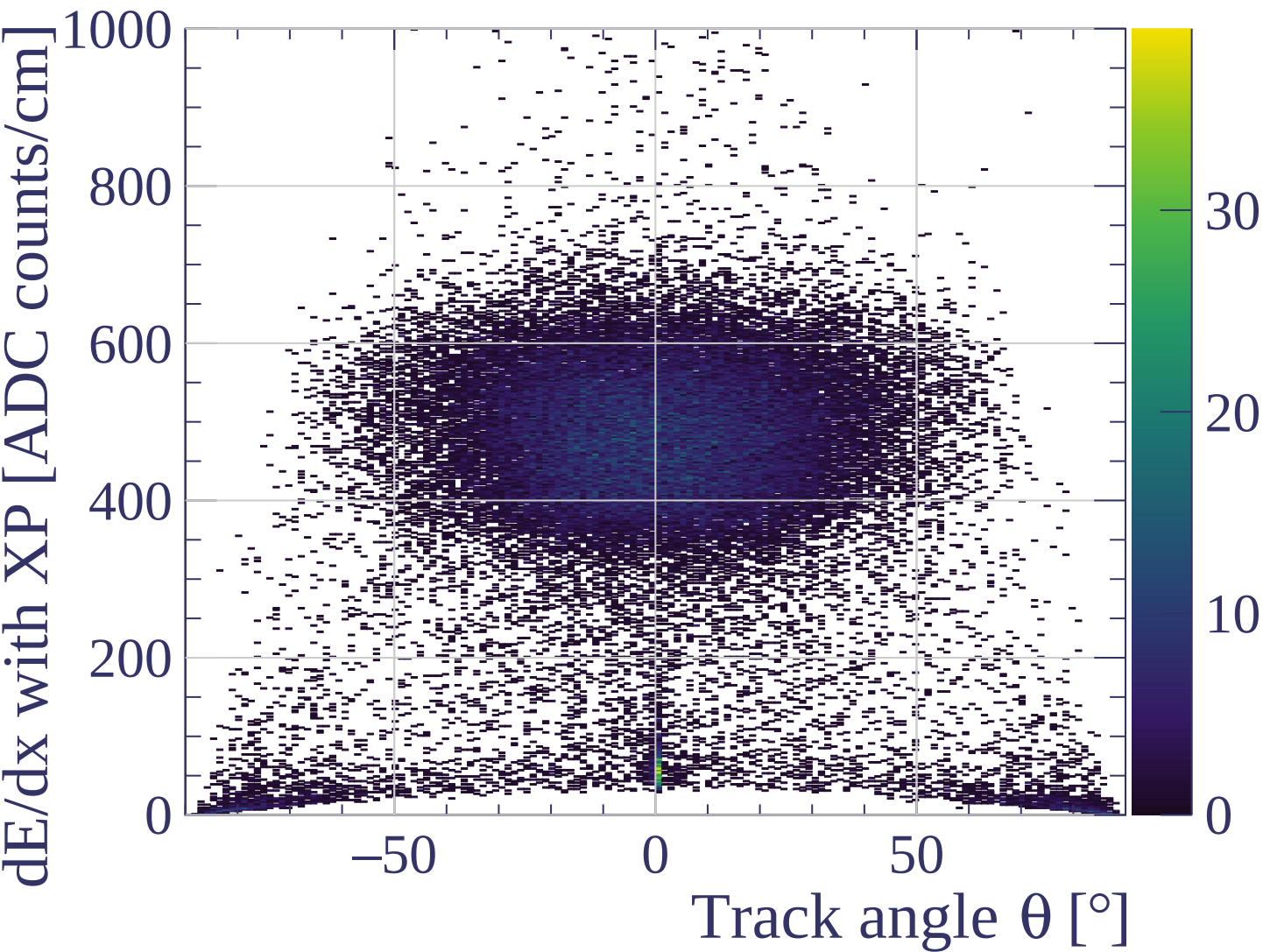
Track angle ϕ [°] angle

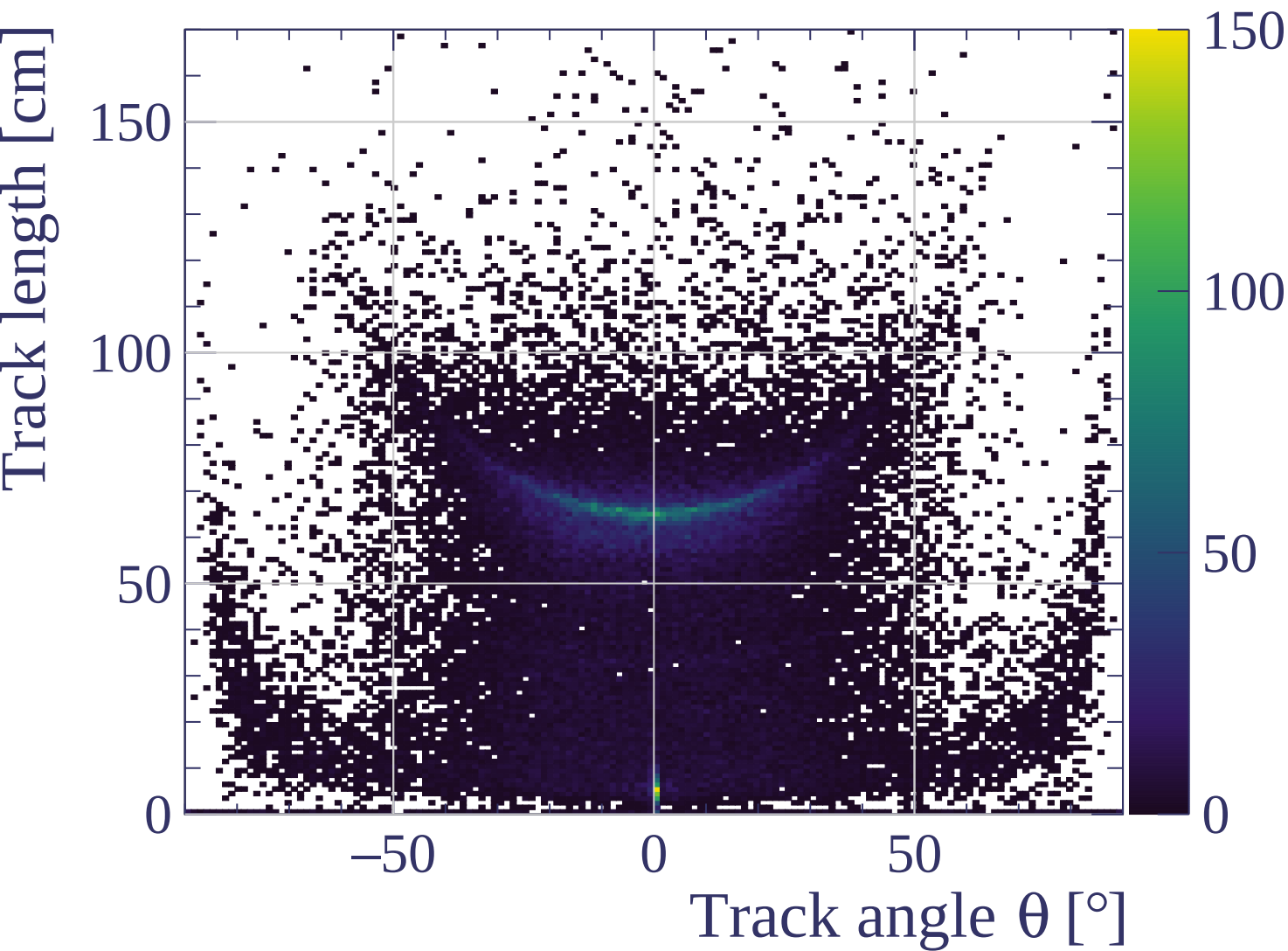


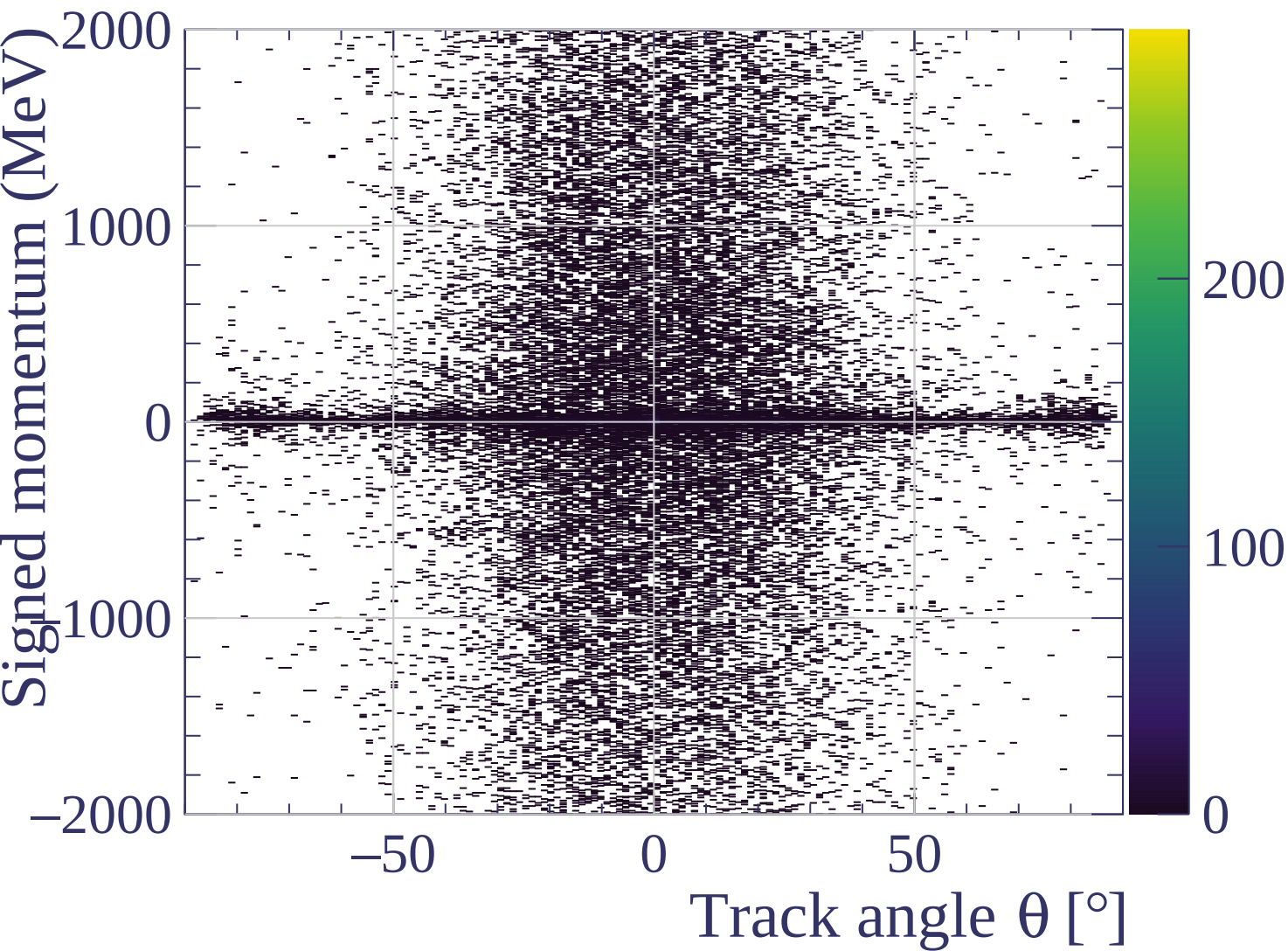


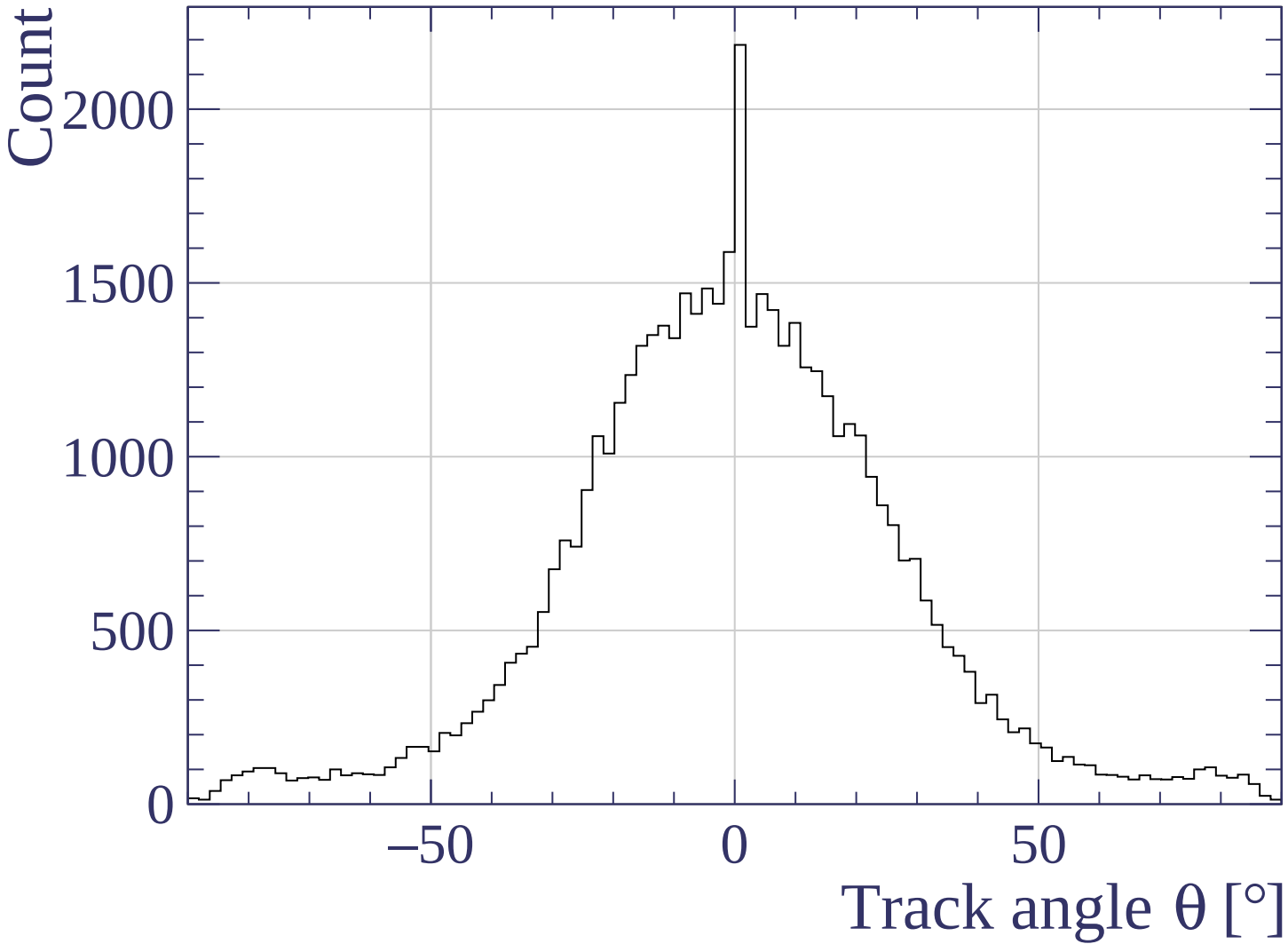




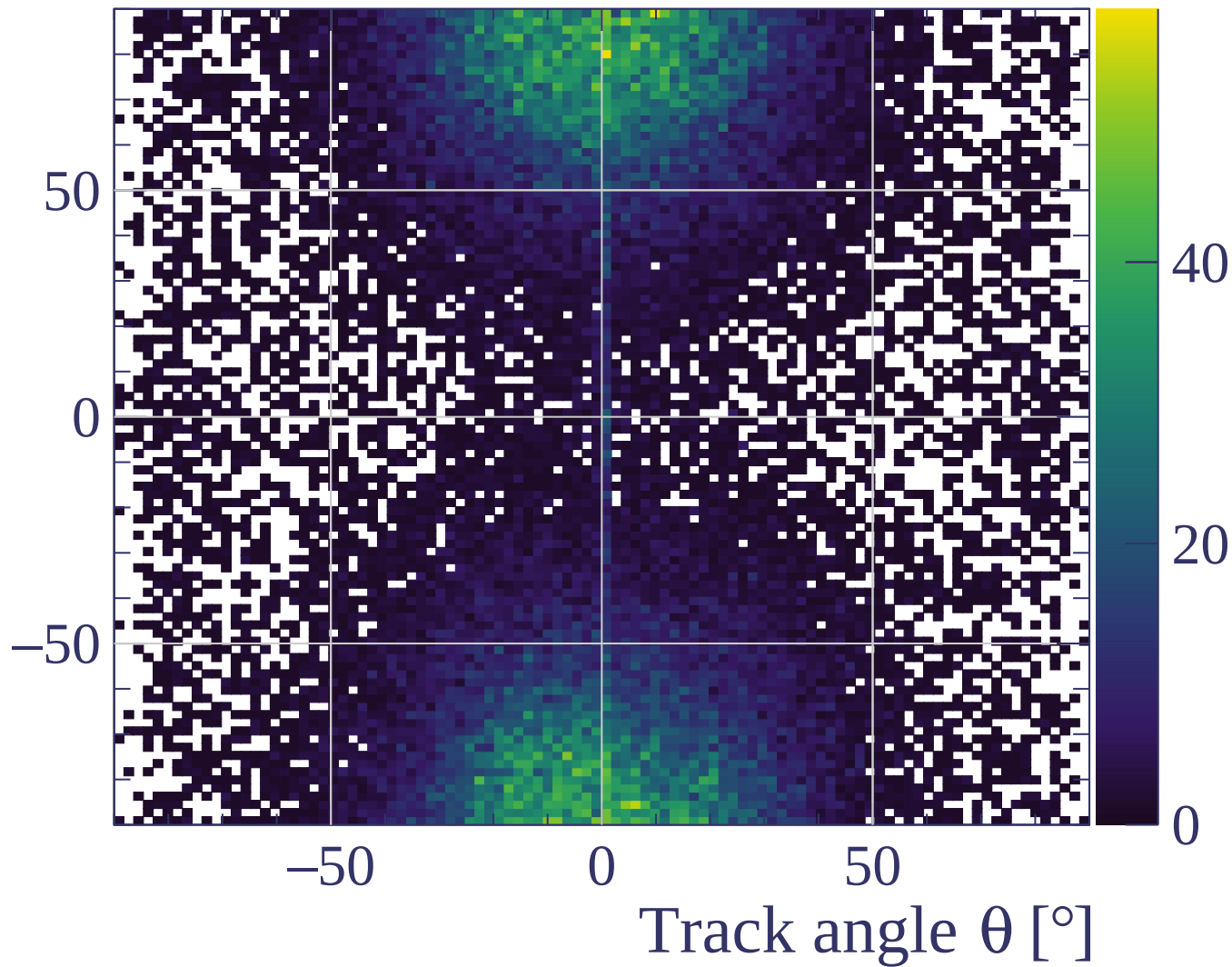




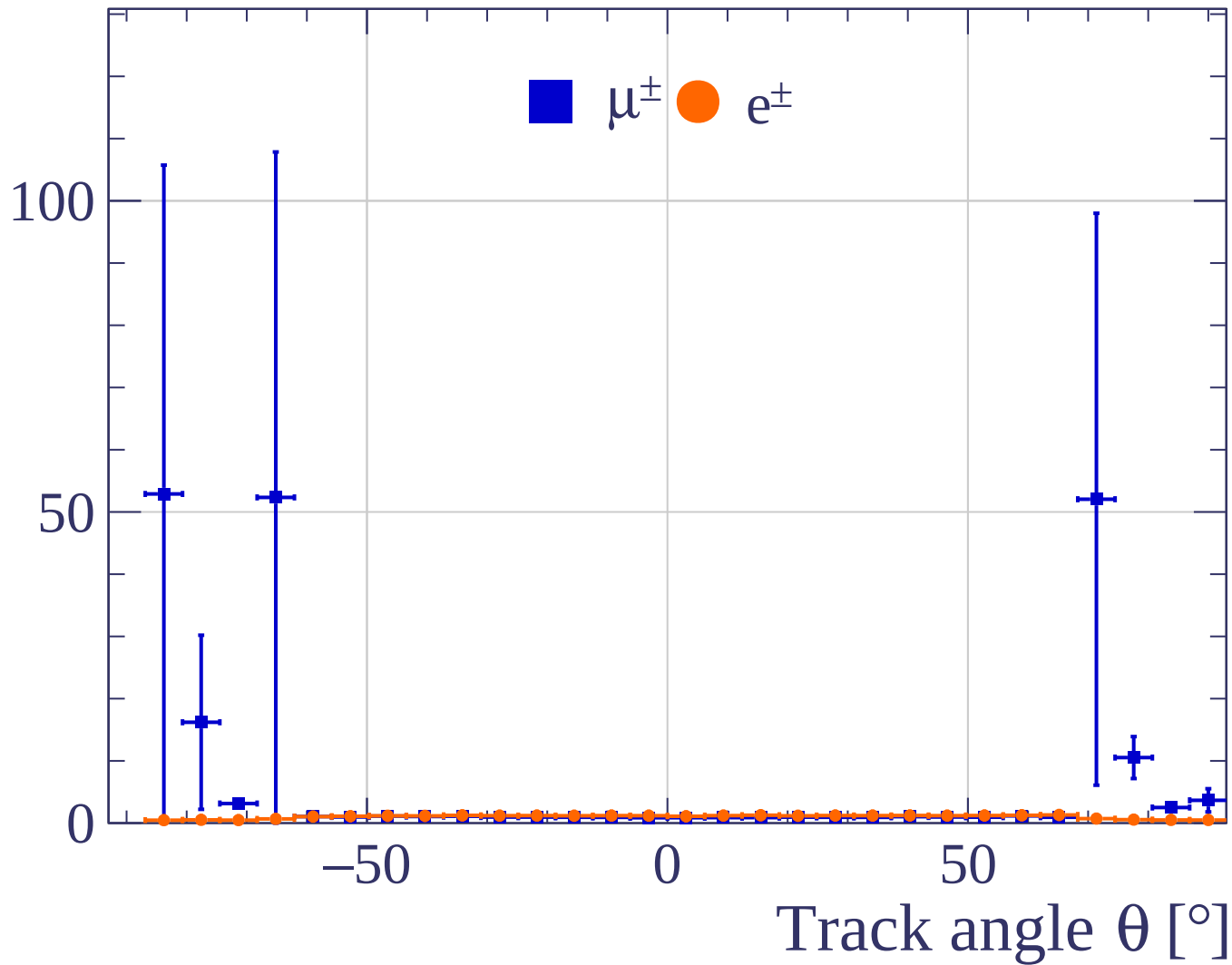




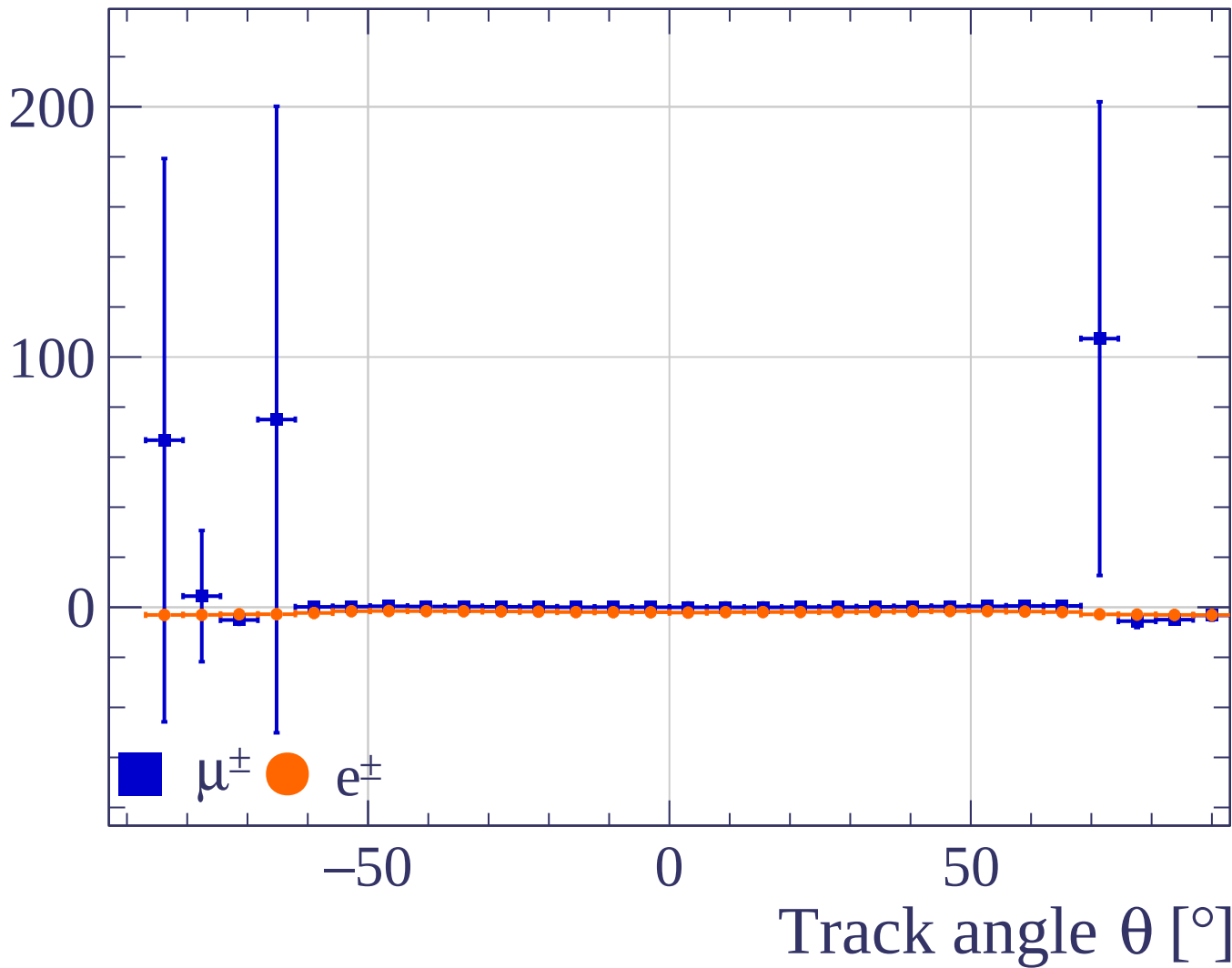
Track angle ϕ [°]



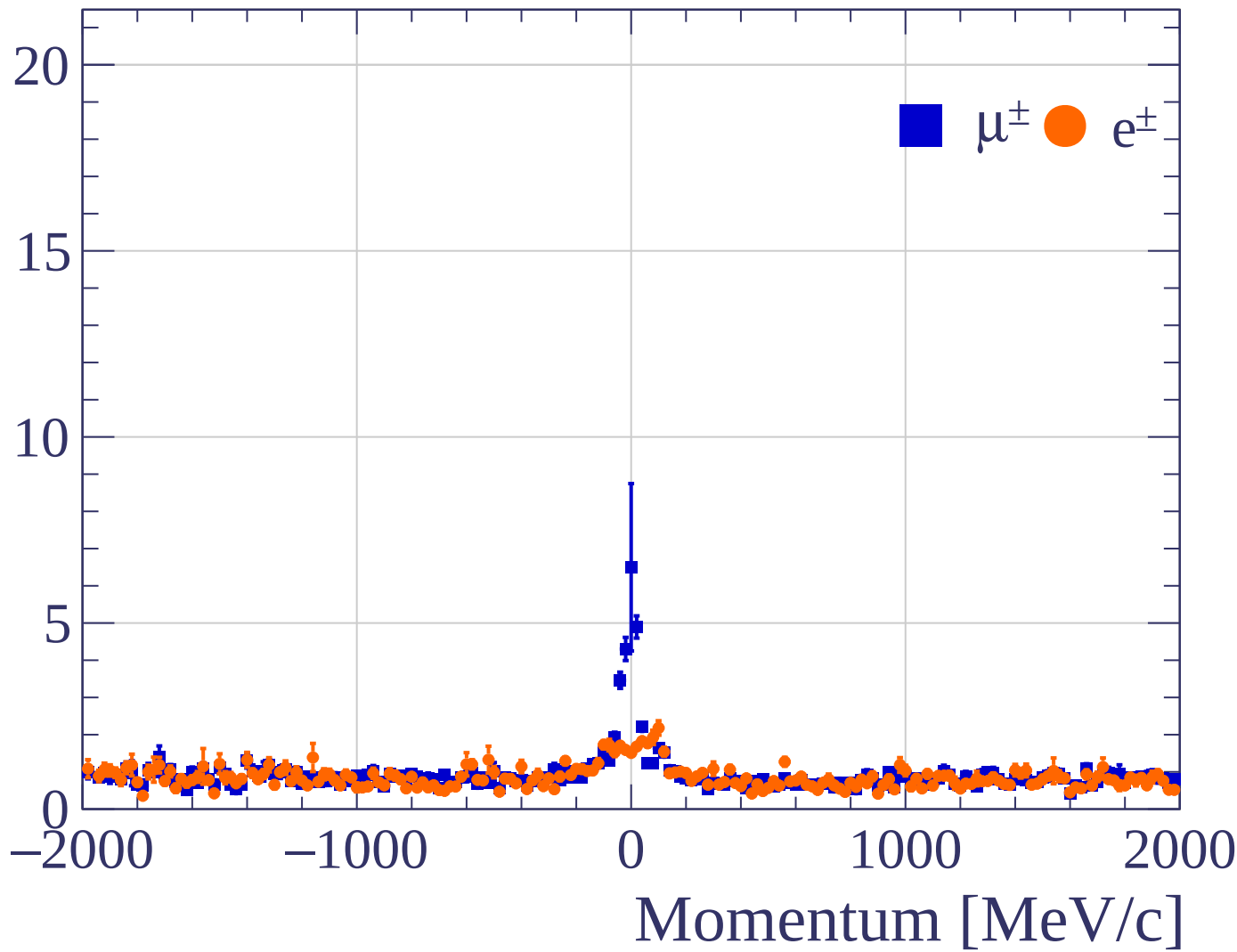
Pulls standard deviation



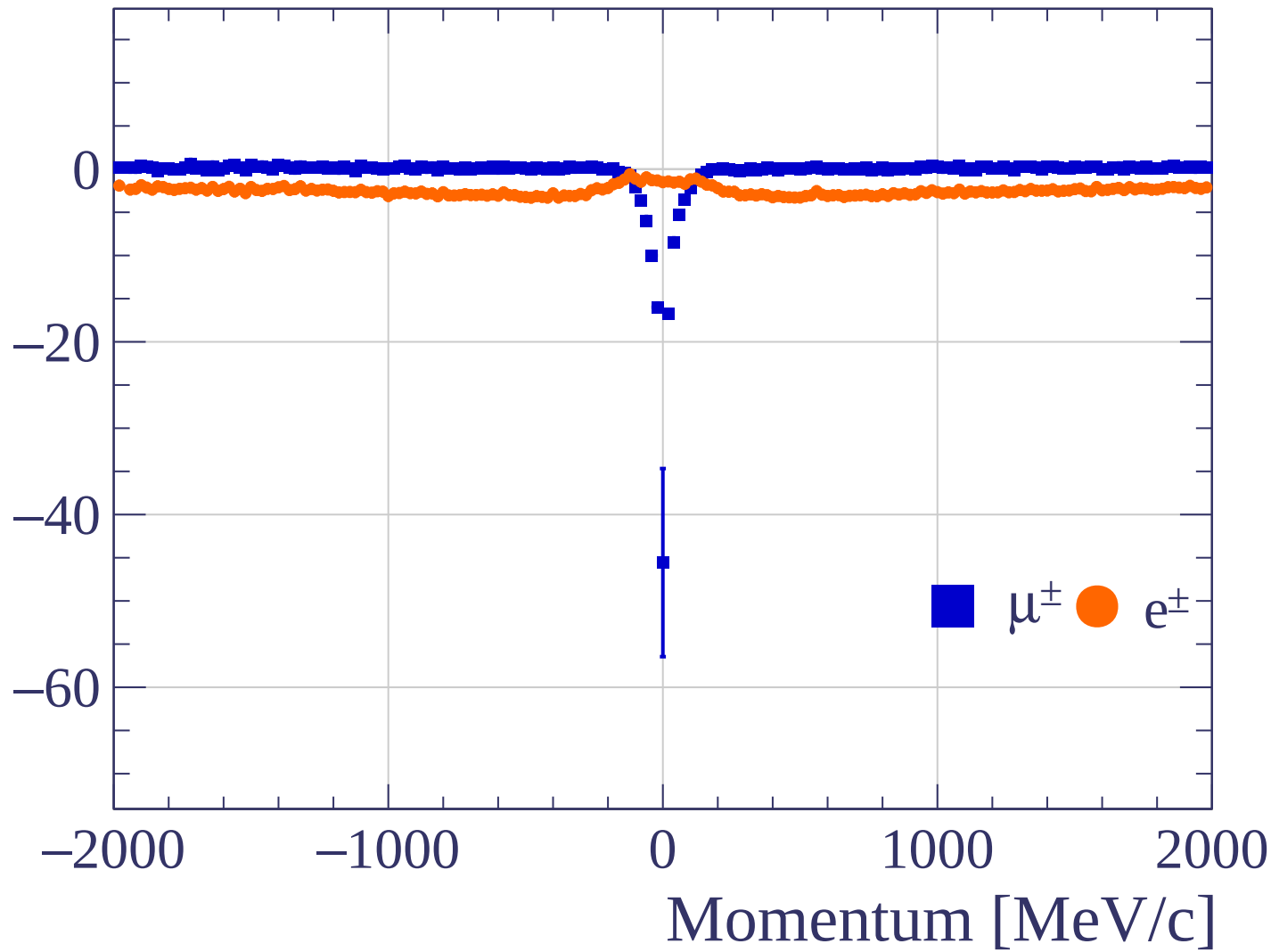
Pulls mean



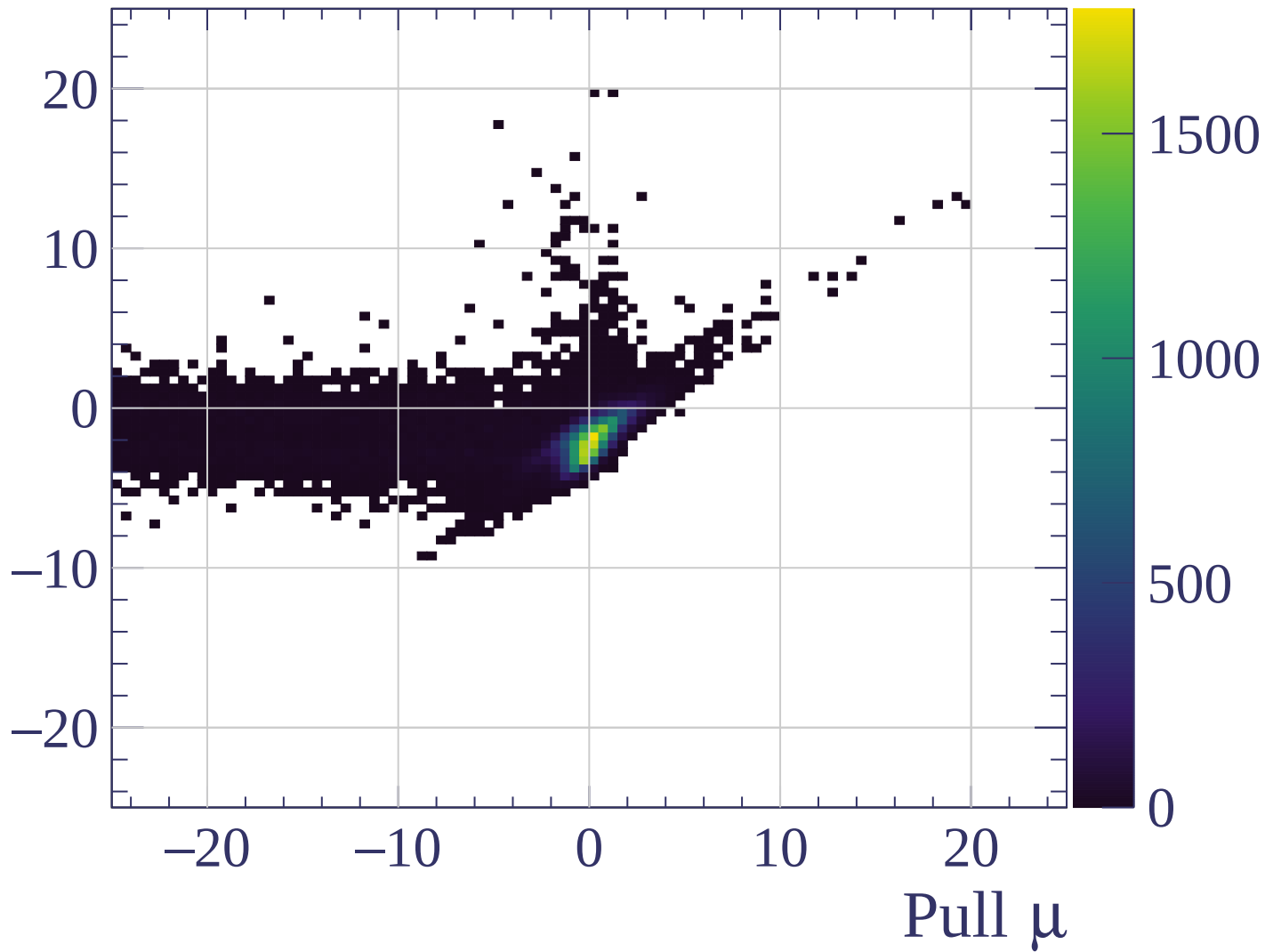
Pulls standard deviation

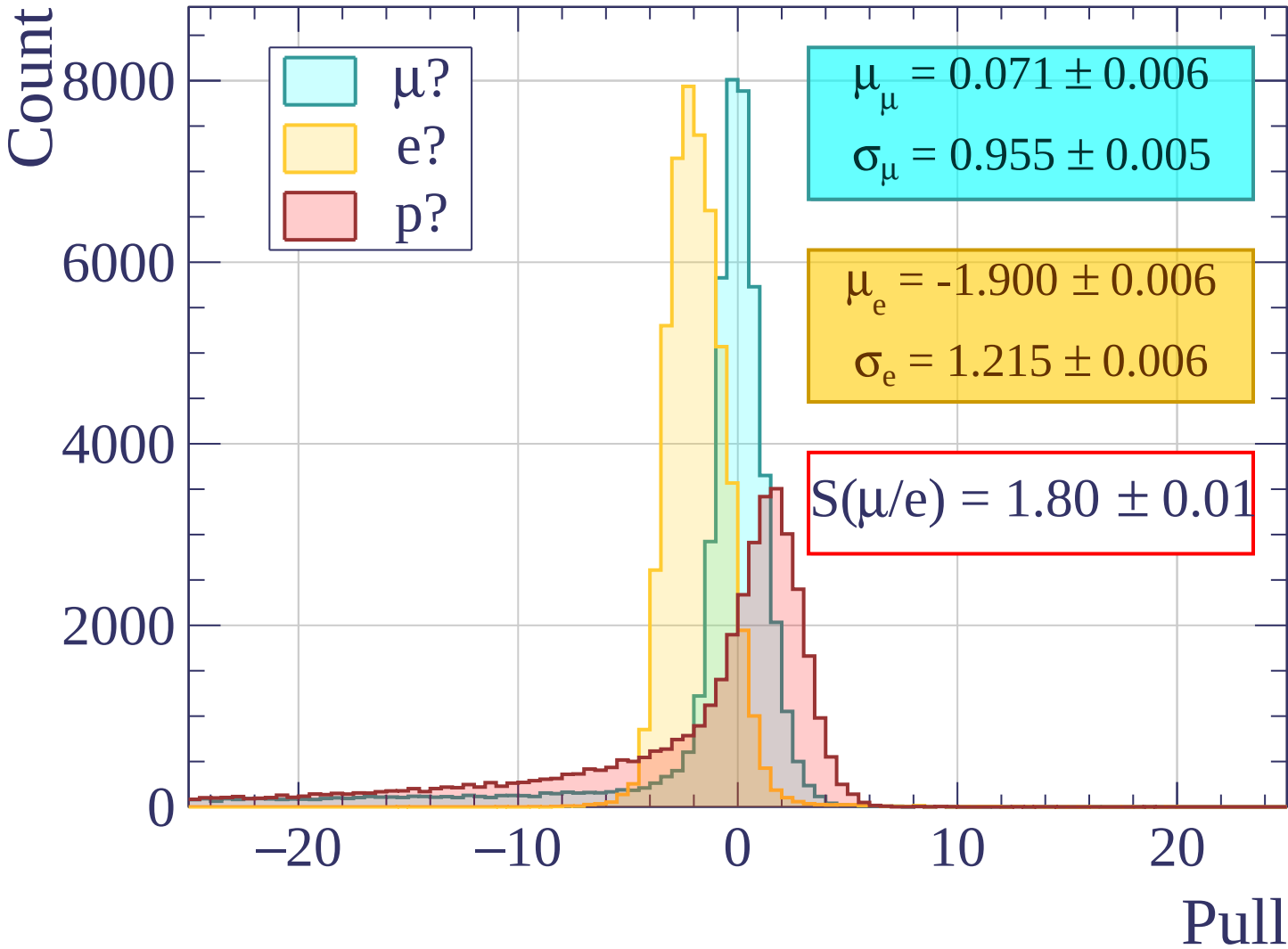


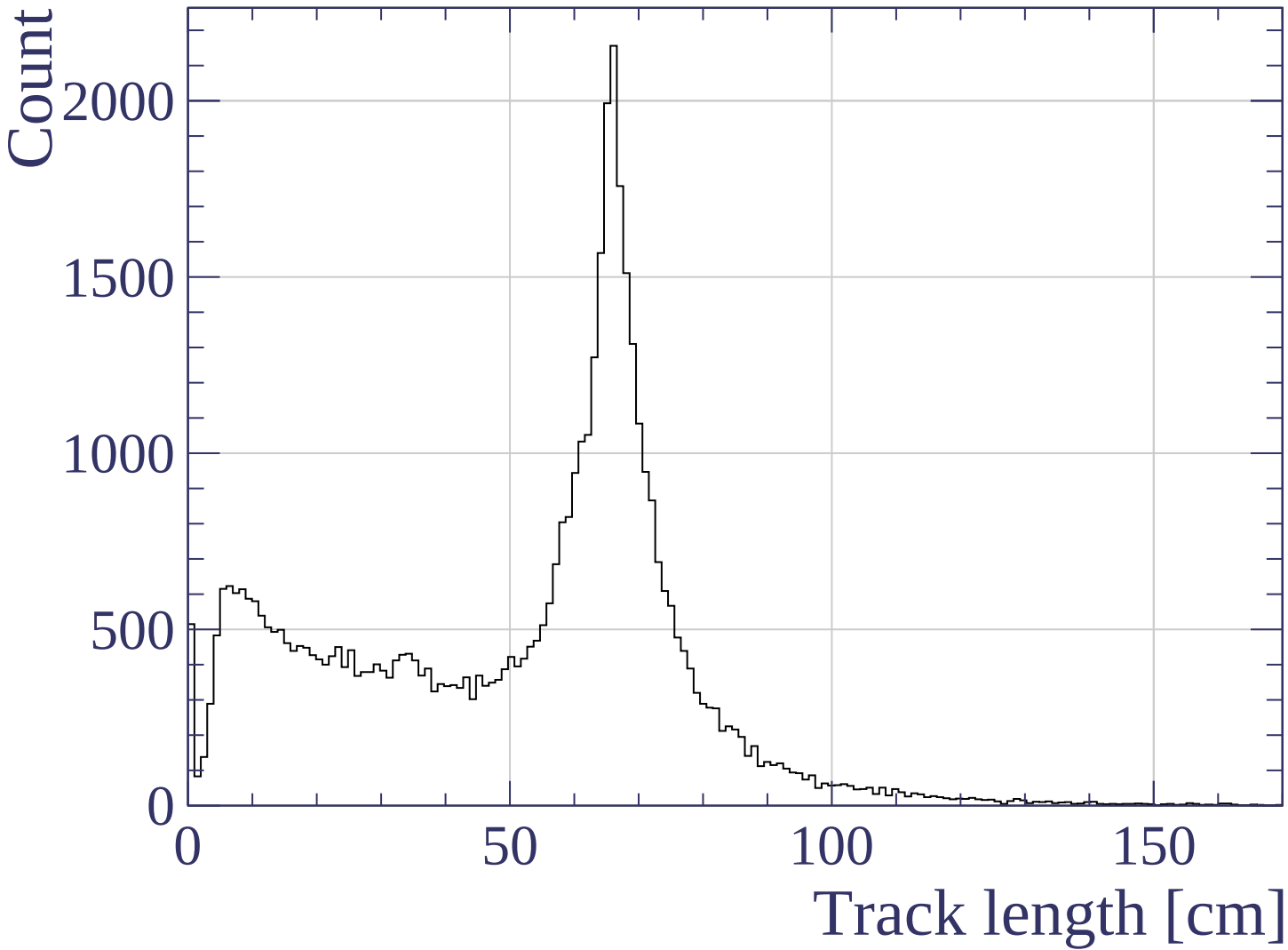
Pulls mean



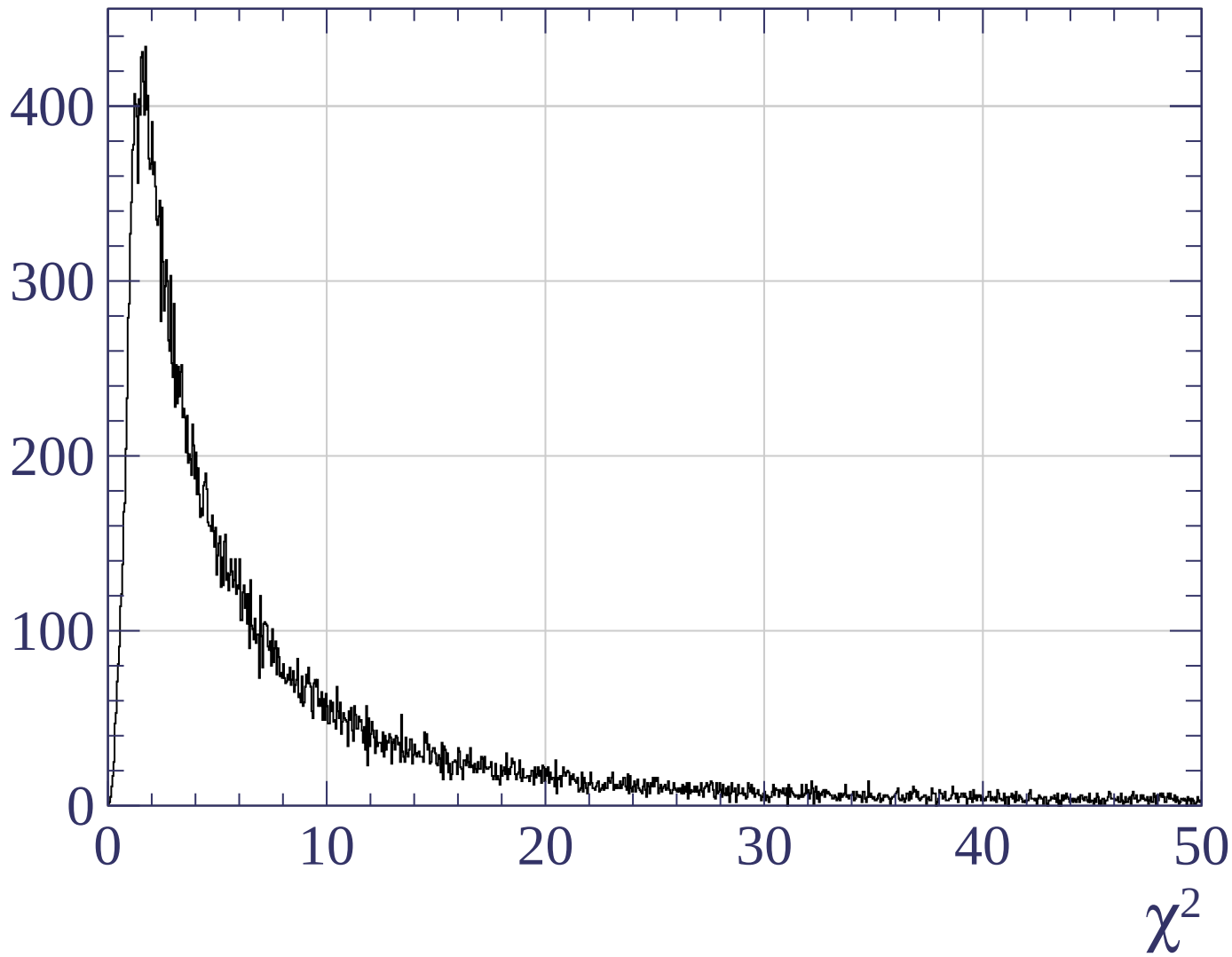
Pull e







Count



Count

1000

500

0

-1000

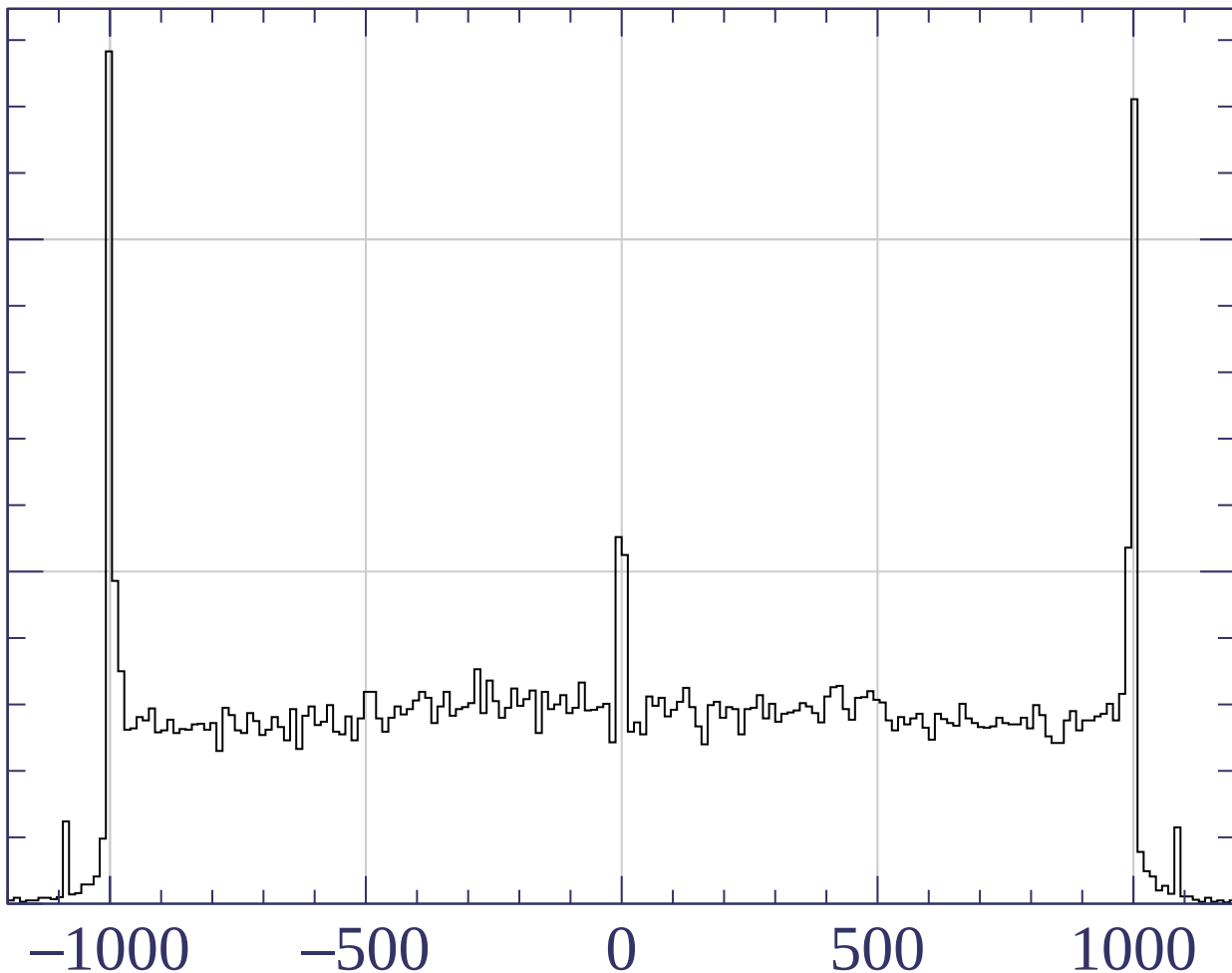
-500

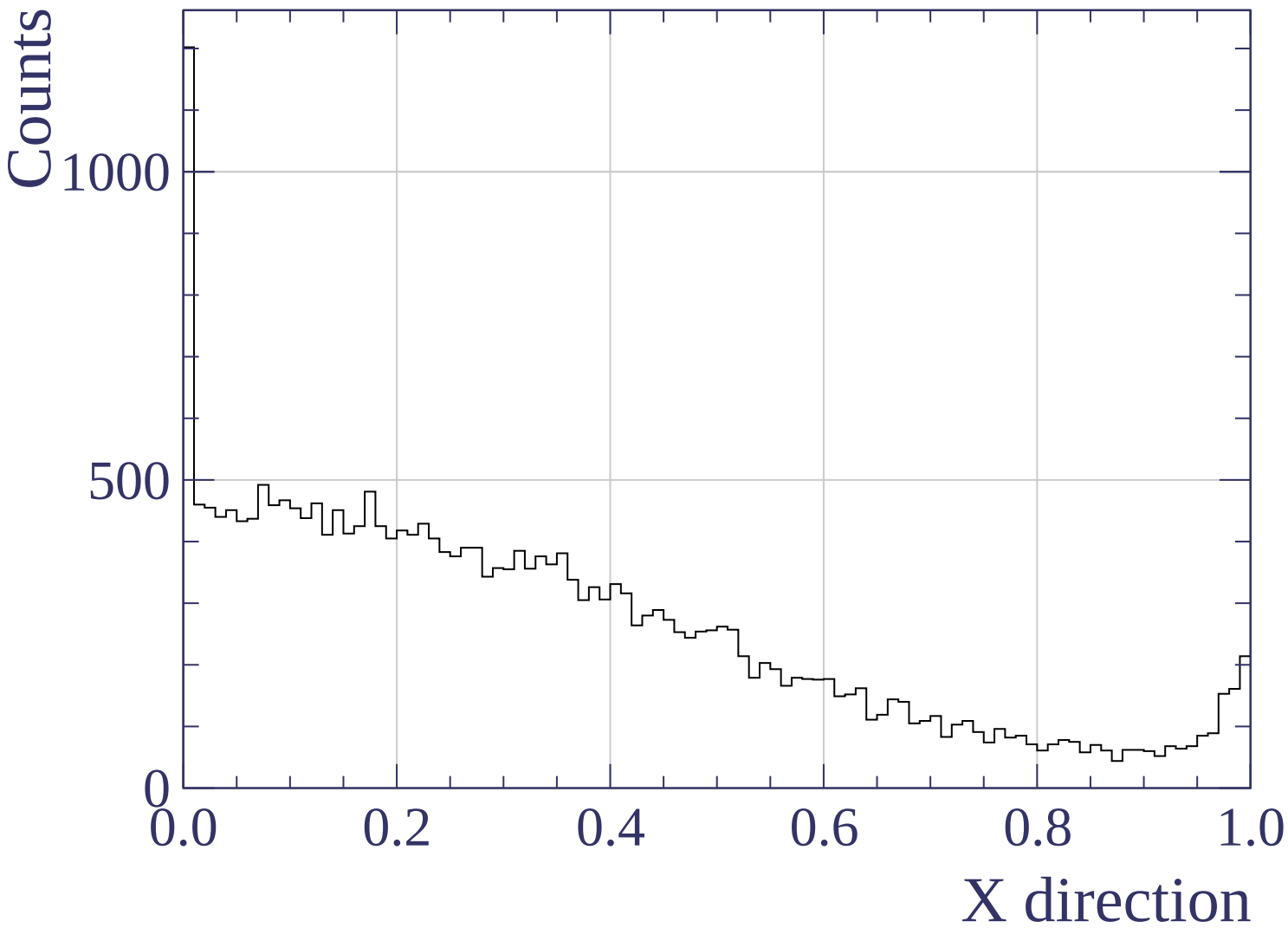
0

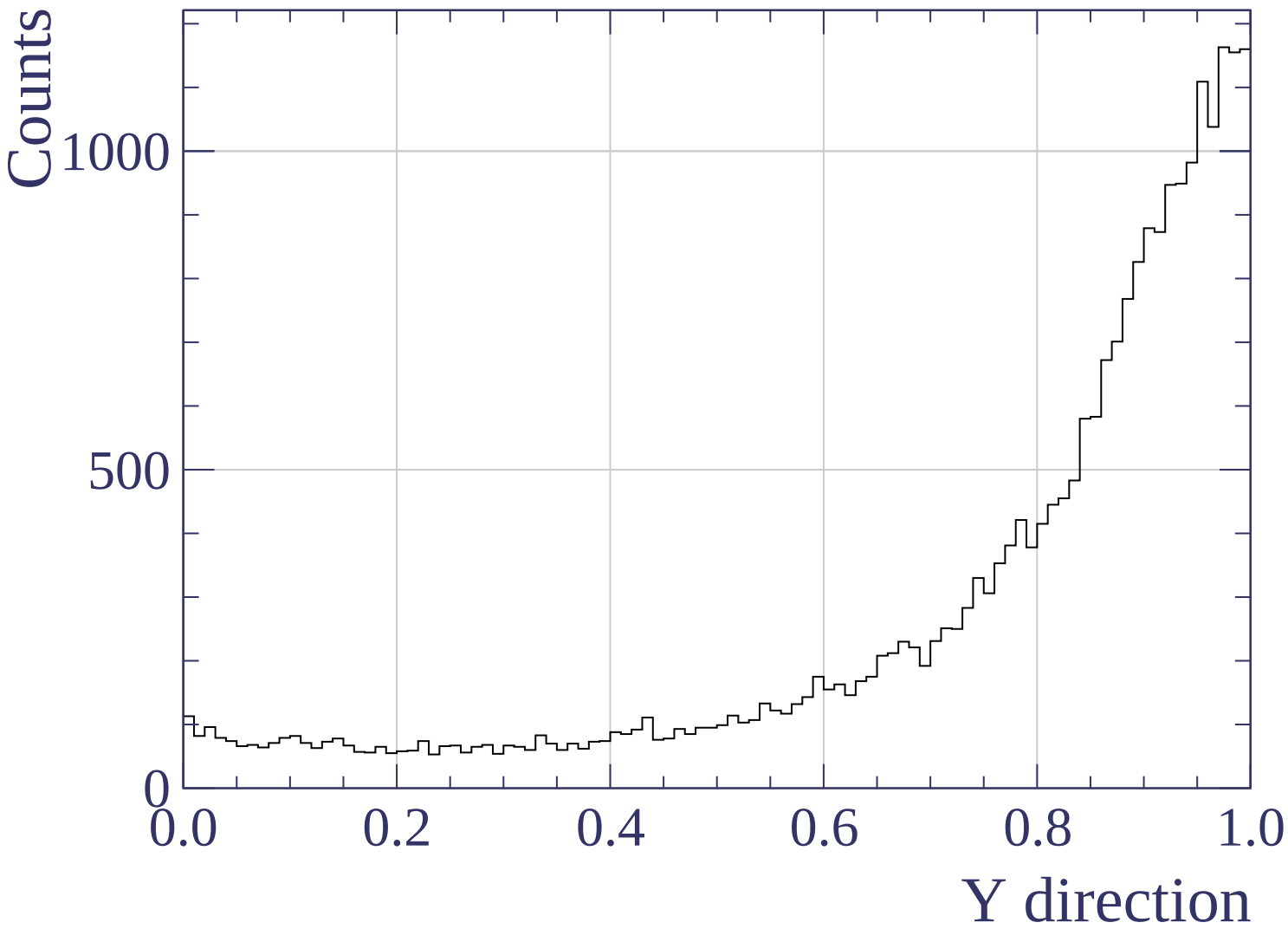
500

1000

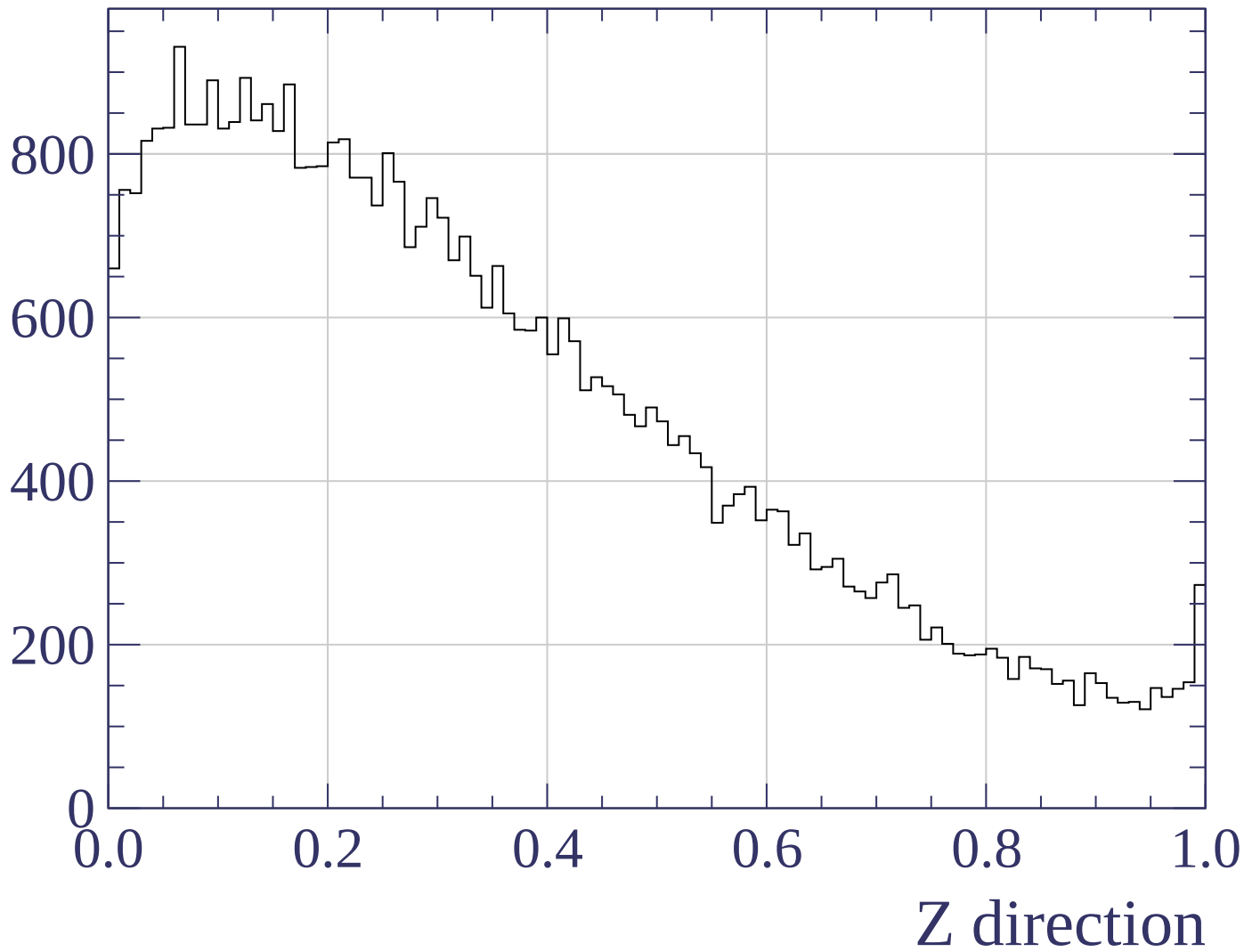
Track X position [mm]

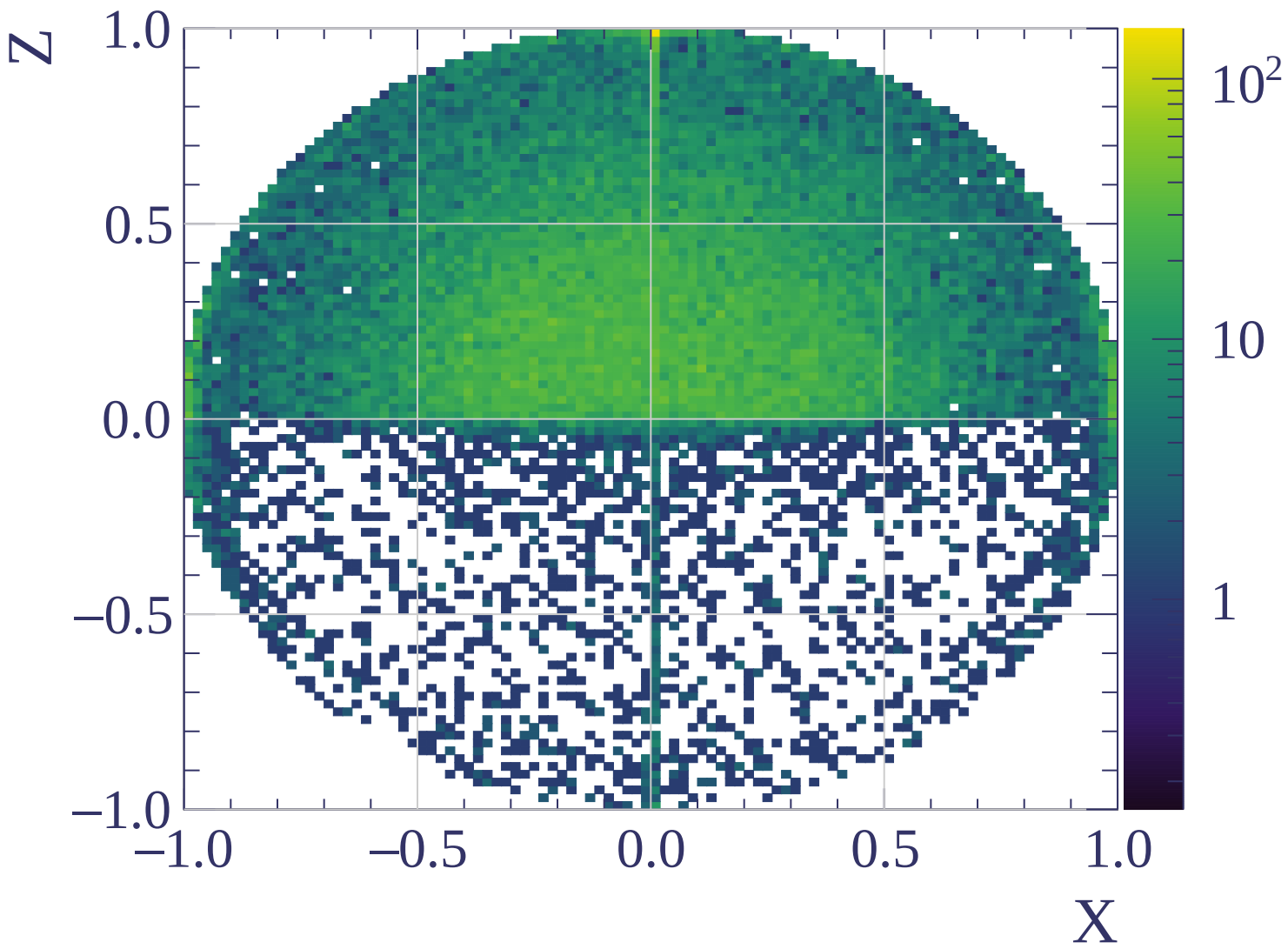


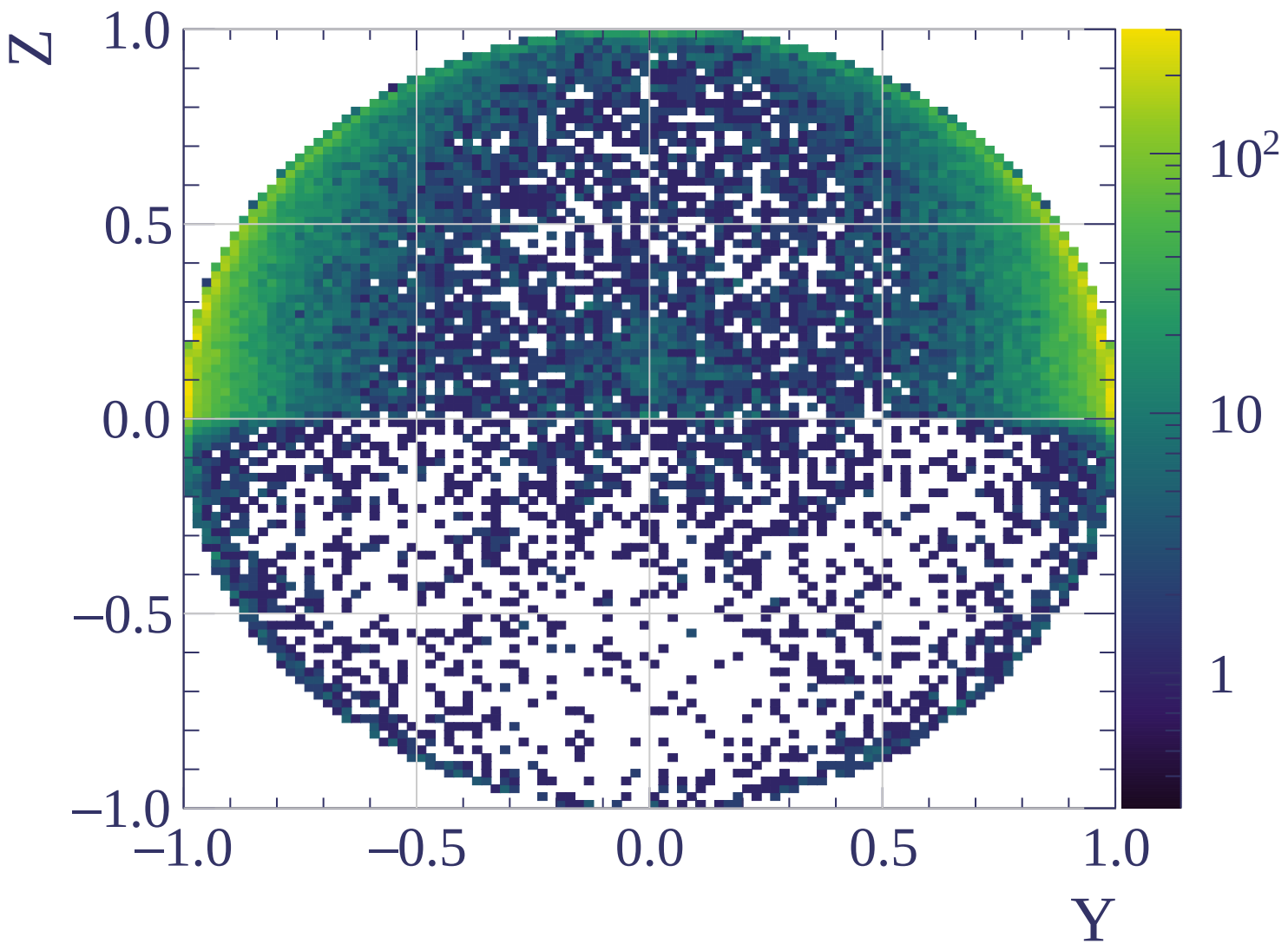


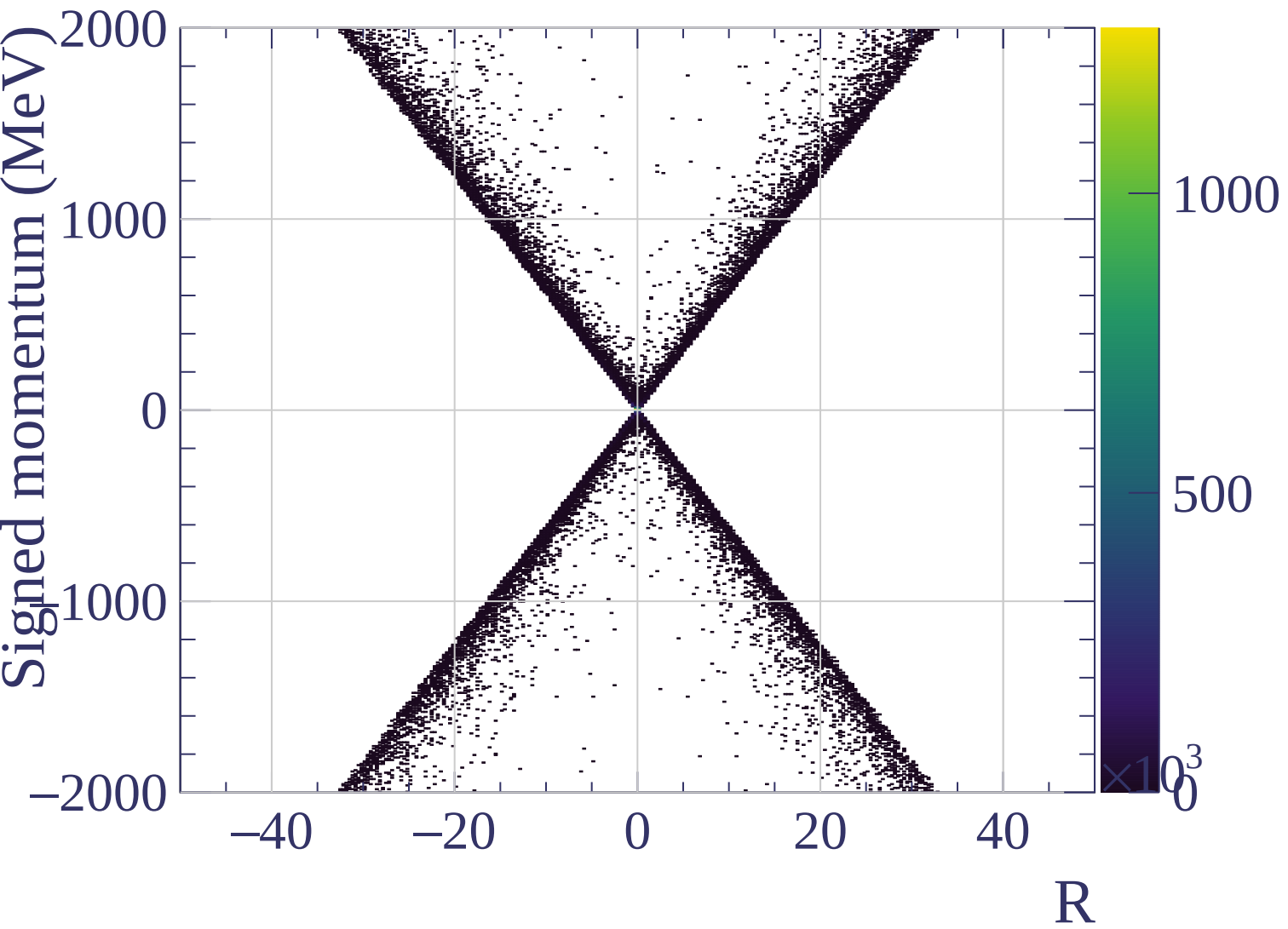


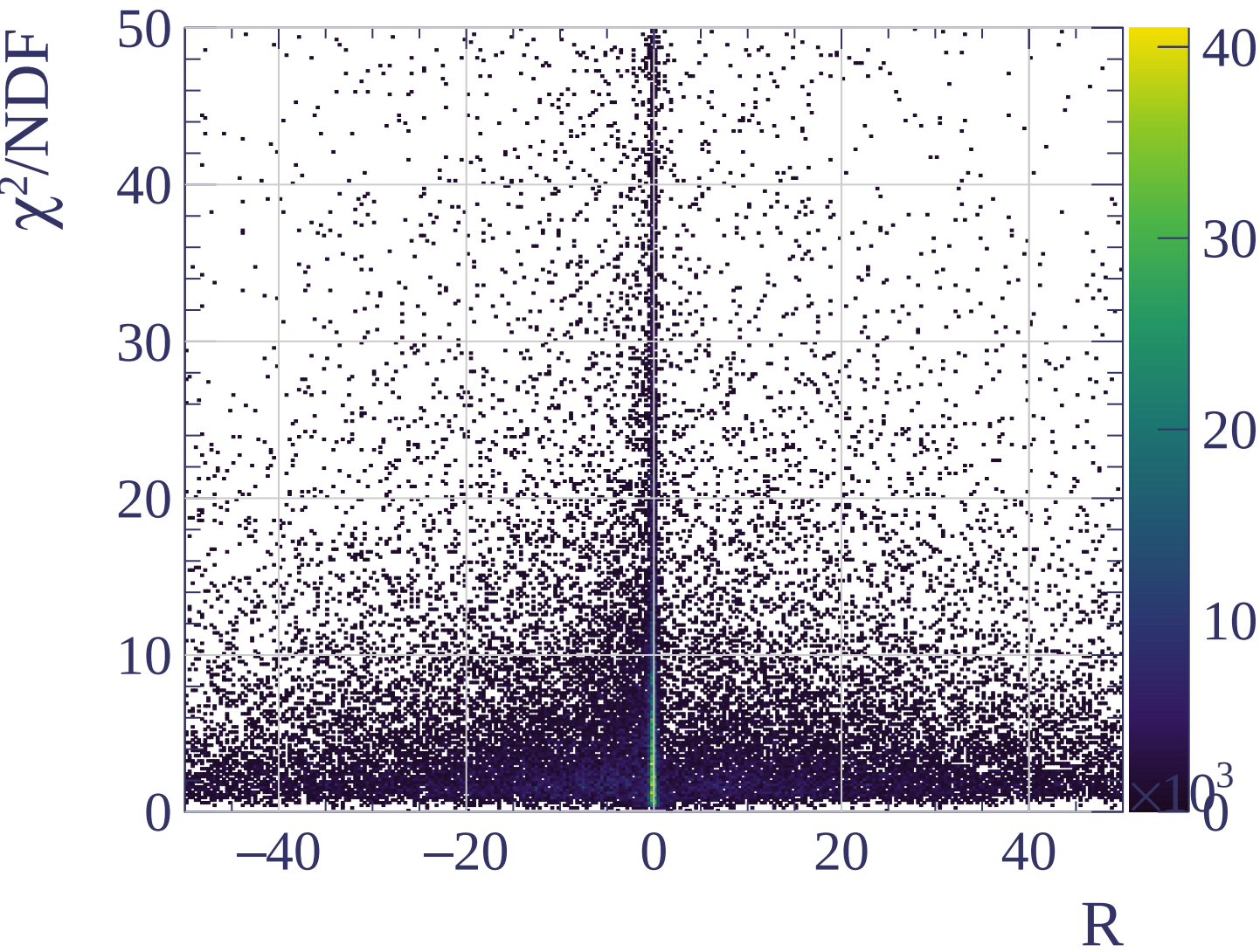
Counts



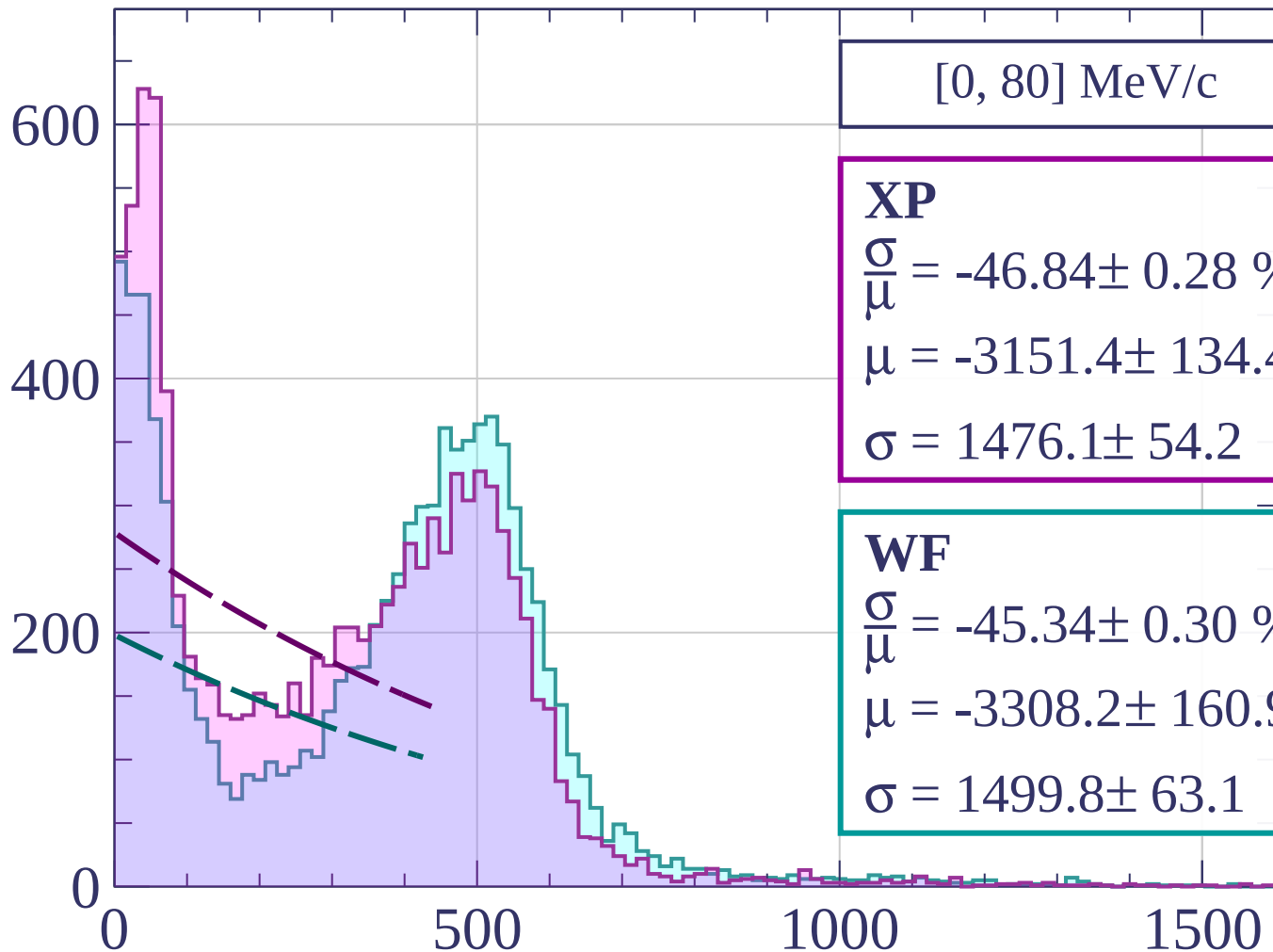








Count



[0, 80] MeV/c

XP

$$\frac{\sigma}{\mu} = -46.84 \pm 0.28 \%$$

$$\mu = -3151.4 \pm 134.4$$

$$\sigma = 1476.1 \pm 54.2$$

WF

$$\frac{\sigma}{\mu} = -45.34 \pm 0.30 \%$$

$$\mu = -3308.2 \pm 160.9$$

$$\sigma = 1499.8 \pm 63.1$$

dE/dx [ADC counts/cm]

Count

[80, 160] MeV/c

XP

$$\frac{\sigma}{\mu} = 26.88 \pm 1.09 \%$$

$$\mu = 493.2 \pm 4.1$$

$$\sigma = 132.6 \pm 4.3$$

WF

$$\frac{\sigma}{\mu} = 23.70 \pm 1.02 \%$$

$$\mu = 515.9 \pm 3.8$$

$$\sigma = 122.3 \pm 4.3$$

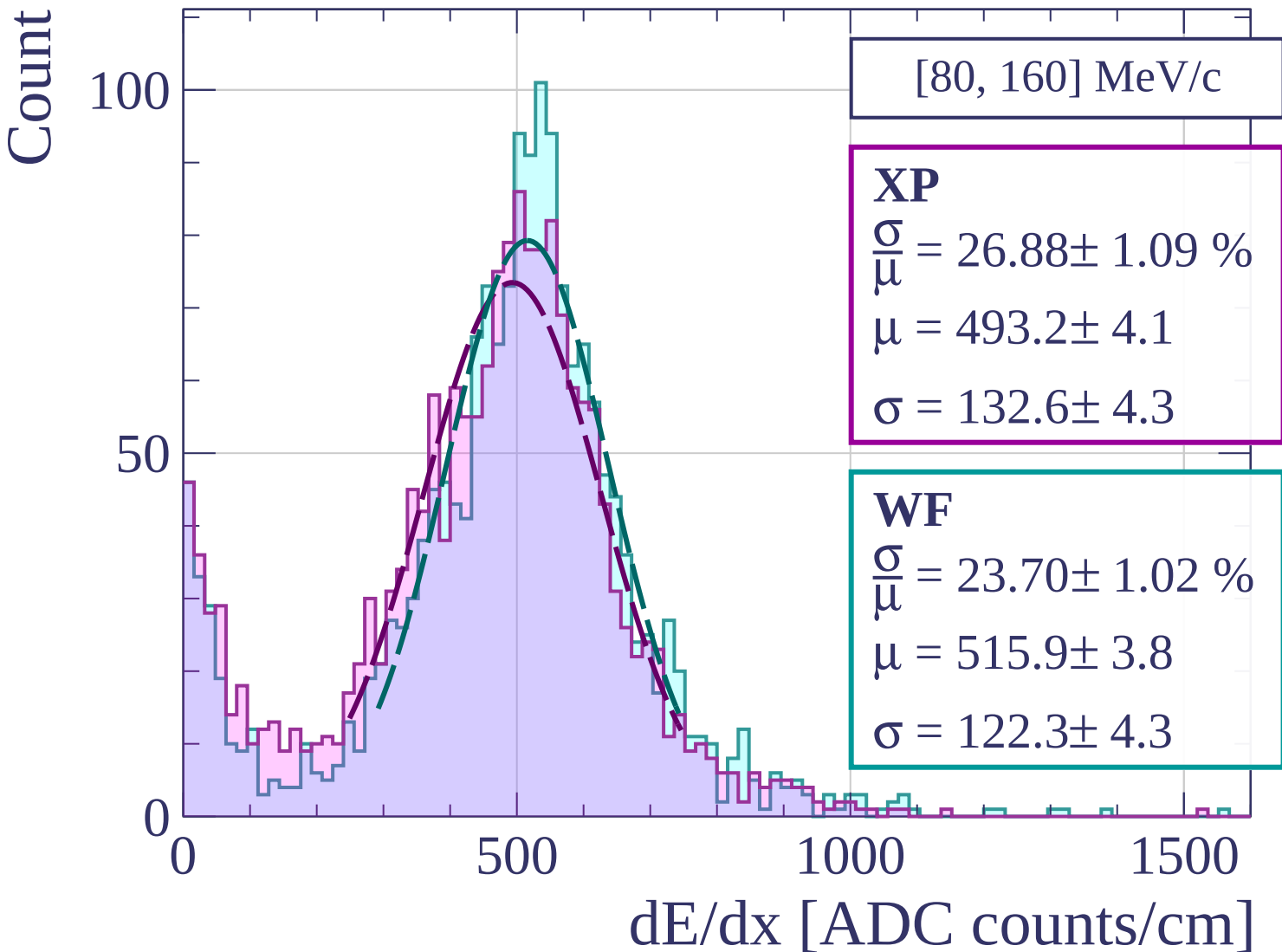
0

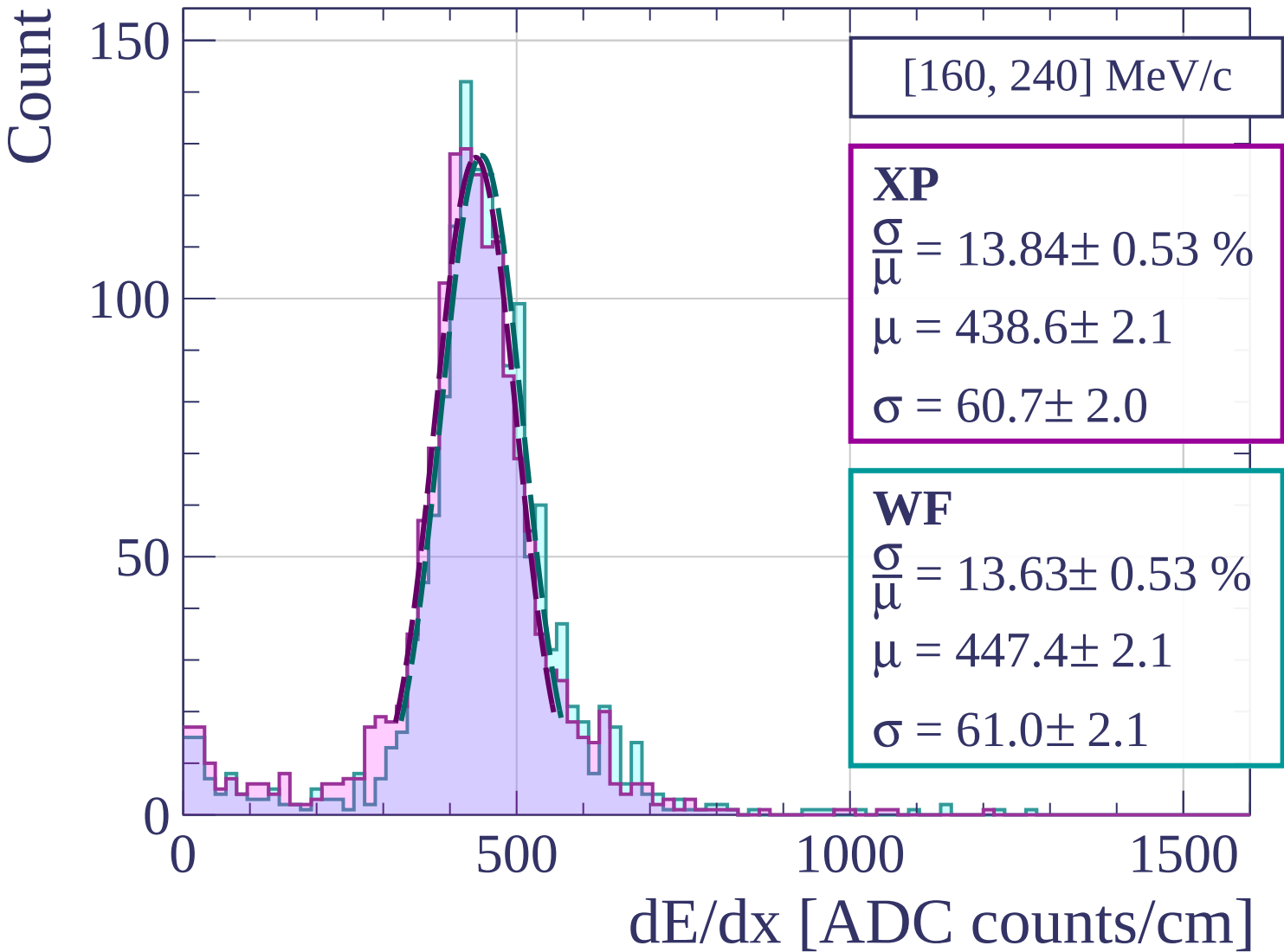
500

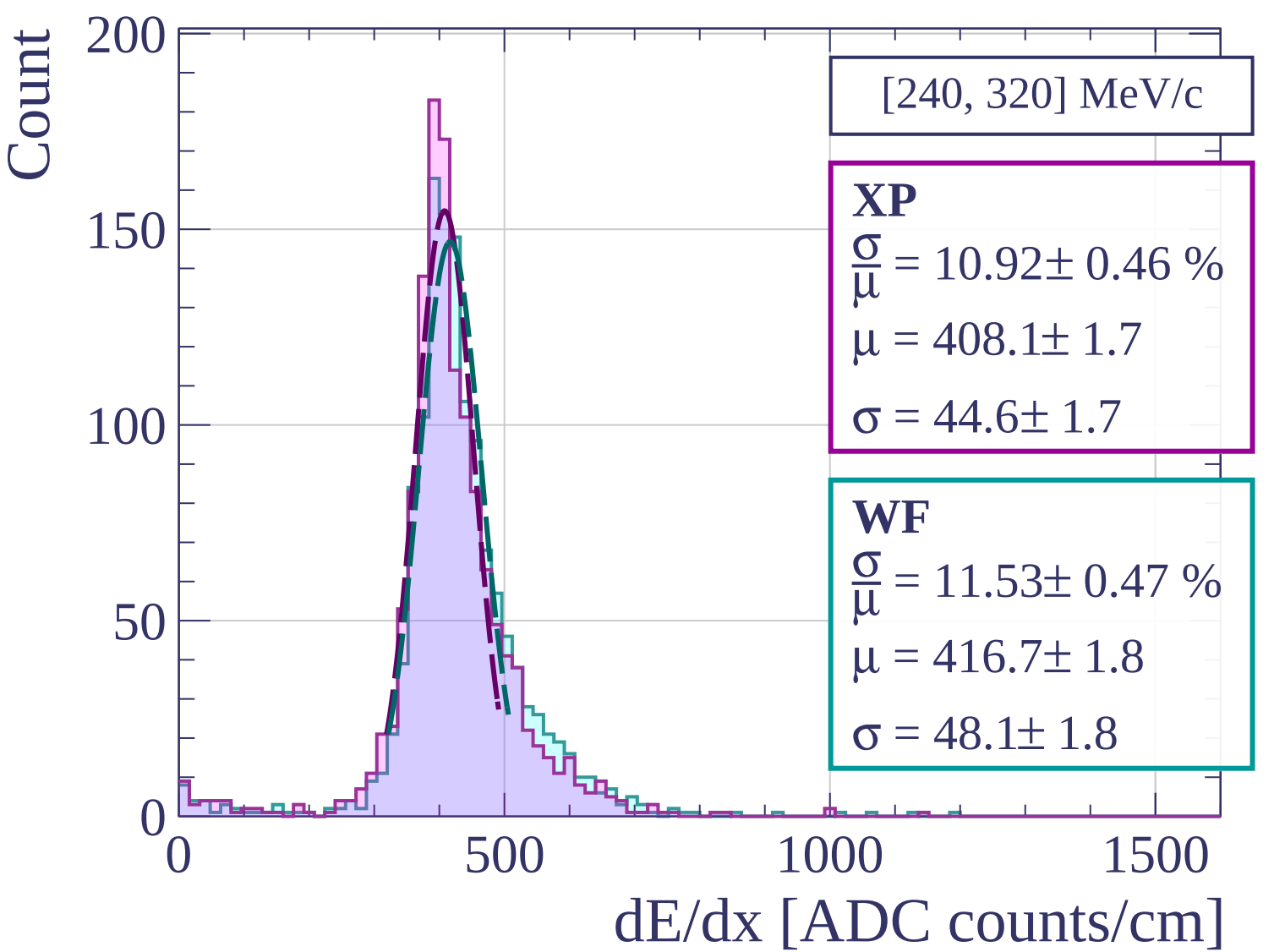
1000

1500

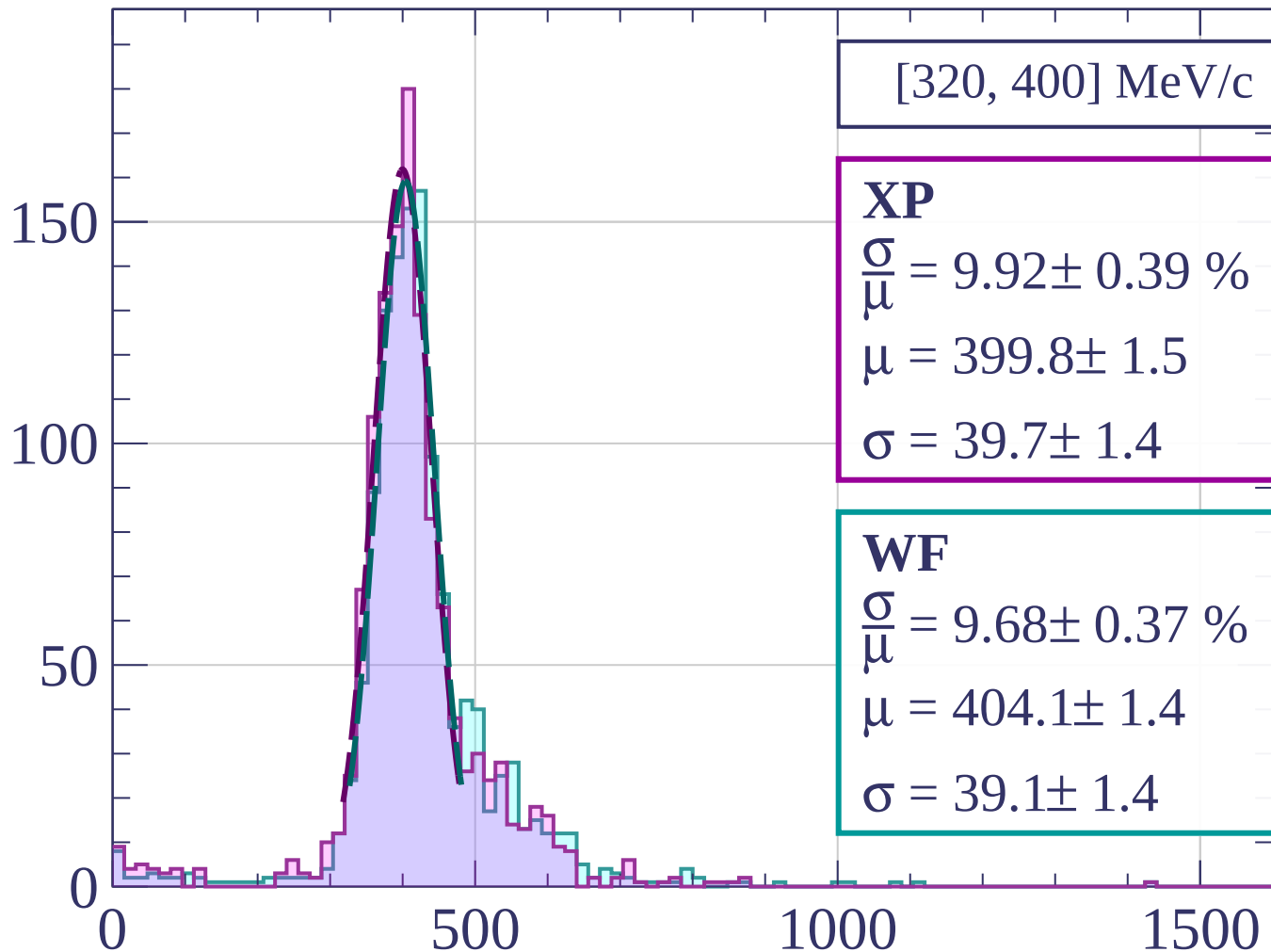
dE/dx [ADC counts/cm]





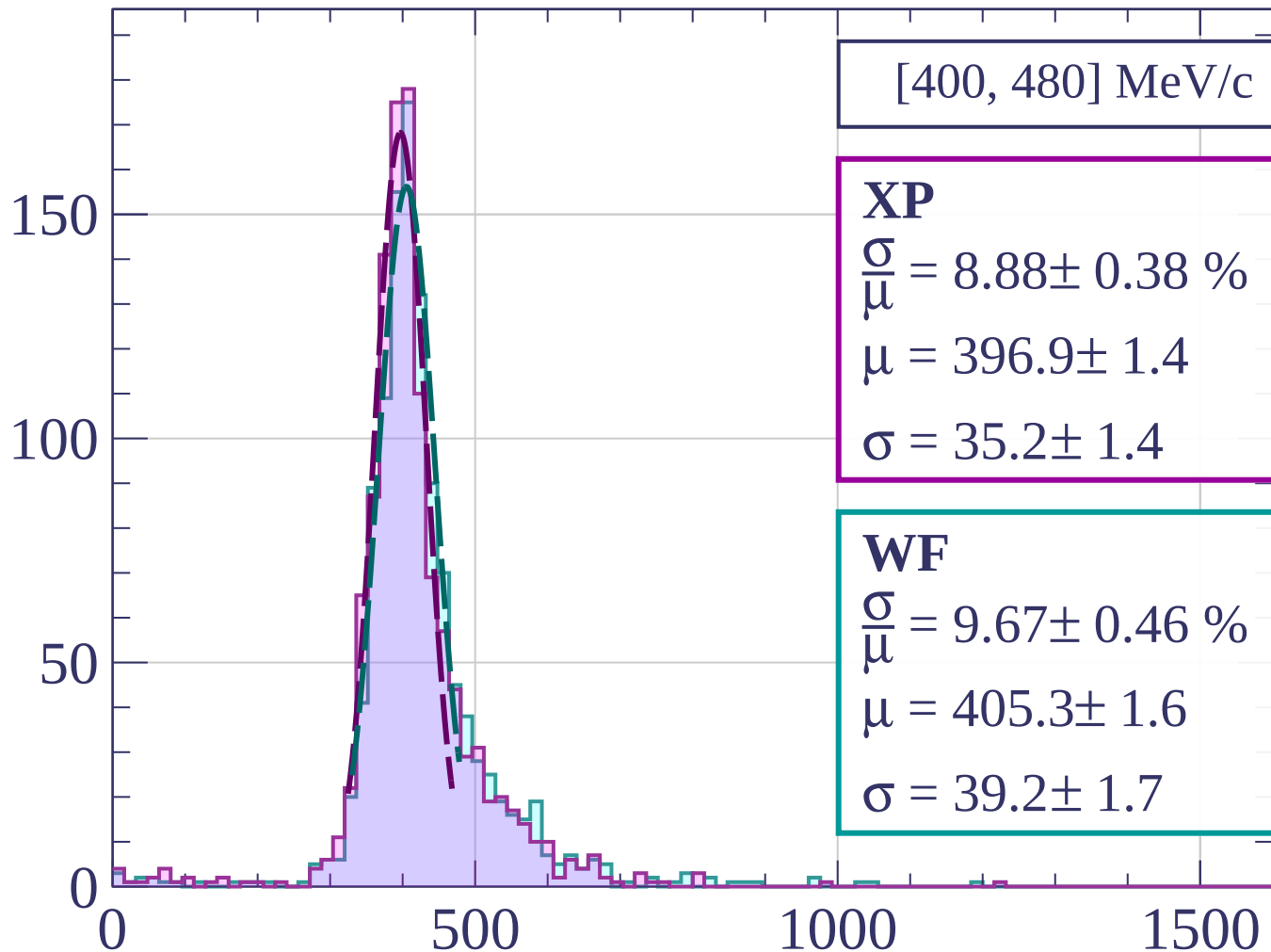


Count



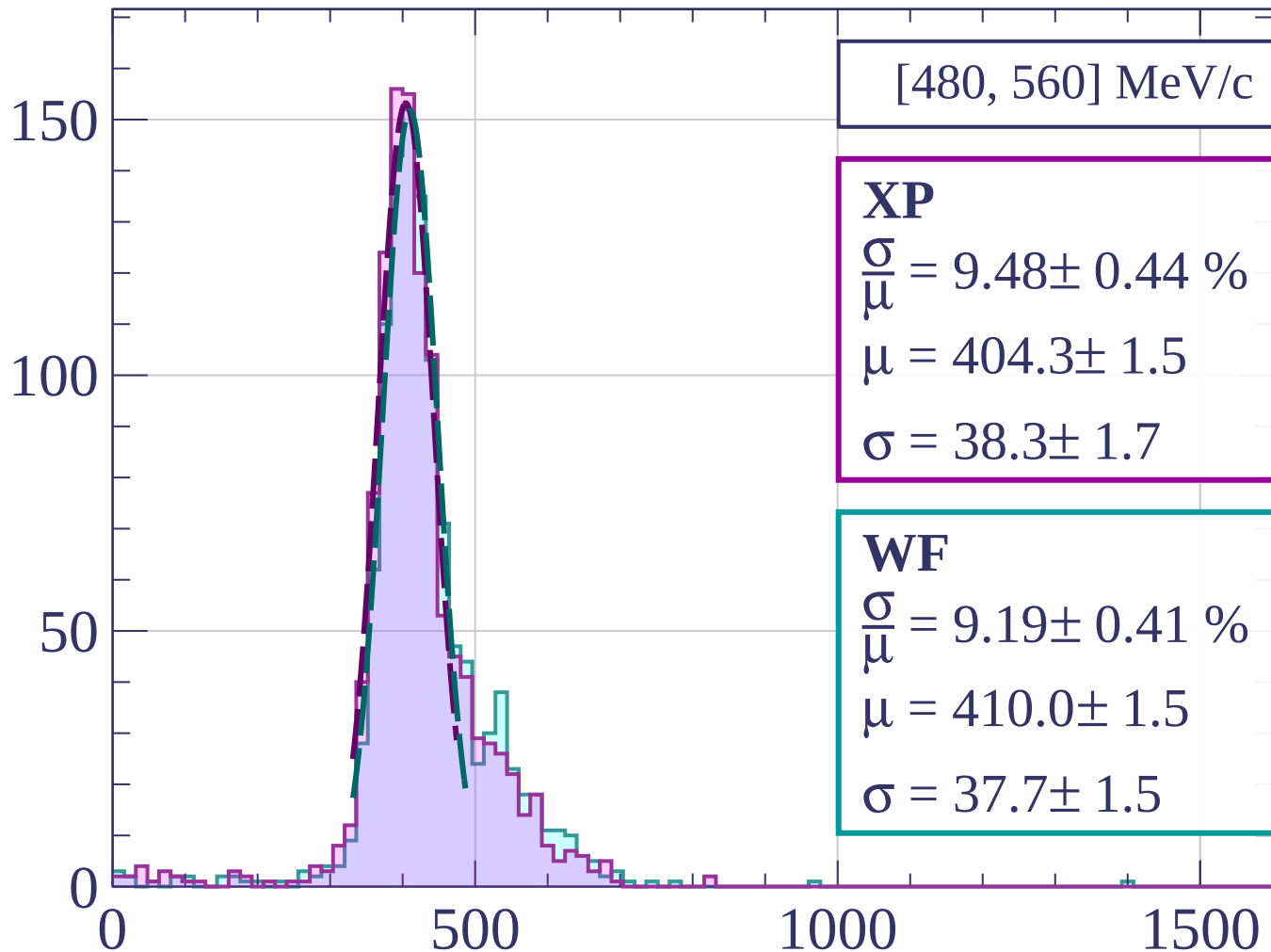
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[480, 560] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.48 \pm 0.44 \%$$

$$\mu = 404.3 \pm 1.5$$

$$\sigma = 38.3 \pm 1.7$$

WF

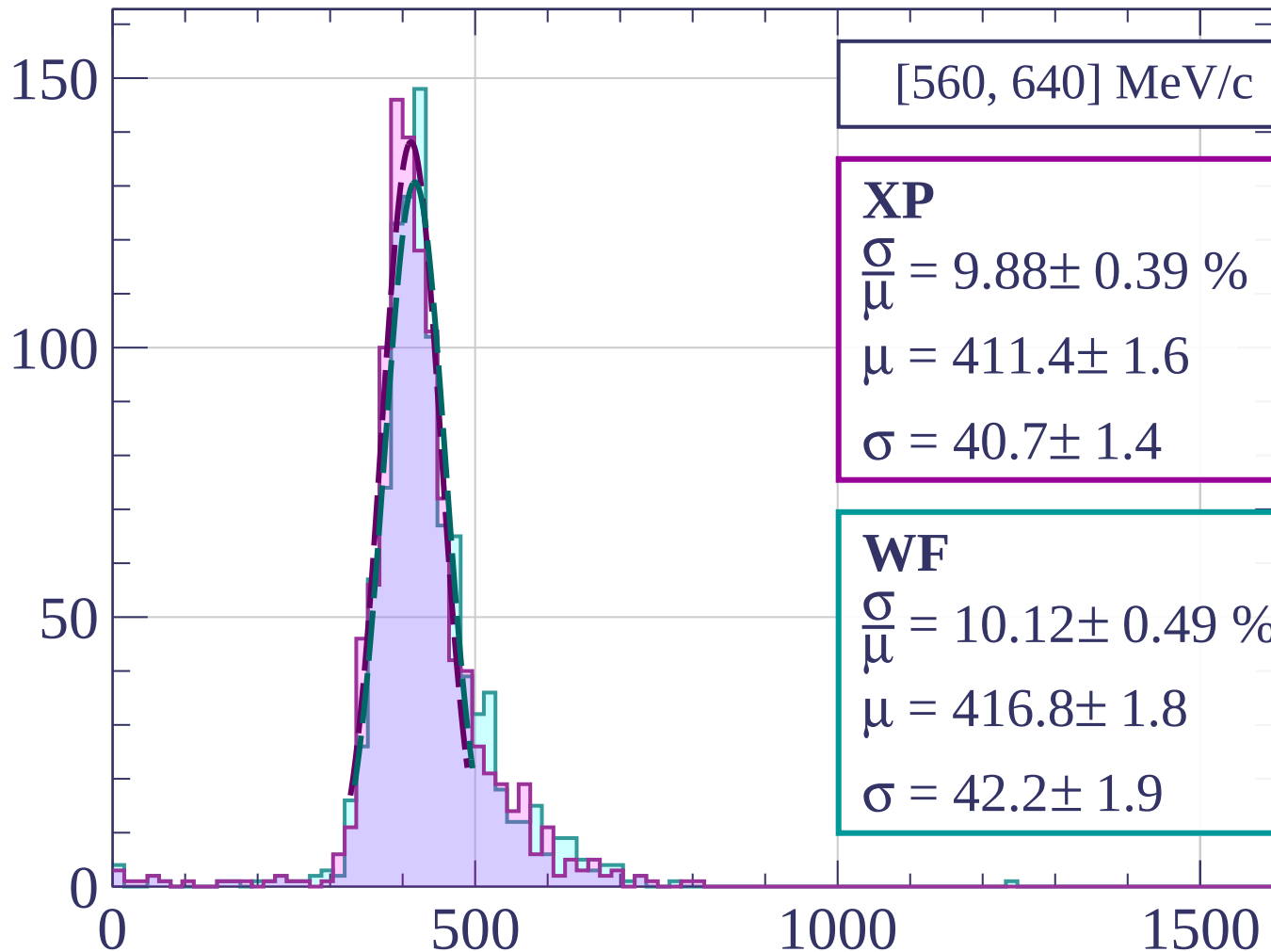
$$\frac{\sigma}{\mu} = 9.19 \pm 0.41 \%$$

$$\mu = 410.0 \pm 1.5$$

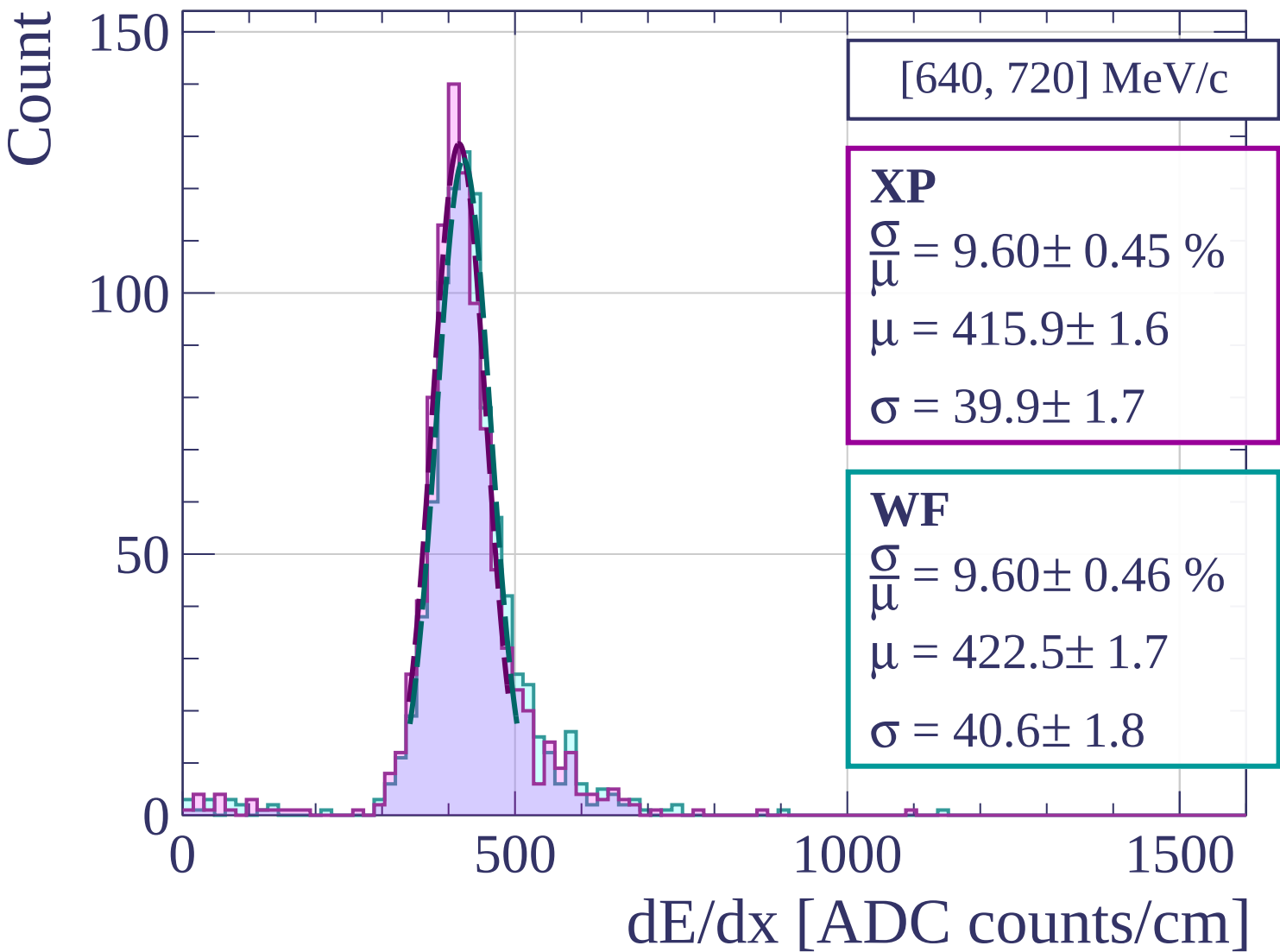
$$\sigma = 37.7 \pm 1.5$$

dE/dx [ADC counts/cm]

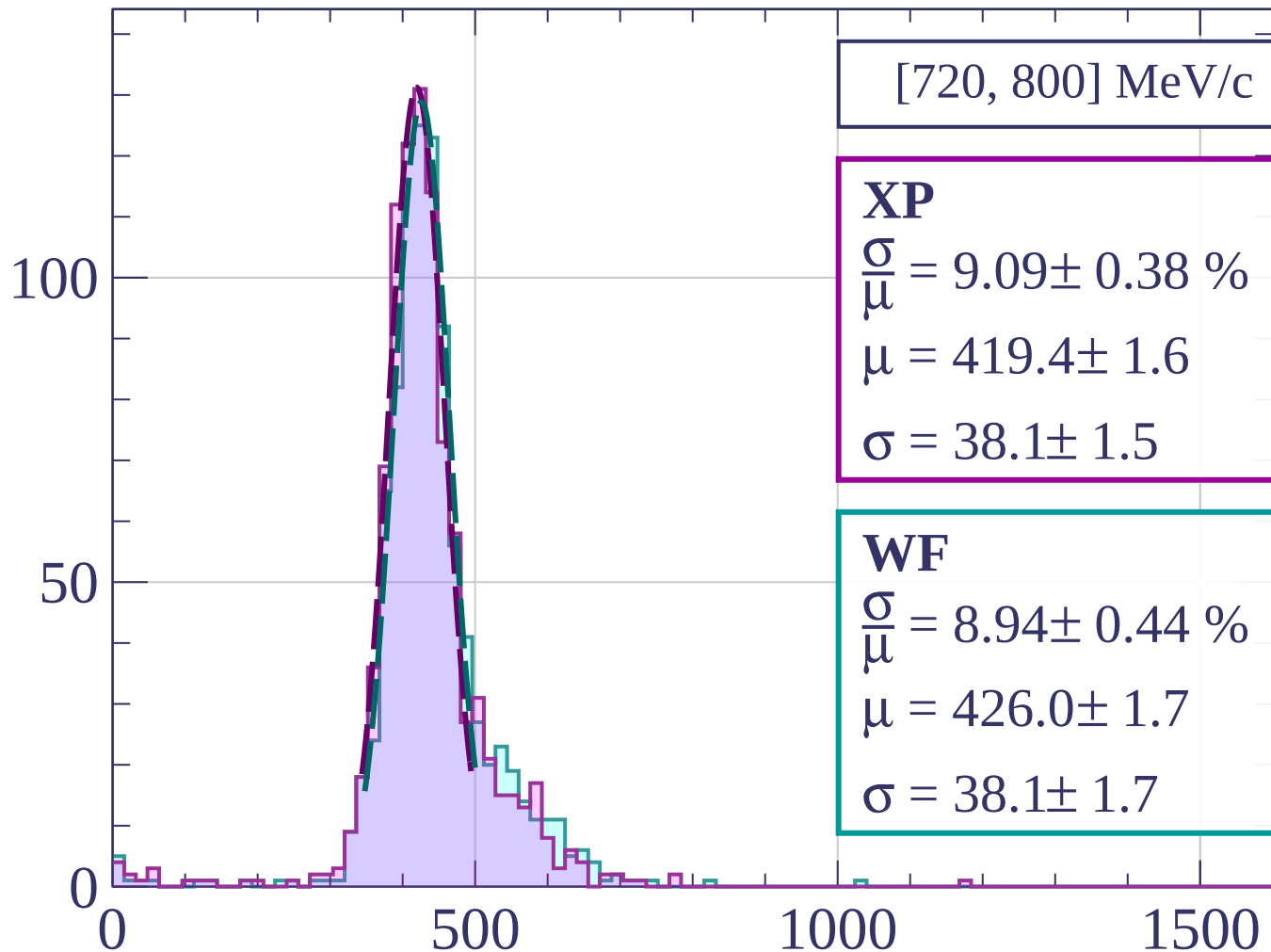
Count



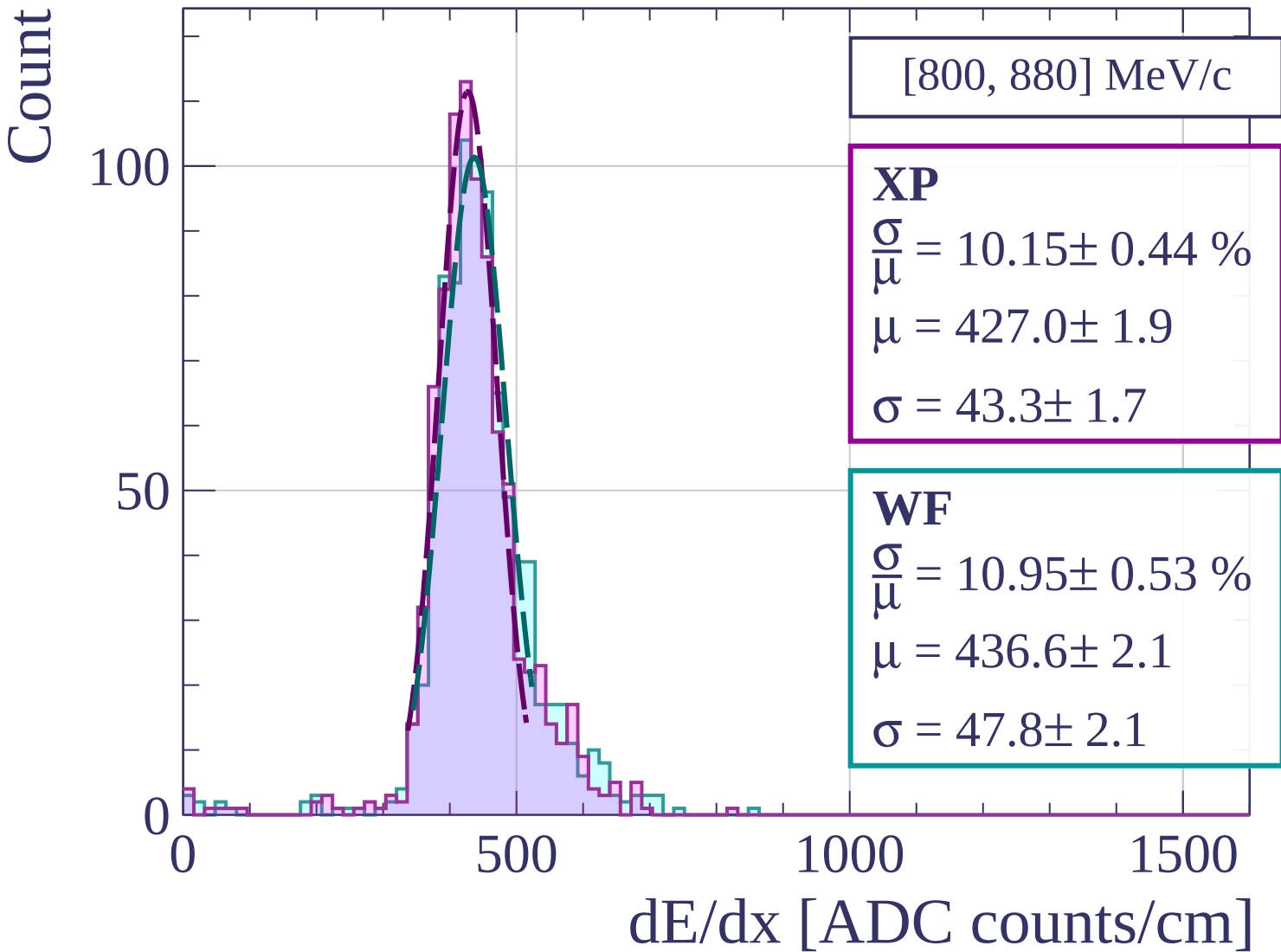
dE/dx [ADC counts/cm]



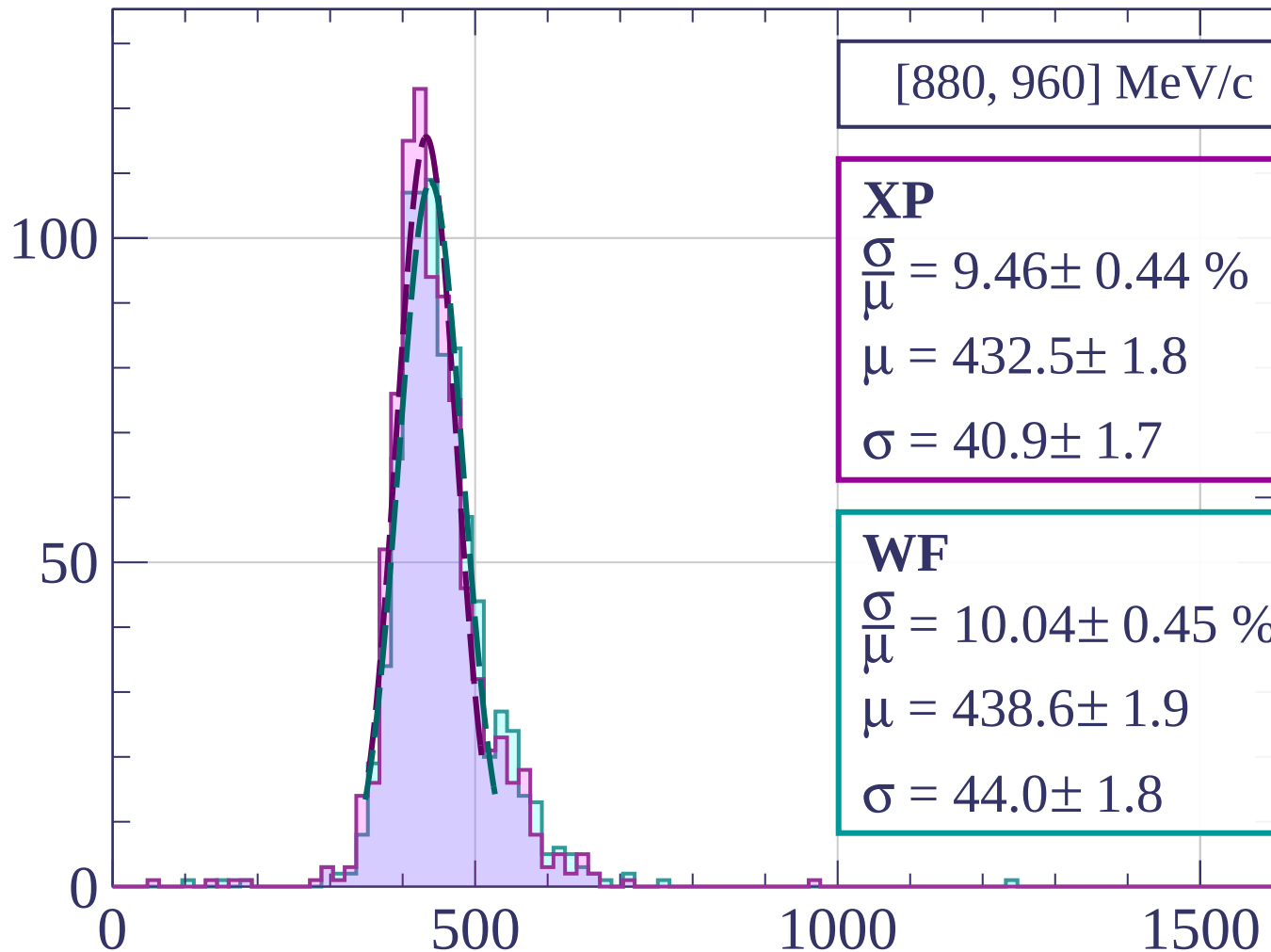
Count



dE/dx [ADC counts/cm]



Count



[880, 960] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.46 \pm 0.44 \%$$

$$\mu = 432.5 \pm 1.8$$

$$\sigma = 40.9 \pm 1.7$$

WF

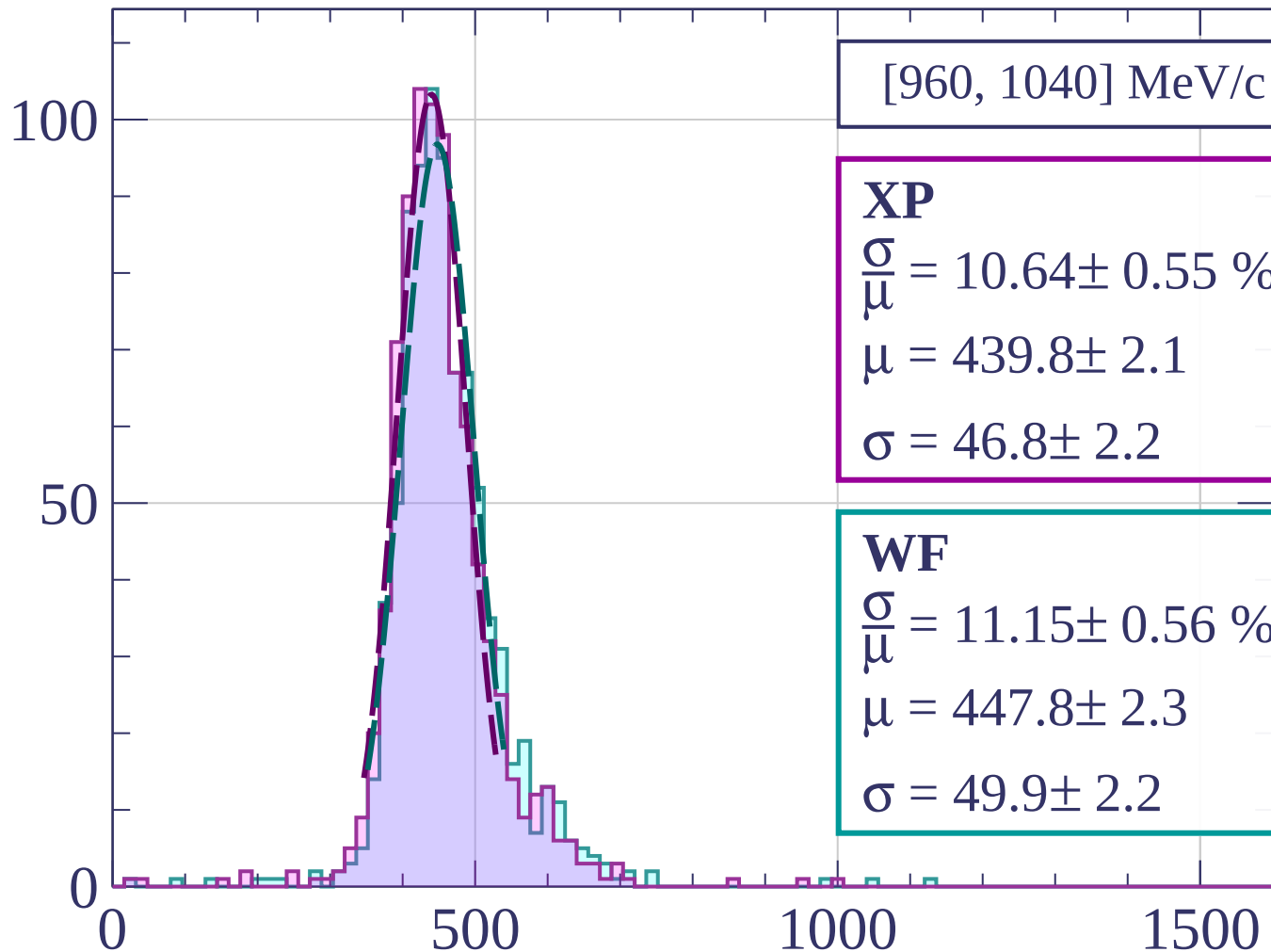
$$\frac{\sigma}{\mu} = 10.04 \pm 0.45 \%$$

$$\mu = 438.6 \pm 1.9$$

$$\sigma = 44.0 \pm 1.8$$

dE/dx [ADC counts/cm]

Count



[960, 1040] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.64 \pm 0.55 \%$$

$$\mu = 439.8 \pm 2.1$$

$$\sigma = 46.8 \pm 2.2$$

WF

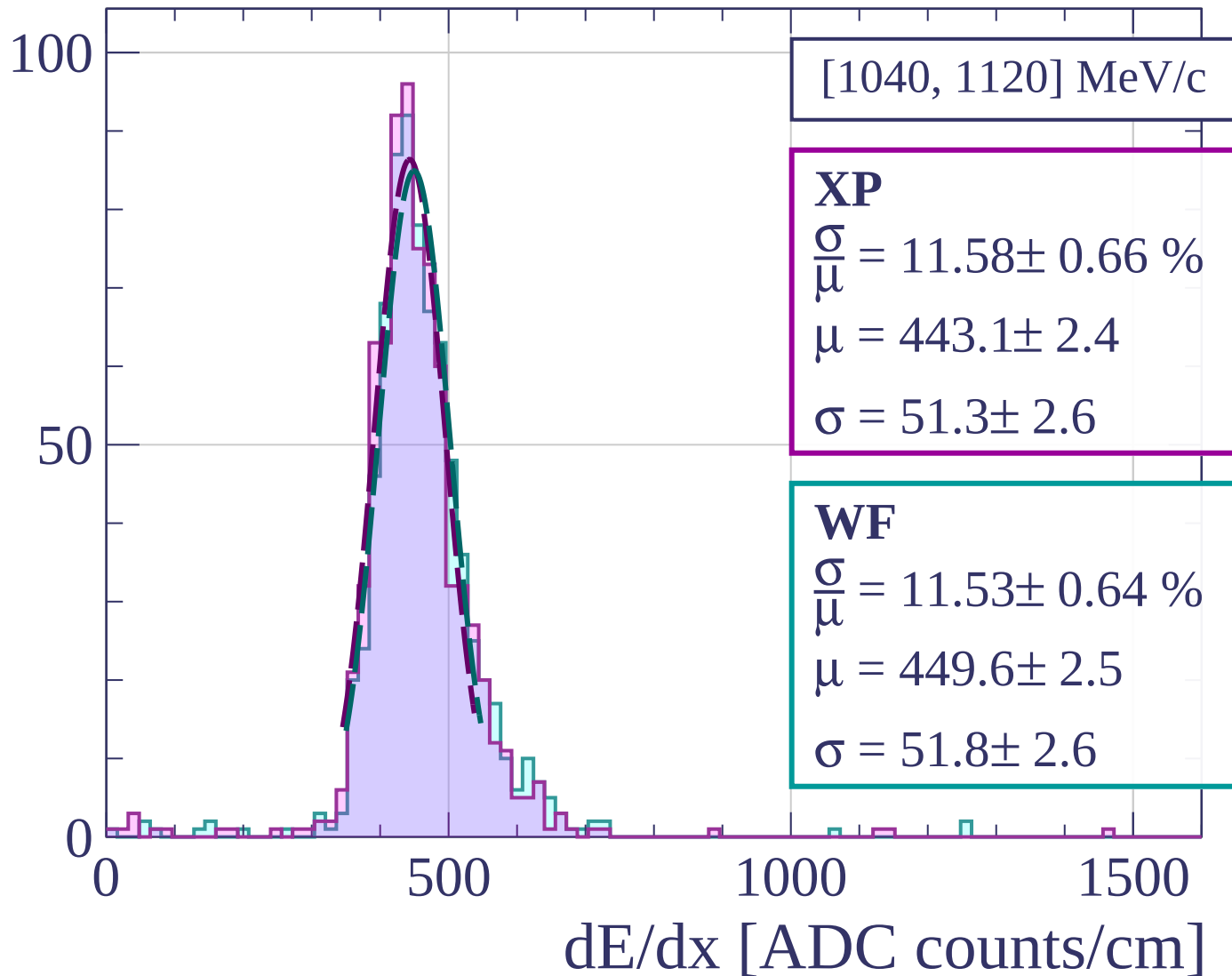
$$\frac{\sigma}{\mu} = 11.15 \pm 0.56 \%$$

$$\mu = 447.8 \pm 2.3$$

$$\sigma = 49.9 \pm 2.2$$

dE/dx [ADC counts/cm]

Count



Count

[1120, 1200] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.78 \pm 0.53 \%$$

$$\mu = 444.6 \pm 2.1$$

$$\sigma = 47.9 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 10.56 \pm 0.55 \%$$

$$\mu = 453.4 \pm 2.3$$

$$\sigma = 47.9 \pm 2.2$$

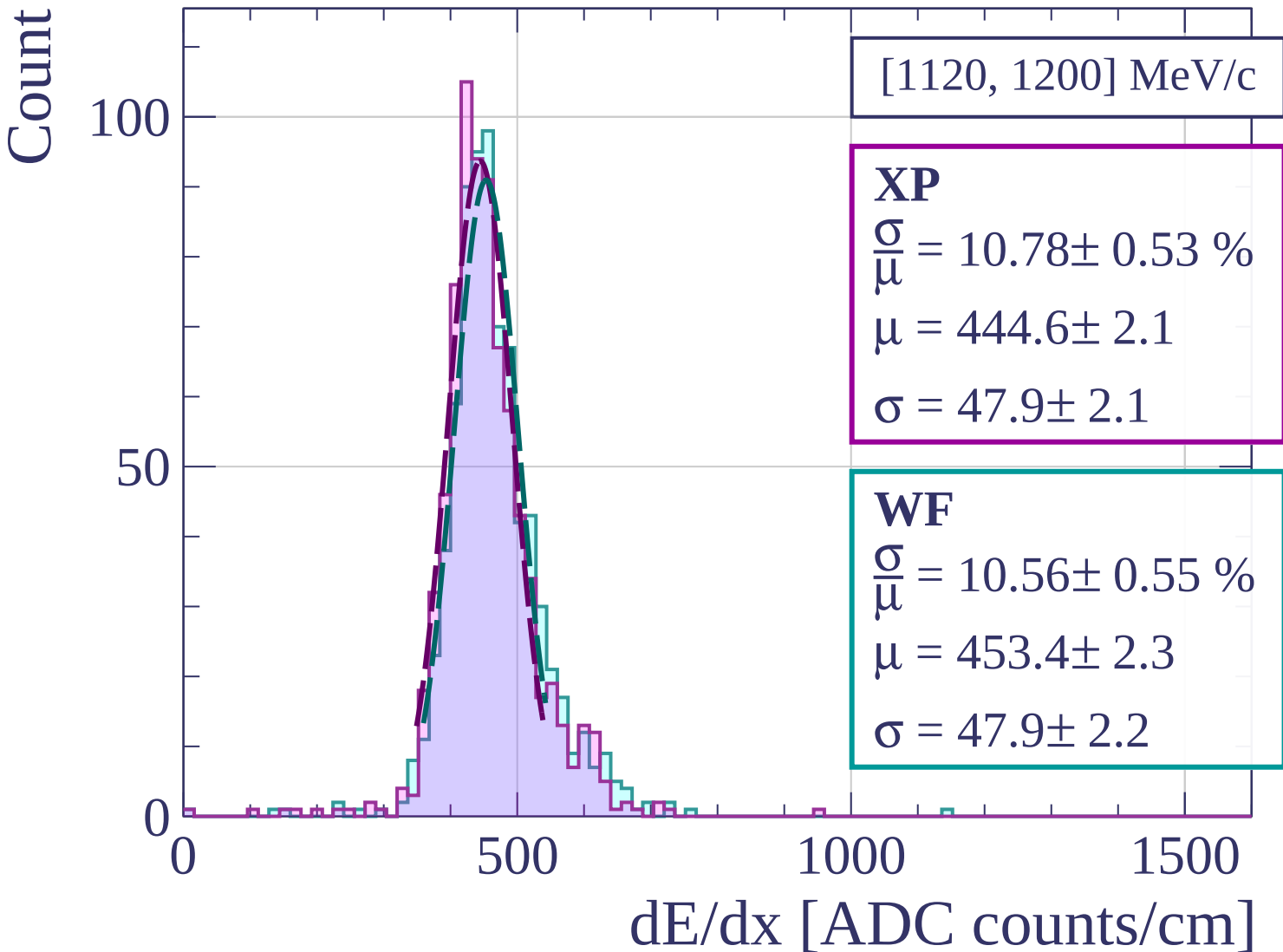
0

500

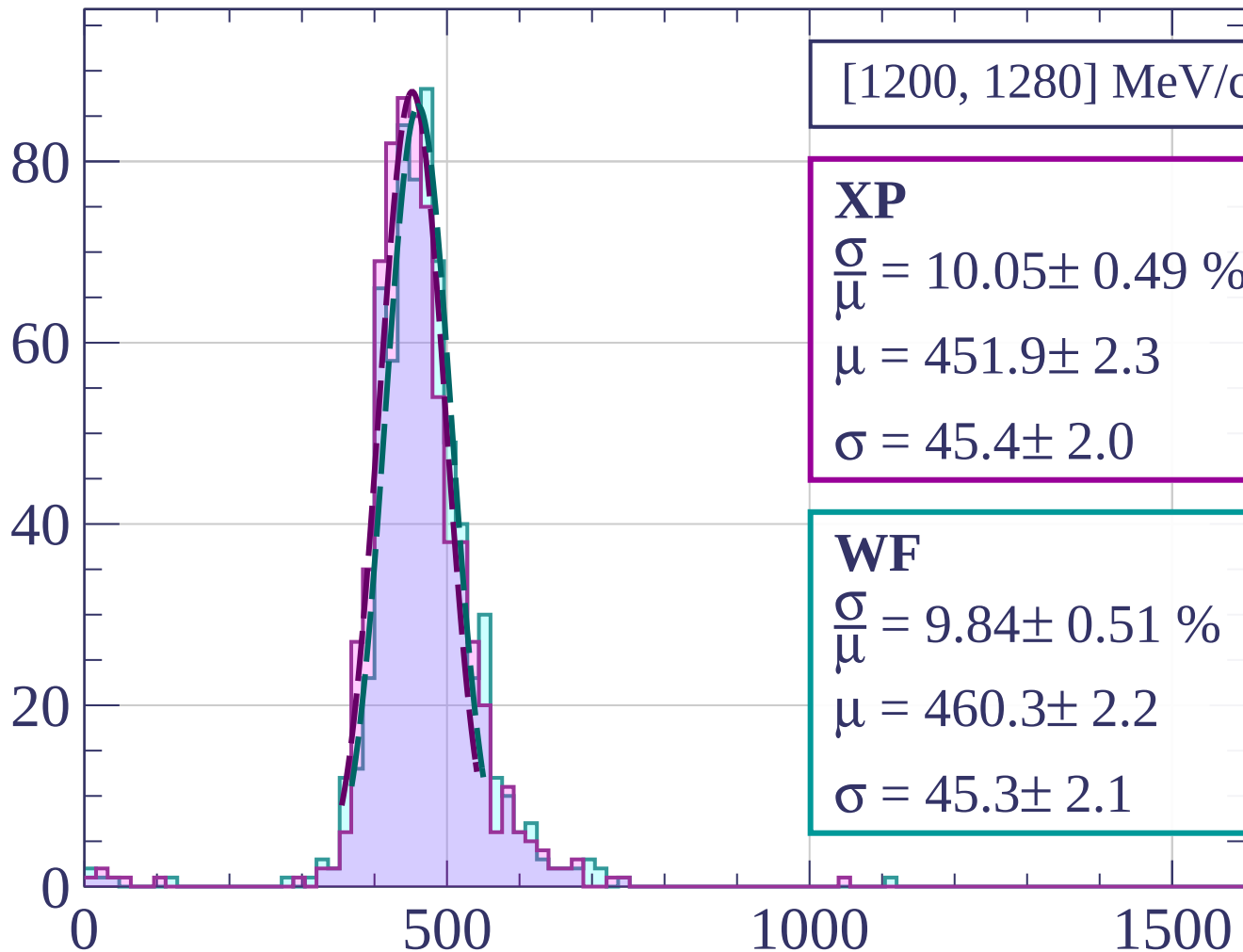
1000

1500

dE/dx [ADC counts/cm]

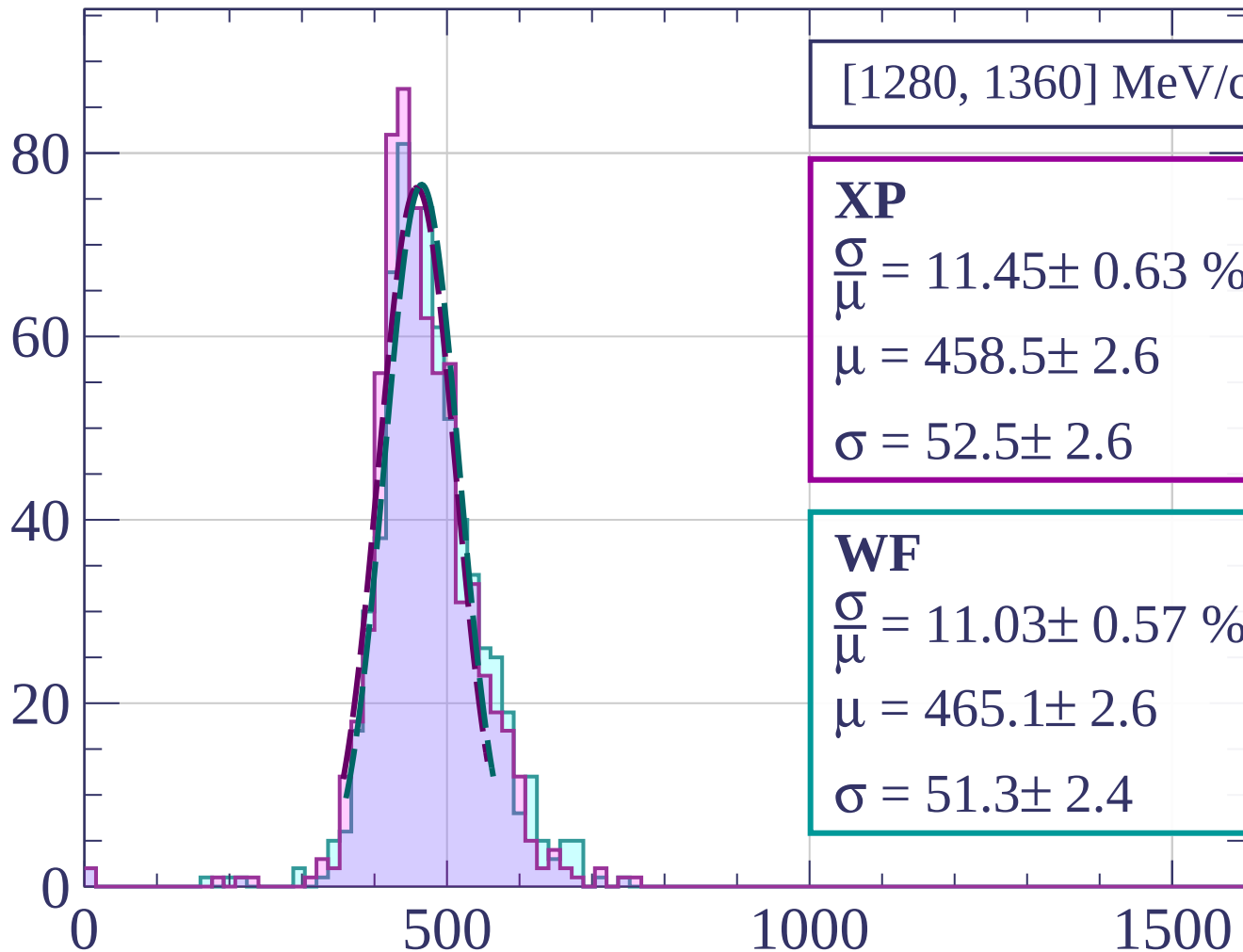


Count



dE/dx [ADC counts/cm]

Count



[1280, 1360] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.45 \pm 0.63 \%$$

$$\mu = 458.5 \pm 2.6$$

$$\sigma = 52.5 \pm 2.6$$

WF

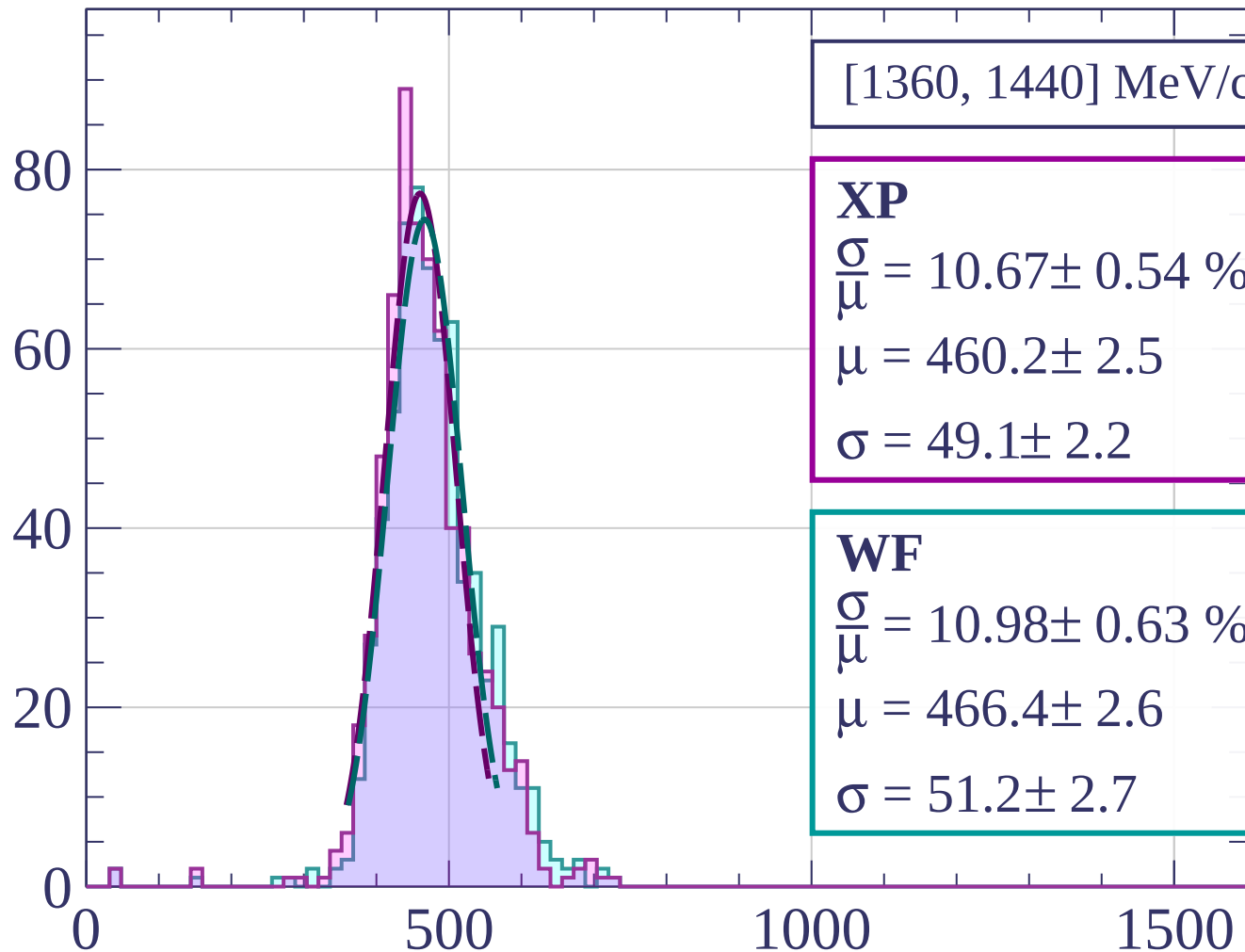
$$\frac{\sigma}{\mu} = 11.03 \pm 0.57 \%$$

$$\mu = 465.1 \pm 2.6$$

$$\sigma = 51.3 \pm 2.4$$

dE/dx [ADC counts/cm]

Count



[1360, 1440] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.67 \pm 0.54 \%$$

$$\mu = 460.2 \pm 2.5$$

$$\sigma = 49.1 \pm 2.2$$

WF

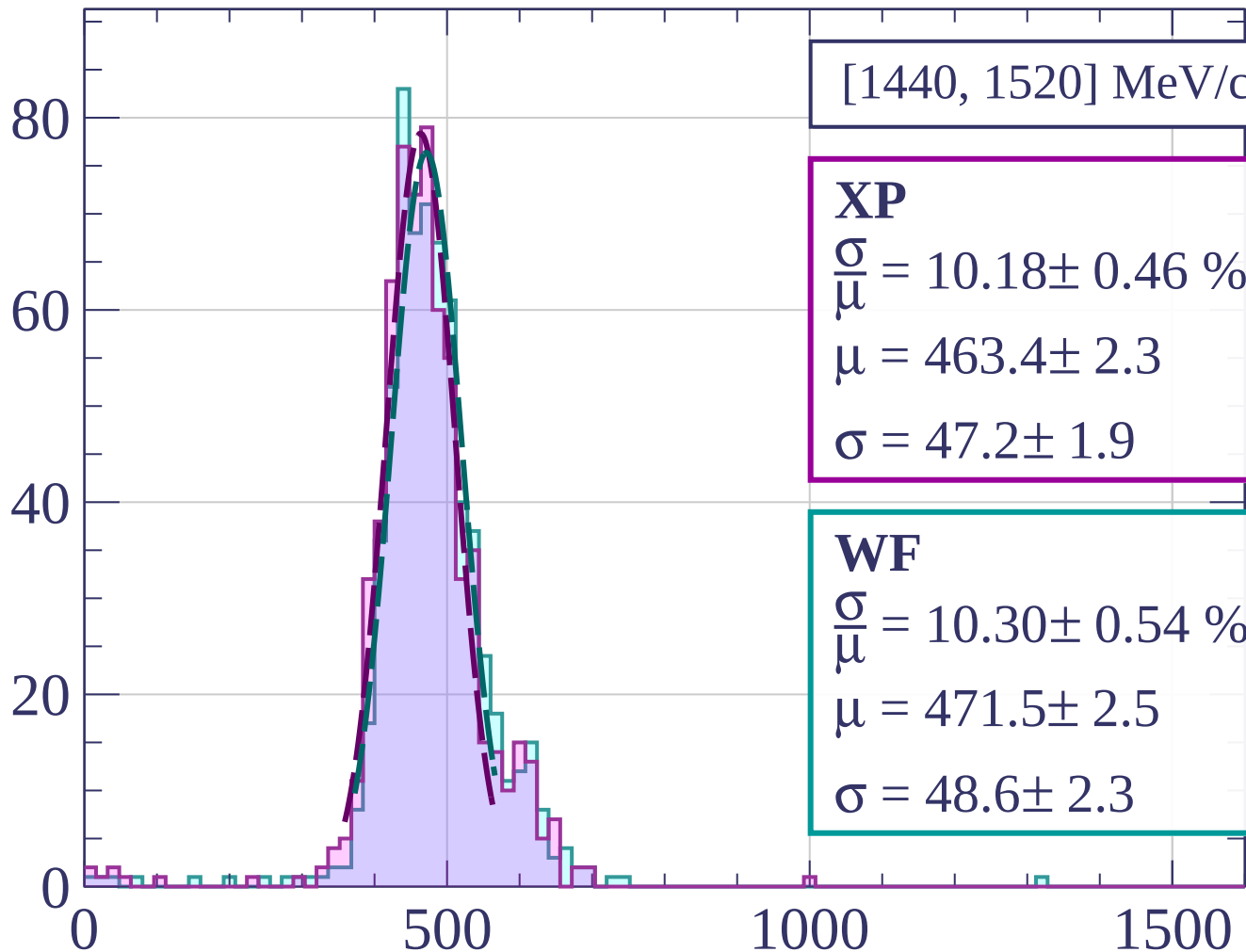
$$\frac{\sigma}{\mu} = 10.98 \pm 0.63 \%$$

$$\mu = 466.4 \pm 2.6$$

$$\sigma = 51.2 \pm 2.7$$

dE/dx [ADC counts/cm]

Count



[1440, 1520] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.18 \pm 0.46 \%$$

$$\mu = 463.4 \pm 2.3$$

$$\sigma = 47.2 \pm 1.9$$

WF

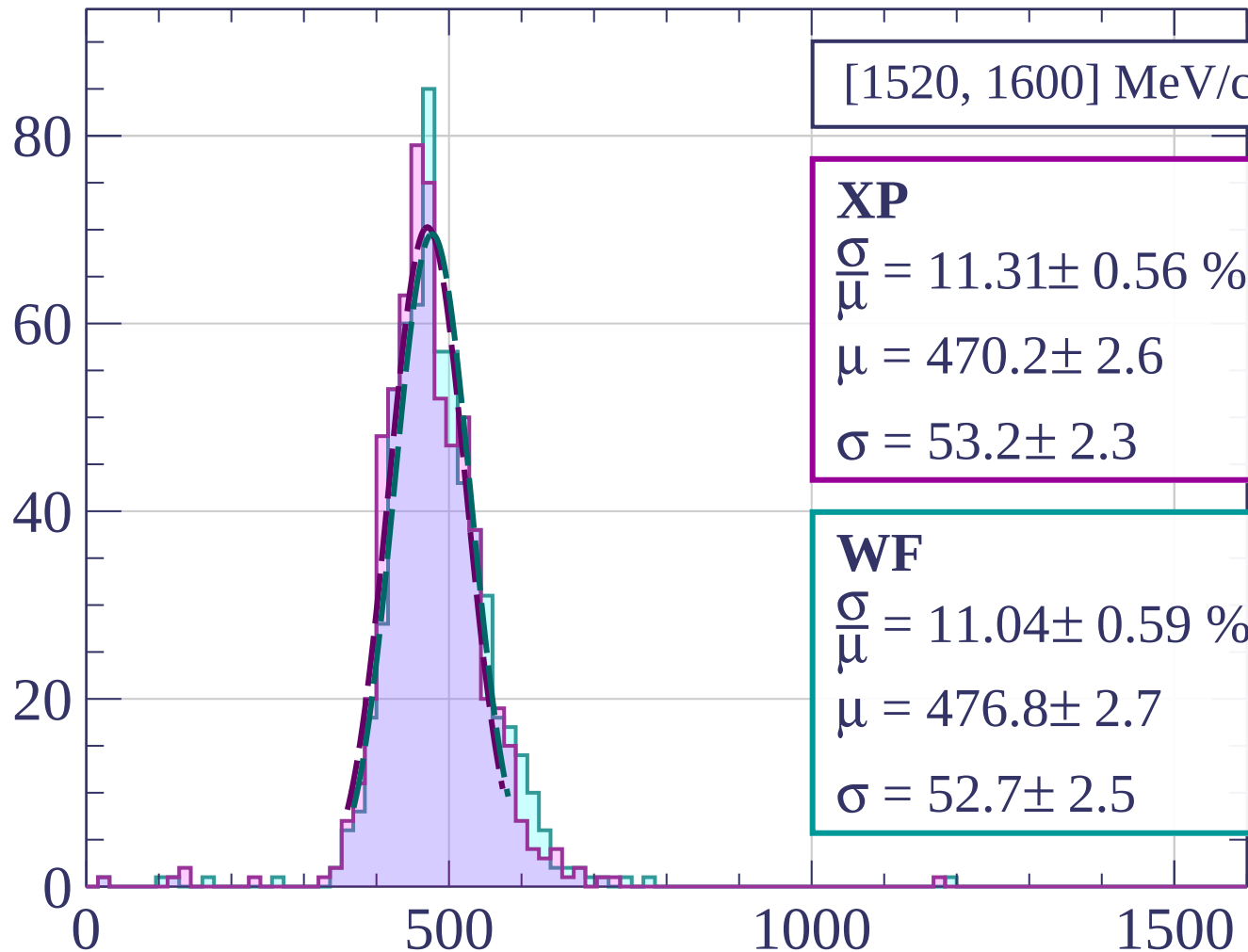
$$\frac{\sigma}{\mu} = 10.30 \pm 0.54 \%$$

$$\mu = 471.5 \pm 2.5$$

$$\sigma = 48.6 \pm 2.3$$

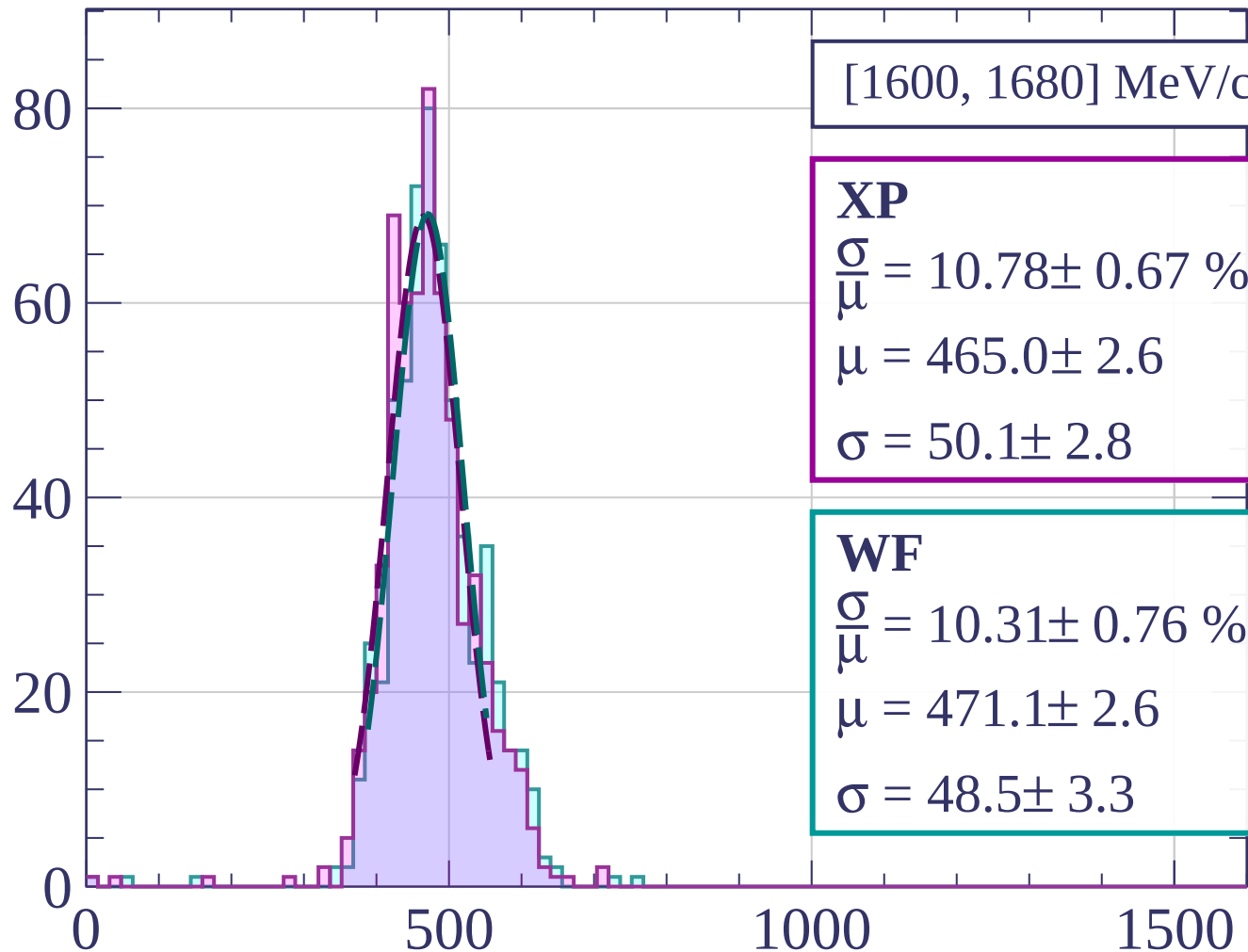
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[1600, 1680] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.78 \pm 0.67 \%$$

$$\mu = 465.0 \pm 2.6$$

$$\sigma = 50.1 \pm 2.8$$

WF

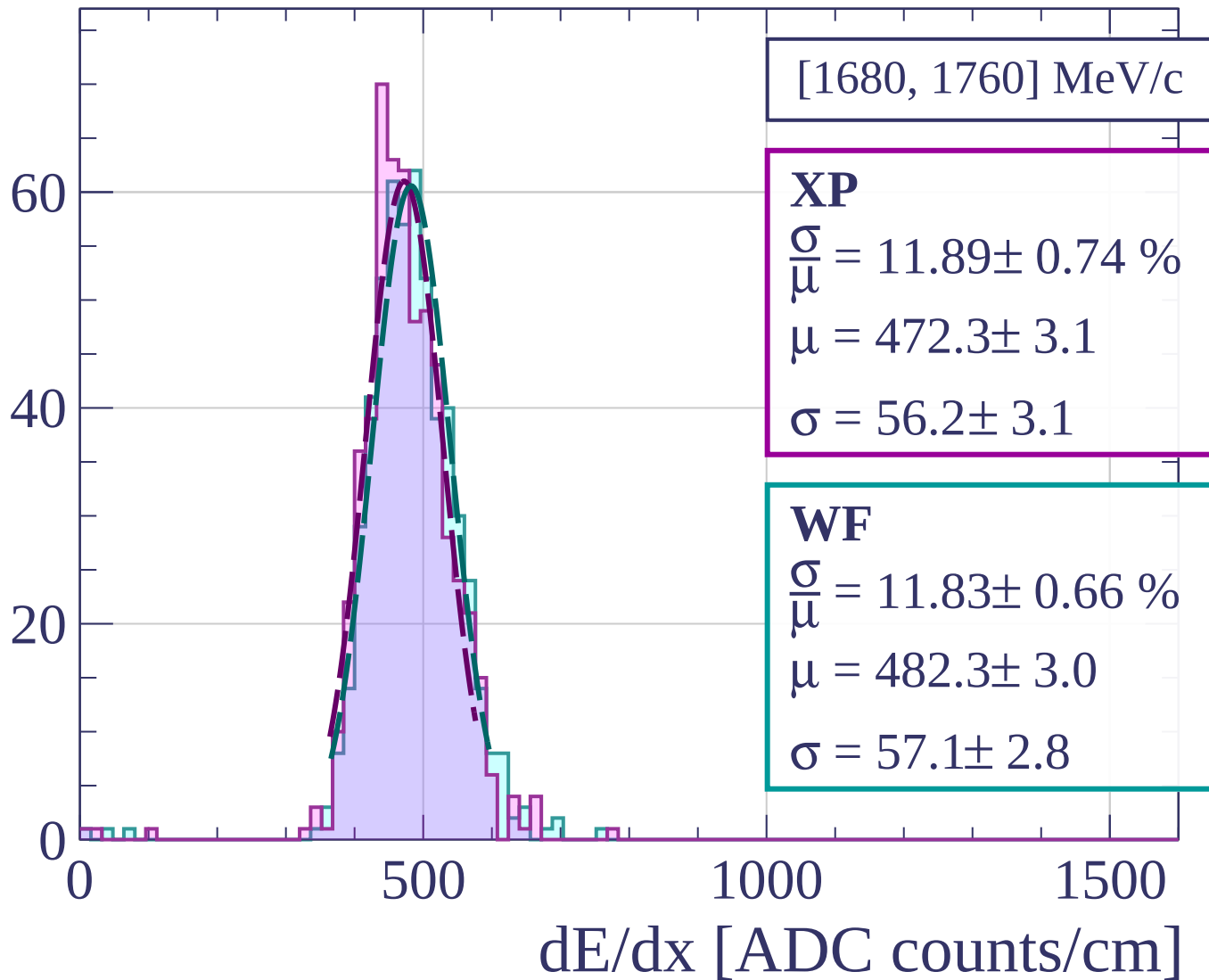
$$\frac{\sigma}{\mu} = 10.31 \pm 0.76 \%$$

$$\mu = 471.1 \pm 2.6$$

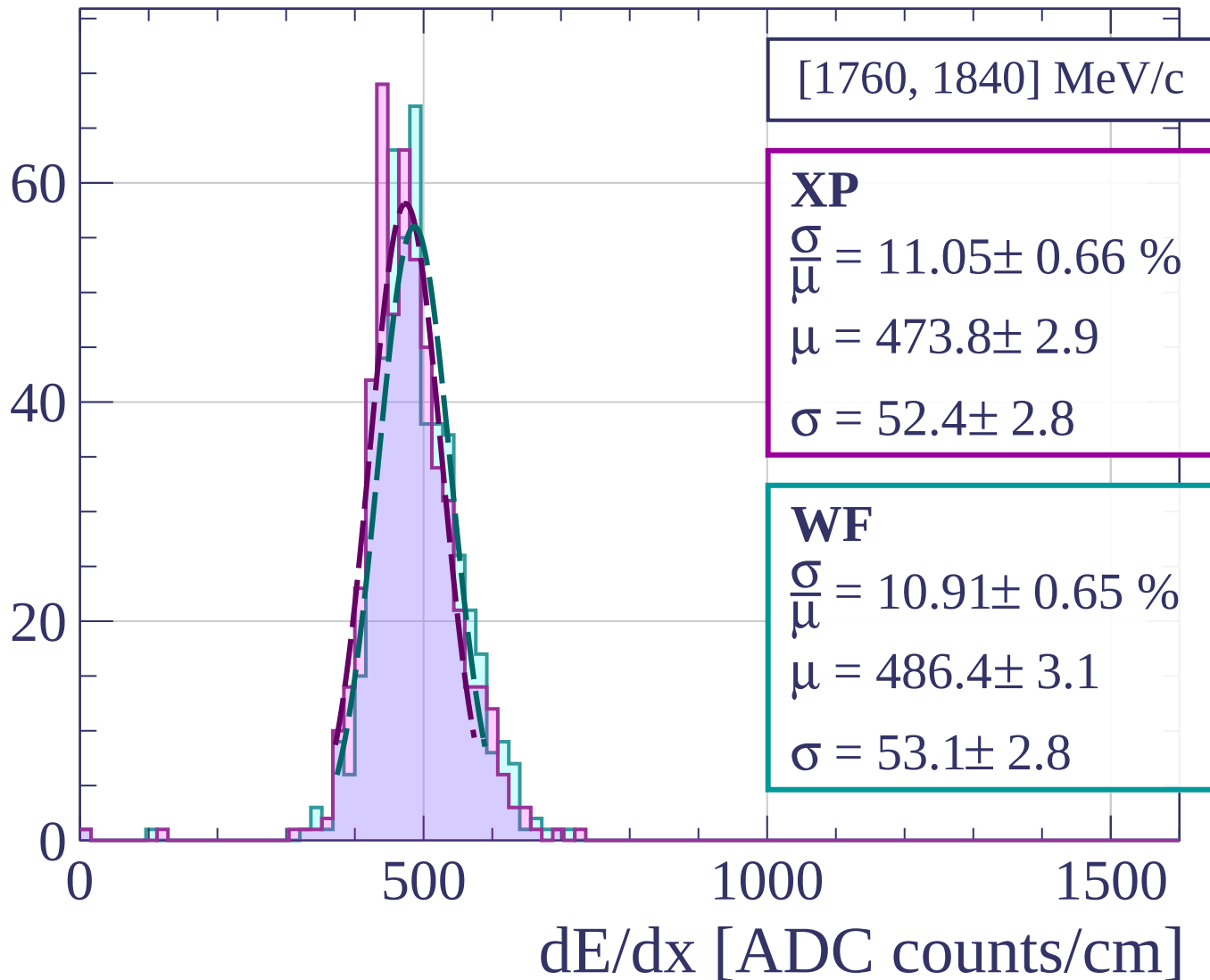
$$\sigma = 48.5 \pm 3.3$$

dE/dx [ADC counts/cm]

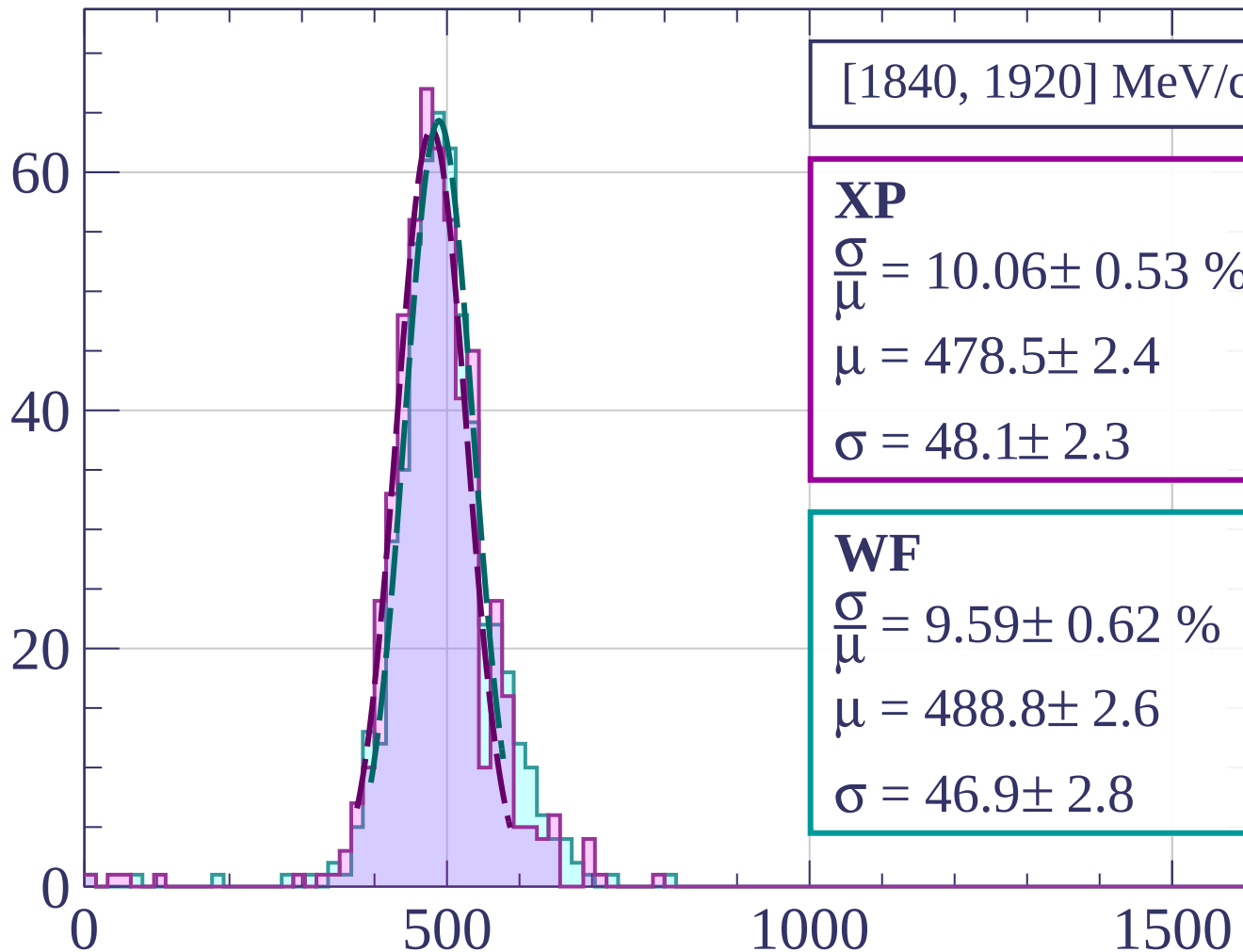
Count



Count



Count



[1840, 1920] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.06 \pm 0.53 \%$$

$$\mu = 478.5 \pm 2.4$$

$$\sigma = 48.1 \pm 2.3$$

WF

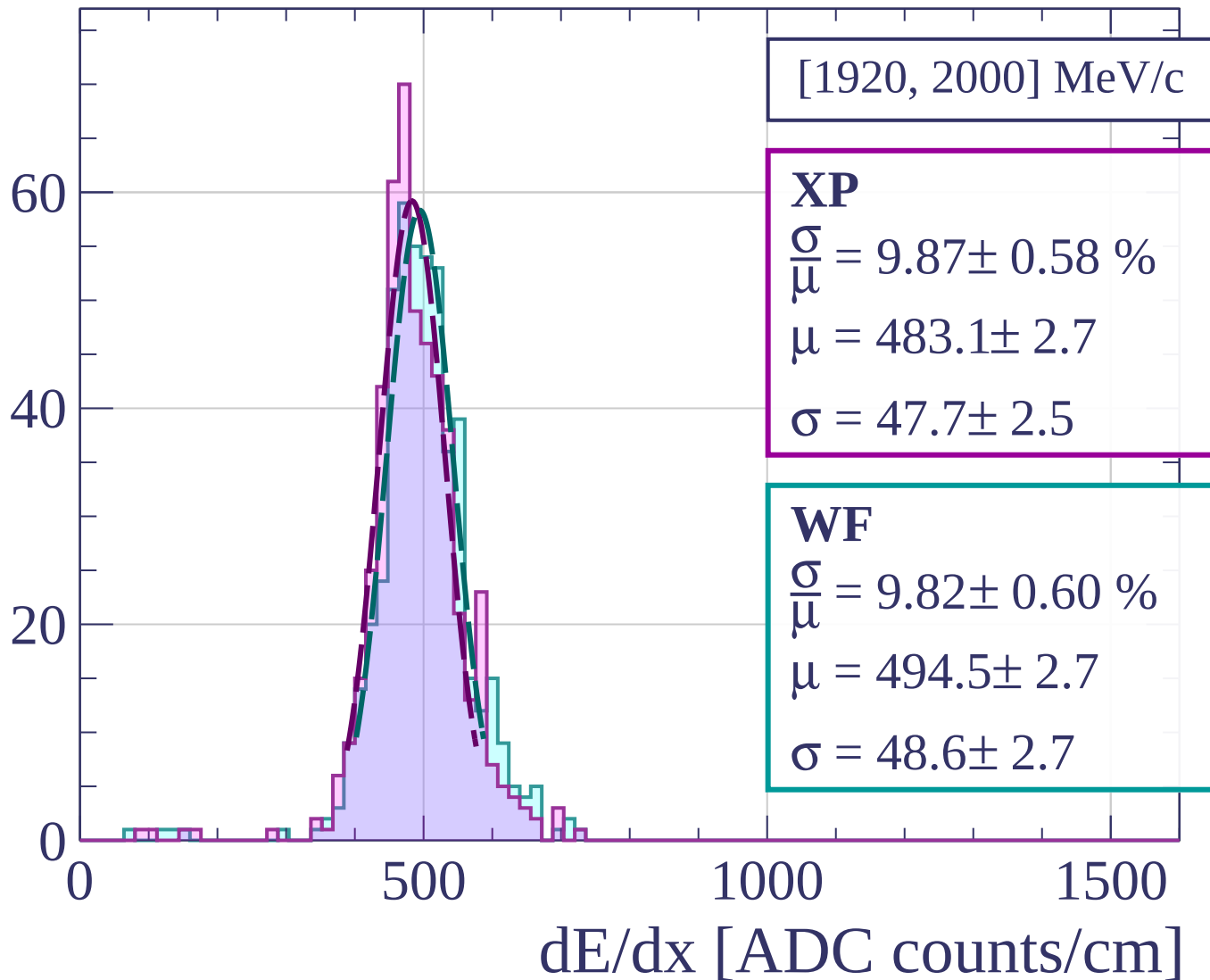
$$\frac{\sigma}{\mu} = 9.59 \pm 0.62 \%$$

$$\mu = 488.8 \pm 2.6$$

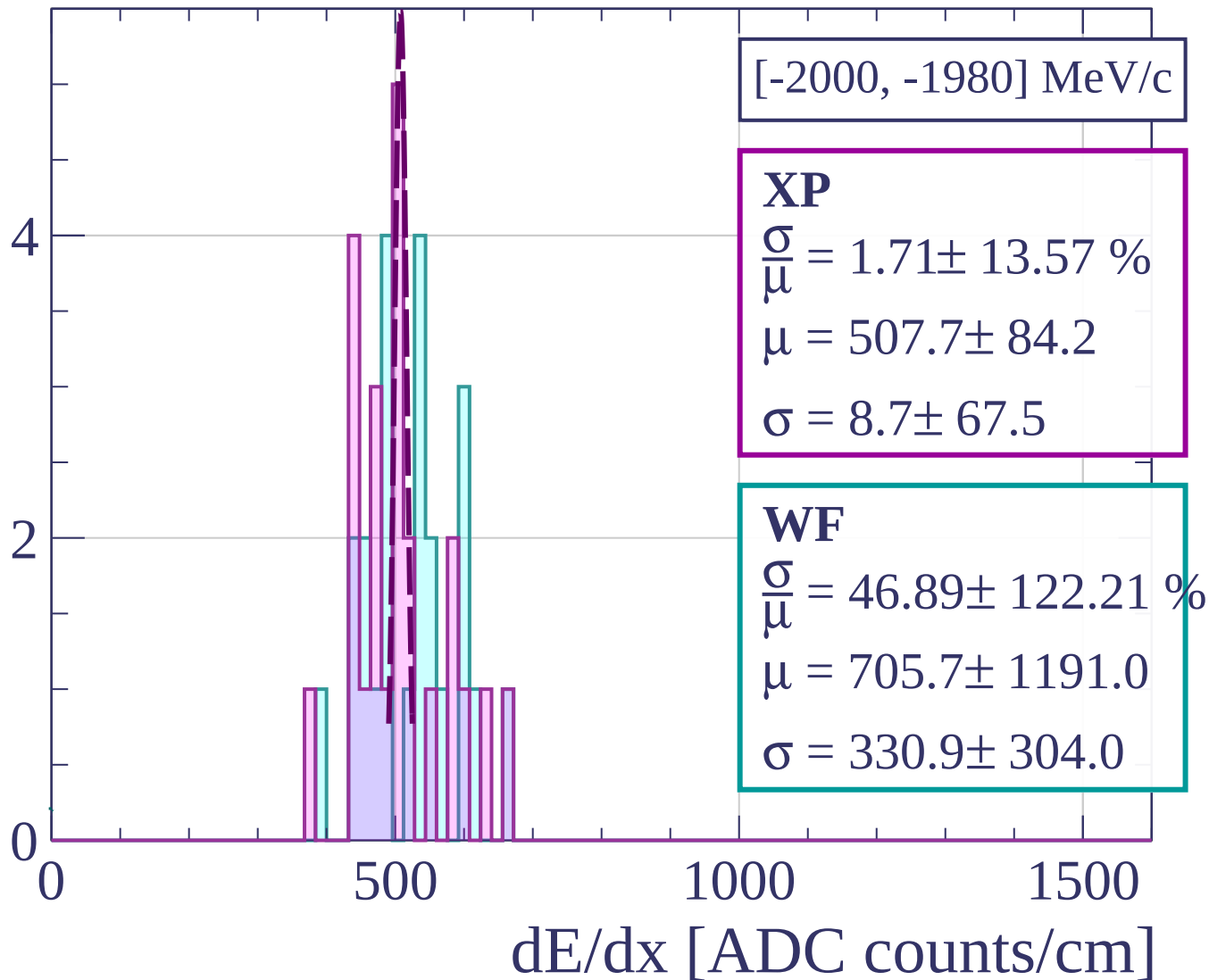
$$\sigma = 46.9 \pm 2.8$$

dE/dx [ADC counts/cm]

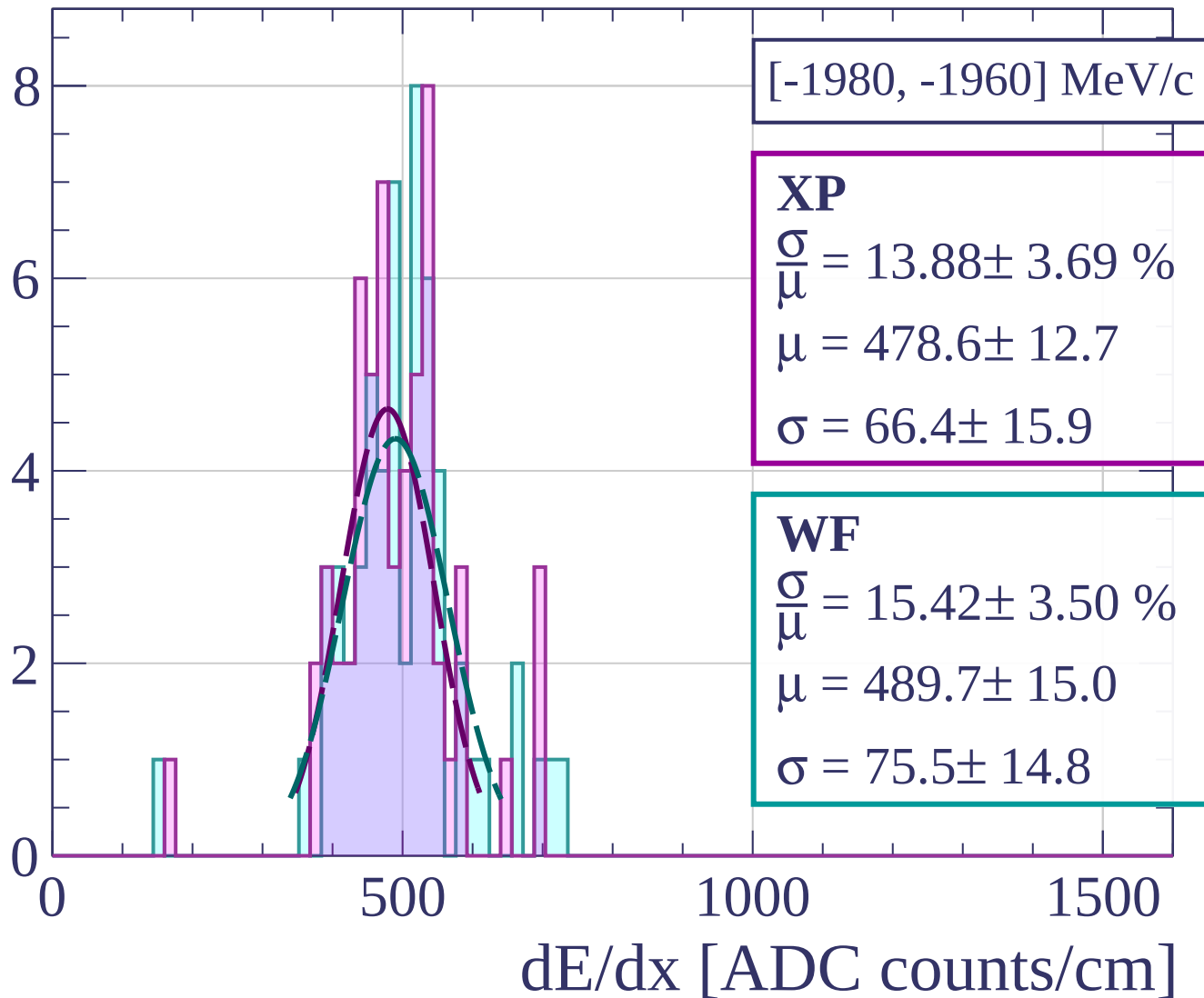
Count



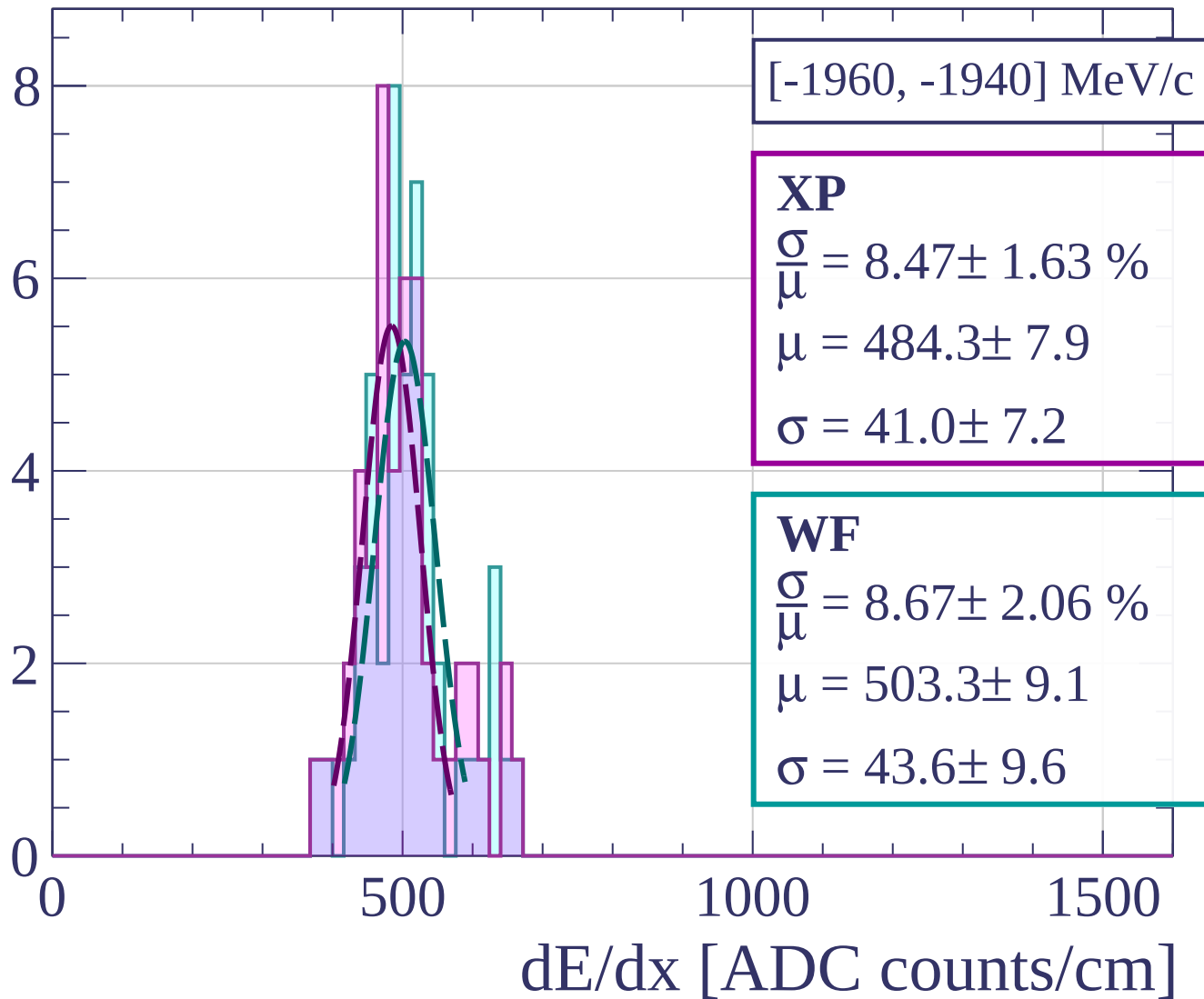
Count



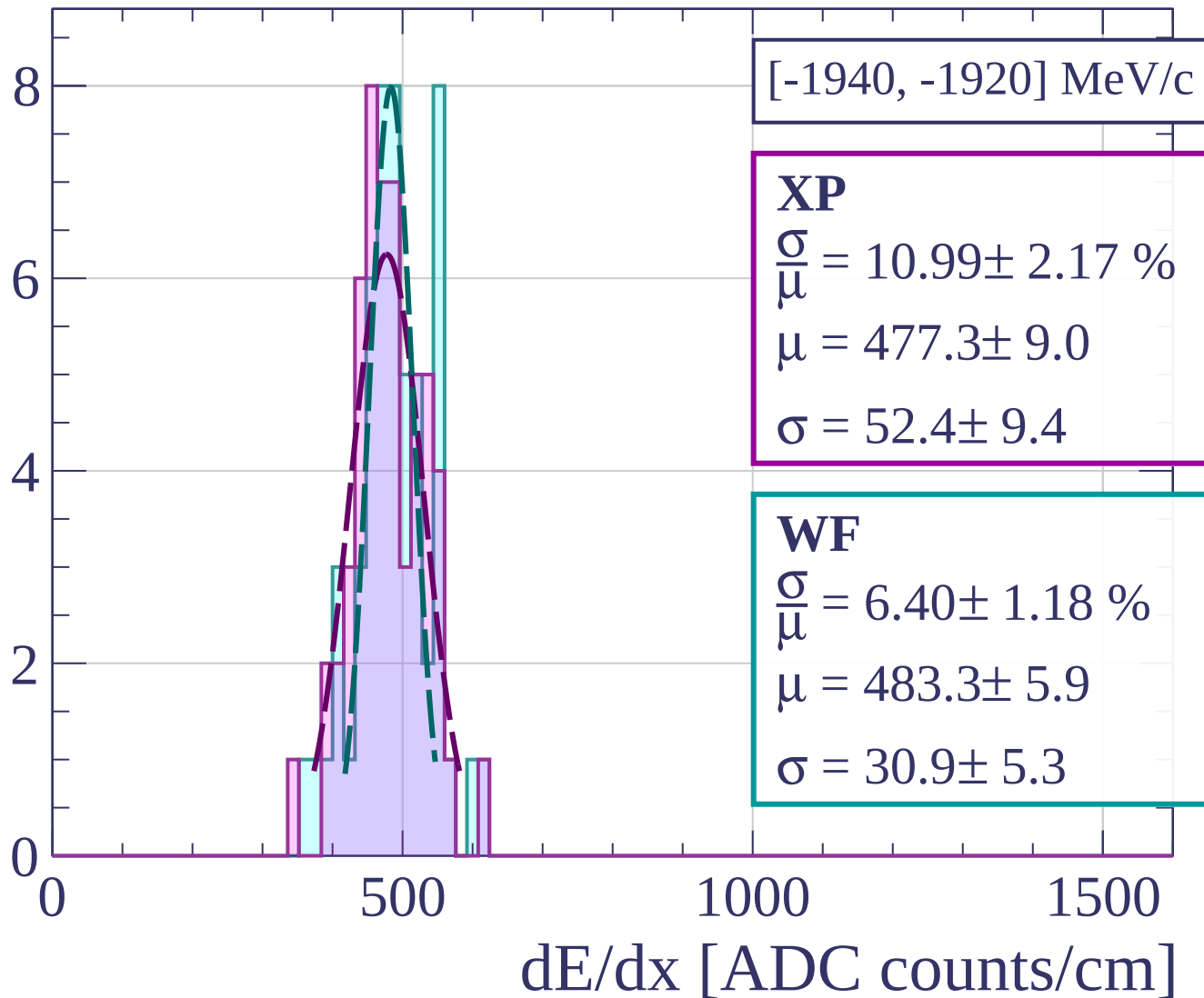
Count



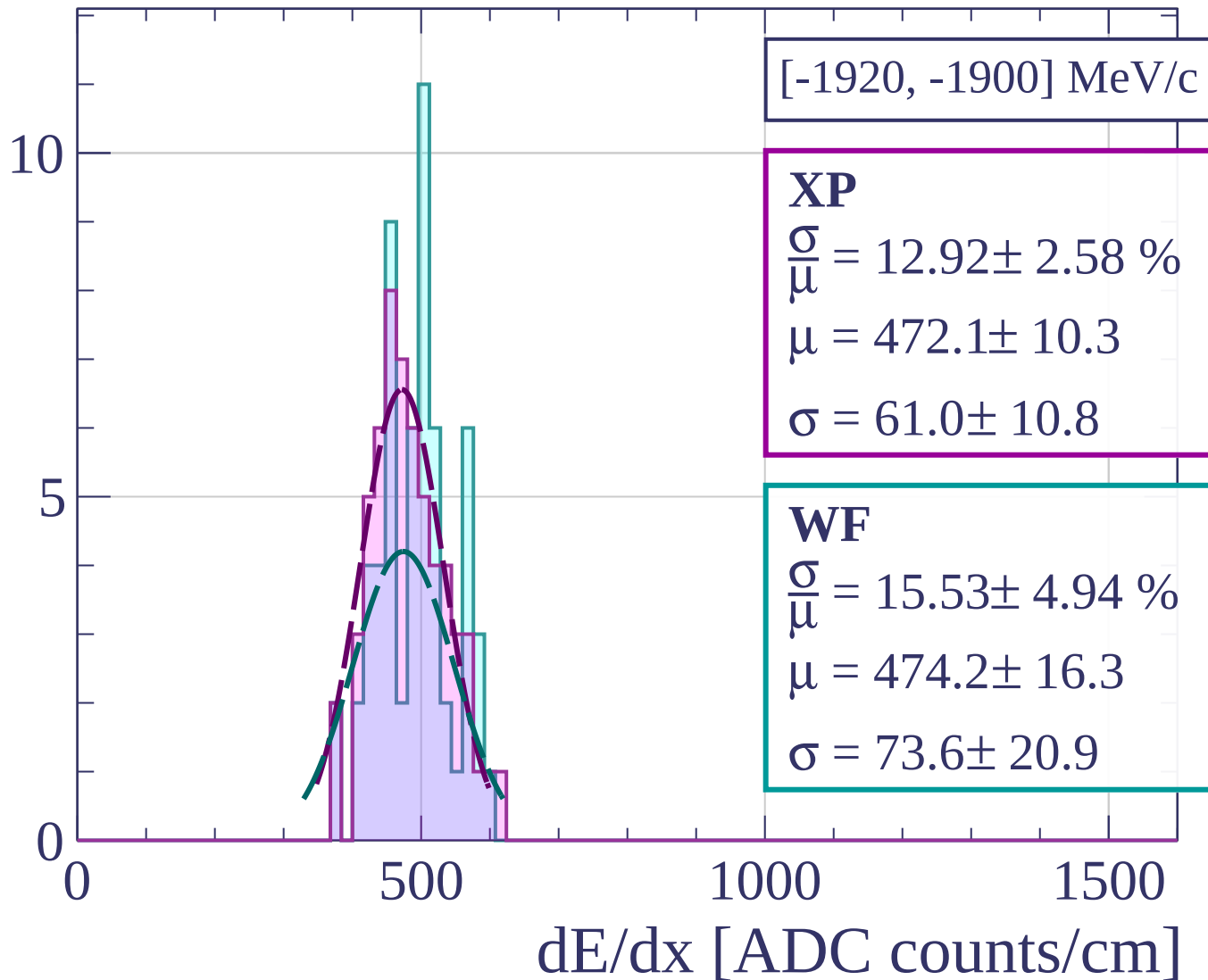
Count



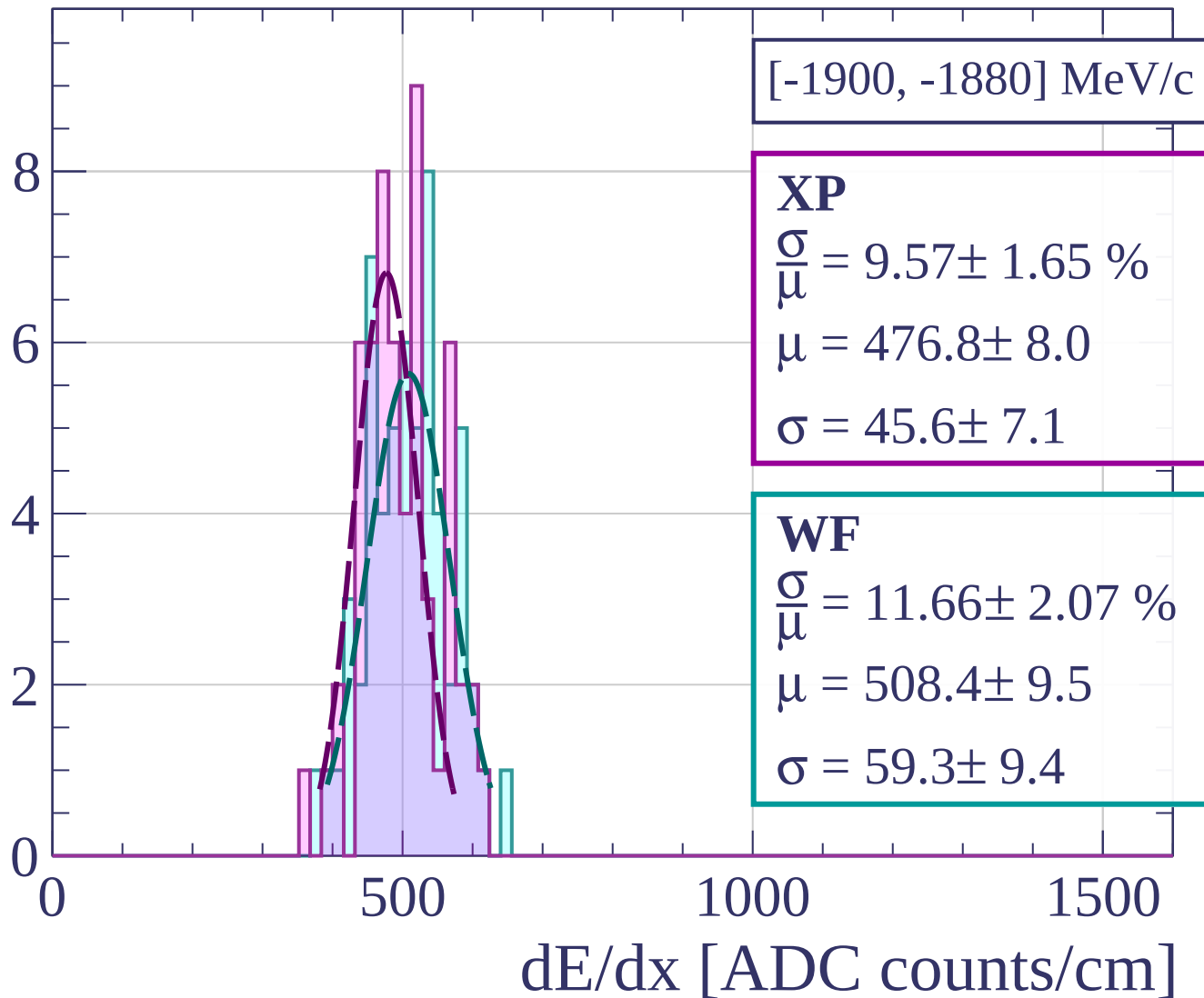
Count



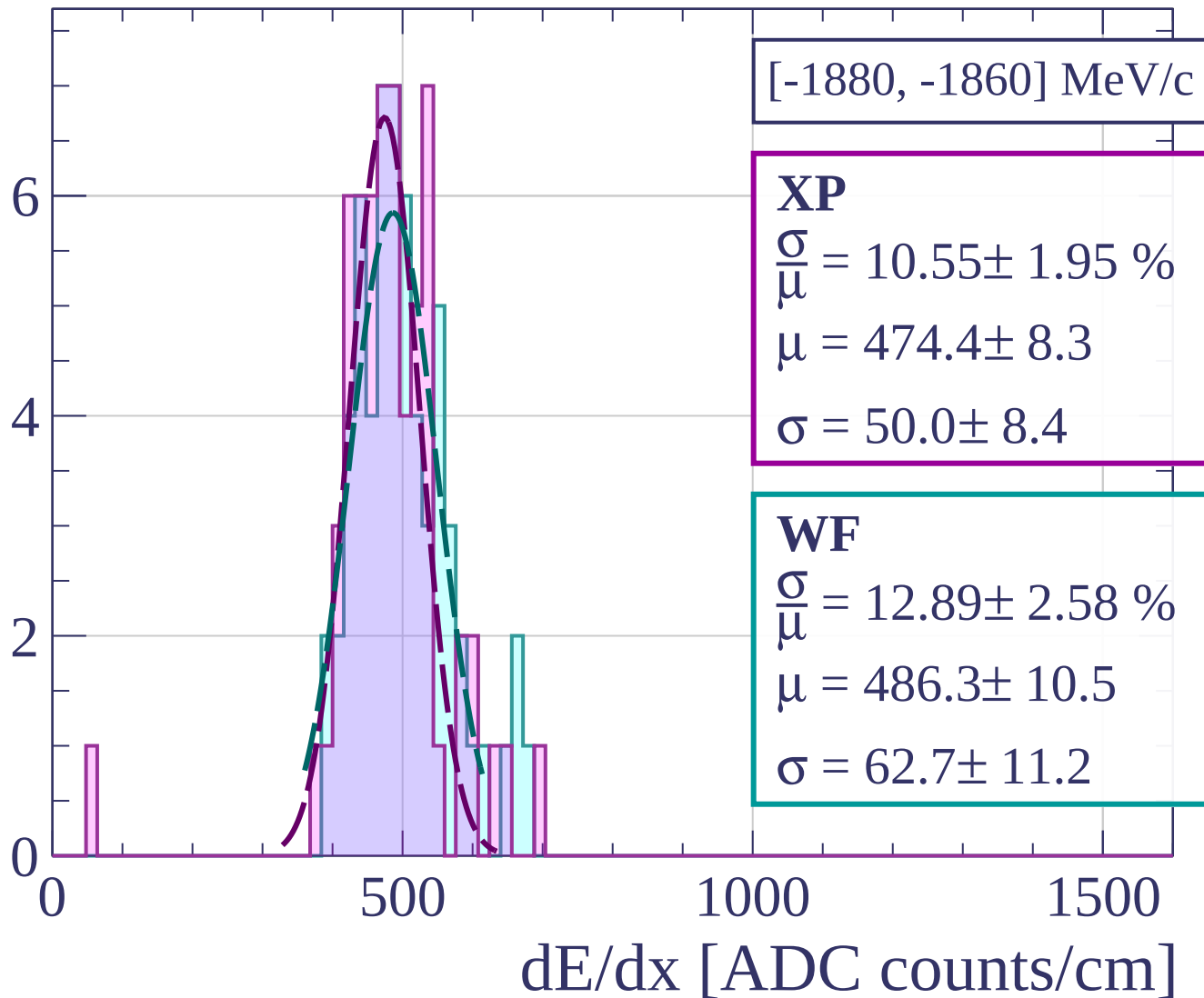
Count



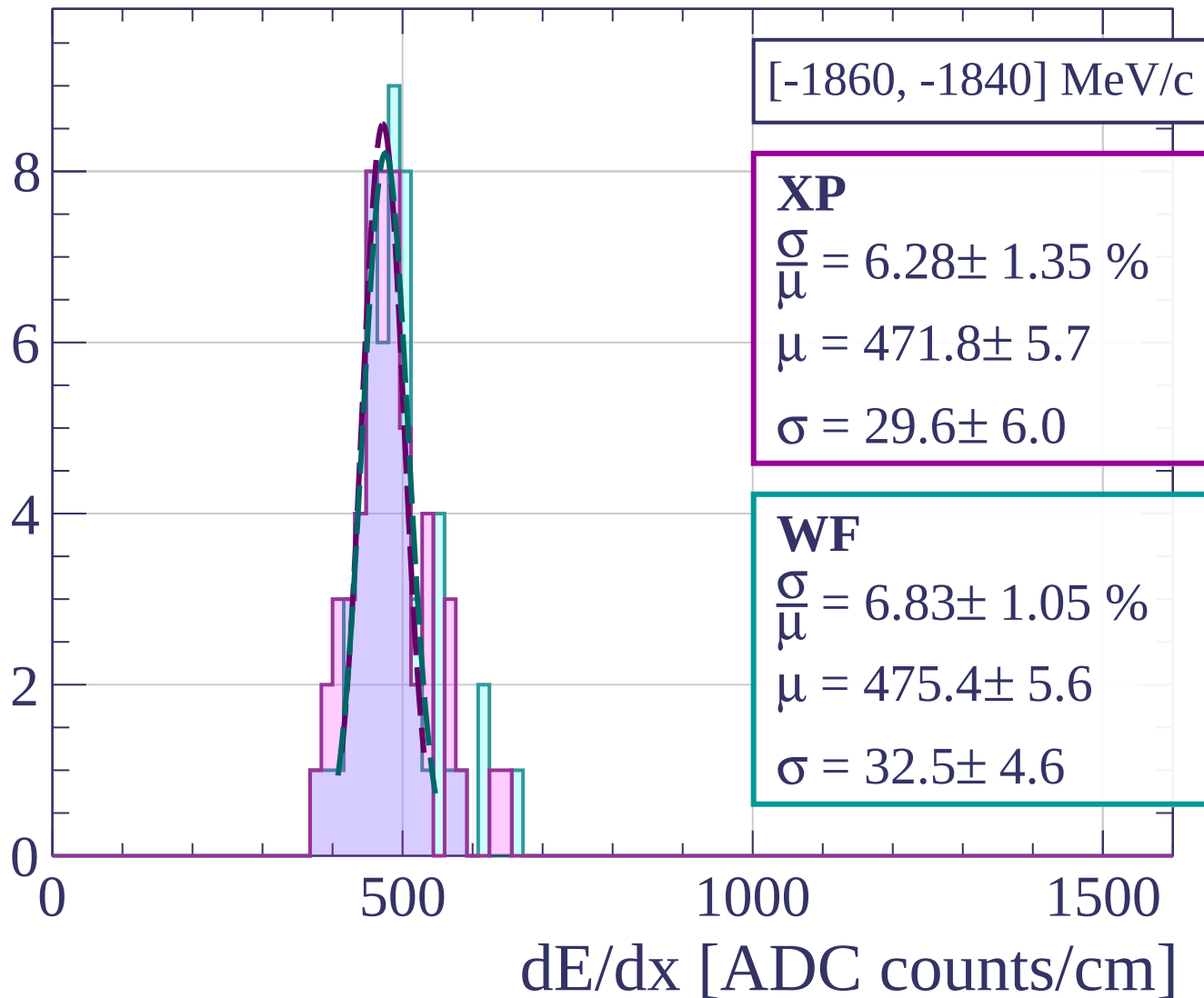
Count



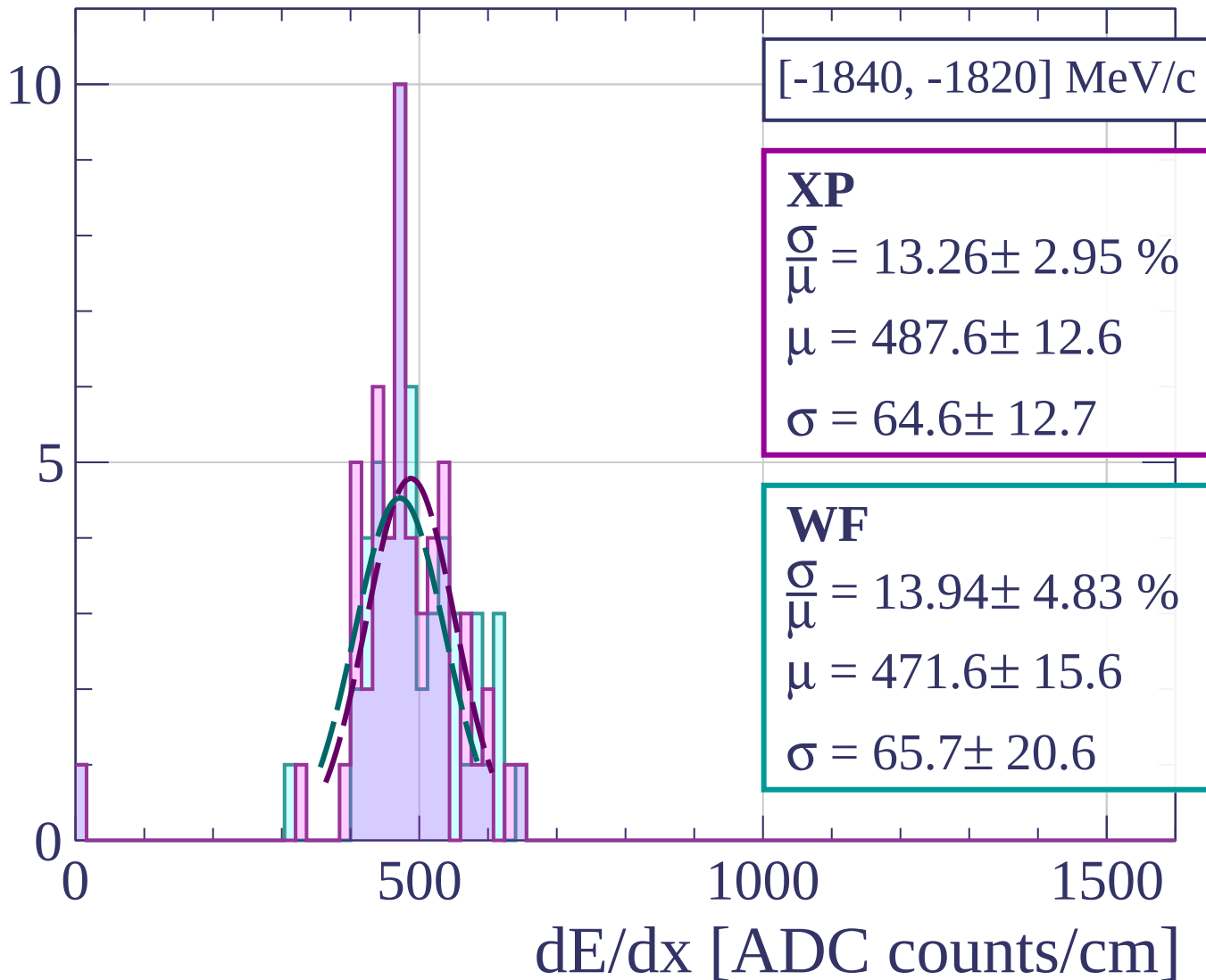
Count



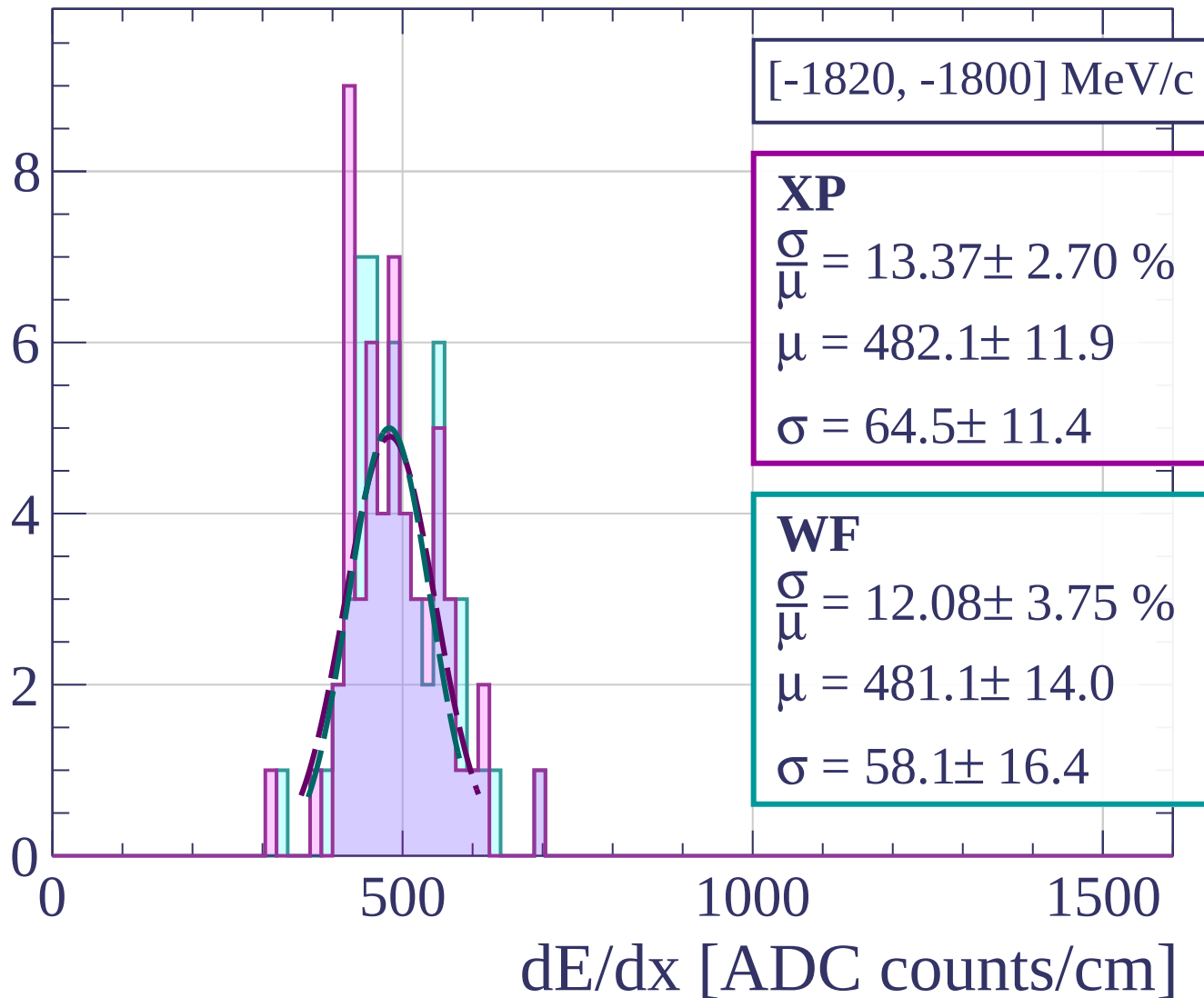
Count



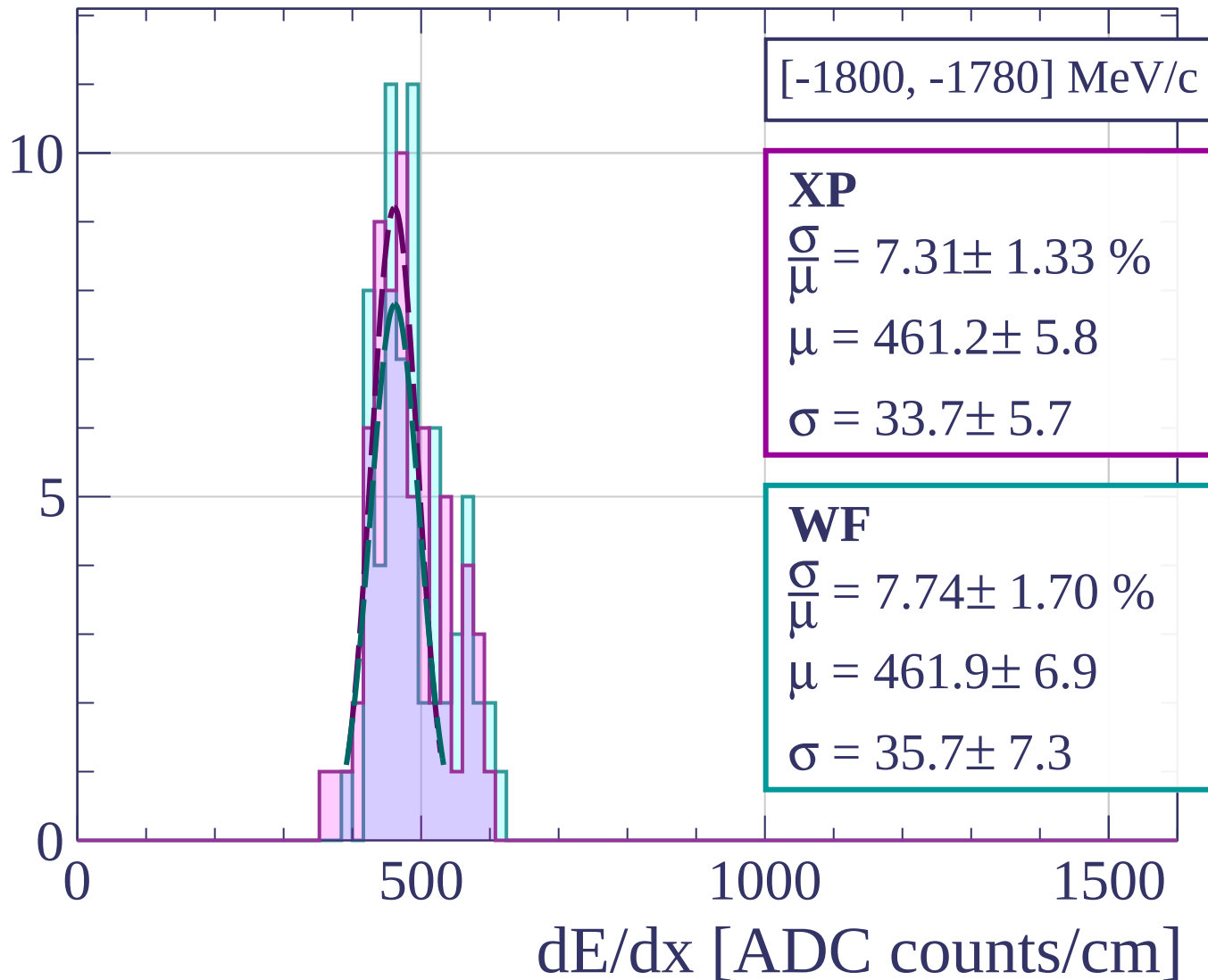
Count



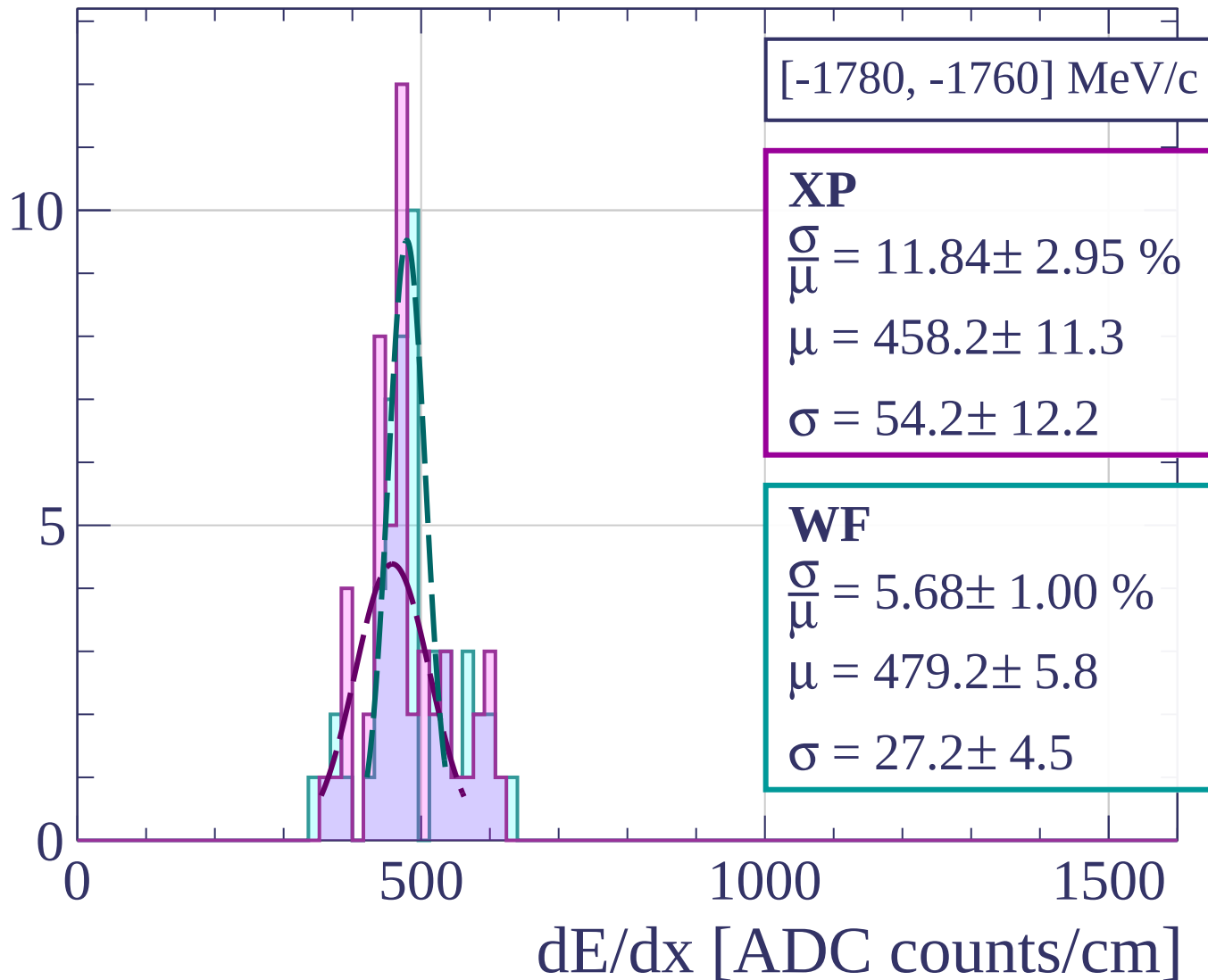
Count



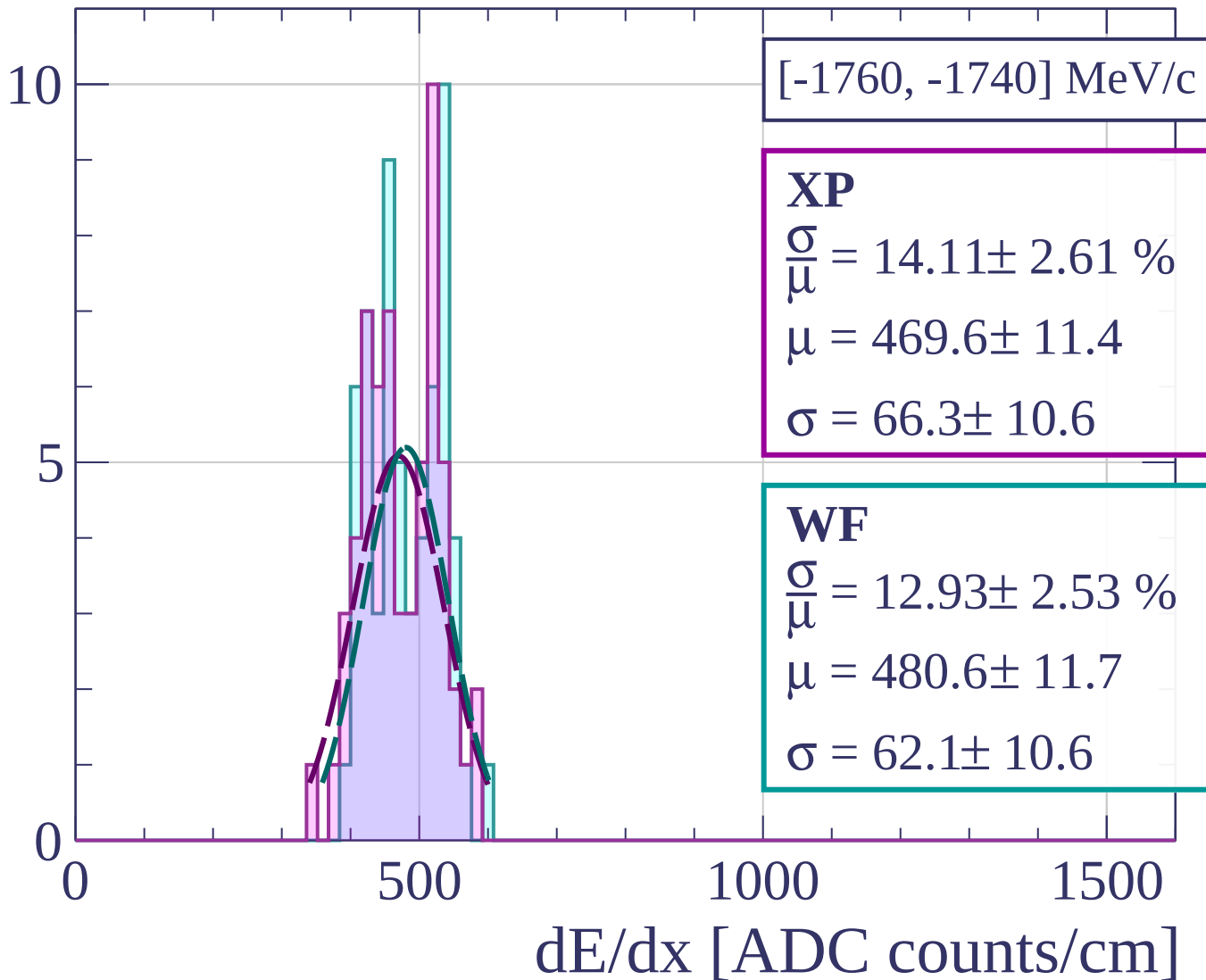
Count



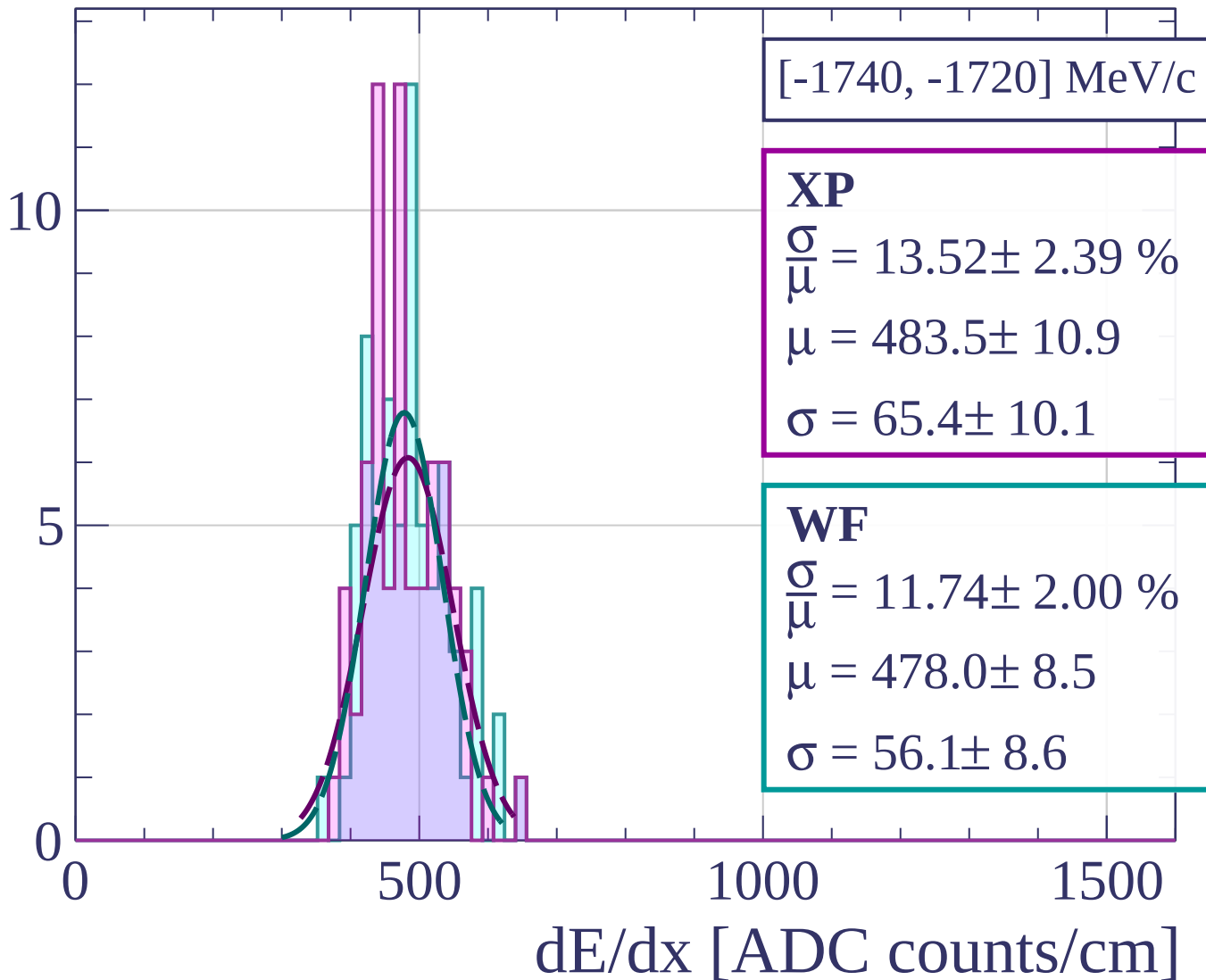
Count



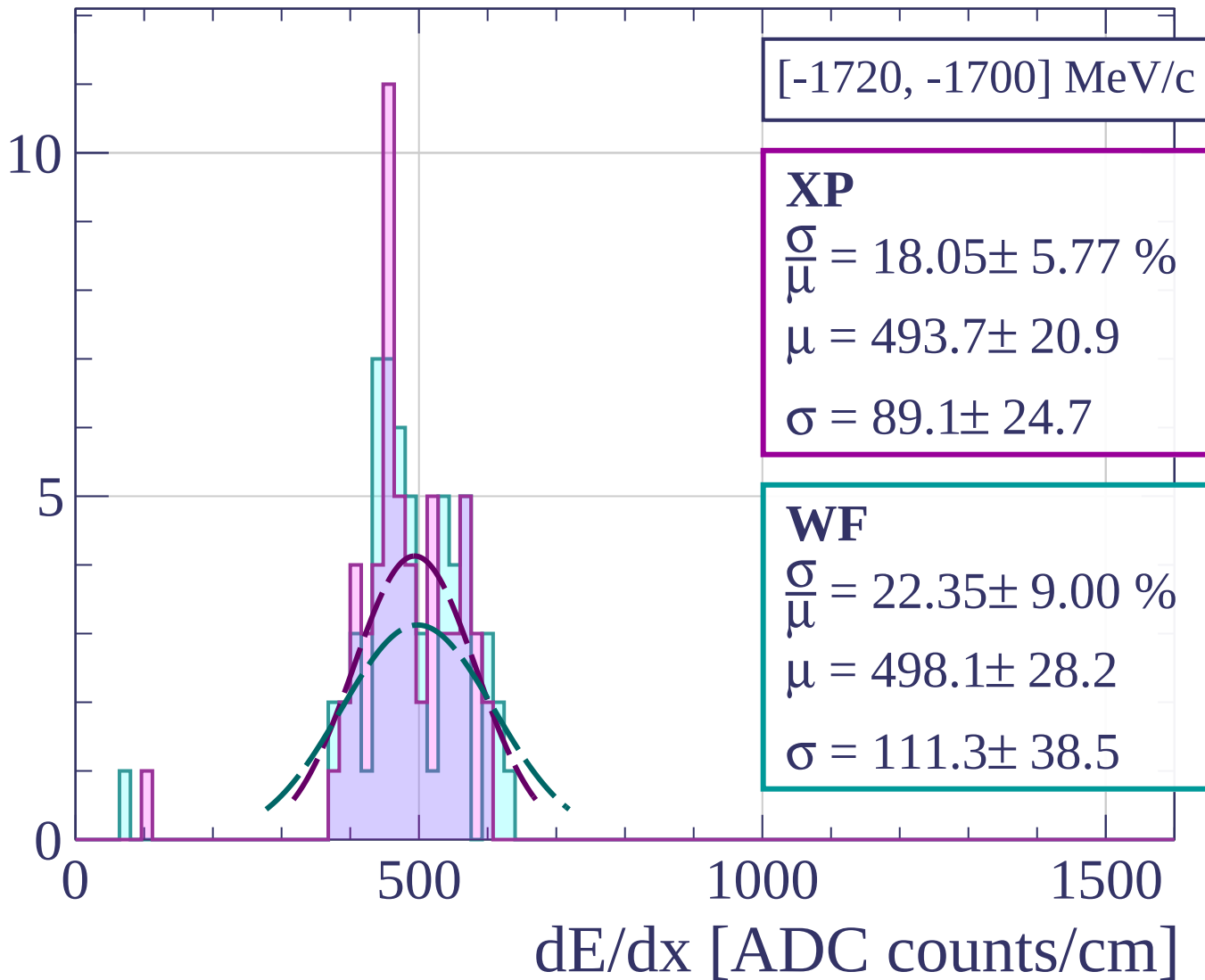
Count



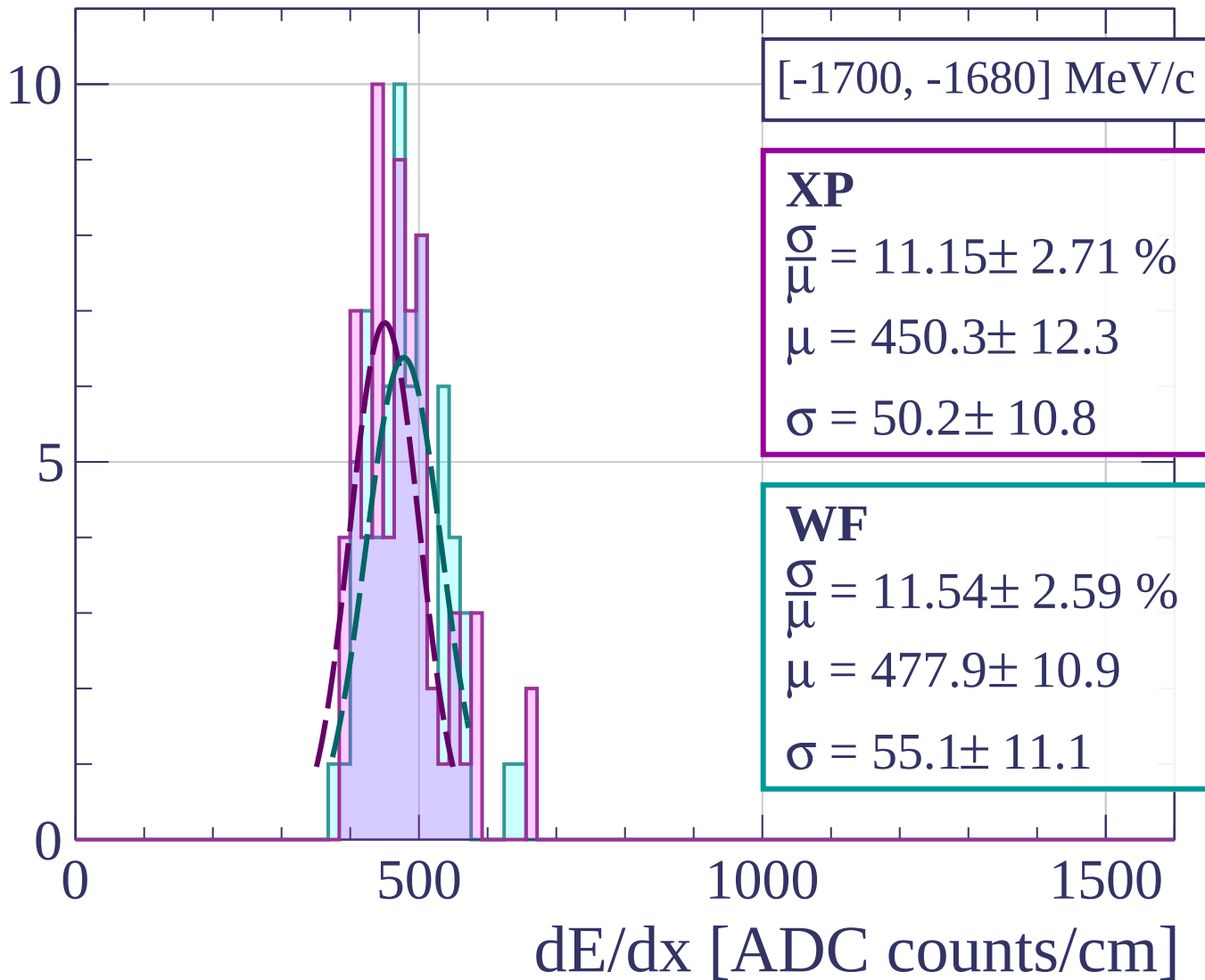
Count



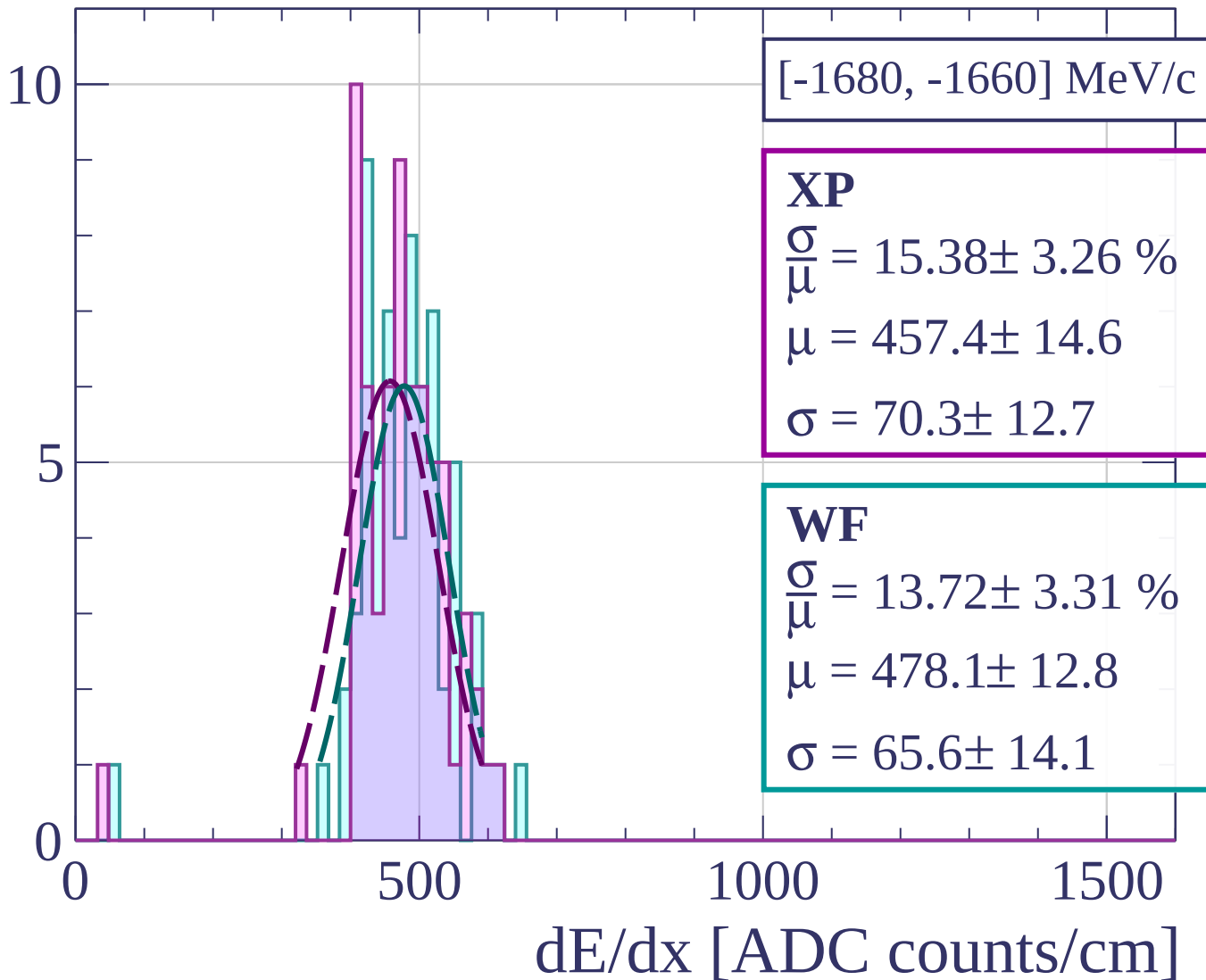
Count



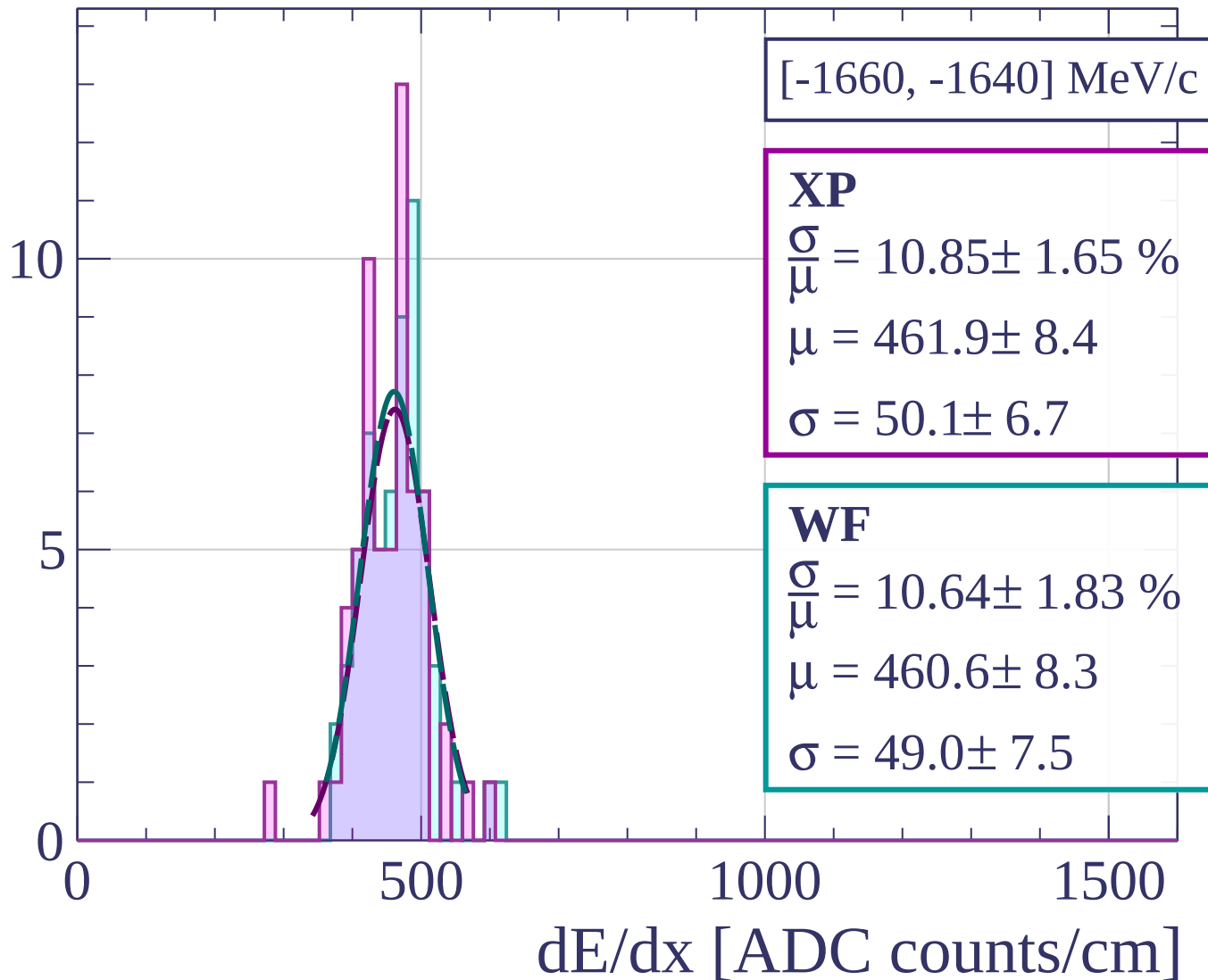
Count



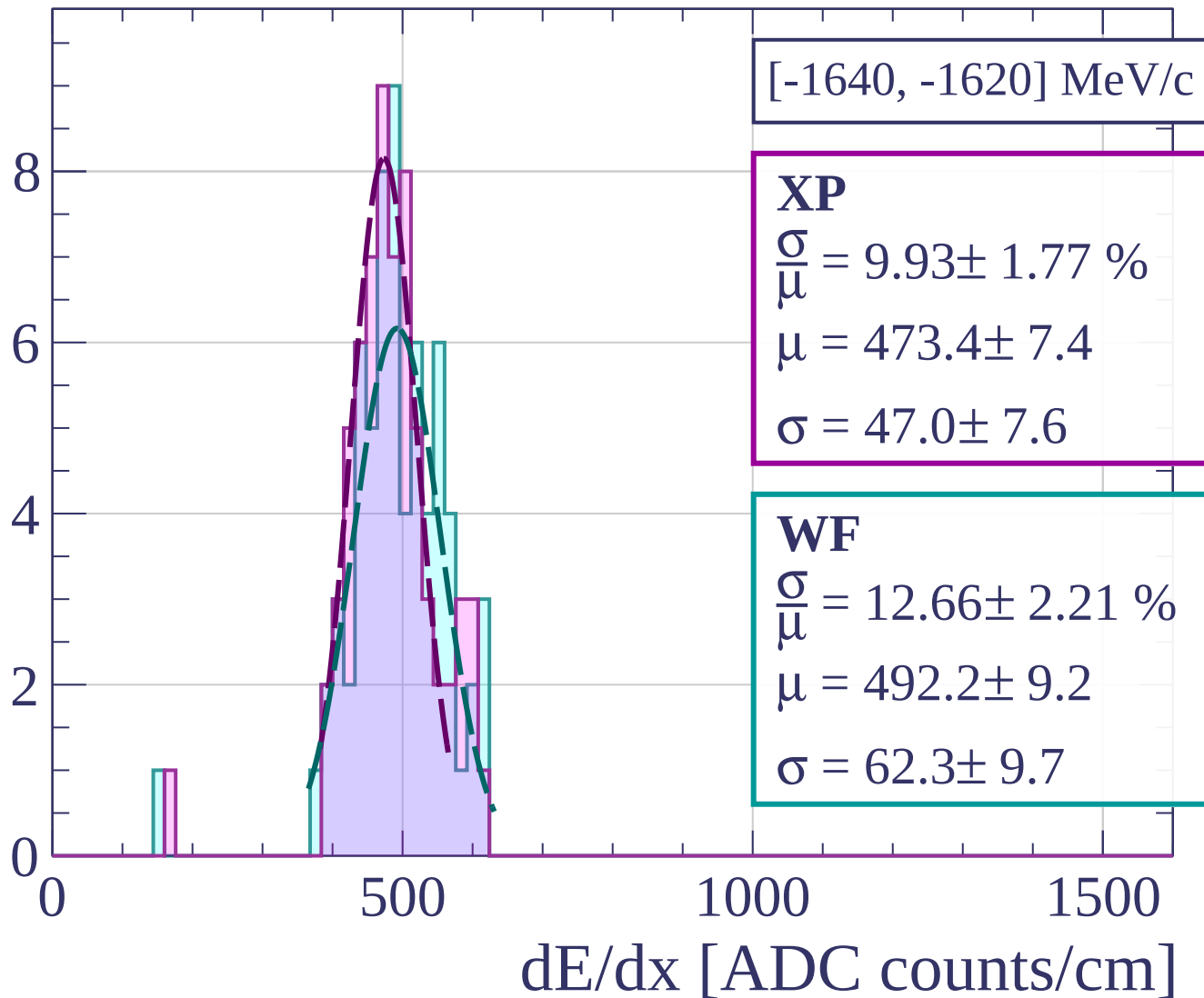
Count



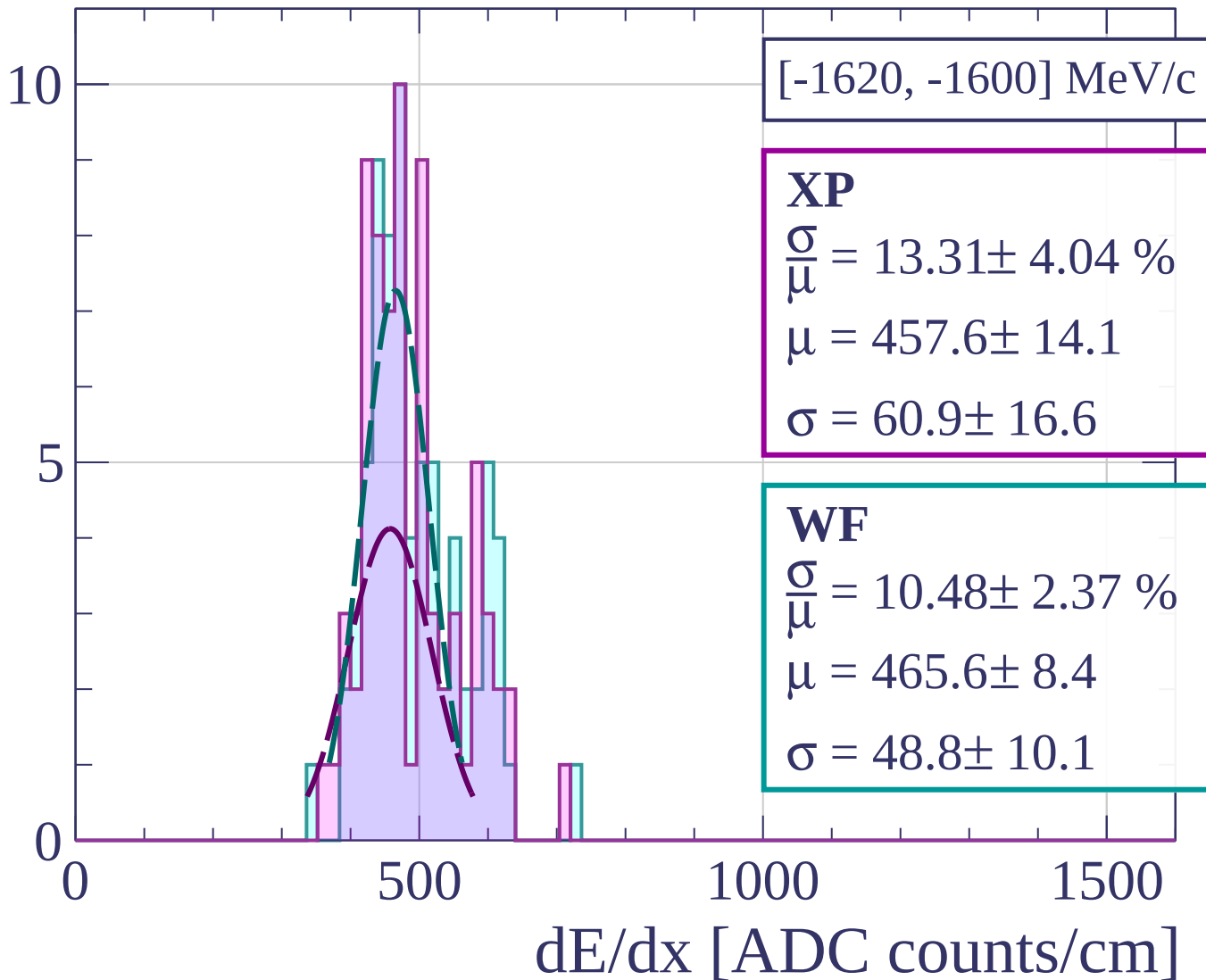
Count



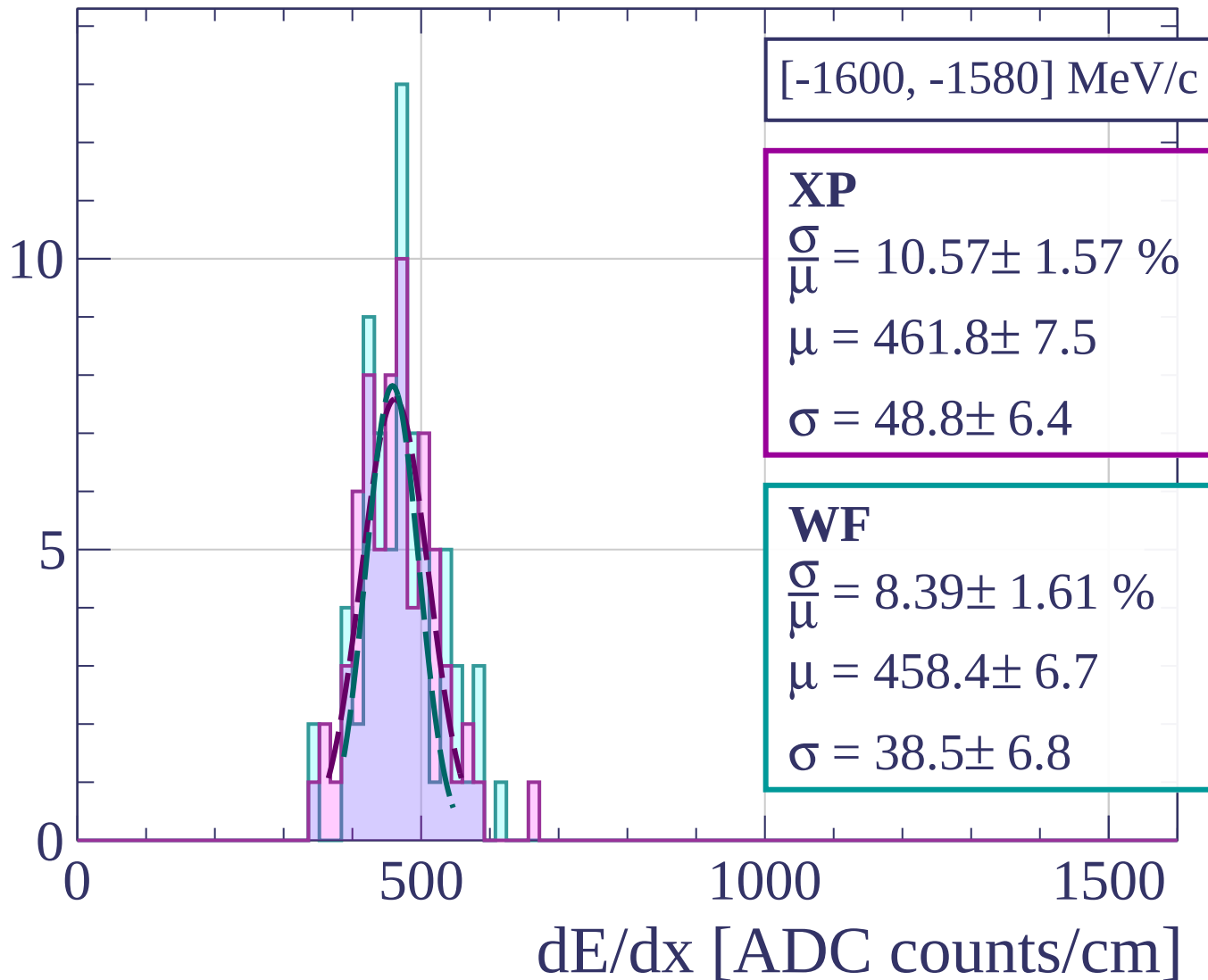
Count



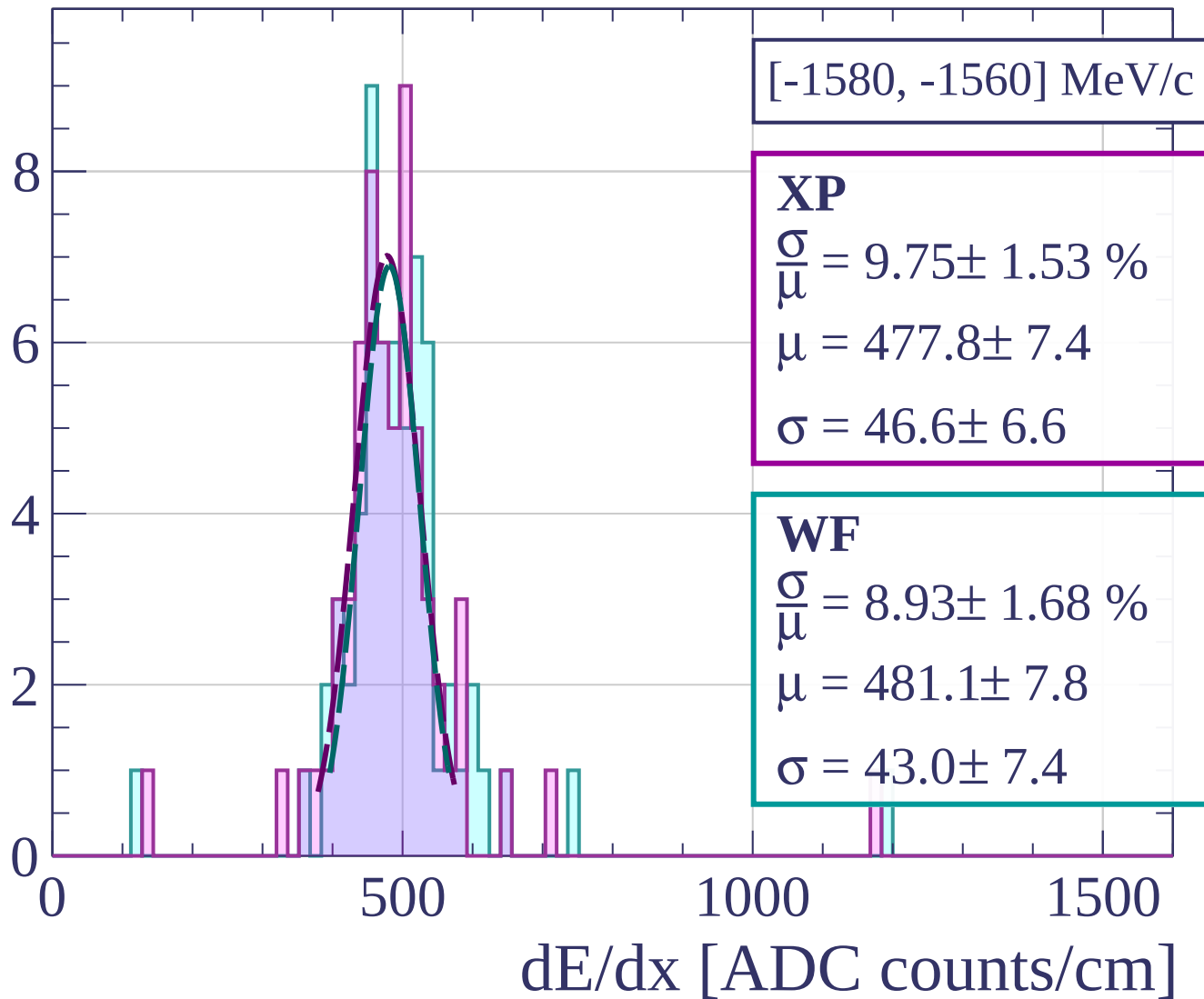
Count



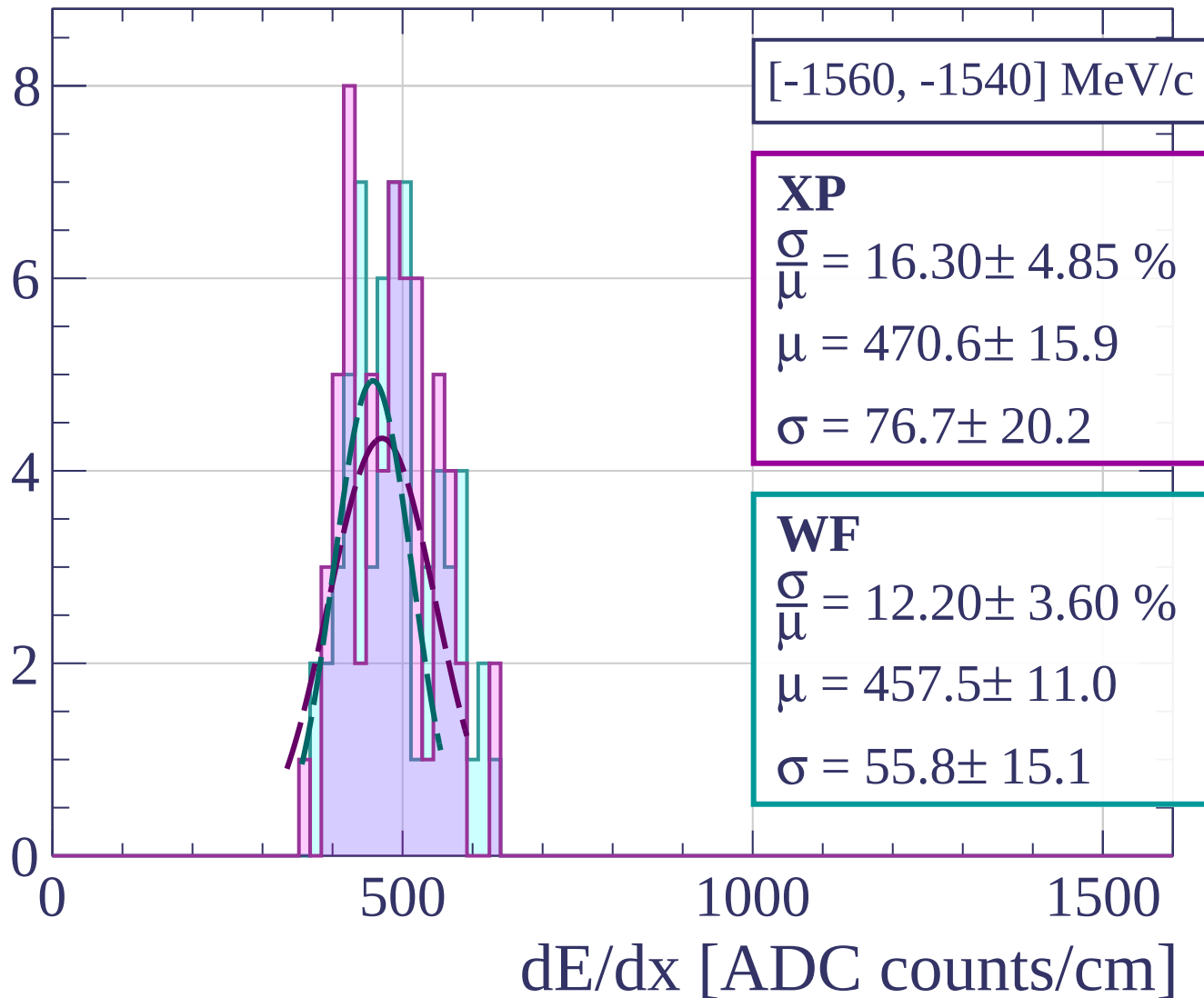
Count



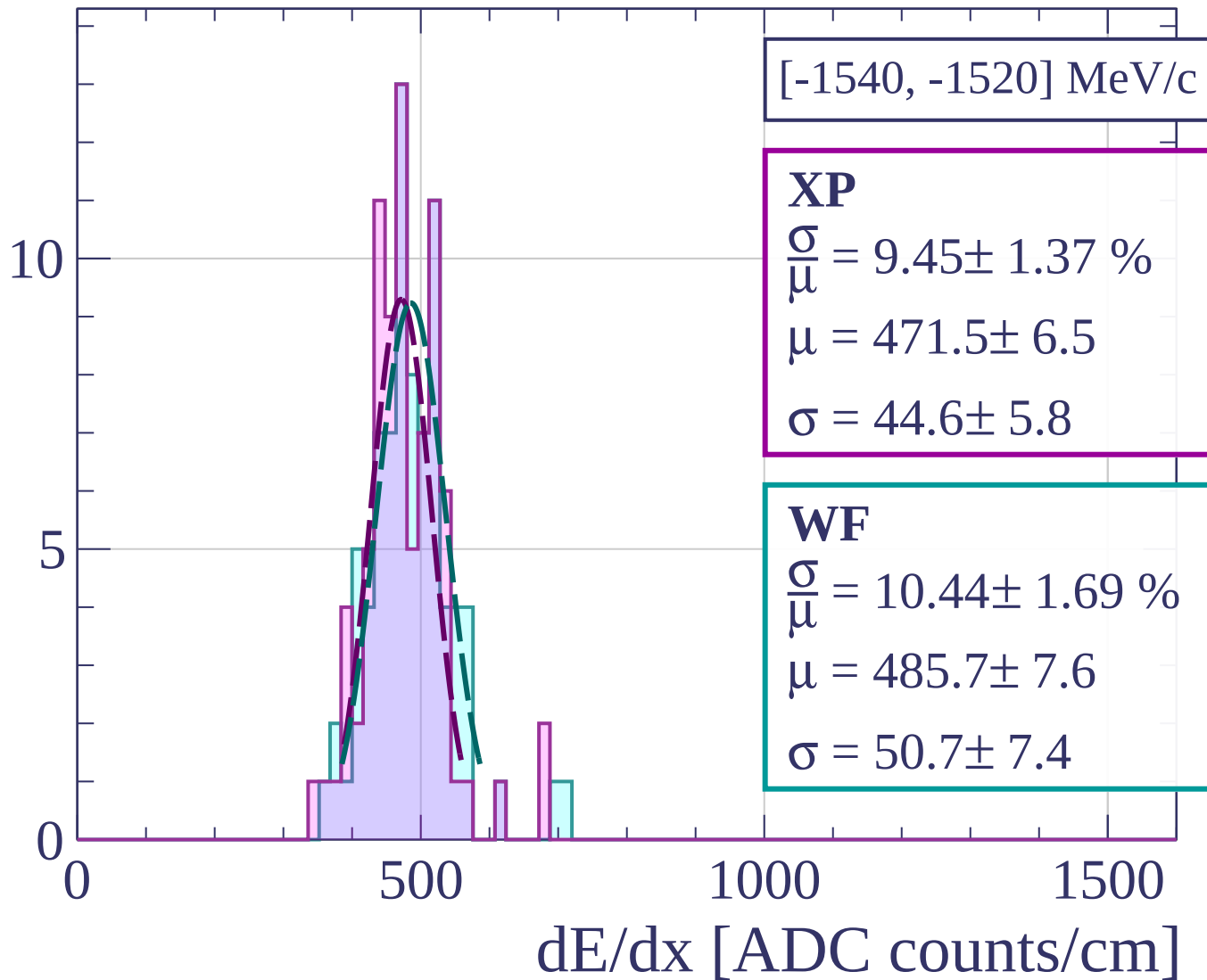
Count



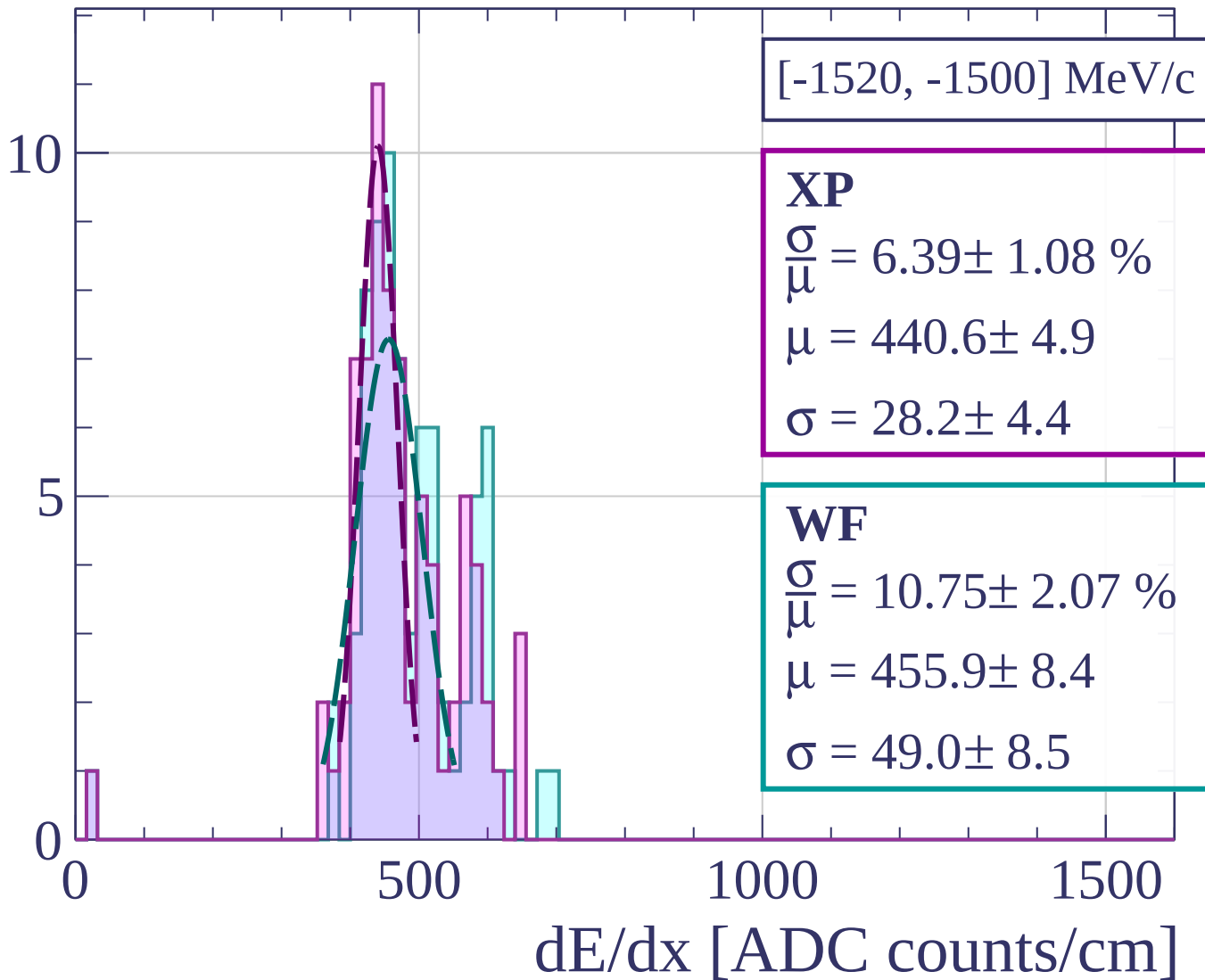
Count



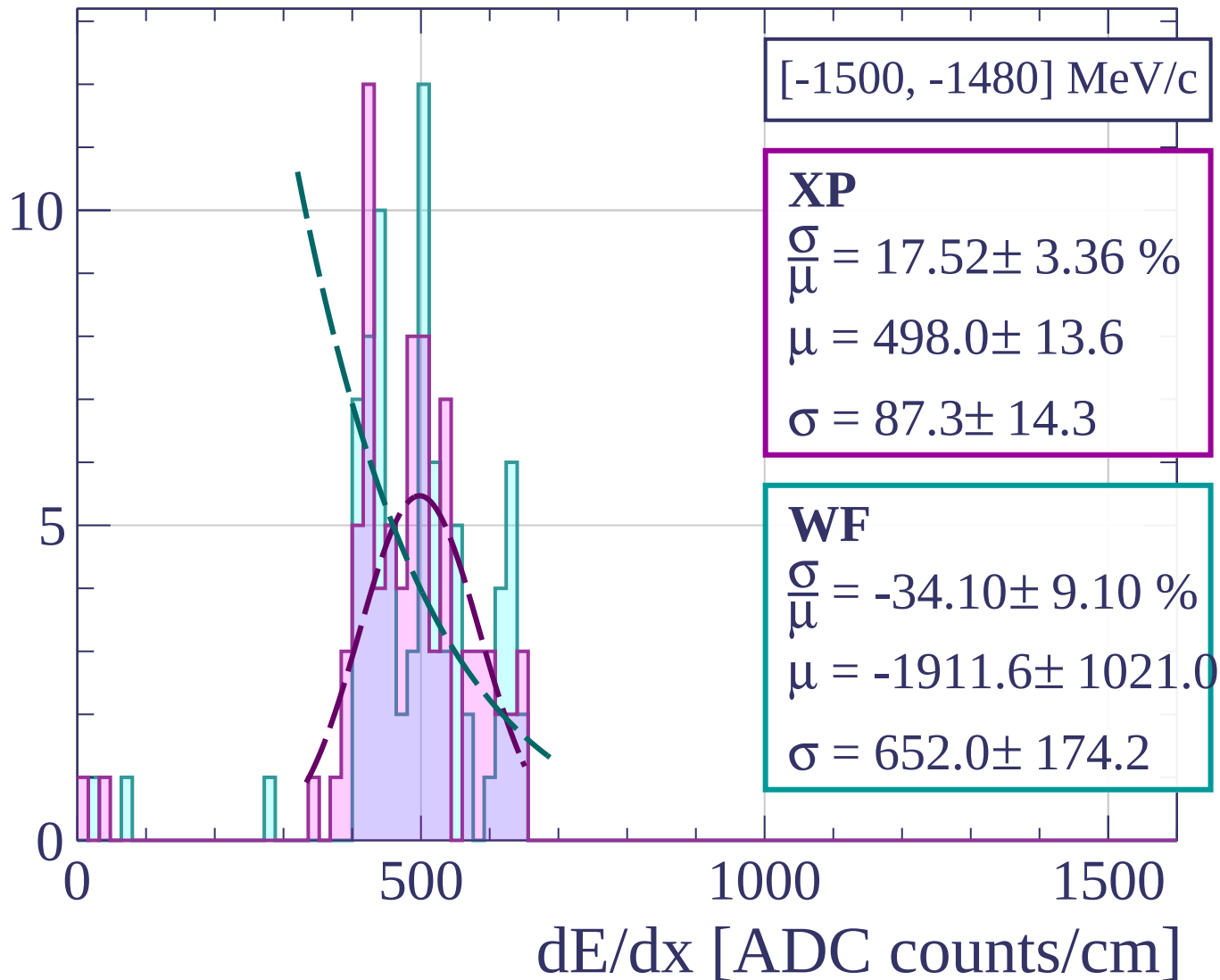
Count



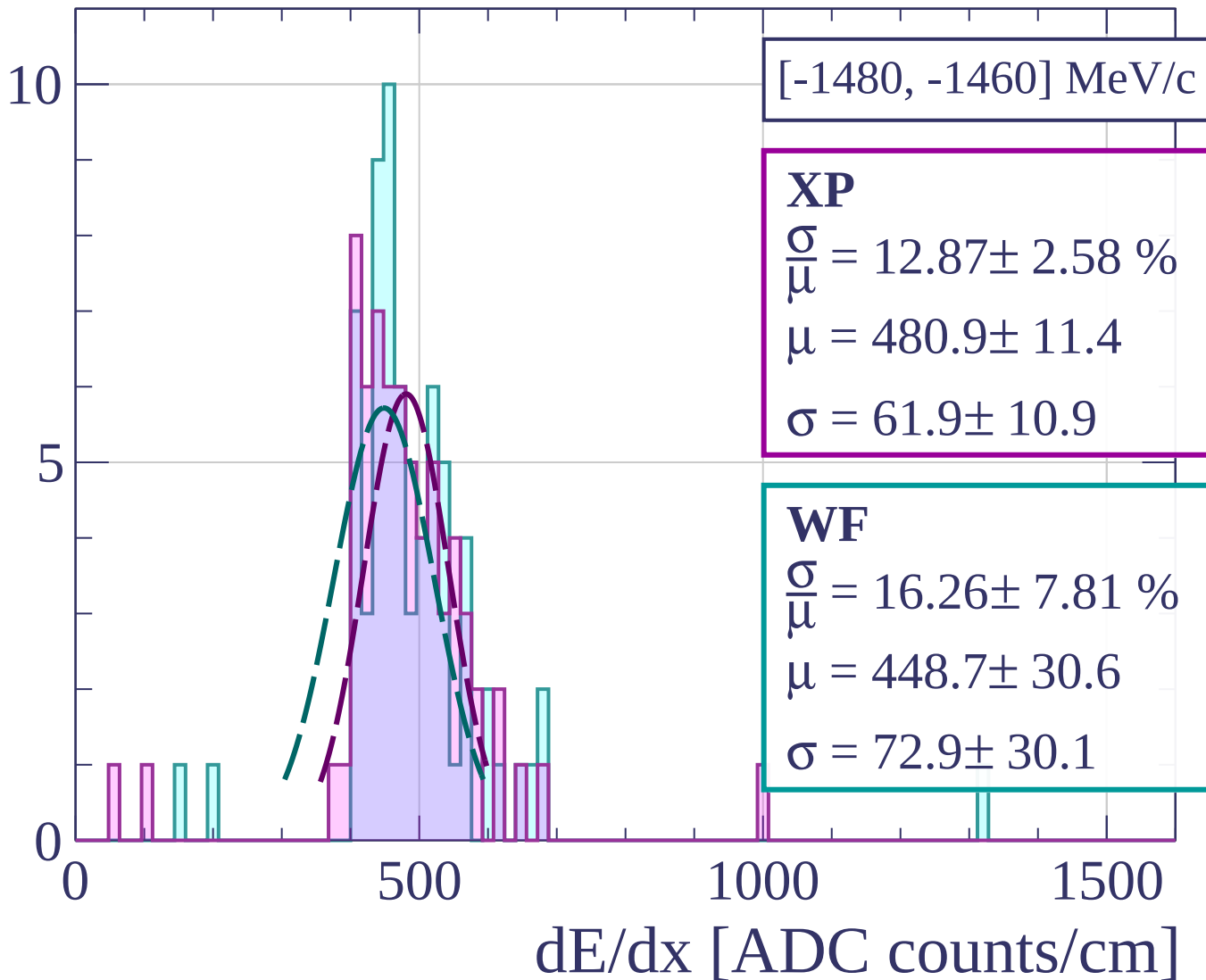
Count



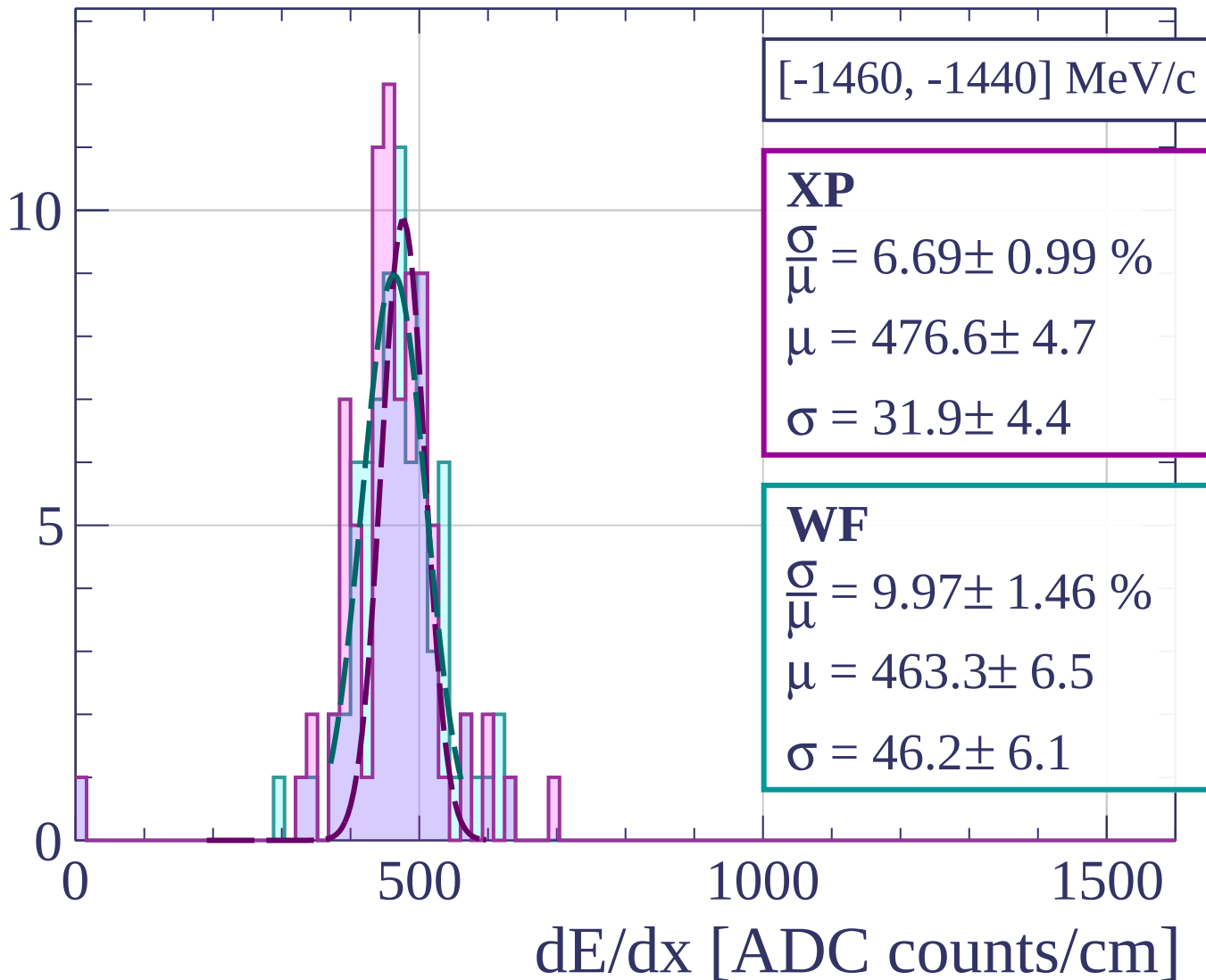
Count



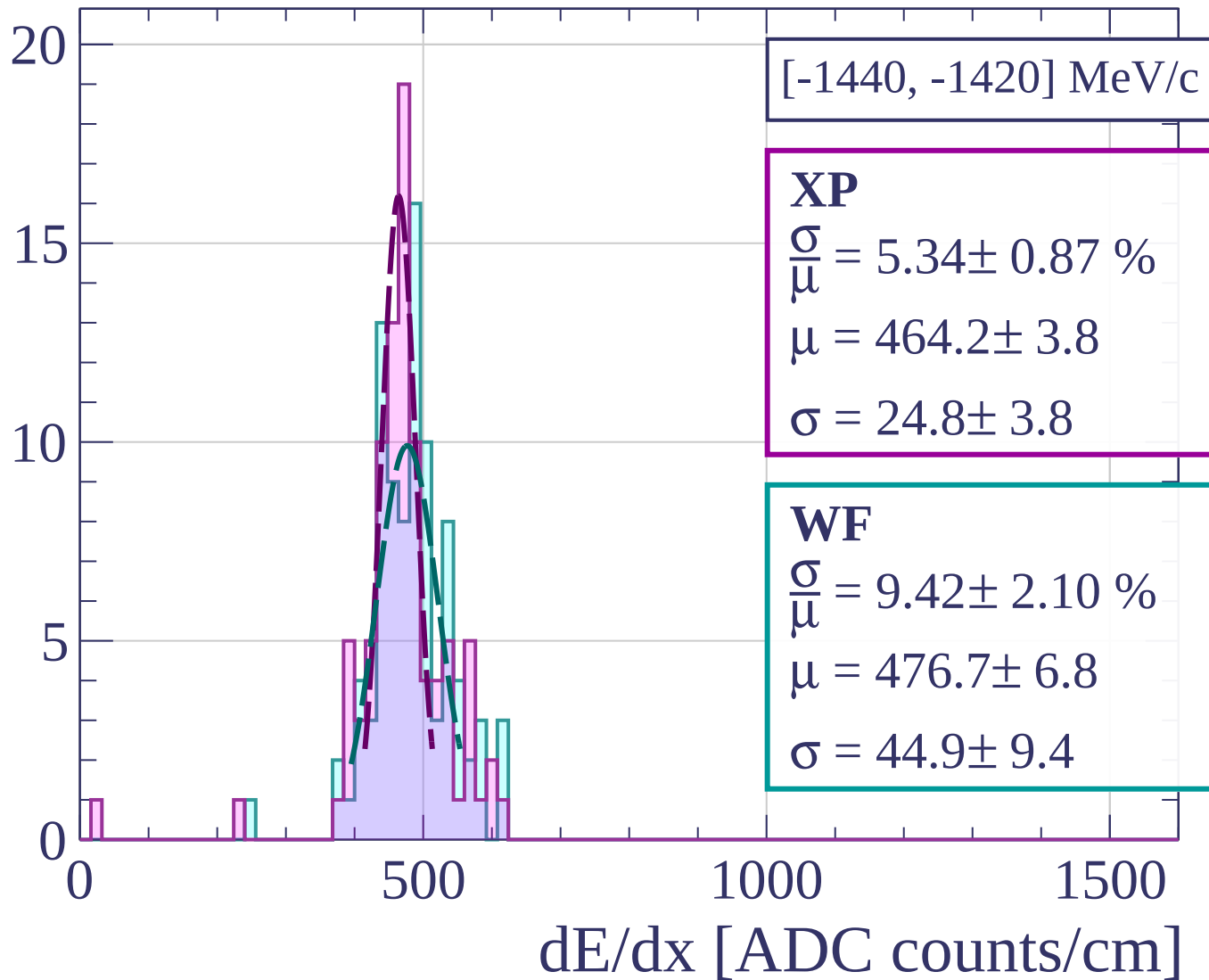
Count



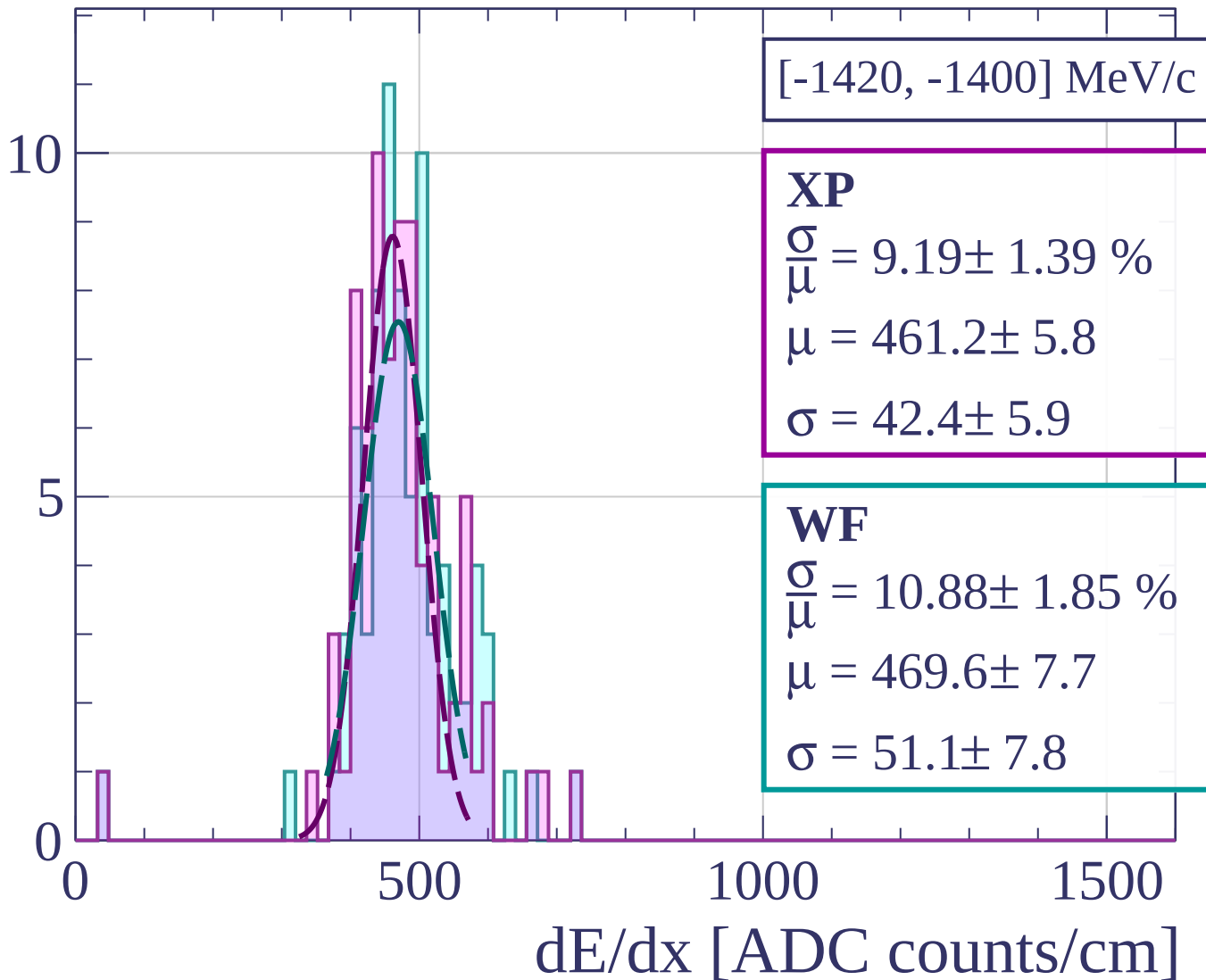
Count



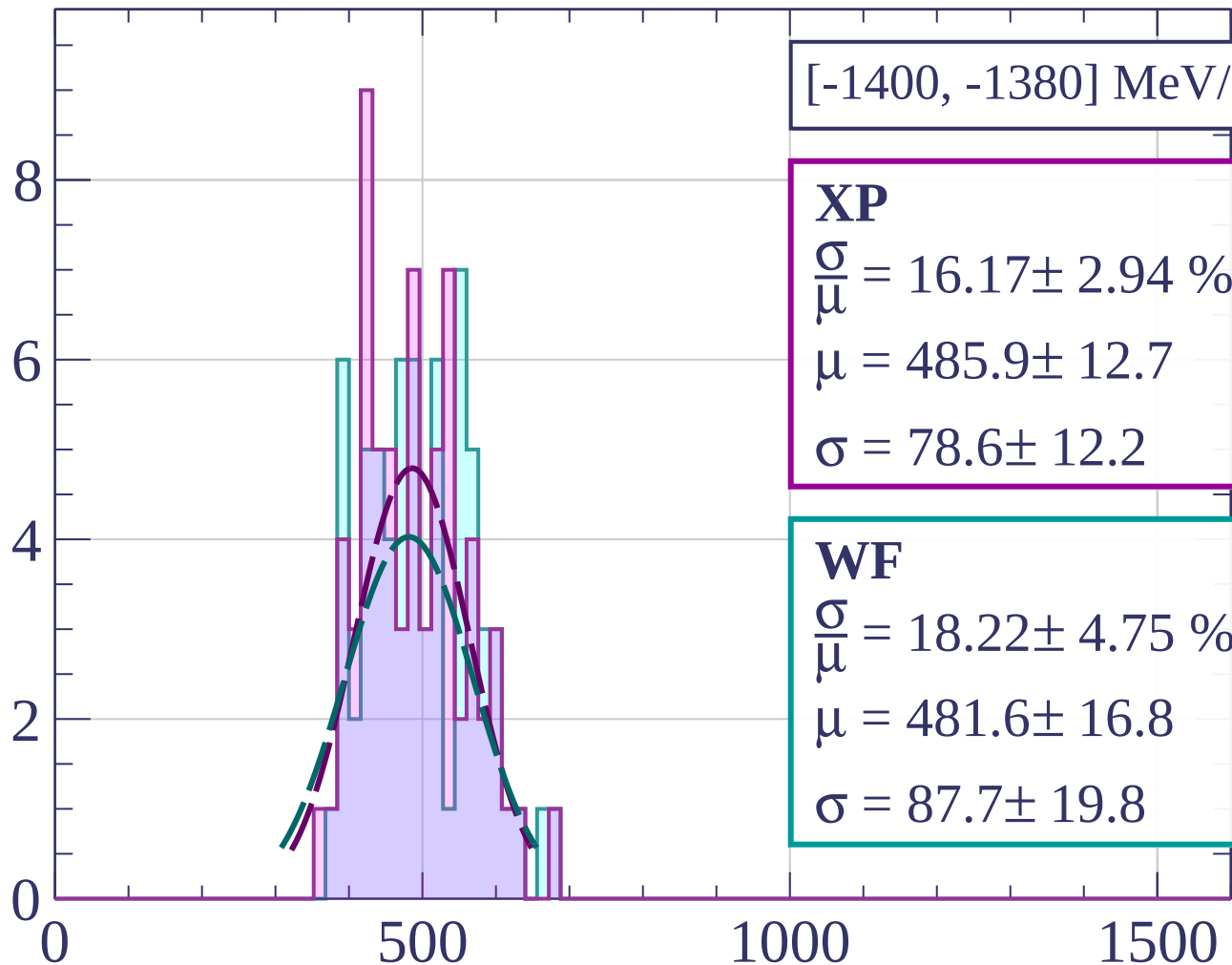
Count



Count



Count



[-1400, -1380] MeV/c

XP

$$\frac{\sigma}{\mu} = 16.17 \pm 2.94 \%$$

$$\mu = 485.9 \pm 12.7$$

$$\sigma = 78.6 \pm 12.2$$

WF

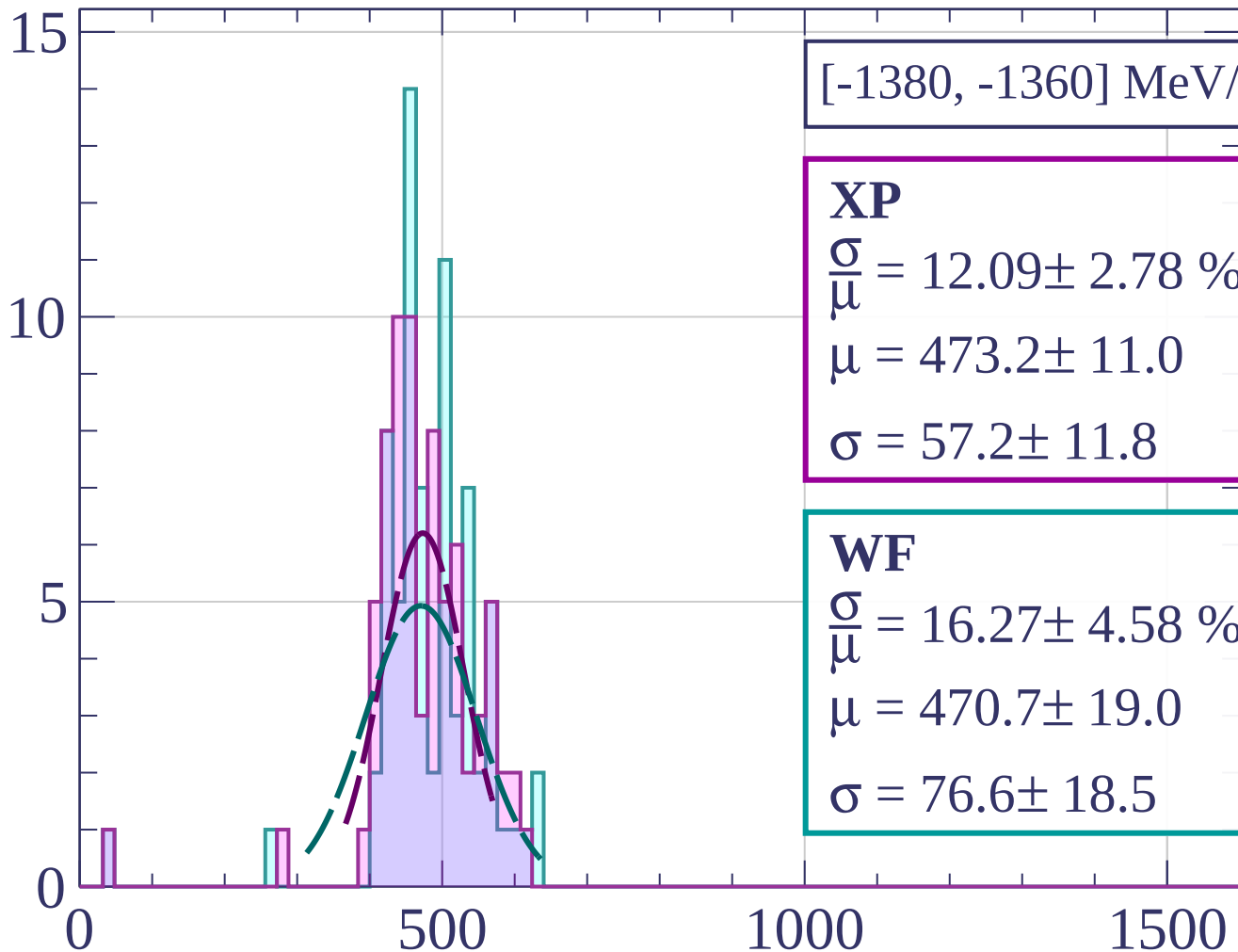
$$\frac{\sigma}{\mu} = 18.22 \pm 4.75 \%$$

$$\mu = 481.6 \pm 16.8$$

$$\sigma = 87.7 \pm 19.8$$

dE/dx [ADC counts/cm]

Count



[-1380, -1360] MeV/c

XP

$$\frac{\sigma}{\mu} = 12.09 \pm 2.78 \%$$

$$\mu = 473.2 \pm 11.0$$

$$\sigma = 57.2 \pm 11.8$$

WF

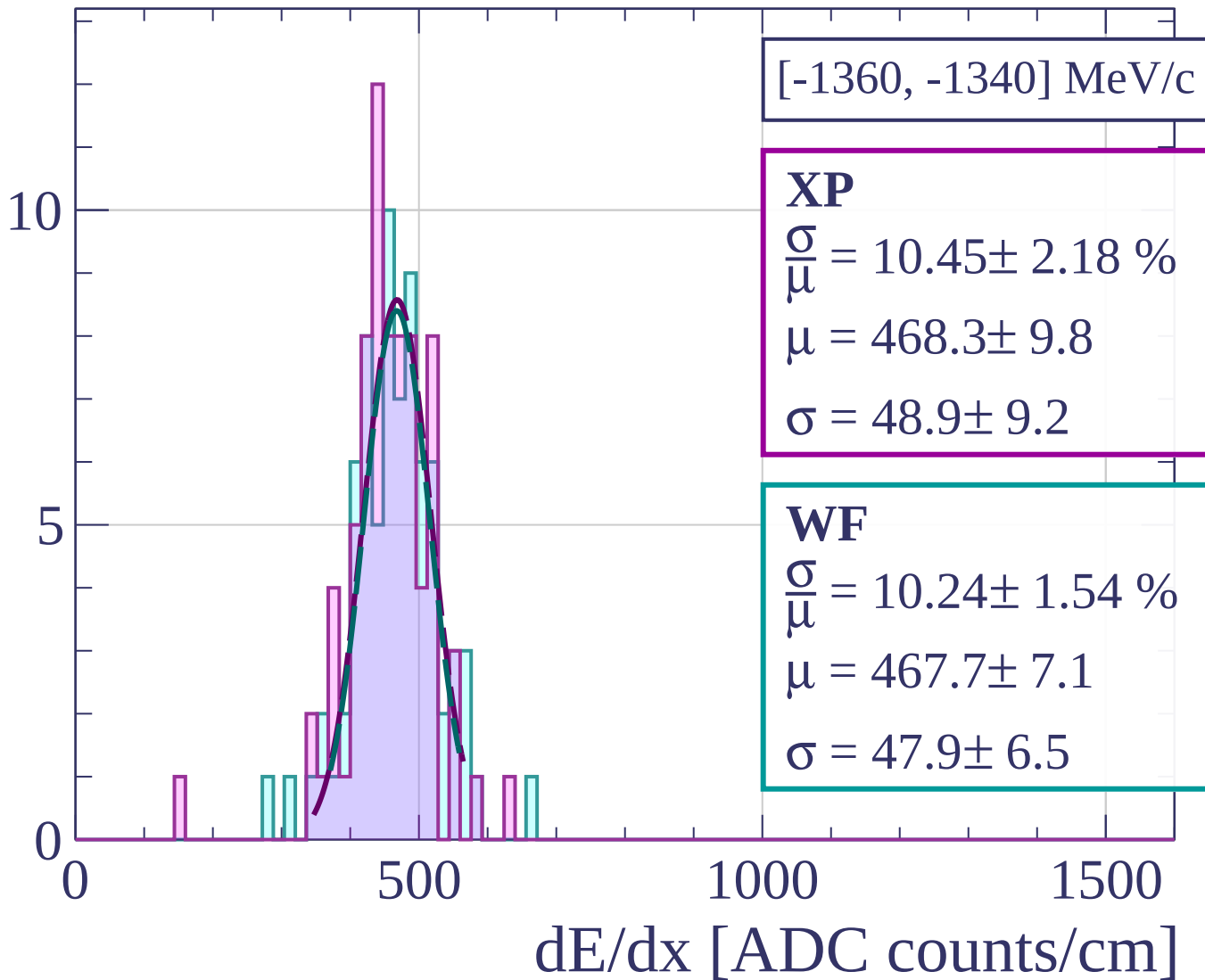
$$\frac{\sigma}{\mu} = 16.27 \pm 4.58 \%$$

$$\mu = 470.7 \pm 19.0$$

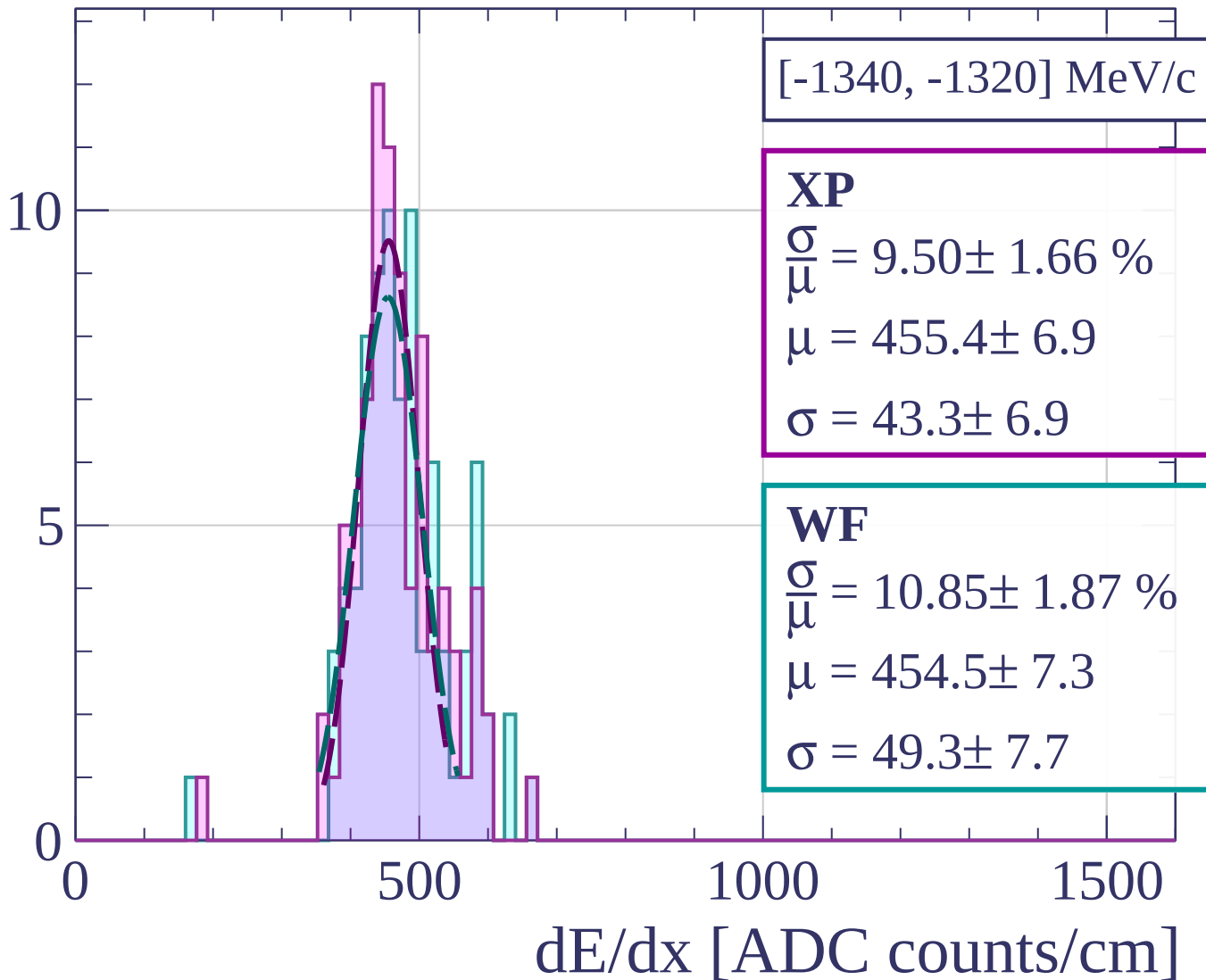
$$\sigma = 76.6 \pm 18.5$$

dE/dx [ADC counts/cm]

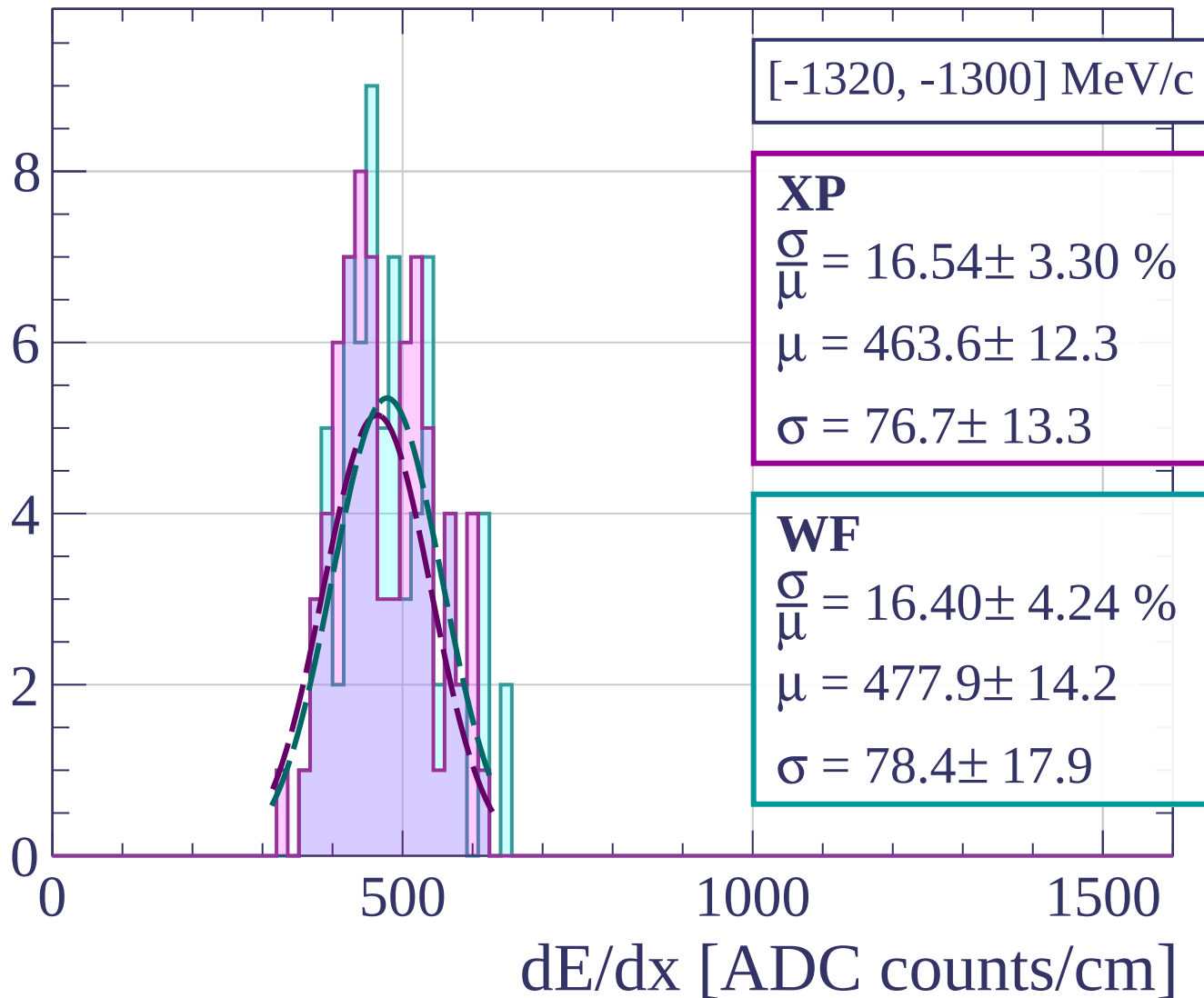
Count



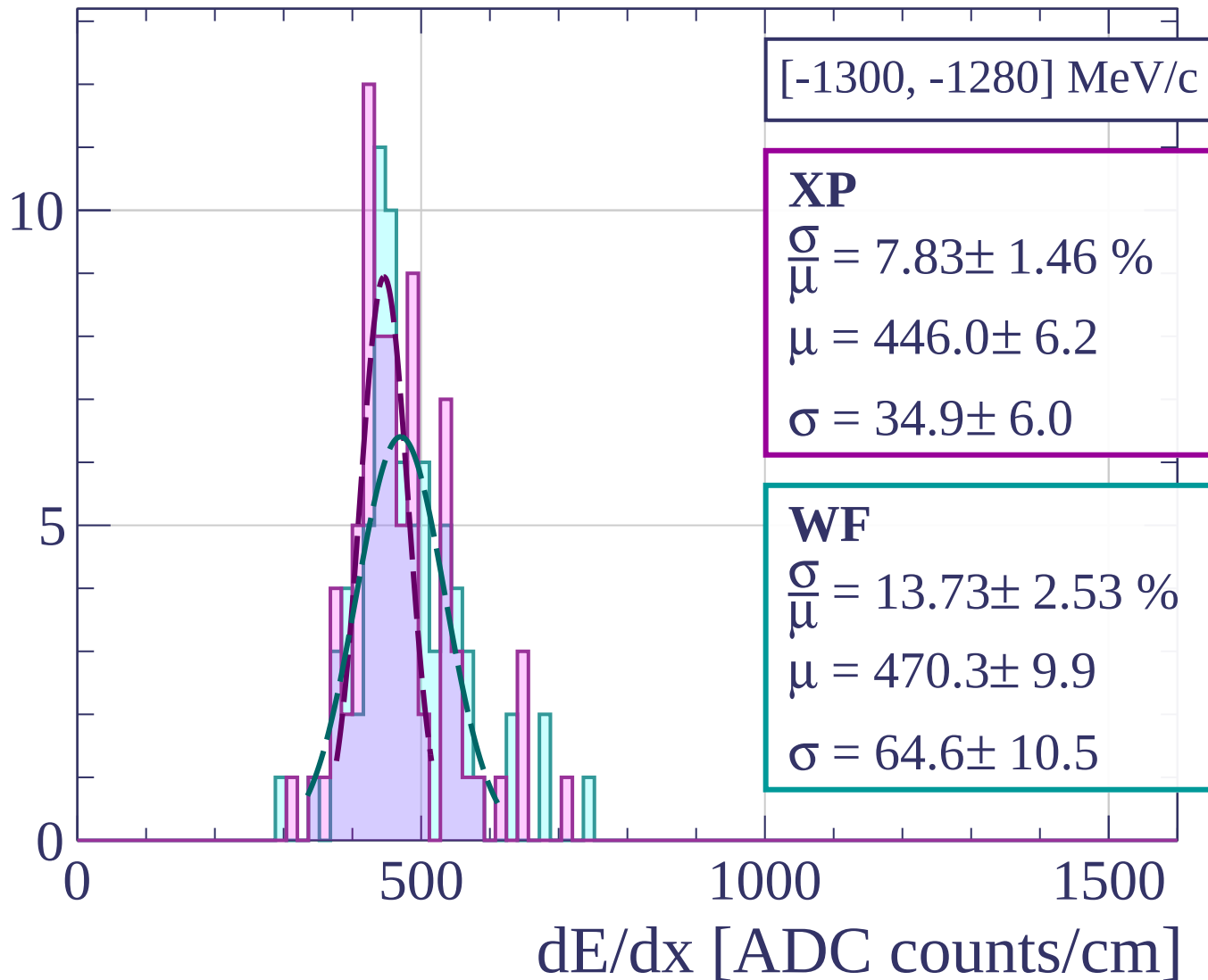
Count



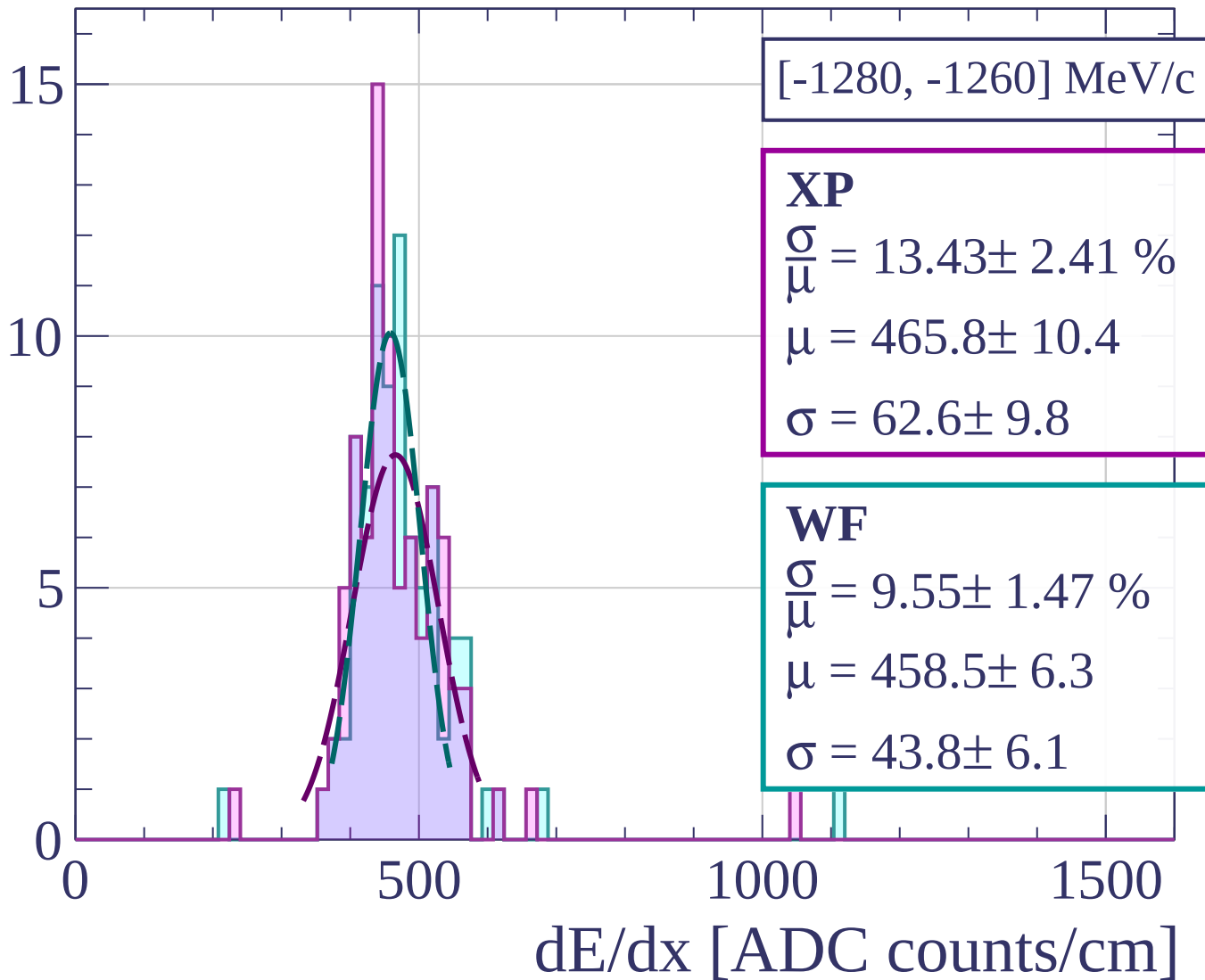
Count



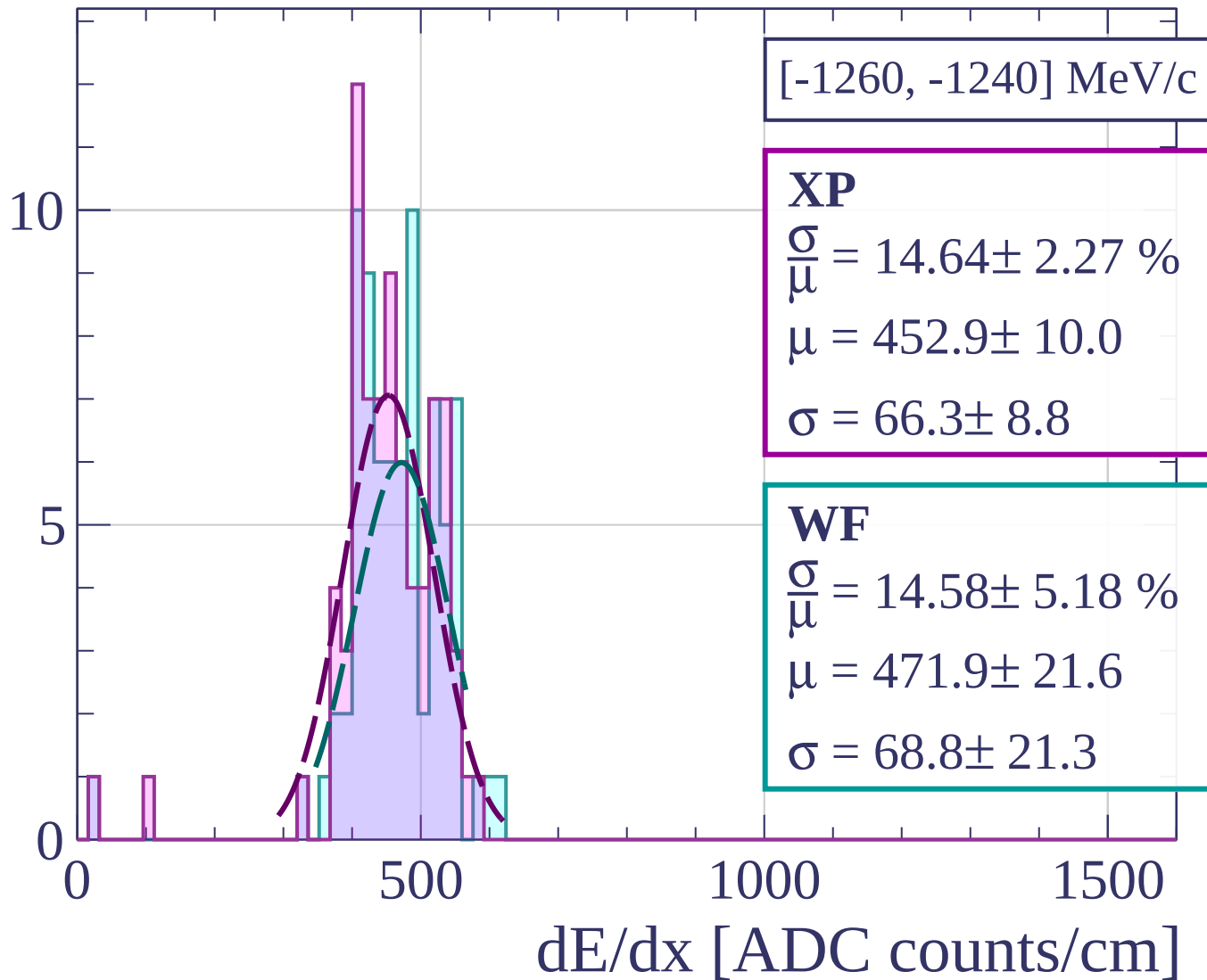
Count



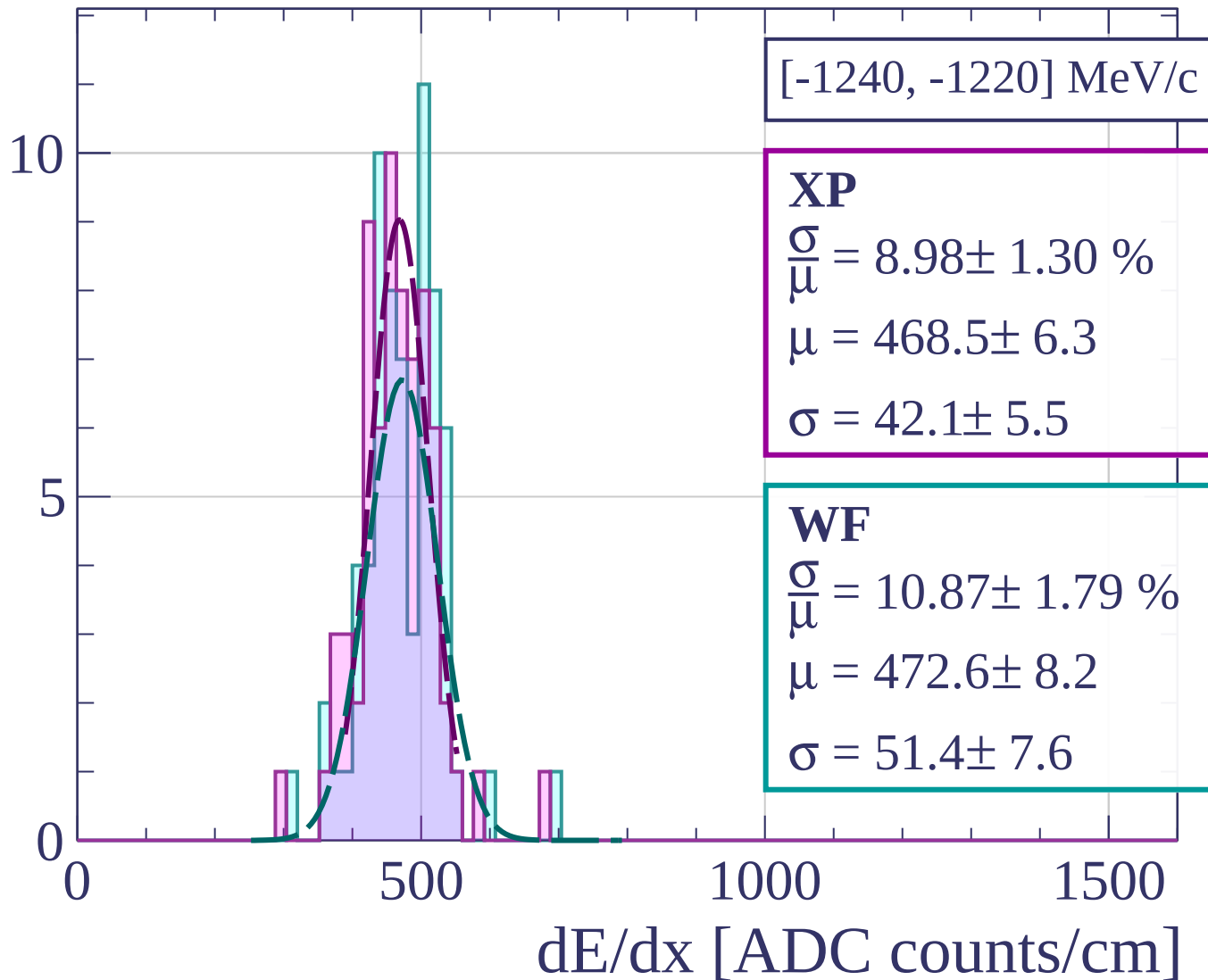
Count



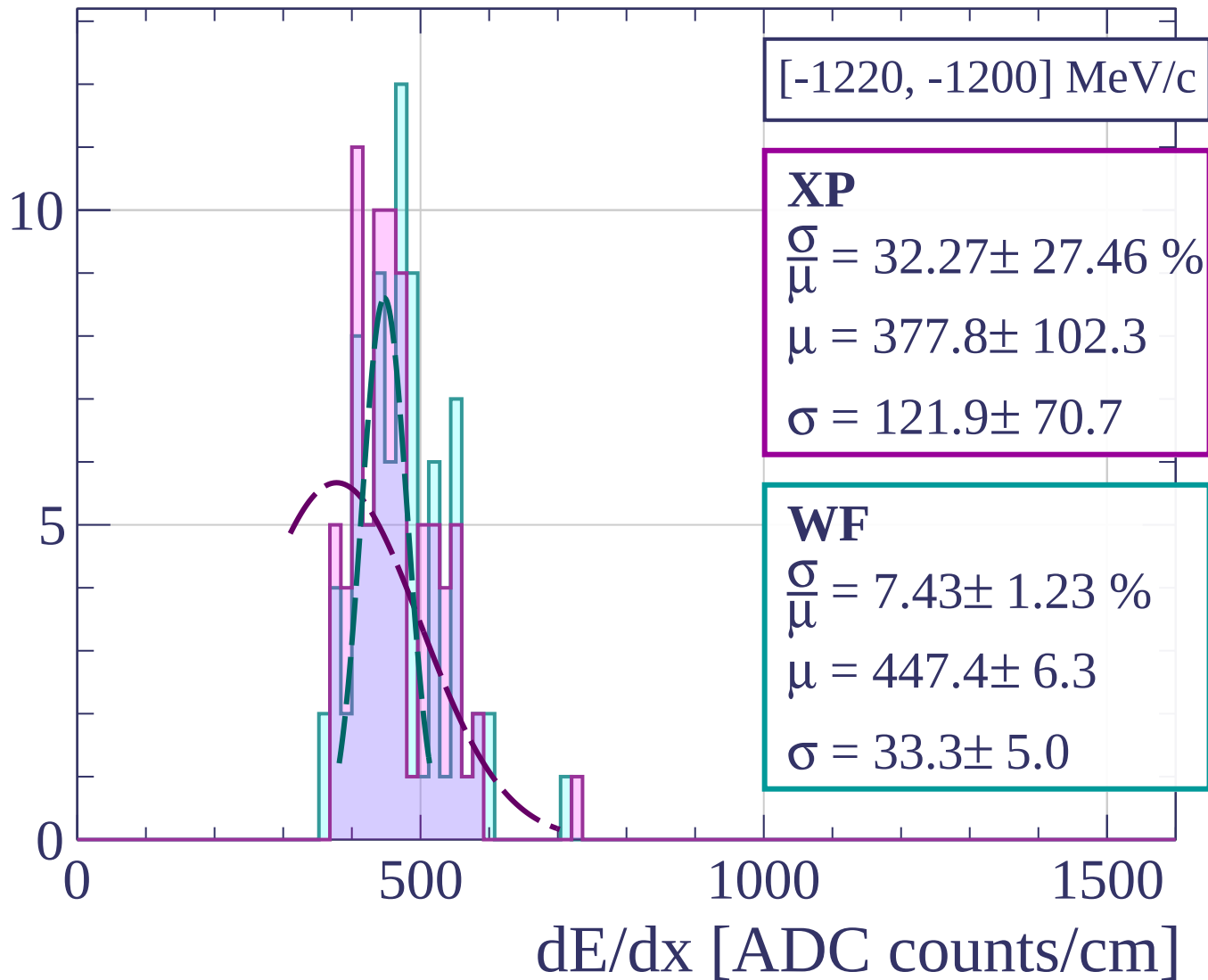
Count



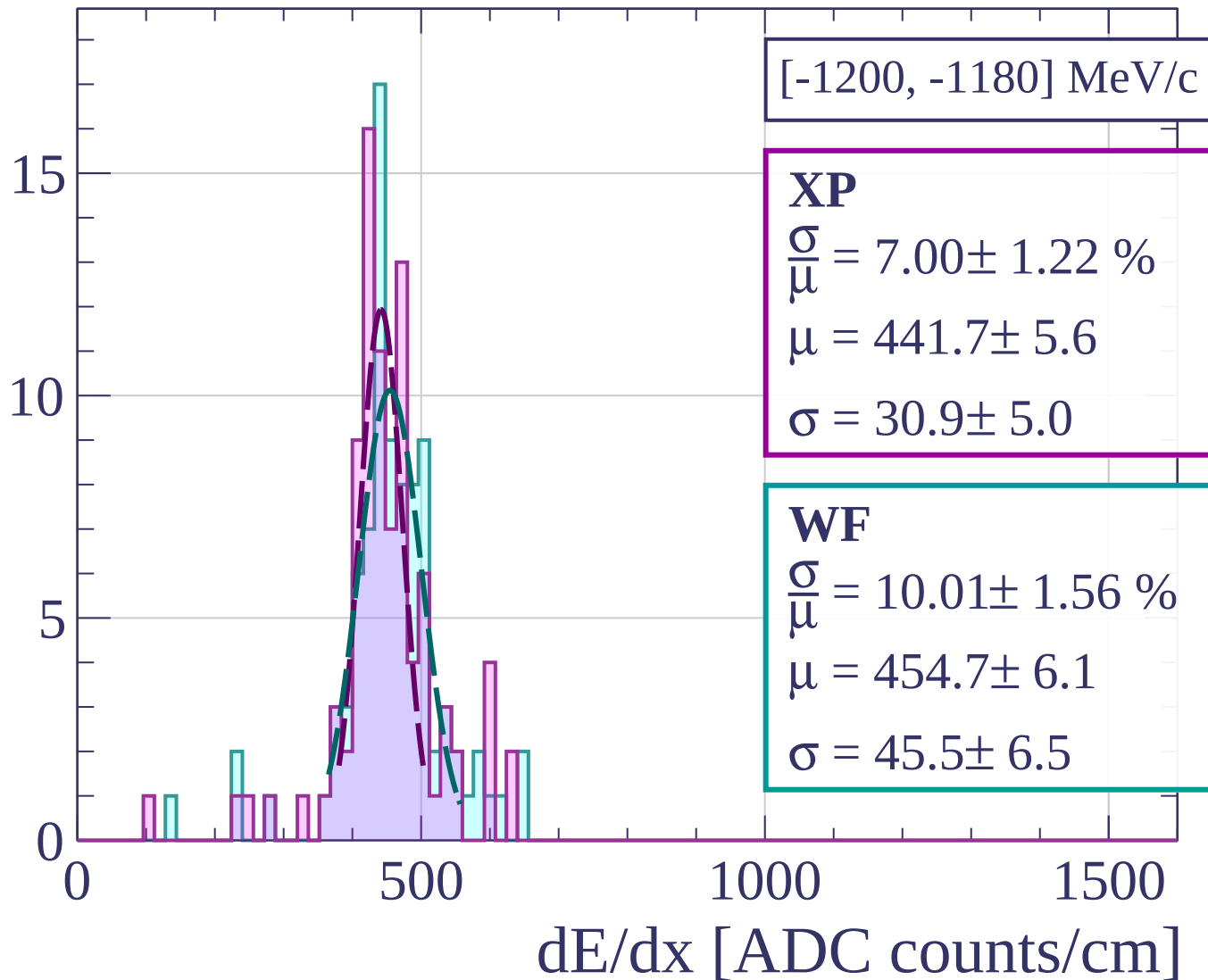
Count



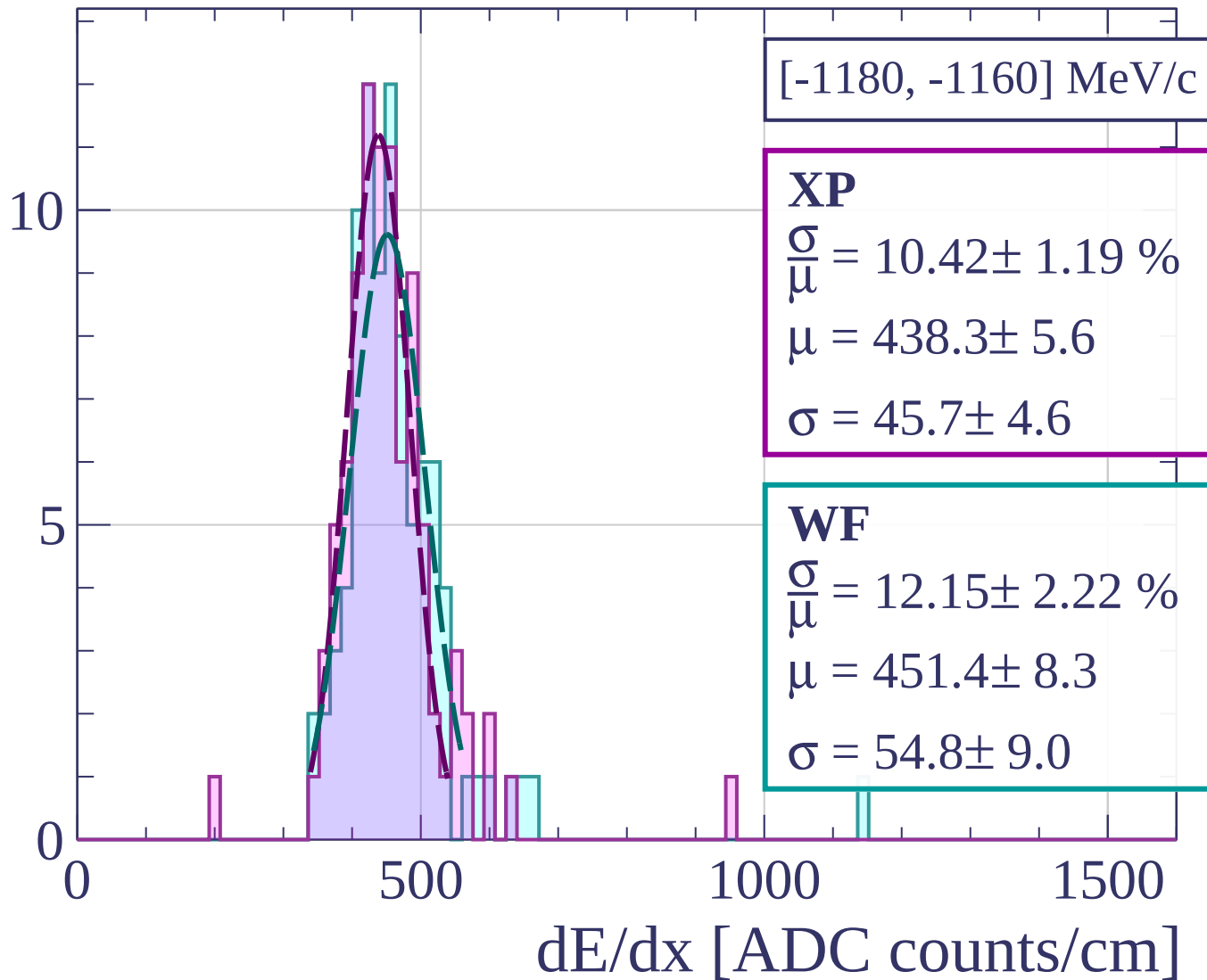
Count



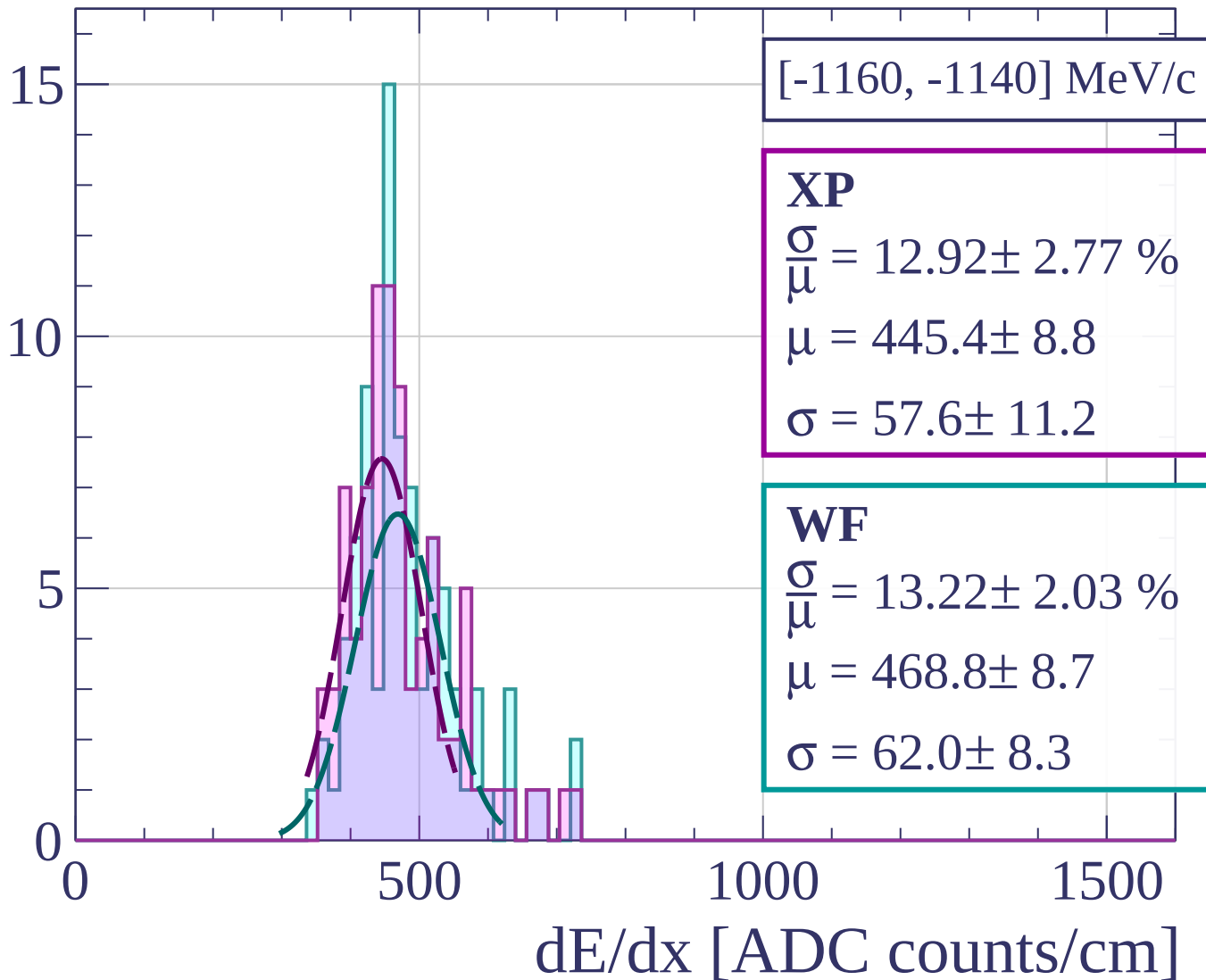
Count



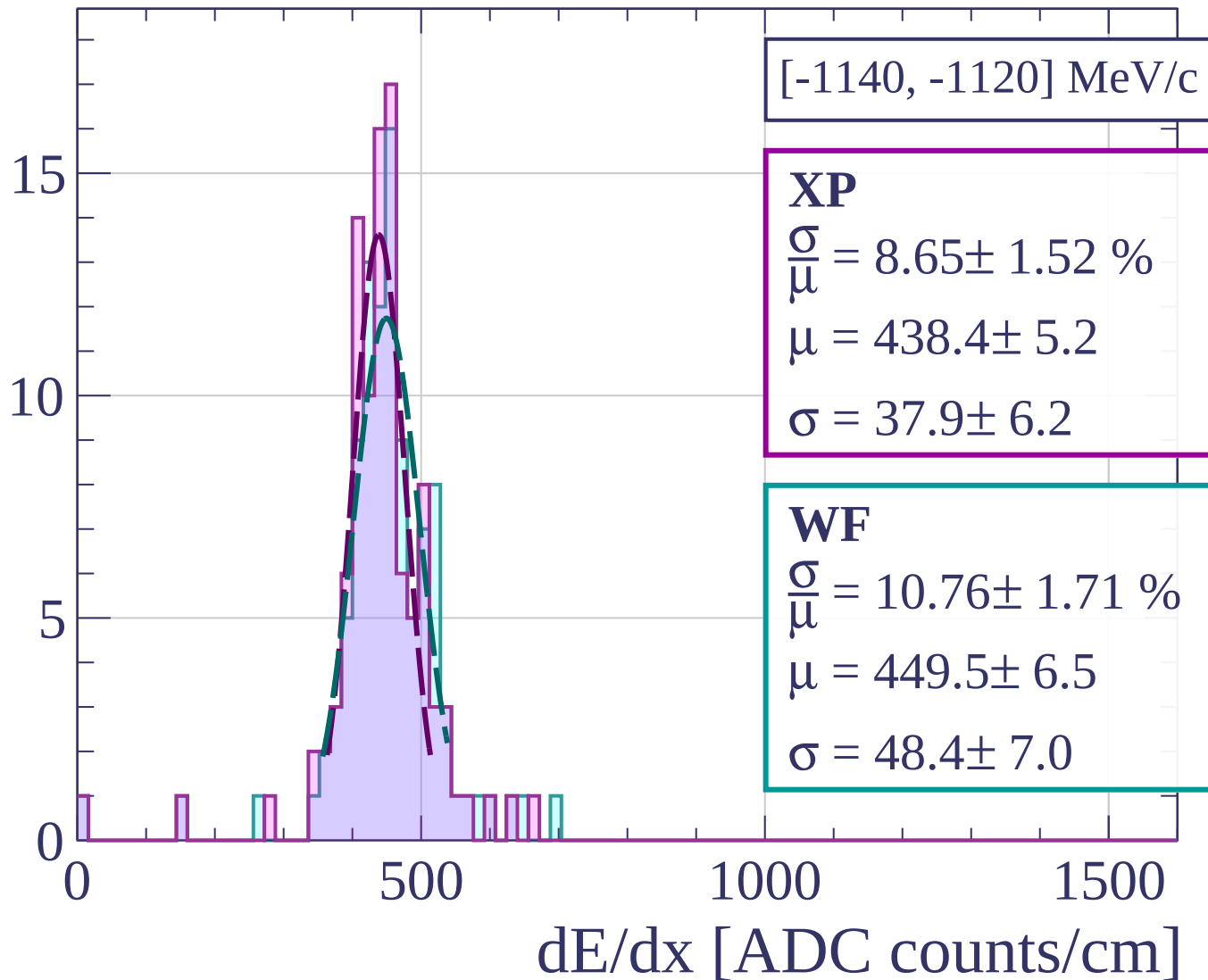
Count



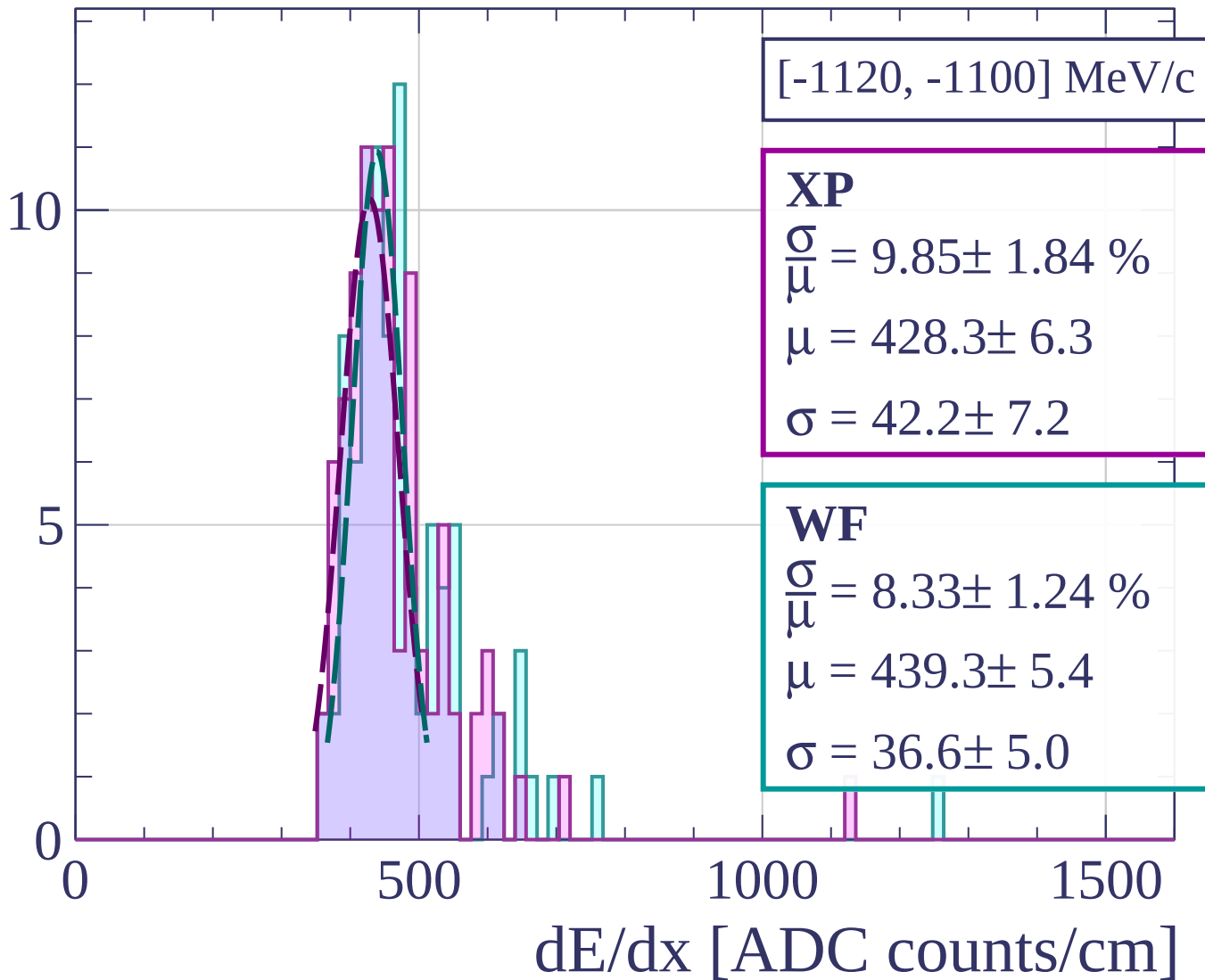
Count



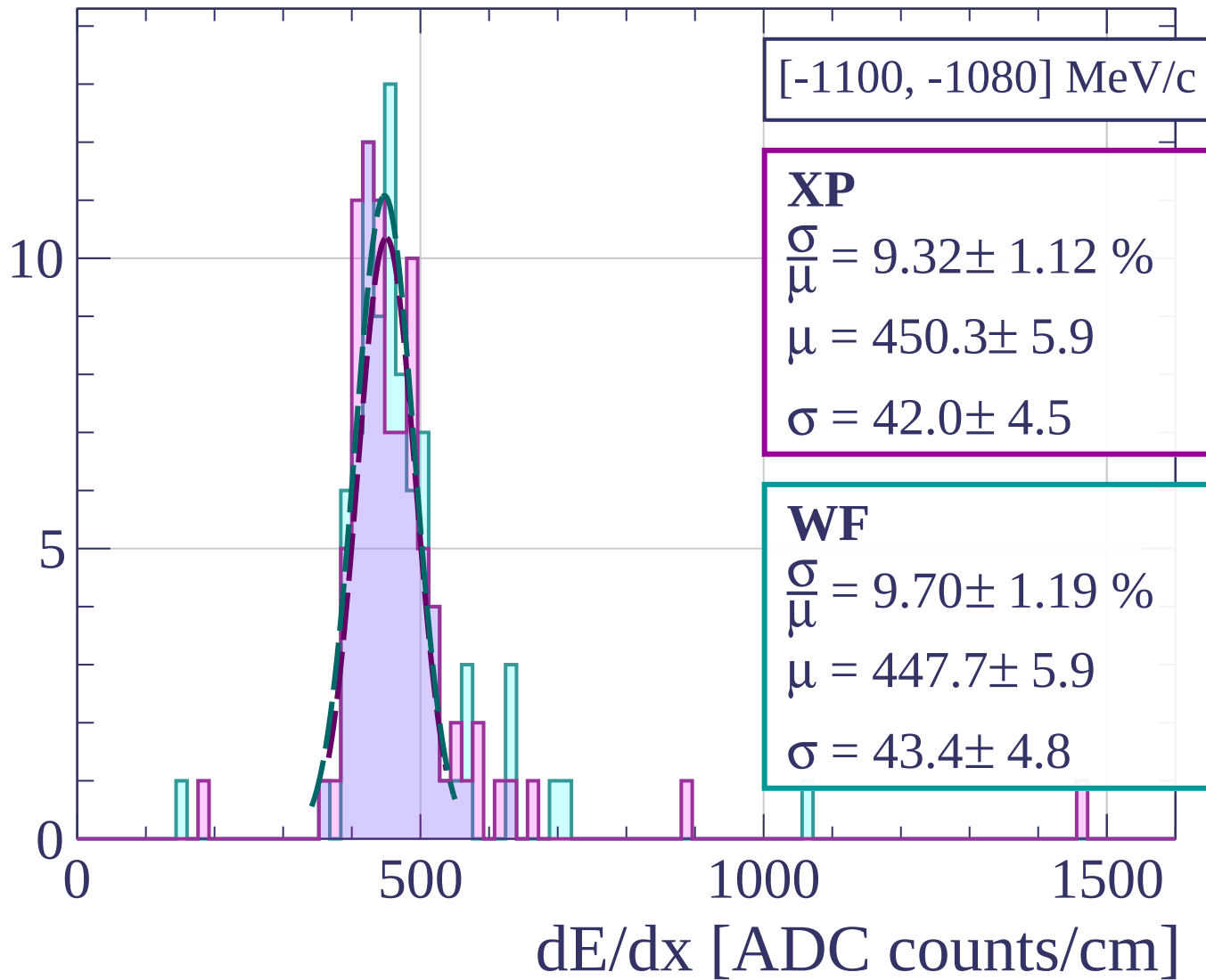
Count



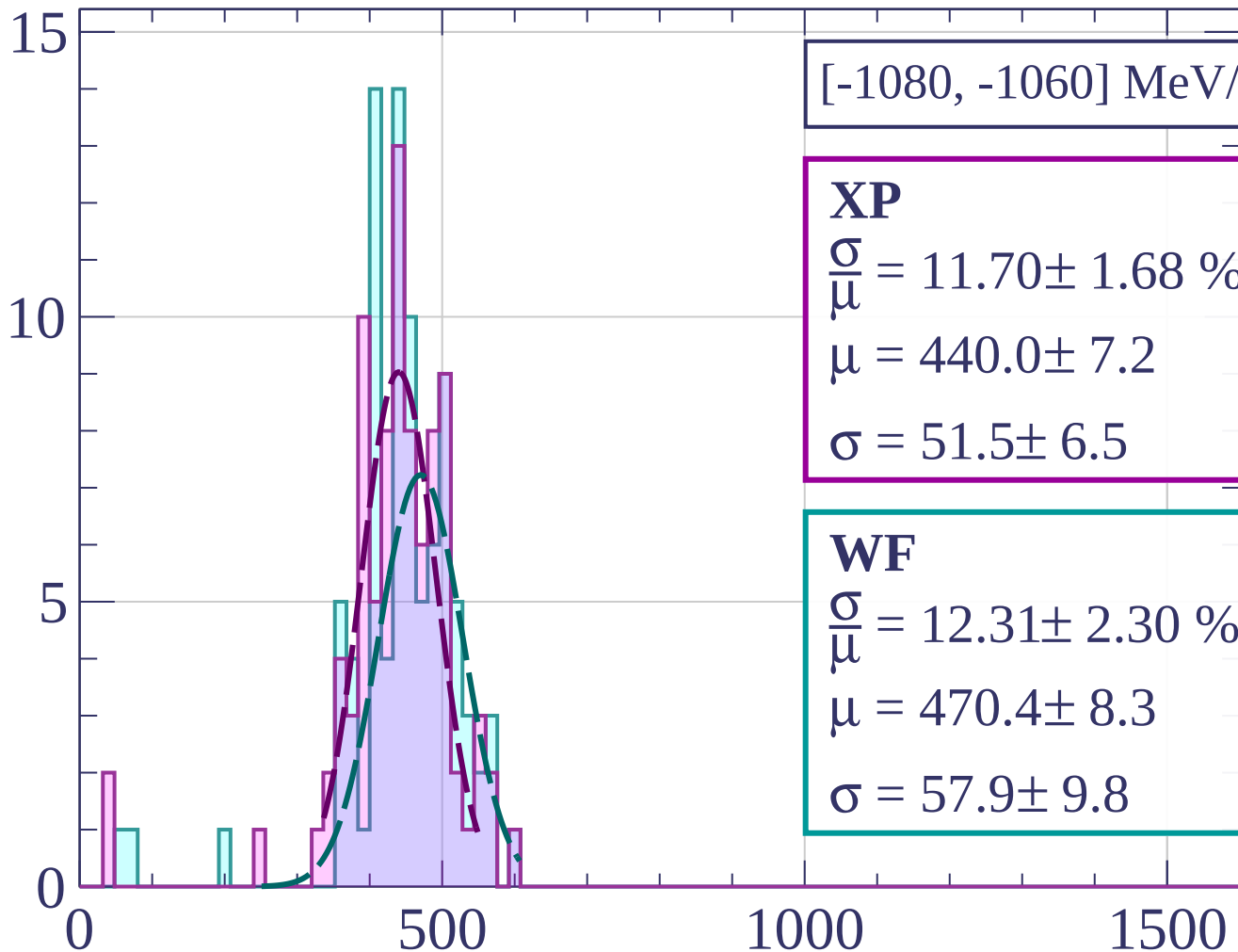
Count



Count



Count



[-1080, -1060] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.70 \pm 1.68 \%$$

$$\mu = 440.0 \pm 7.2$$

$$\sigma = 51.5 \pm 6.5$$

WF

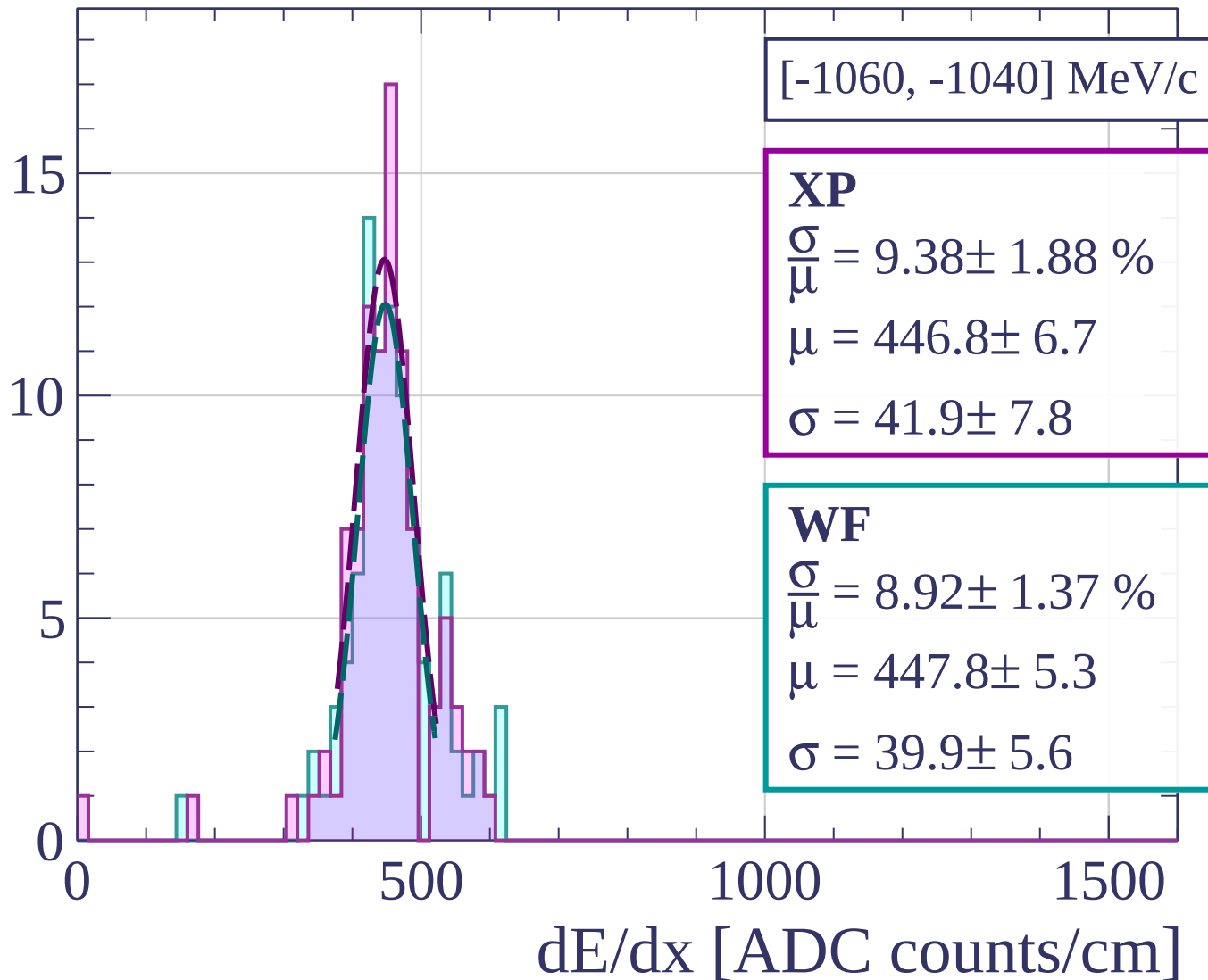
$$\frac{\sigma}{\mu} = 12.31 \pm 2.30 \%$$

$$\mu = 470.4 \pm 8.3$$

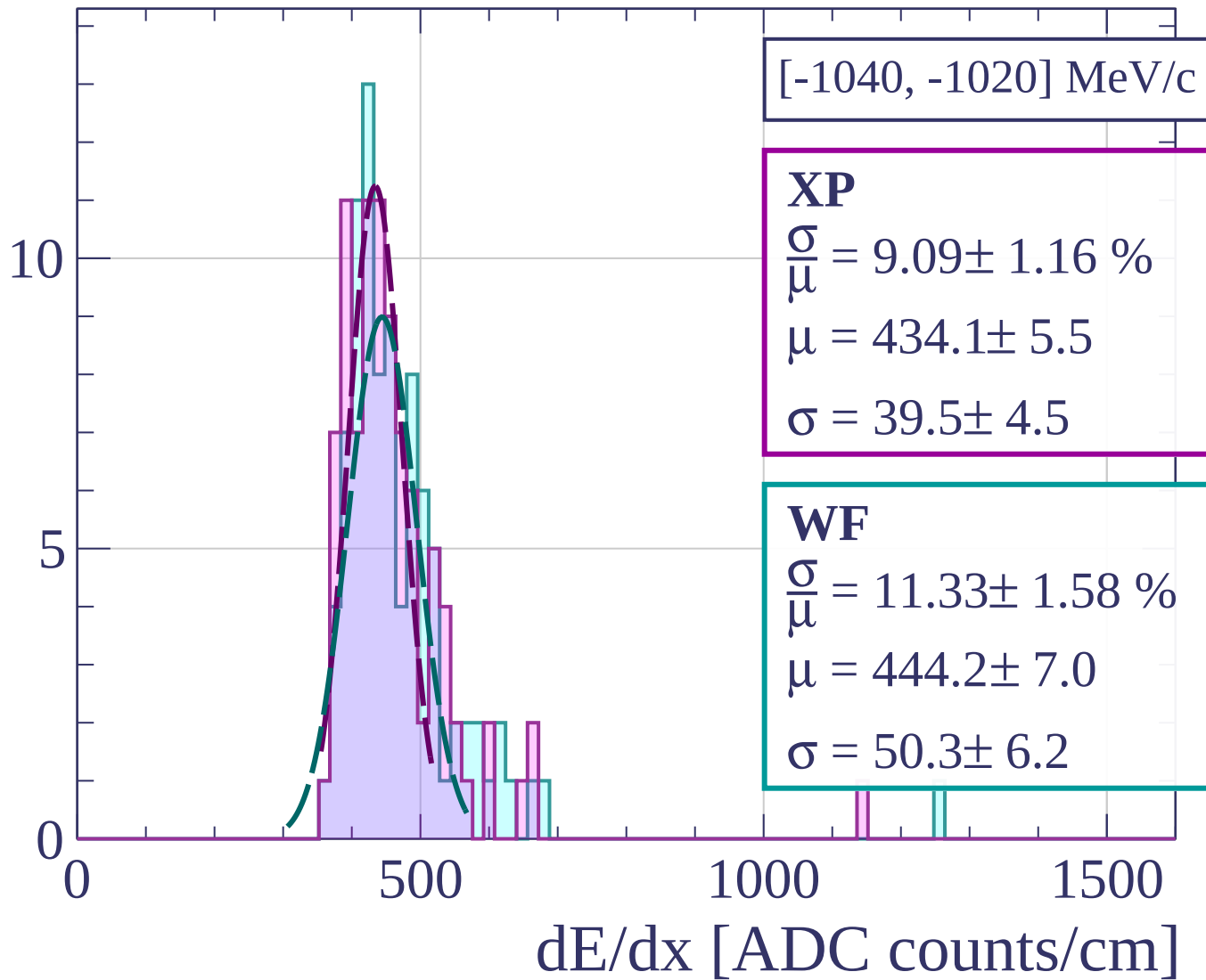
$$\sigma = 57.9 \pm 9.8$$

dE/dx [ADC counts/cm]

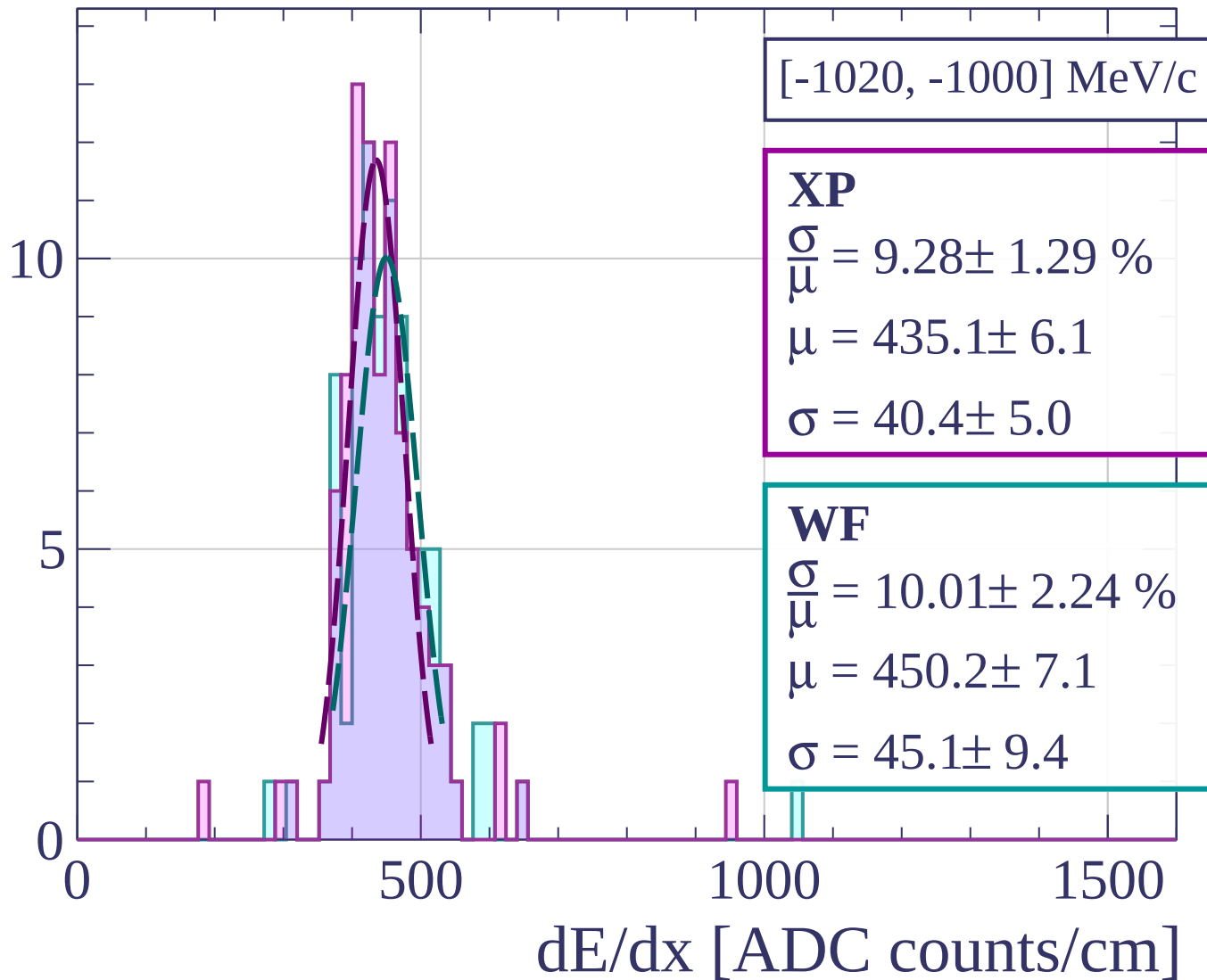
Count



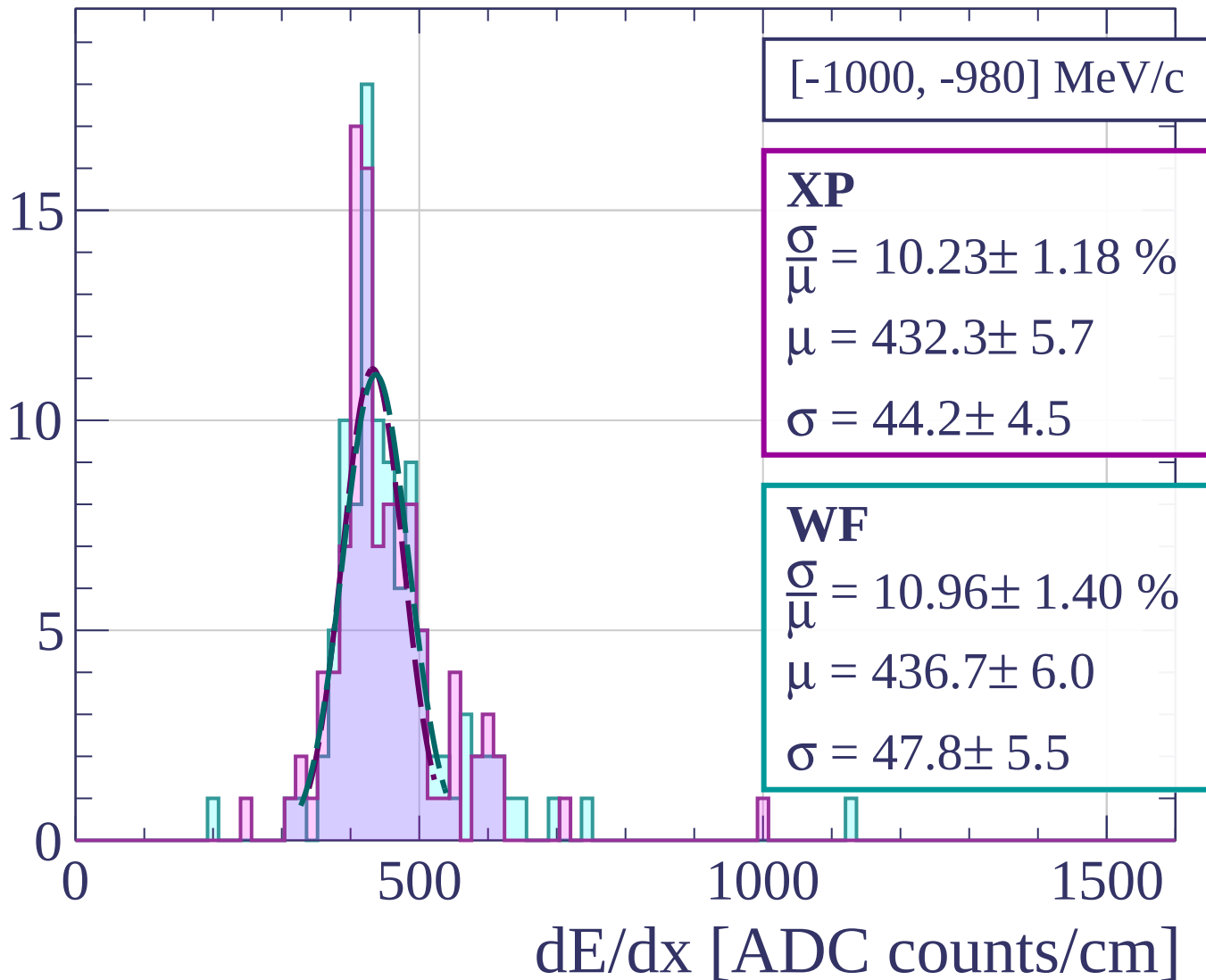
Count



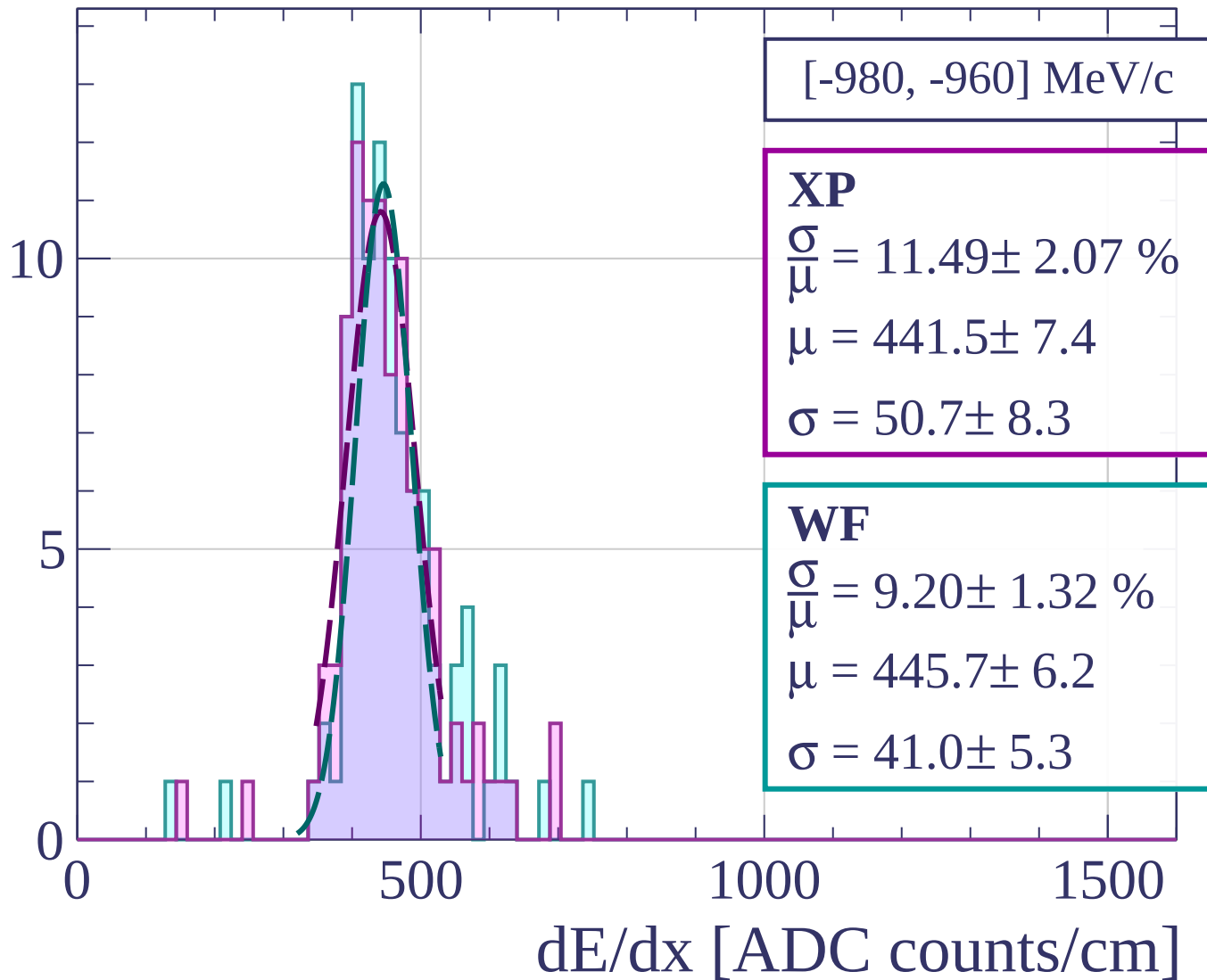
Count



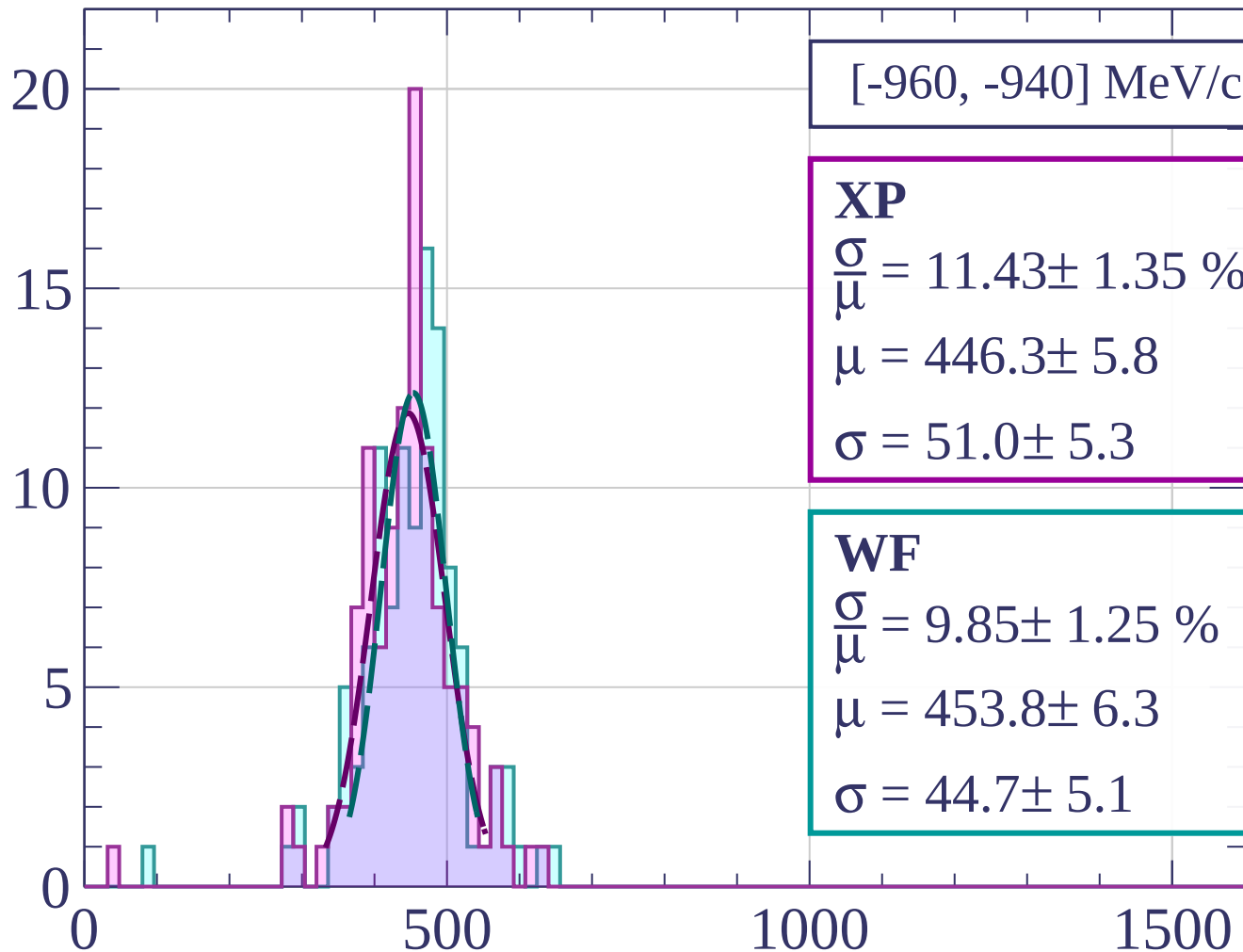
Count



Count



Count



[-960, -940] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.43 \pm 1.35 \%$$

$$\mu = 446.3 \pm 5.8$$

$$\sigma = 51.0 \pm 5.3$$

WF

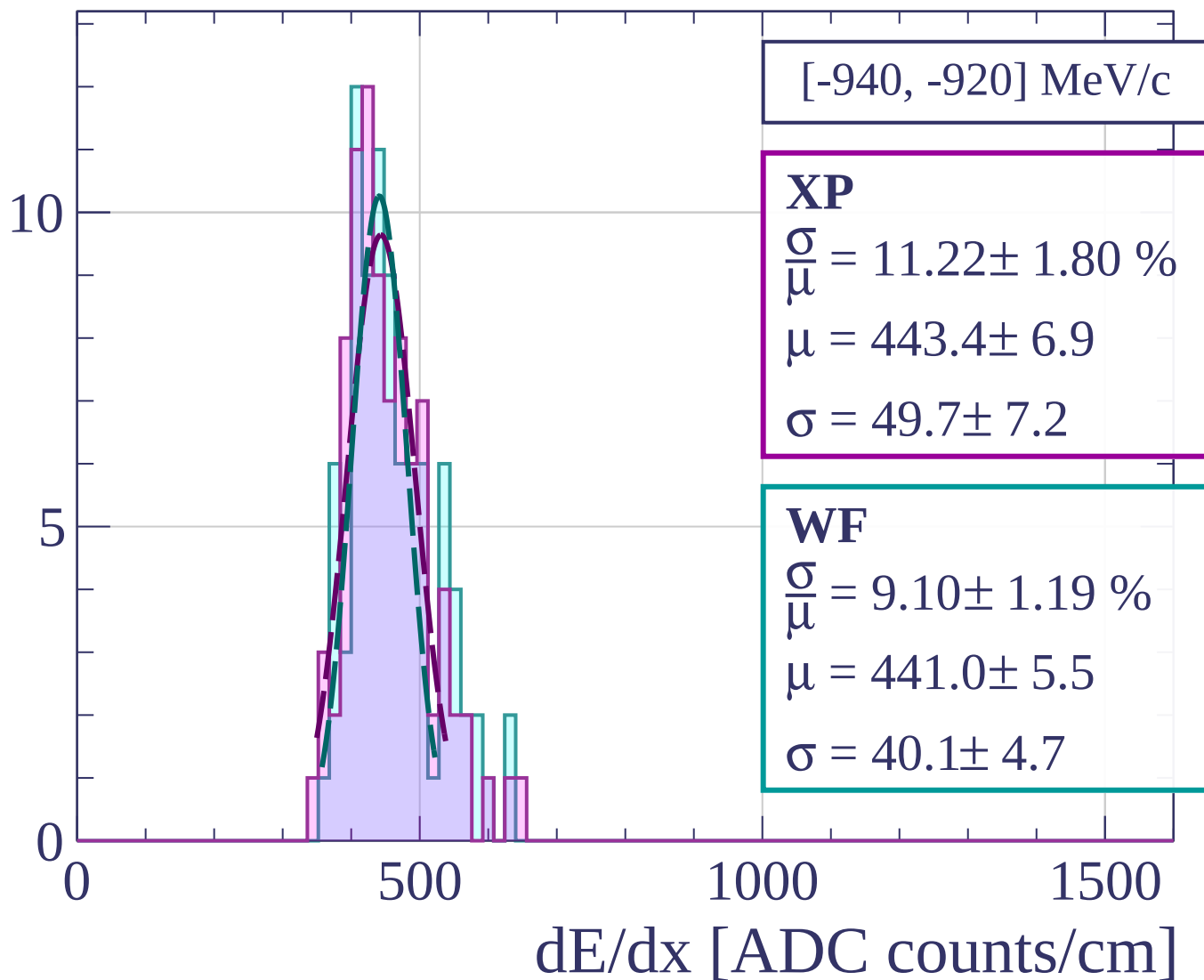
$$\frac{\sigma}{\mu} = 9.85 \pm 1.25 \%$$

$$\mu = 453.8 \pm 6.3$$

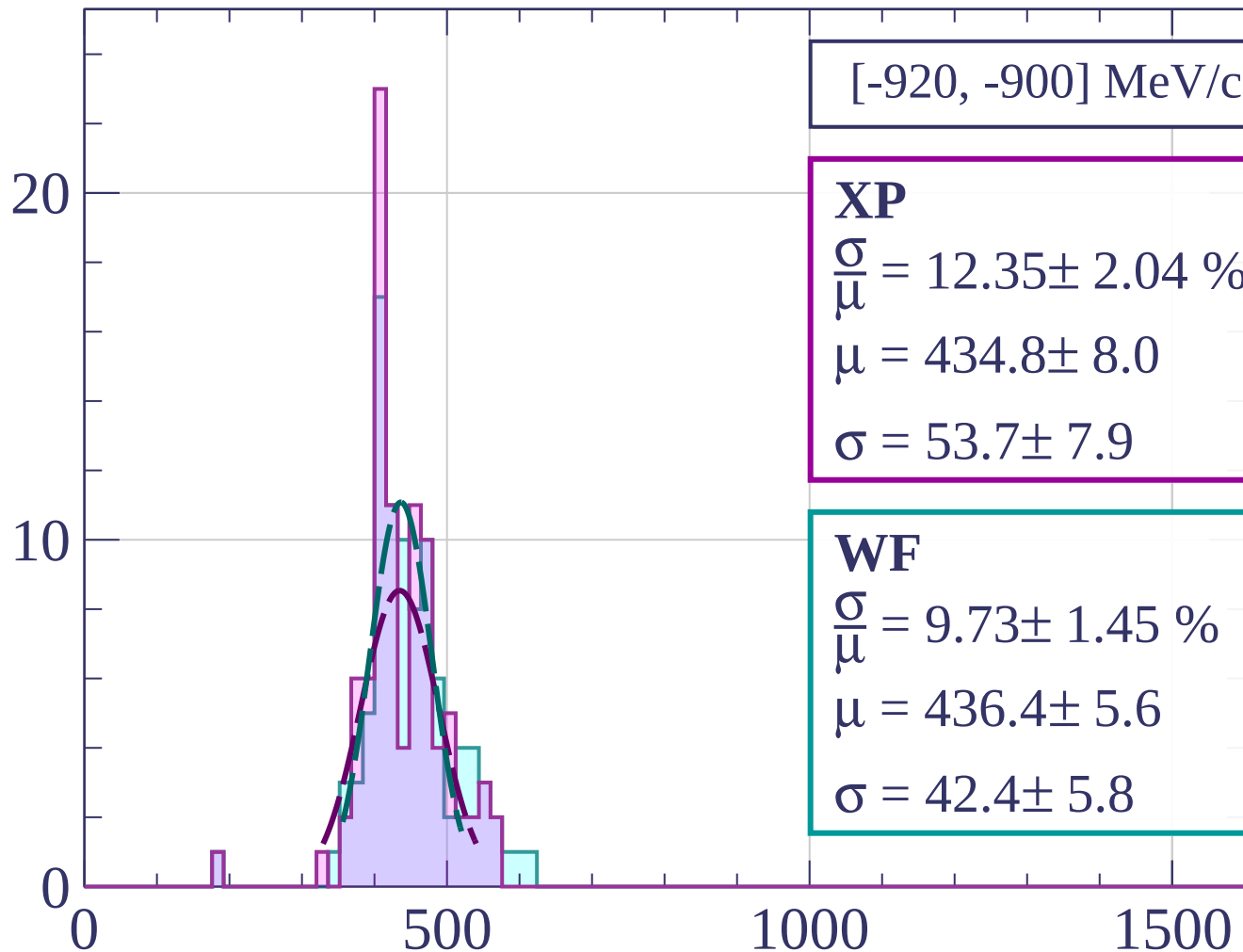
$$\sigma = 44.7 \pm 5.1$$

dE/dx [ADC counts/cm]

Count

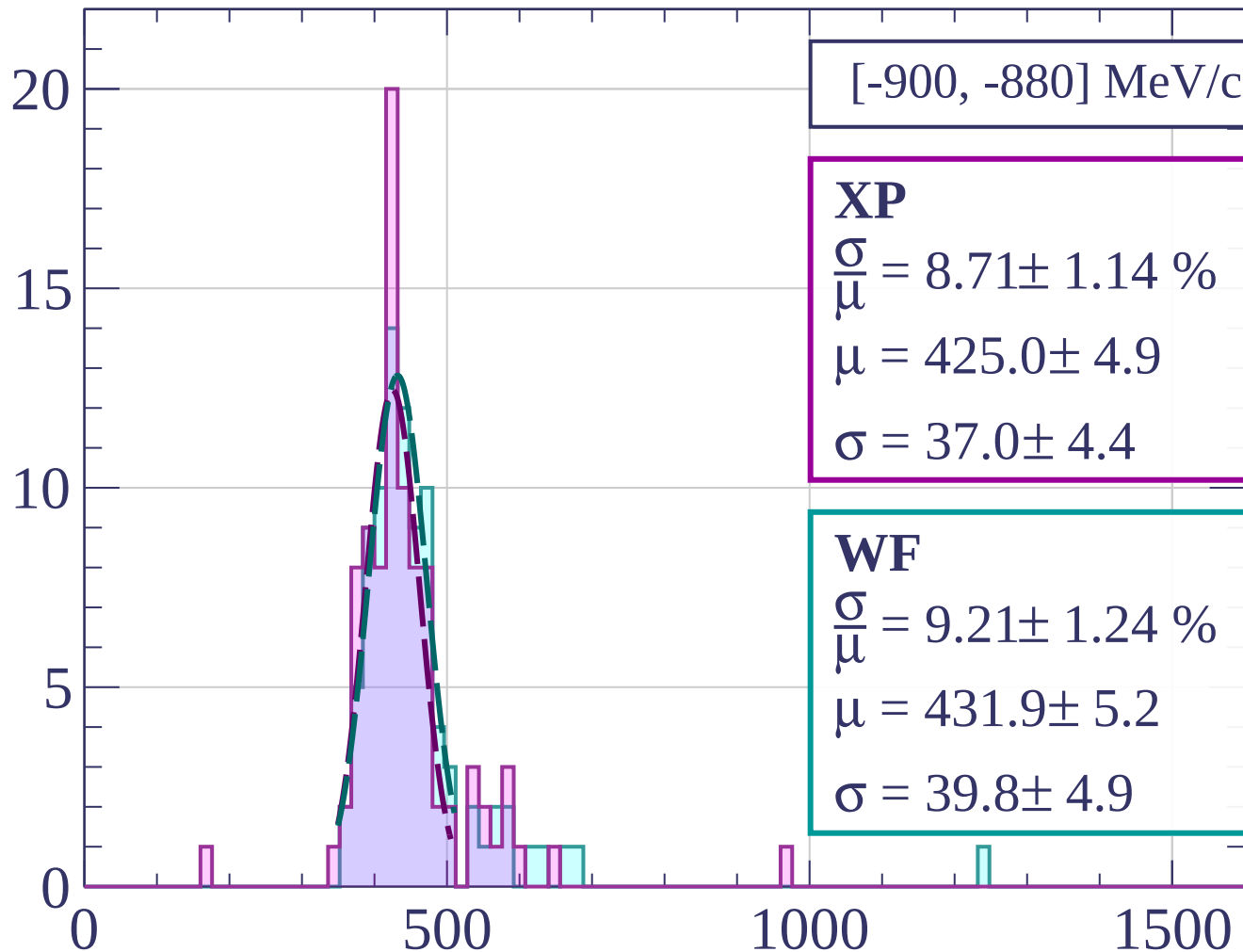


Count



dE/dx [ADC counts/cm]

Count



[-900, -880] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.71 \pm 1.14 \%$$

$$\mu = 425.0 \pm 4.9$$

$$\sigma = 37.0 \pm 4.4$$

WF

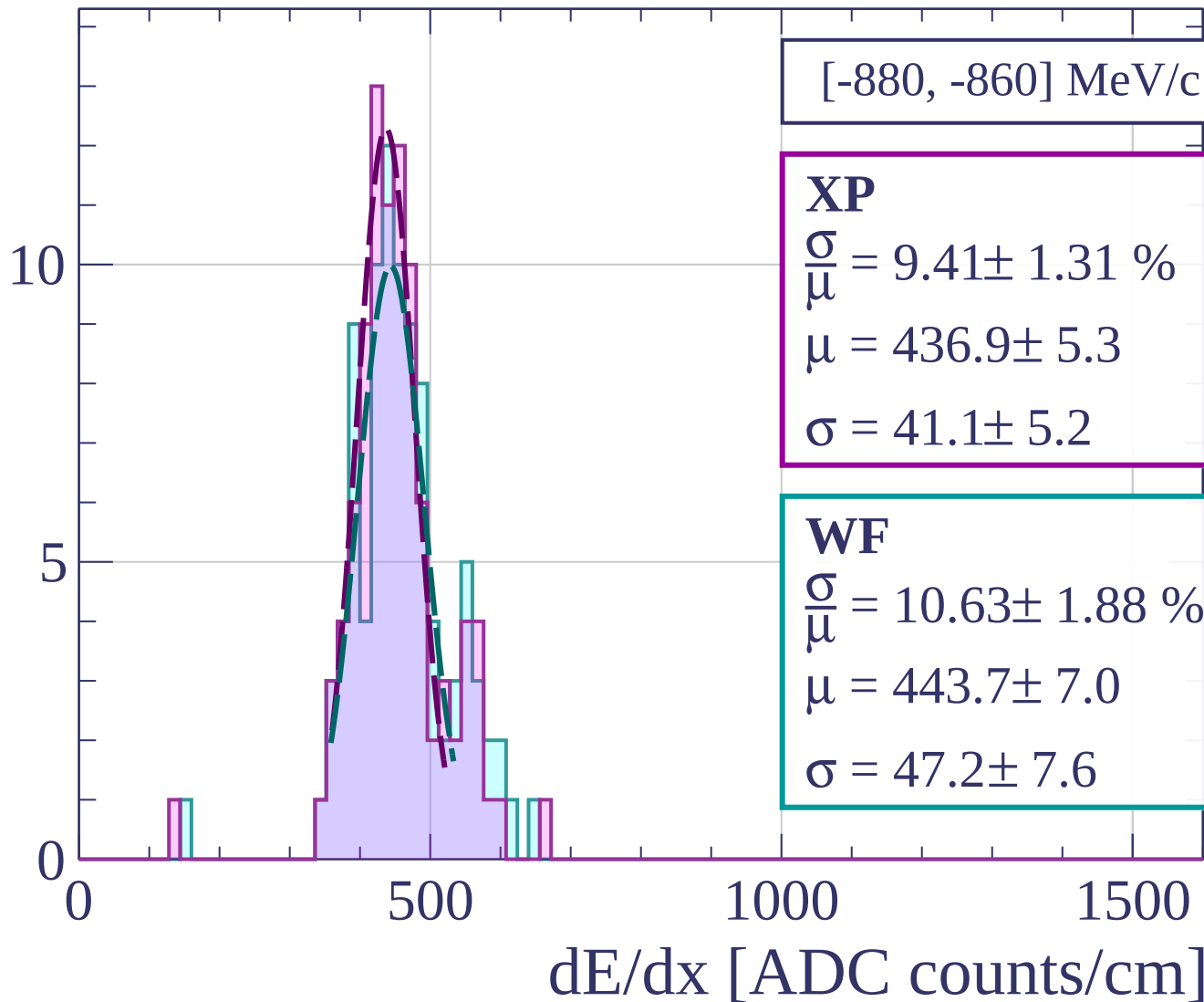
$$\frac{\sigma}{\mu} = 9.21 \pm 1.24 \%$$

$$\mu = 431.9 \pm 5.2$$

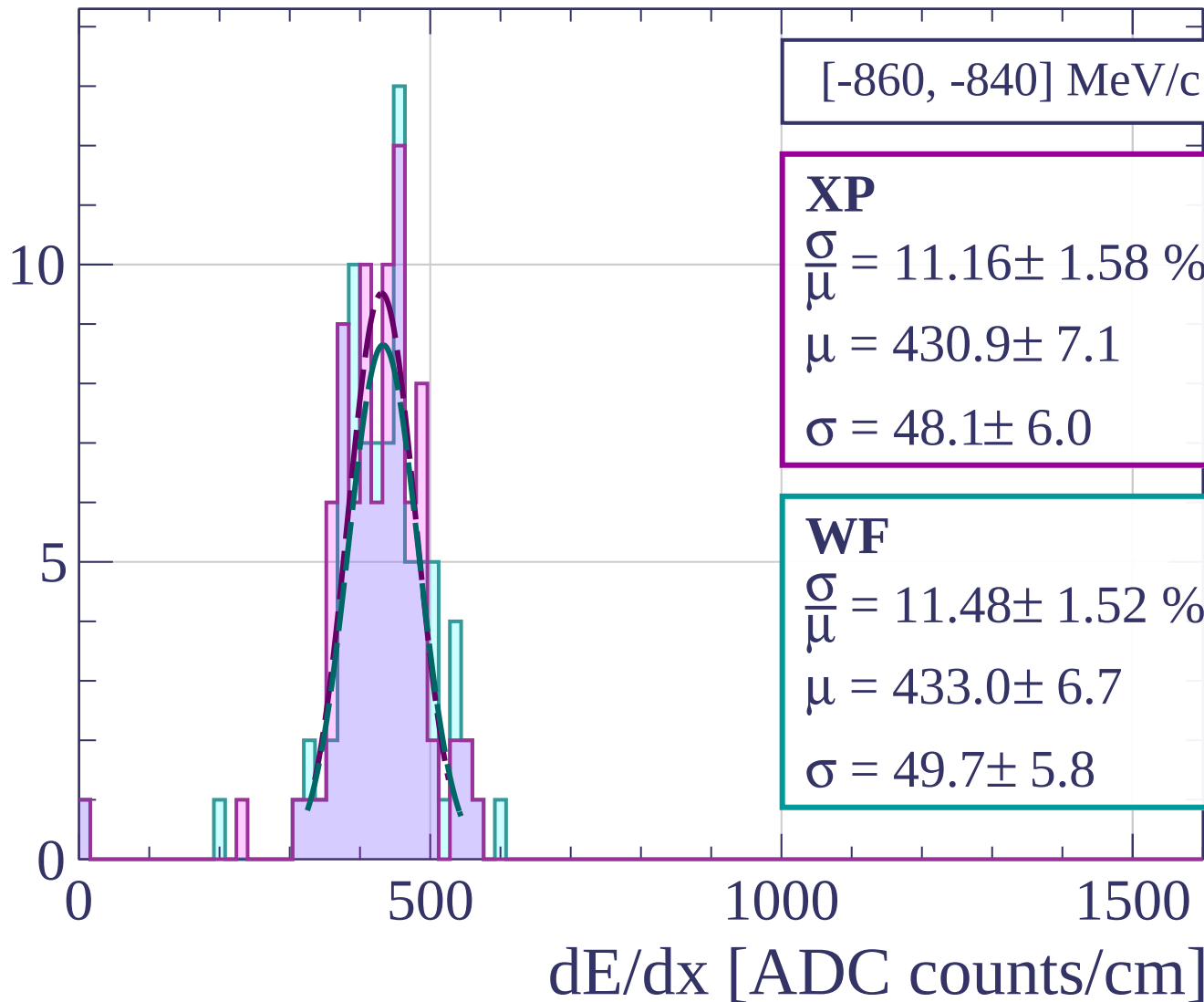
$$\sigma = 39.8 \pm 4.9$$

dE/dx [ADC counts/cm]

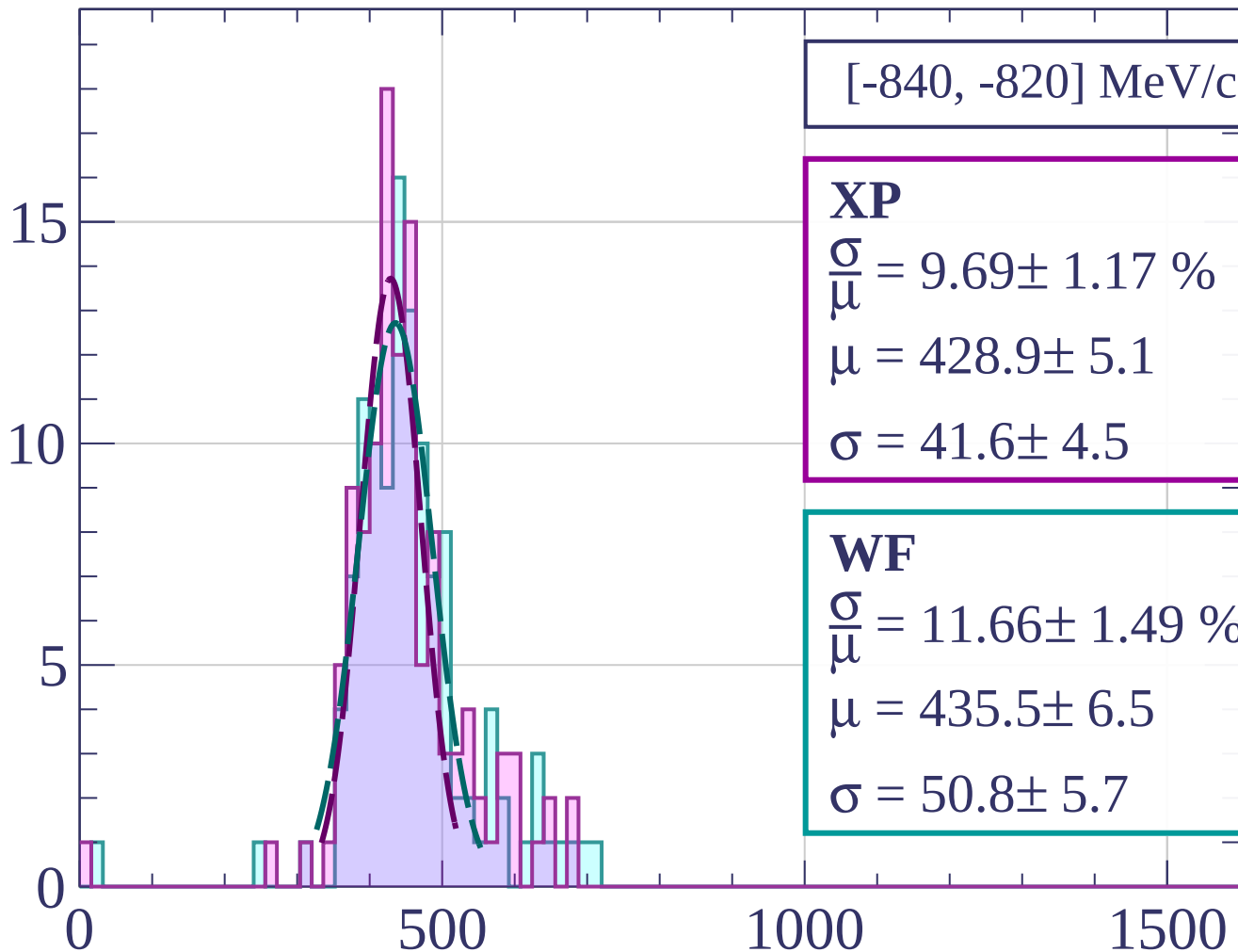
Count



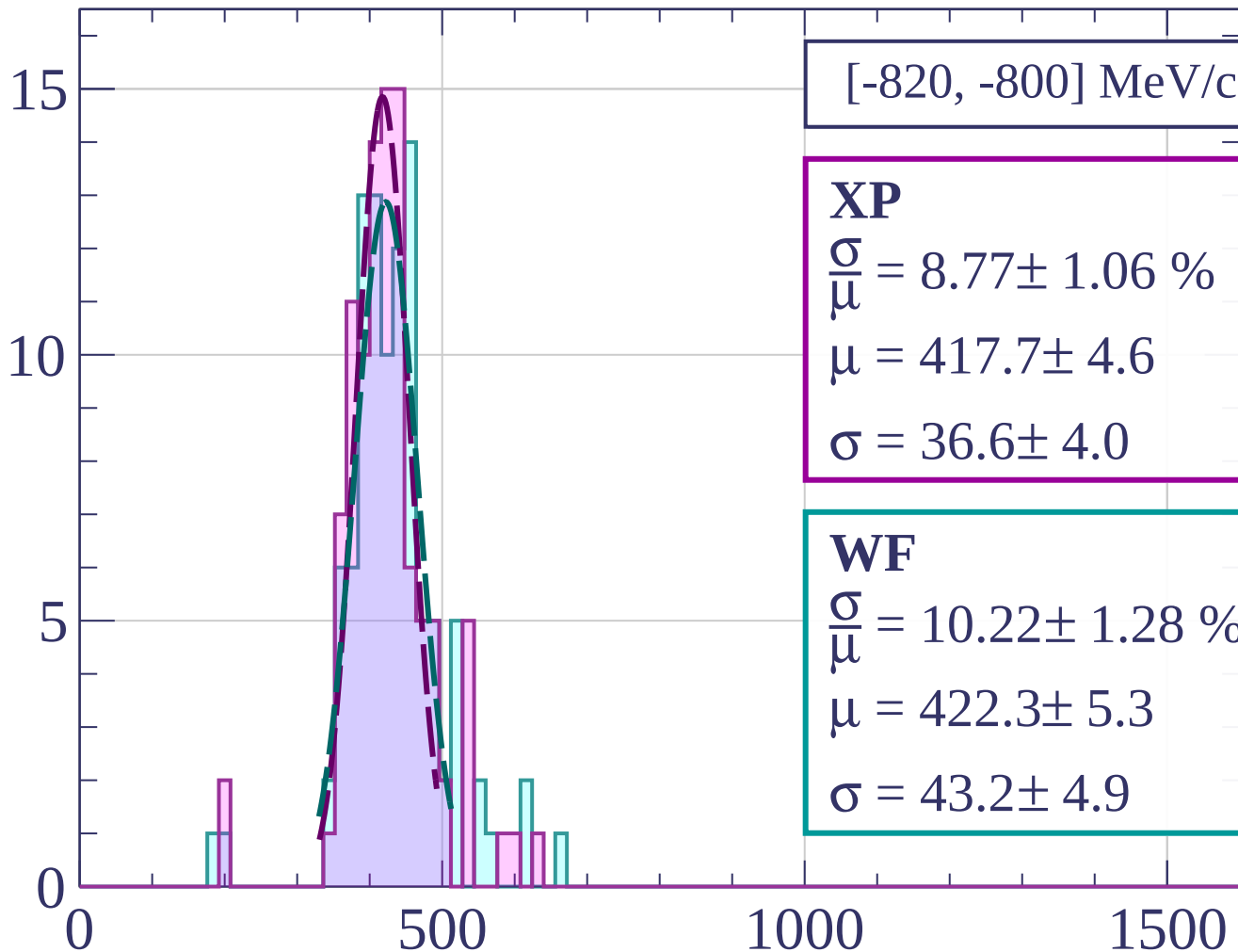
Count



Count

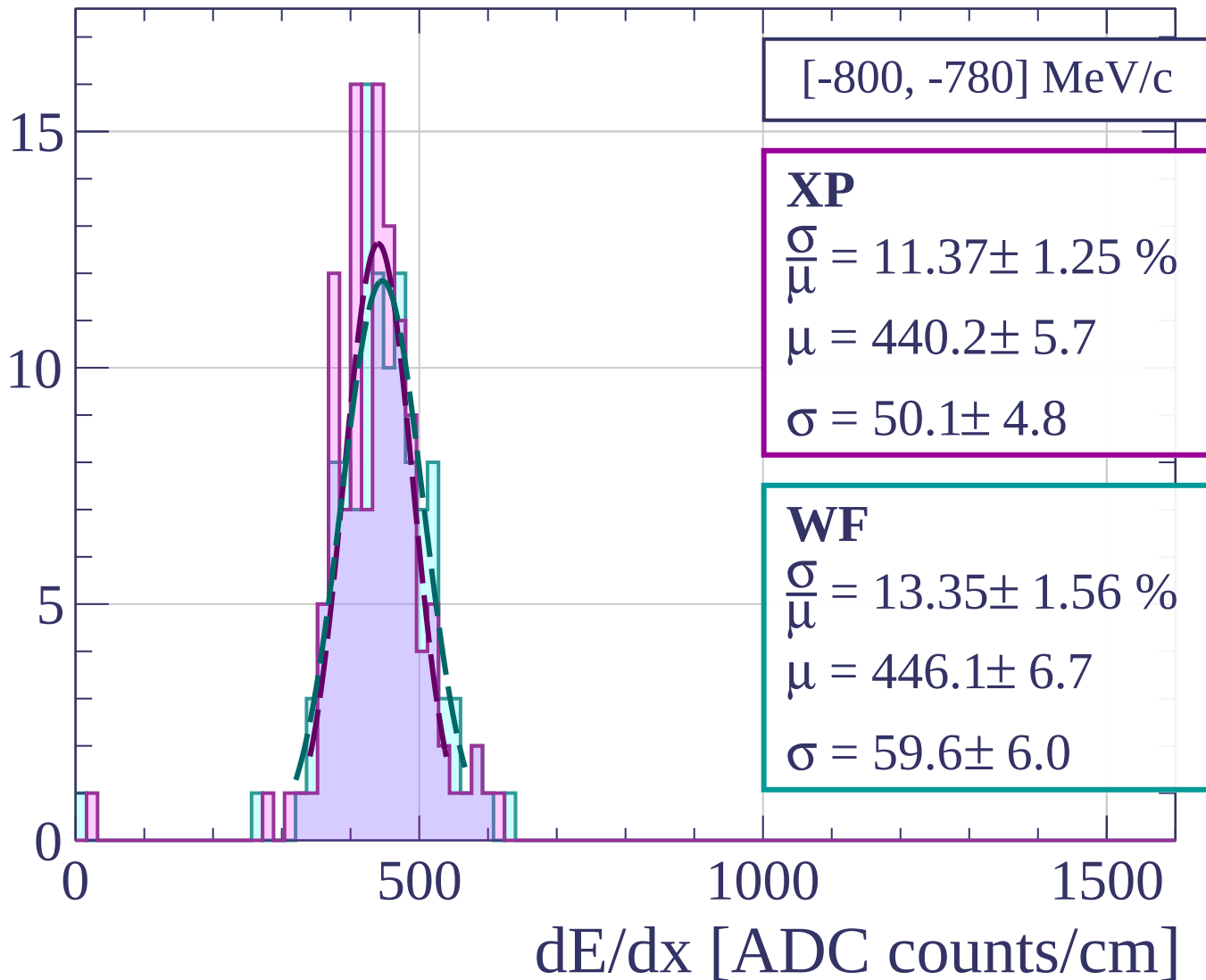


Count

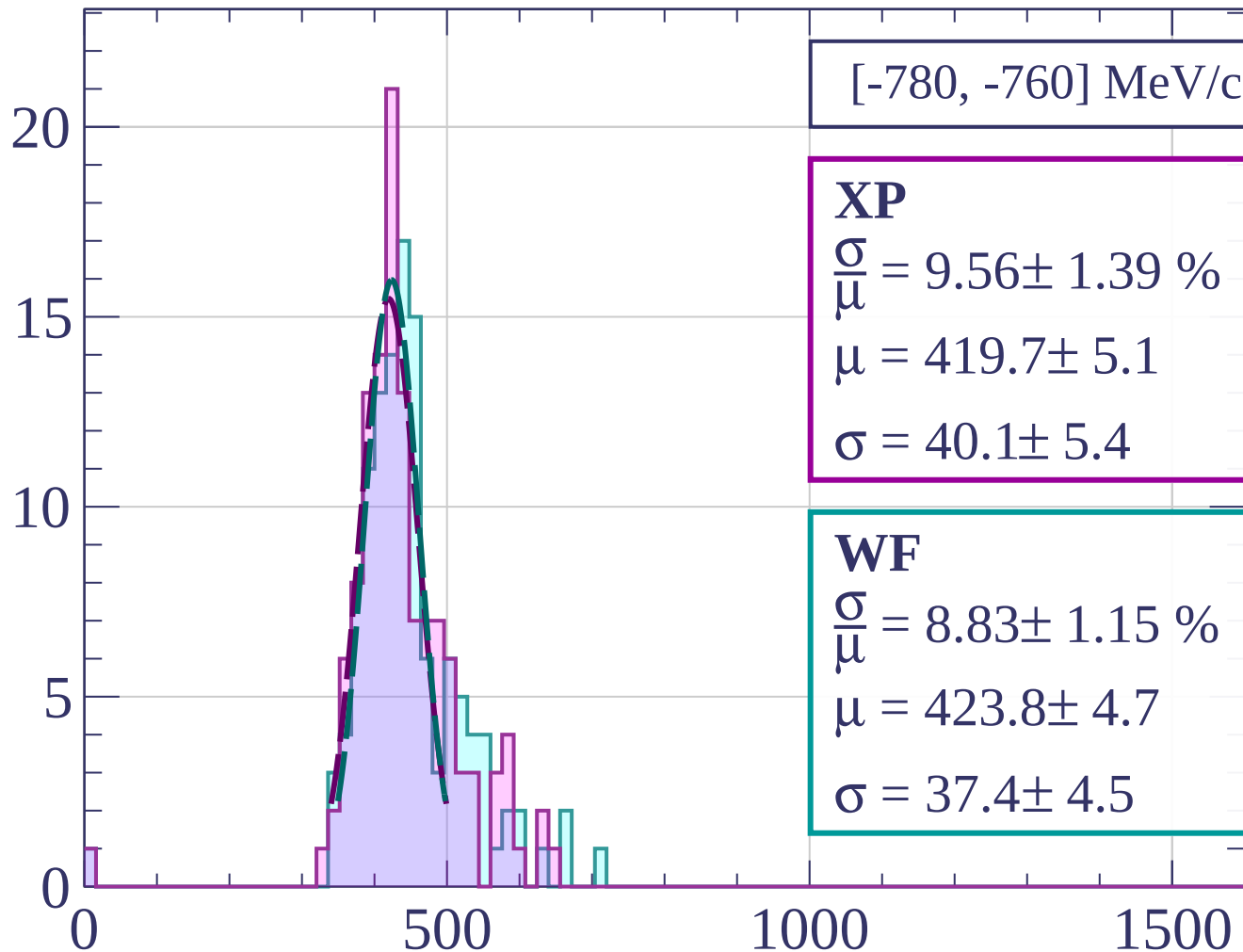


dE/dx [ADC counts/cm]

Count



Count



[-780, -760] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.56 \pm 1.39 \%$$

$$\mu = 419.7 \pm 5.1$$

$$\sigma = 40.1 \pm 5.4$$

WF

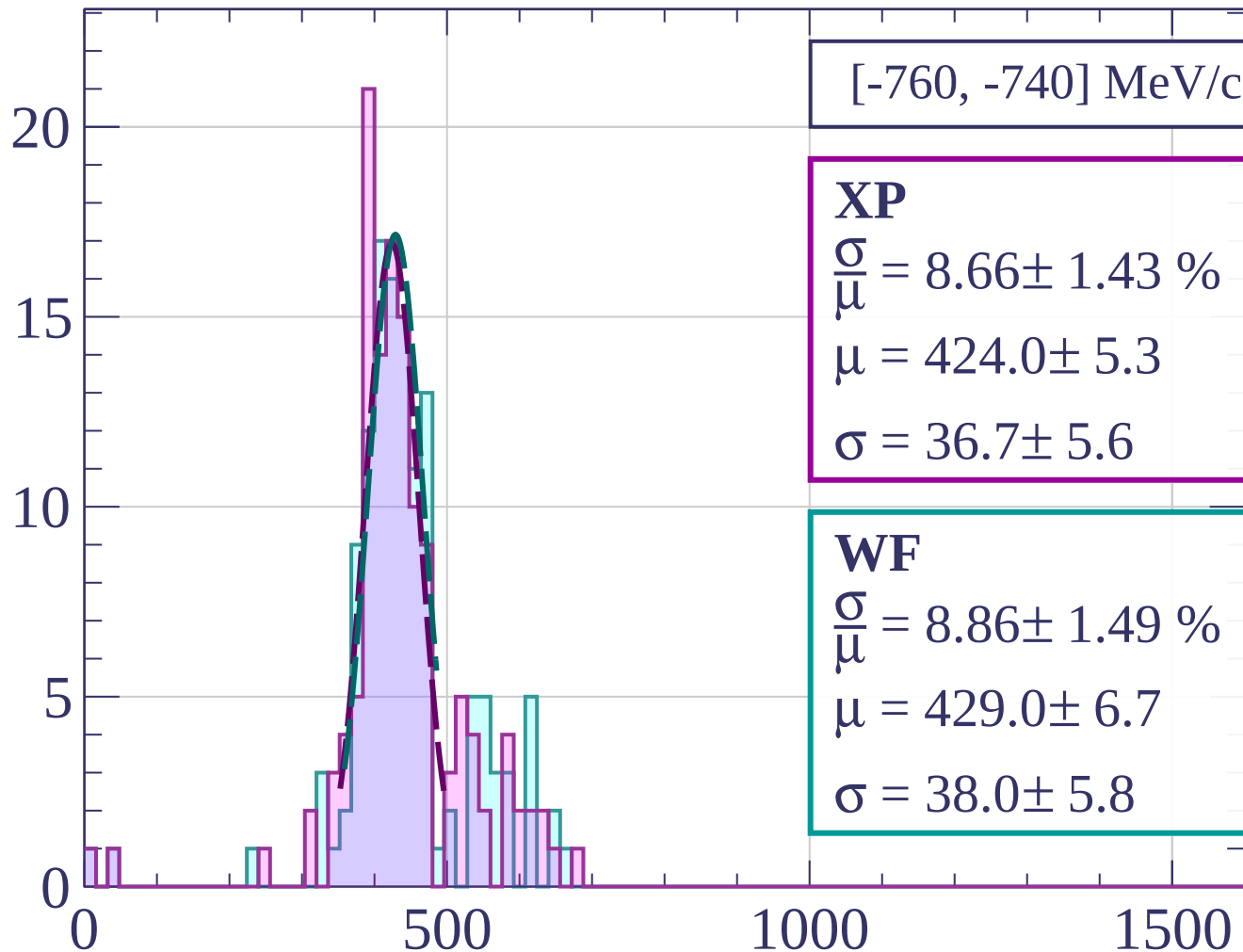
$$\frac{\sigma}{\mu} = 8.83 \pm 1.15 \%$$

$$\mu = 423.8 \pm 4.7$$

$$\sigma = 37.4 \pm 4.5$$

dE/dx [ADC counts/cm]

Count



[-760, -740] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.66 \pm 1.43 \%$$

$$\mu = 424.0 \pm 5.3$$

$$\sigma = 36.7 \pm 5.6$$

WF

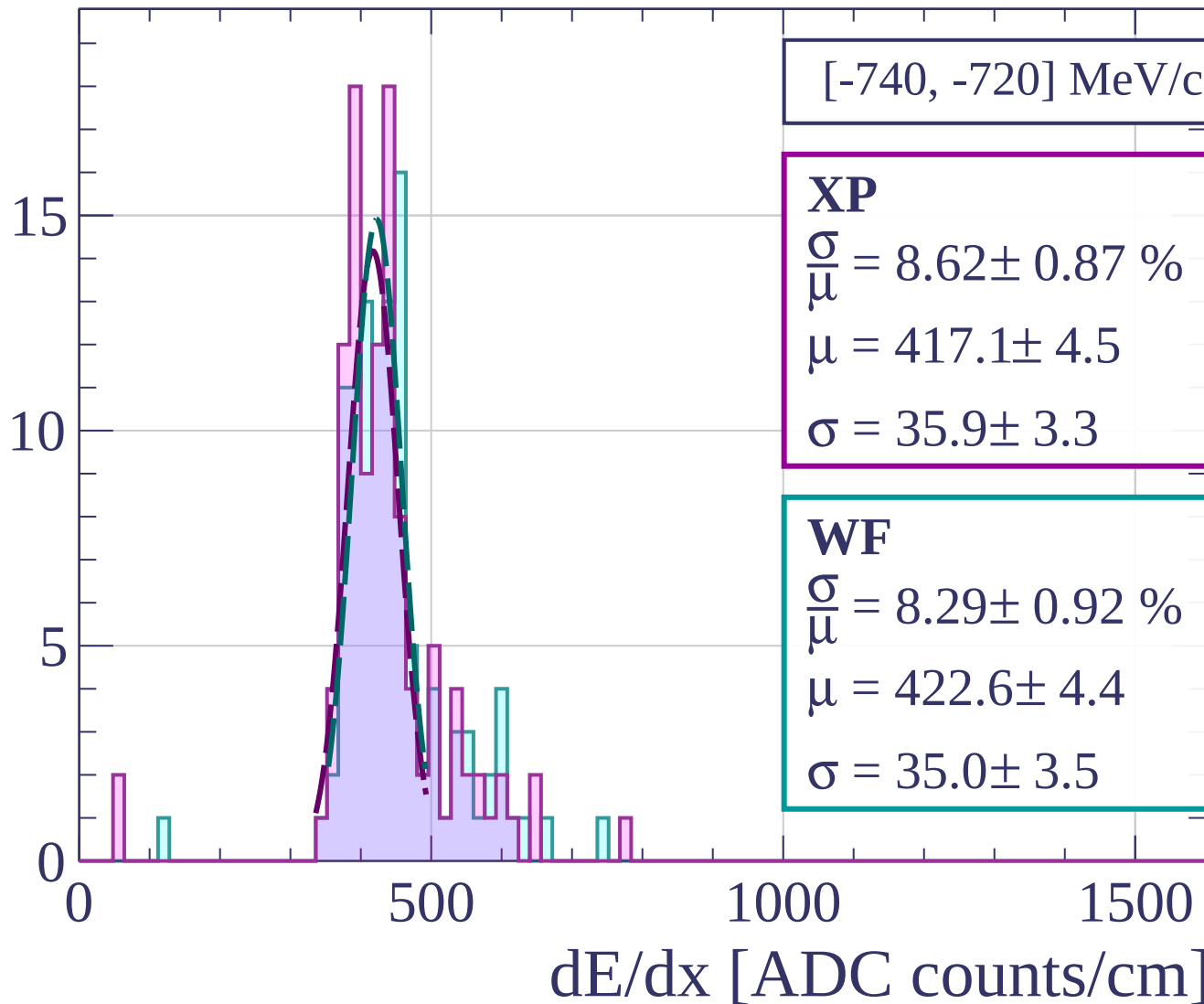
$$\frac{\sigma}{\mu} = 8.86 \pm 1.49 \%$$

$$\mu = 429.0 \pm 6.7$$

$$\sigma = 38.0 \pm 5.8$$

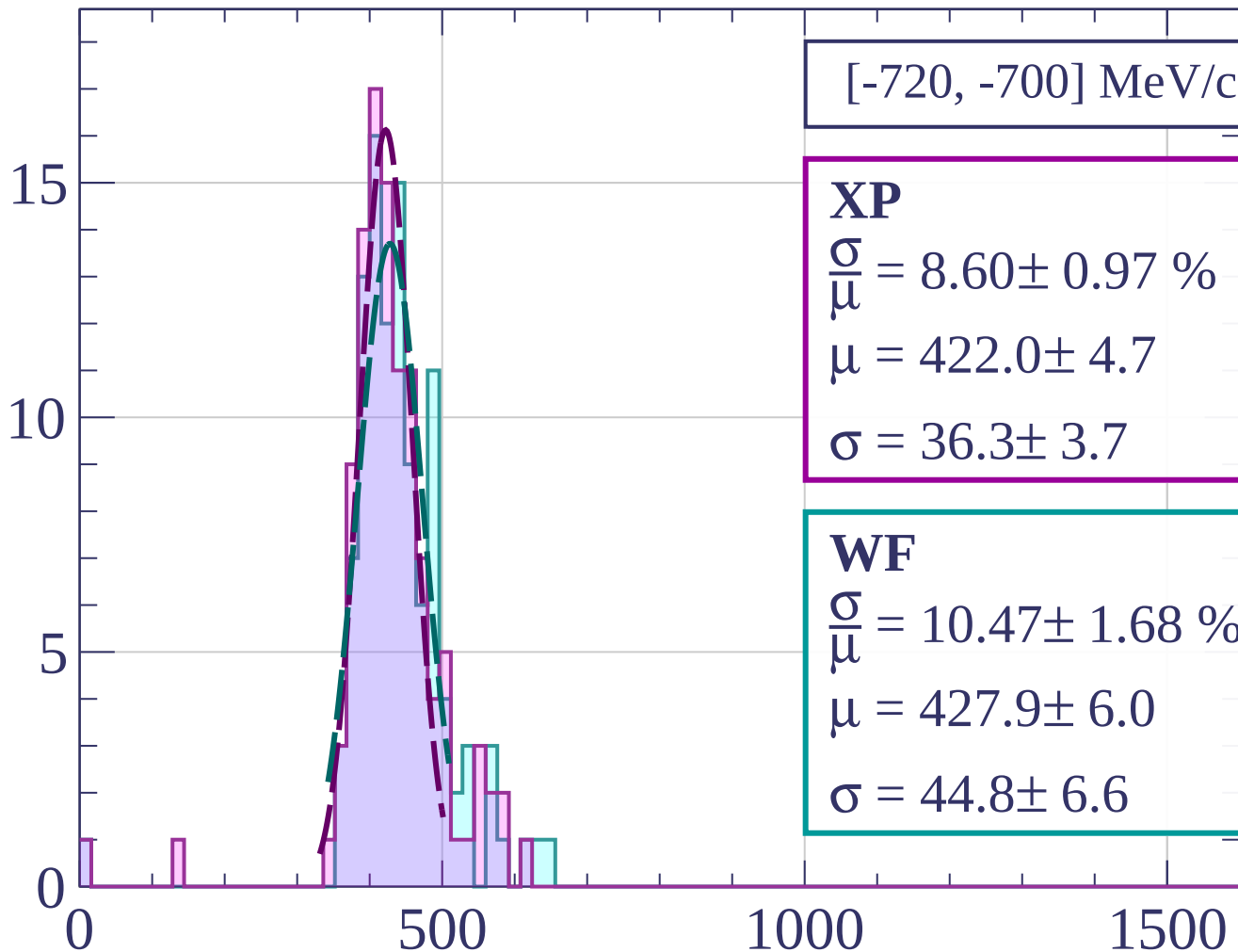
dE/dx [ADC counts/cm]

Count



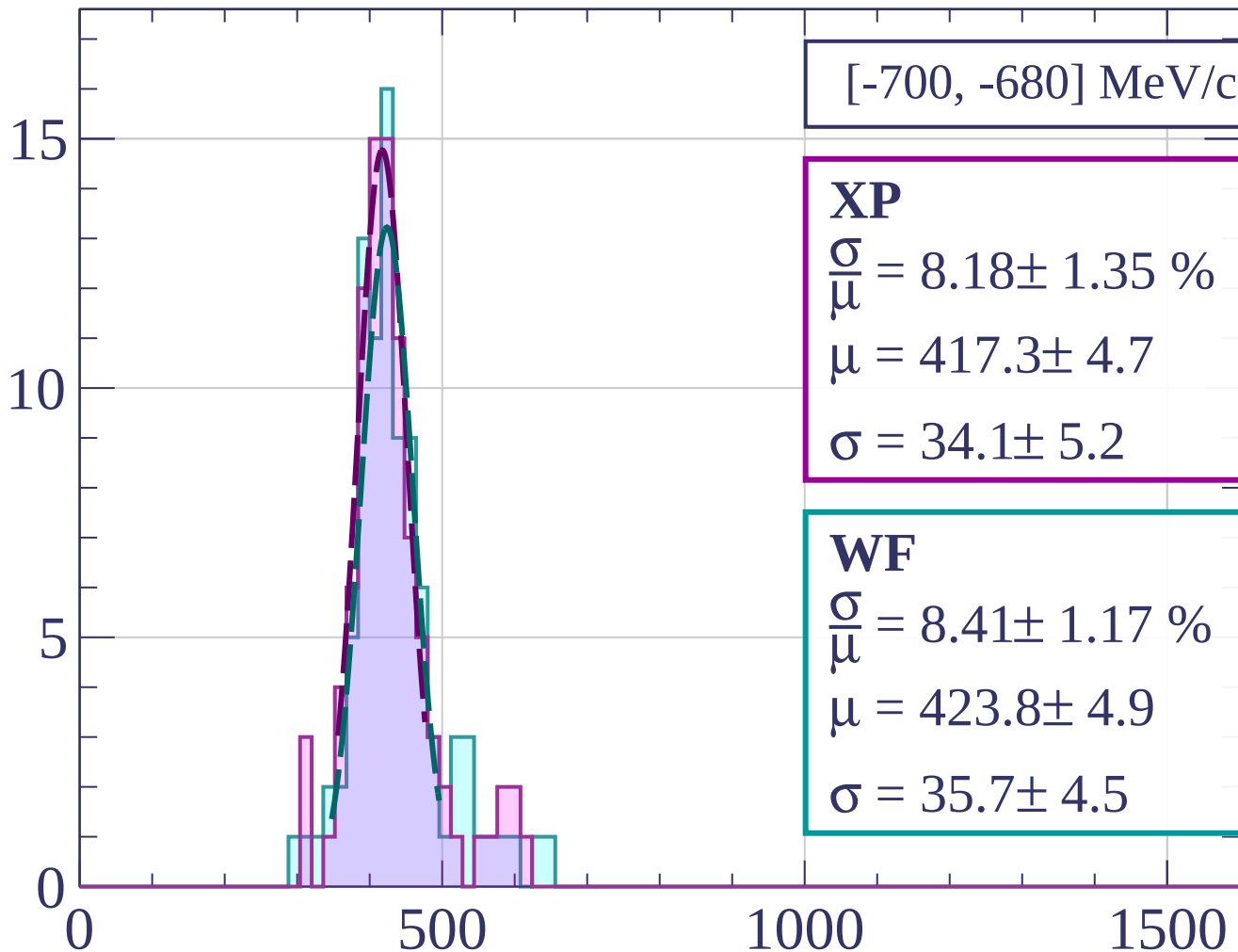
[-740, -720] MeV/c

Count



dE/dx [ADC counts/cm]

Count



[-700, -680] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.18 \pm 1.35 \%$$

$$\mu = 417.3 \pm 4.7$$

$$\sigma = 34.1 \pm 5.2$$

WF

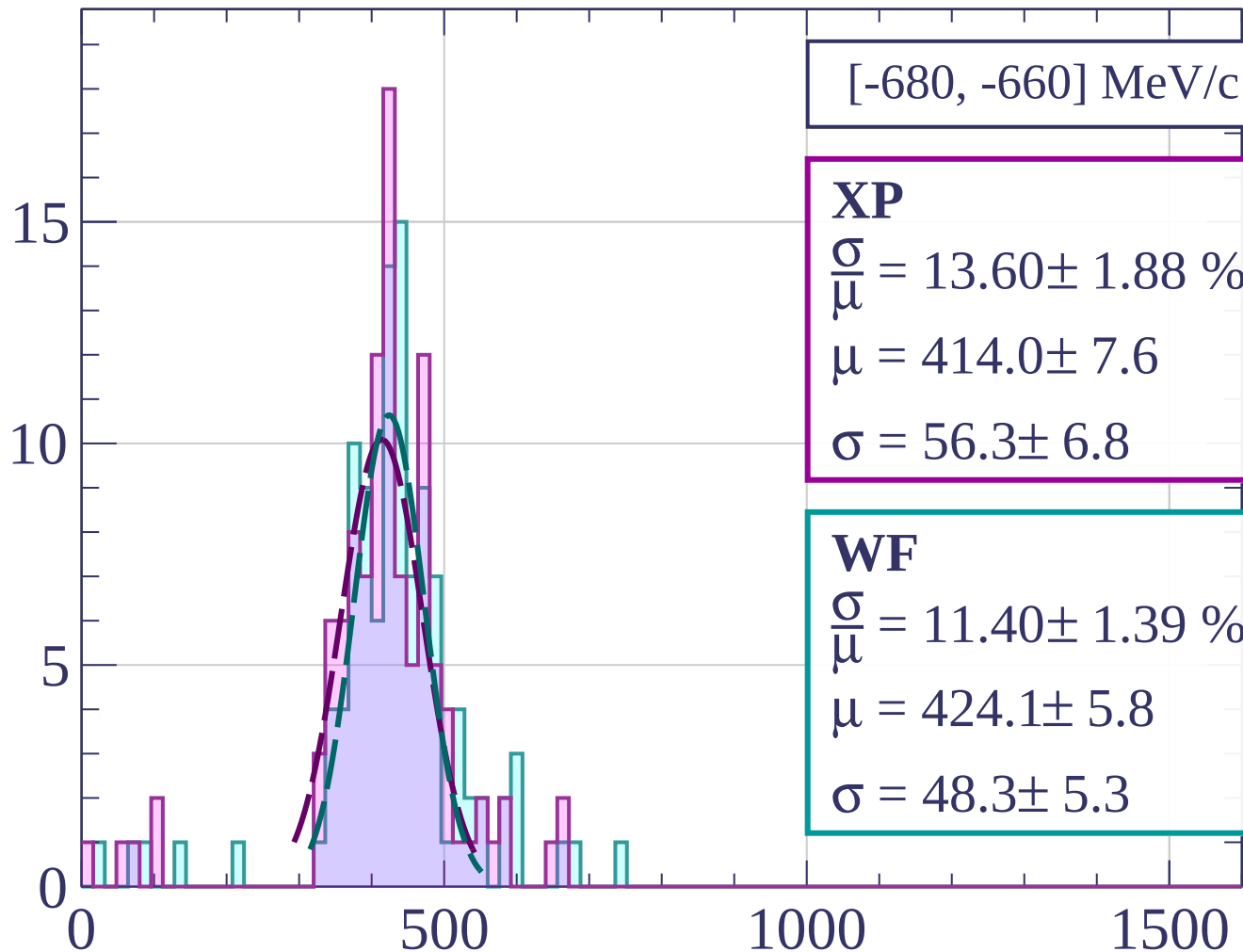
$$\frac{\sigma}{\mu} = 8.41 \pm 1.17 \%$$

$$\mu = 423.8 \pm 4.9$$

$$\sigma = 35.7 \pm 4.5$$

dE/dx [ADC counts/cm]

Count



[-680, -660] MeV/c

XP

$$\frac{\sigma}{\mu} = 13.60 \pm 1.88 \%$$

$$\mu = 414.0 \pm 7.6$$

$$\sigma = 56.3 \pm 6.8$$

WF

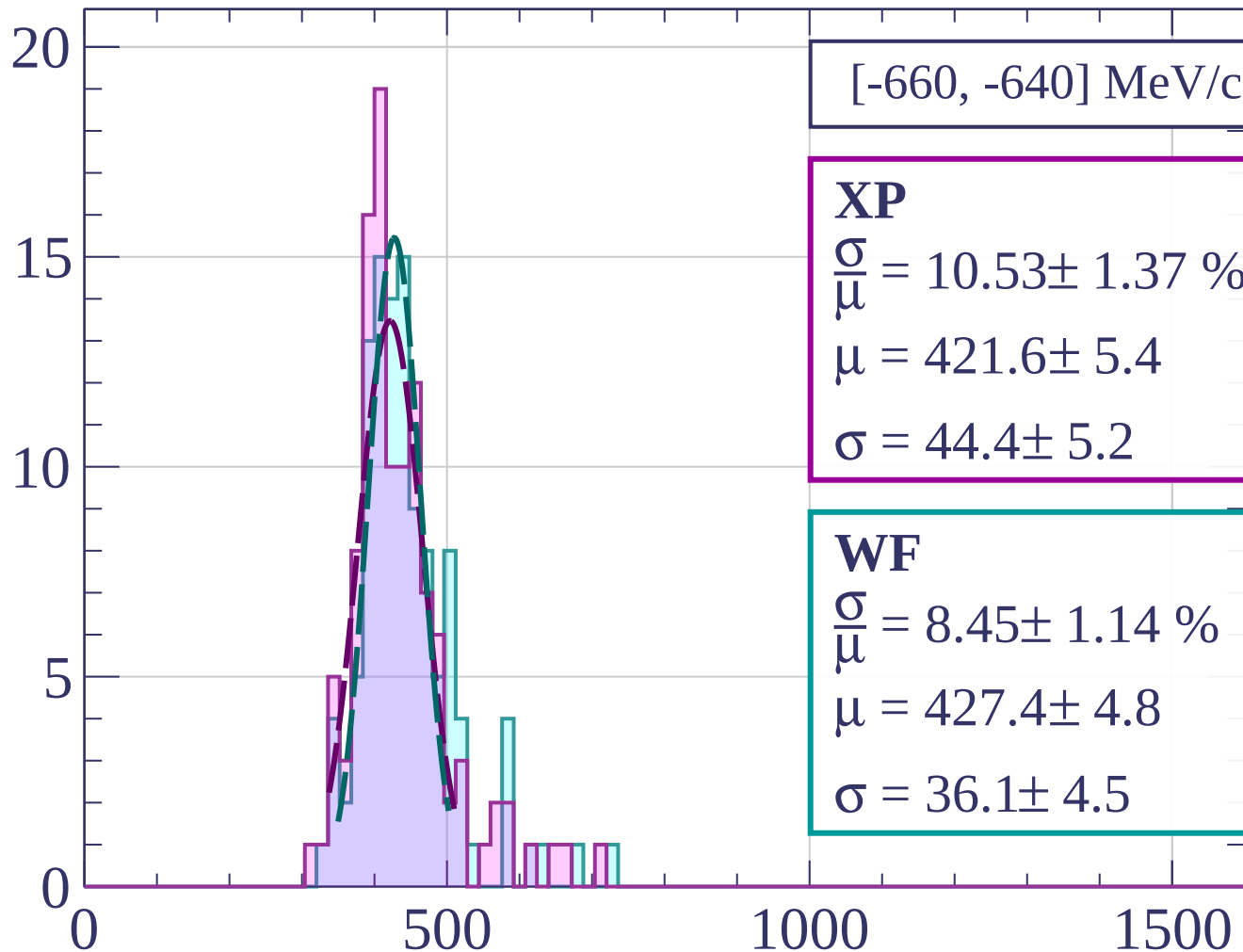
$$\frac{\sigma}{\mu} = 11.40 \pm 1.39 \%$$

$$\mu = 424.1 \pm 5.8$$

$$\sigma = 48.3 \pm 5.3$$

dE/dx [ADC counts/cm]

Count



[-660, -640] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.53 \pm 1.37 \%$$

$$\mu = 421.6 \pm 5.4$$

$$\sigma = 44.4 \pm 5.2$$

WF

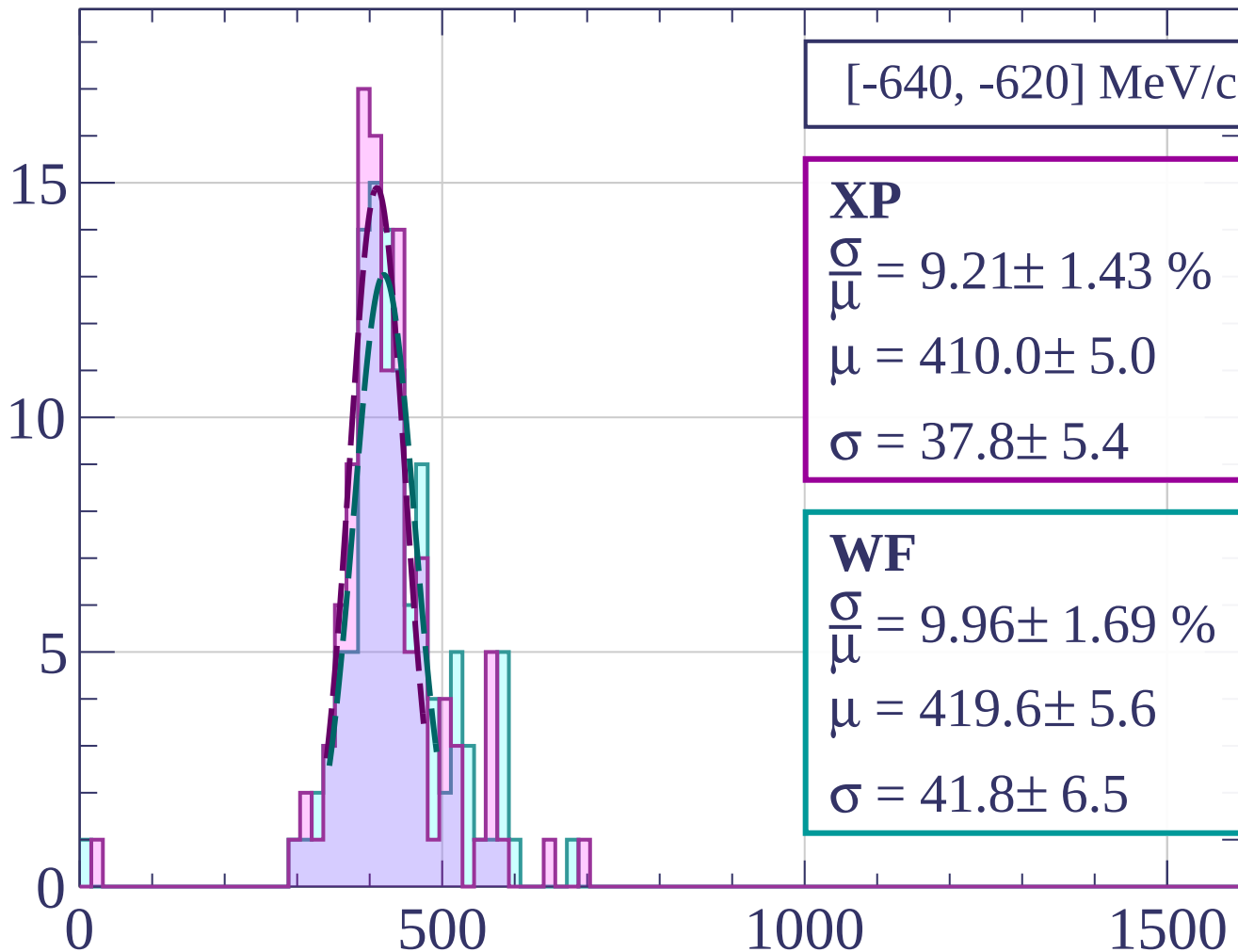
$$\frac{\sigma}{\mu} = 8.45 \pm 1.14 \%$$

$$\mu = 427.4 \pm 4.8$$

$$\sigma = 36.1 \pm 4.5$$

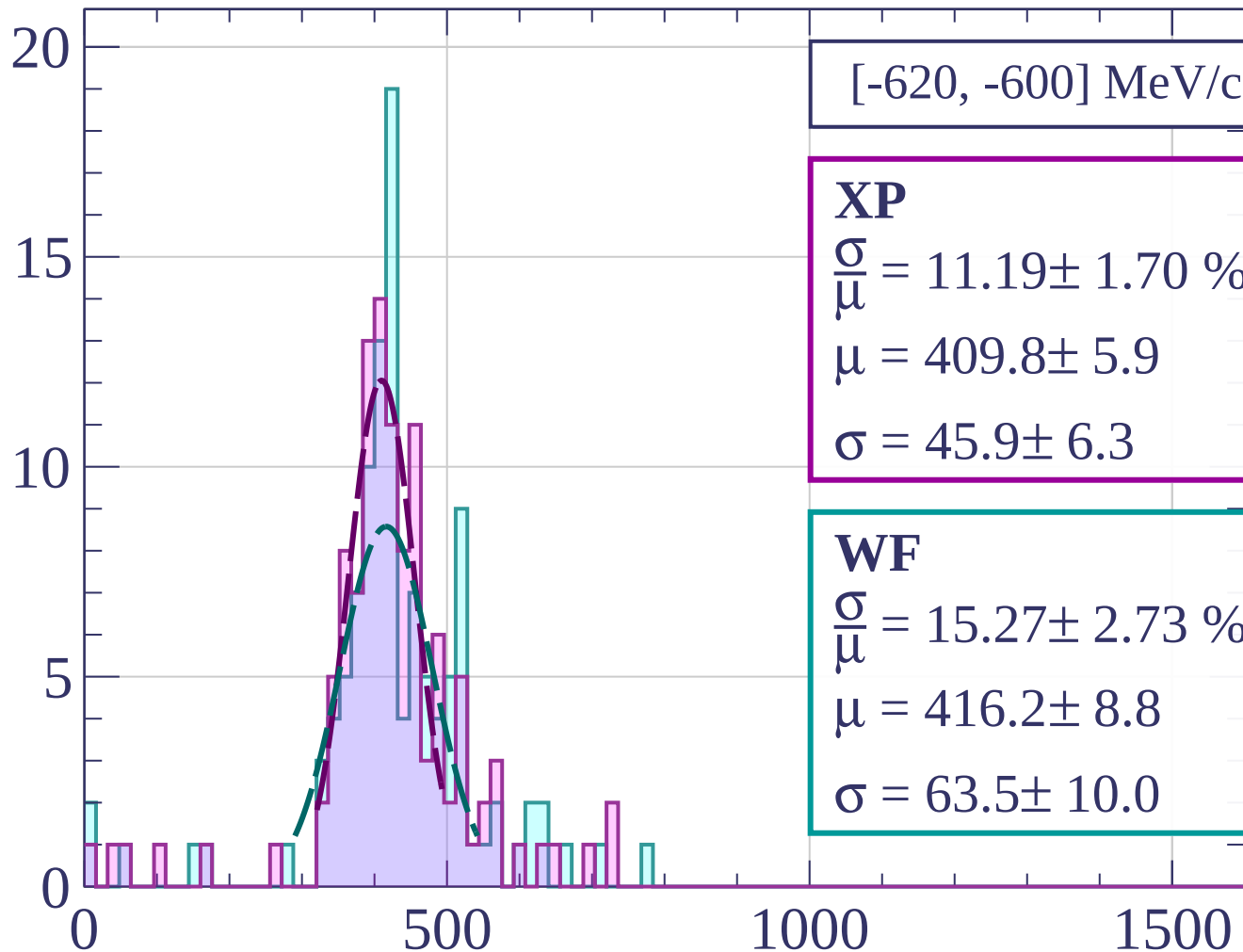
dE/dx [ADC counts/cm]

Count



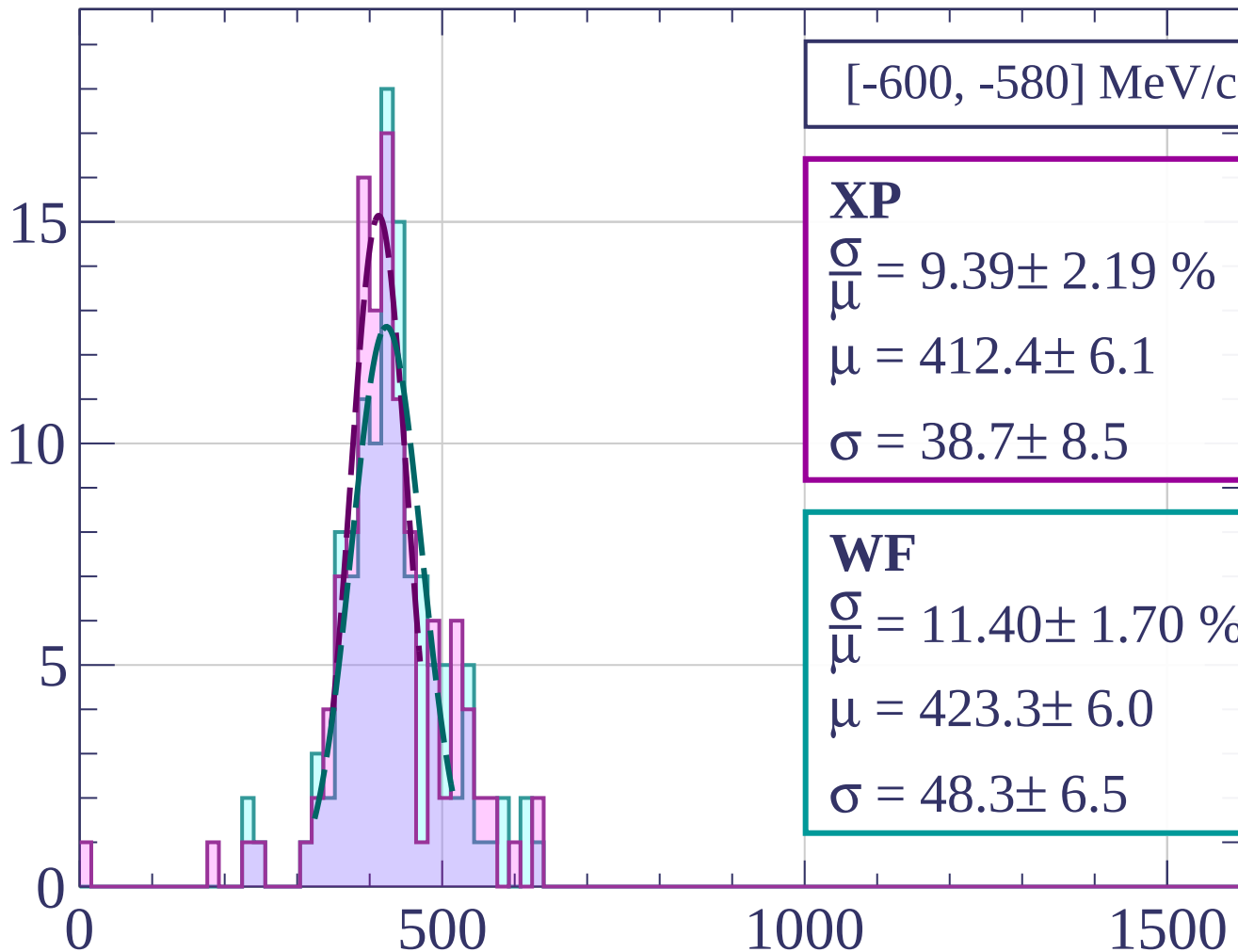
dE/dx [ADC counts/cm]

Count



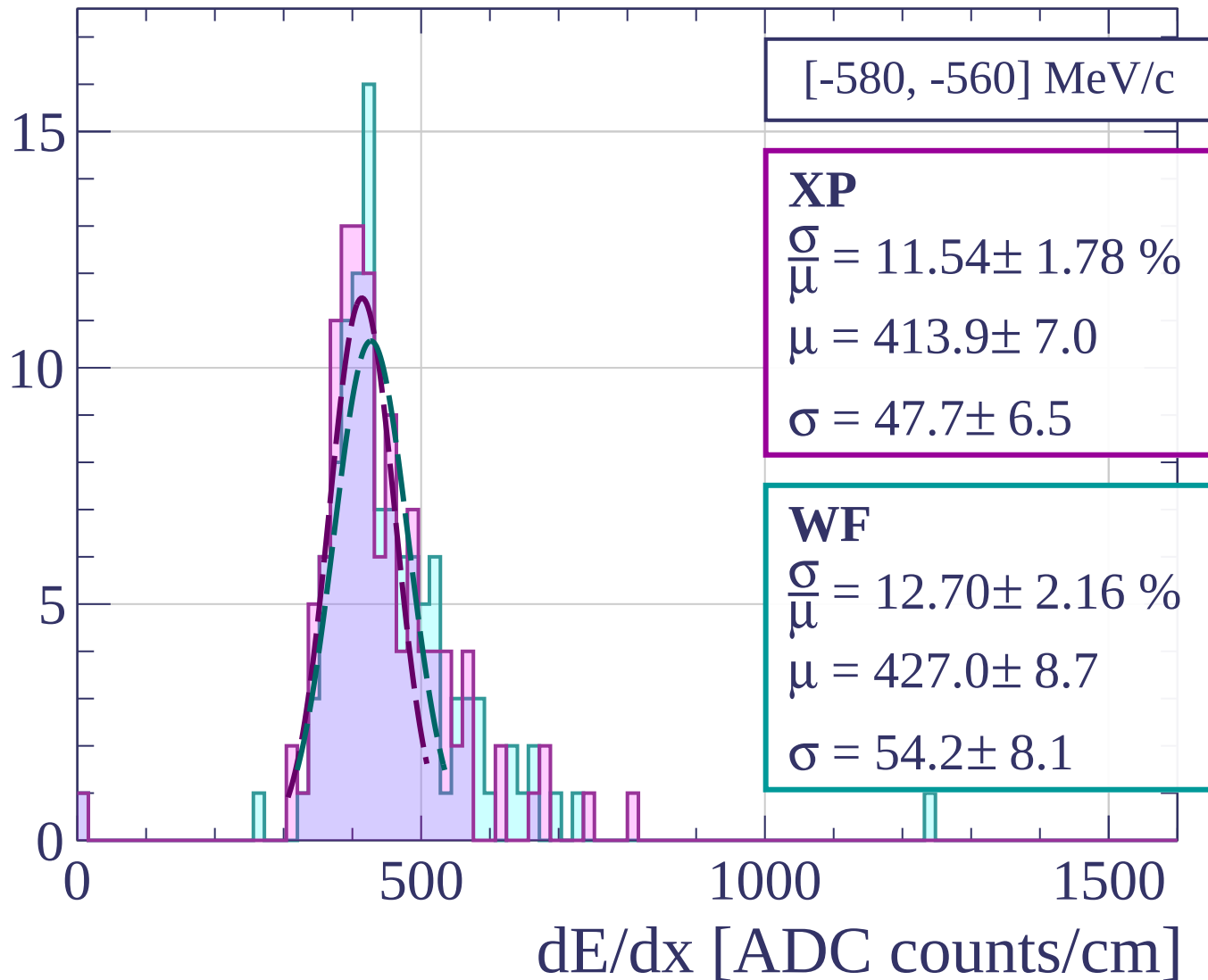
dE/dx [ADC counts/cm]

Count

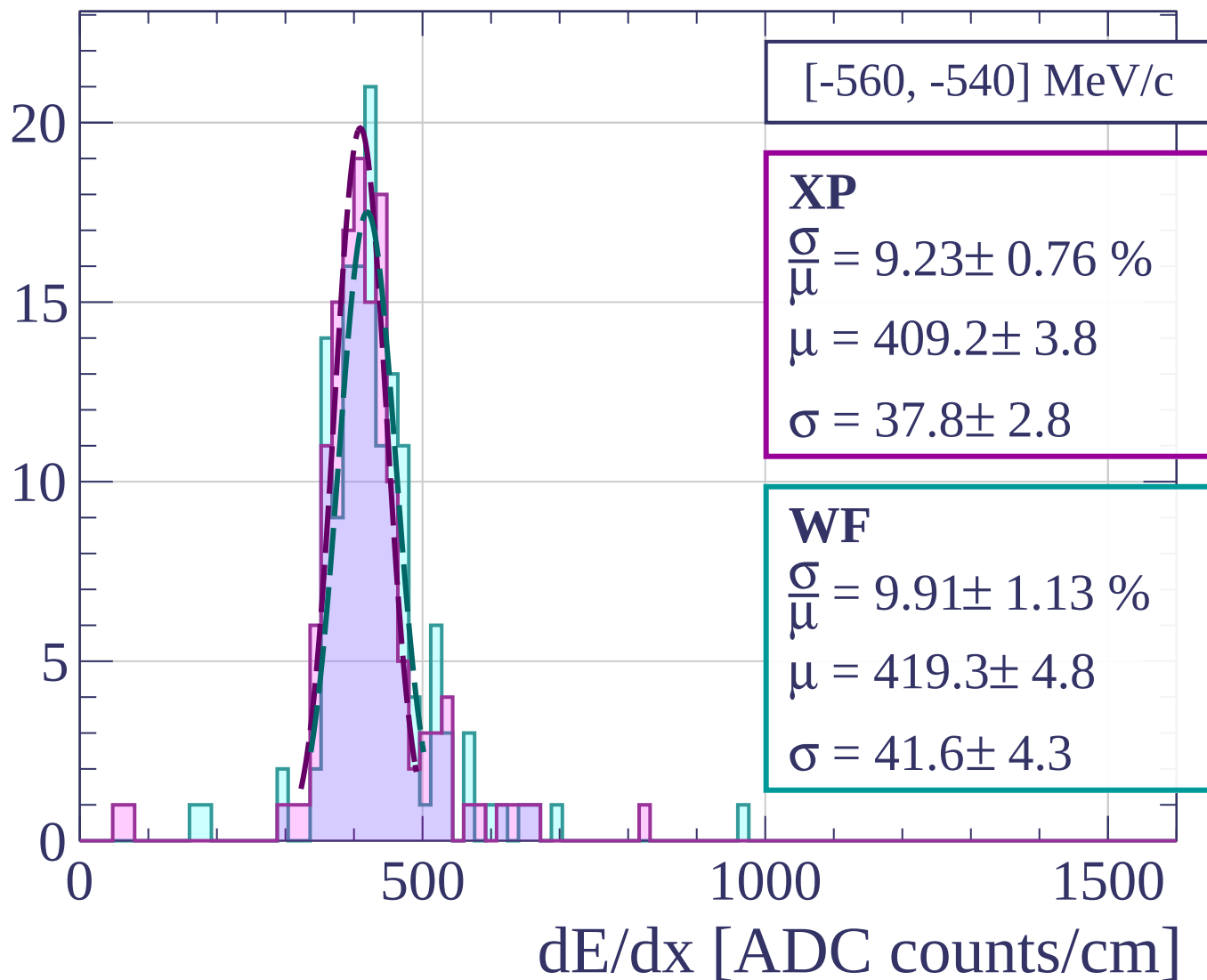


dE/dx [ADC counts/cm]

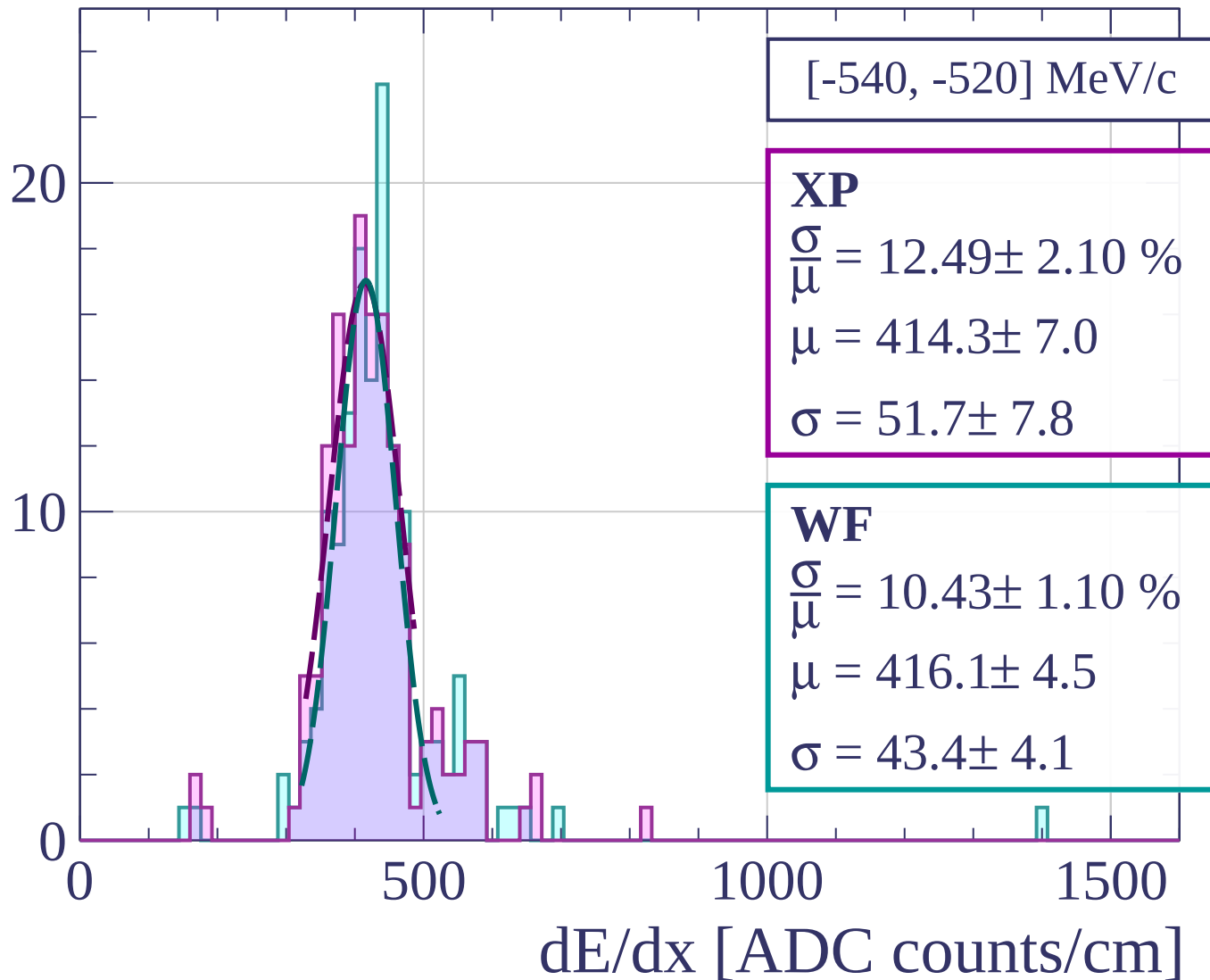
Count



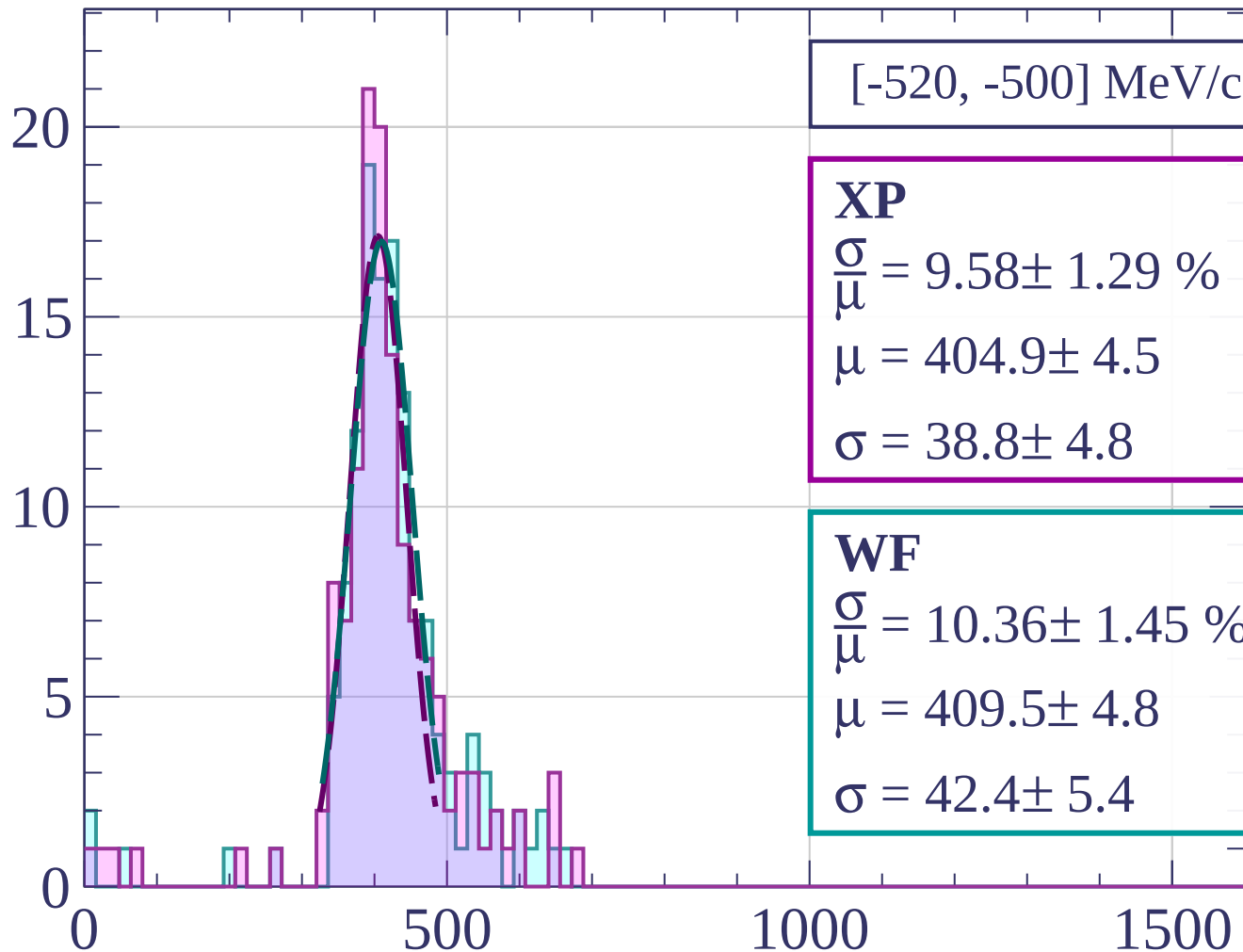
Count



Count

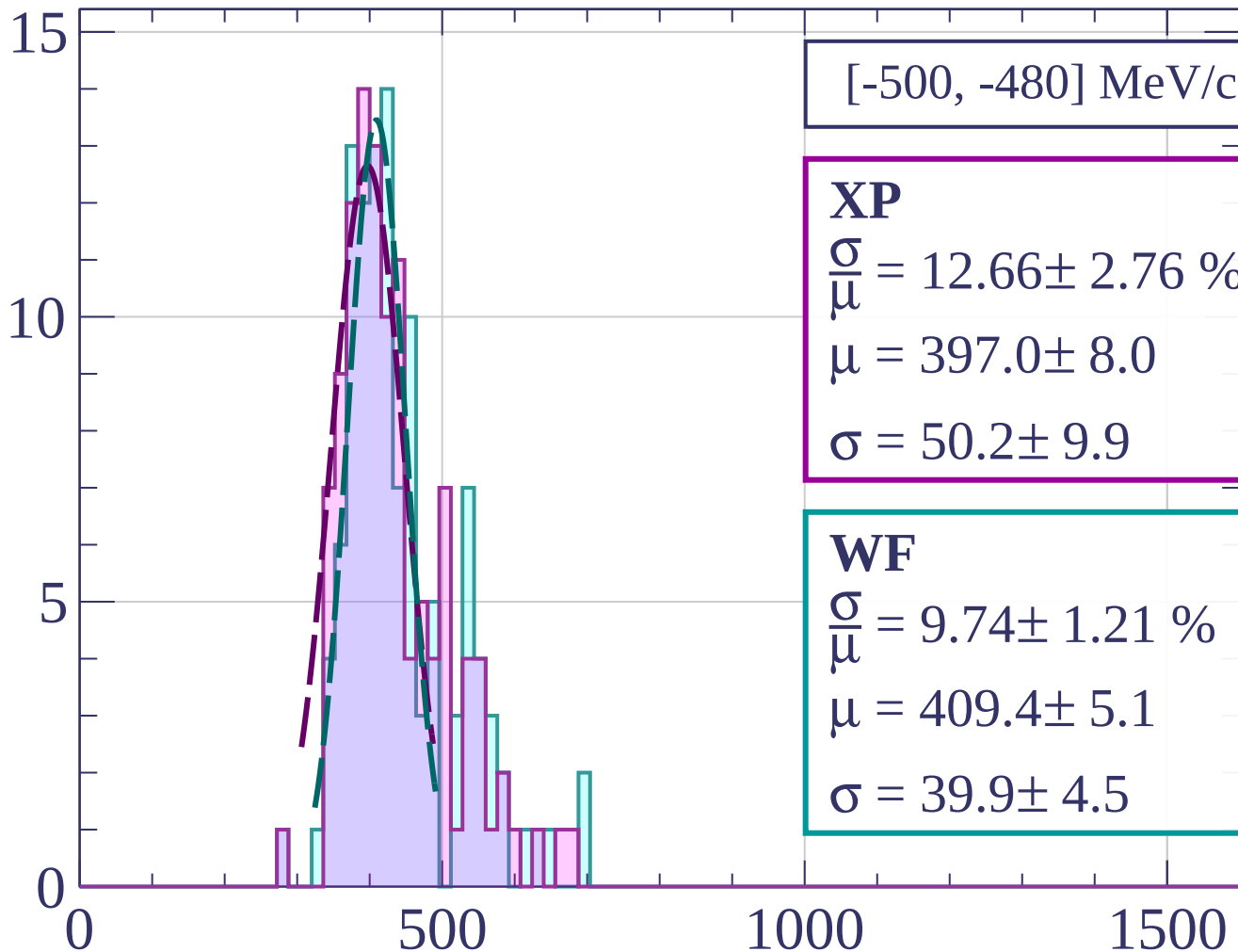


Count



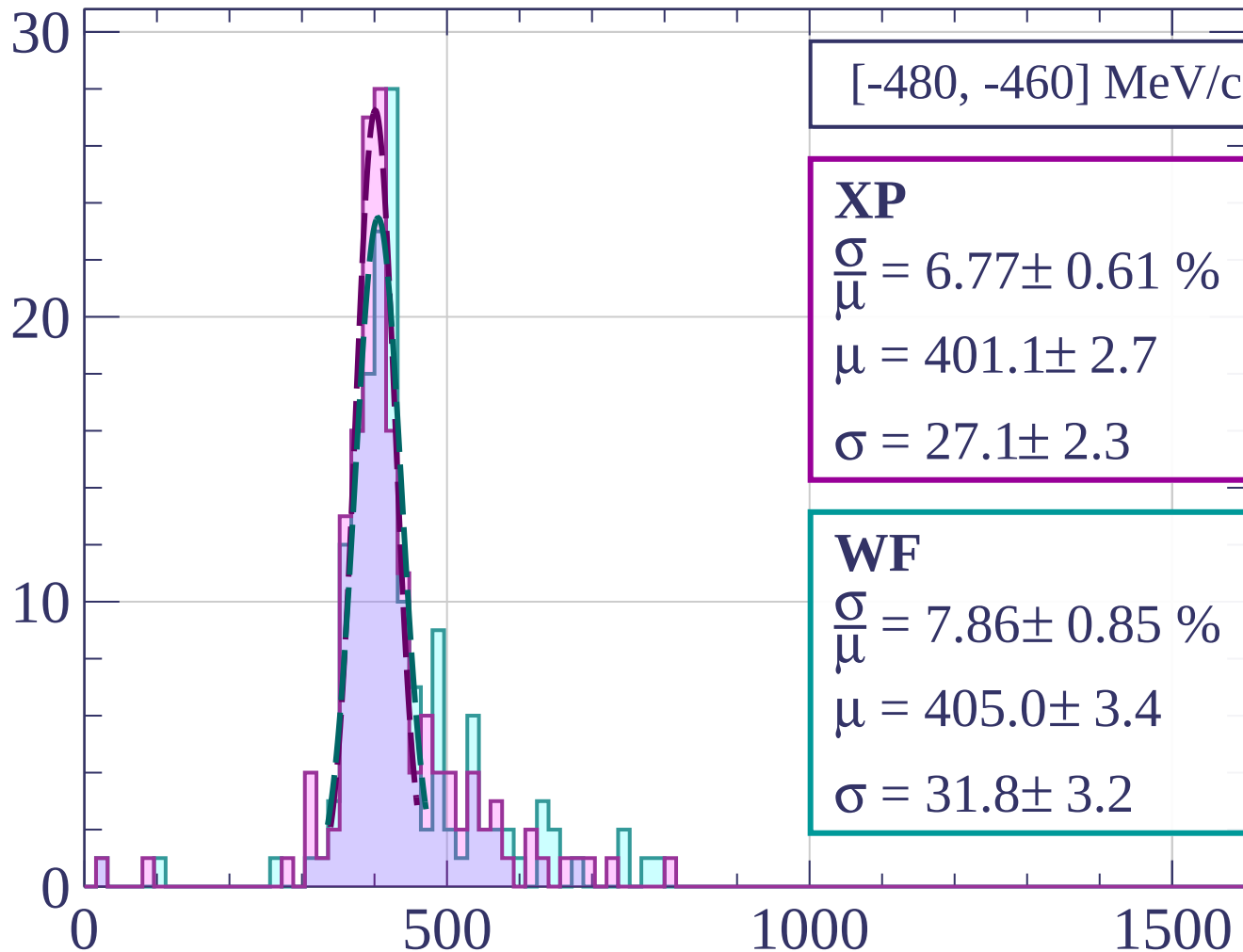
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[-480, -460] MeV/c

XP

$$\frac{\sigma}{\bar{\mu}} = 6.77 \pm 0.61 \%$$

$$\mu = 401.1 \pm 2.7$$

$$\sigma = 27.1 \pm 2.3$$

WF

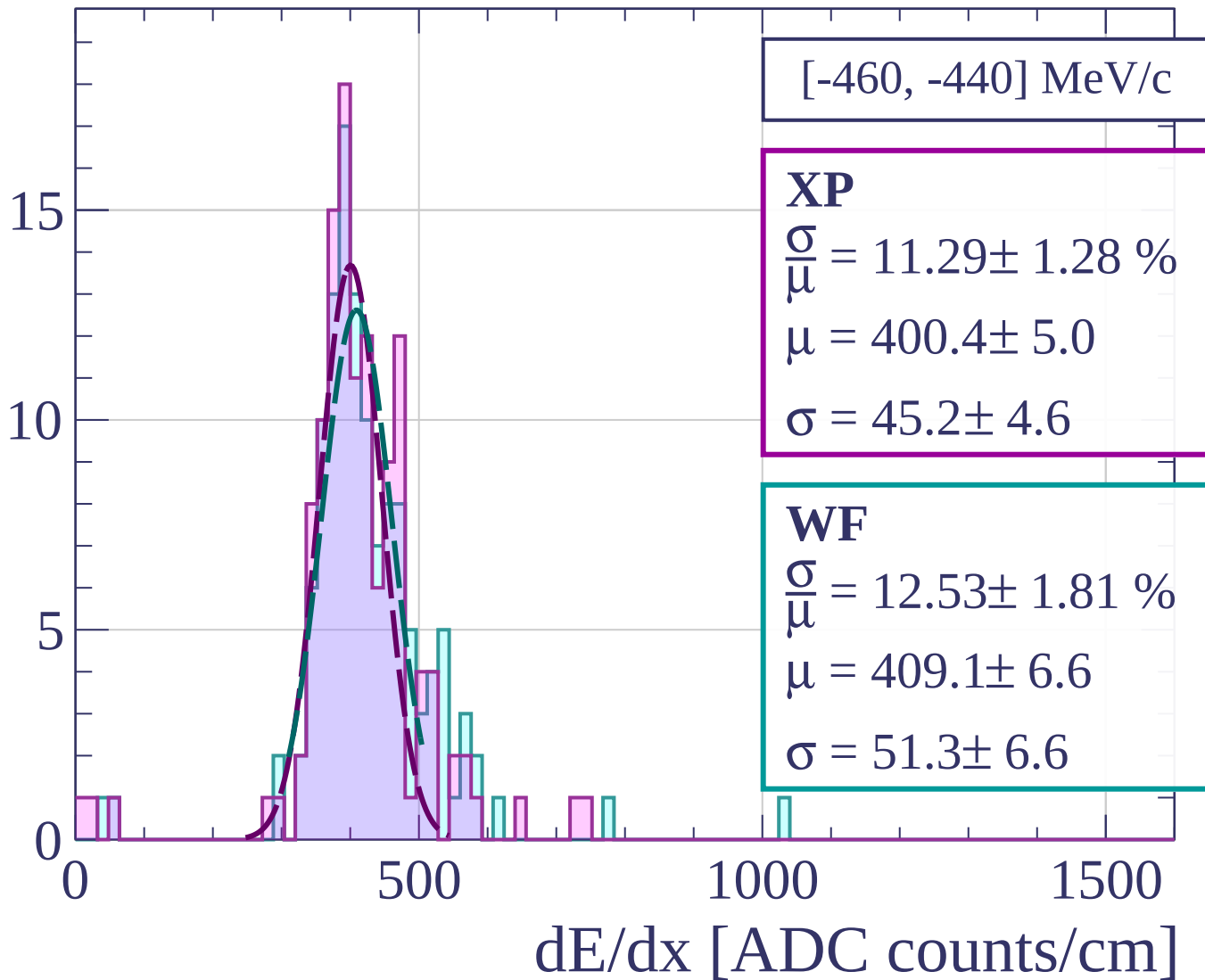
$$\frac{\sigma}{\bar{\mu}} = 7.86 \pm 0.85 \%$$

$$\mu = 405.0 \pm 3.4$$

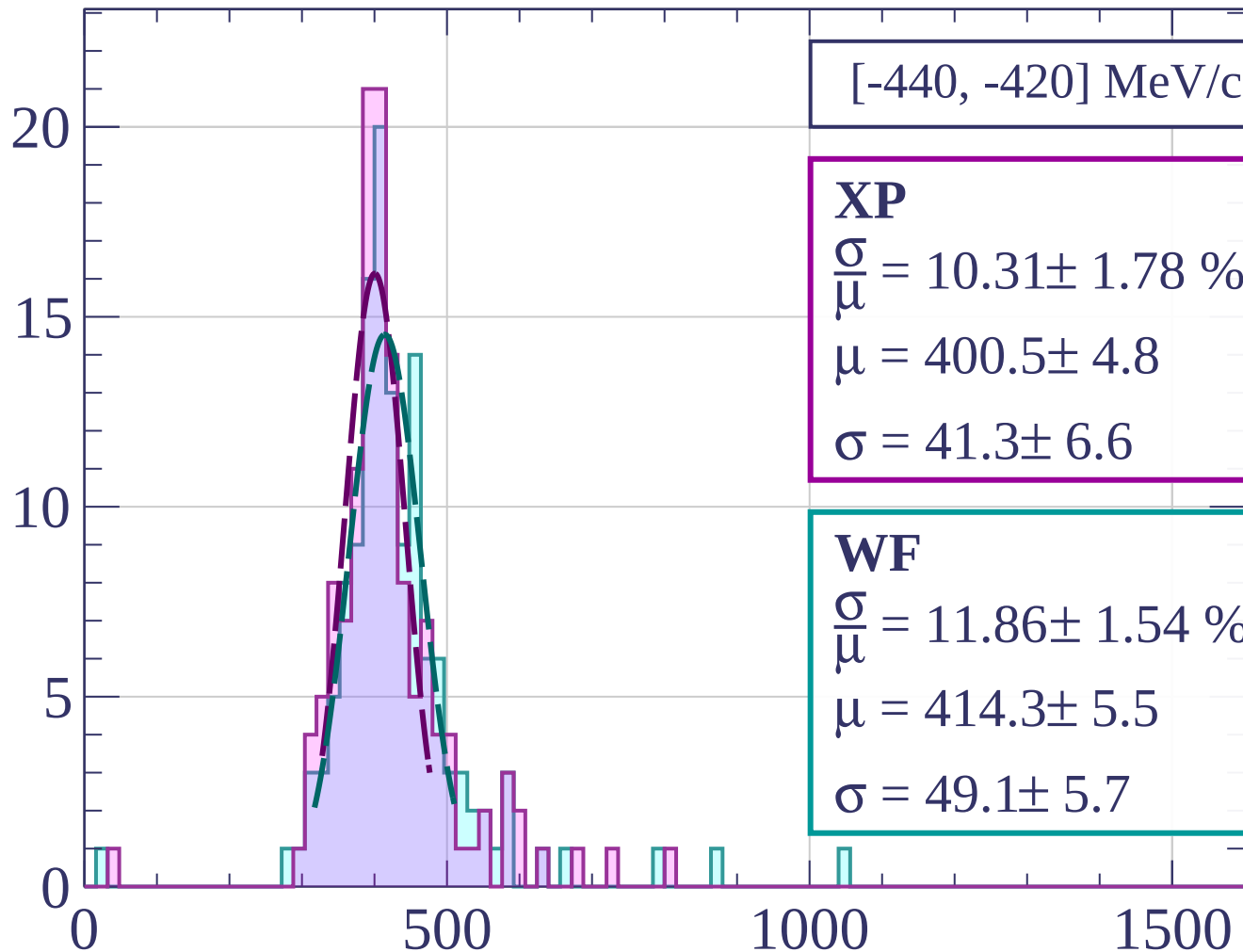
$$\sigma = 31.8 \pm 3.2$$

dE/dx [ADC counts/cm]

Count



Count



[-440, -420] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.31 \pm 1.78 \%$$

$$\mu = 400.5 \pm 4.8$$

$$\sigma = 41.3 \pm 6.6$$

WF

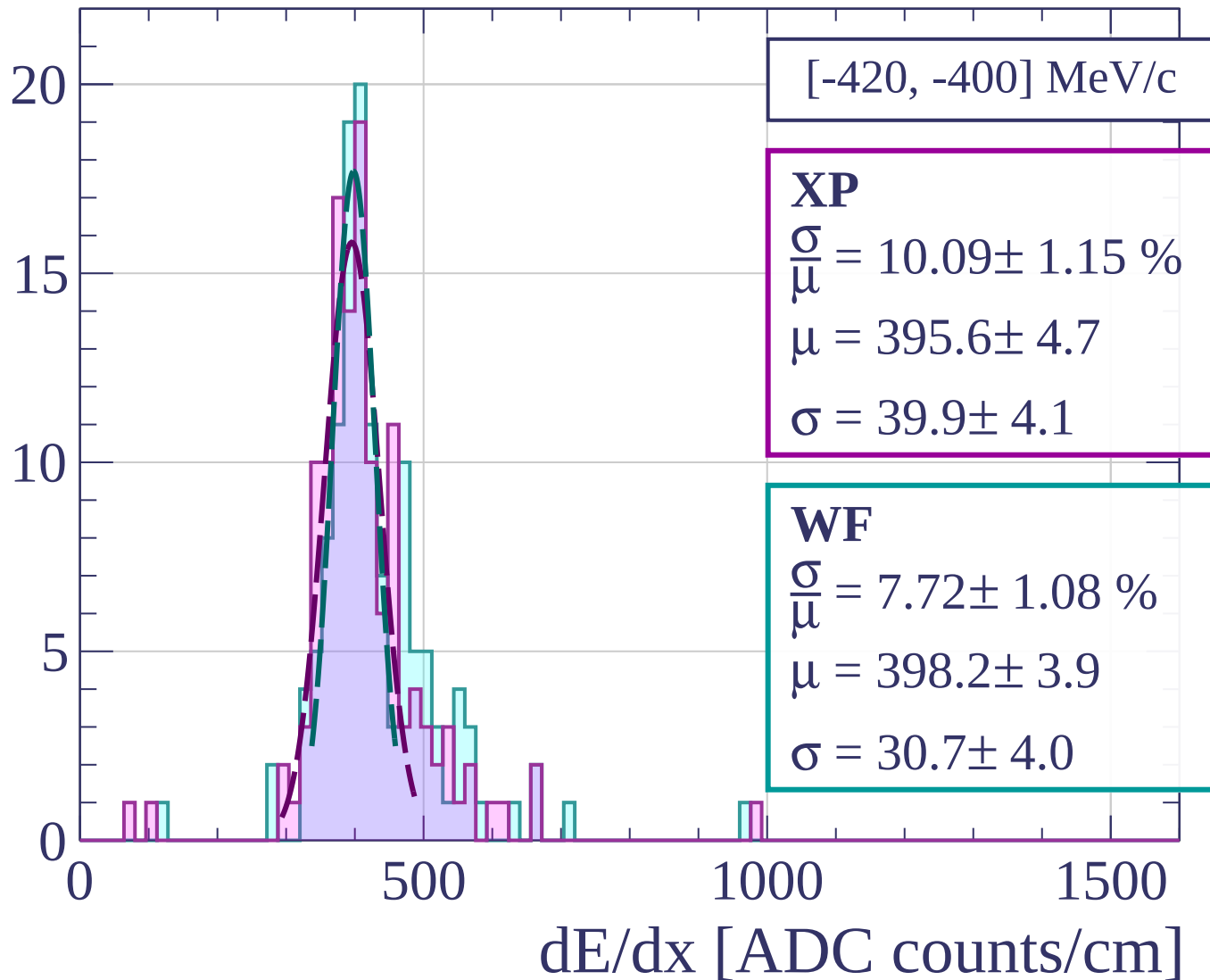
$$\frac{\sigma}{\mu} = 11.86 \pm 1.54 \%$$

$$\mu = 414.3 \pm 5.5$$

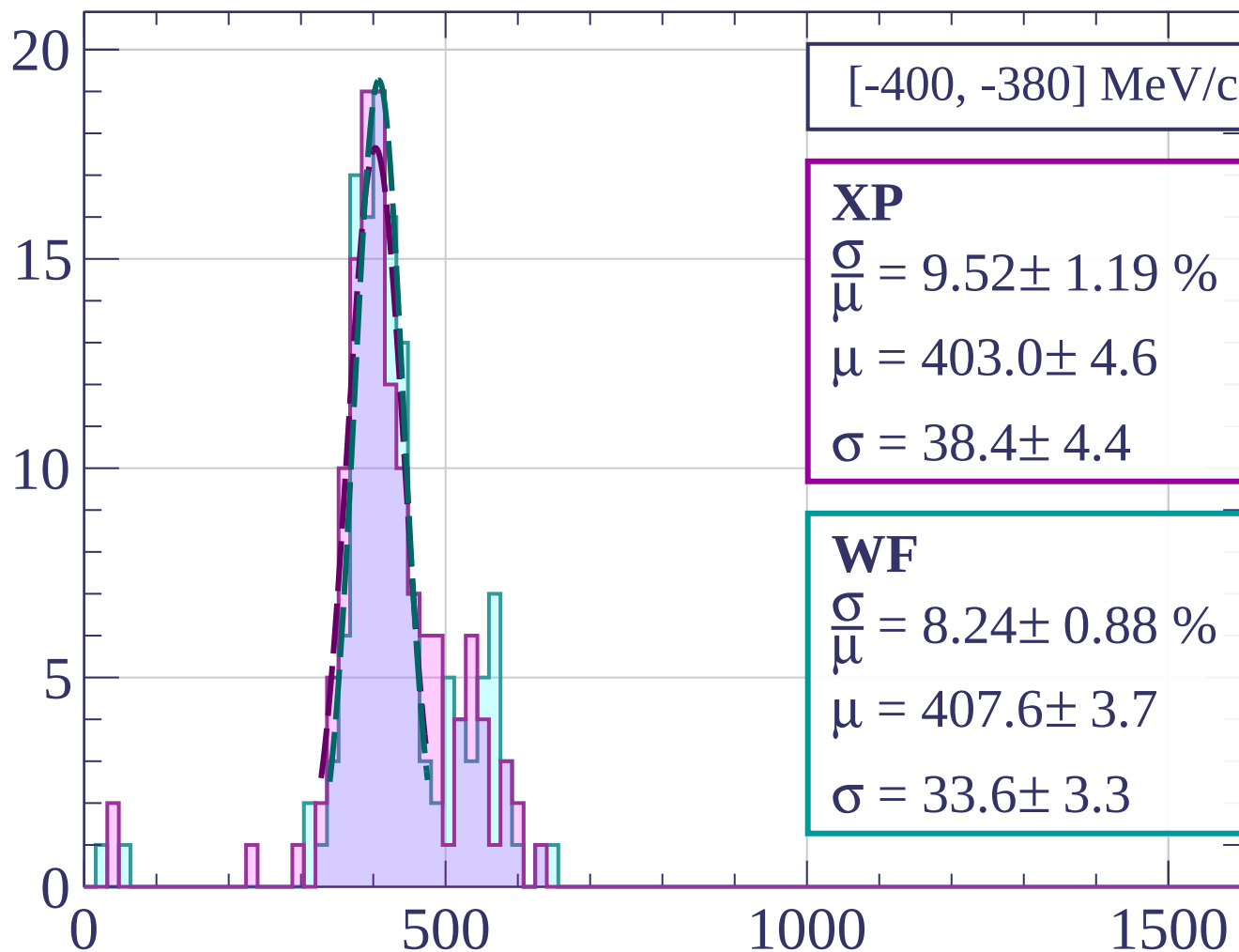
$$\sigma = 49.1 \pm 5.7$$

dE/dx [ADC counts/cm]

Count



Count



[-400, -380] MeV/c

XP

$$\frac{\sigma}{\bar{\mu}} = 9.52 \pm 1.19 \%$$

$$\mu = 403.0 \pm 4.6$$

$$\sigma = 38.4 \pm 4.4$$

WF

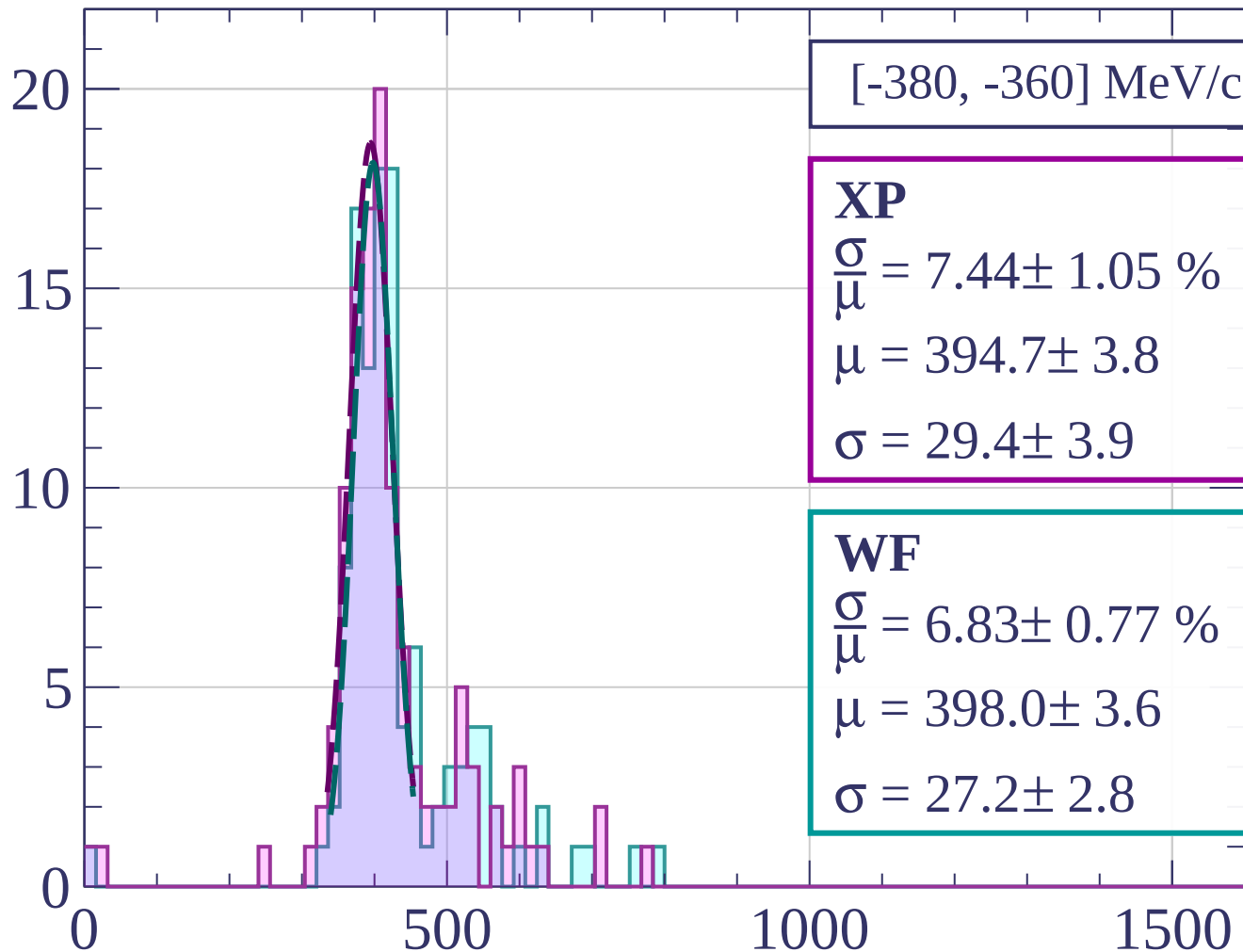
$$\frac{\sigma}{\bar{\mu}} = 8.24 \pm 0.88 \%$$

$$\mu = 407.6 \pm 3.7$$

$$\sigma = 33.6 \pm 3.3$$

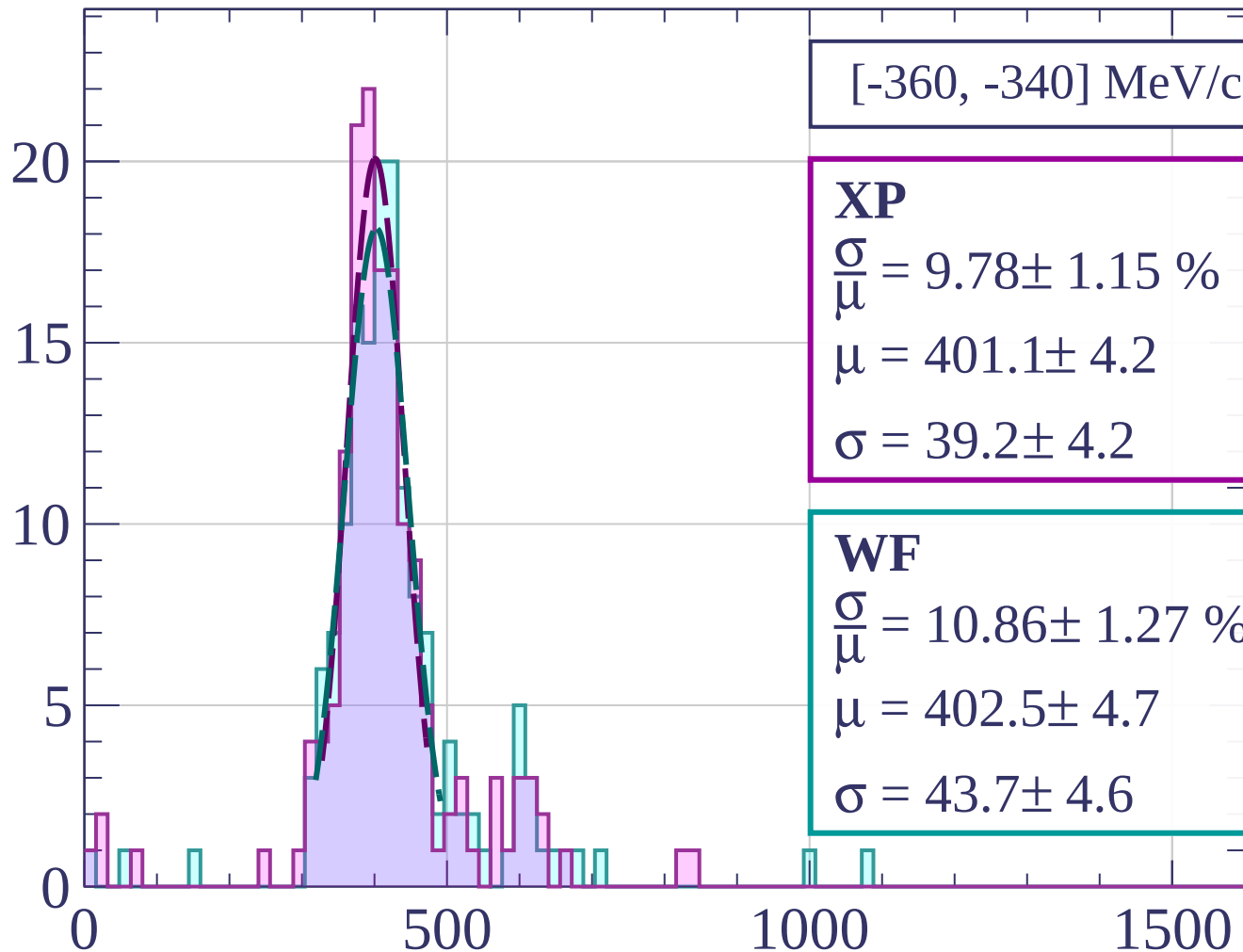
dE/dx [ADC counts/cm]

Count



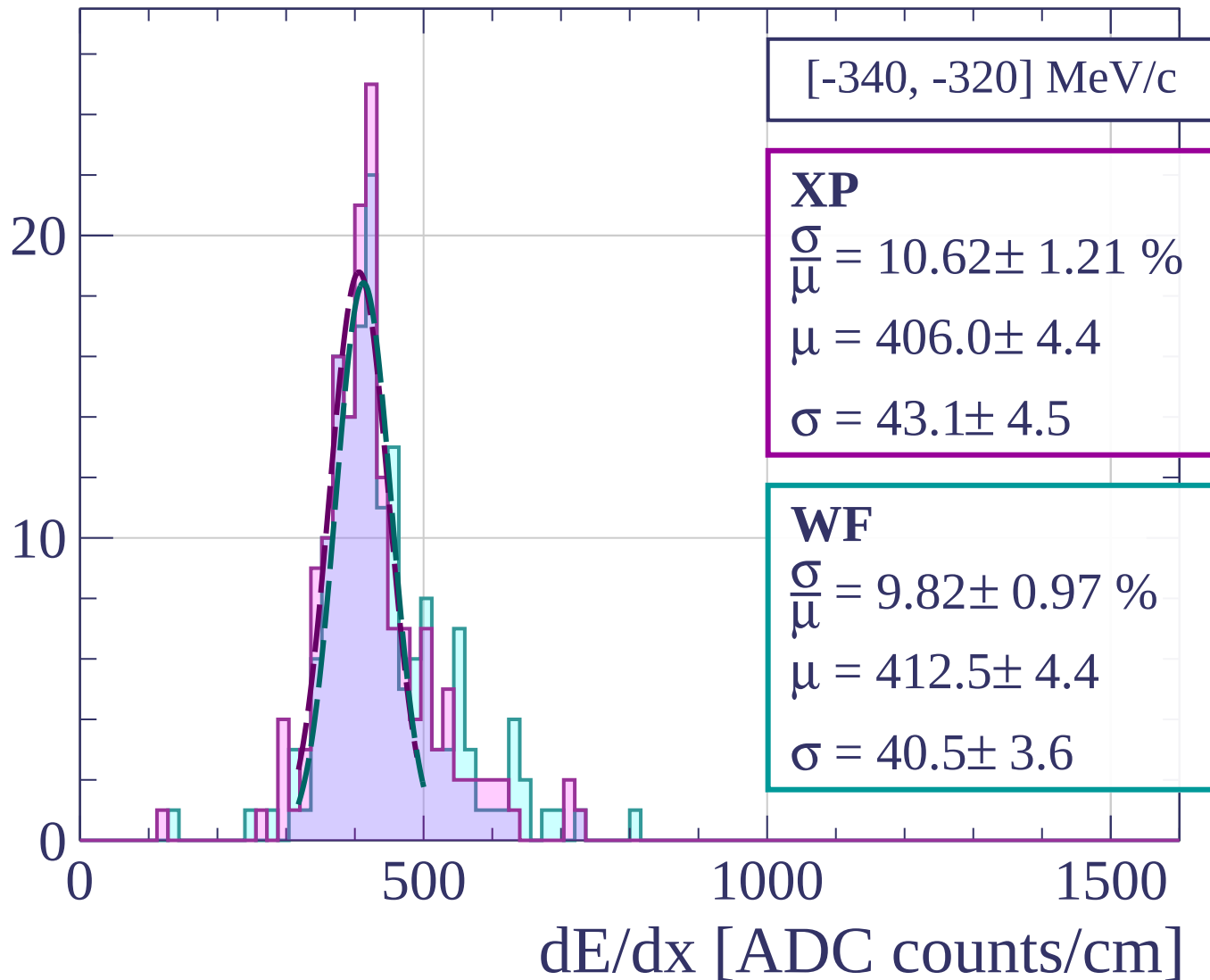
dE/dx [ADC counts/cm]

Count

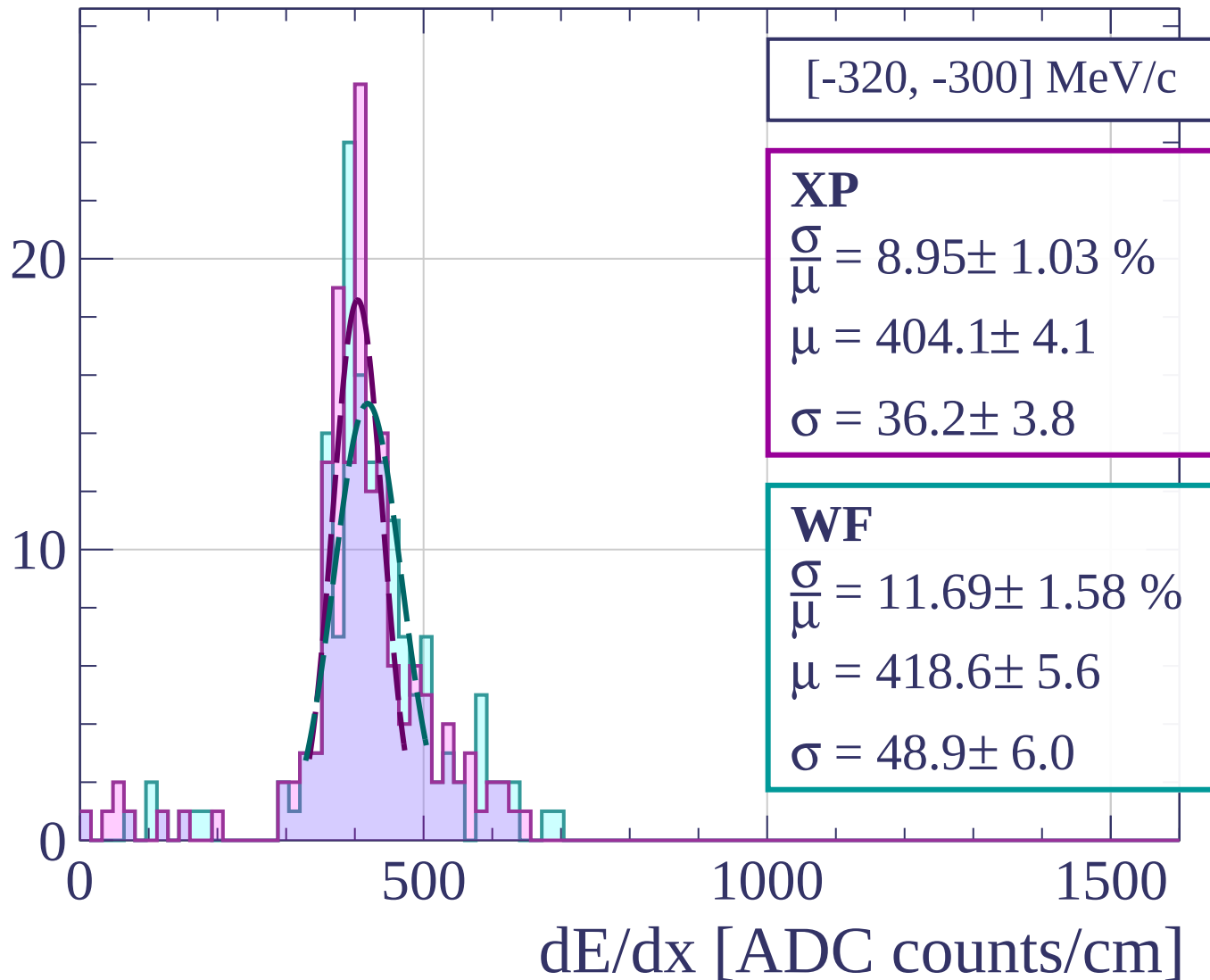


dE/dx [ADC counts/cm]

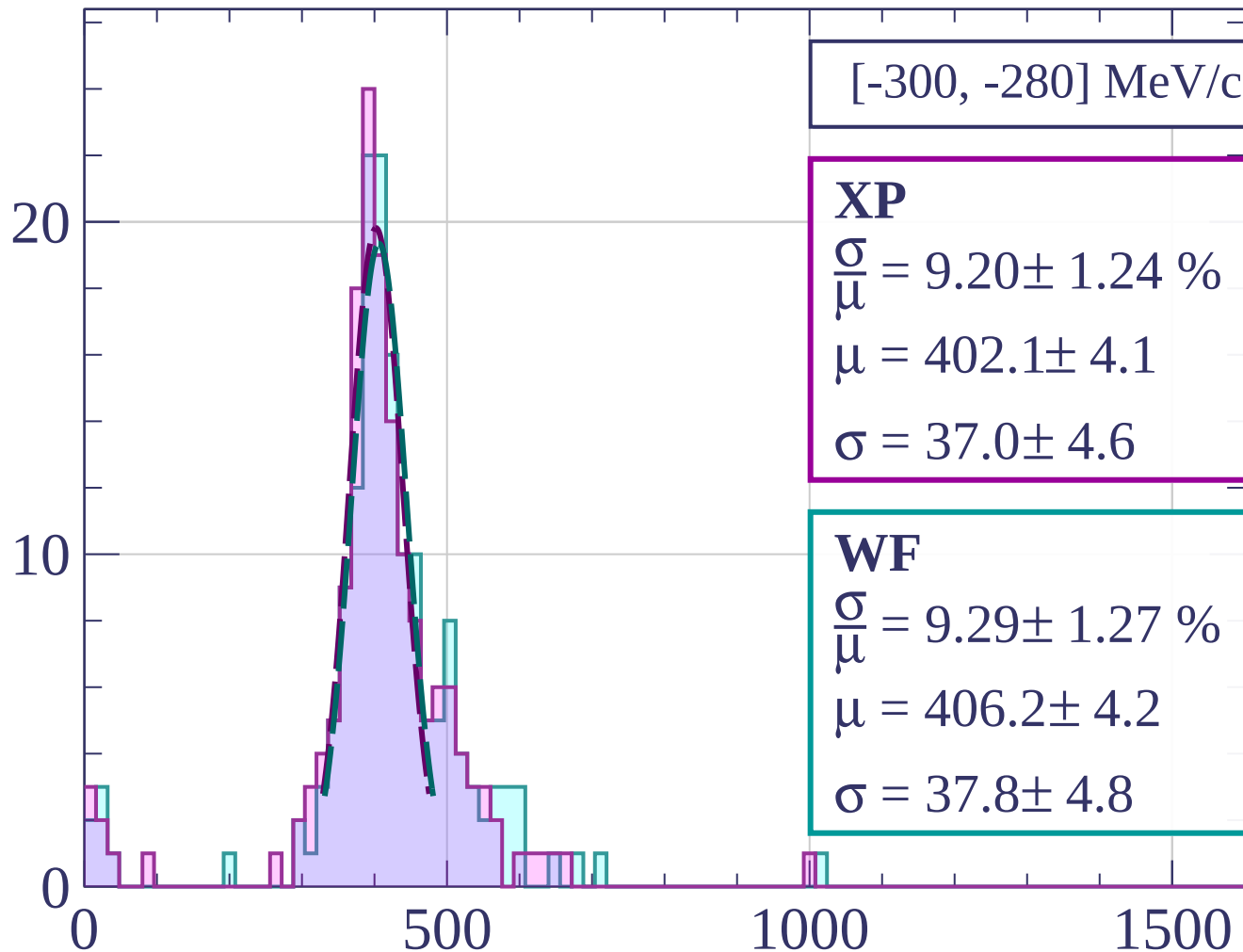
Count



Count



Count



[-300, -280] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.20 \pm 1.24 \%$$

$$\mu = 402.1 \pm 4.1$$

$$\sigma = 37.0 \pm 4.6$$

WF

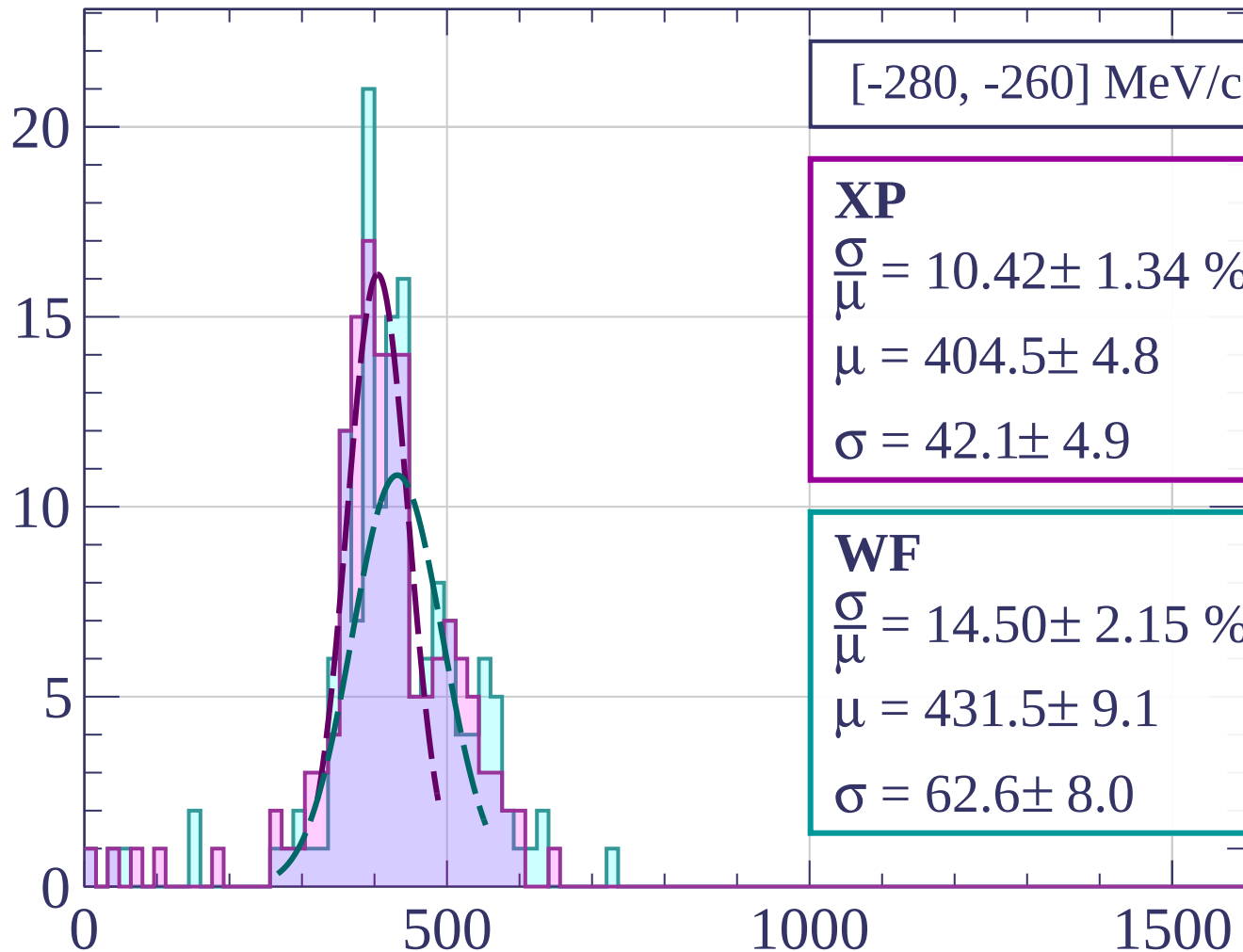
$$\frac{\sigma}{\mu} = 9.29 \pm 1.27 \%$$

$$\mu = 406.2 \pm 4.2$$

$$\sigma = 37.8 \pm 4.8$$

dE/dx [ADC counts/cm]

Count



[-280, -260] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.42 \pm 1.34 \%$$

$$\mu = 404.5 \pm 4.8$$

$$\sigma = 42.1 \pm 4.9$$

WF

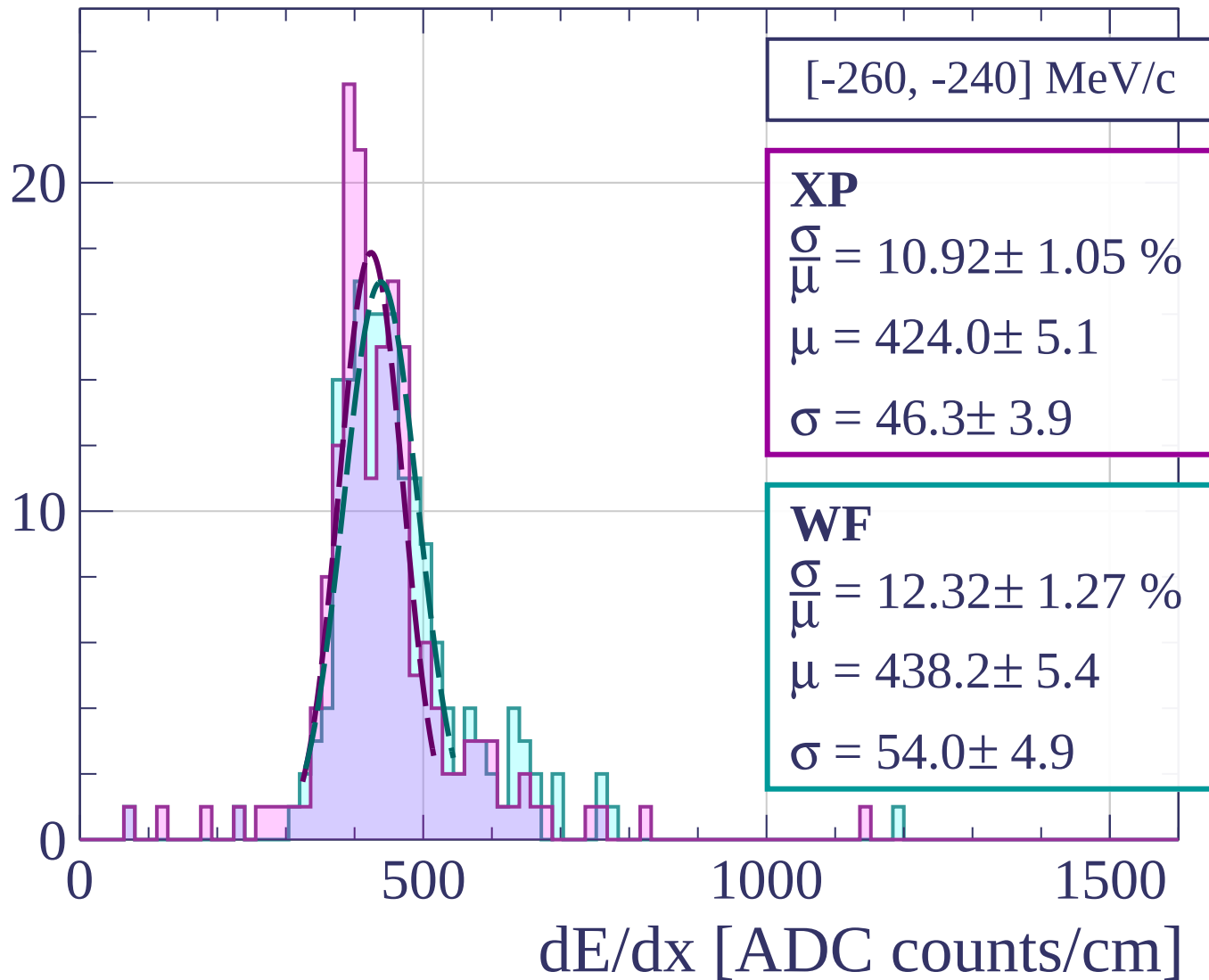
$$\frac{\sigma}{\mu} = 14.50 \pm 2.15 \%$$

$$\mu = 431.5 \pm 9.1$$

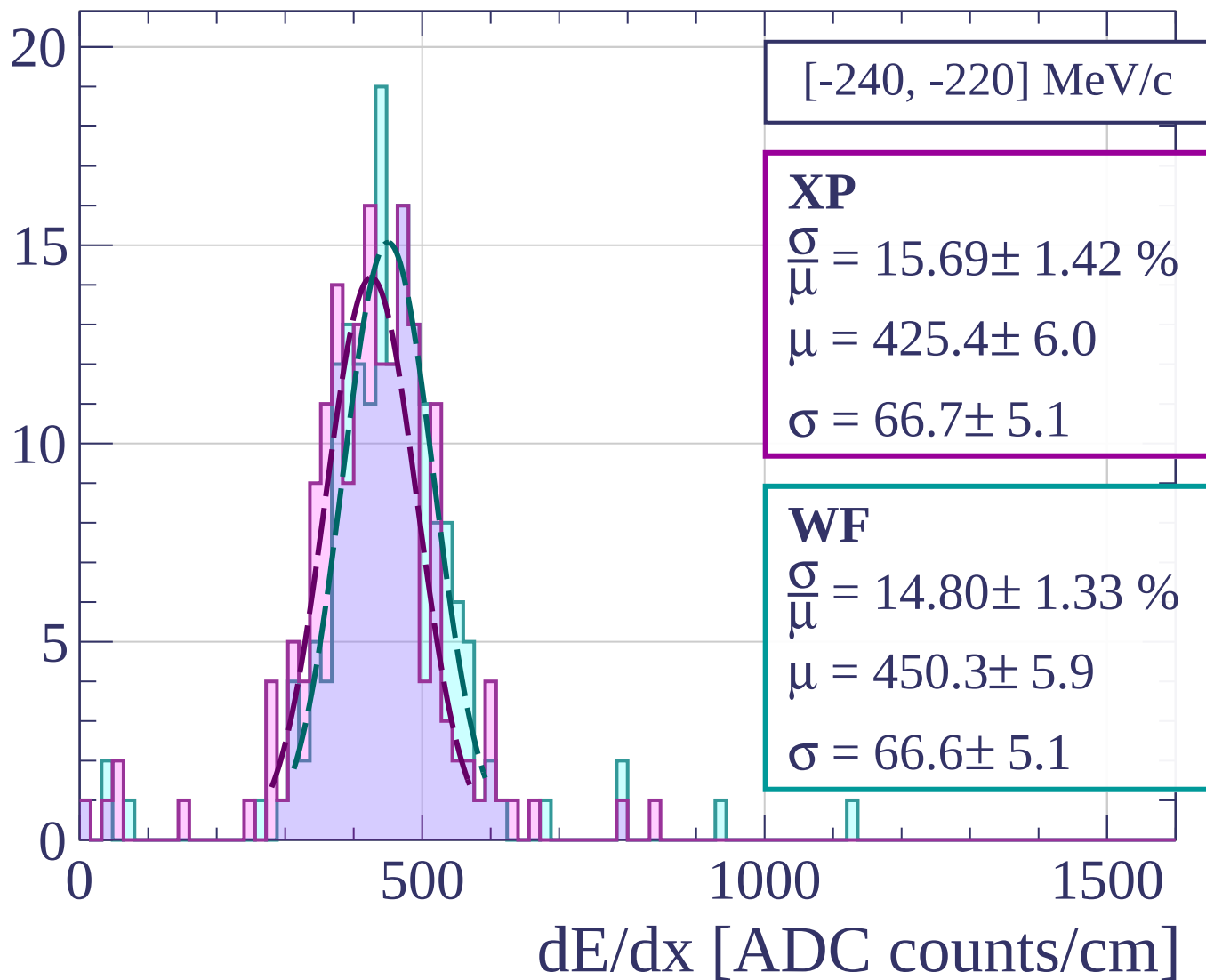
$$\sigma = 62.6 \pm 8.0$$

dE/dx [ADC counts/cm]

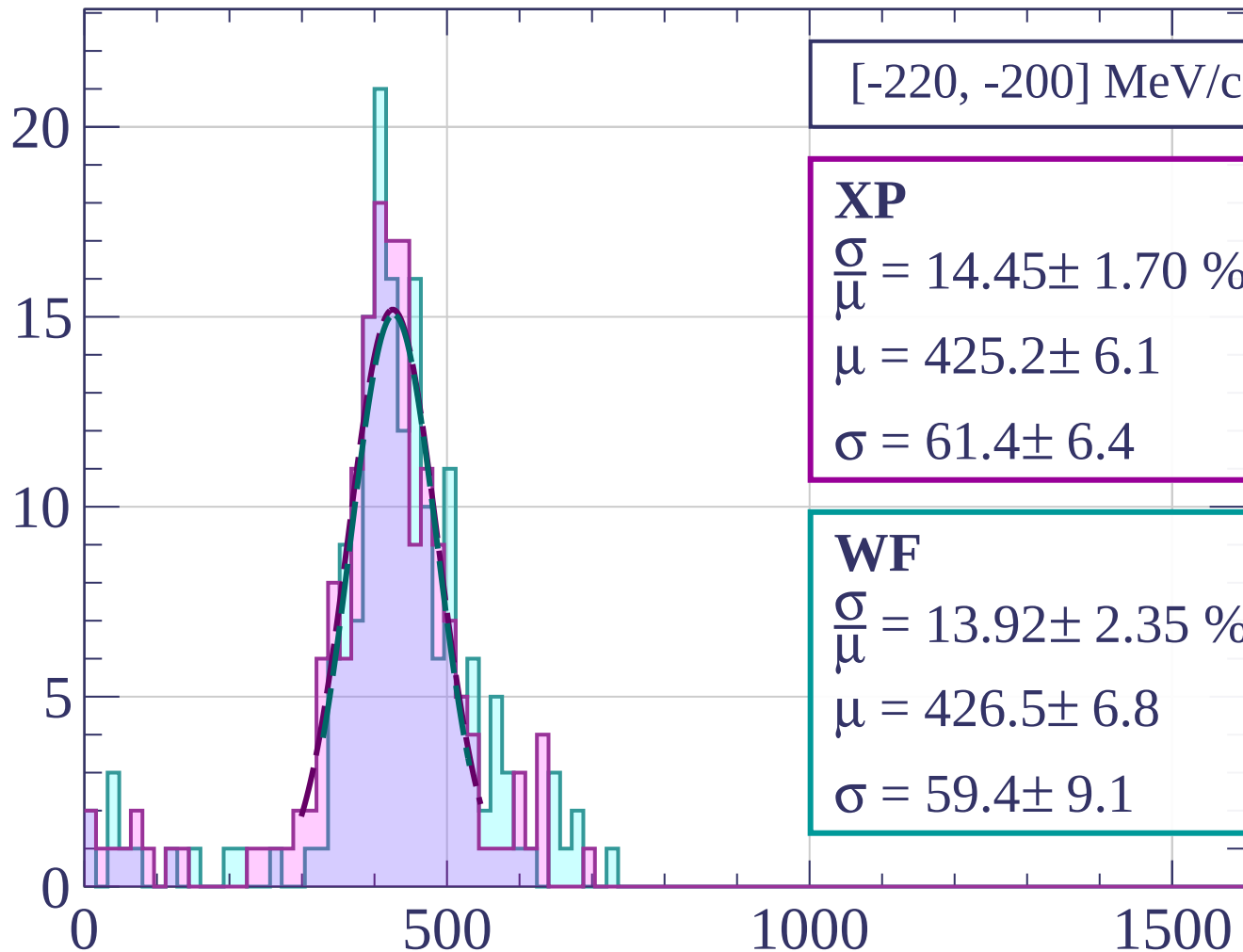
Count



Count



Count



[-220, -200] MeV/c

XP

$$\frac{\sigma}{\mu} = 14.45 \pm 1.70 \%$$

$$\mu = 425.2 \pm 6.1$$

$$\sigma = 61.4 \pm 6.4$$

WF

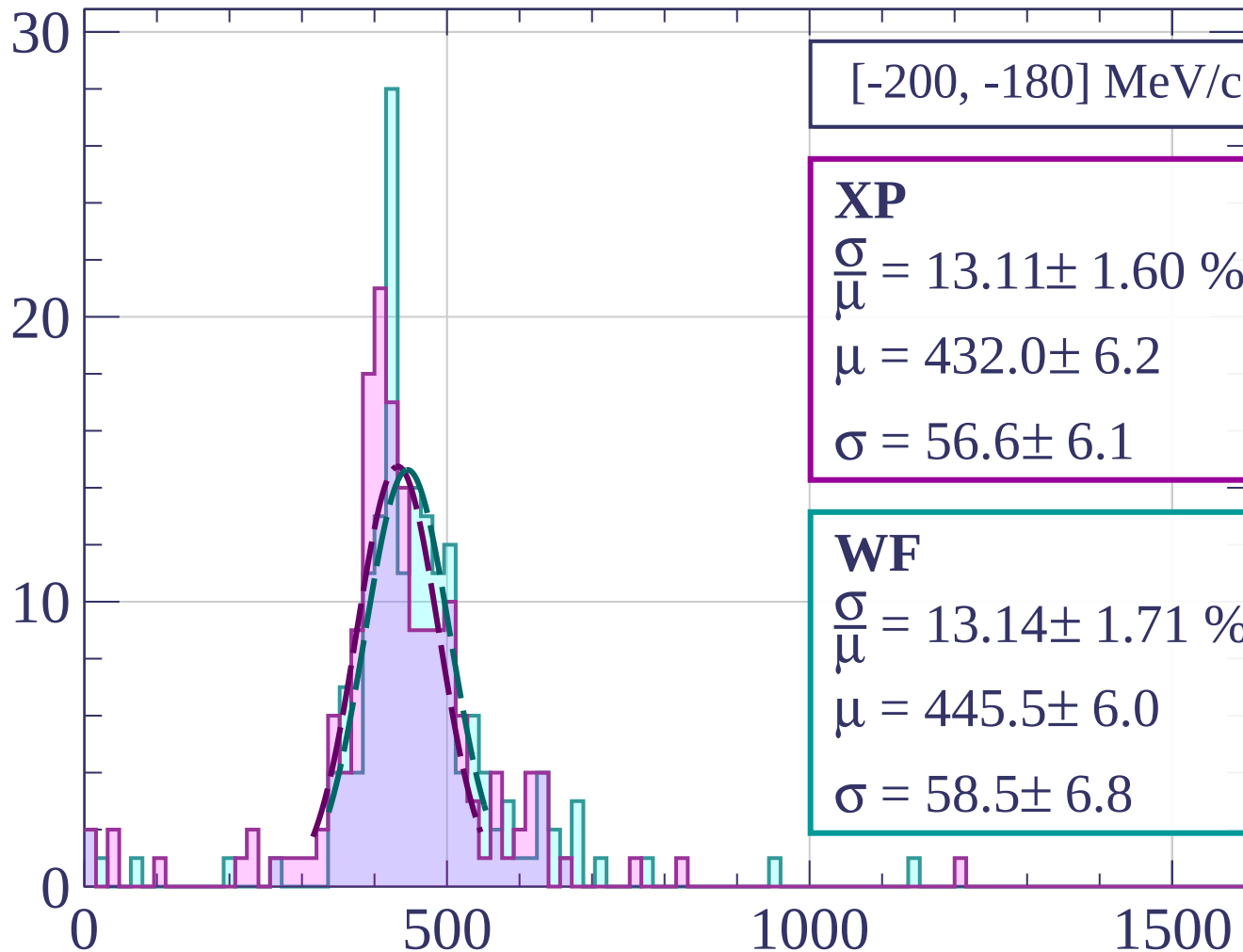
$$\frac{\sigma}{\mu} = 13.92 \pm 2.35 \%$$

$$\mu = 426.5 \pm 6.8$$

$$\sigma = 59.4 \pm 9.1$$

dE/dx [ADC counts/cm]

Count



[-200, -180] MeV/c

XP

$$\frac{\sigma}{\mu} = 13.11 \pm 1.60 \%$$

$$\mu = 432.0 \pm 6.2$$

$$\sigma = 56.6 \pm 6.1$$

WF

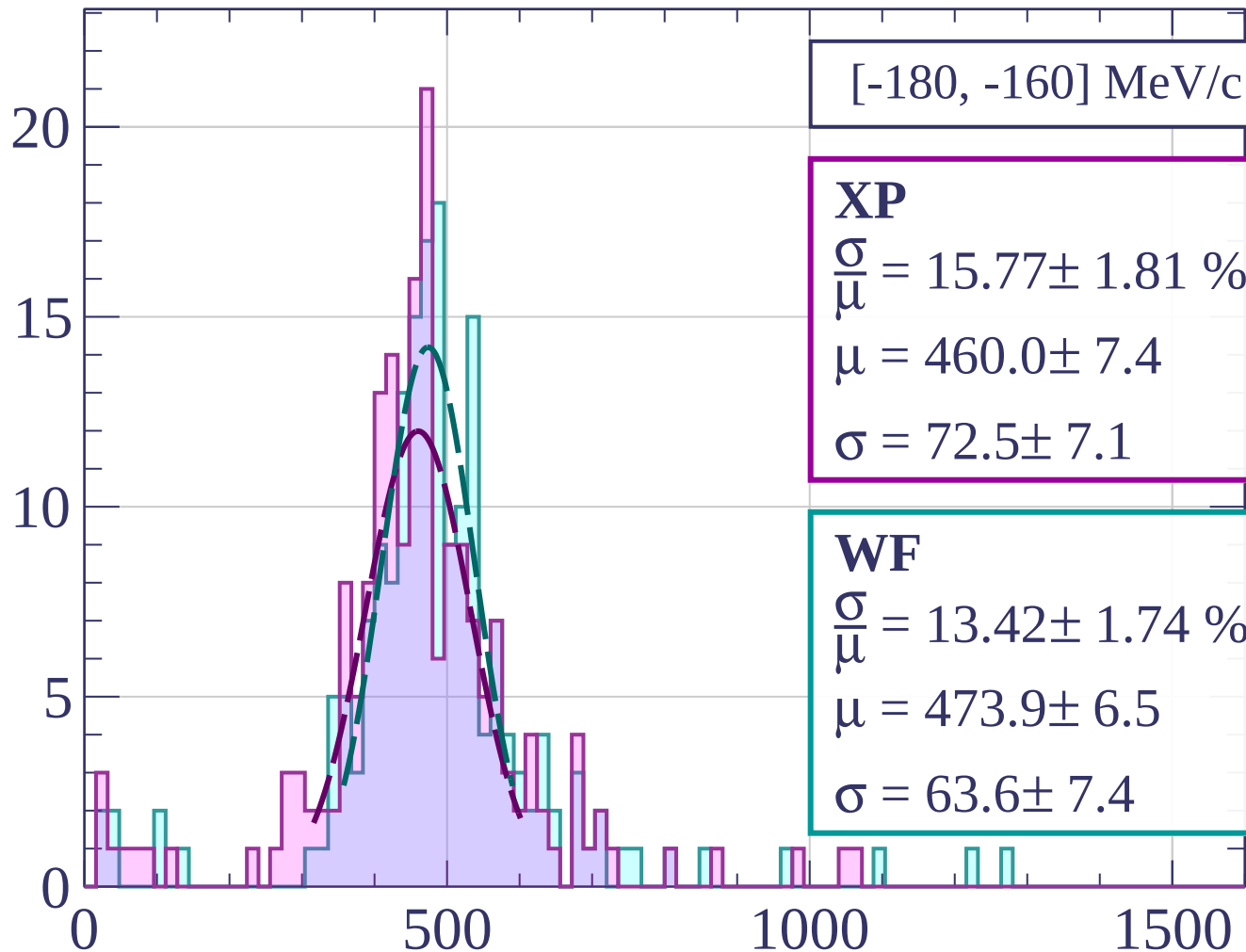
$$\frac{\sigma}{\mu} = 13.14 \pm 1.71 \%$$

$$\mu = 445.5 \pm 6.0$$

$$\sigma = 58.5 \pm 6.8$$

dE/dx [ADC counts/cm]

Count



[-180, -160] MeV/c

XP

$$\frac{\sigma}{\mu} = 15.77 \pm 1.81 \%$$

$$\mu = 460.0 \pm 7.4$$

$$\sigma = 72.5 \pm 7.1$$

WF

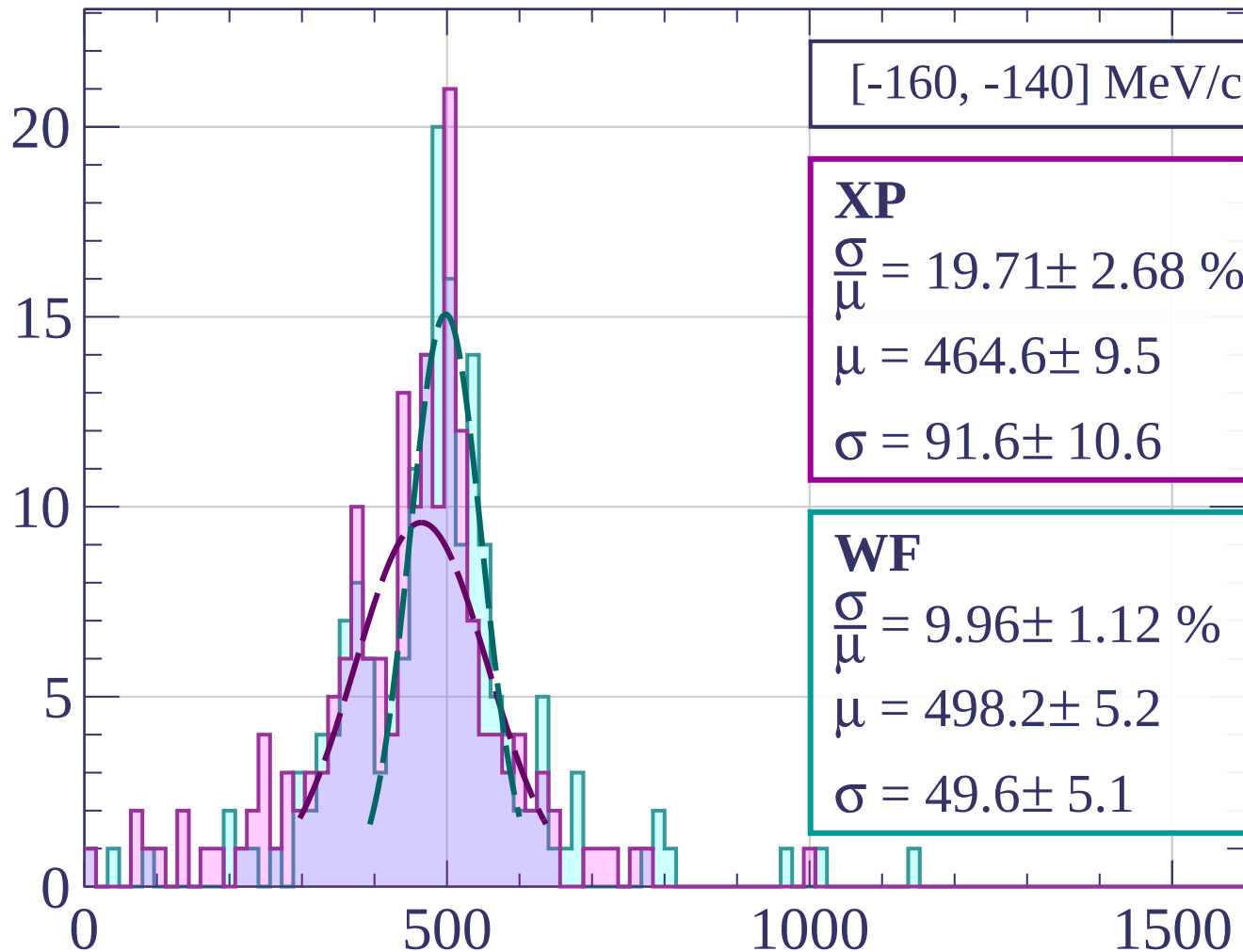
$$\frac{\sigma}{\mu} = 13.42 \pm 1.74 \%$$

$$\mu = 473.9 \pm 6.5$$

$$\sigma = 63.6 \pm 7.4$$

dE/dx [ADC counts/cm]

Count



[-160, -140] MeV/c

XP

$$\frac{\sigma}{\mu} = 19.71 \pm 2.68 \%$$

$$\mu = 464.6 \pm 9.5$$

$$\sigma = 91.6 \pm 10.6$$

WF

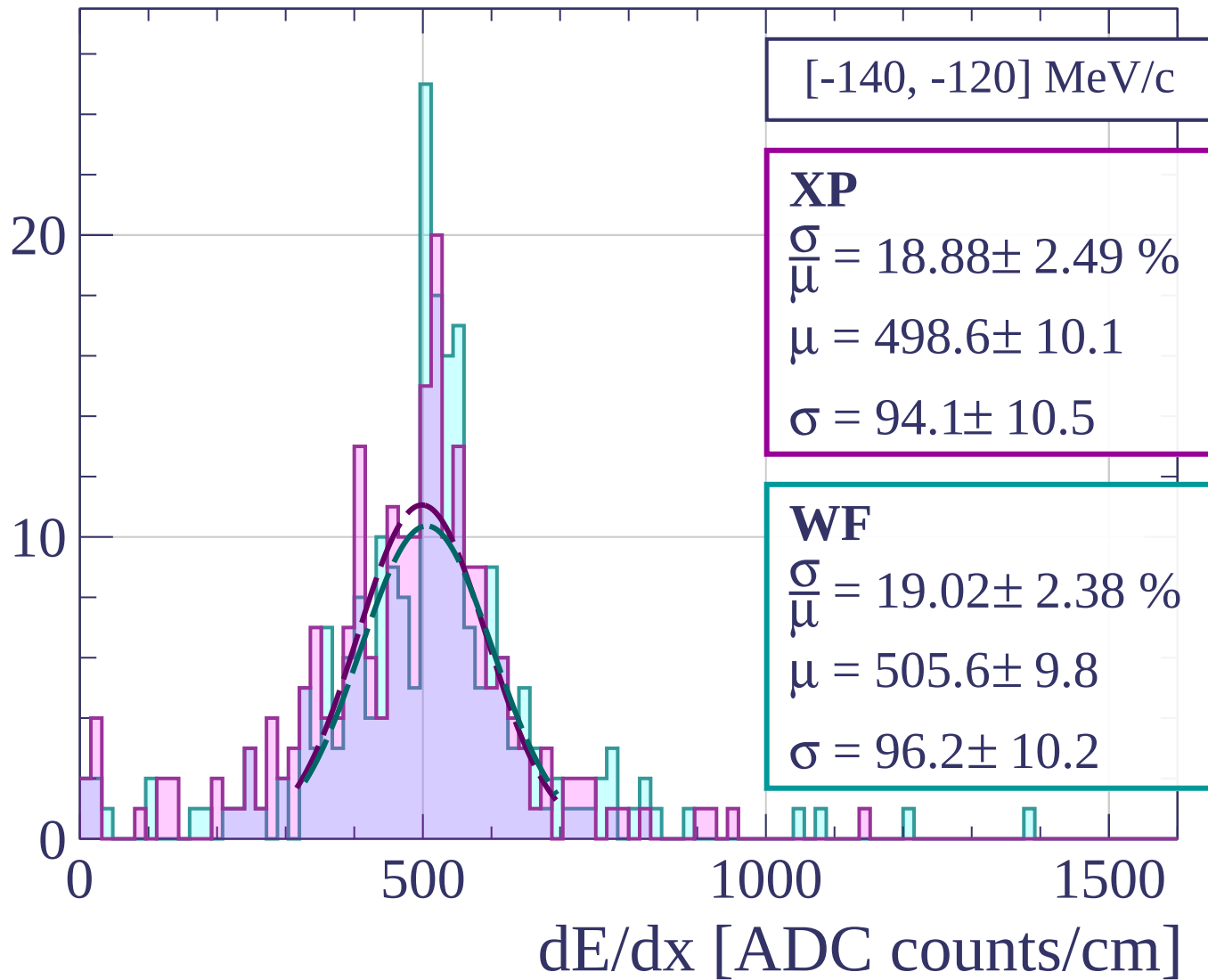
$$\frac{\sigma}{\mu} = 9.96 \pm 1.12 \%$$

$$\mu = 498.2 \pm 5.2$$

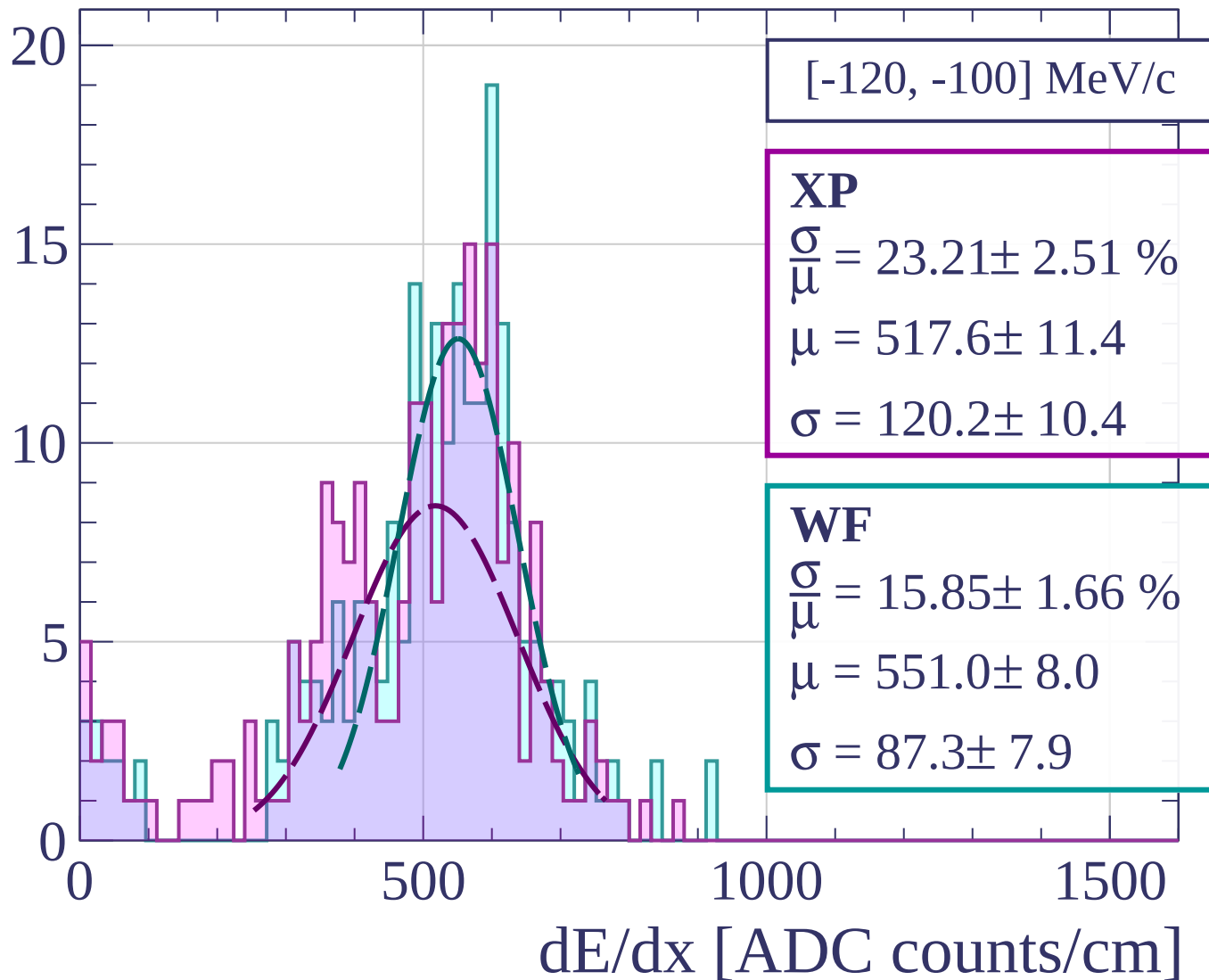
$$\sigma = 49.6 \pm 5.1$$

dE/dx [ADC counts/cm]

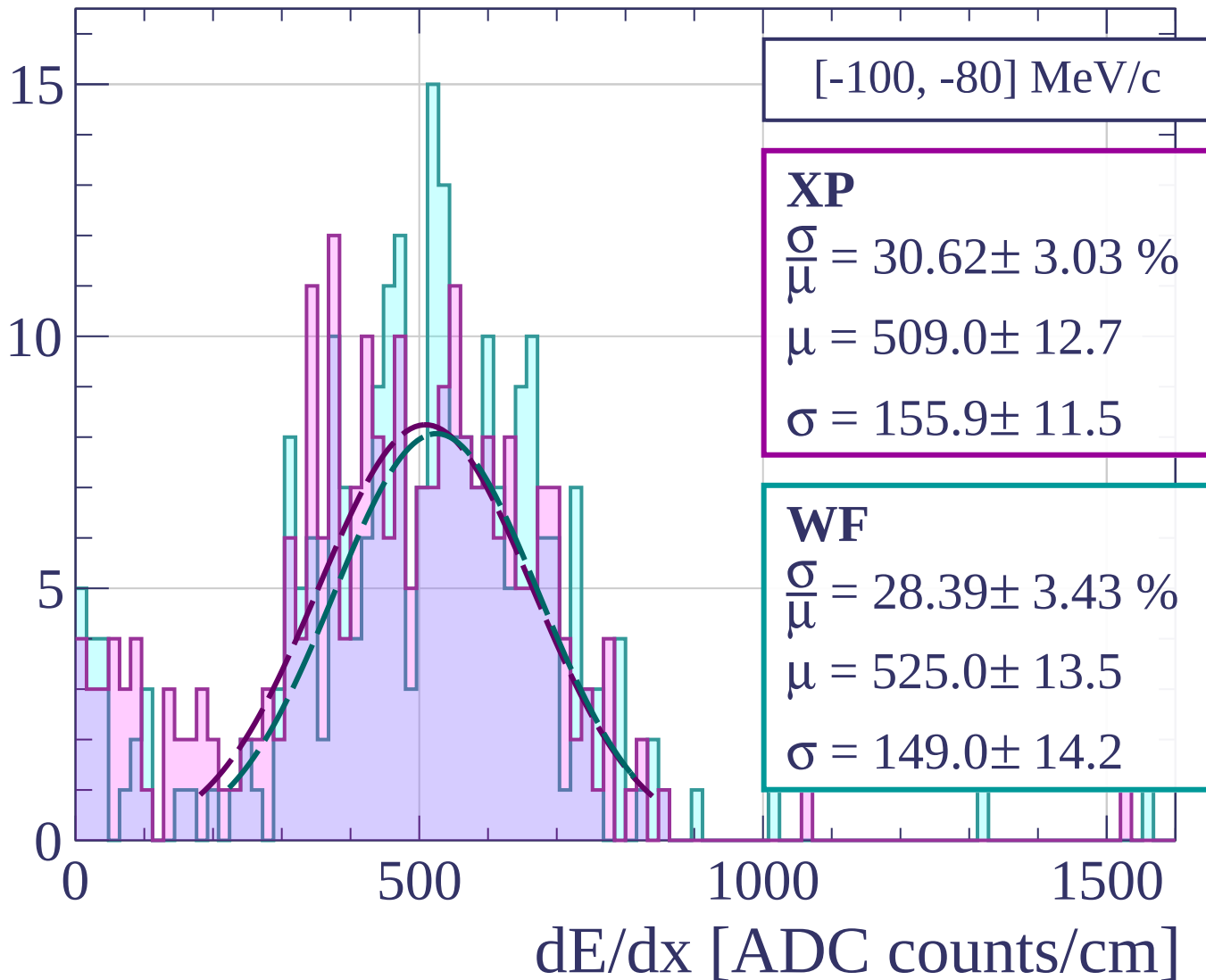
Count



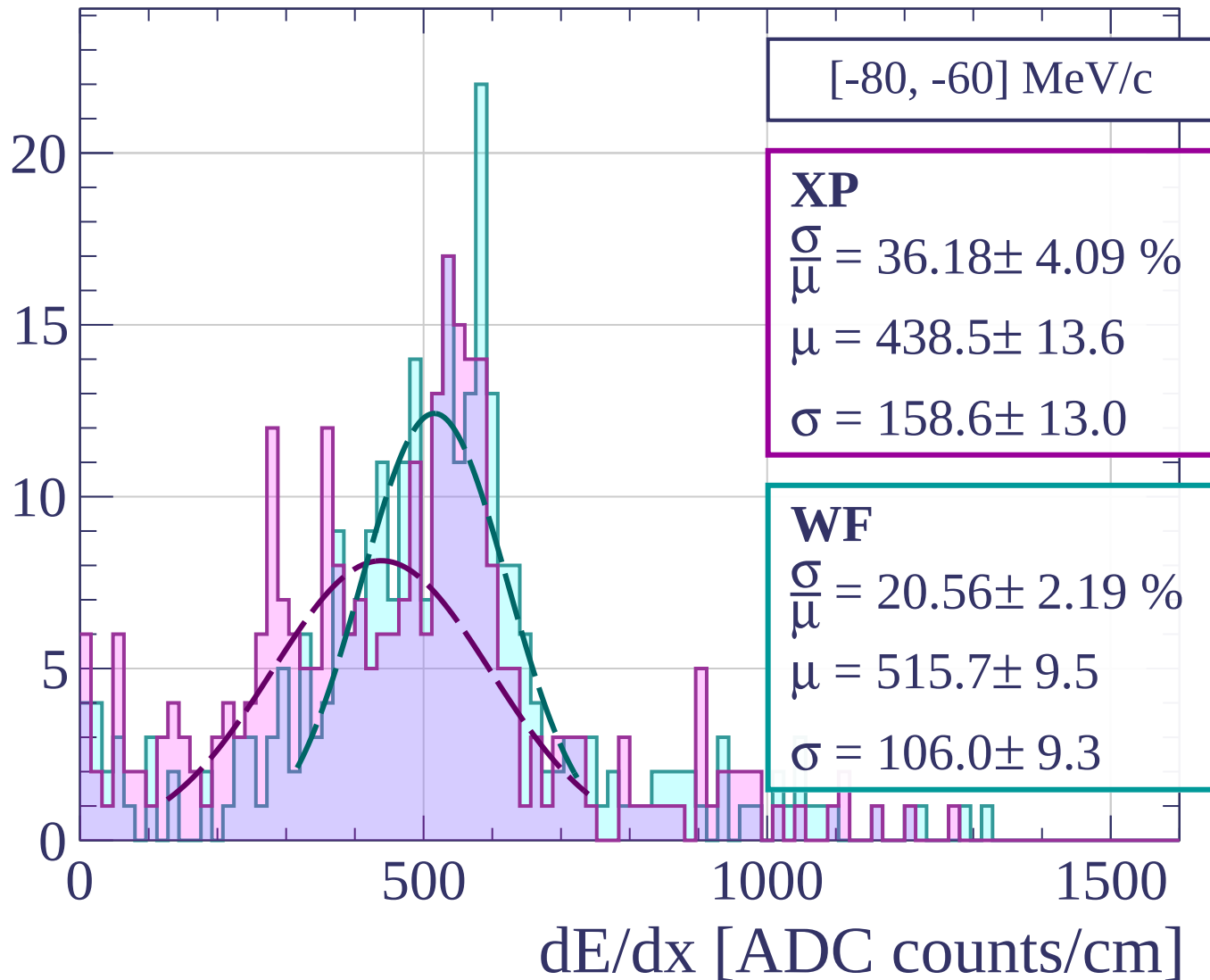
Count



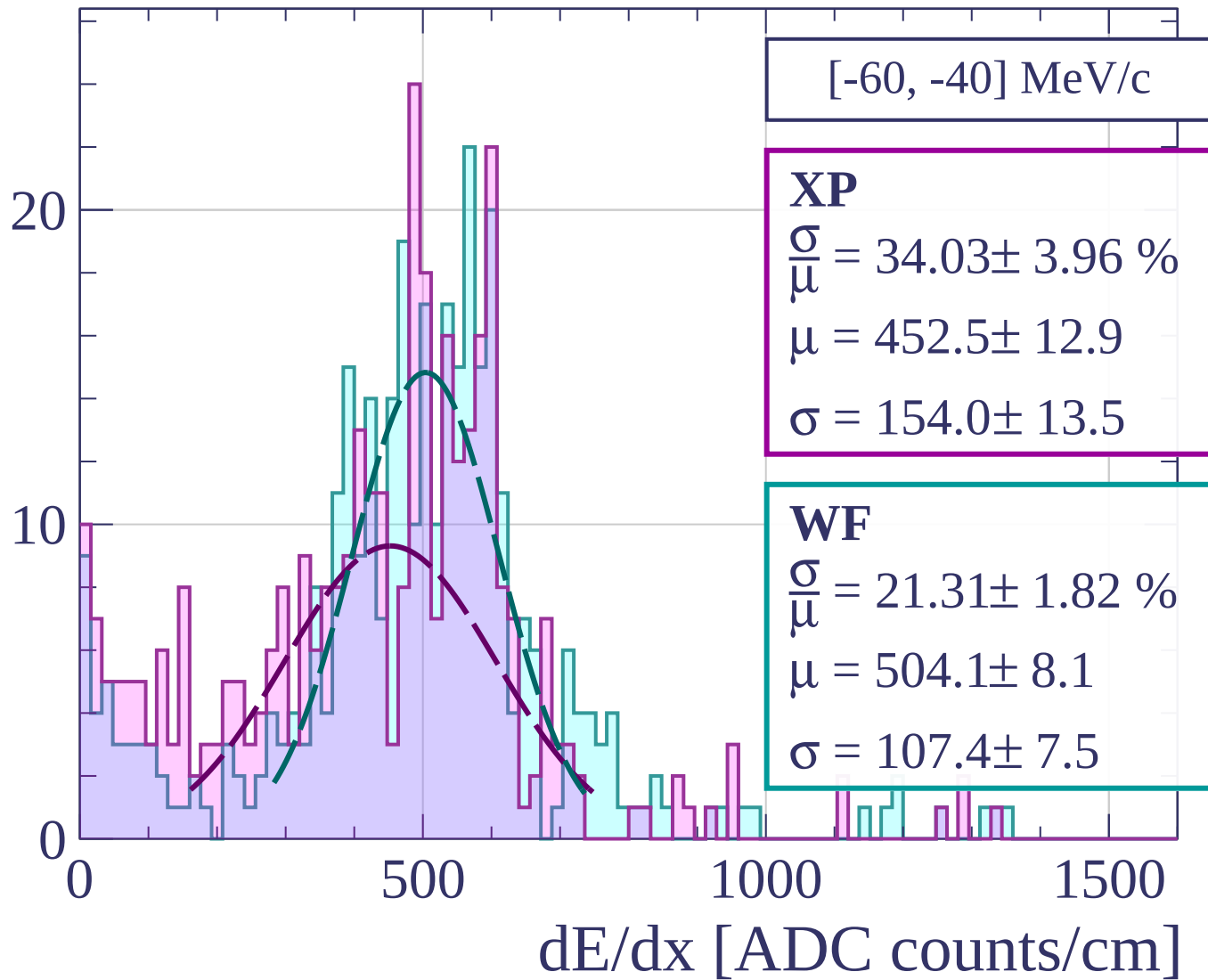
Count



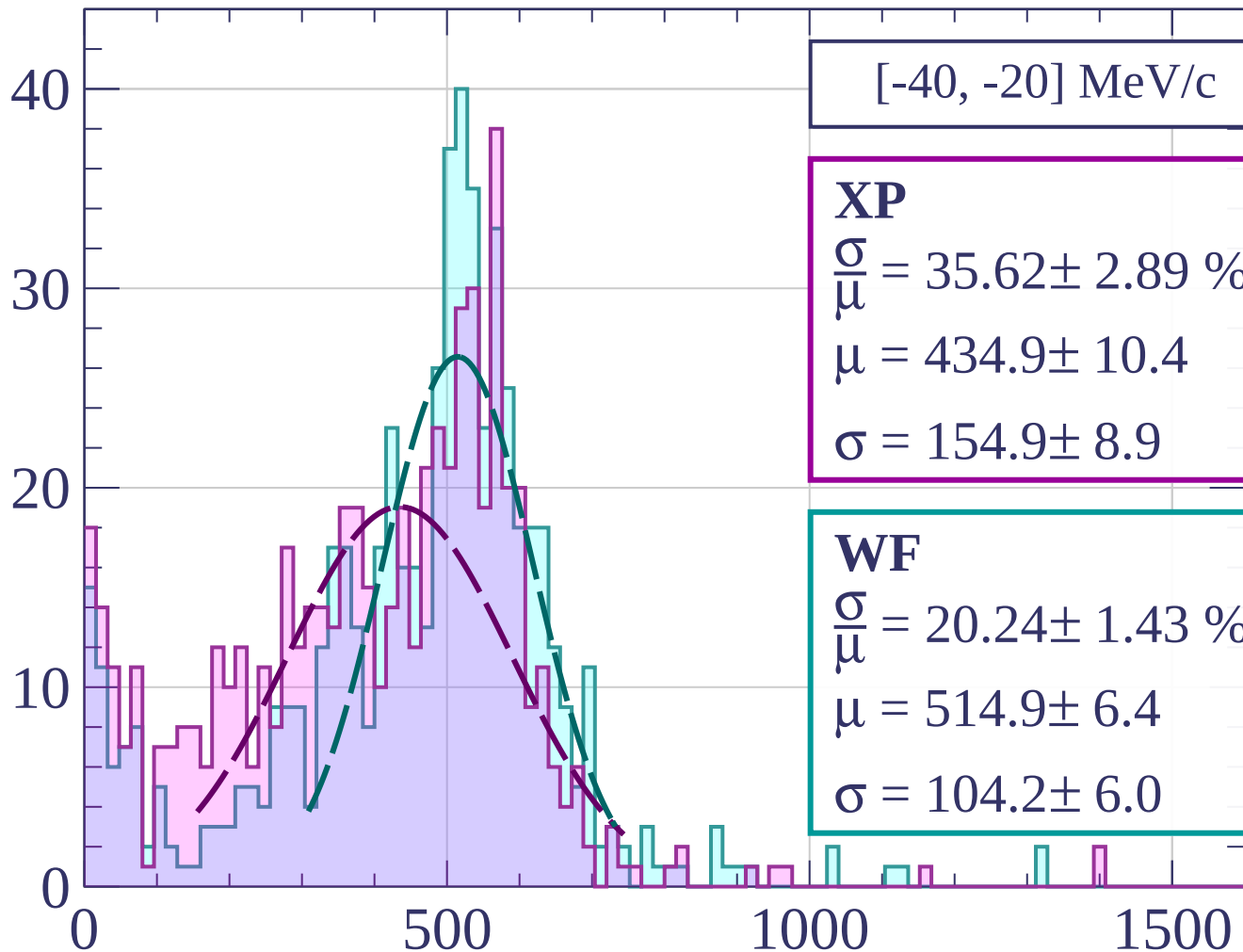
Count



Count



Count



[-40, -20] MeV/c

XP

$$\frac{\sigma}{\mu} = 35.62 \pm 2.89 \%$$

$$\mu = 434.9 \pm 10.4$$

$$\sigma = 154.9 \pm 8.9$$

WF

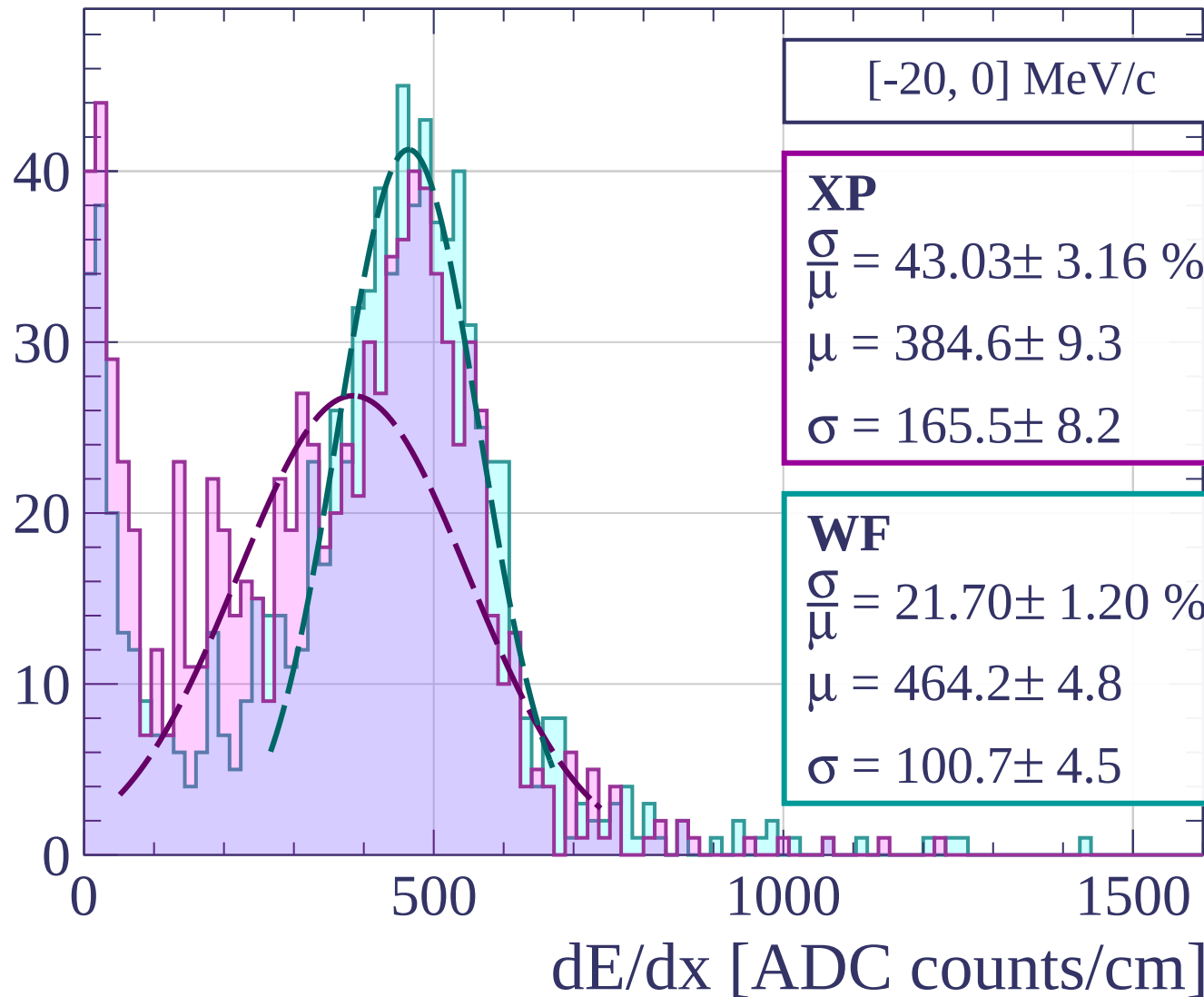
$$\frac{\sigma}{\mu} = 20.24 \pm 1.43 \%$$

$$\mu = 514.9 \pm 6.4$$

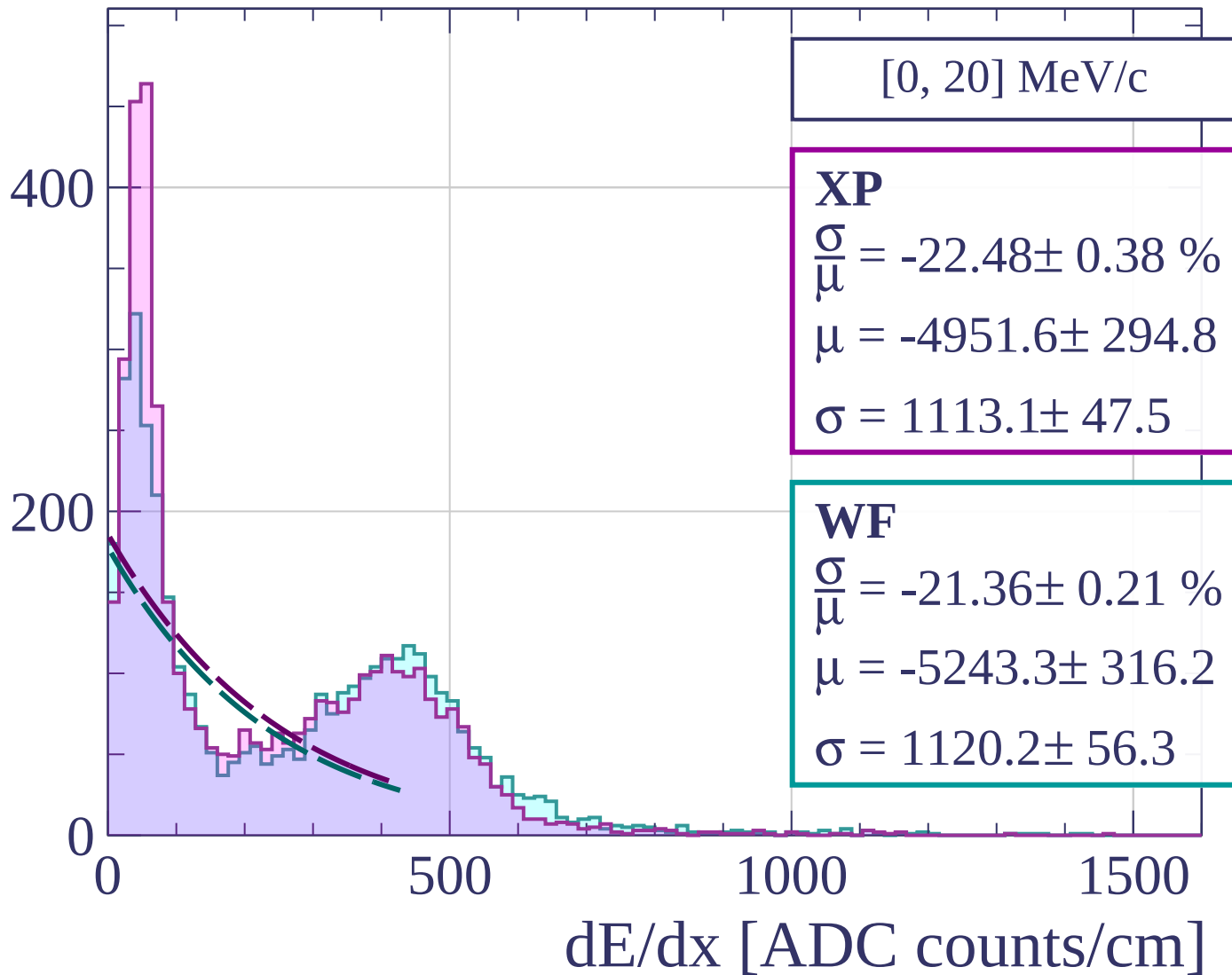
$$\sigma = 104.2 \pm 6.0$$

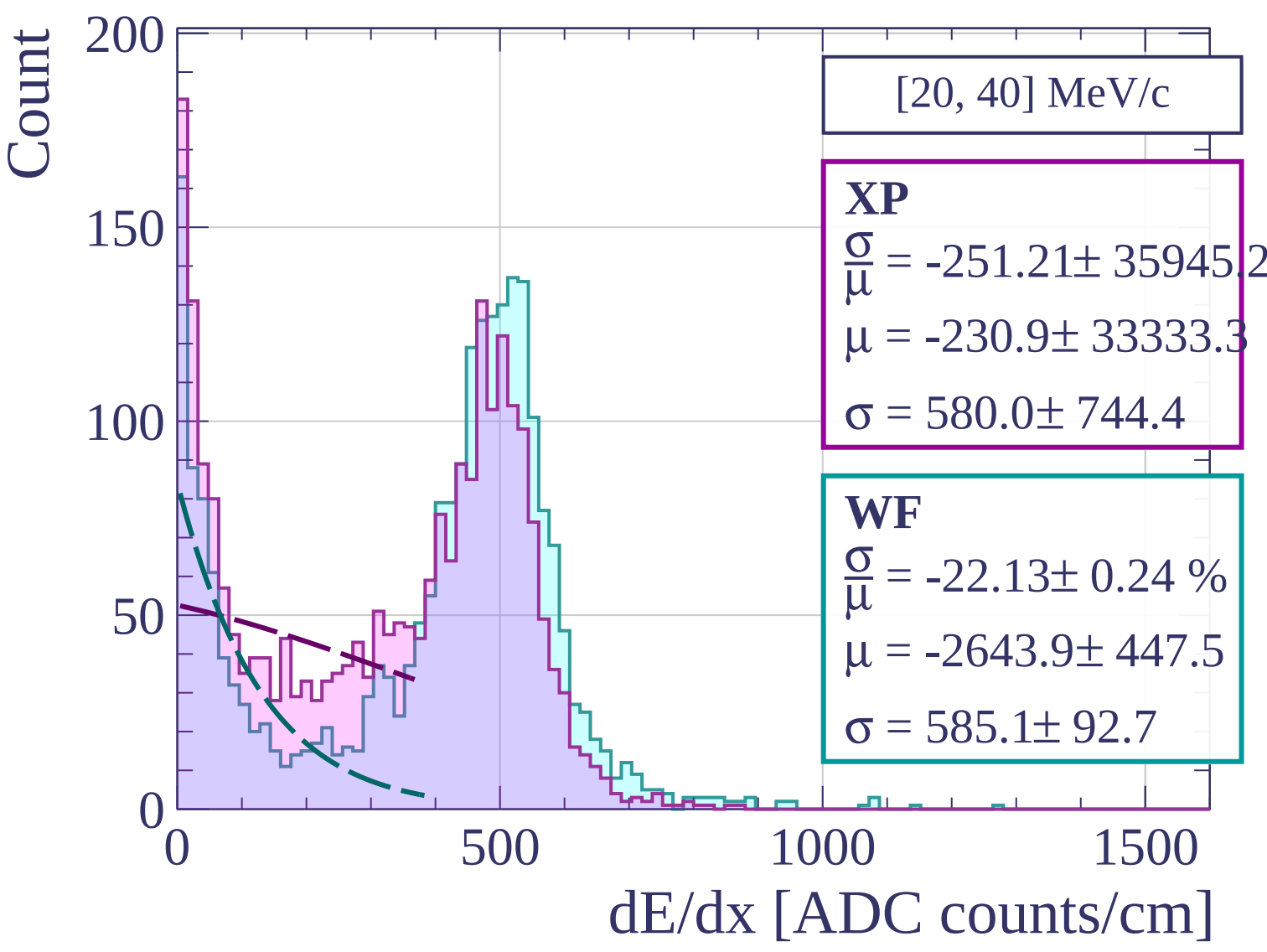
dE/dx [ADC counts/cm]

Count

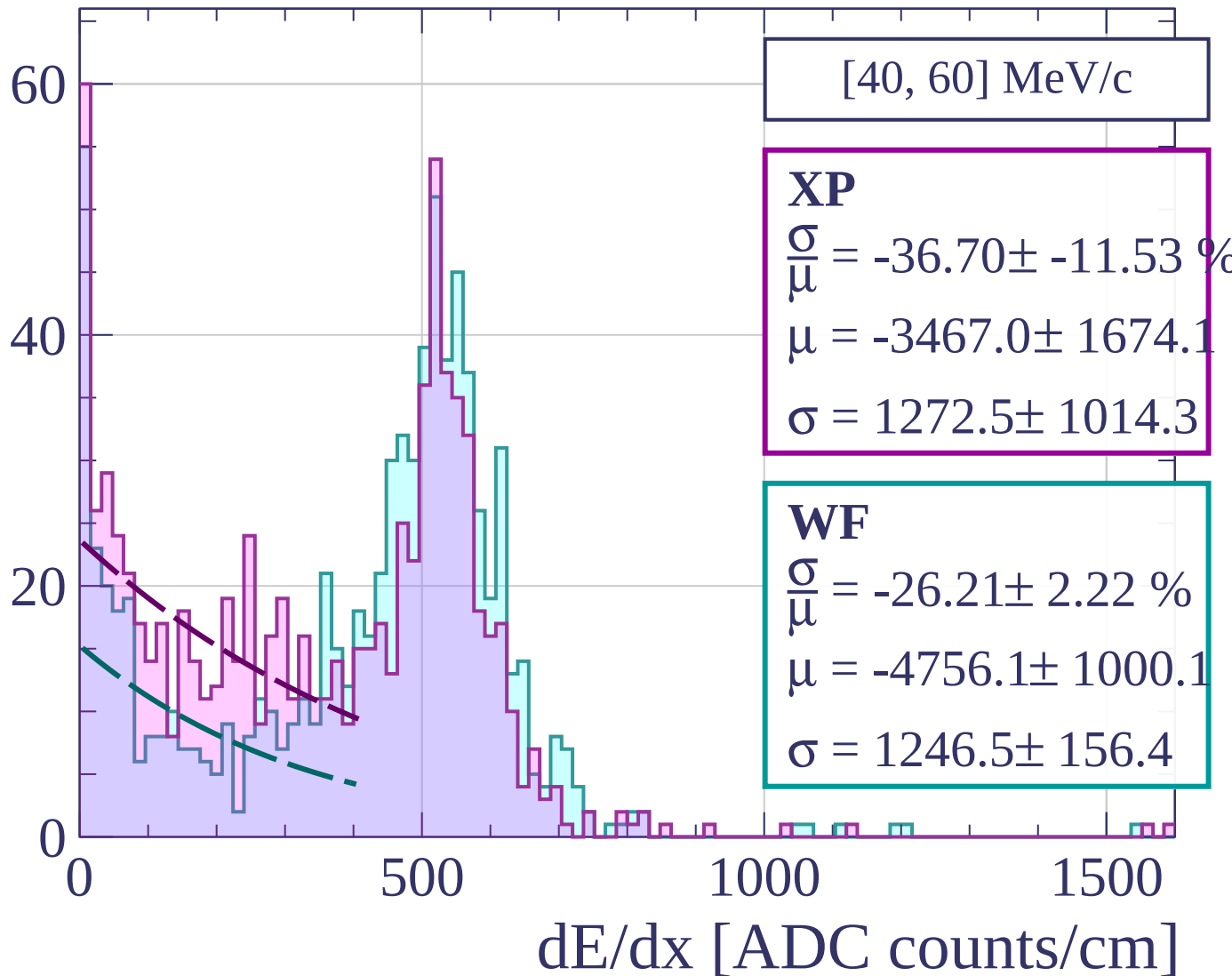


Count

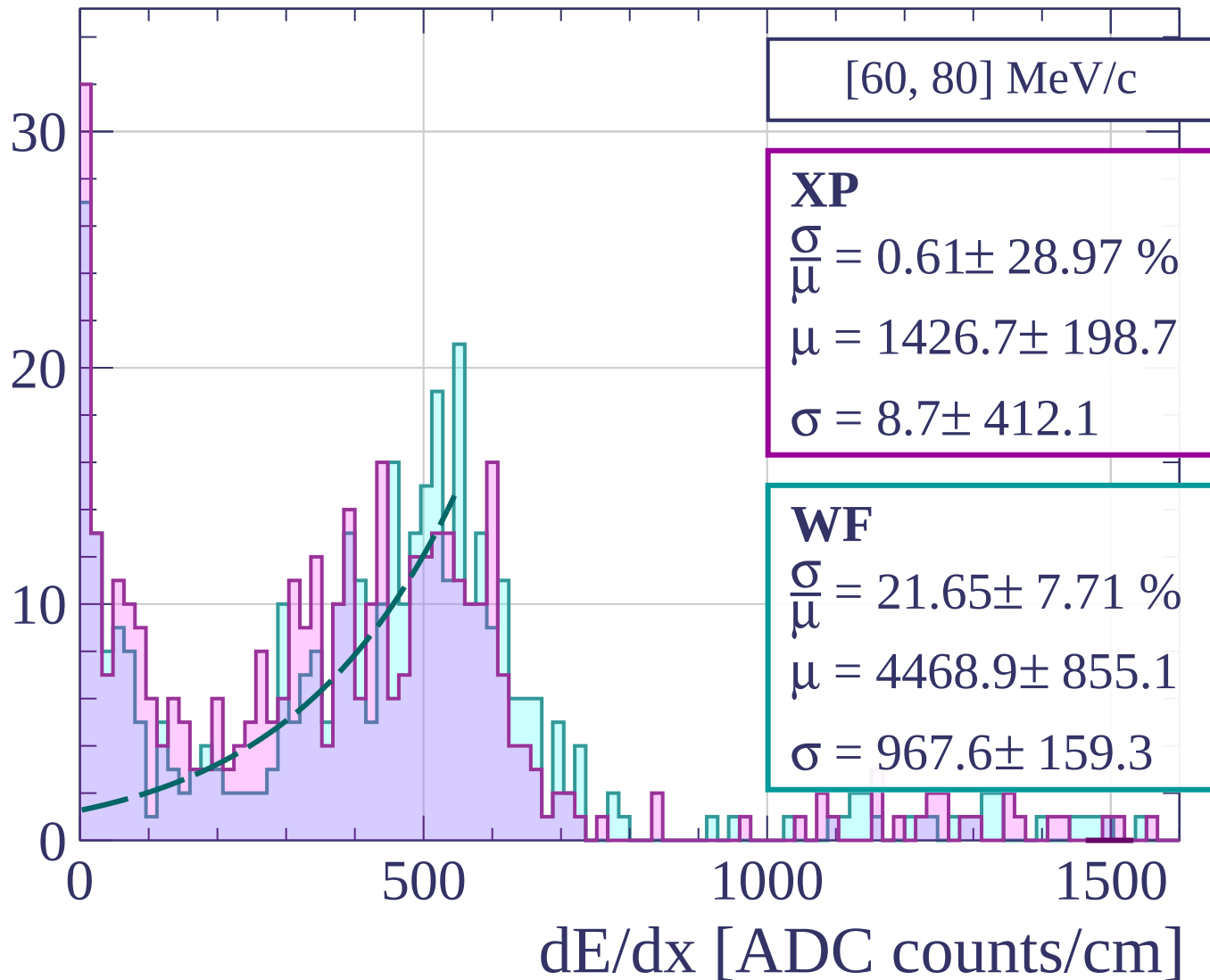




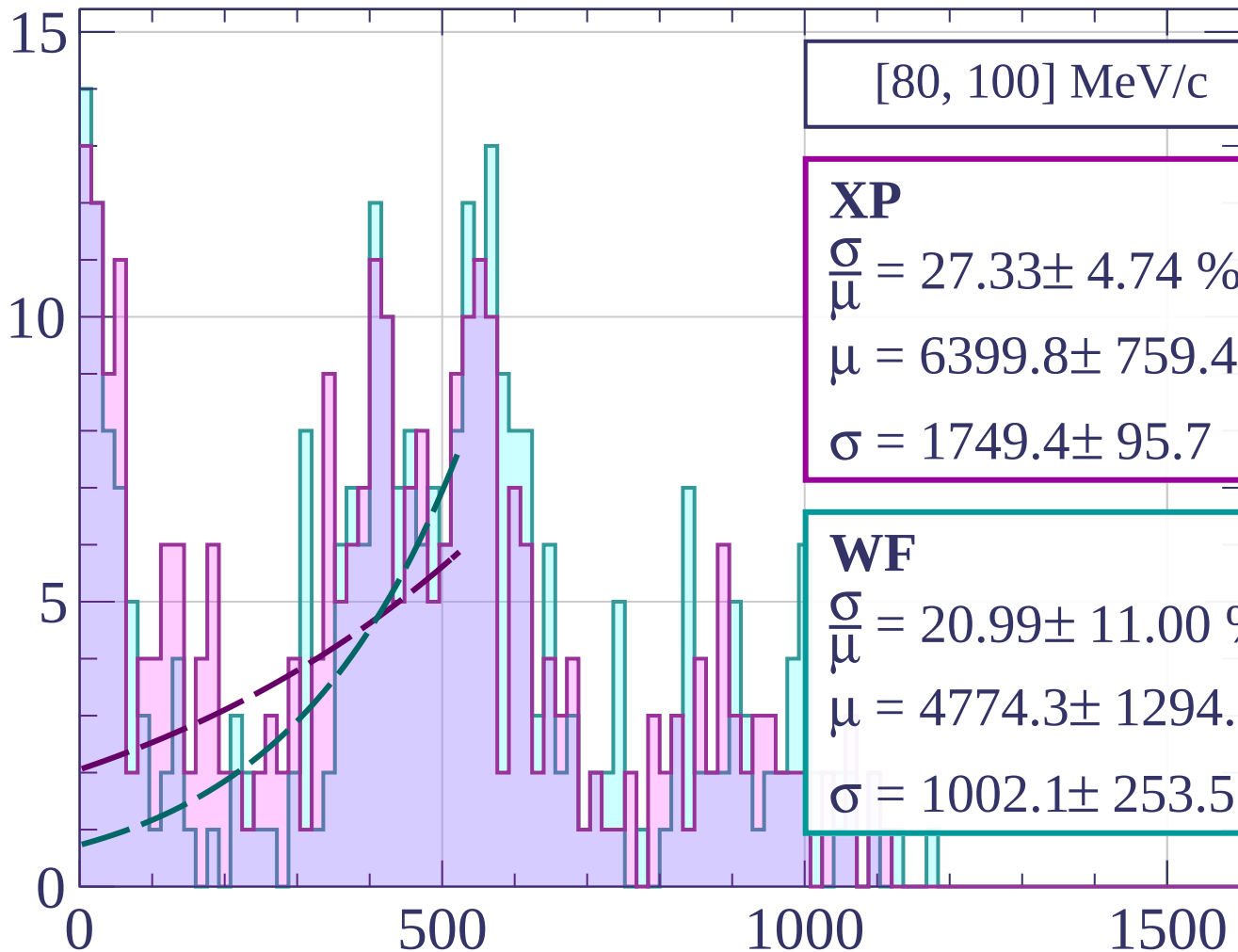
Count



Count

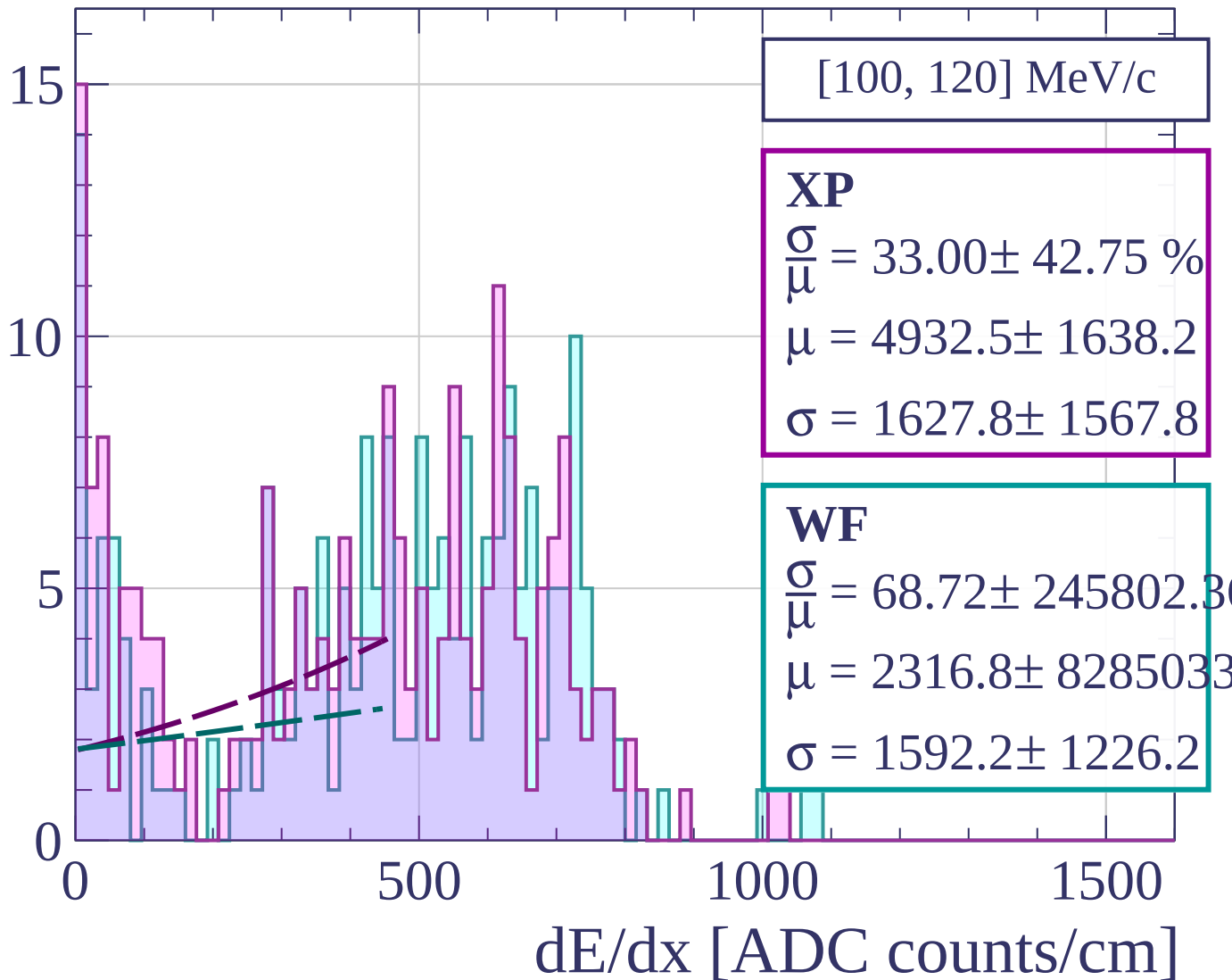


Count

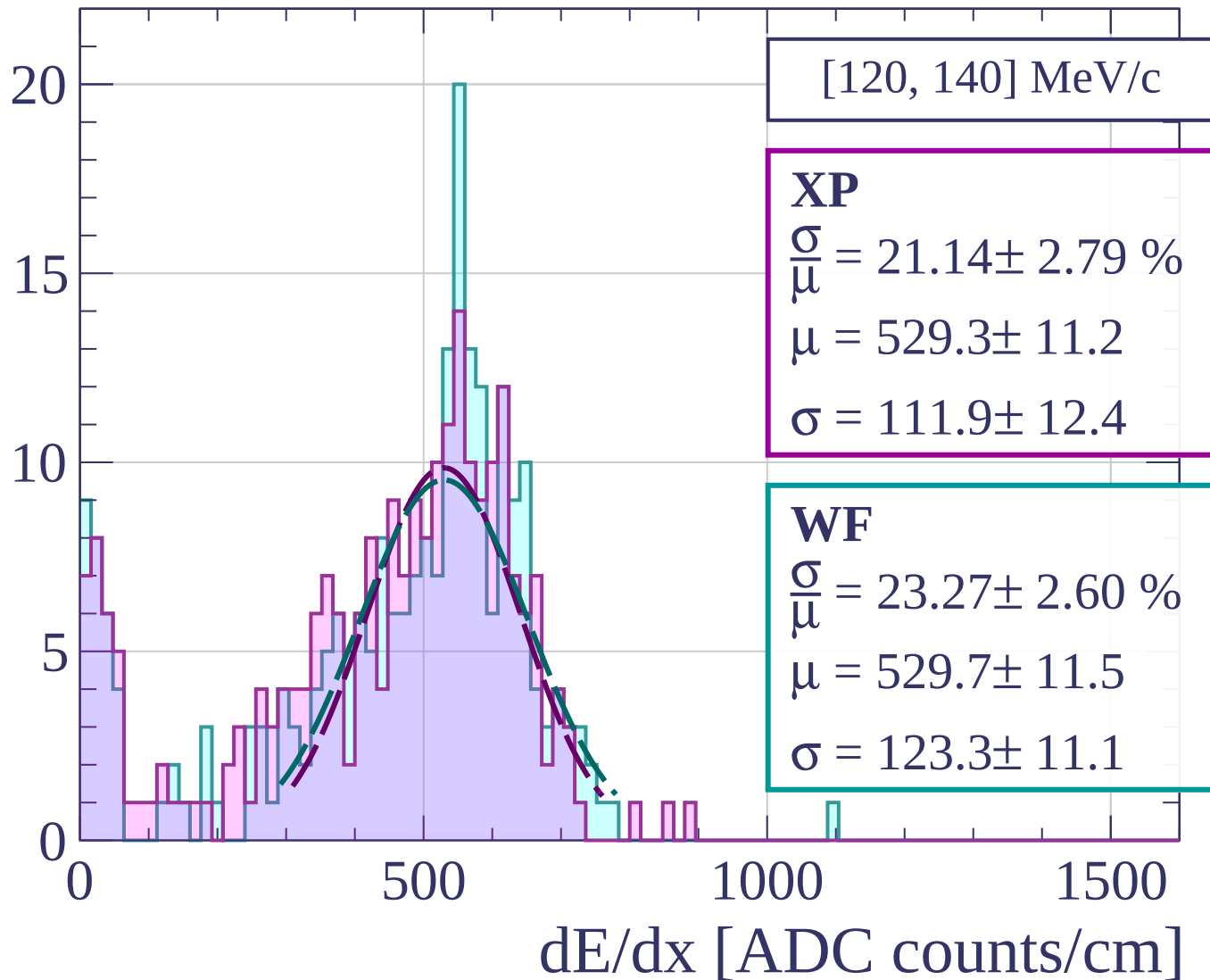


dE/dx [ADC counts/cm]

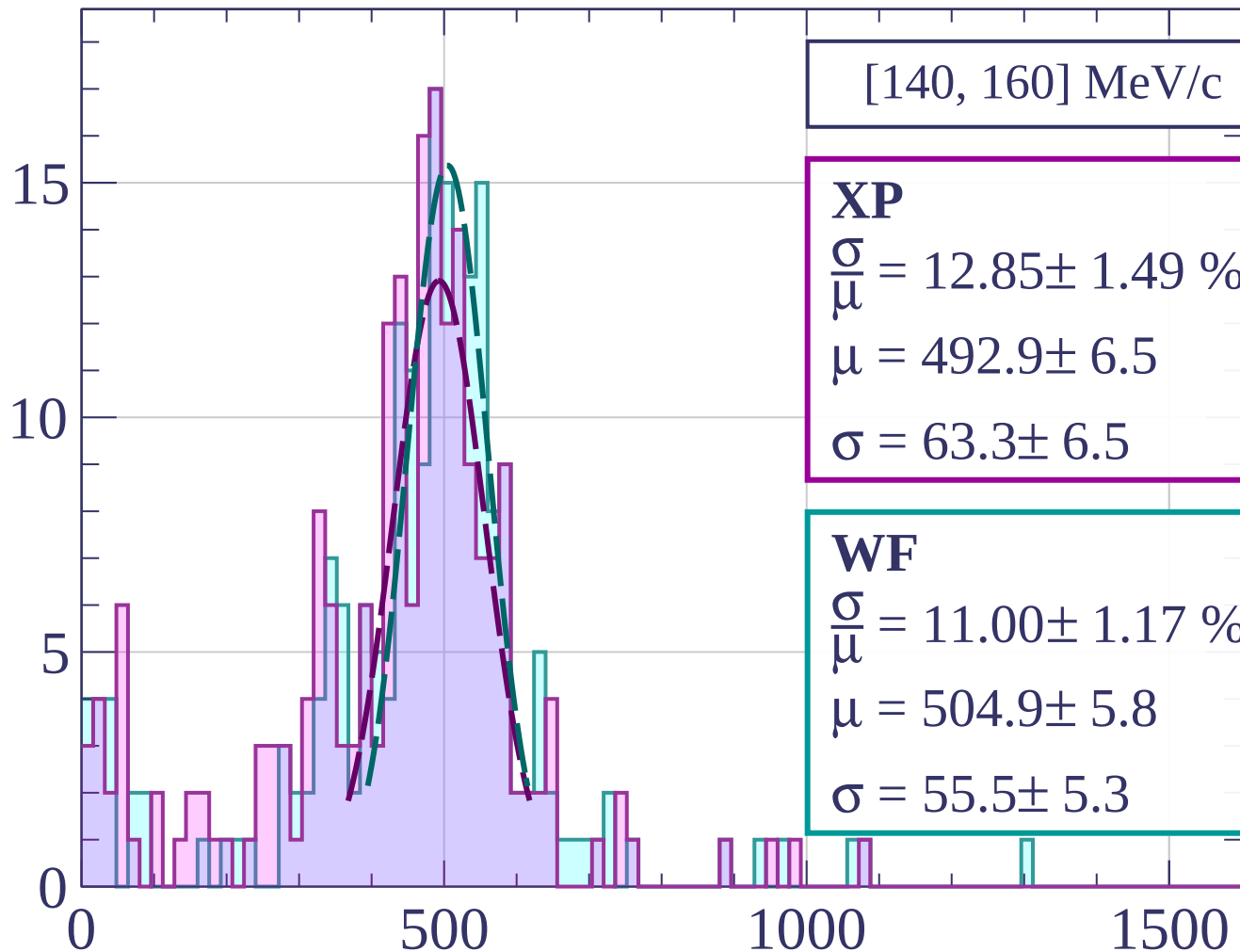
Count



Count



Count



[140, 160] MeV/c

XP

$$\frac{\sigma}{\mu} = 12.85 \pm 1.49 \%$$

$$\mu = 492.9 \pm 6.5$$

$$\sigma = 63.3 \pm 6.5$$

WF

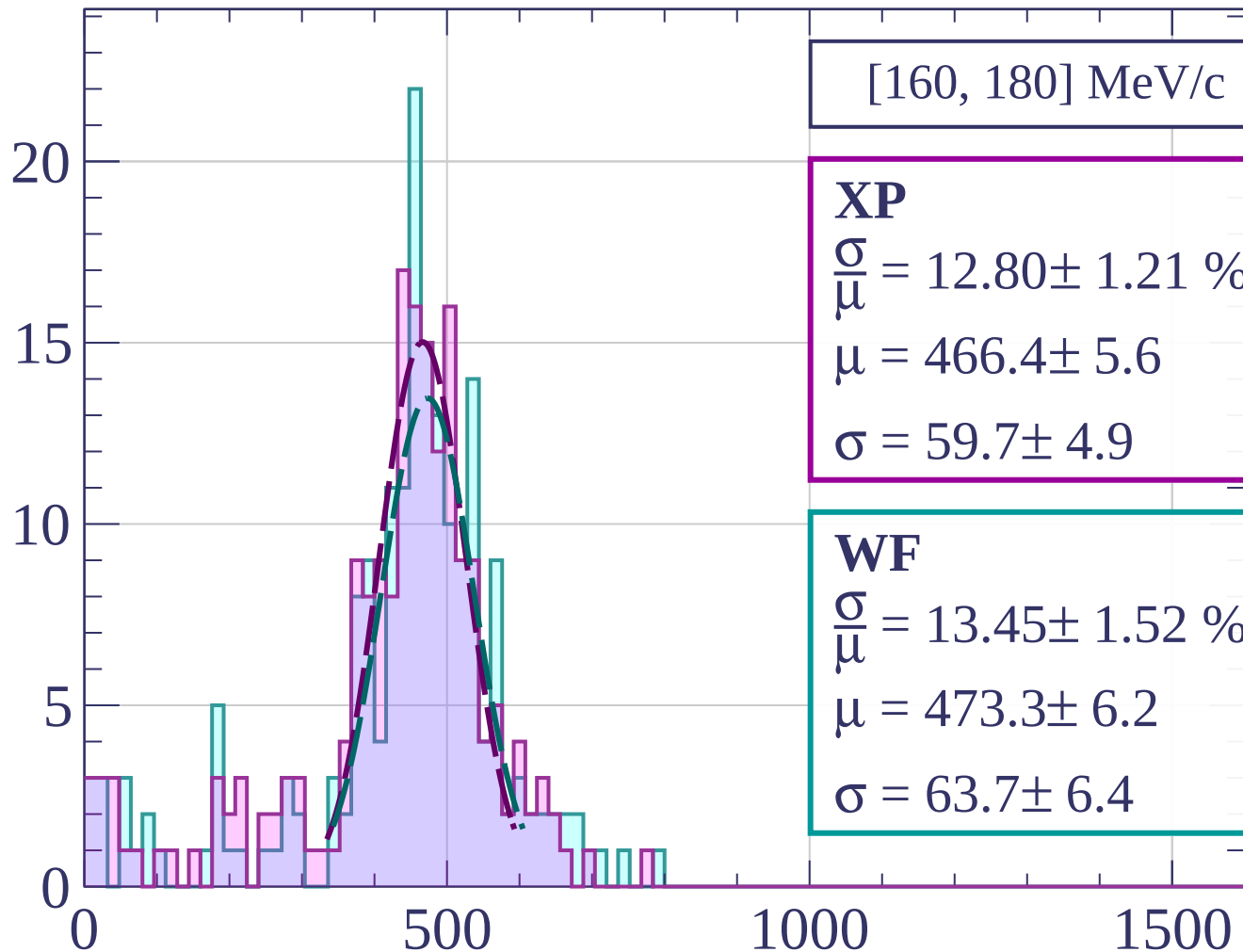
$$\frac{\sigma}{\mu} = 11.00 \pm 1.17 \%$$

$$\mu = 504.9 \pm 5.8$$

$$\sigma = 55.5 \pm 5.3$$

dE/dx [ADC counts/cm]

Count



[160, 180] MeV/c

XP

$$\frac{\sigma}{\mu} = 12.80 \pm 1.21 \%$$

$$\mu = 466.4 \pm 5.6$$

$$\sigma = 59.7 \pm 4.9$$

WF

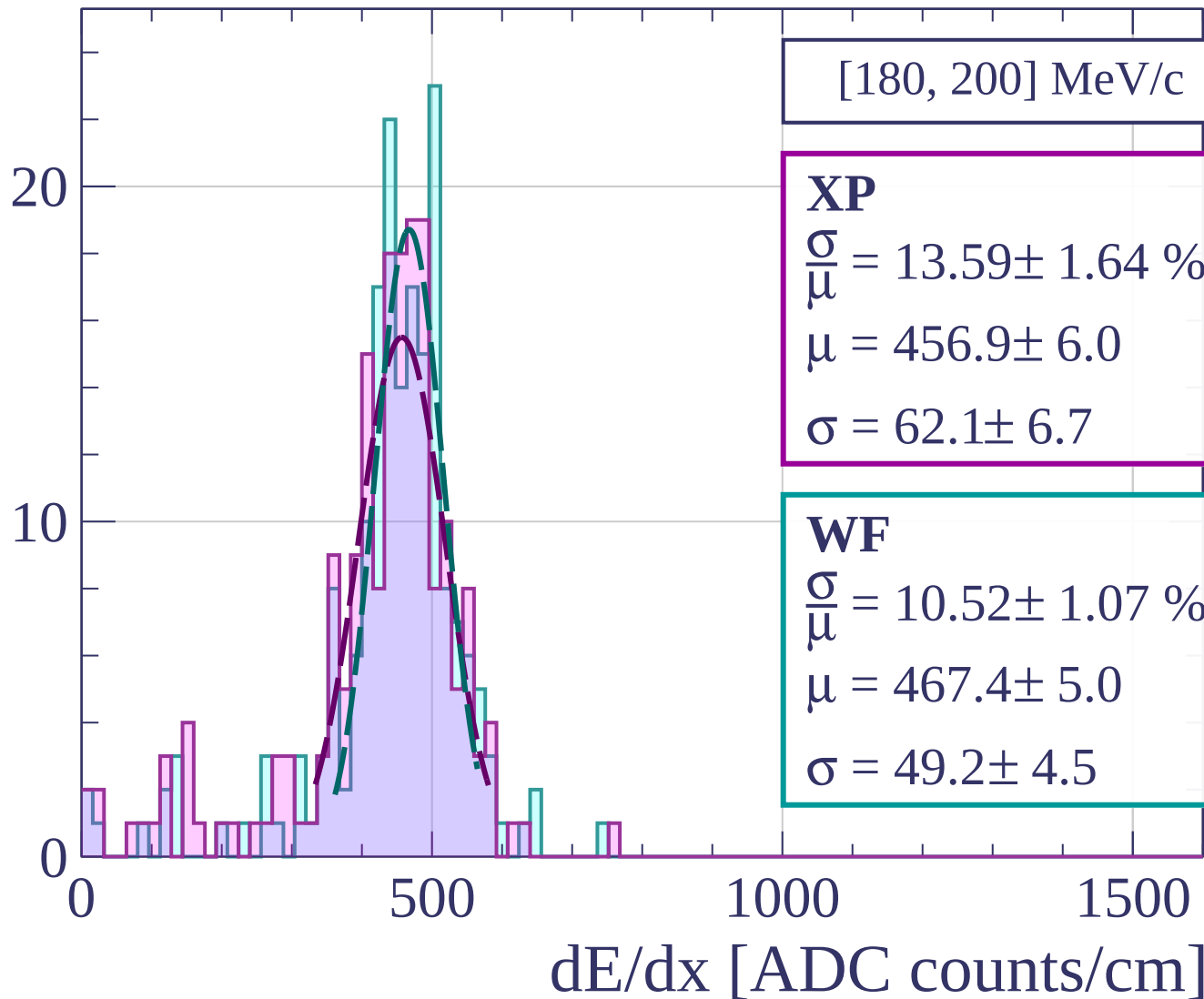
$$\frac{\sigma}{\mu} = 13.45 \pm 1.52 \%$$

$$\mu = 473.3 \pm 6.2$$

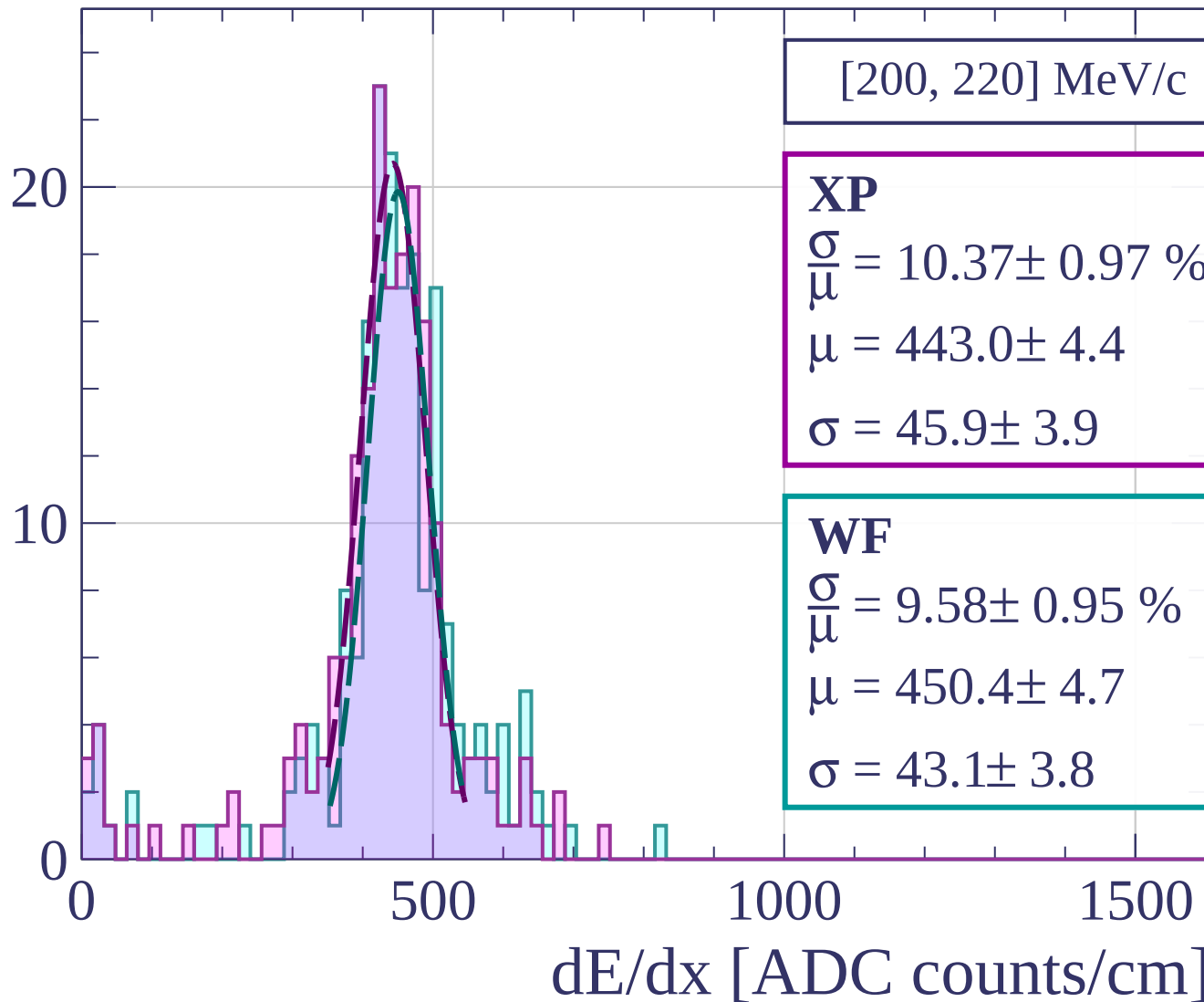
$$\sigma = 63.7 \pm 6.4$$

dE/dx [ADC counts/cm]

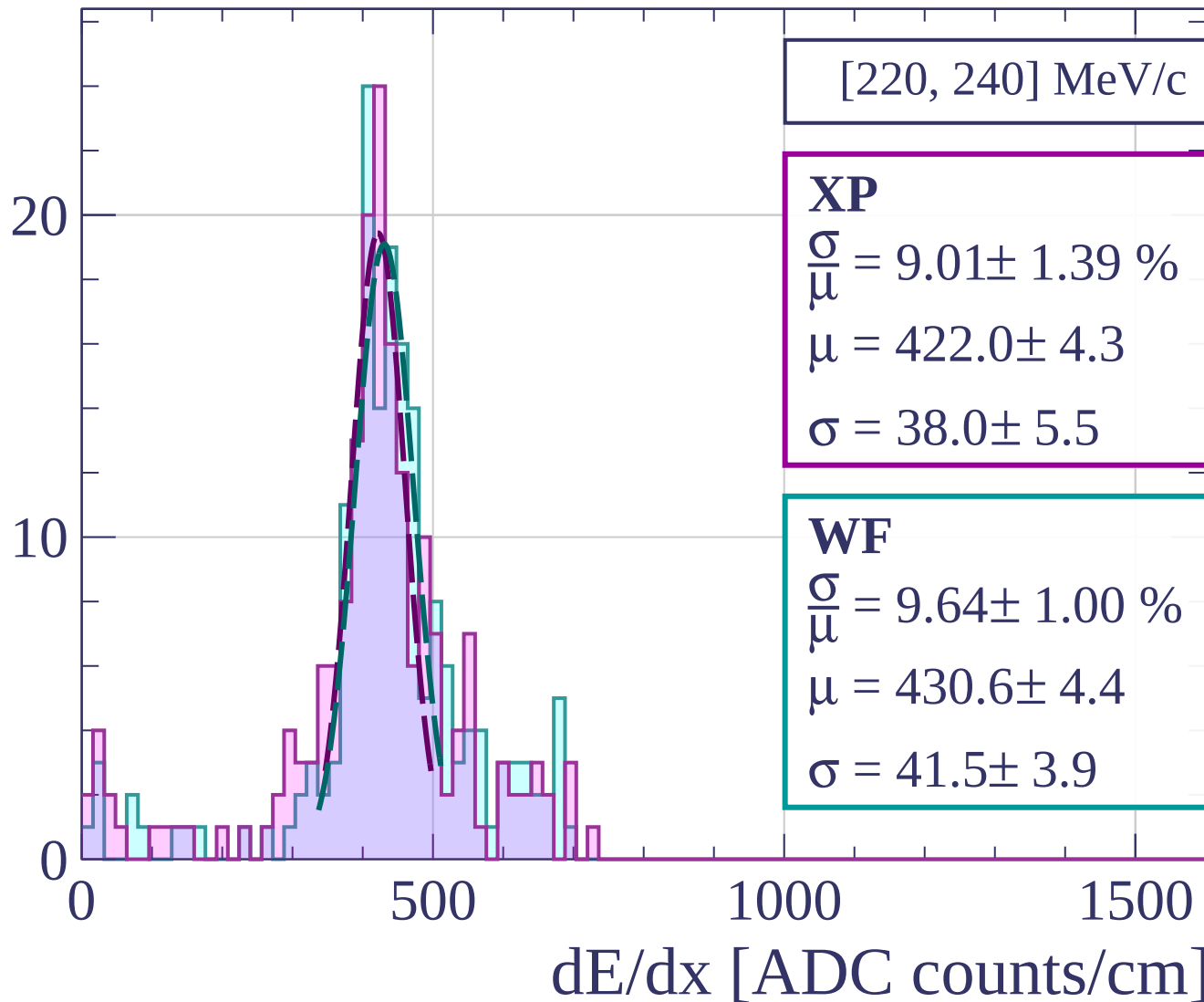
Count



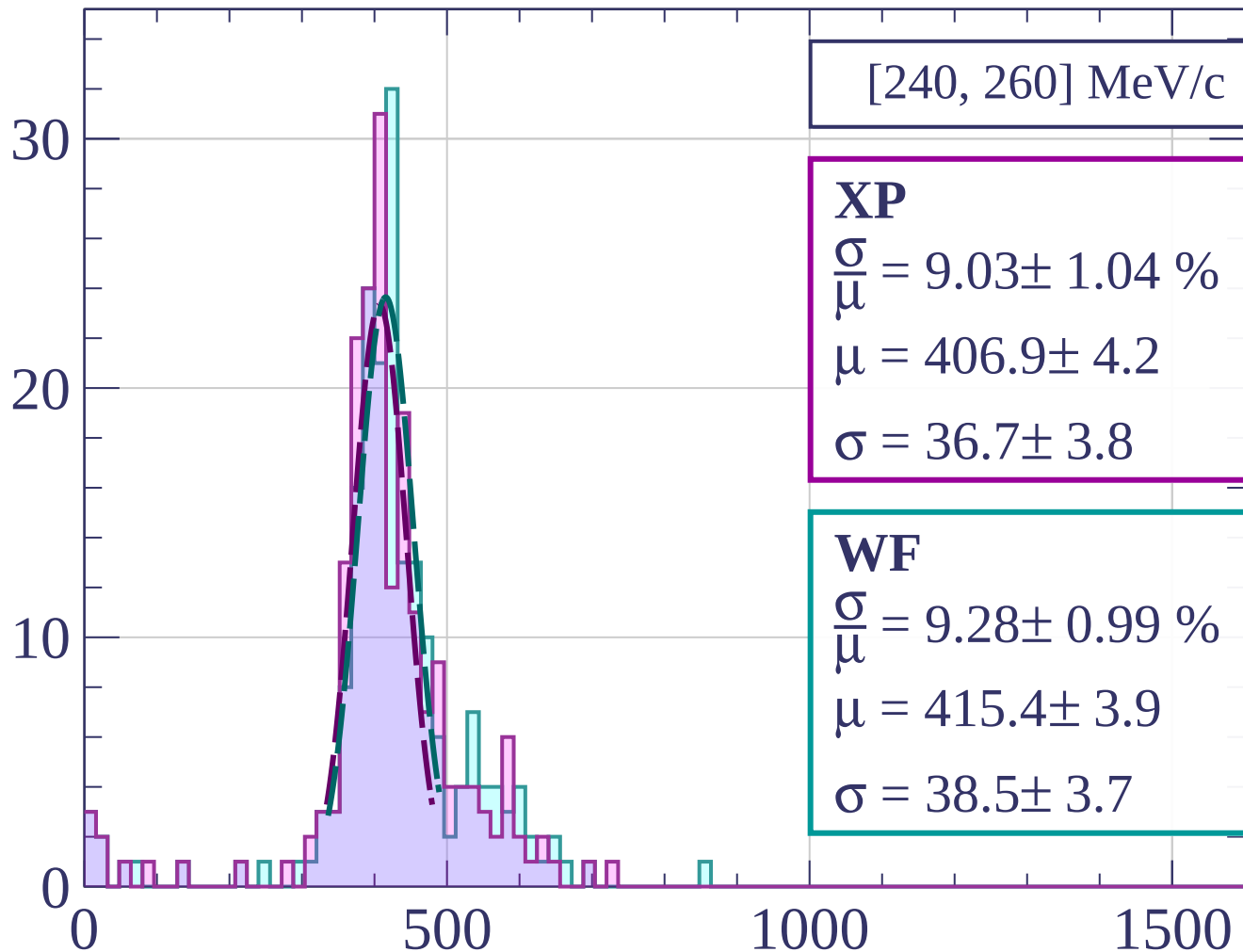
Count



Count



Count



[240, 260] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.03 \pm 1.04 \%$$

$$\mu = 406.9 \pm 4.2$$

$$\sigma = 36.7 \pm 3.8$$

WF

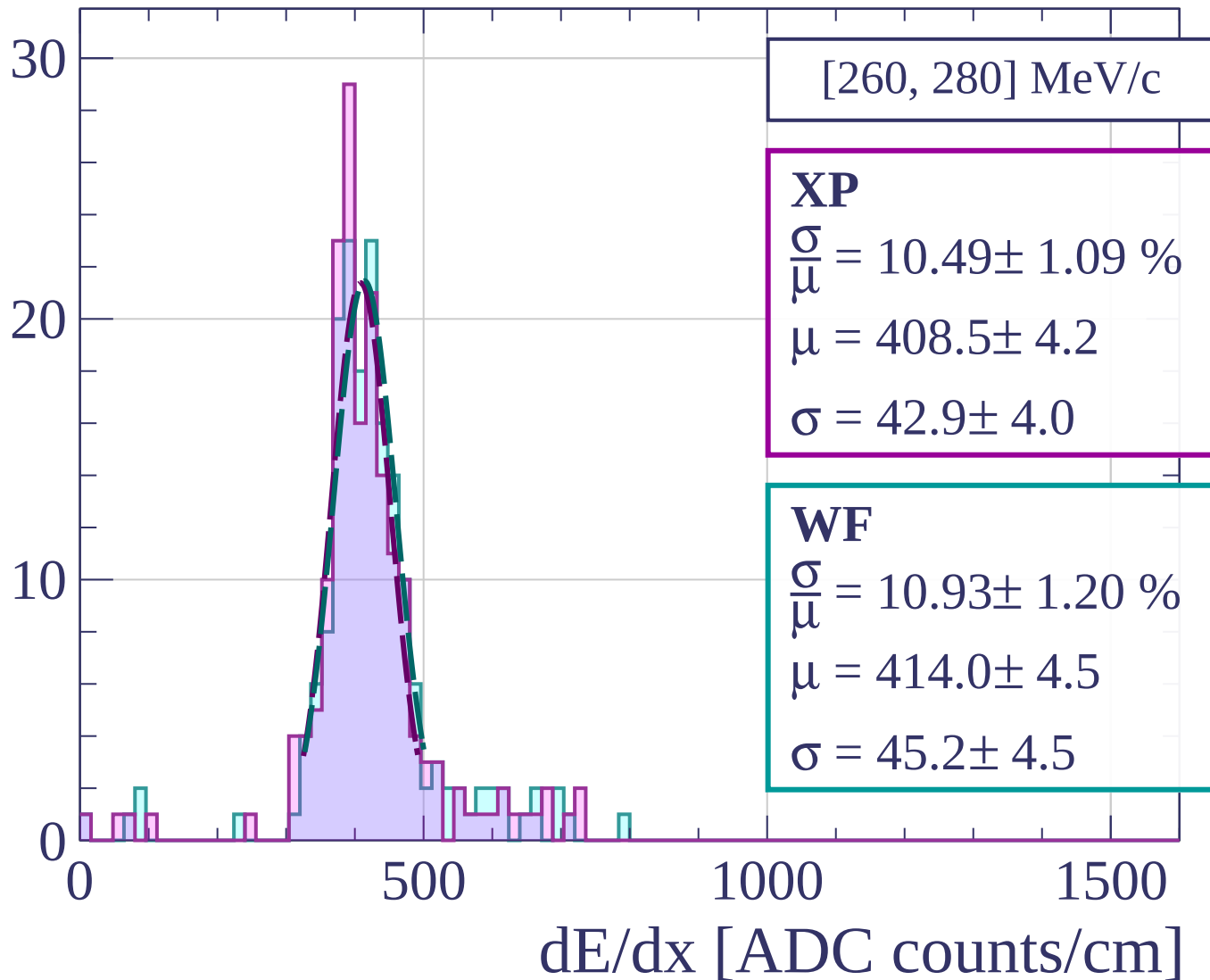
$$\frac{\sigma}{\mu} = 9.28 \pm 0.99 \%$$

$$\mu = 415.4 \pm 3.9$$

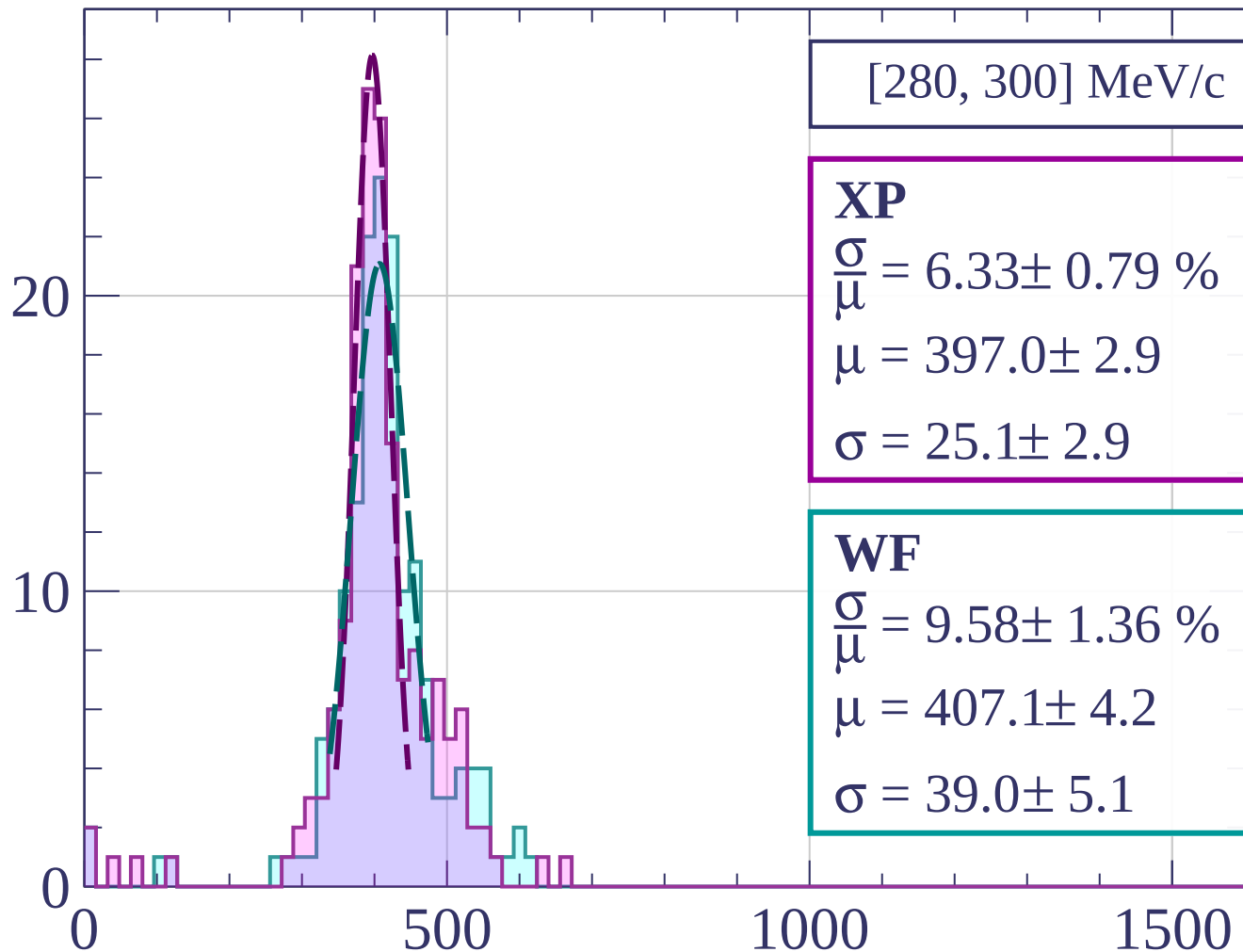
$$\sigma = 38.5 \pm 3.7$$

dE/dx [ADC counts/cm]

Count

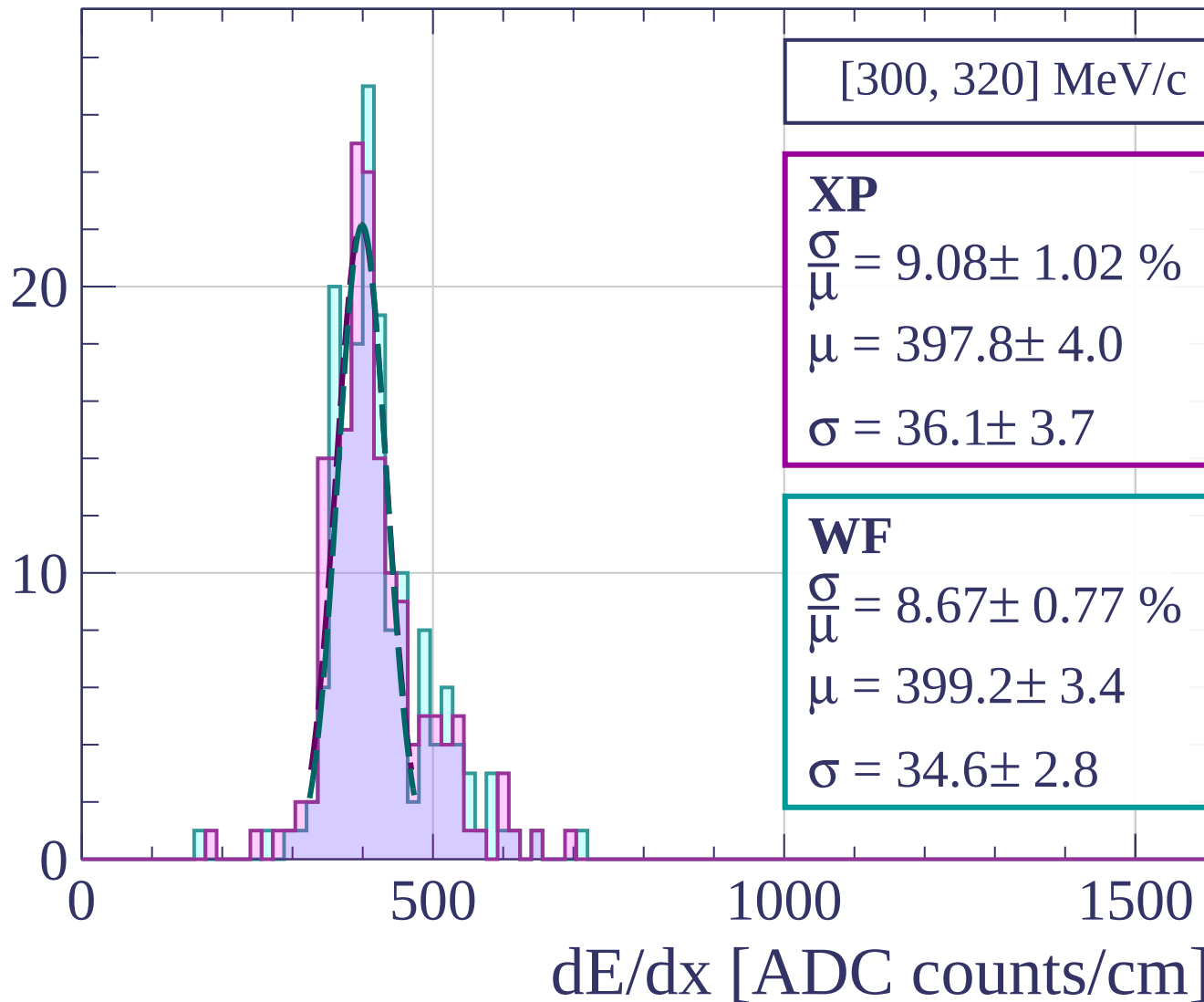


Count

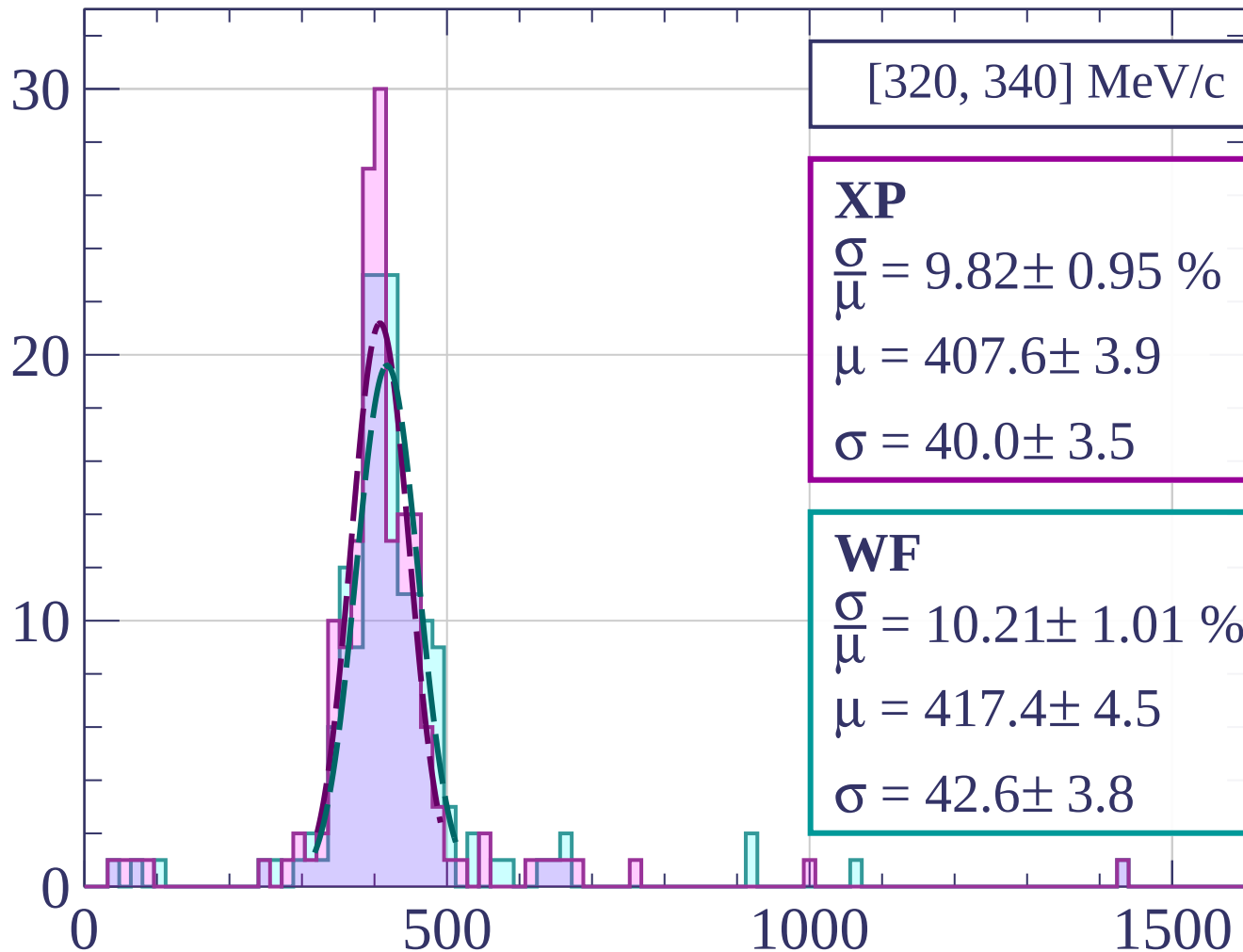


dE/dx [ADC counts/cm]

Count



Count



[320, 340] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.82 \pm 0.95 \%$$

$$\mu = 407.6 \pm 3.9$$

$$\sigma = 40.0 \pm 3.5$$

WF

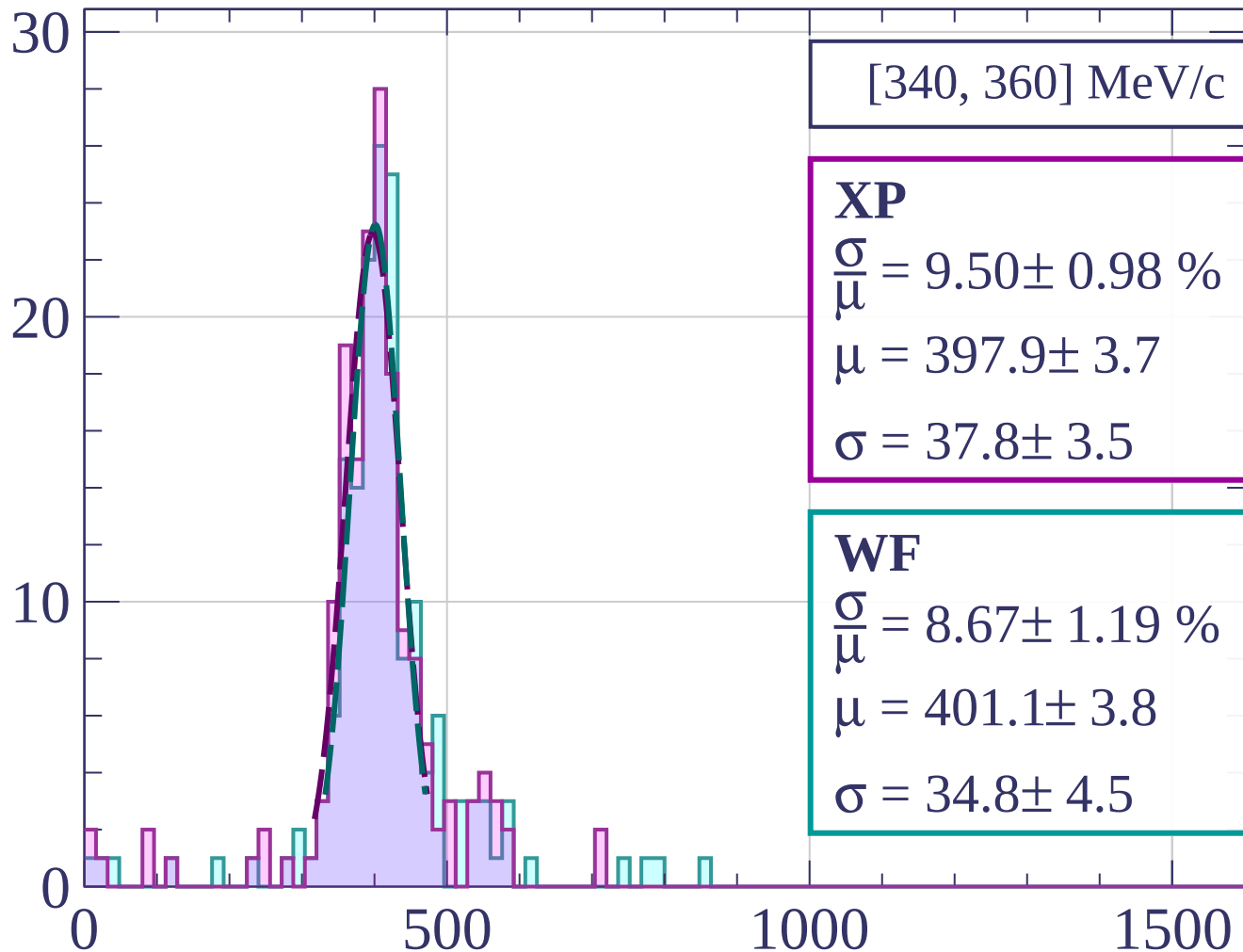
$$\frac{\sigma}{\mu} = 10.21 \pm 1.01 \%$$

$$\mu = 417.4 \pm 4.5$$

$$\sigma = 42.6 \pm 3.8$$

dE/dx [ADC counts/cm]

Count



[340, 360] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.50 \pm 0.98 \%$$

$$\mu = 397.9 \pm 3.7$$

$$\sigma = 37.8 \pm 3.5$$

WF

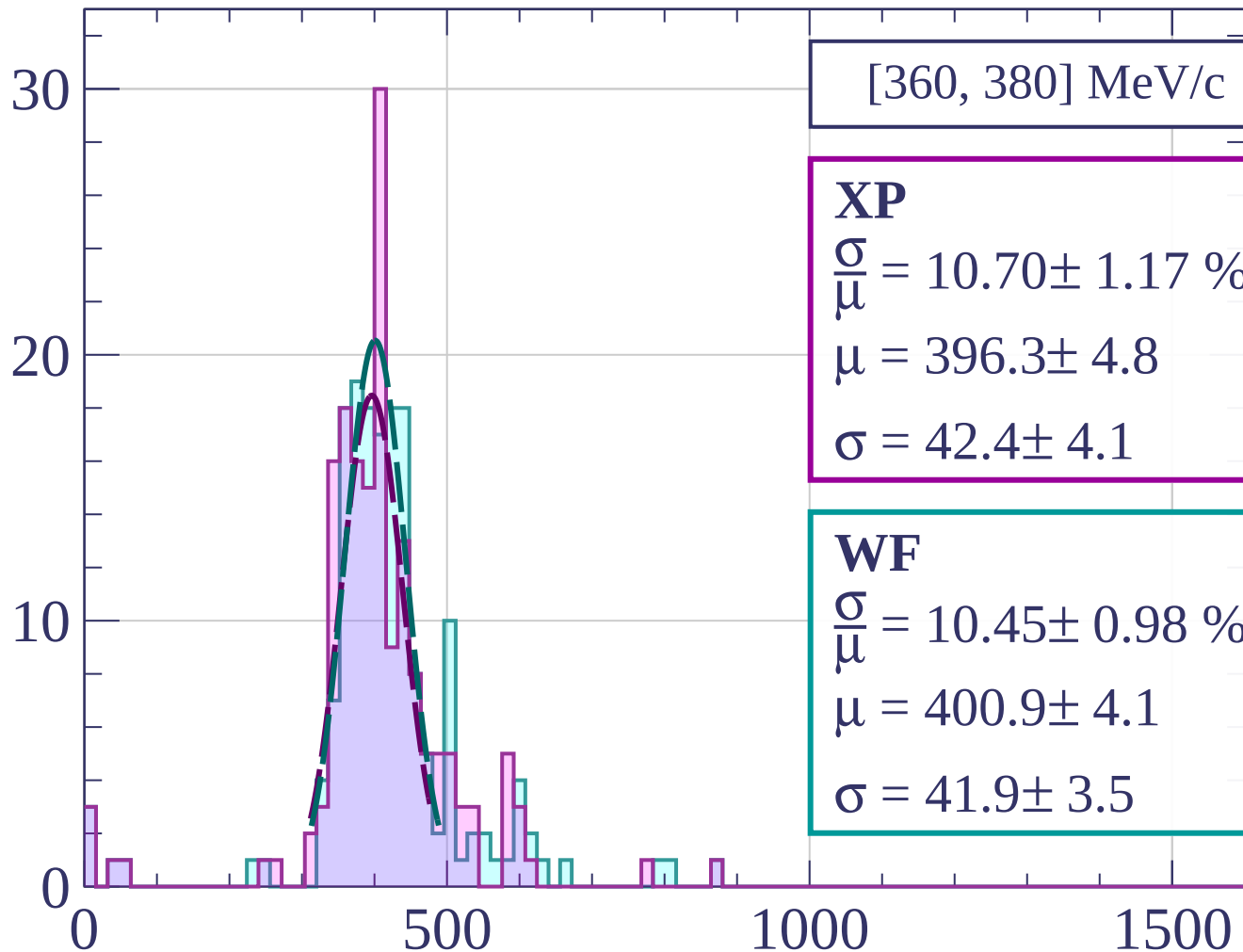
$$\frac{\sigma}{\mu} = 8.67 \pm 1.19 \%$$

$$\mu = 401.1 \pm 3.8$$

$$\sigma = 34.8 \pm 4.5$$

dE/dx [ADC counts/cm]

Count



[360, 380] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.70 \pm 1.17 \%$$

$$\mu = 396.3 \pm 4.8$$

$$\sigma = 42.4 \pm 4.1$$

WF

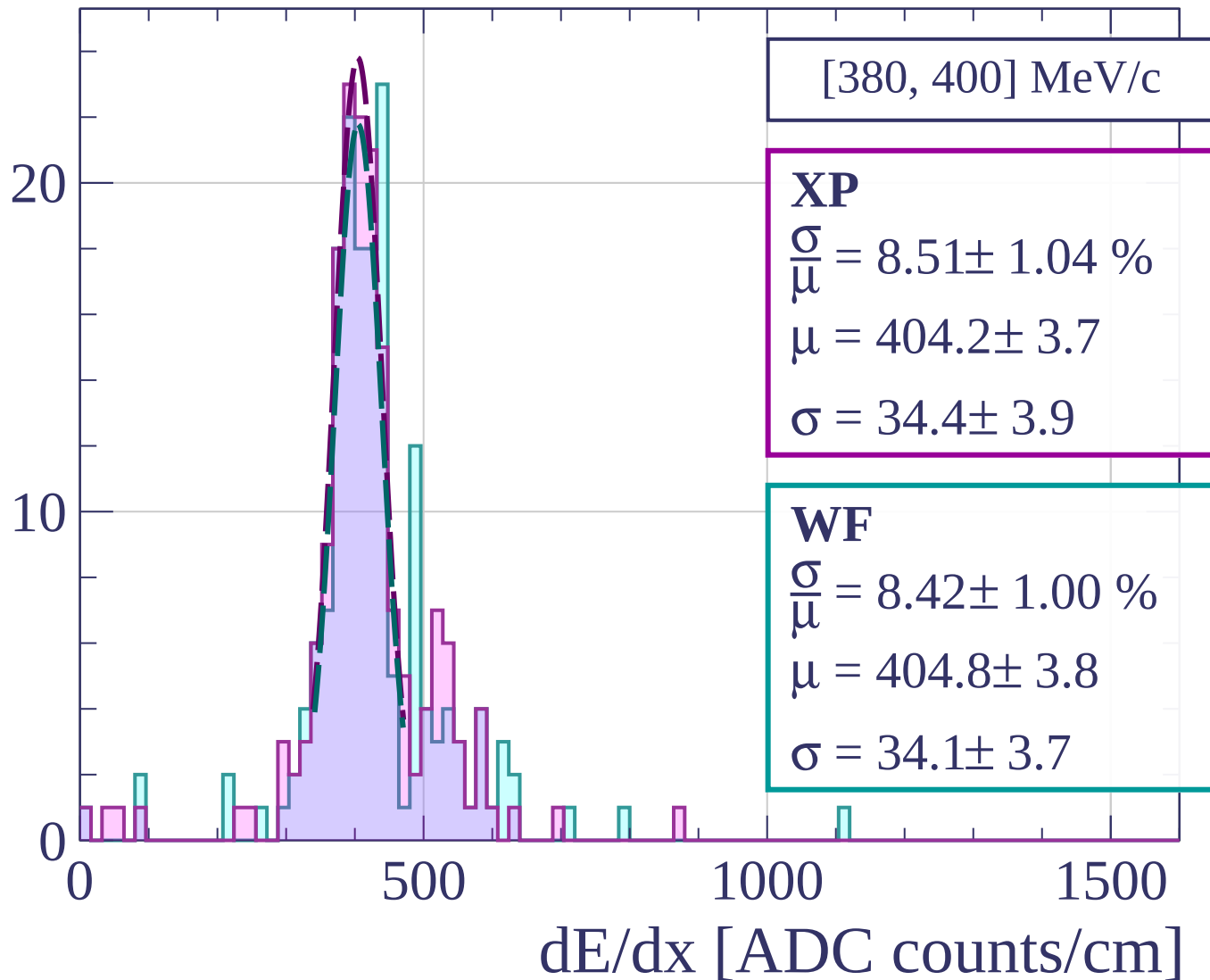
$$\frac{\sigma}{\mu} = 10.45 \pm 0.98 \%$$

$$\mu = 400.9 \pm 4.1$$

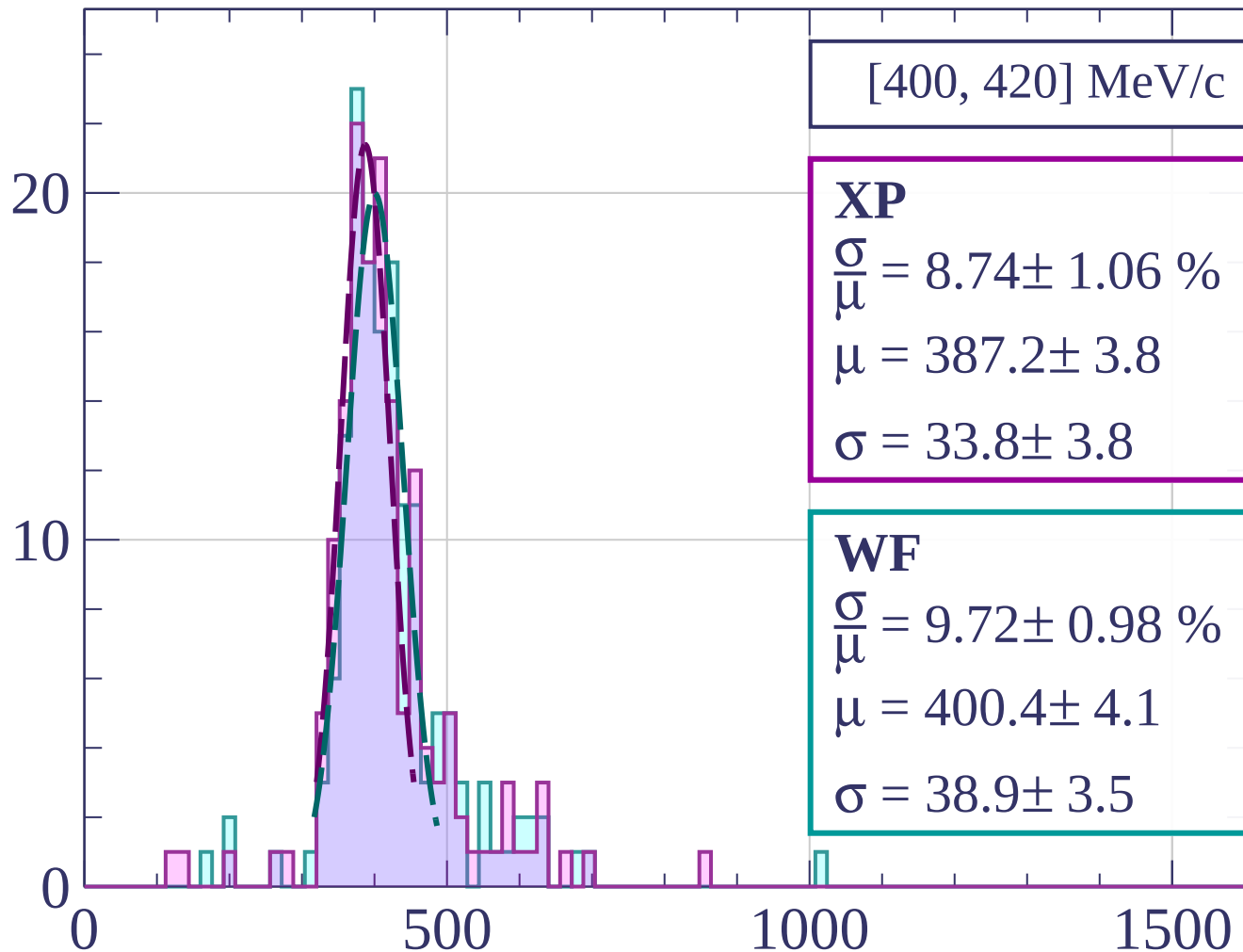
$$\sigma = 41.9 \pm 3.5$$

dE/dx [ADC counts/cm]

Count



Count



[400, 420] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.74 \pm 1.06 \%$$

$$\mu = 387.2 \pm 3.8$$

$$\sigma = 33.8 \pm 3.8$$

WF

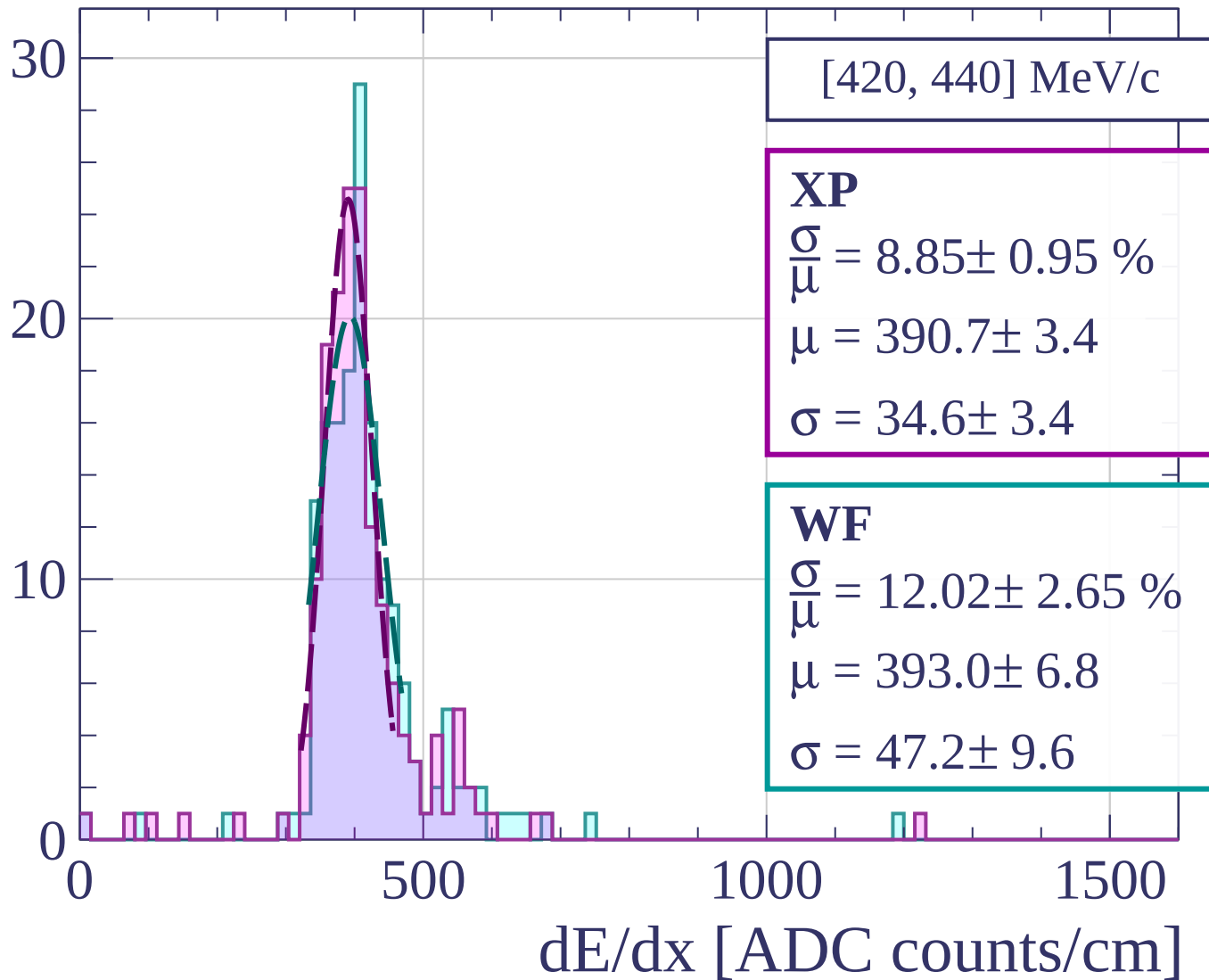
$$\frac{\sigma}{\mu} = 9.72 \pm 0.98 \%$$

$$\mu = 400.4 \pm 4.1$$

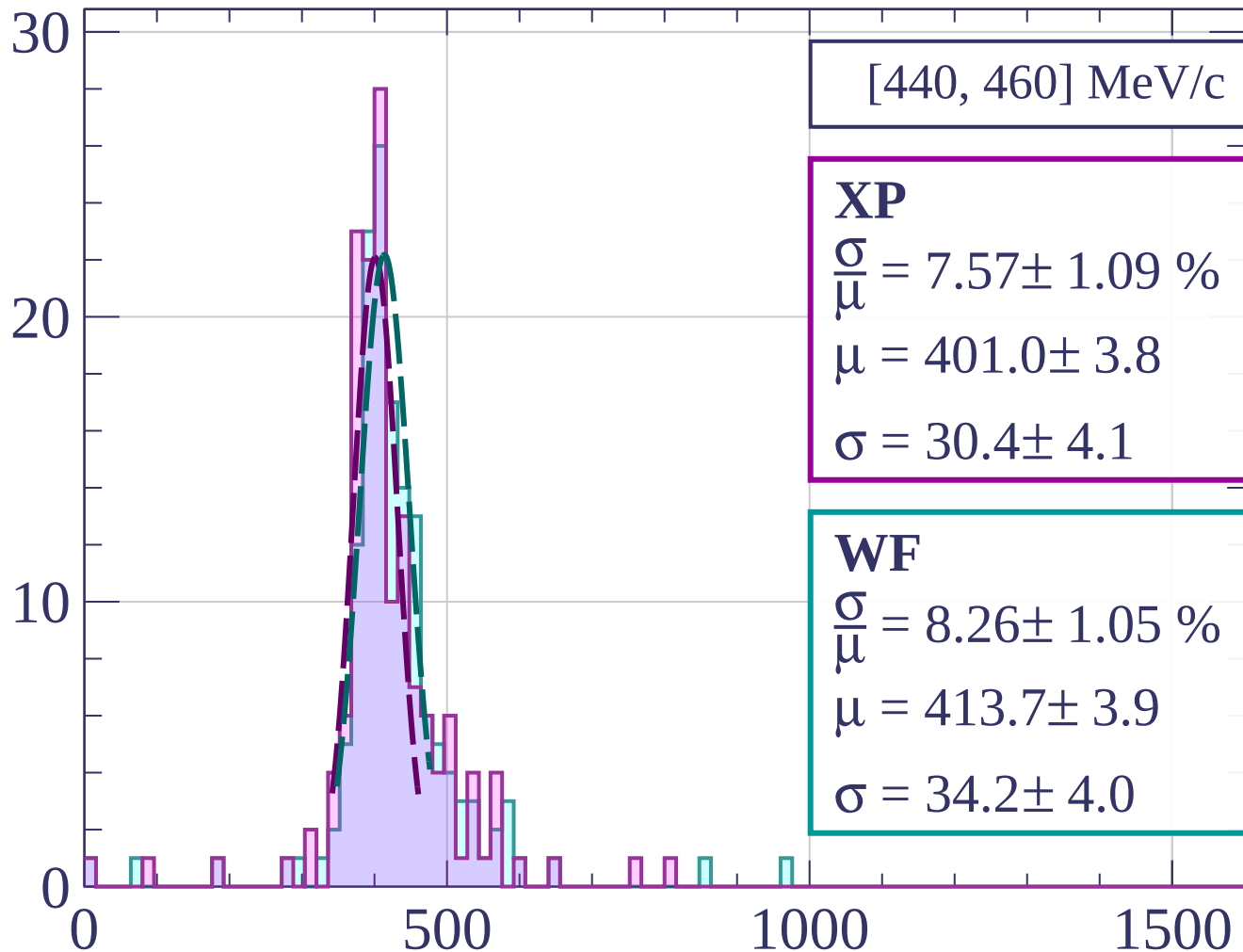
$$\sigma = 38.9 \pm 3.5$$

dE/dx [ADC counts/cm]

Count



Count



[440, 460] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.57 \pm 1.09 \%$$

$$\mu = 401.0 \pm 3.8$$

$$\sigma = 30.4 \pm 4.1$$

WF

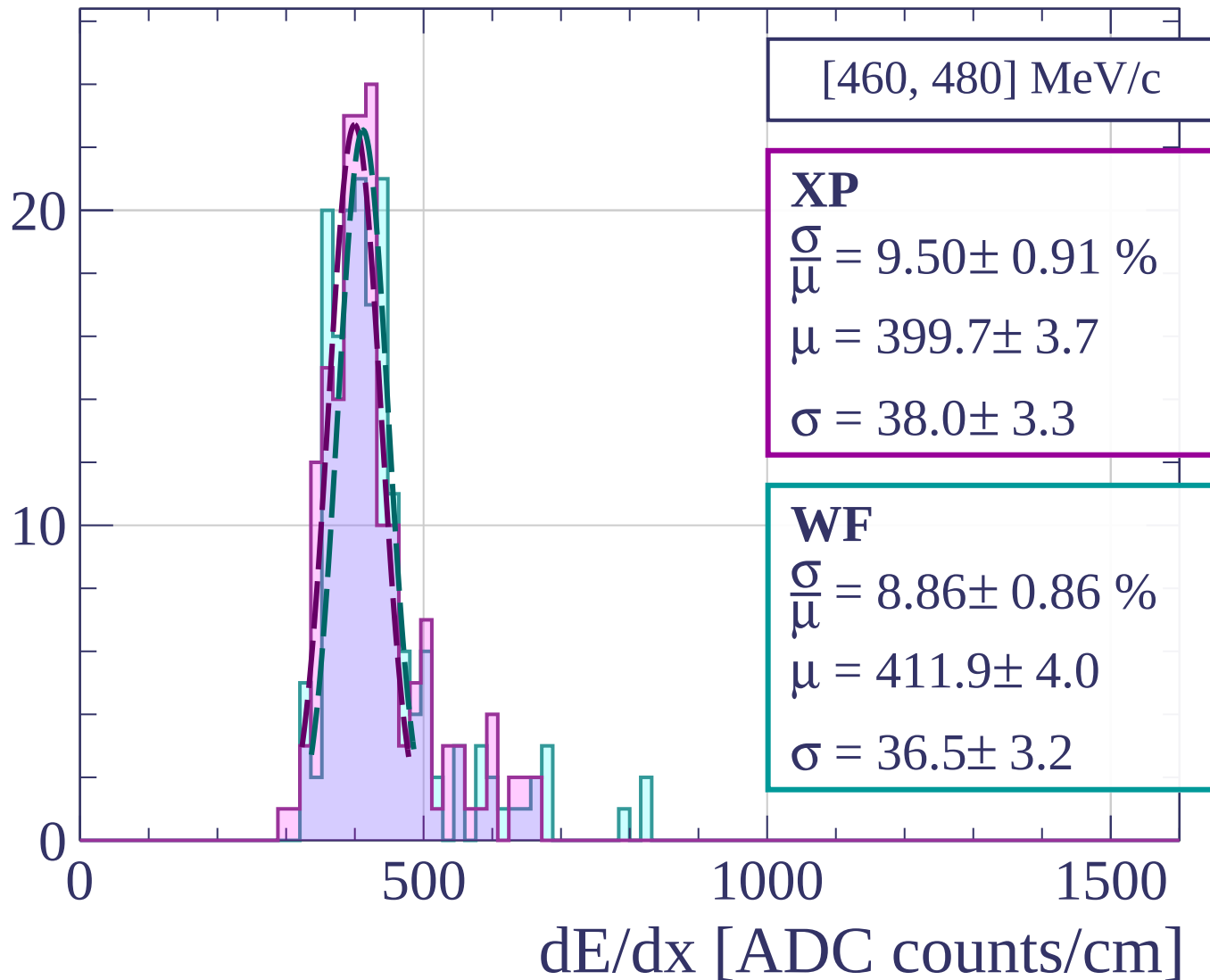
$$\frac{\sigma}{\mu} = 8.26 \pm 1.05 \%$$

$$\mu = 413.7 \pm 3.9$$

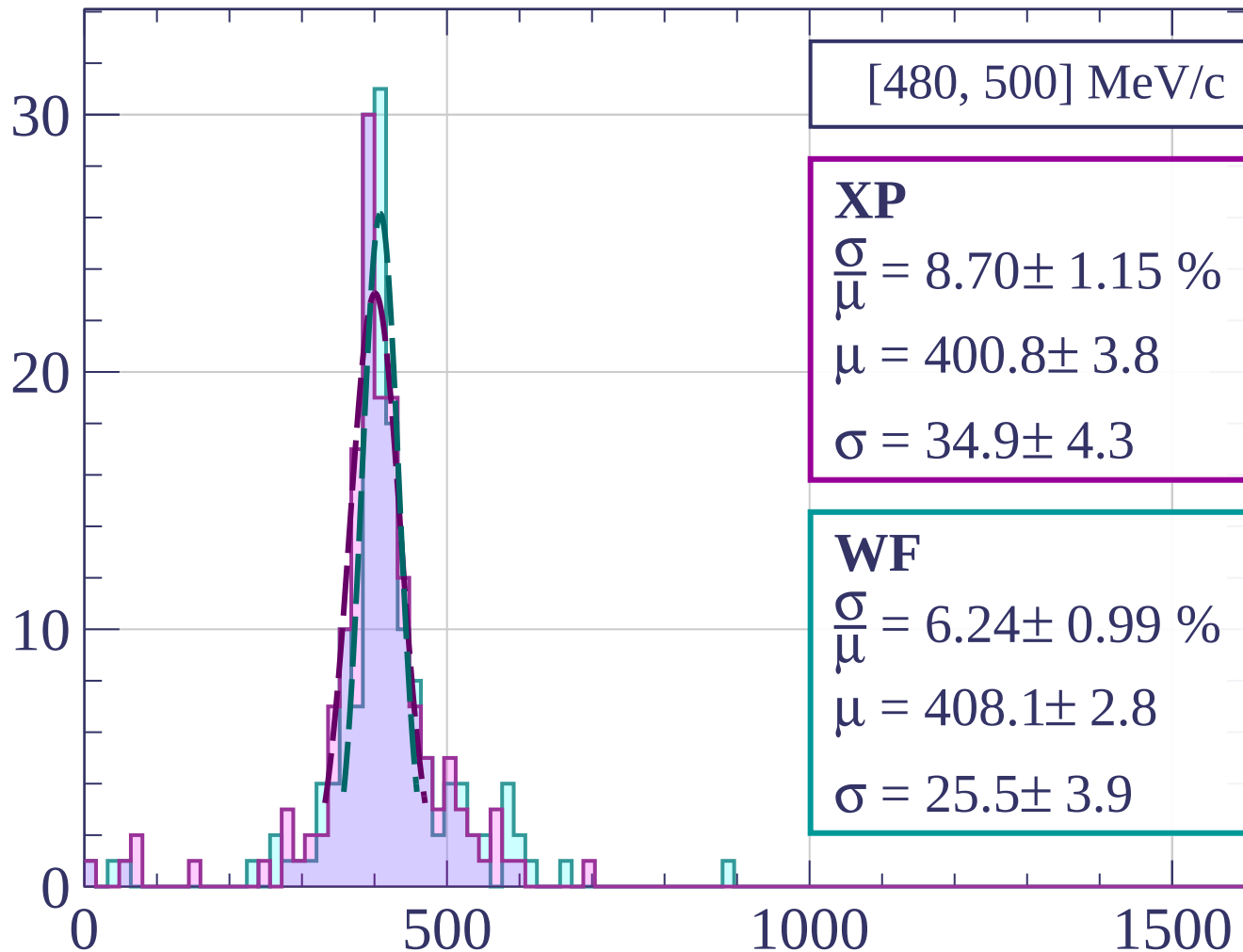
$$\sigma = 34.2 \pm 4.0$$

dE/dx [ADC counts/cm]

Count

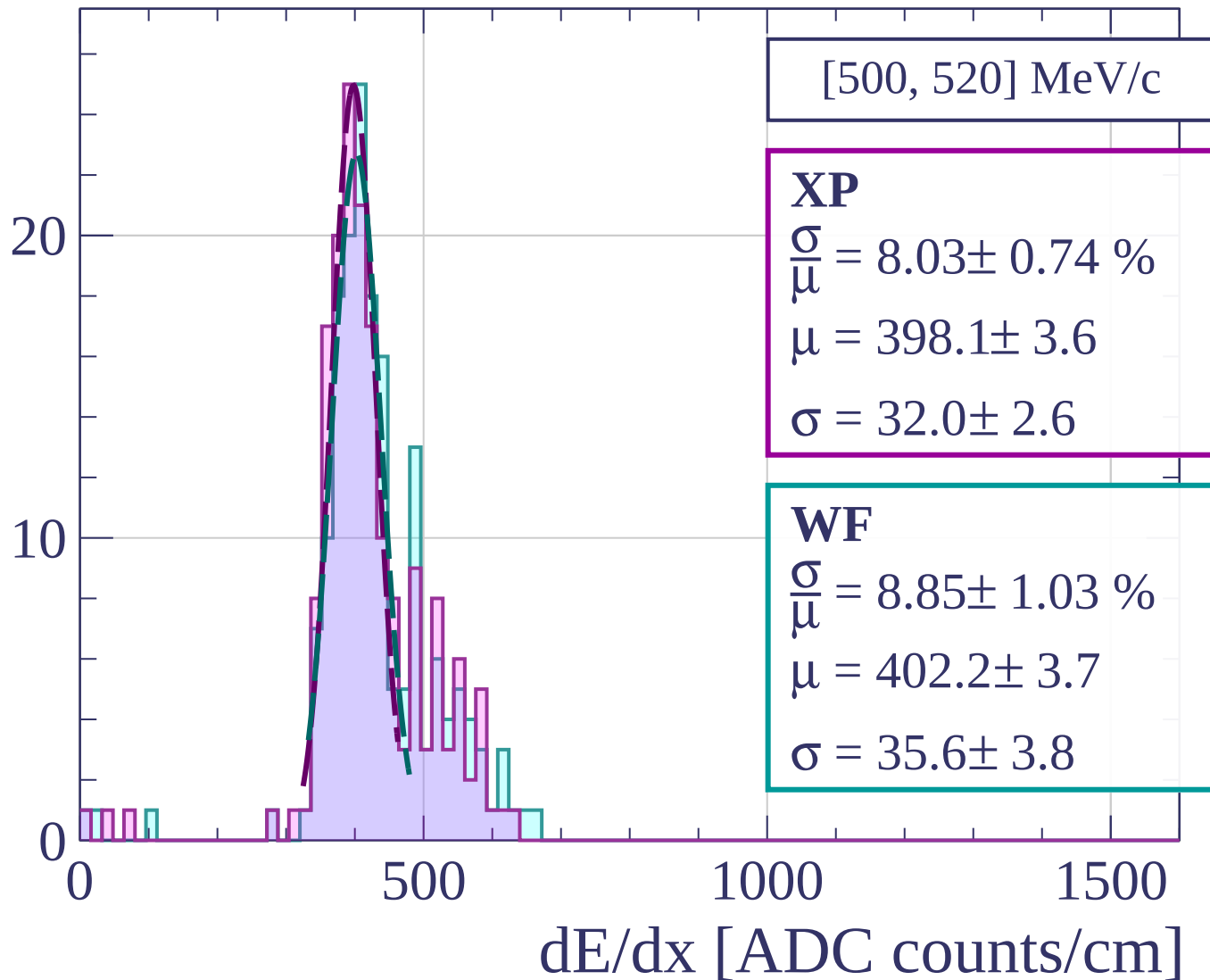


Count

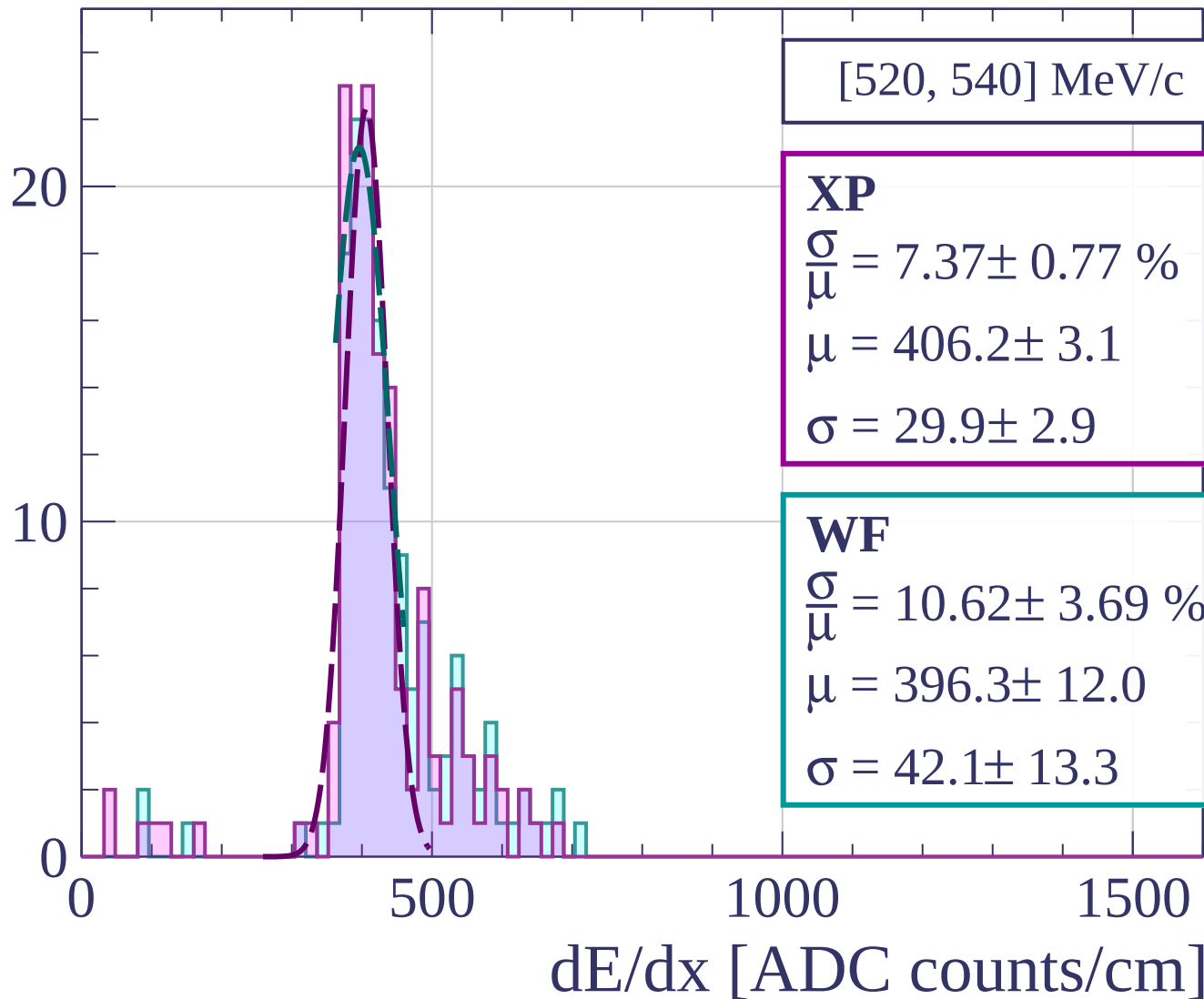


dE/dx [ADC counts/cm]

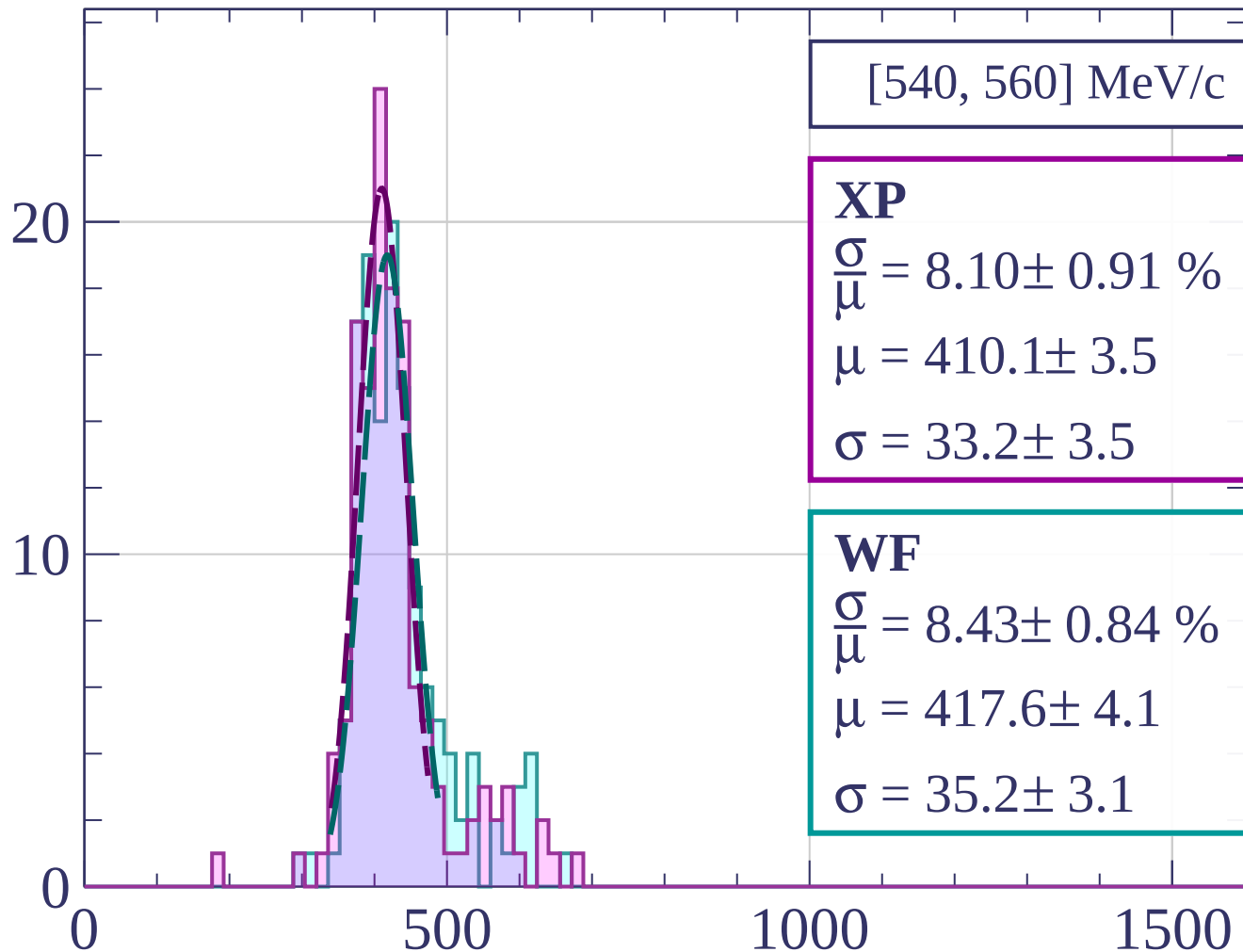
Count



Count



Count



[540, 560] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.10 \pm 0.91 \%$$

$$\mu = 410.1 \pm 3.5$$

$$\sigma = 33.2 \pm 3.5$$

WF

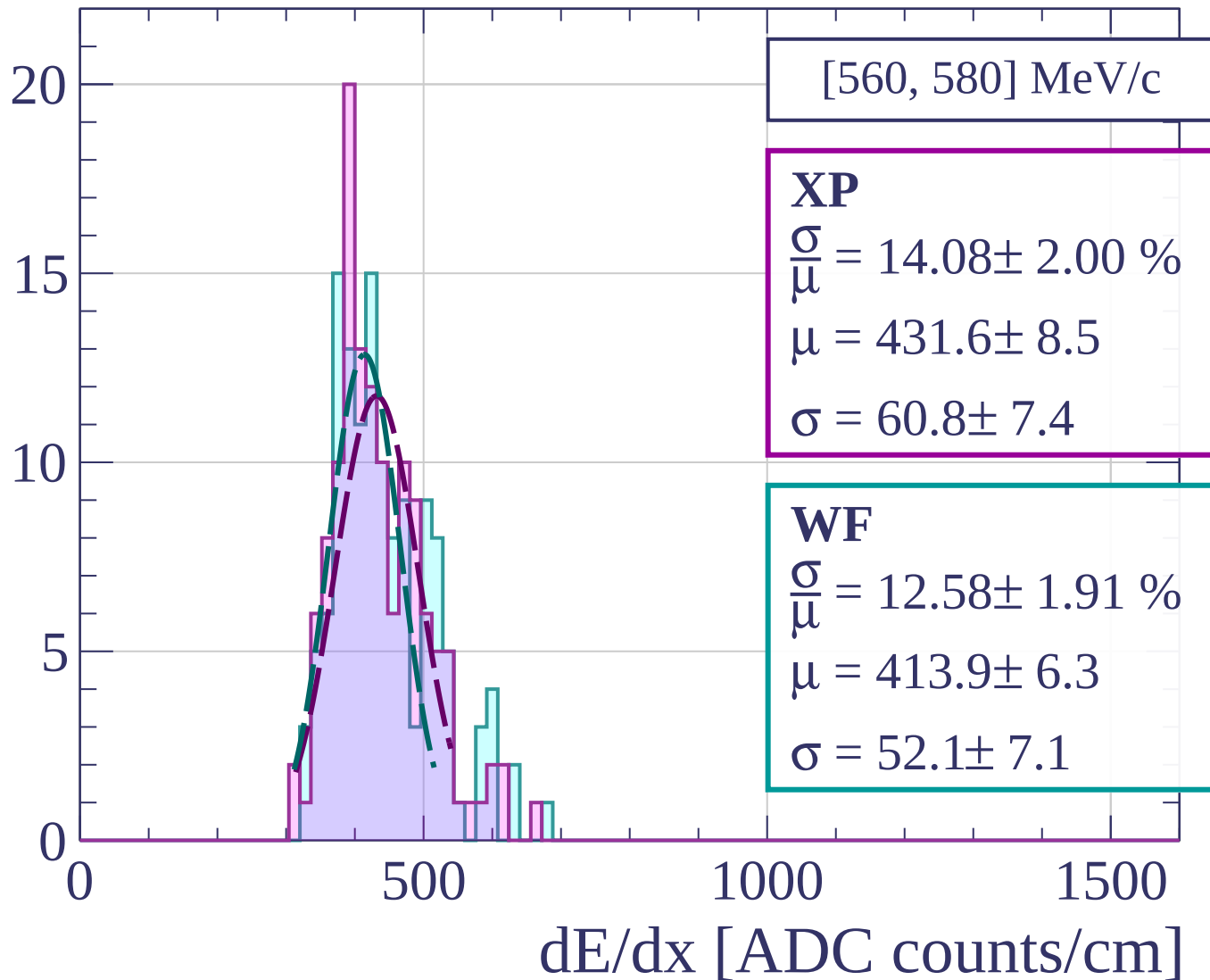
$$\frac{\sigma}{\mu} = 8.43 \pm 0.84 \%$$

$$\mu = 417.6 \pm 4.1$$

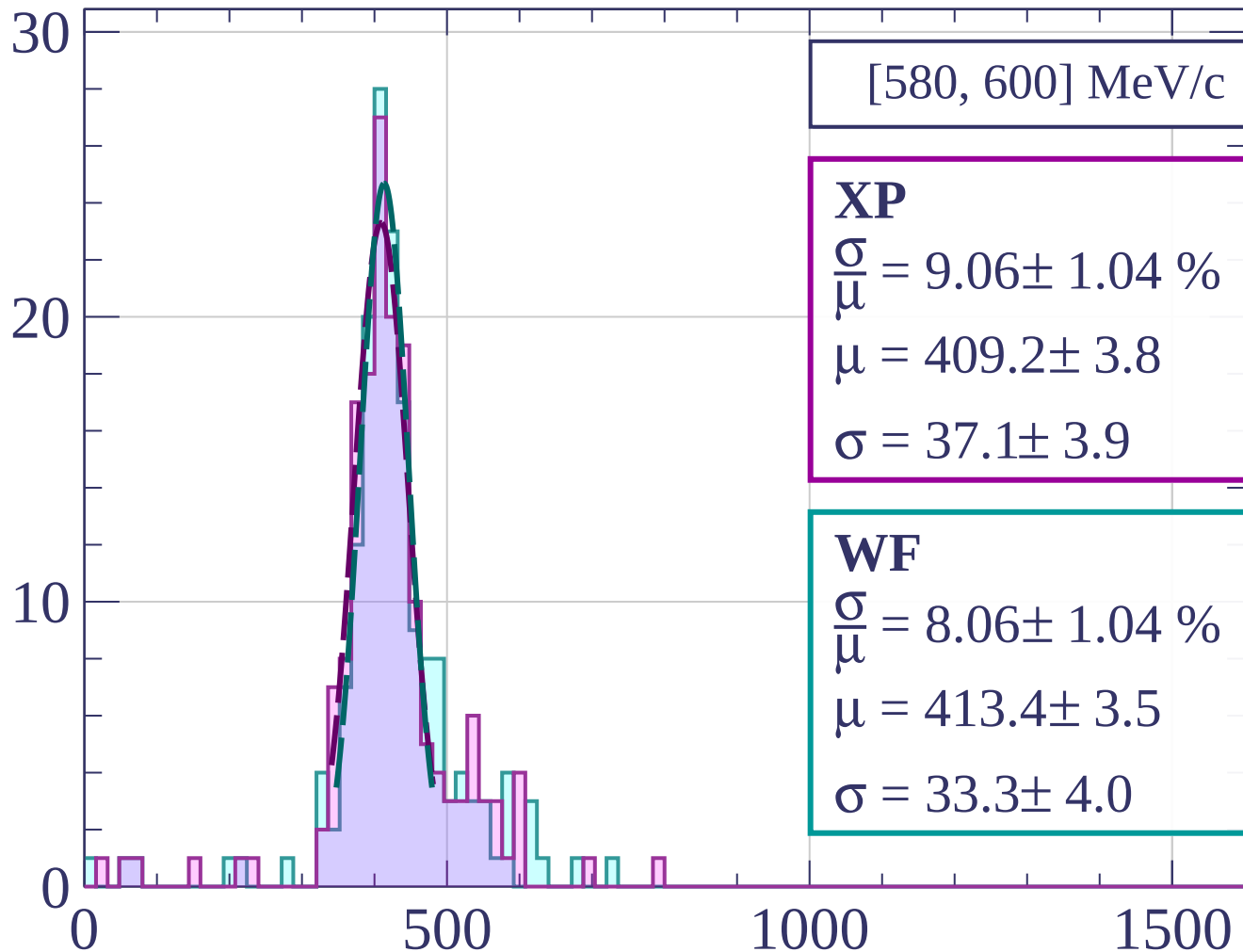
$$\sigma = 35.2 \pm 3.1$$

dE/dx [ADC counts/cm]

Count



Count



[580, 600] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.06 \pm 1.04 \%$$

$$\mu = 409.2 \pm 3.8$$

$$\sigma = 37.1 \pm 3.9$$

WF

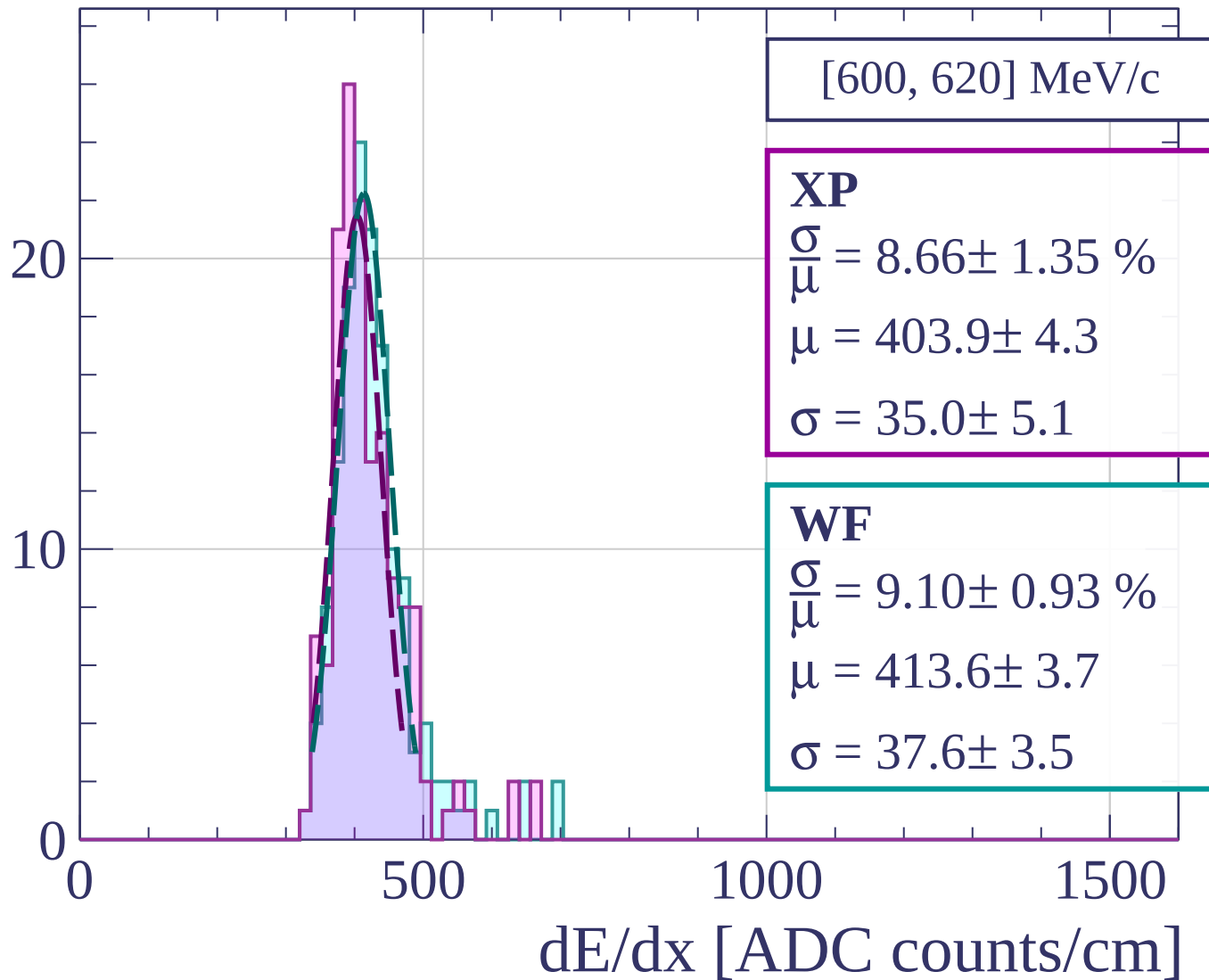
$$\frac{\sigma}{\mu} = 8.06 \pm 1.04 \%$$

$$\mu = 413.4 \pm 3.5$$

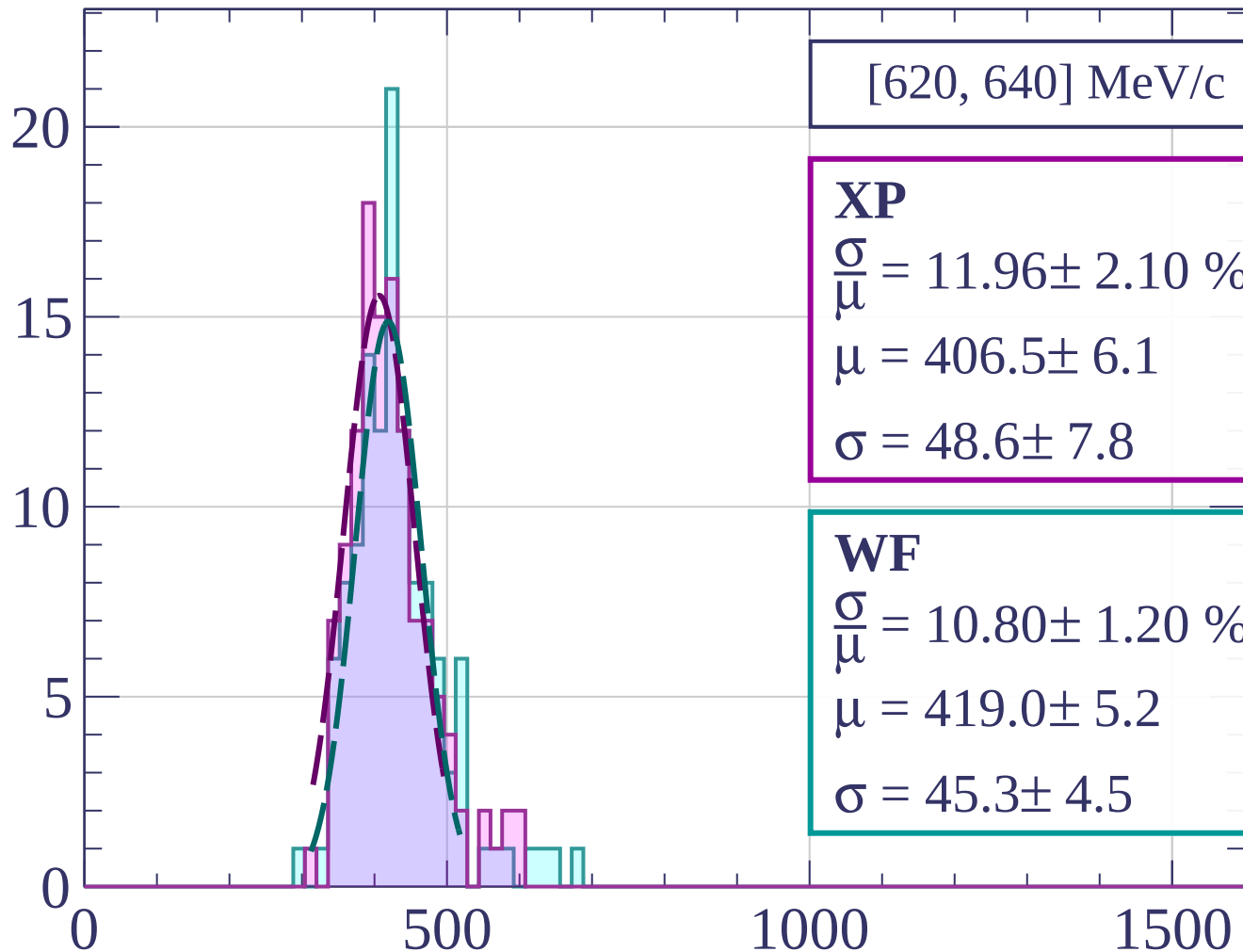
$$\sigma = 33.3 \pm 4.0$$

dE/dx [ADC counts/cm]

Count

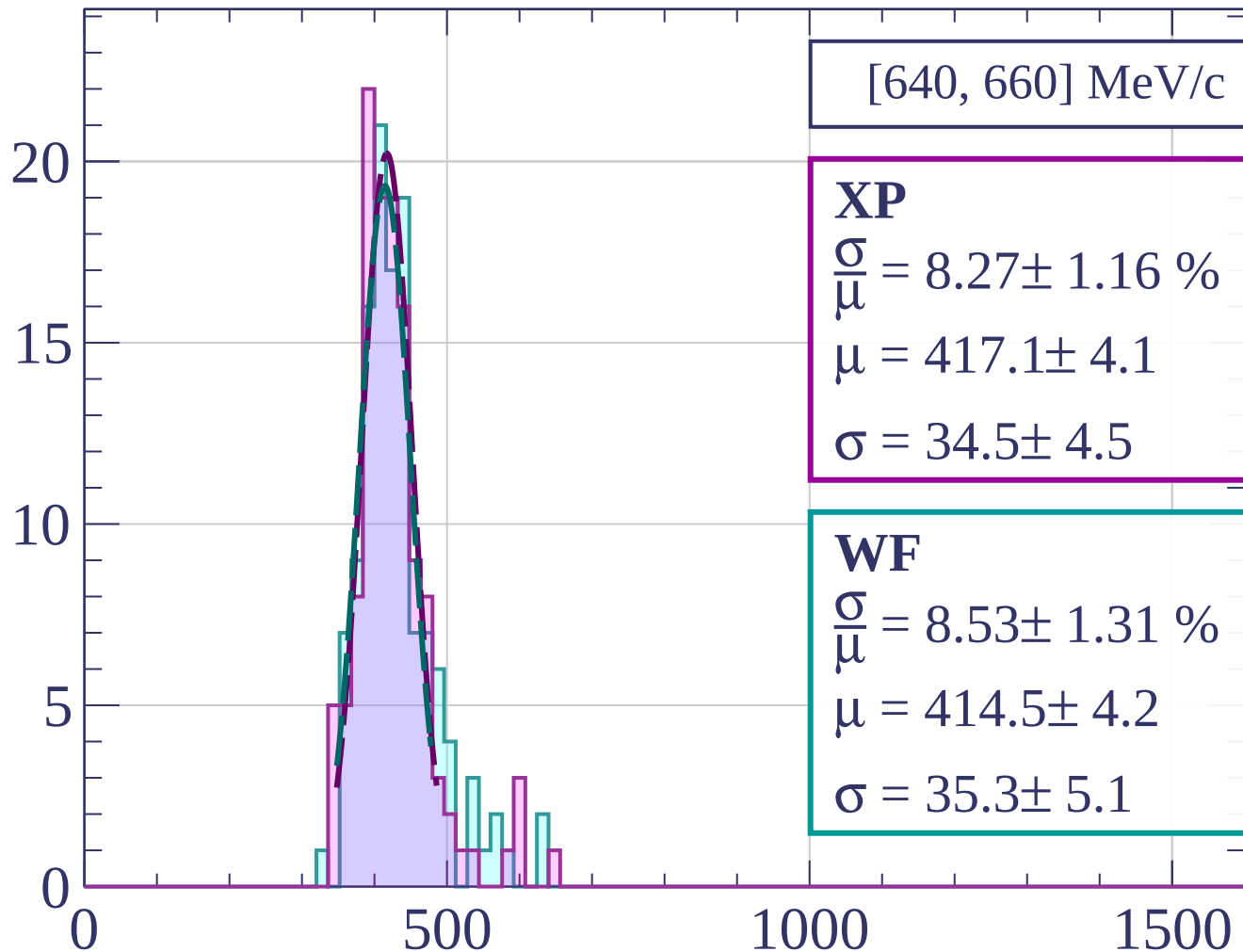


Count



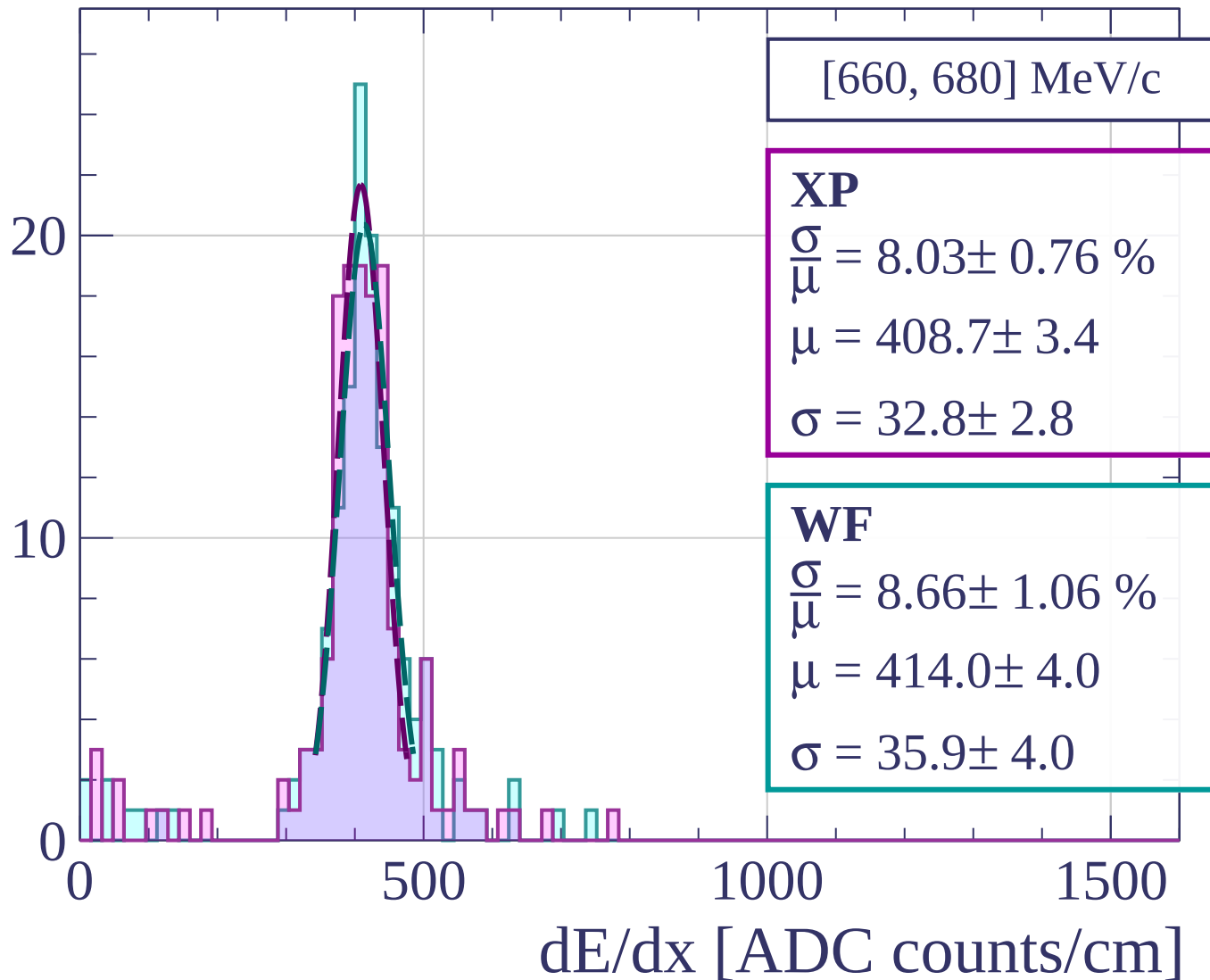
dE/dx [ADC counts/cm]

Count

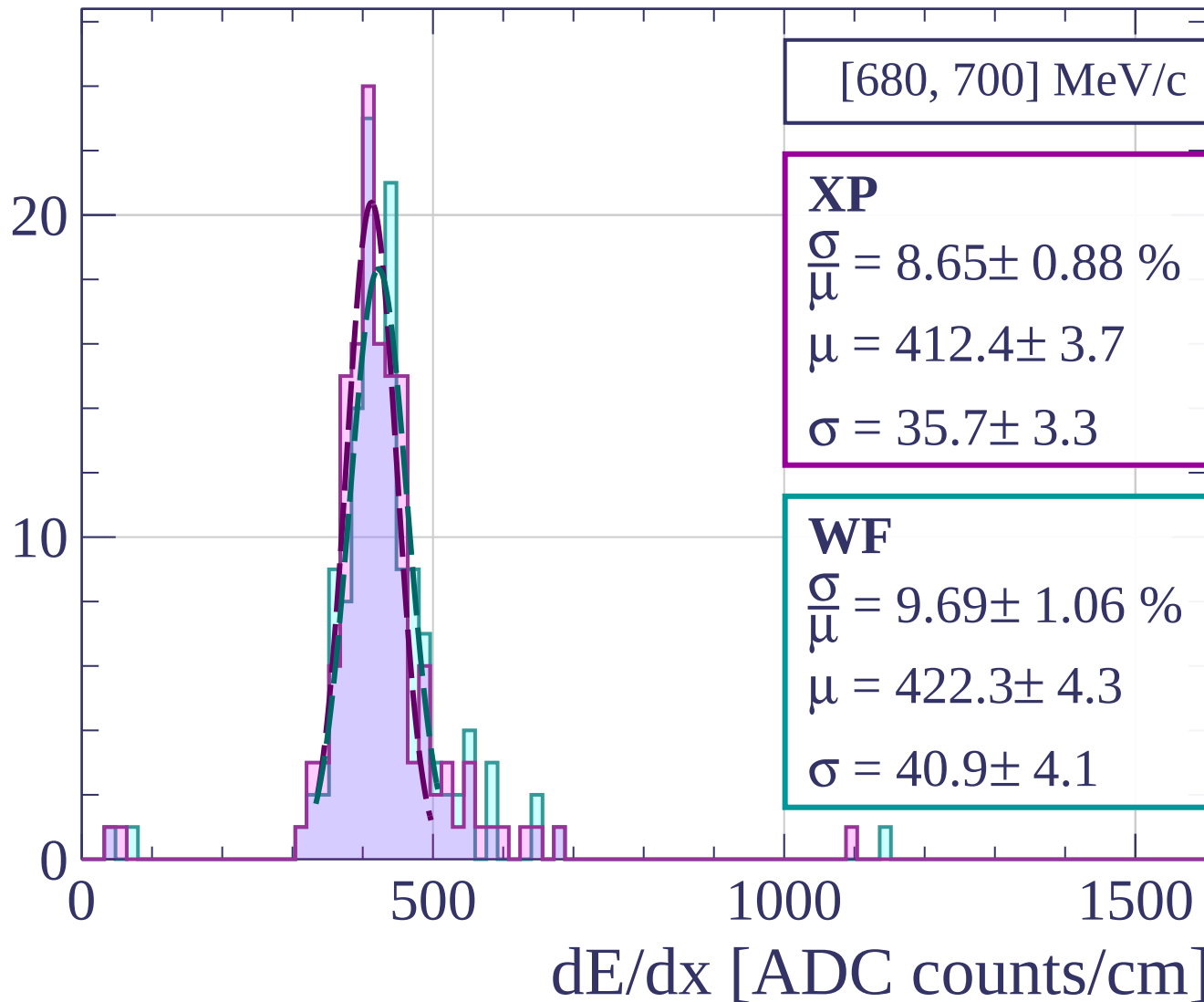


dE/dx [ADC counts/cm]

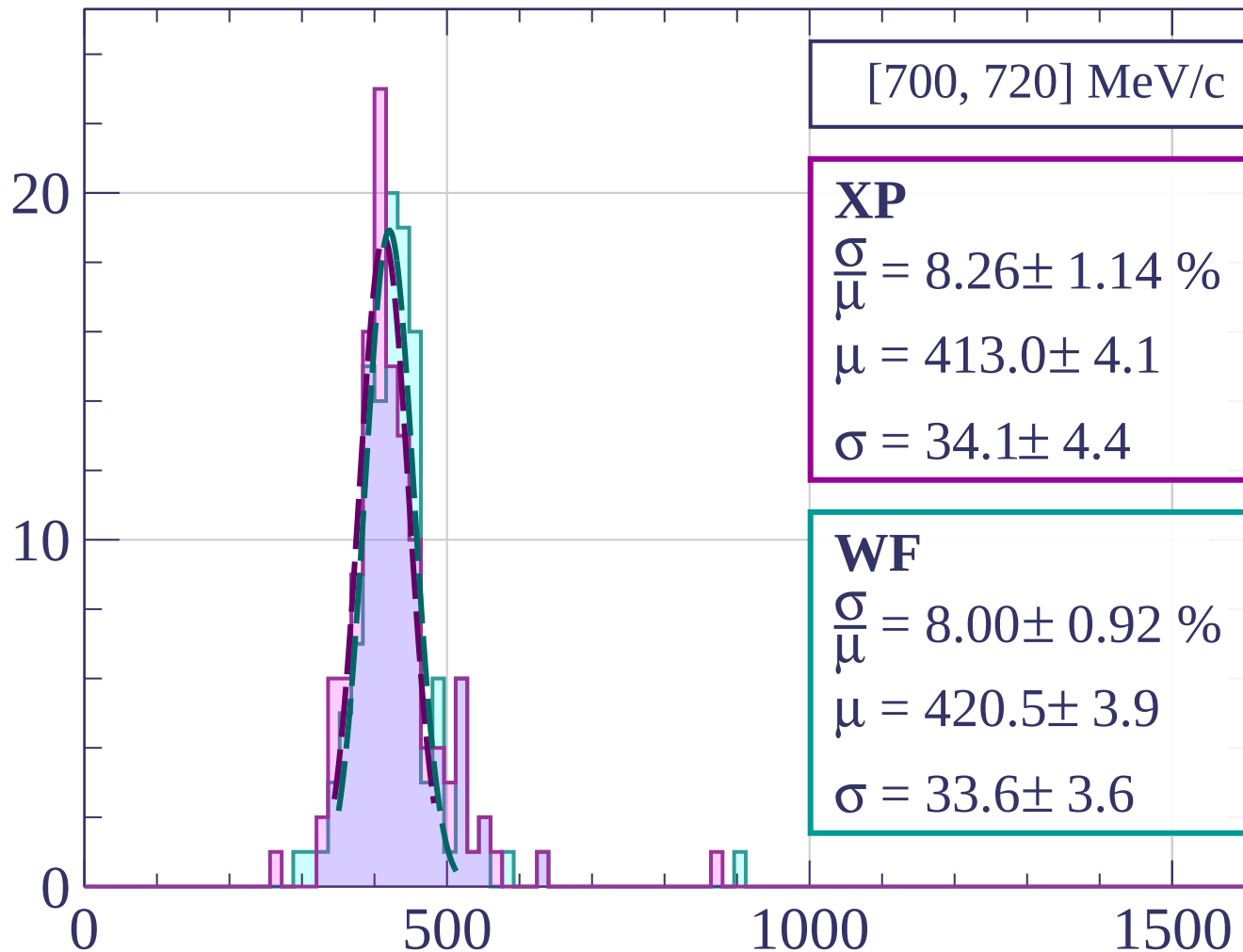
Count



Count

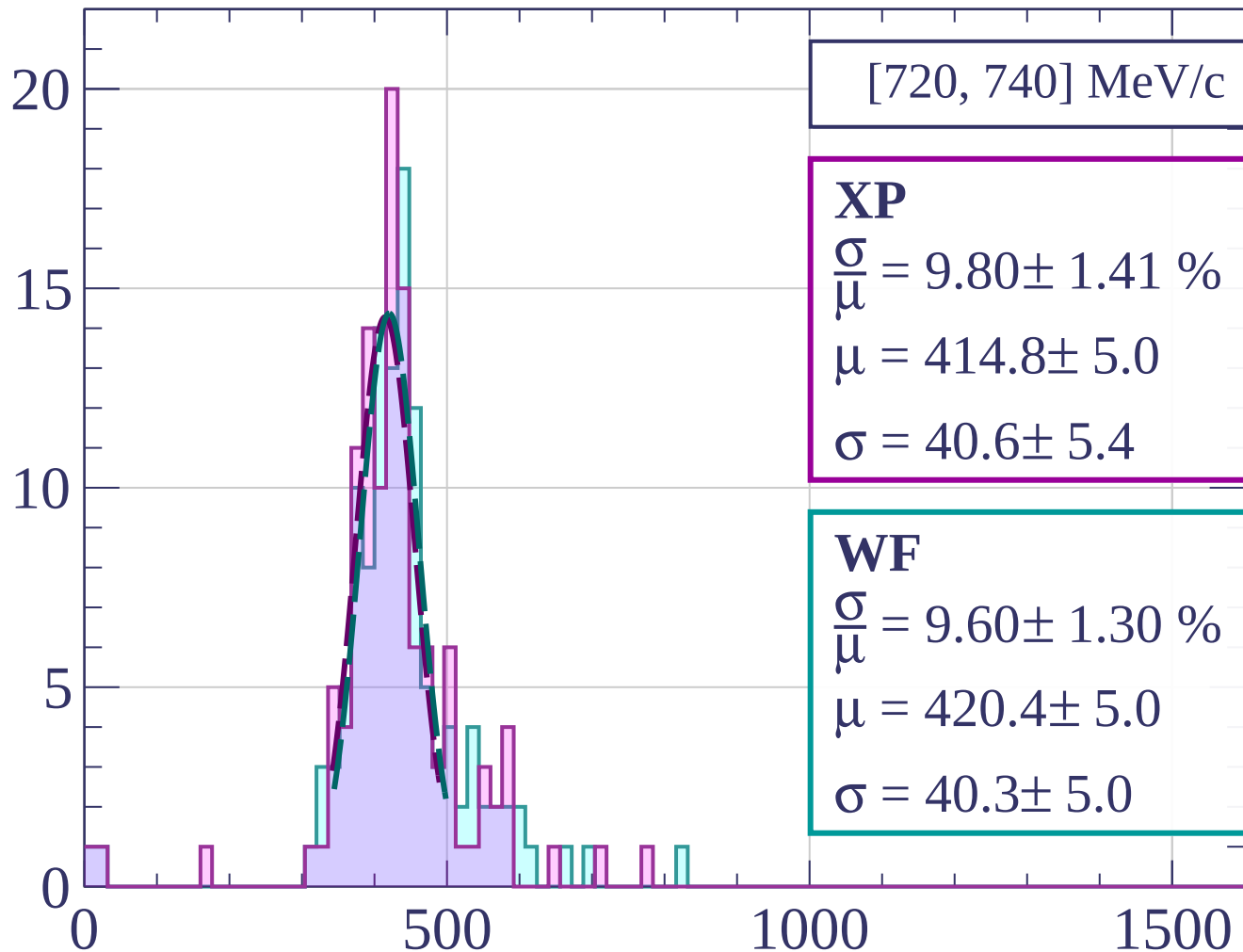


Count



dE/dx [ADC counts/cm]

Count



[720, 740] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.80 \pm 1.41 \%$$

$$\mu = 414.8 \pm 5.0$$

$$\sigma = 40.6 \pm 5.4$$

WF

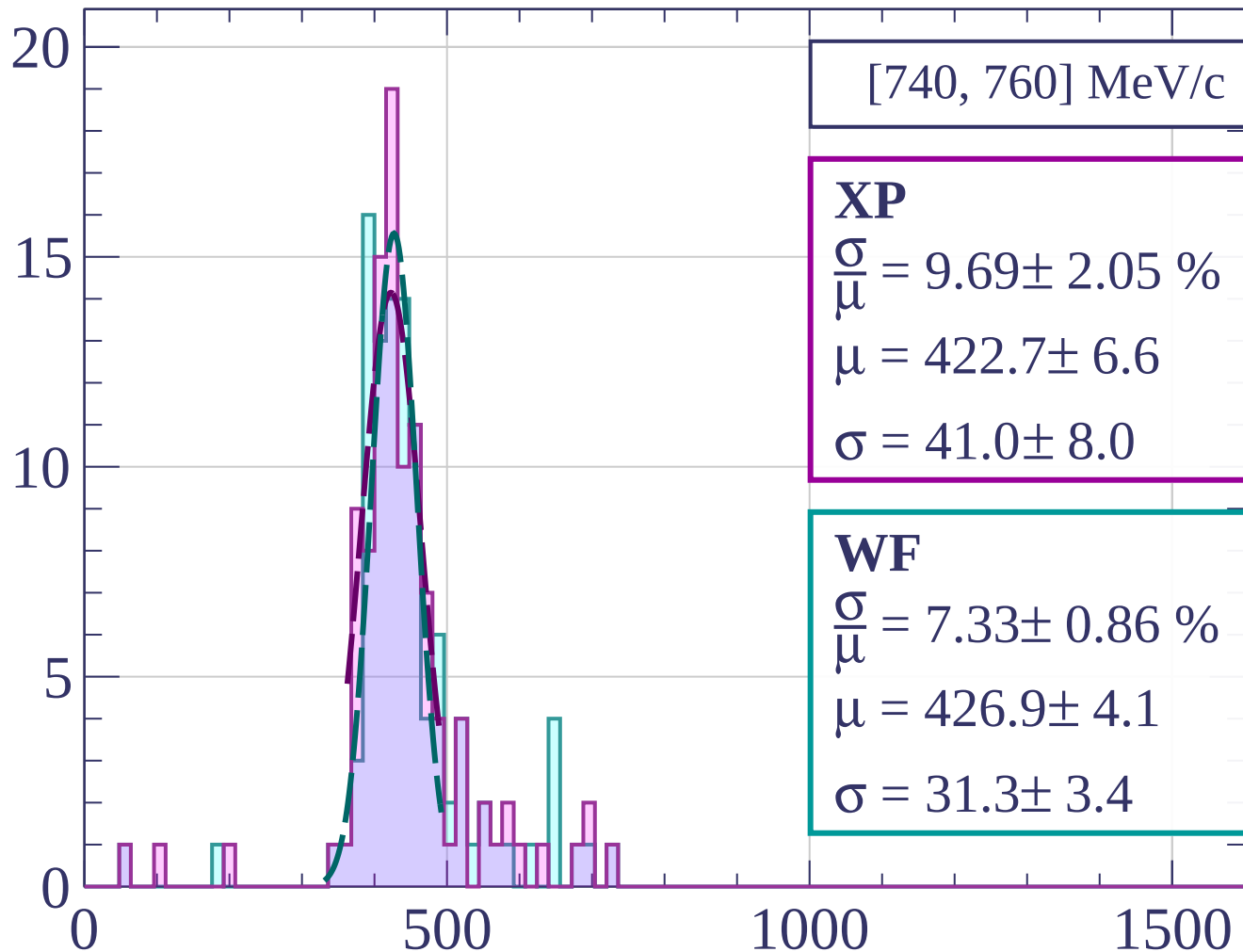
$$\frac{\sigma}{\mu} = 9.60 \pm 1.30 \%$$

$$\mu = 420.4 \pm 5.0$$

$$\sigma = 40.3 \pm 5.0$$

dE/dx [ADC counts/cm]

Count



[740, 760] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.69 \pm 2.05 \%$$

$$\mu = 422.7 \pm 6.6$$

$$\sigma = 41.0 \pm 8.0$$

WF

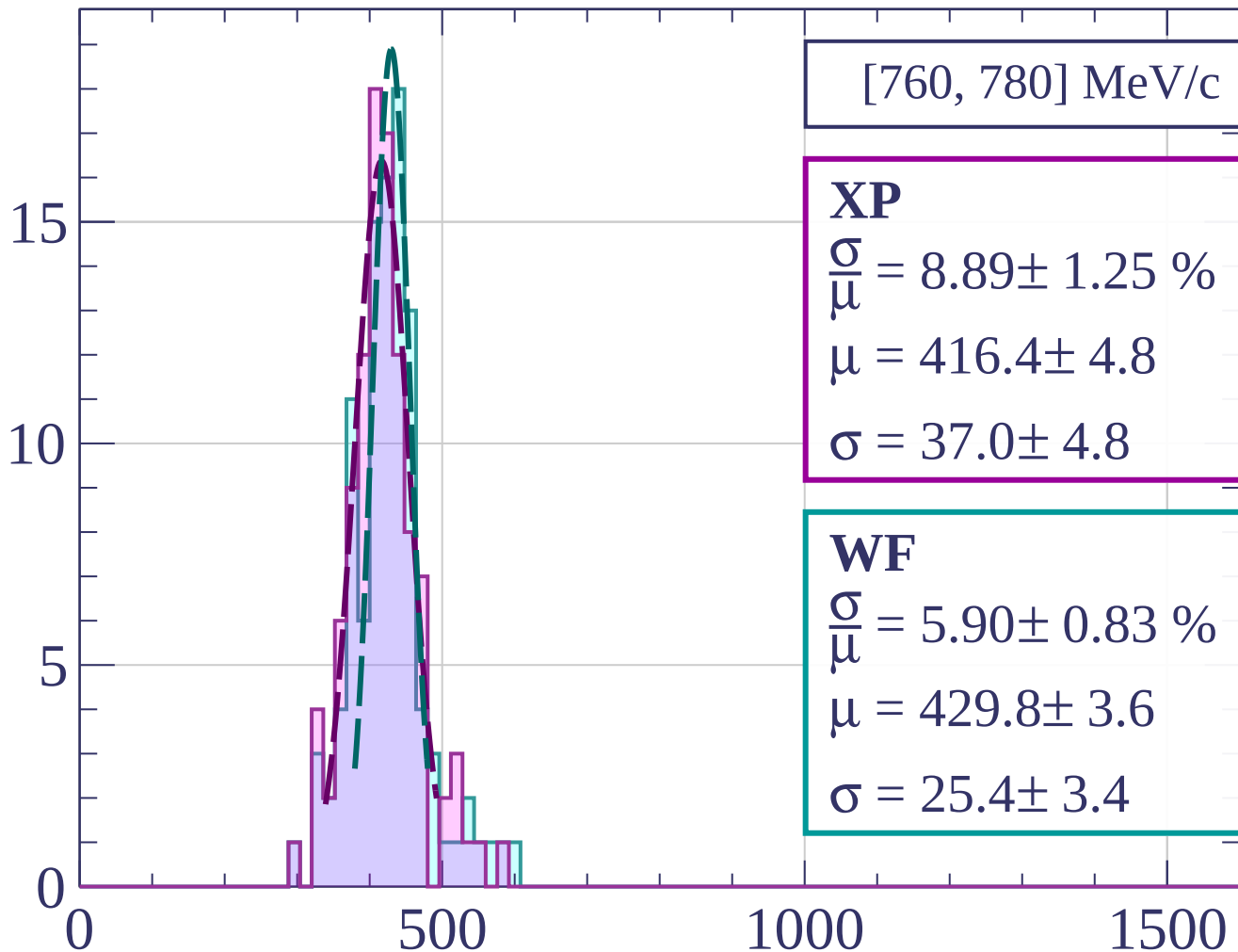
$$\frac{\sigma}{\mu} = 7.33 \pm 0.86 \%$$

$$\mu = 426.9 \pm 4.1$$

$$\sigma = 31.3 \pm 3.4$$

dE/dx [ADC counts/cm]

Count



[760, 780] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.89 \pm 1.25 \%$$

$$\mu = 416.4 \pm 4.8$$

$$\sigma = 37.0 \pm 4.8$$

WF

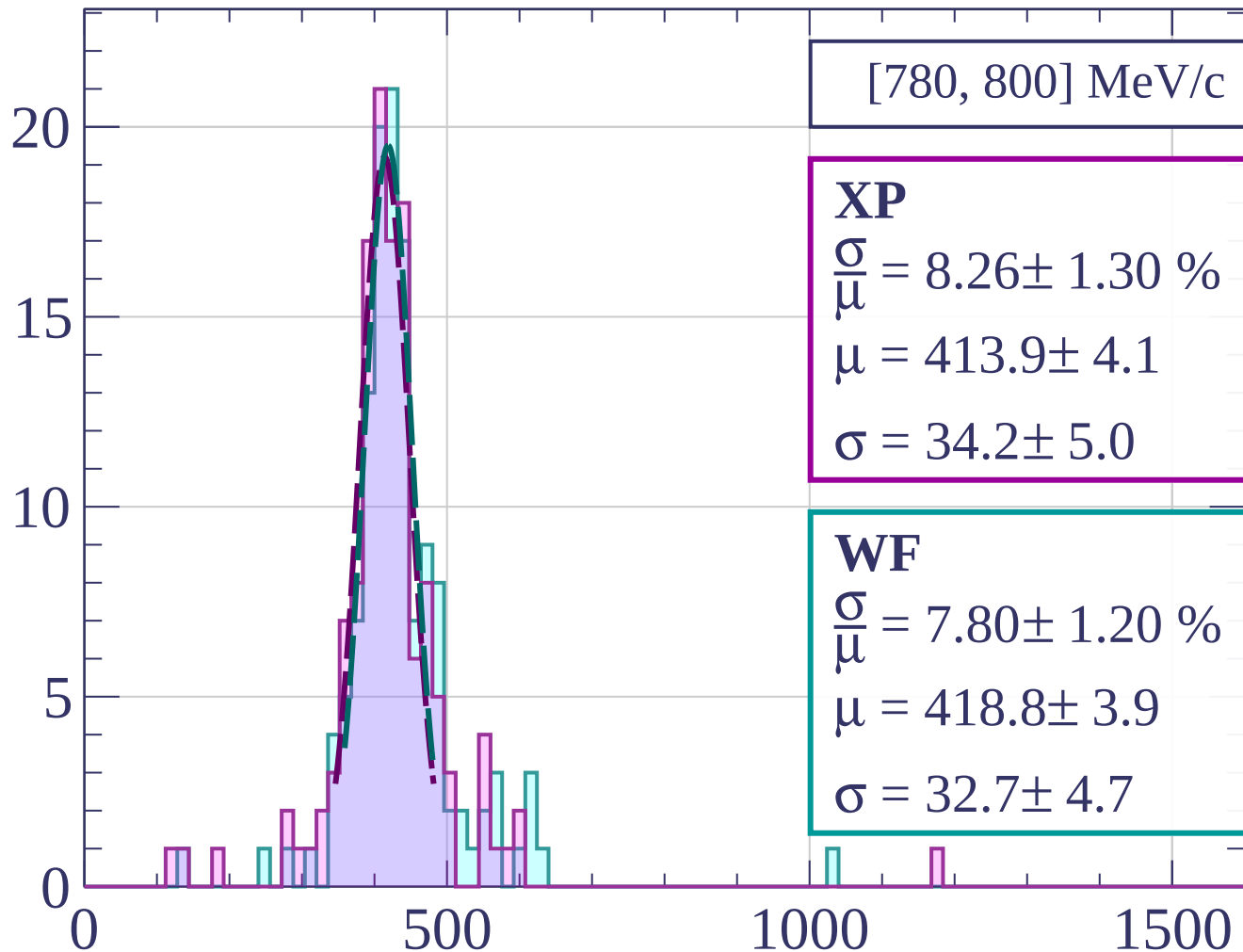
$$\frac{\sigma}{\mu} = 5.90 \pm 0.83 \%$$

$$\mu = 429.8 \pm 3.6$$

$$\sigma = 25.4 \pm 3.4$$

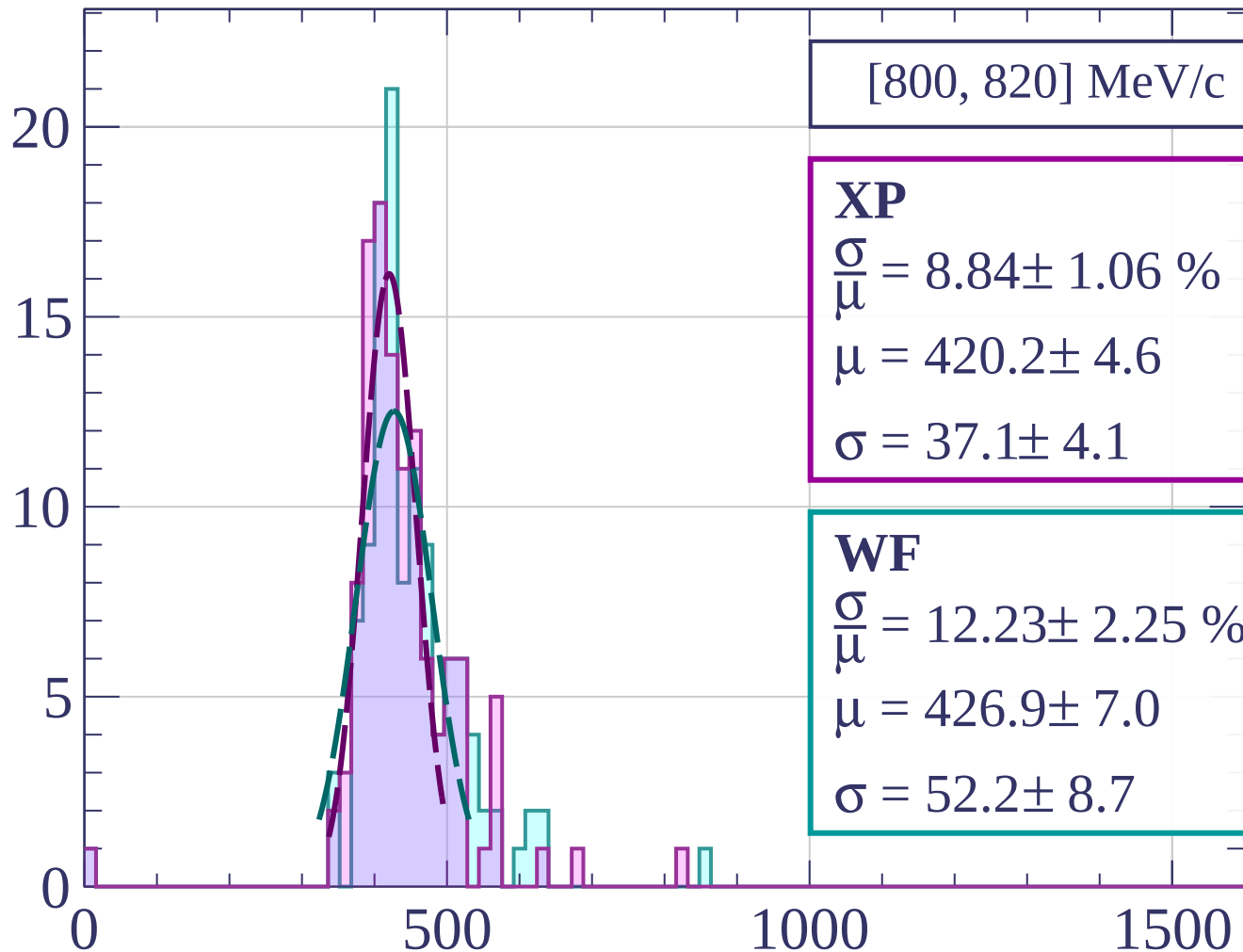
dE/dx [ADC counts/cm]

Count



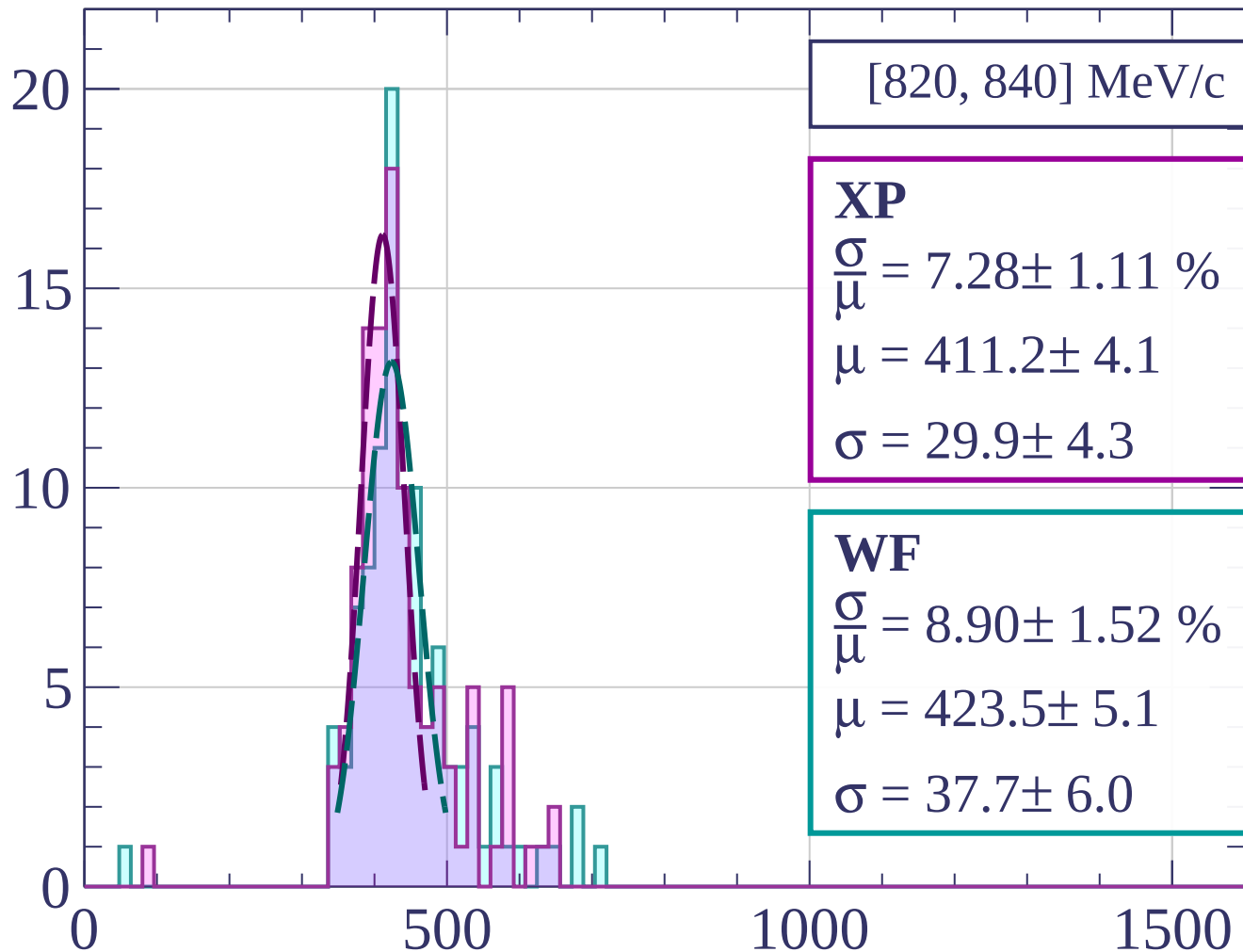
dE/dx [ADC counts/cm]

Count

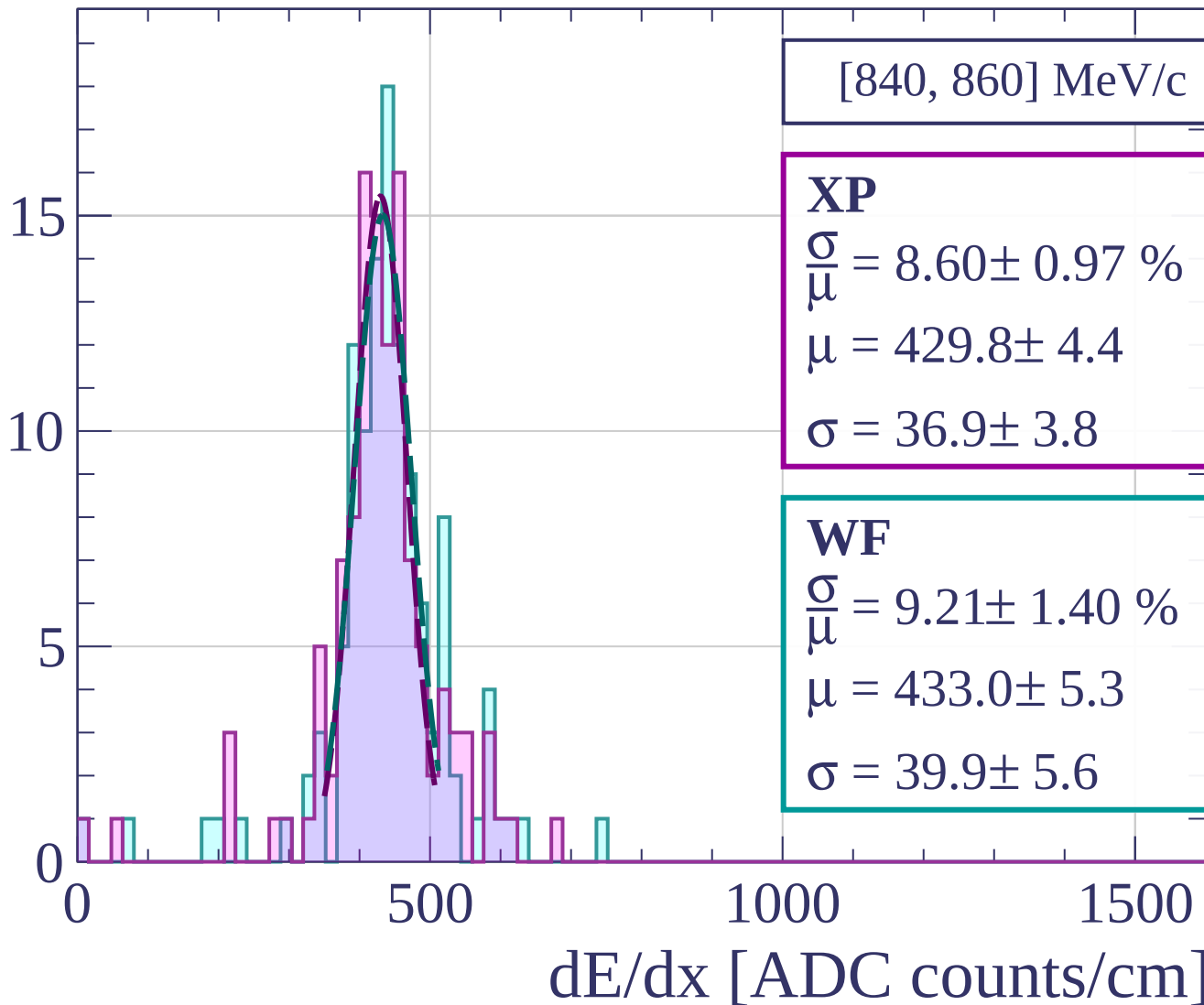


dE/dx [ADC counts/cm]

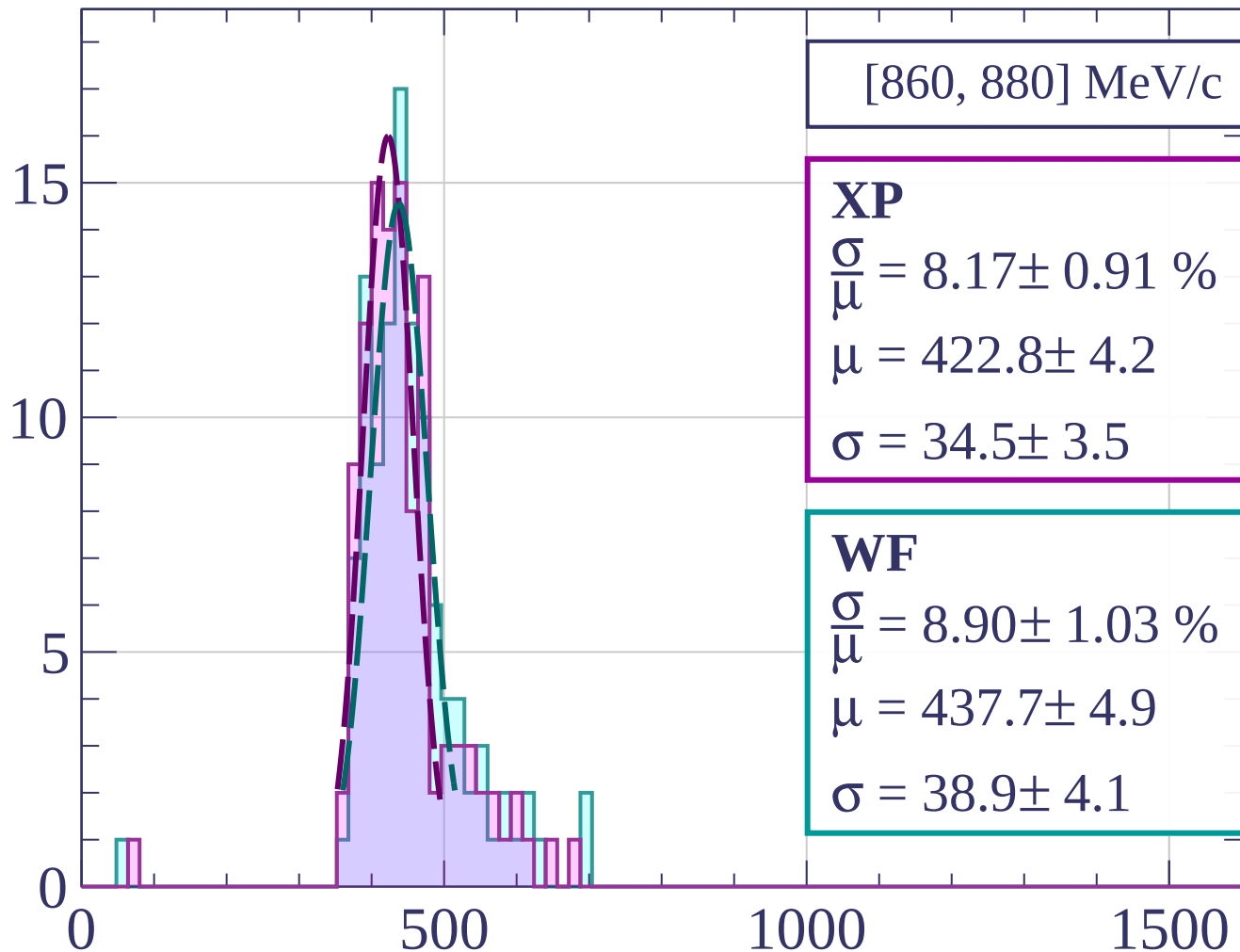
Count



Count



Count



[860, 880] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.17 \pm 0.91 \%$$

$$\mu = 422.8 \pm 4.2$$

$$\sigma = 34.5 \pm 3.5$$

WF

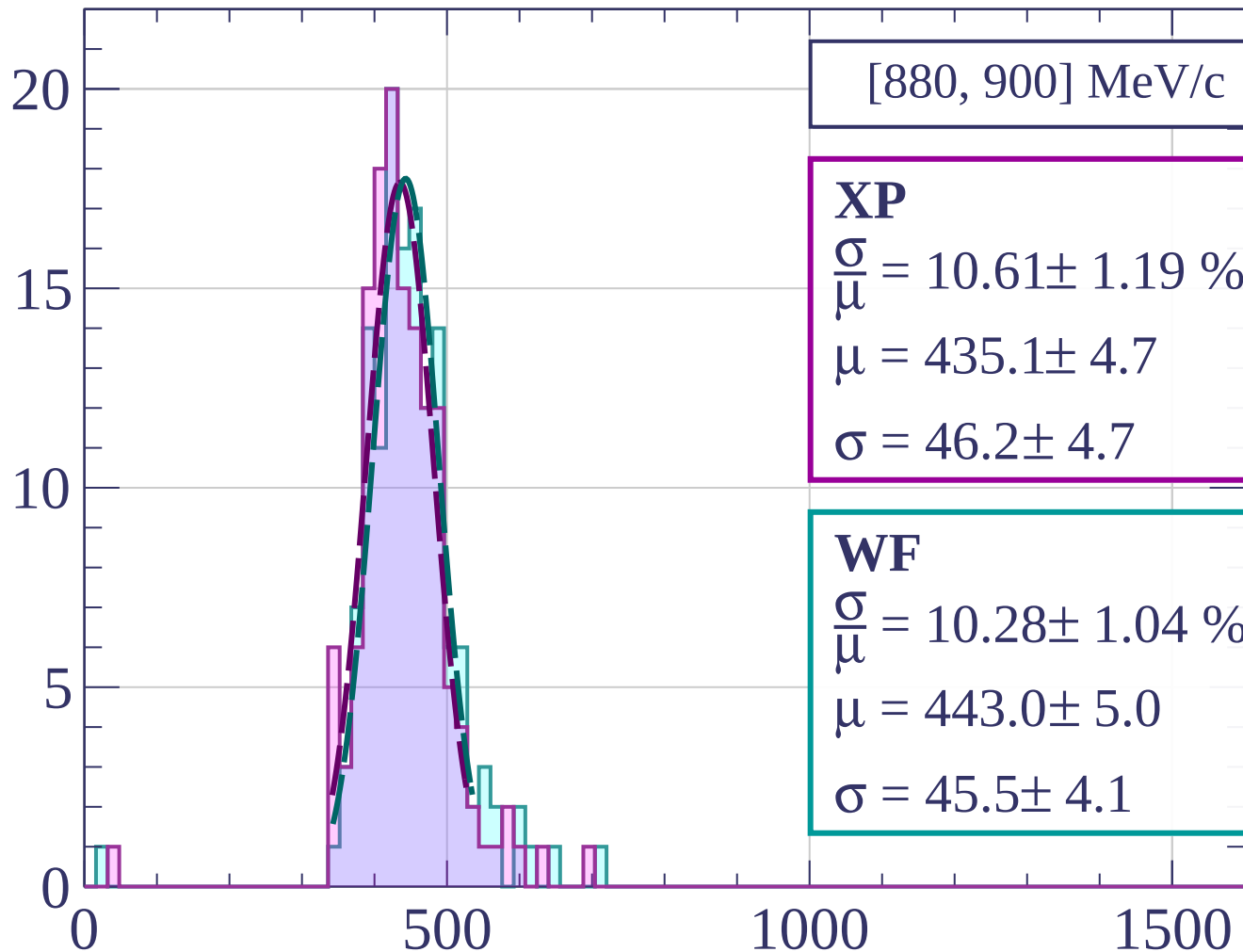
$$\frac{\sigma}{\mu} = 8.90 \pm 1.03 \%$$

$$\mu = 437.7 \pm 4.9$$

$$\sigma = 38.9 \pm 4.1$$

dE/dx [ADC counts/cm]

Count



[880, 900] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.61 \pm 1.19 \%$$

$$\mu = 435.1 \pm 4.7$$

$$\sigma = 46.2 \pm 4.7$$

WF

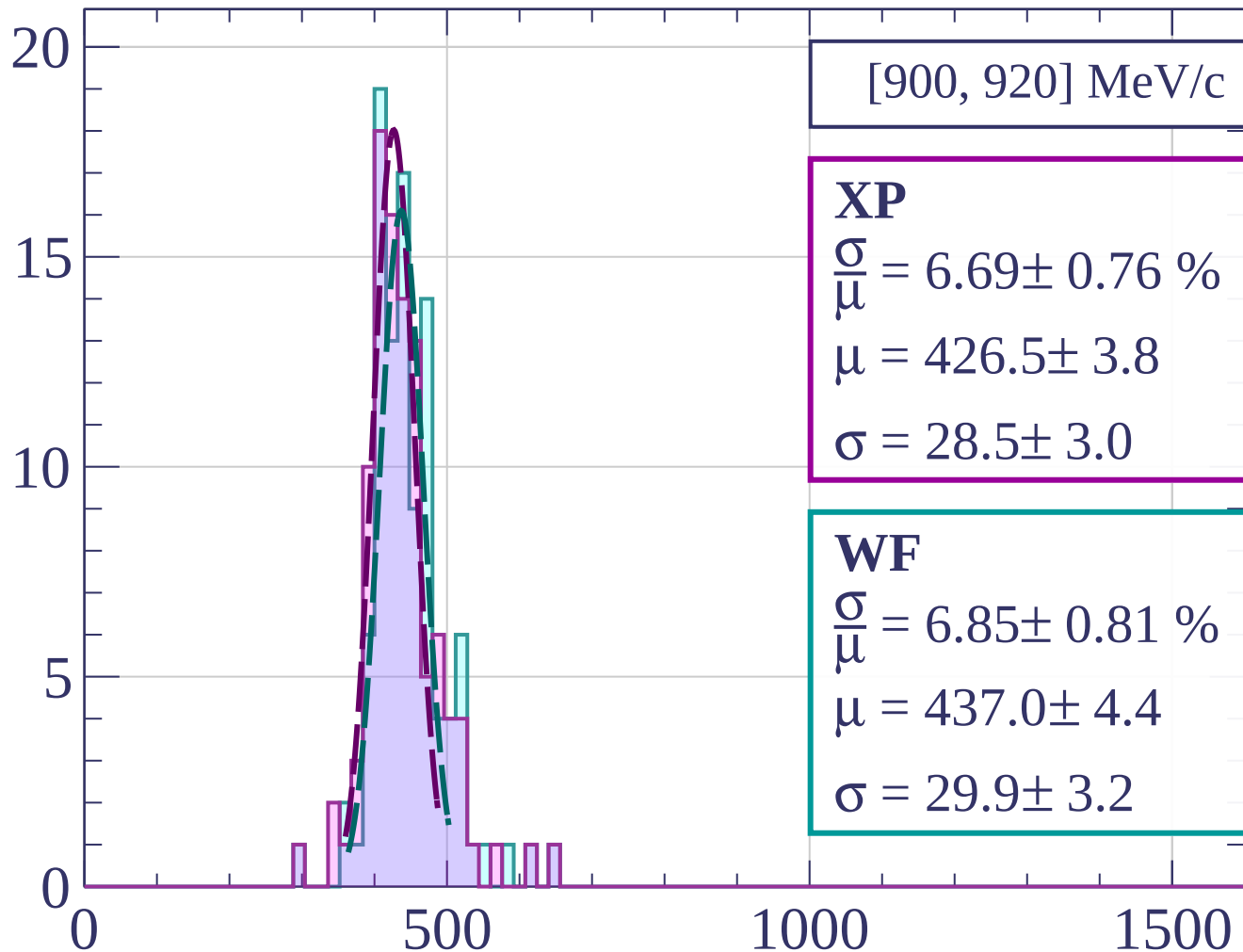
$$\frac{\sigma}{\mu} = 10.28 \pm 1.04 \%$$

$$\mu = 443.0 \pm 5.0$$

$$\sigma = 45.5 \pm 4.1$$

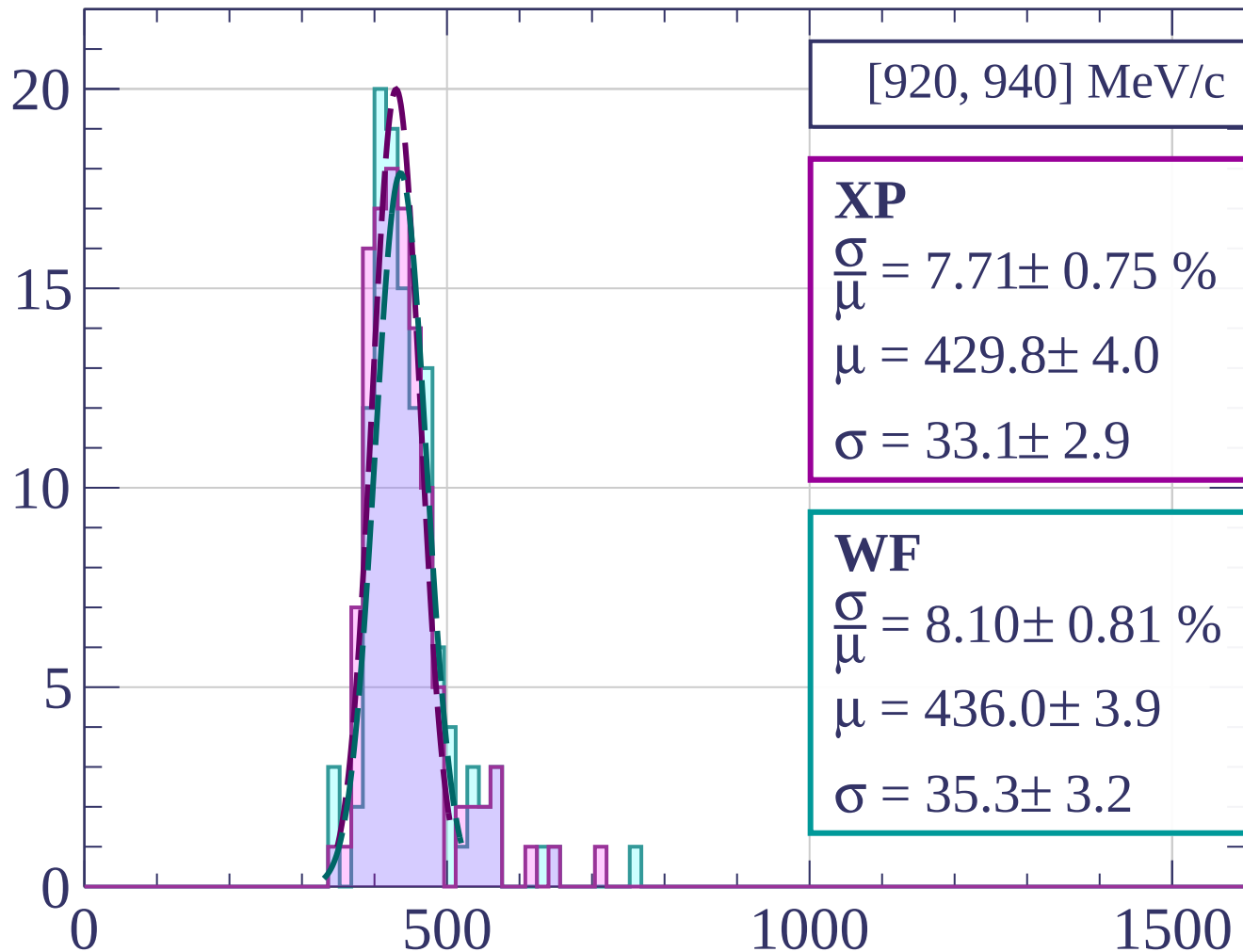
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[920, 940] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.71 \pm 0.75 \%$$

$$\mu = 429.8 \pm 4.0$$

$$\sigma = 33.1 \pm 2.9$$

WF

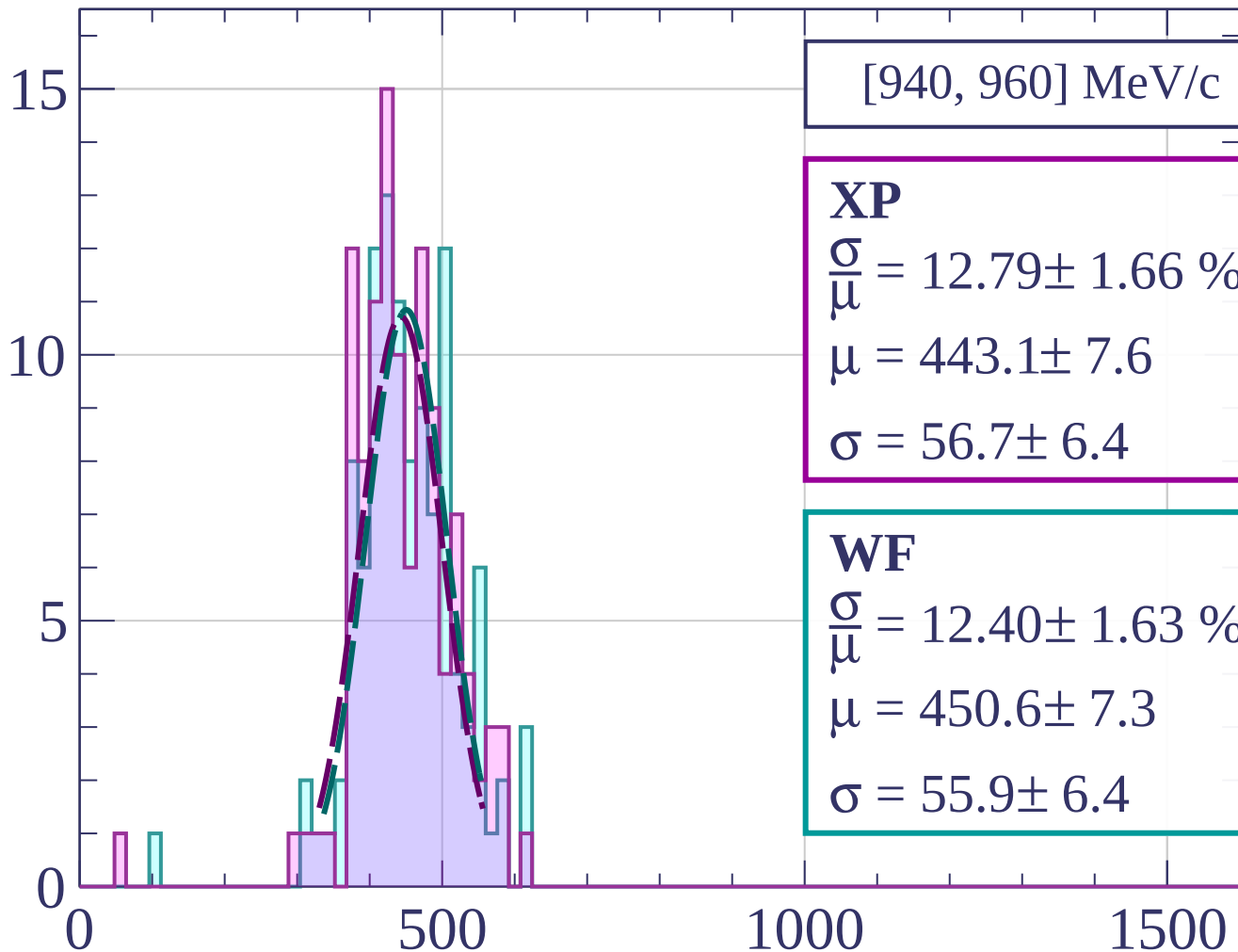
$$\frac{\sigma}{\mu} = 8.10 \pm 0.81 \%$$

$$\mu = 436.0 \pm 3.9$$

$$\sigma = 35.3 \pm 3.2$$

dE/dx [ADC counts/cm]

Count



[940, 960] MeV/c

XP

$$\frac{\sigma}{\mu} = 12.79 \pm 1.66 \%$$

$$\mu = 443.1 \pm 7.6$$

$$\sigma = 56.7 \pm 6.4$$

WF

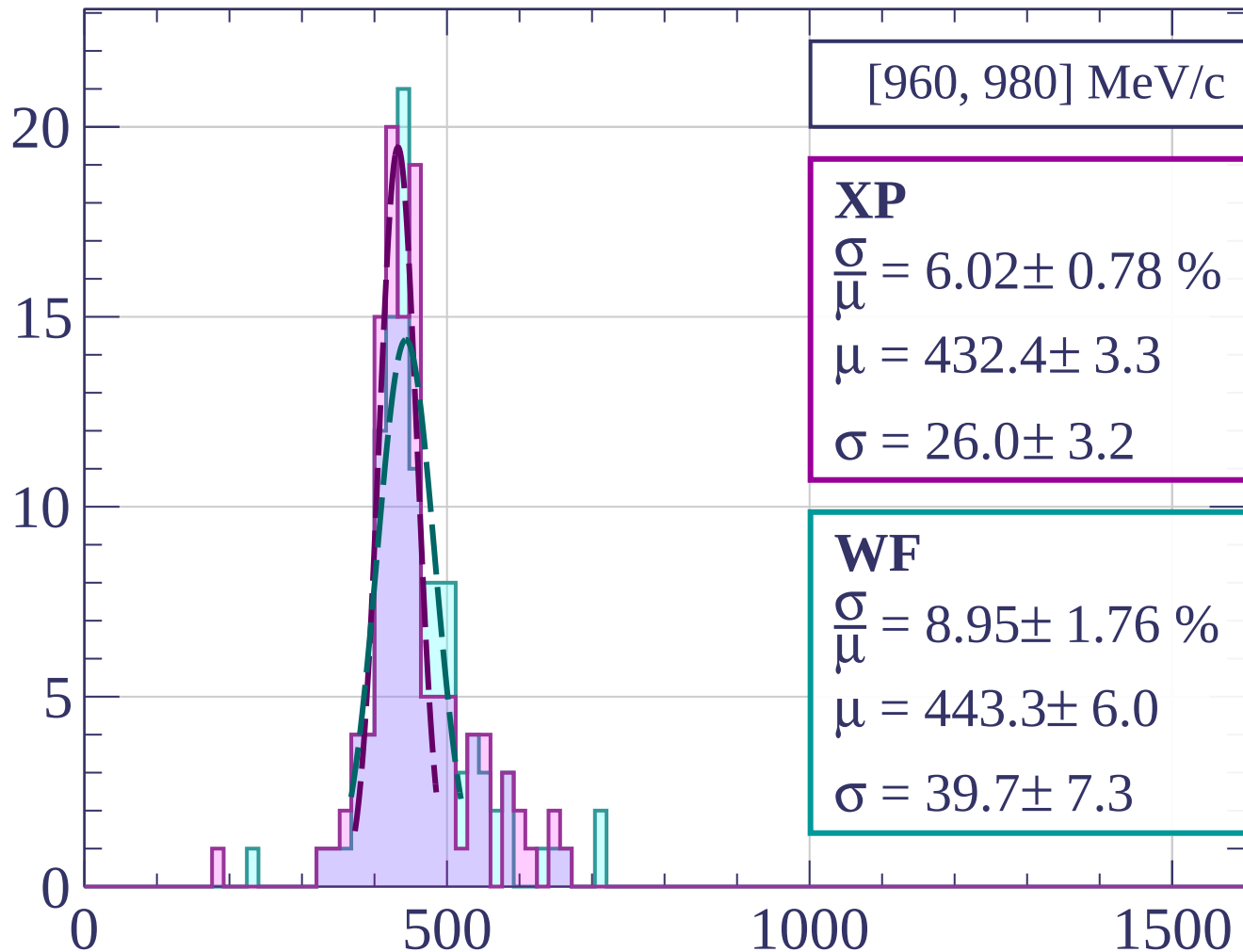
$$\frac{\sigma}{\mu} = 12.40 \pm 1.63 \%$$

$$\mu = 450.6 \pm 7.3$$

$$\sigma = 55.9 \pm 6.4$$

dE/dx [ADC counts/cm]

Count



[960, 980] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.02 \pm 0.78 \%$$

$$\mu = 432.4 \pm 3.3$$

$$\sigma = 26.0 \pm 3.2$$

WF

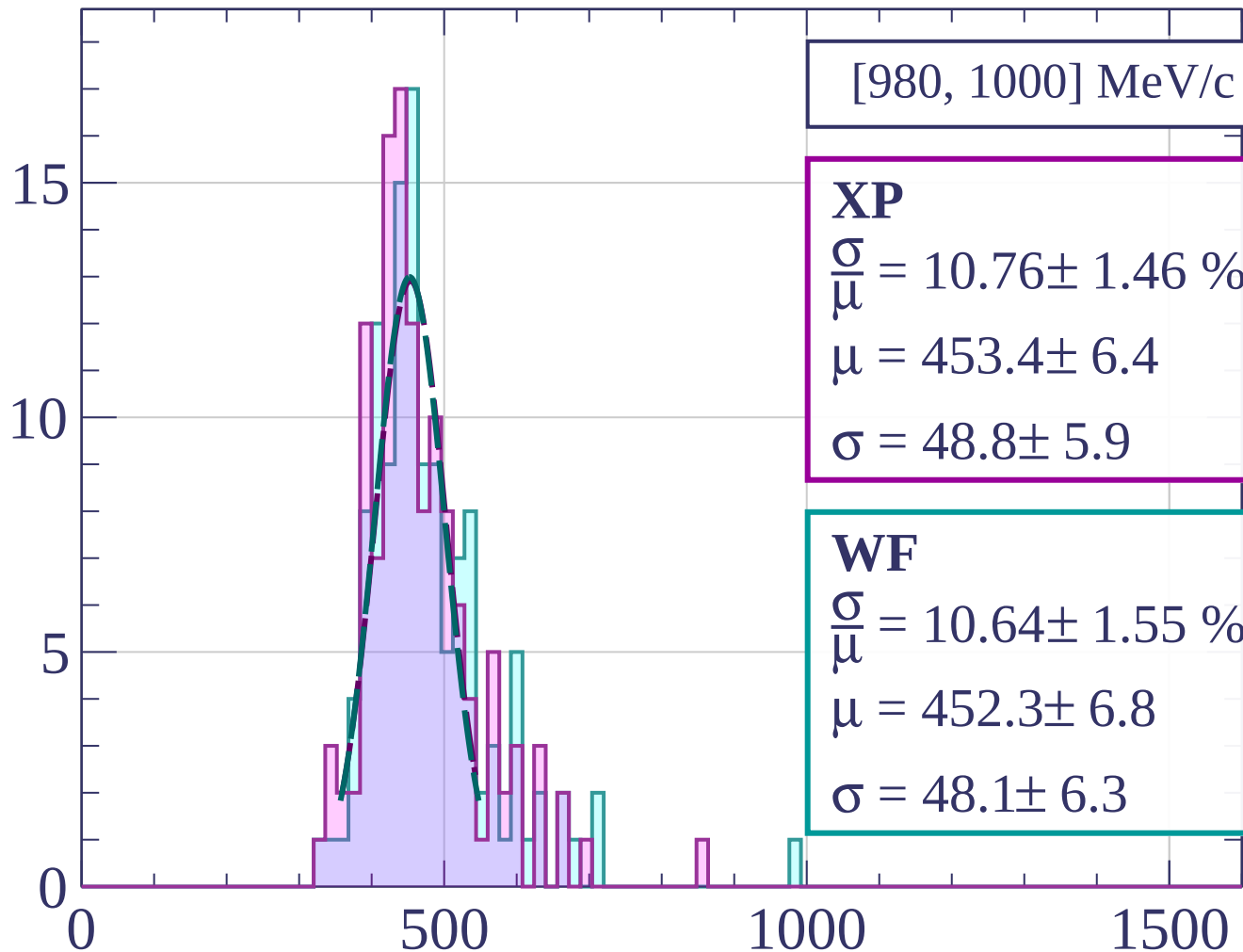
$$\frac{\sigma}{\mu} = 8.95 \pm 1.76 \%$$

$$\mu = 443.3 \pm 6.0$$

$$\sigma = 39.7 \pm 7.3$$

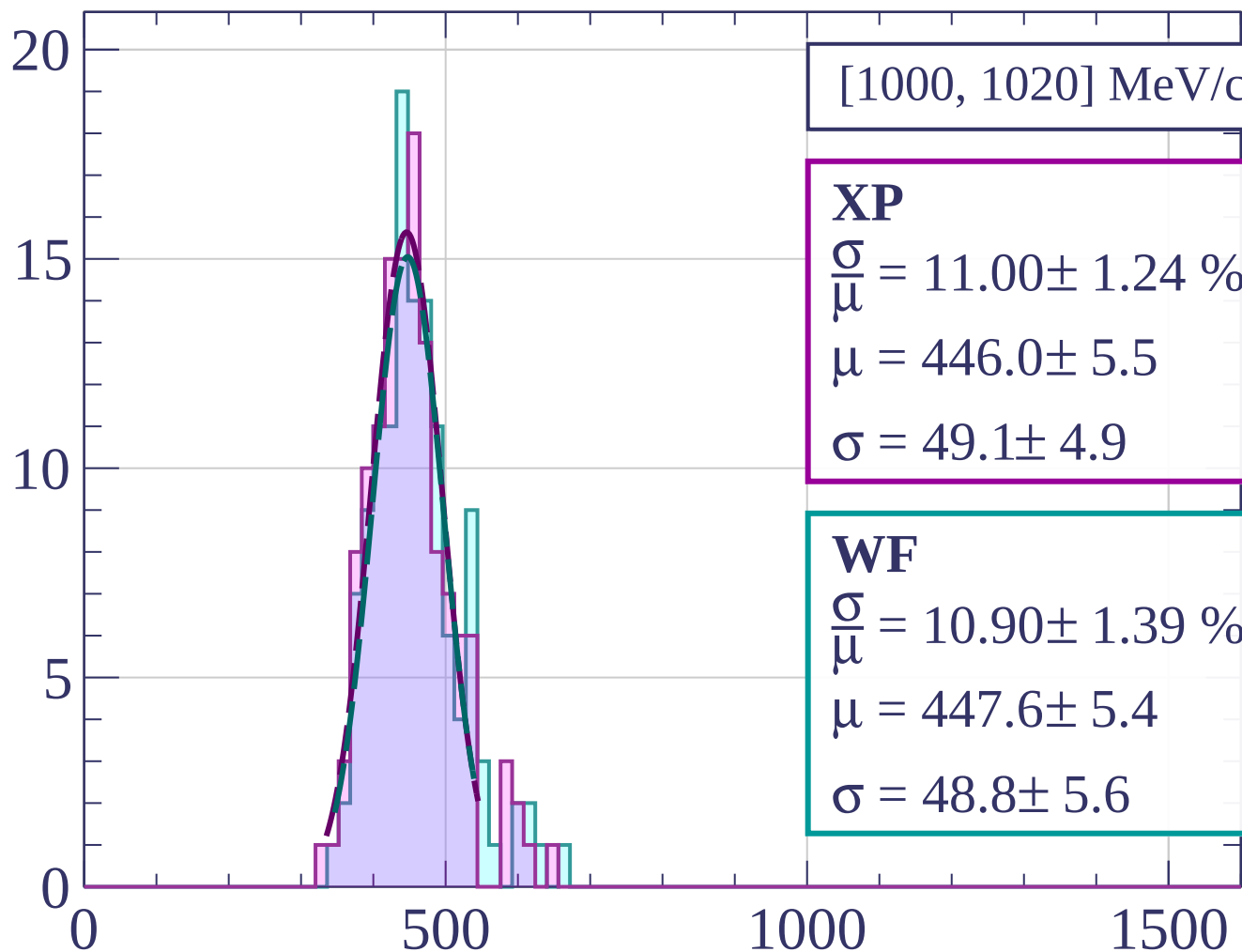
dE/dx [ADC counts/cm]

Count



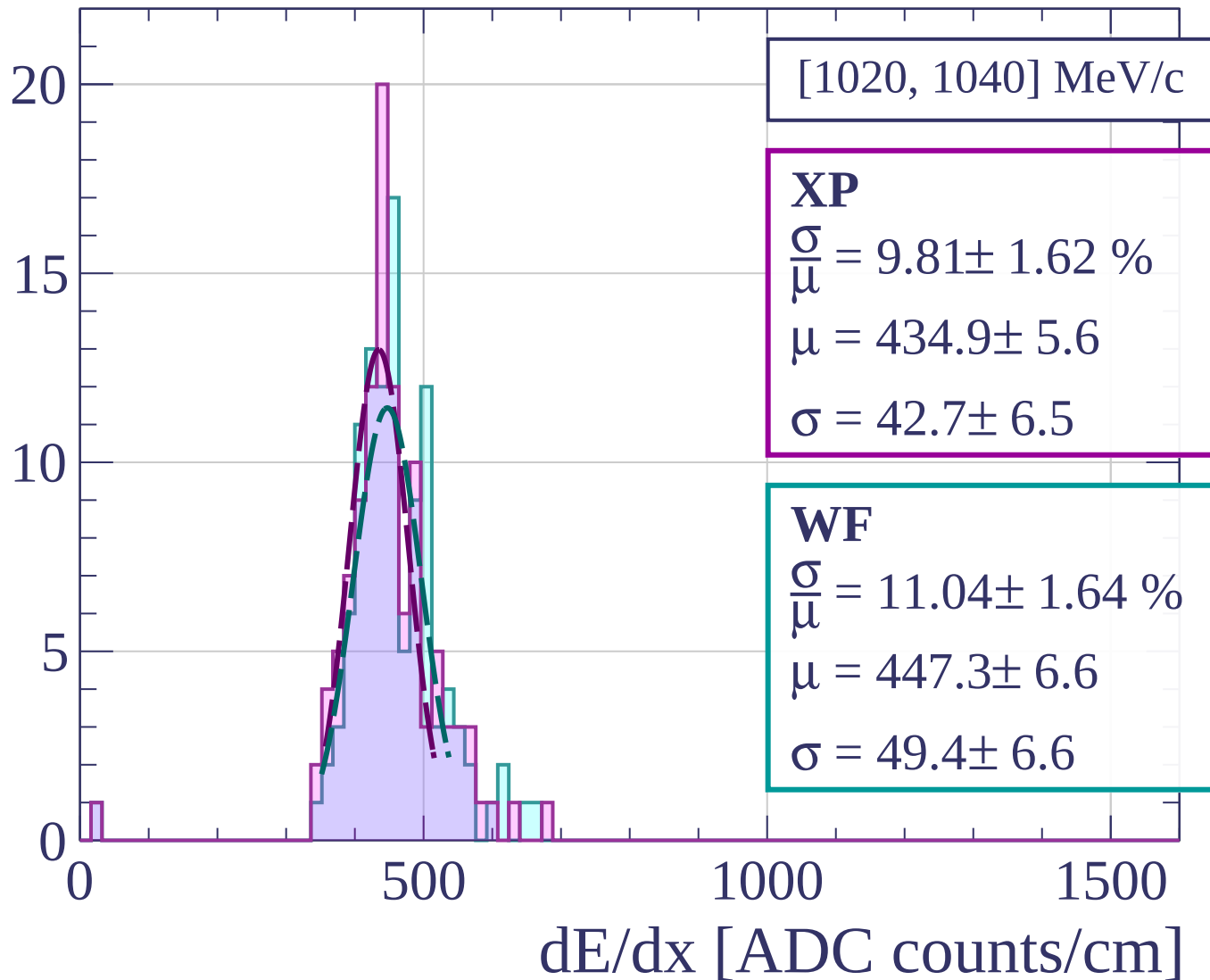
dE/dx [ADC counts/cm]

Count

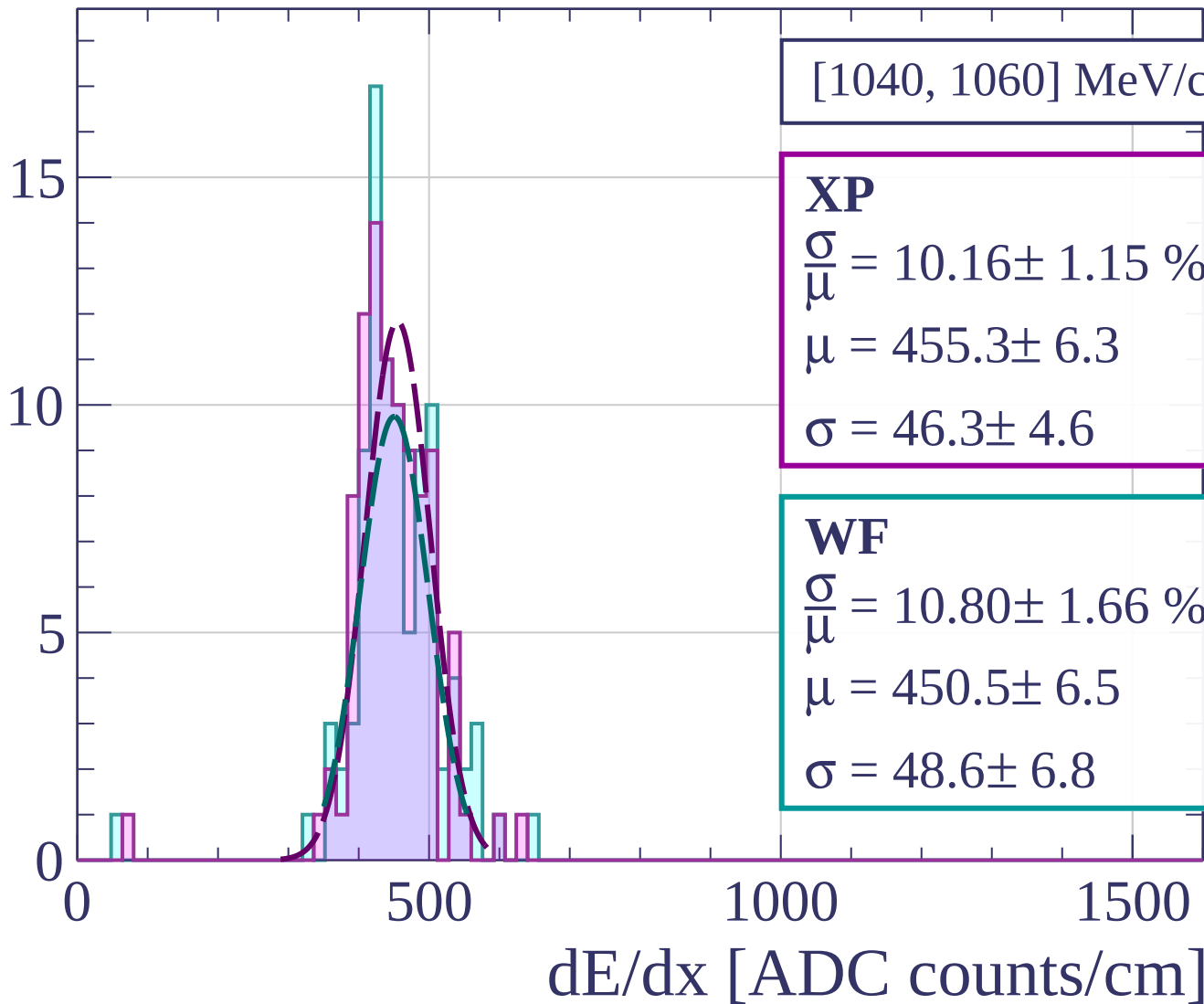


dE/dx [ADC counts/cm]

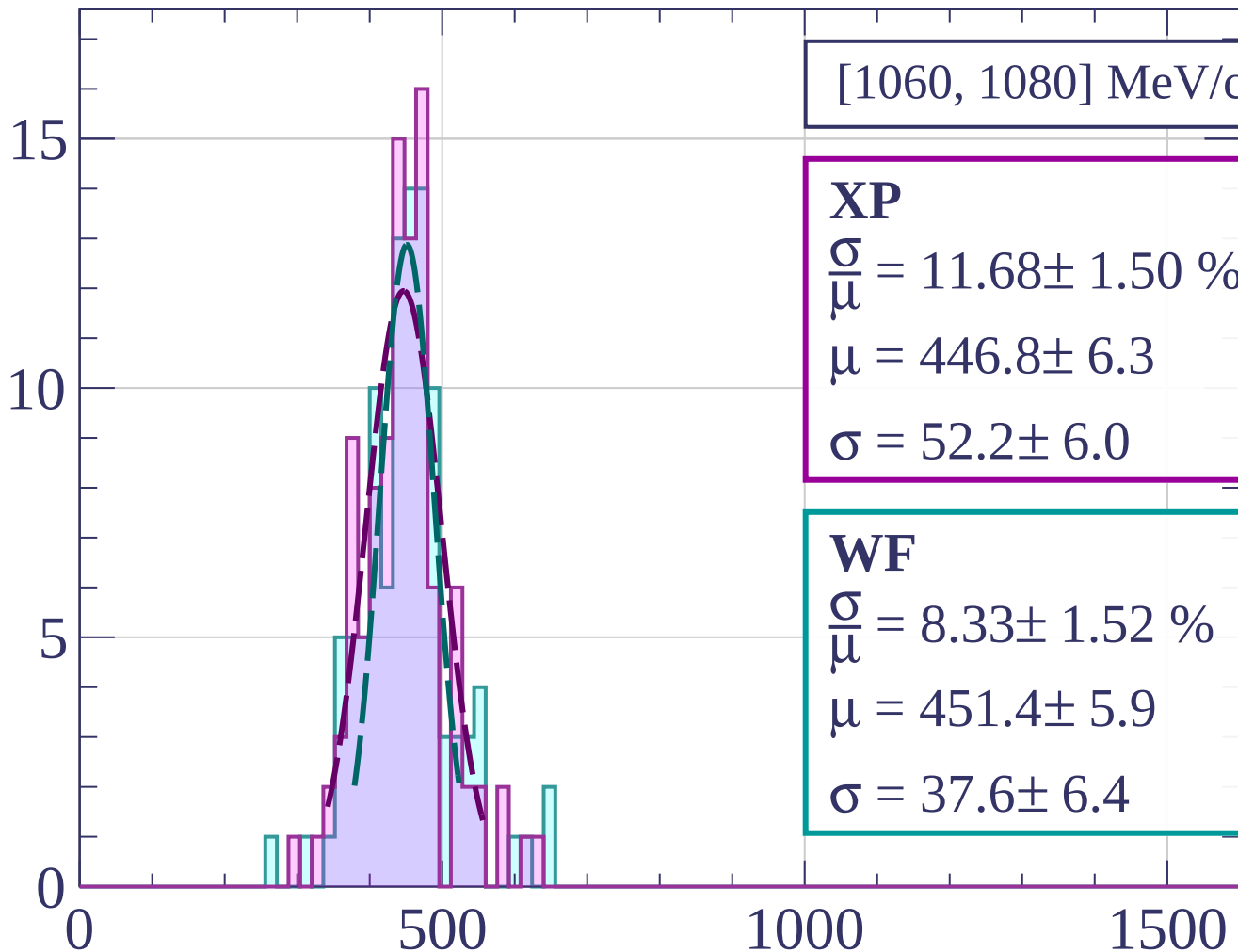
Count



Count



Count



[1060, 1080] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.68 \pm 1.50 \%$$

$$\mu = 446.8 \pm 6.3$$

$$\sigma = 52.2 \pm 6.0$$

WF

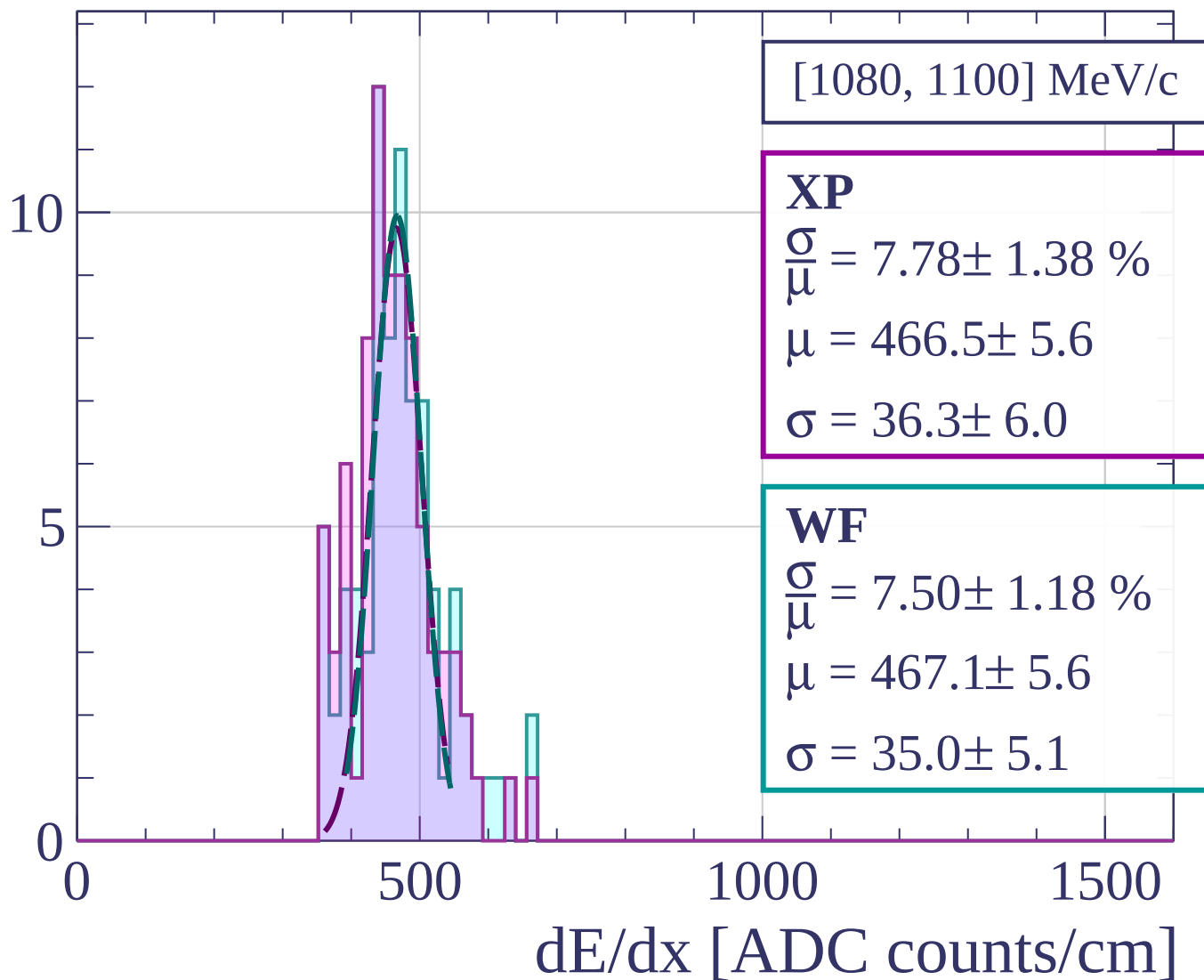
$$\frac{\sigma}{\mu} = 8.33 \pm 1.52 \%$$

$$\mu = 451.4 \pm 5.9$$

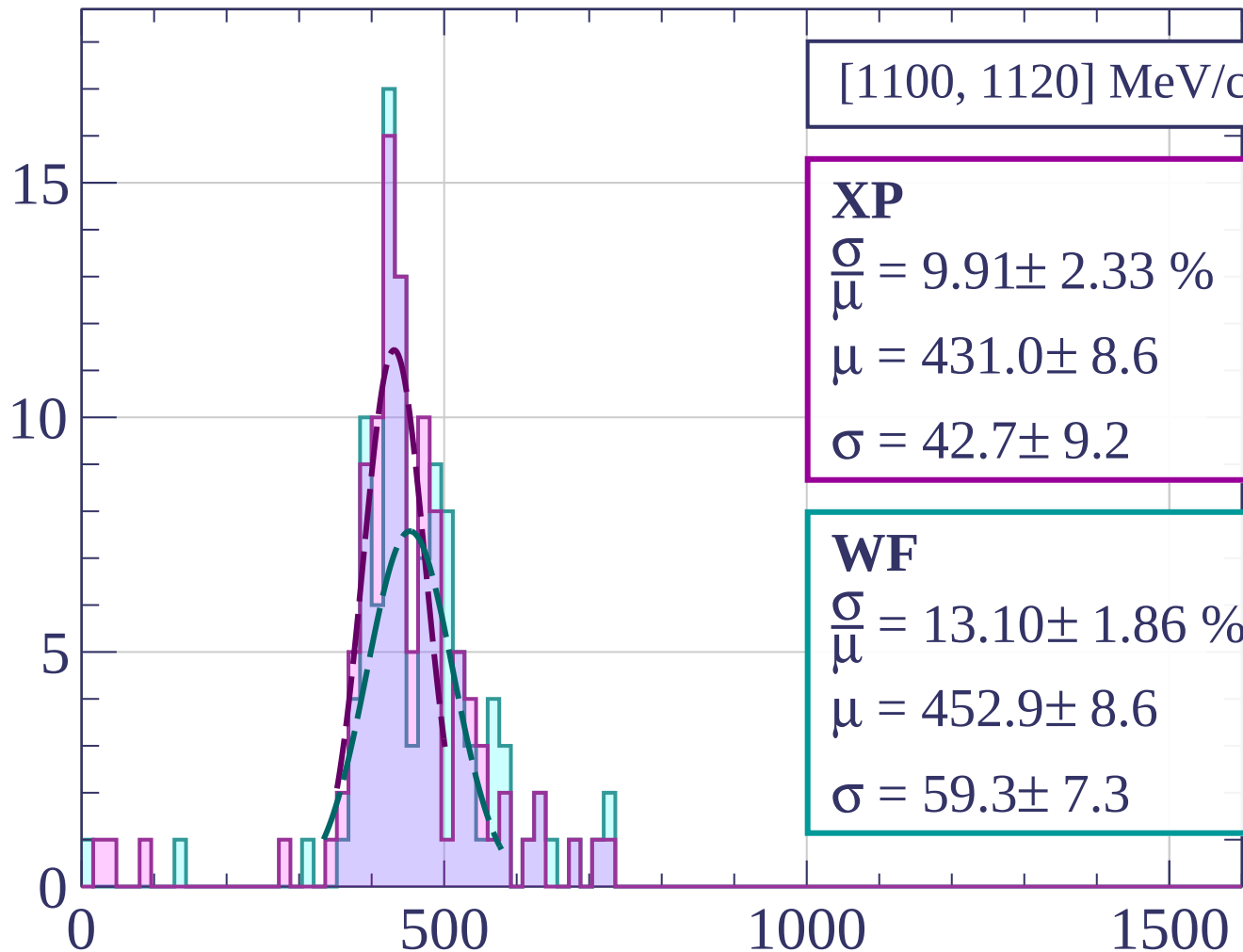
$$\sigma = 37.6 \pm 6.4$$

dE/dx [ADC counts/cm]

Count

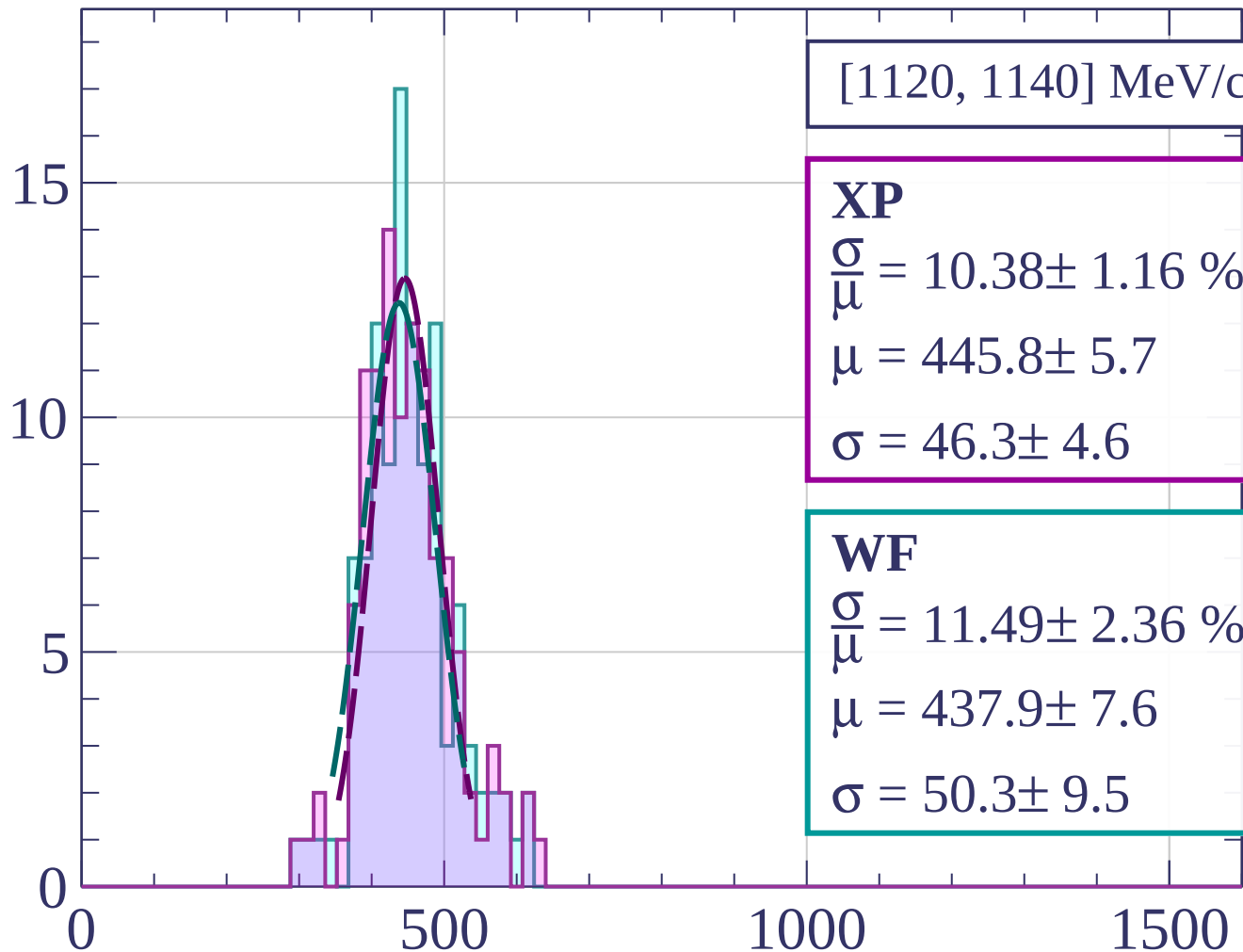


Count



dE/dx [ADC counts/cm]

Count



[1120, 1140] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.38 \pm 1.16 \%$$

$$\mu = 445.8 \pm 5.7$$

$$\sigma = 46.3 \pm 4.6$$

WF

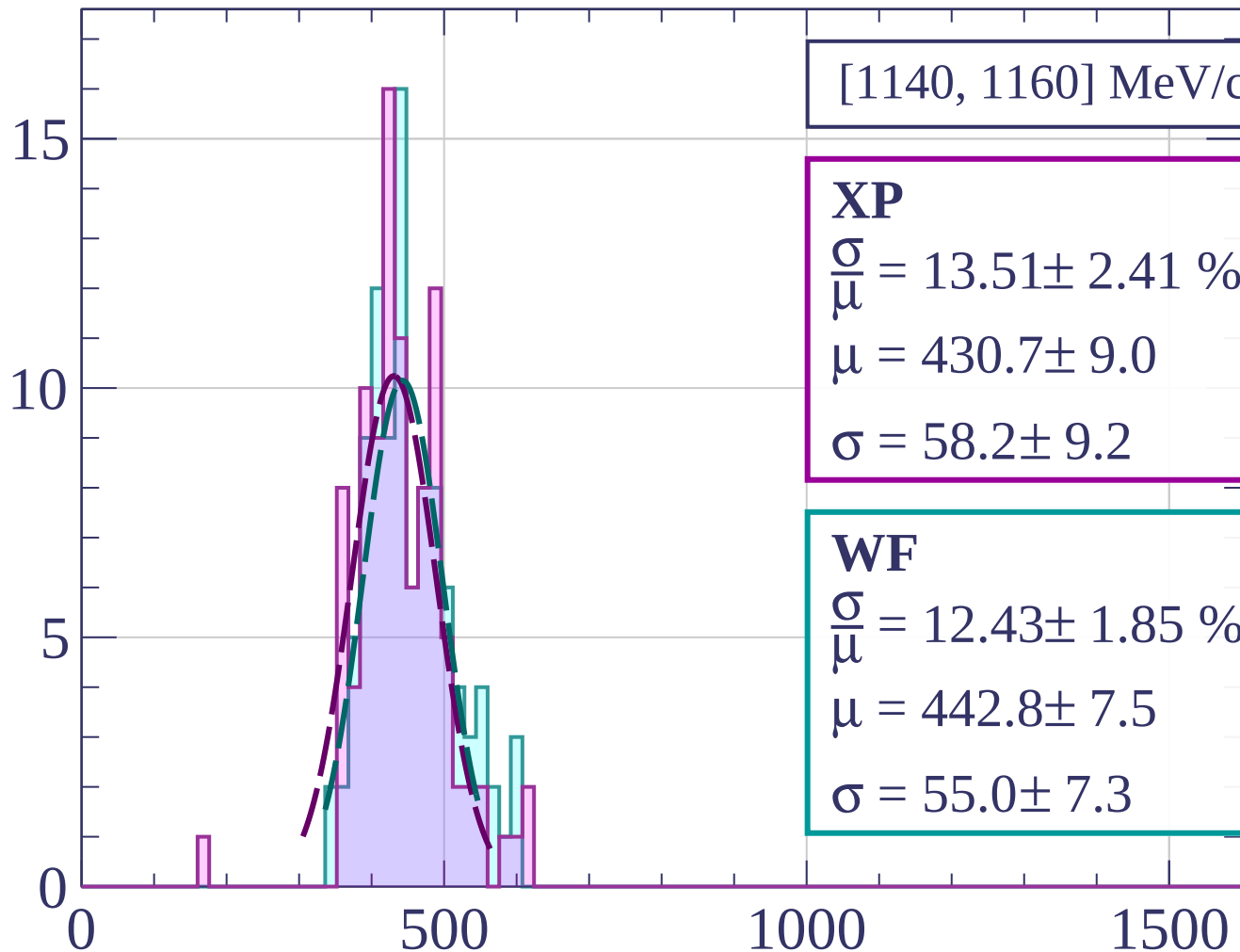
$$\frac{\sigma}{\mu} = 11.49 \pm 2.36 \%$$

$$\mu = 437.9 \pm 7.6$$

$$\sigma = 50.3 \pm 9.5$$

dE/dx [ADC counts/cm]

Count



[1140, 1160] MeV/c

XP

$$\frac{\sigma}{\mu} = 13.51 \pm 2.41 \%$$

$$\mu = 430.7 \pm 9.0$$

$$\sigma = 58.2 \pm 9.2$$

WF

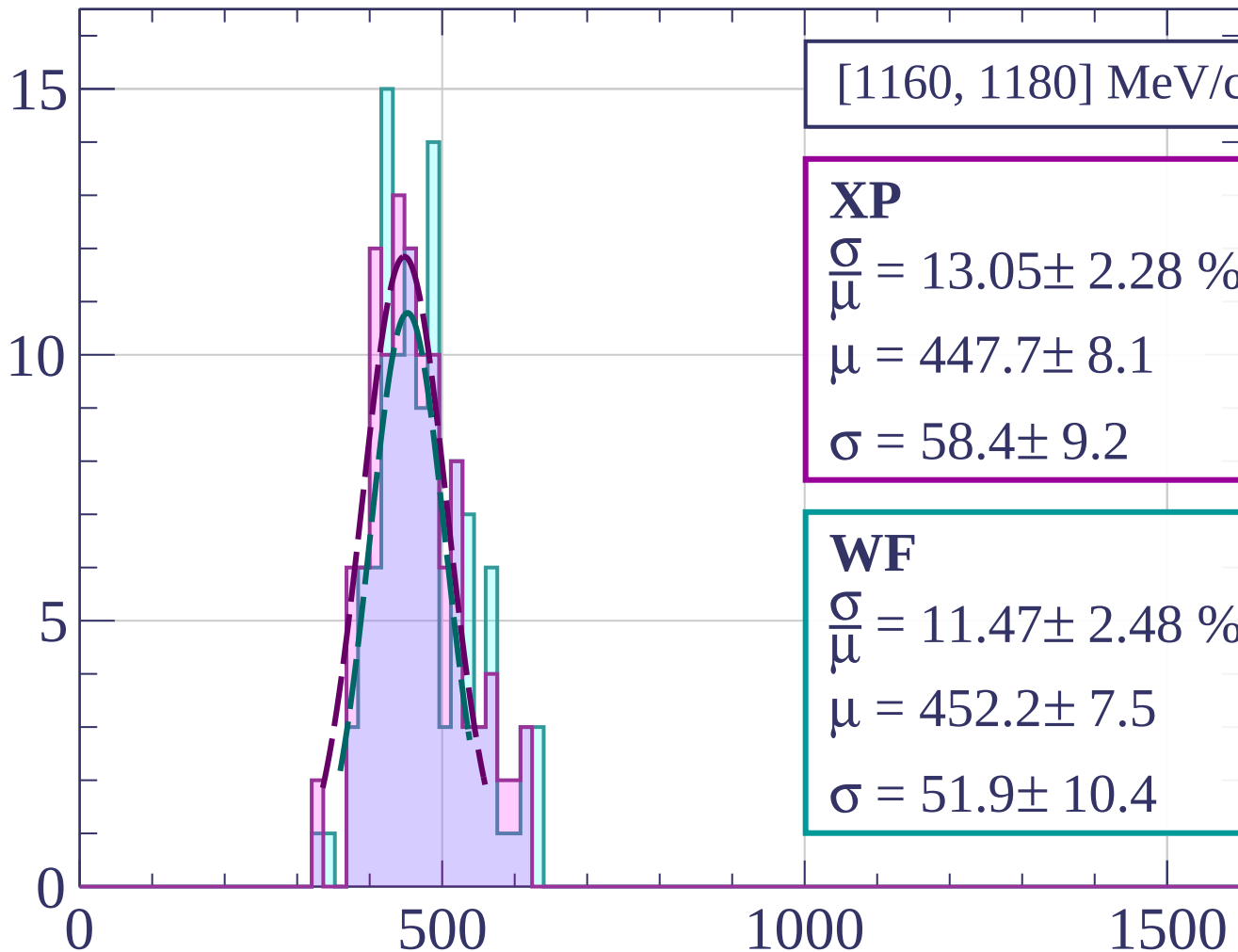
$$\frac{\sigma}{\mu} = 12.43 \pm 1.85 \%$$

$$\mu = 442.8 \pm 7.5$$

$$\sigma = 55.0 \pm 7.3$$

dE/dx [ADC counts/cm]

Count



[1160, 1180] MeV/c

XP

$$\frac{\sigma}{\mu} = 13.05 \pm 2.28 \%$$

$$\mu = 447.7 \pm 8.1$$

$$\sigma = 58.4 \pm 9.2$$

WF

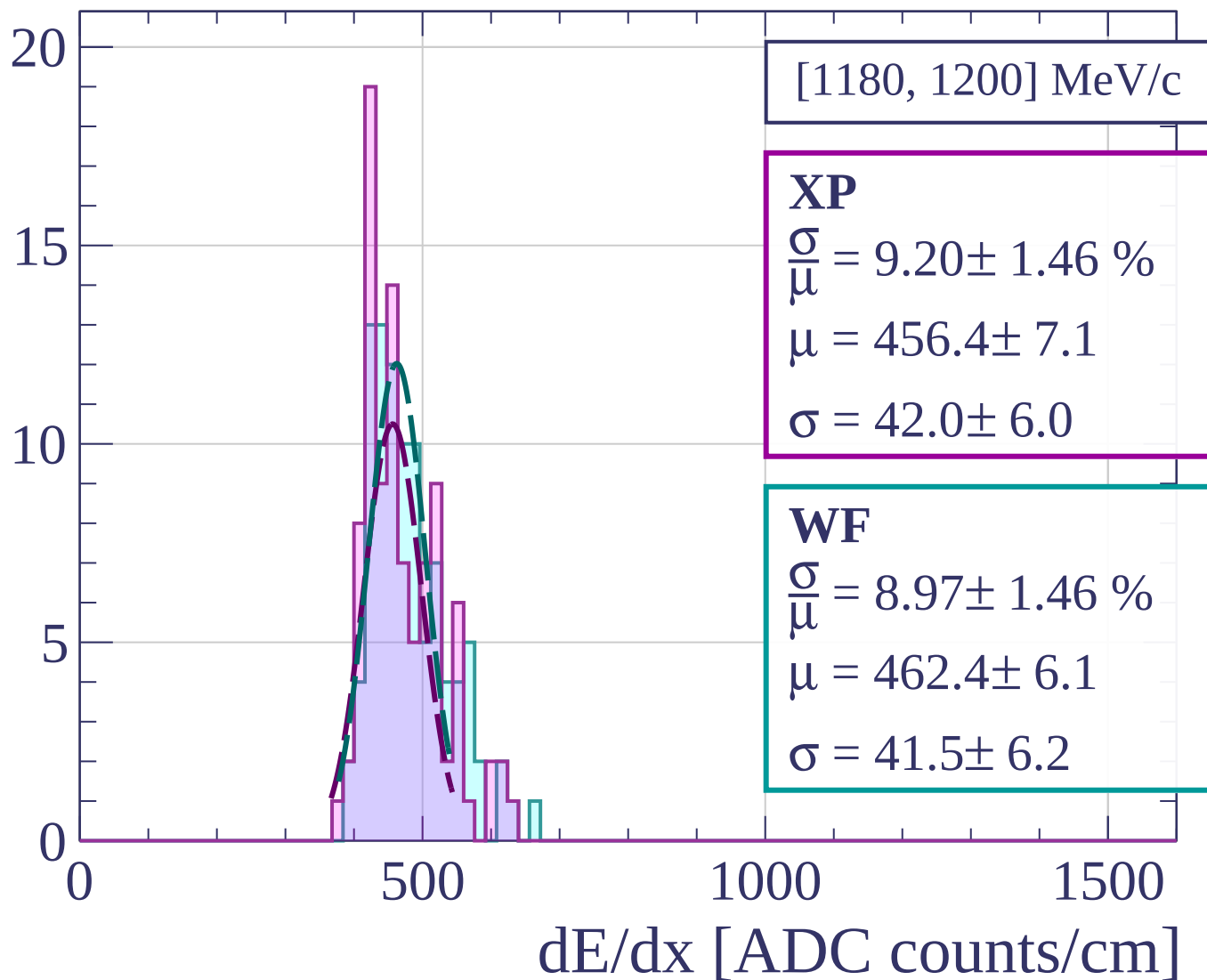
$$\frac{\sigma}{\mu} = 11.47 \pm 2.48 \%$$

$$\mu = 452.2 \pm 7.5$$

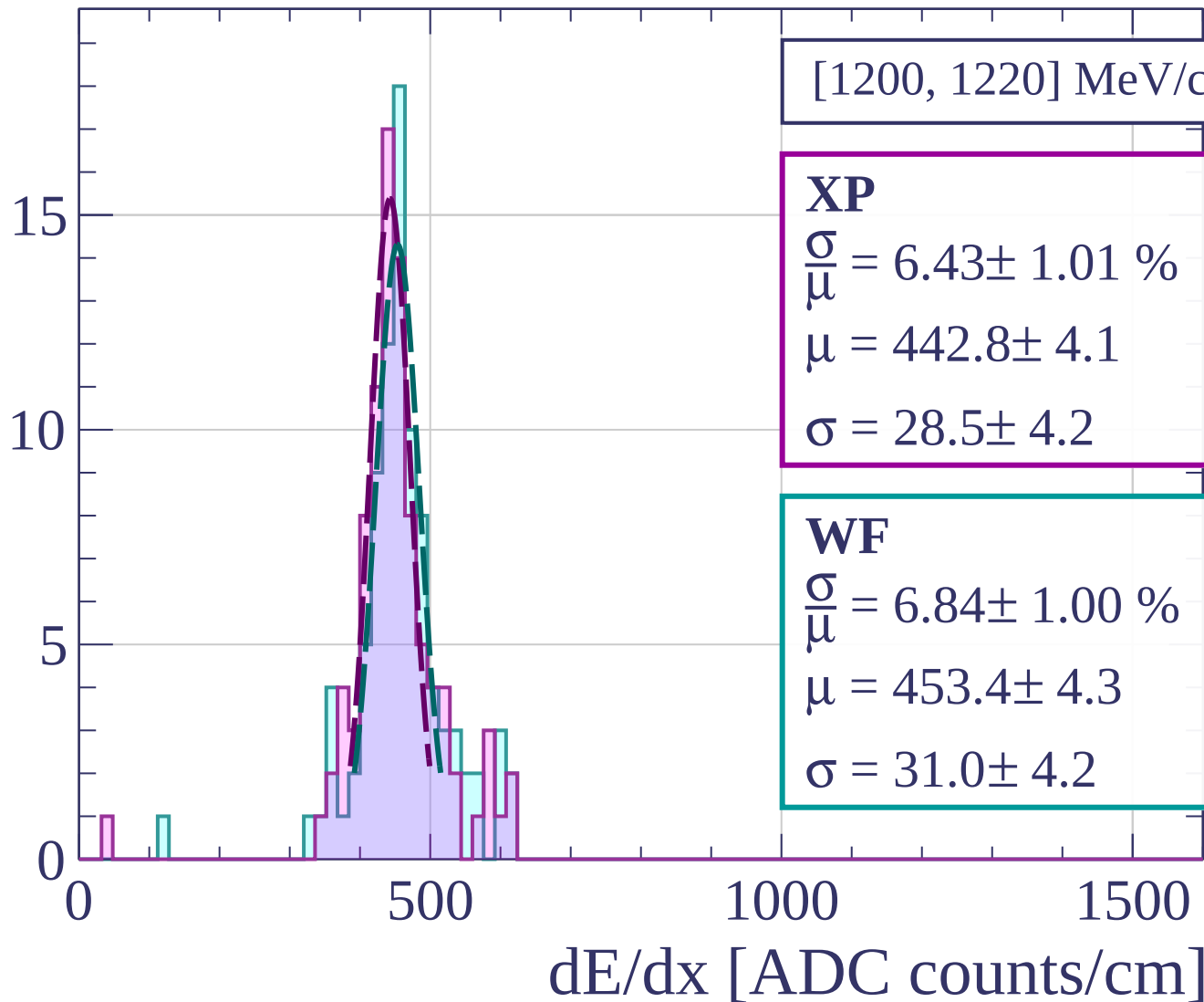
$$\sigma = 51.9 \pm 10.4$$

dE/dx [ADC counts/cm]

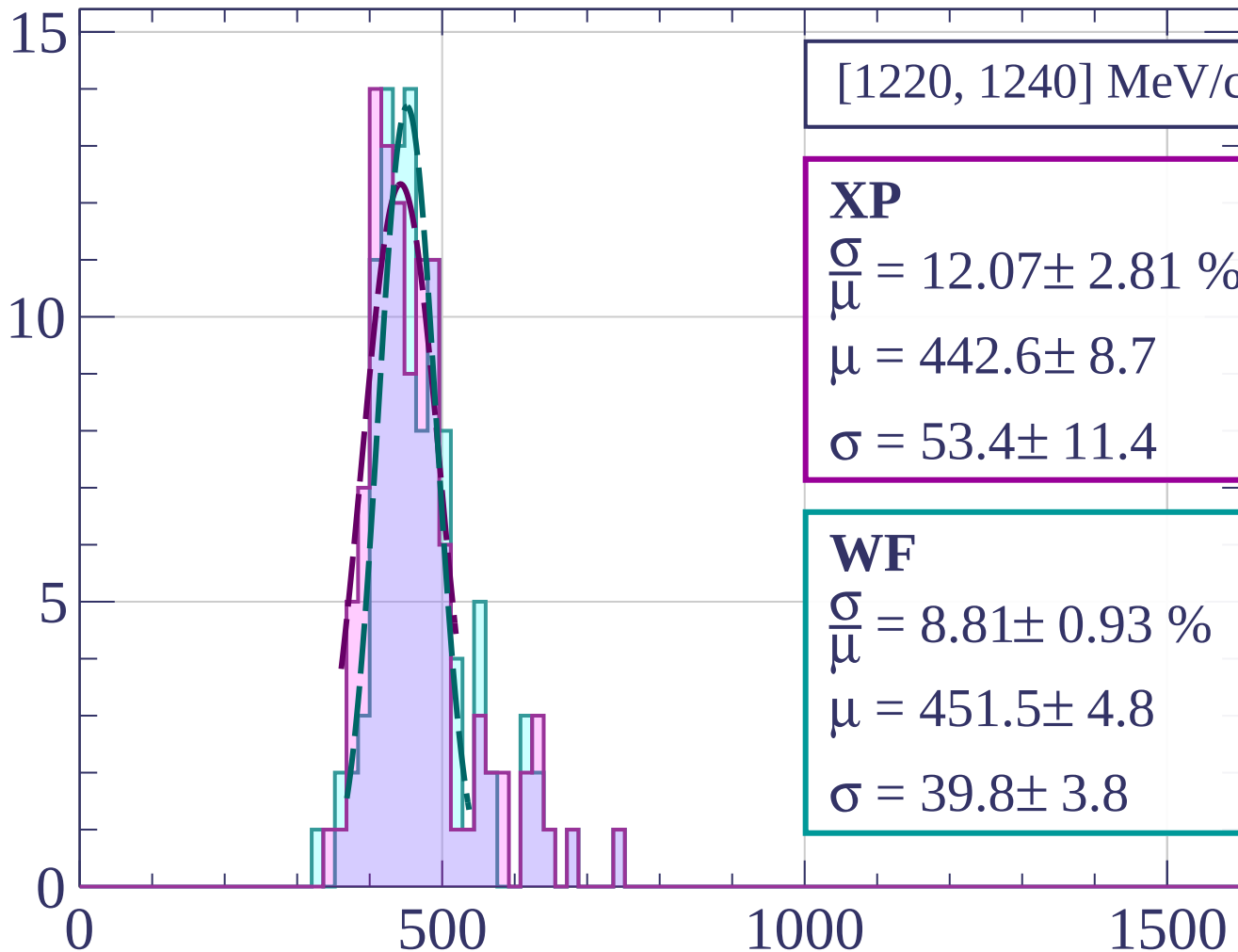
Count



Count

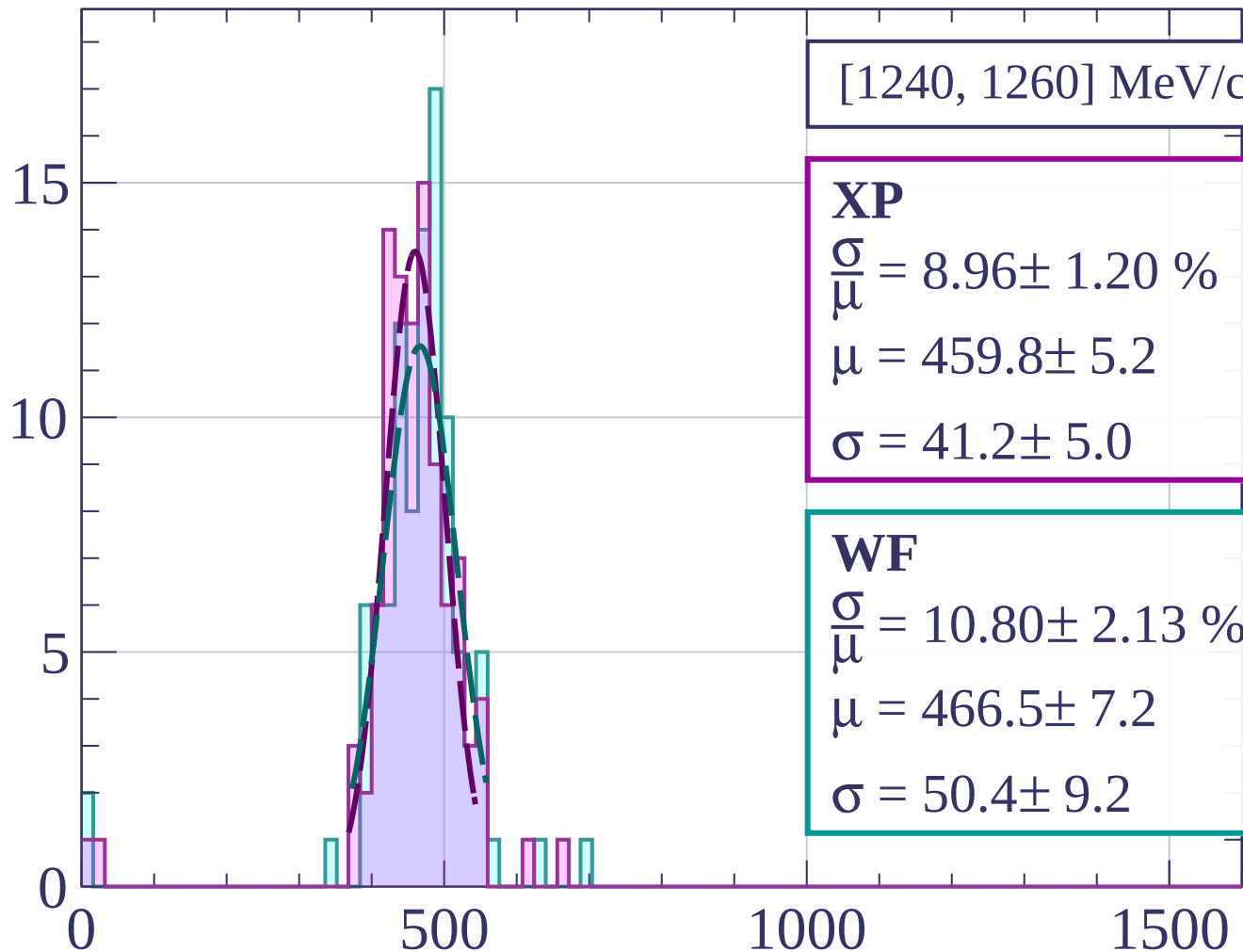


Count



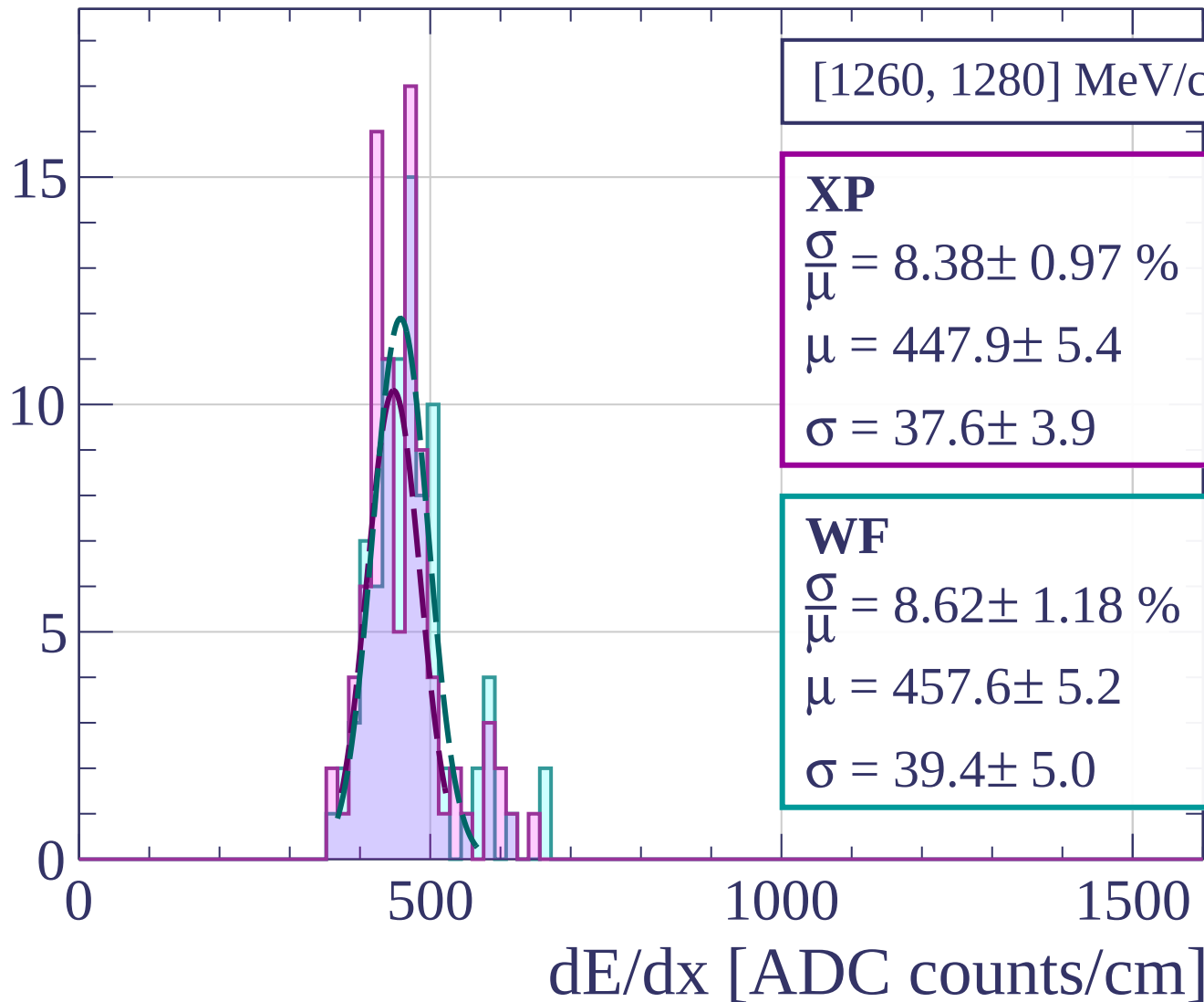
dE/dx [ADC counts/cm]

Count

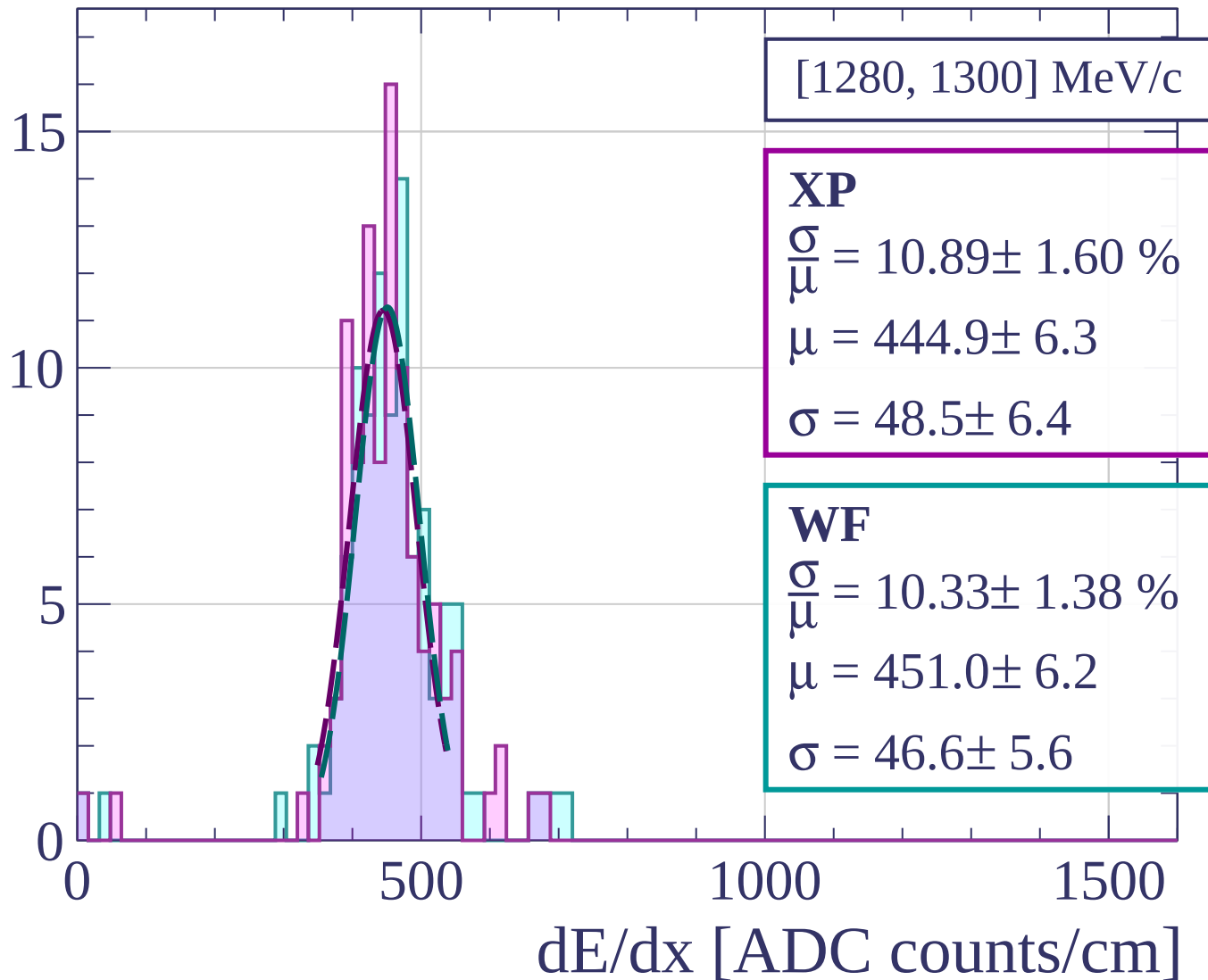


dE/dx [ADC counts/cm]

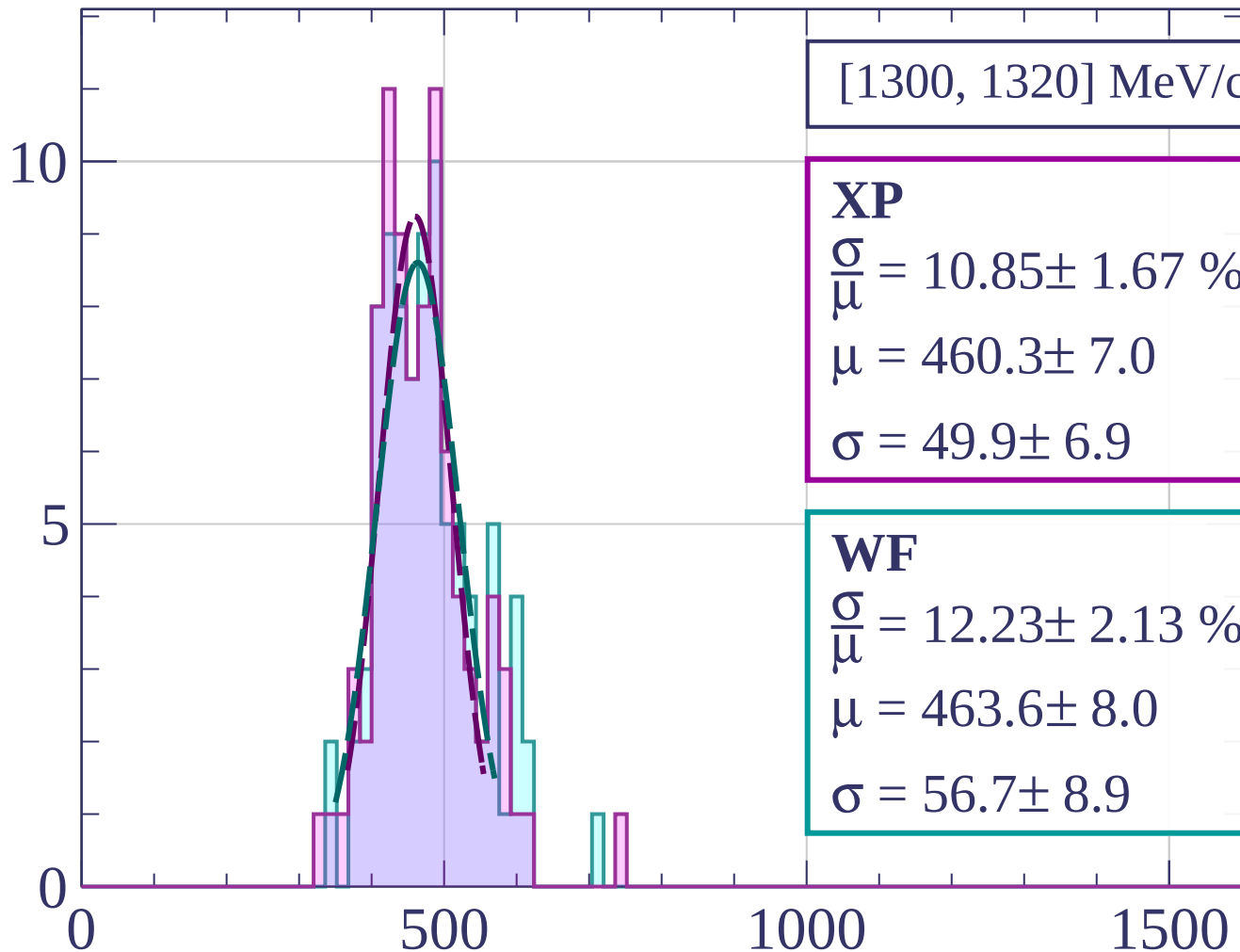
Count



Count



Count



[1300, 1320] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.85 \pm 1.67 \%$$

$$\mu = 460.3 \pm 7.0$$

$$\sigma = 49.9 \pm 6.9$$

WF

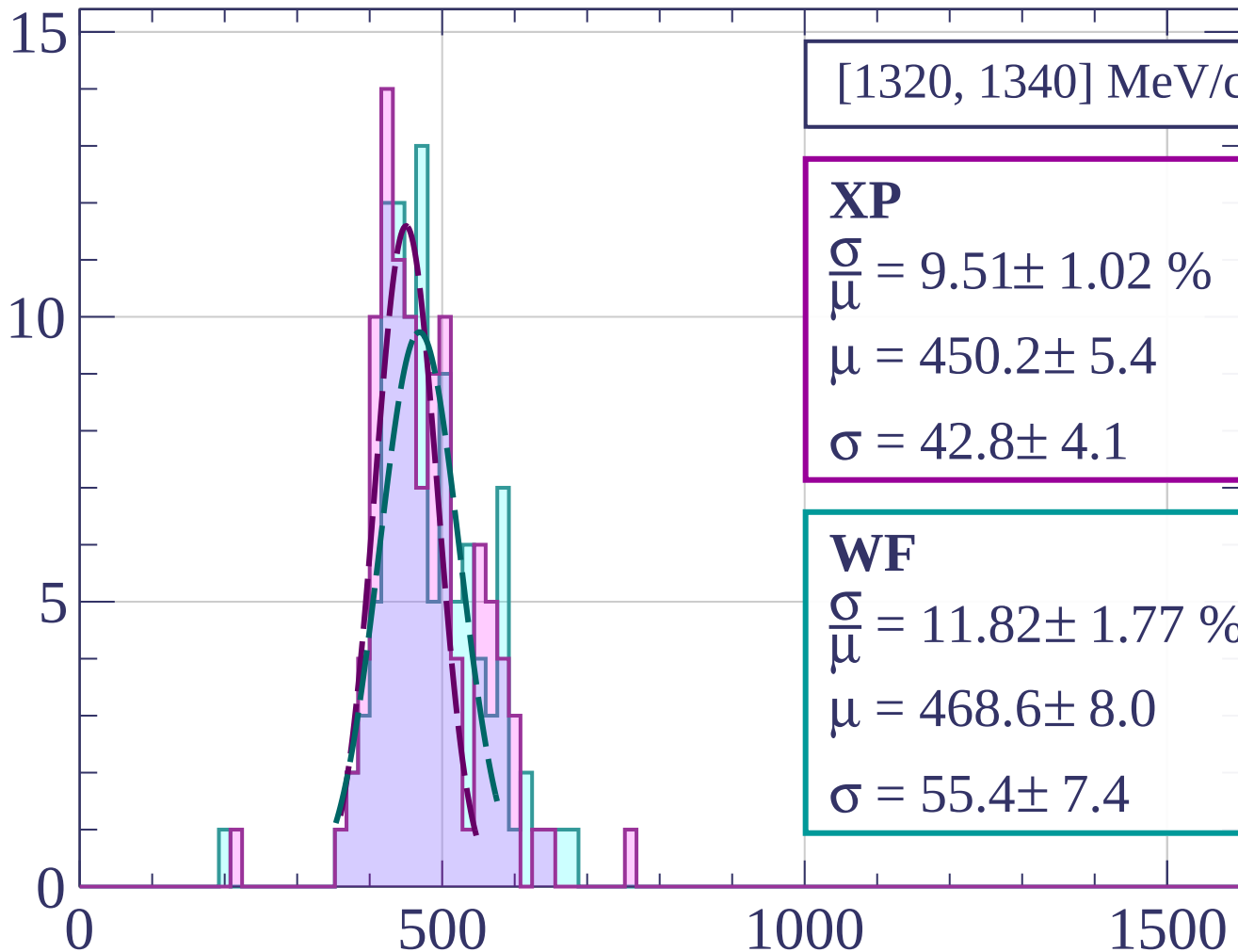
$$\frac{\sigma}{\mu} = 12.23 \pm 2.13 \%$$

$$\mu = 463.6 \pm 8.0$$

$$\sigma = 56.7 \pm 8.9$$

dE/dx [ADC counts/cm]

Count



[1320, 1340] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.51 \pm 1.02 \%$$

$$\mu = 450.2 \pm 5.4$$

$$\sigma = 42.8 \pm 4.1$$

WF

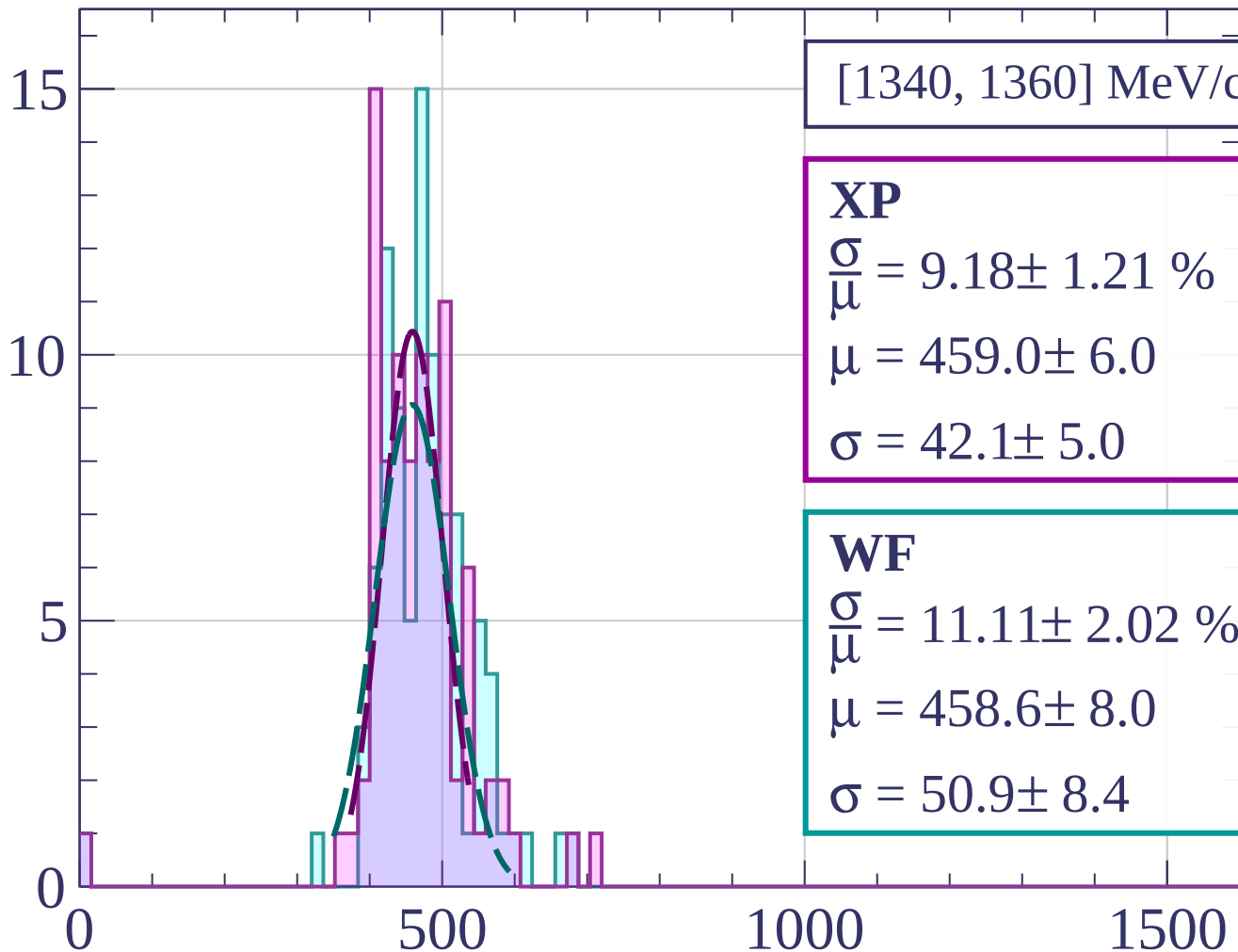
$$\frac{\sigma}{\mu} = 11.82 \pm 1.77 \%$$

$$\mu = 468.6 \pm 8.0$$

$$\sigma = 55.4 \pm 7.4$$

dE/dx [ADC counts/cm]

Count



[1340, 1360] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.18 \pm 1.21 \%$$

$$\mu = 459.0 \pm 6.0$$

$$\sigma = 42.1 \pm 5.0$$

WF

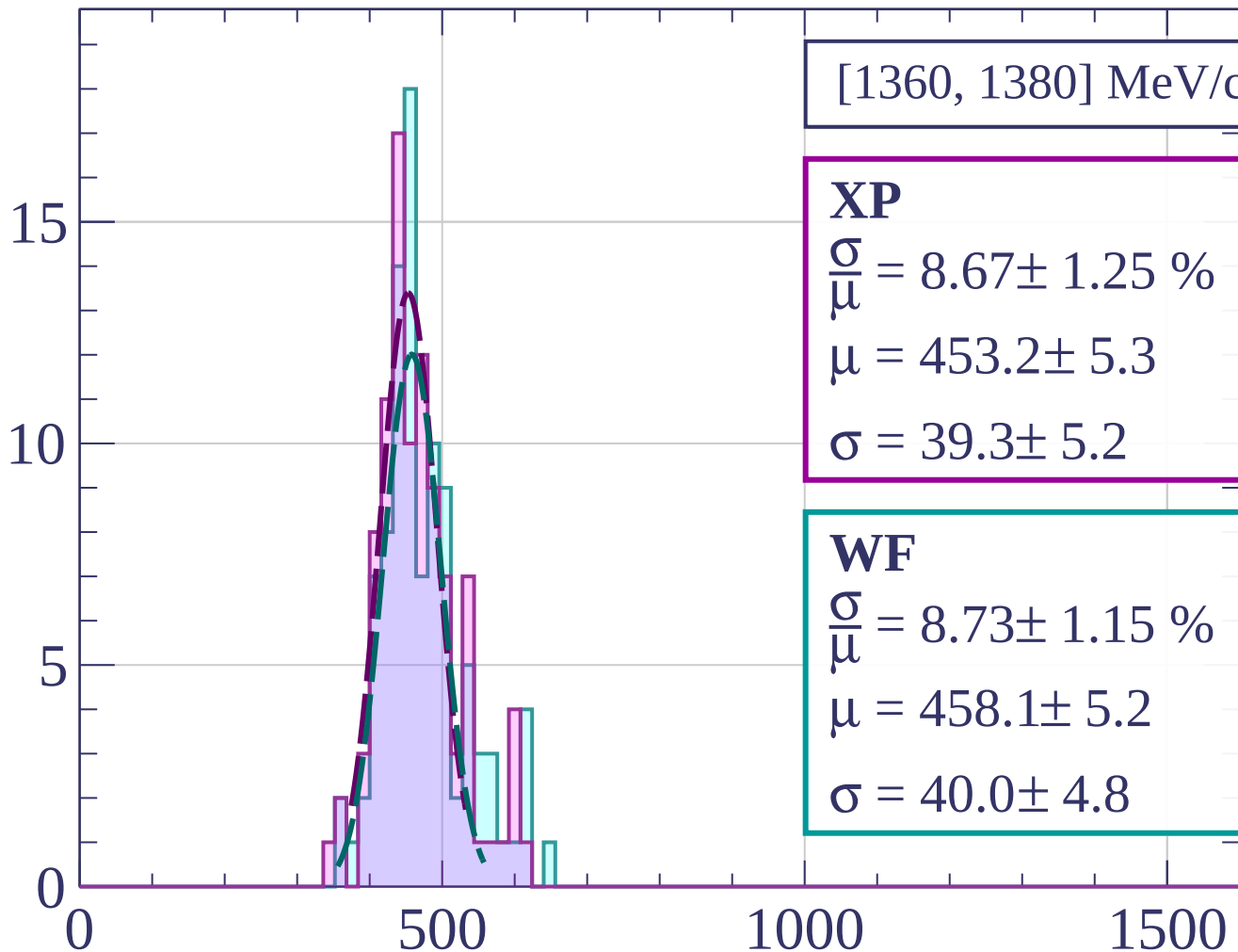
$$\frac{\sigma}{\mu} = 11.11 \pm 2.02 \%$$

$$\mu = 458.6 \pm 8.0$$

$$\sigma = 50.9 \pm 8.4$$

dE/dx [ADC counts/cm]

Count



[1360, 1380] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.67 \pm 1.25 \%$$

$$\mu = 453.2 \pm 5.3$$

$$\sigma = 39.3 \pm 5.2$$

WF

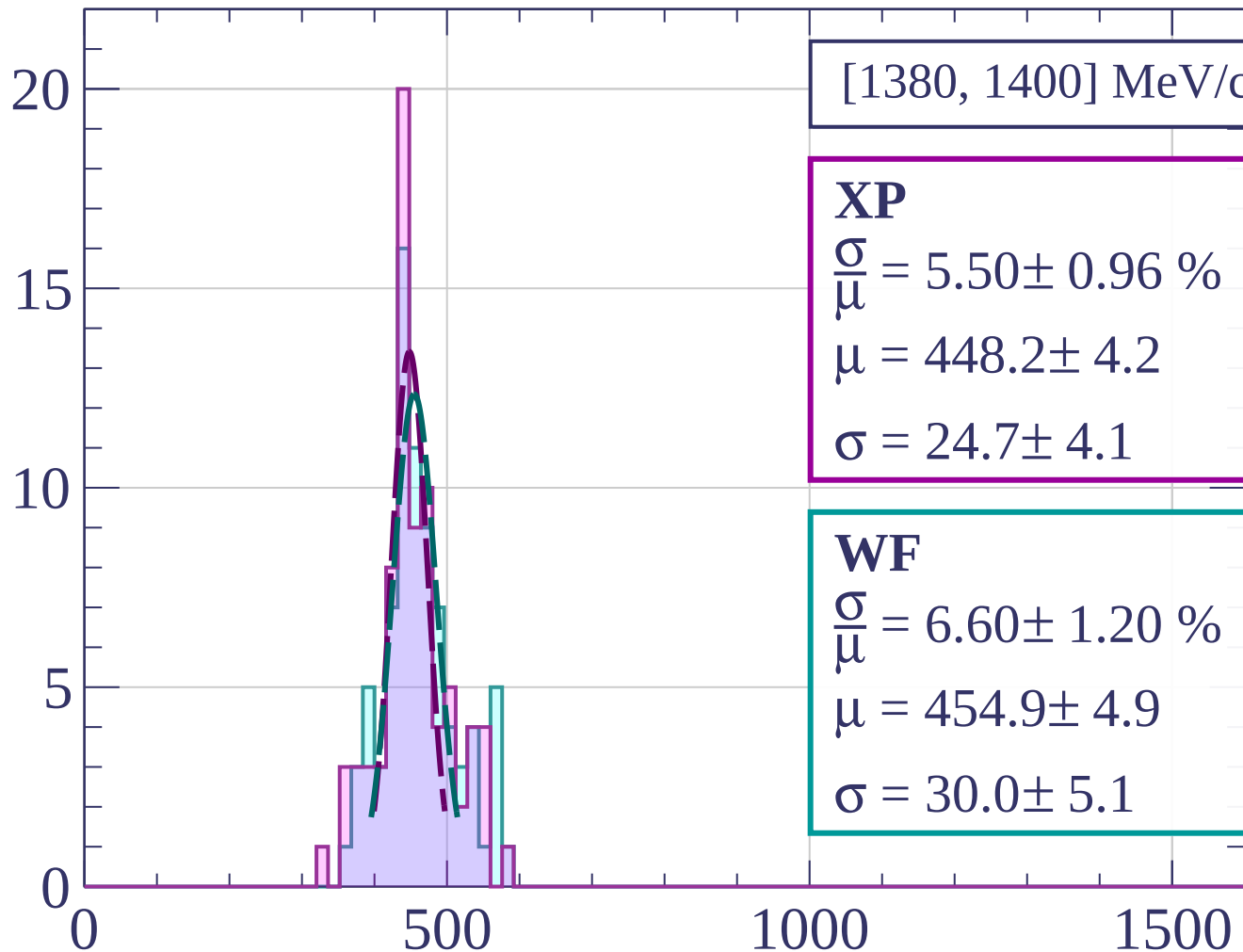
$$\frac{\sigma}{\mu} = 8.73 \pm 1.15 \%$$

$$\mu = 458.1 \pm 5.2$$

$$\sigma = 40.0 \pm 4.8$$

dE/dx [ADC counts/cm]

Count



[1380, 1400] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.50 \pm 0.96 \%$$

$$\mu = 448.2 \pm 4.2$$

$$\sigma = 24.7 \pm 4.1$$

WF

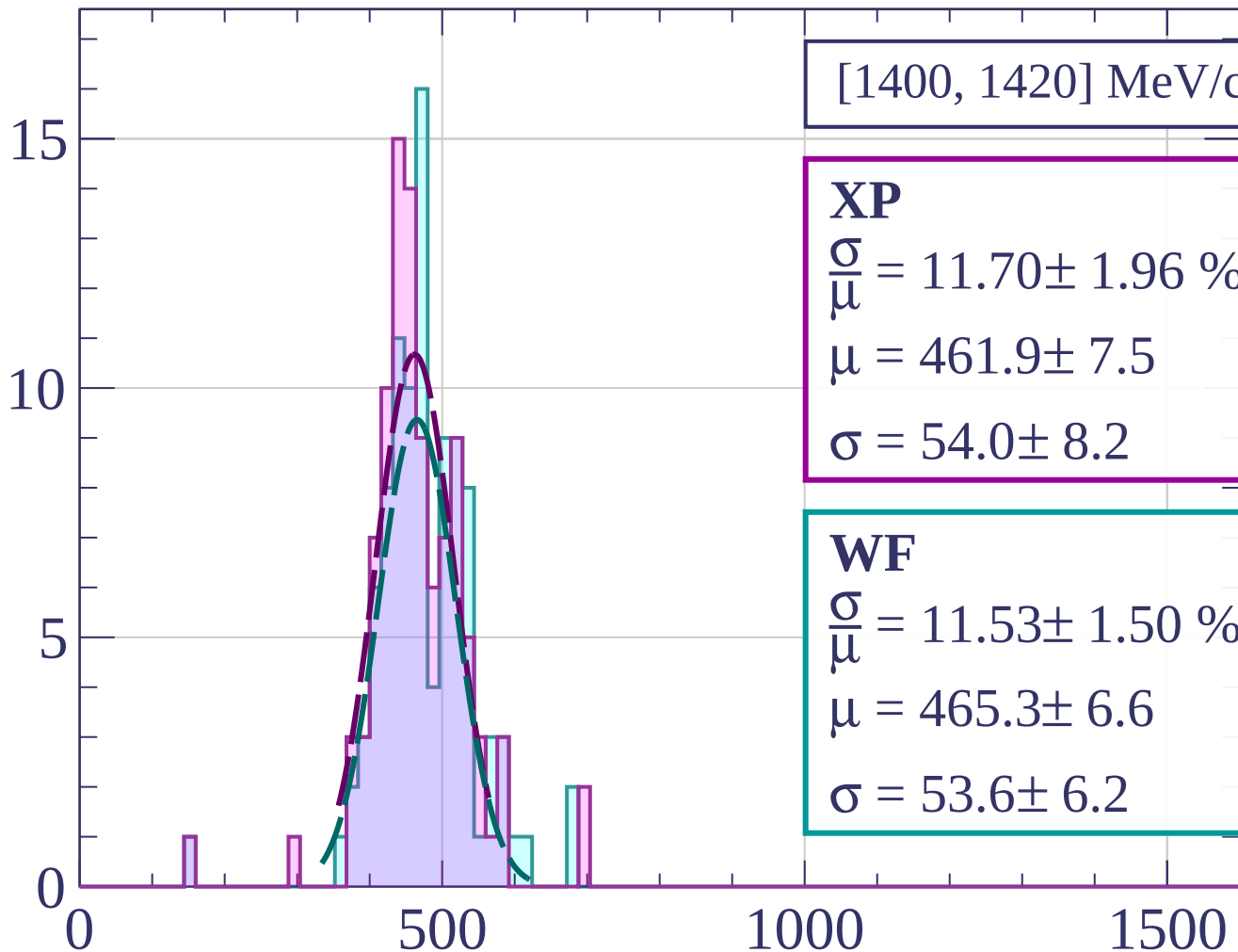
$$\frac{\sigma}{\mu} = 6.60 \pm 1.20 \%$$

$$\mu = 454.9 \pm 4.9$$

$$\sigma = 30.0 \pm 5.1$$

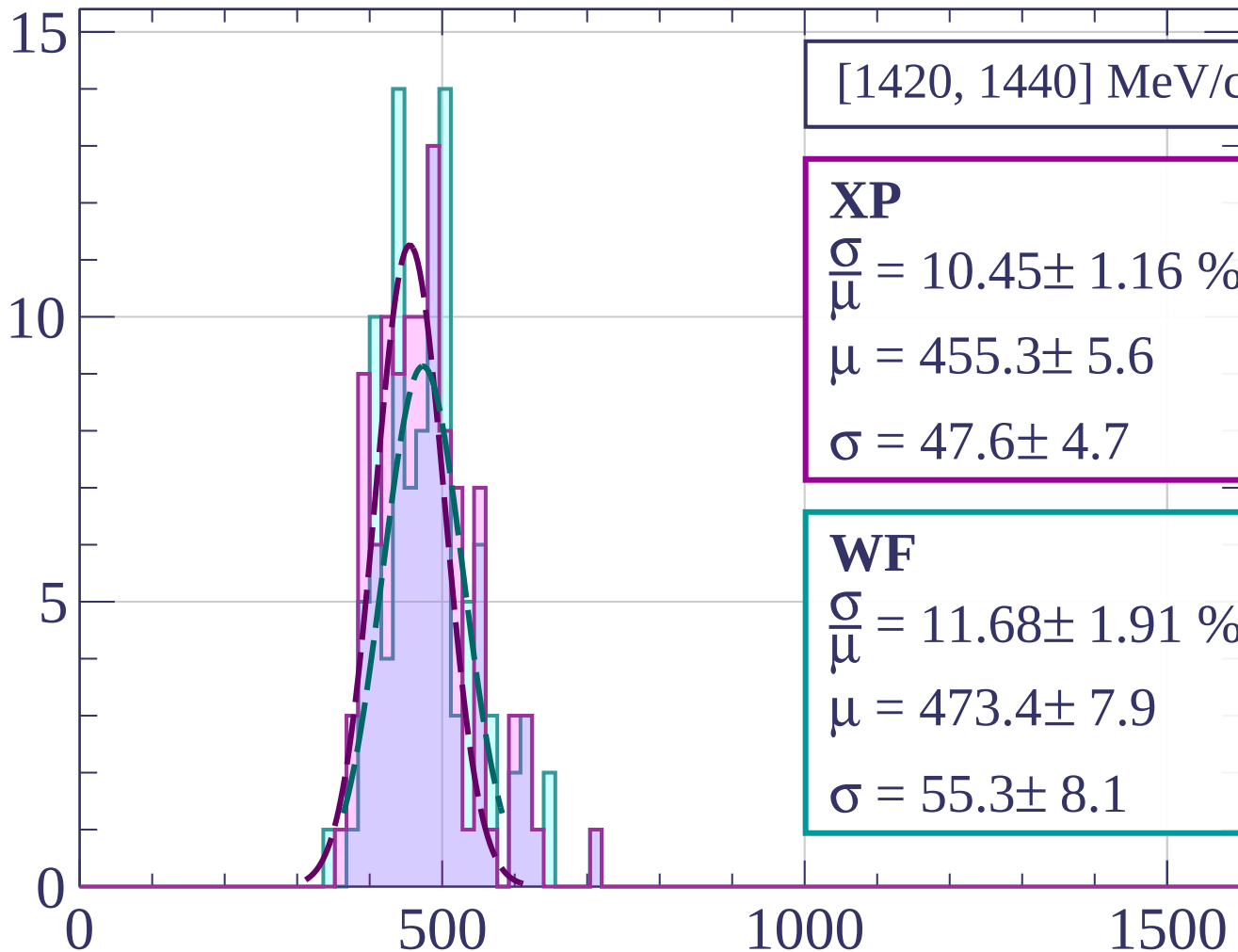
dE/dx [ADC counts/cm]

Count



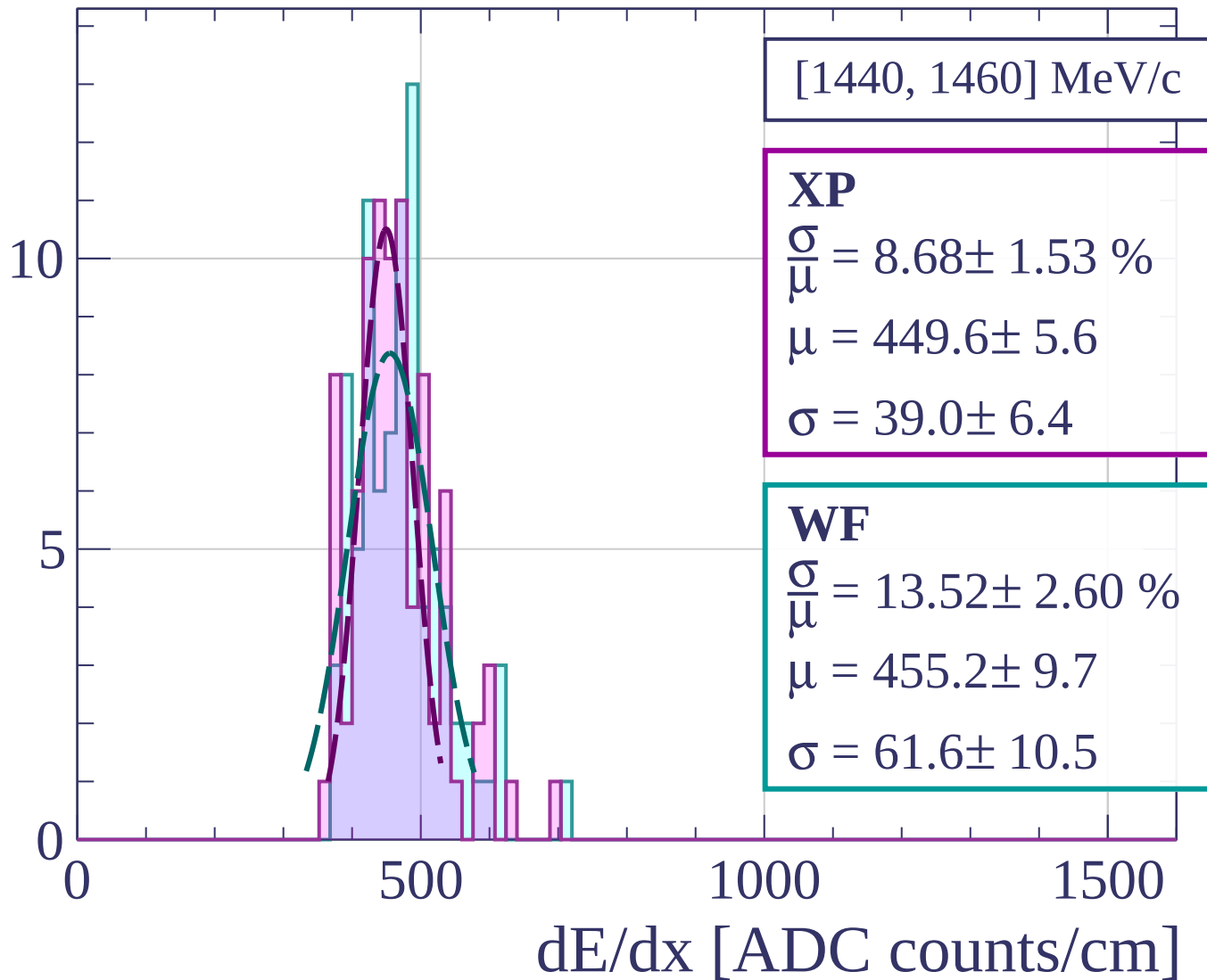
dE/dx [ADC counts/cm]

Count

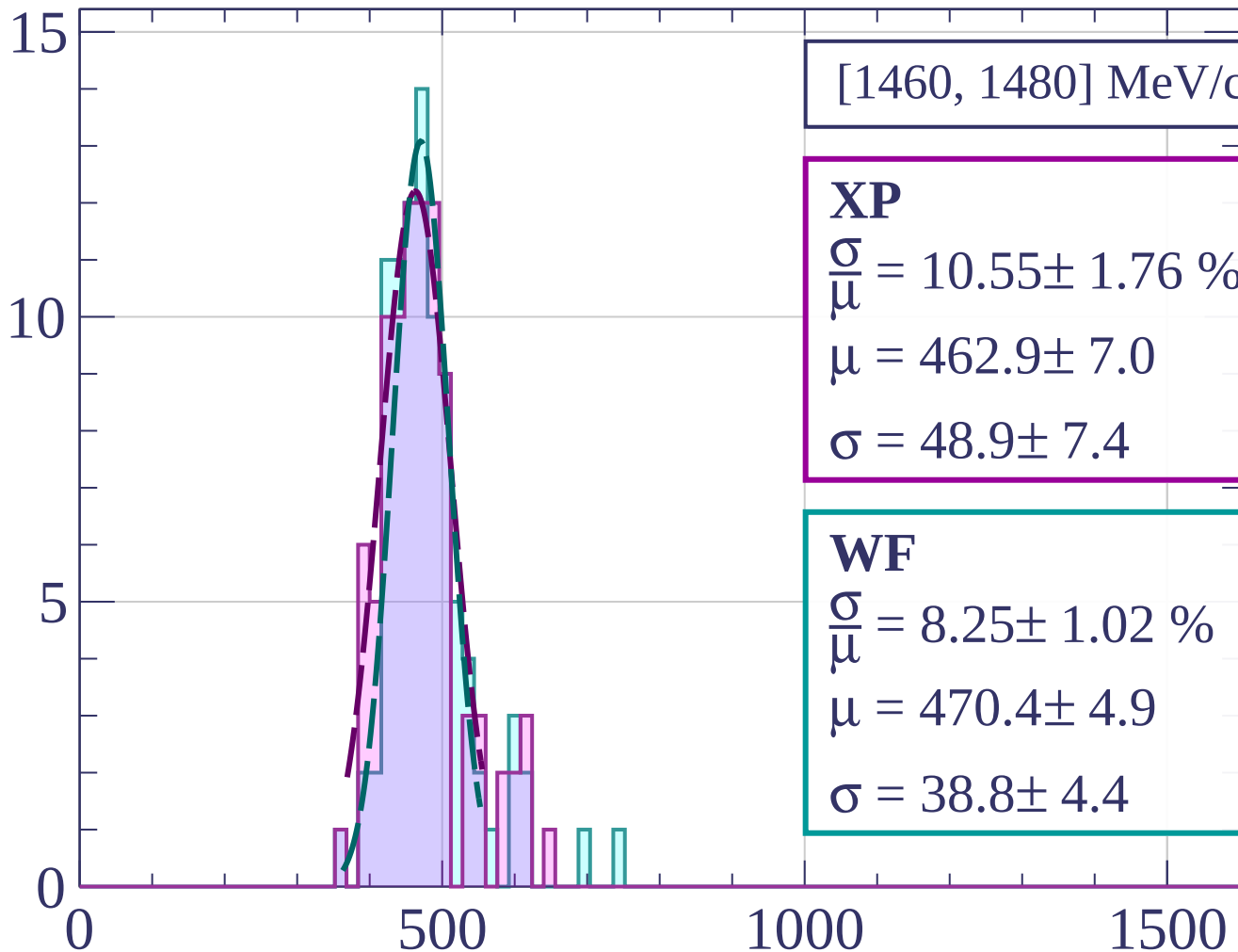


dE/dx [ADC counts/cm]

Count



Count



[1460, 1480] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.55 \pm 1.76 \%$$

$$\mu = 462.9 \pm 7.0$$

$$\sigma = 48.9 \pm 7.4$$

WF

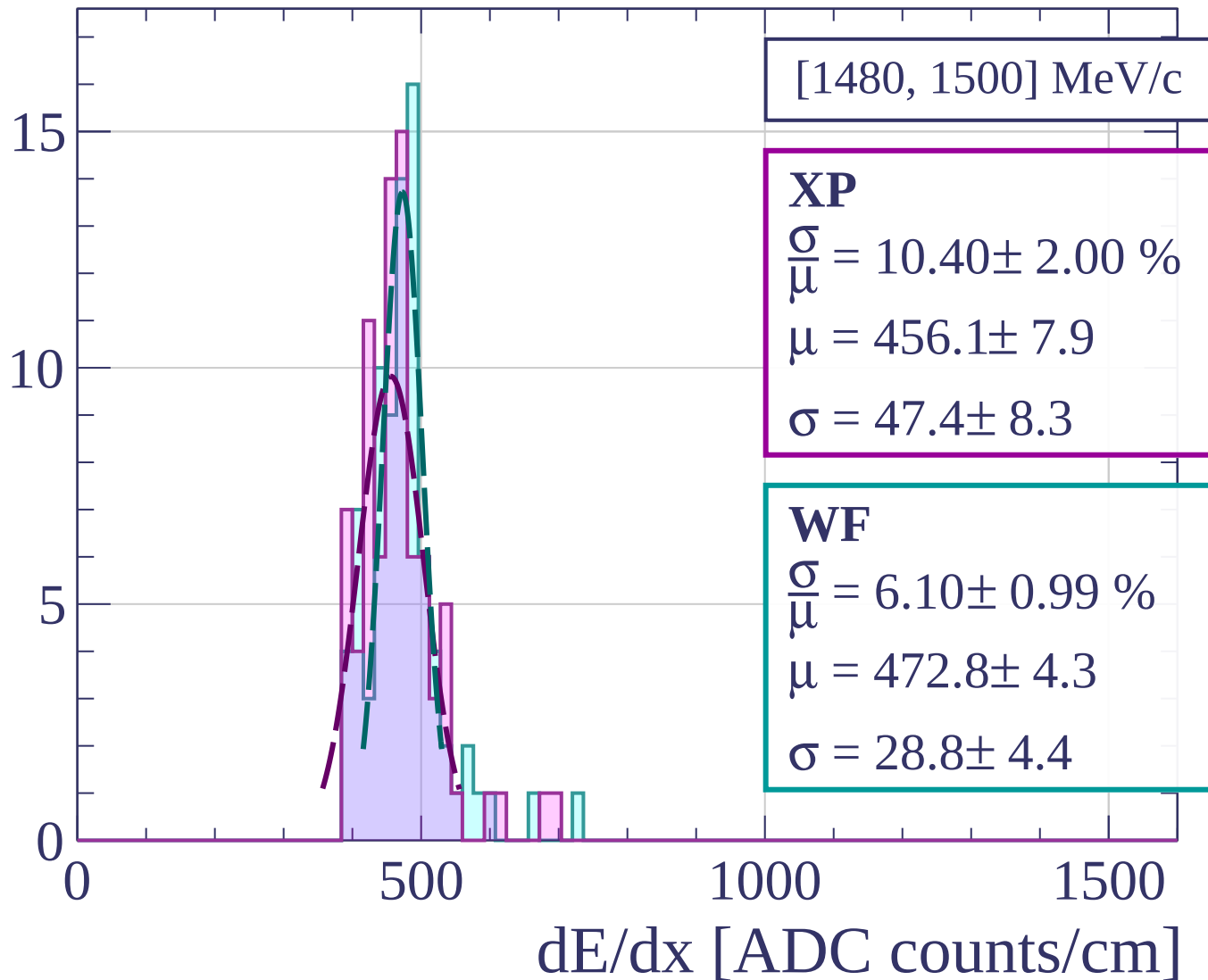
$$\frac{\sigma}{\mu} = 8.25 \pm 1.02 \%$$

$$\mu = 470.4 \pm 4.9$$

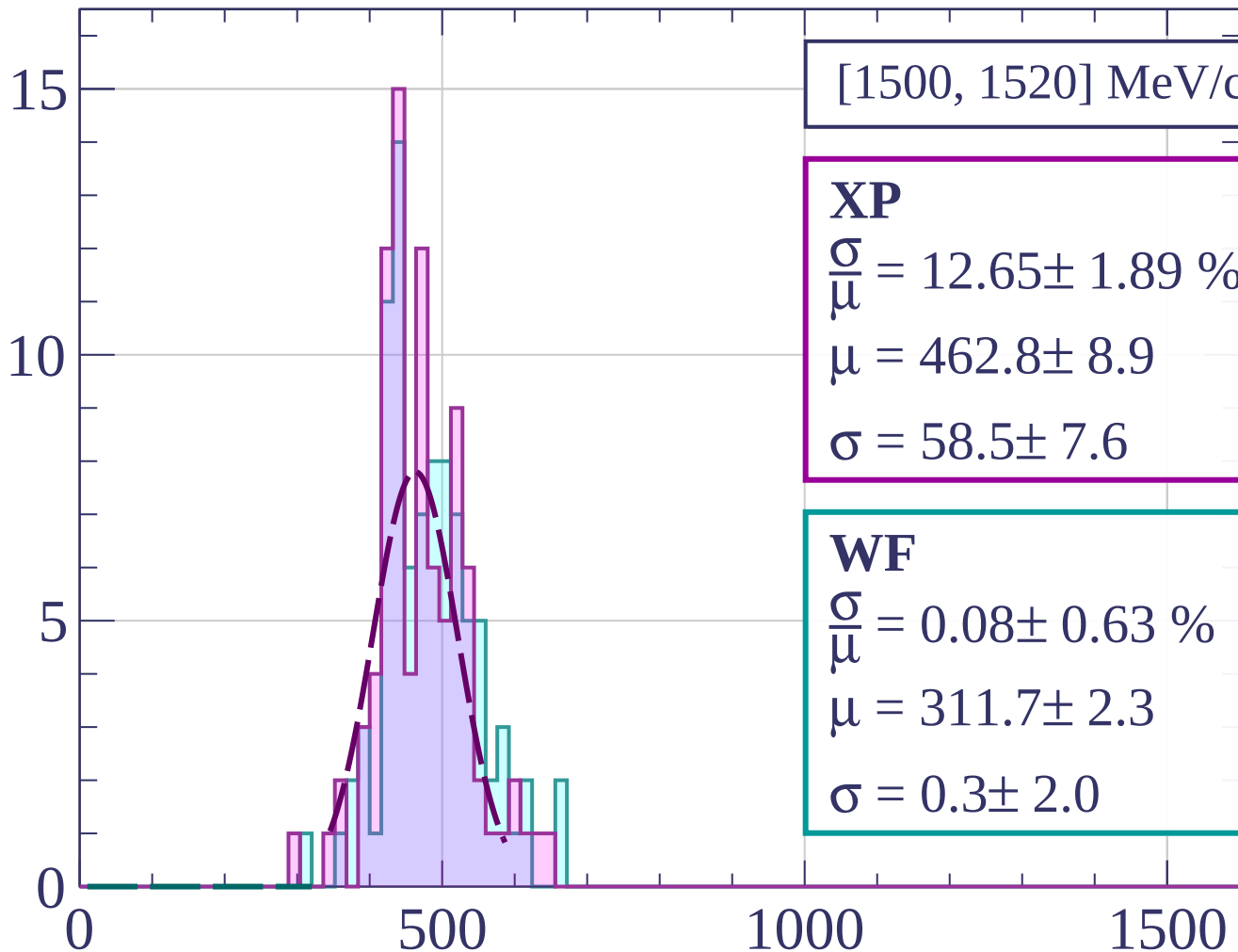
$$\sigma = 38.8 \pm 4.4$$

dE/dx [ADC counts/cm]

Count

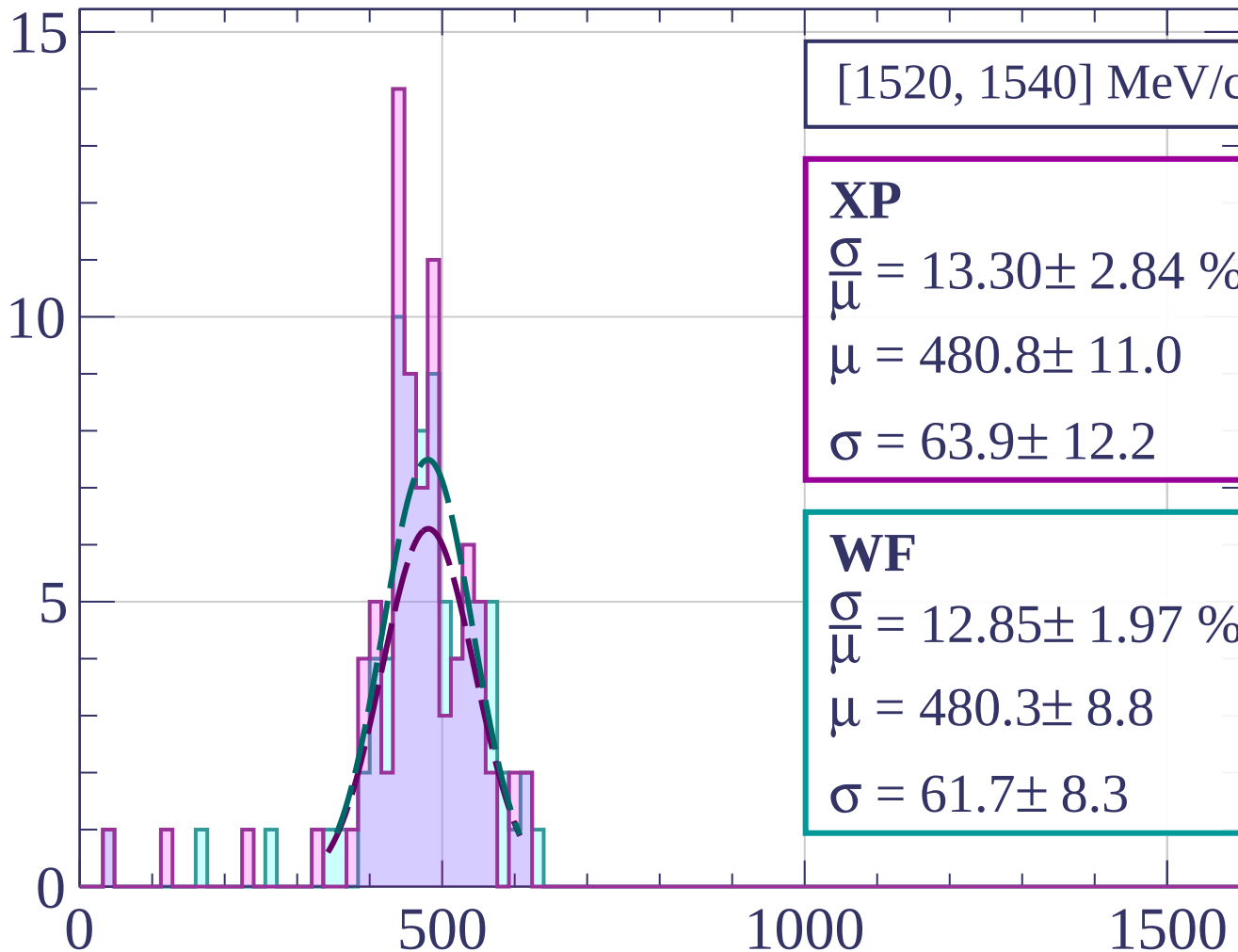


Count



dE/dx [ADC counts/cm]

Count



[1520, 1540] MeV/c

XP

$$\frac{\sigma}{\mu} = 13.30 \pm 2.84 \%$$

$$\mu = 480.8 \pm 11.0$$

$$\sigma = 63.9 \pm 12.2$$

WF

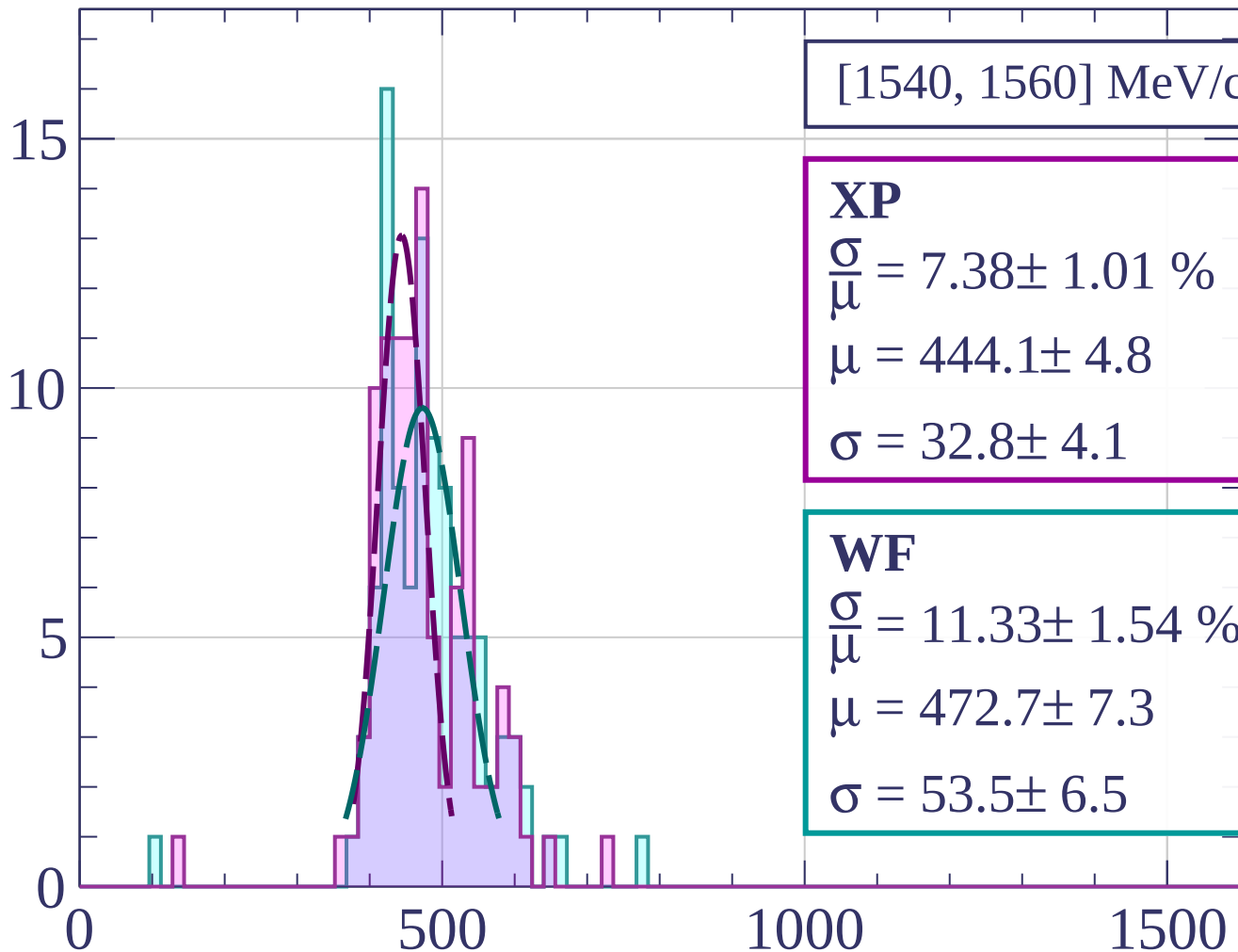
$$\frac{\sigma}{\mu} = 12.85 \pm 1.97 \%$$

$$\mu = 480.3 \pm 8.8$$

$$\sigma = 61.7 \pm 8.3$$

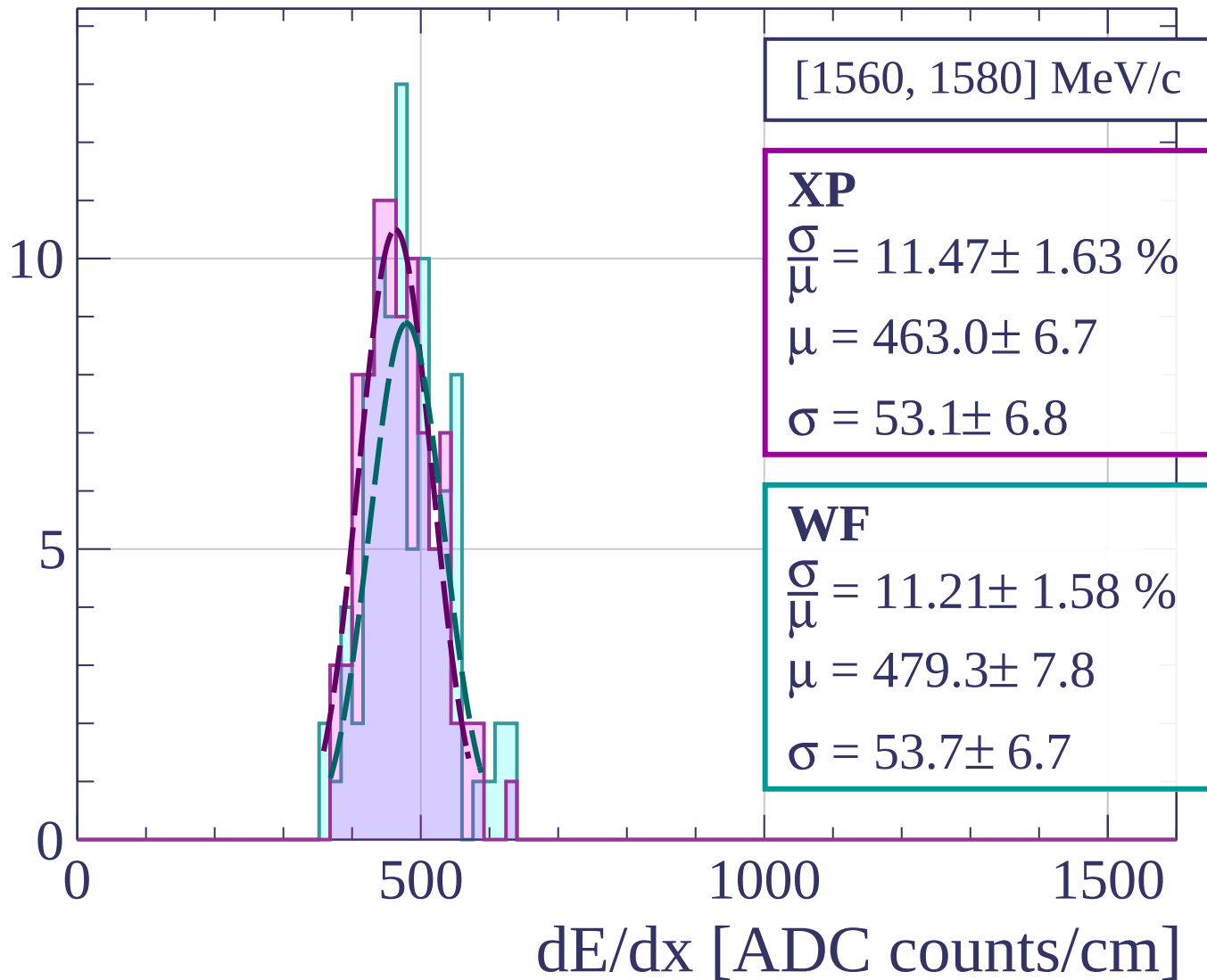
dE/dx [ADC counts/cm]

Count

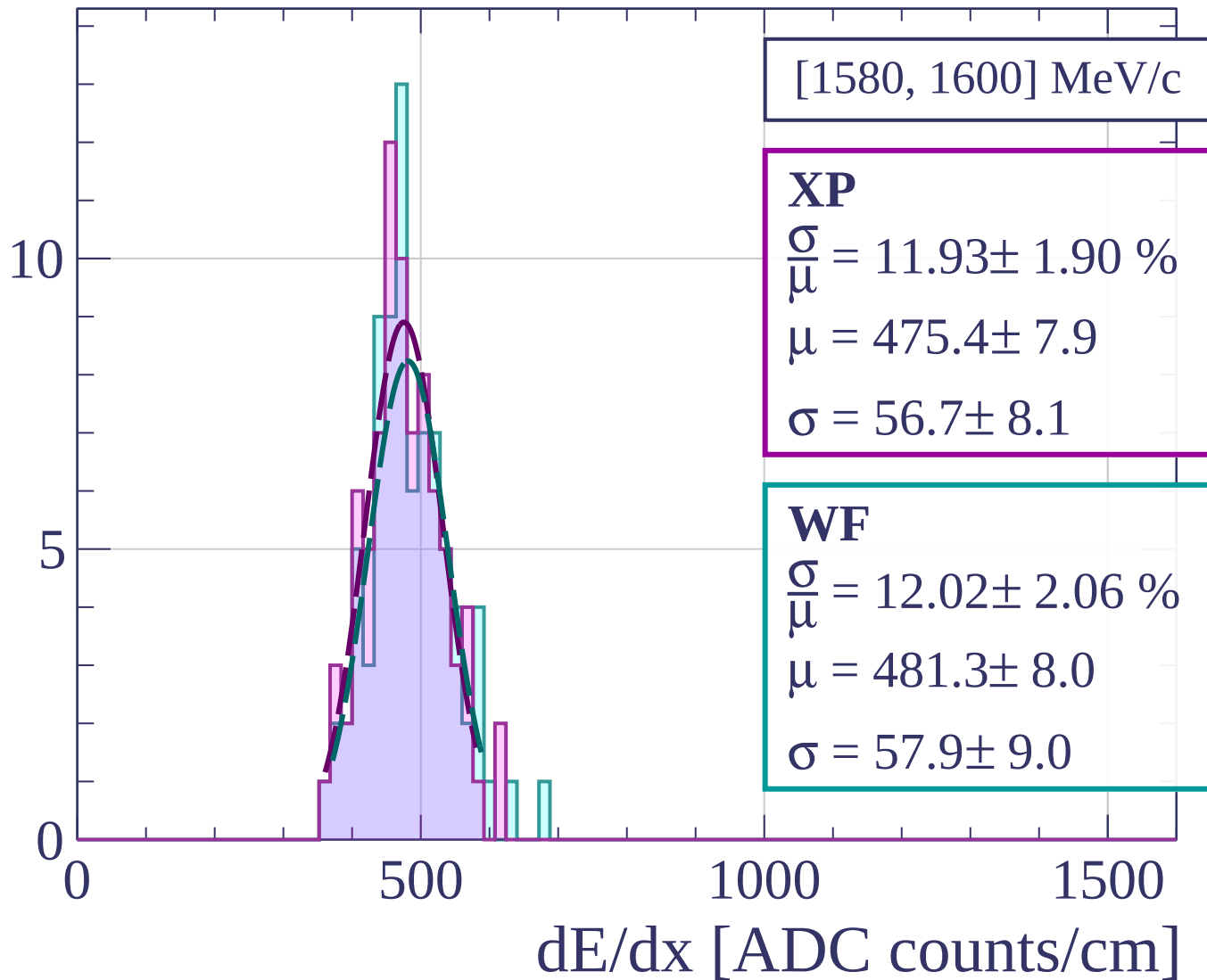


dE/dx [ADC counts/cm]

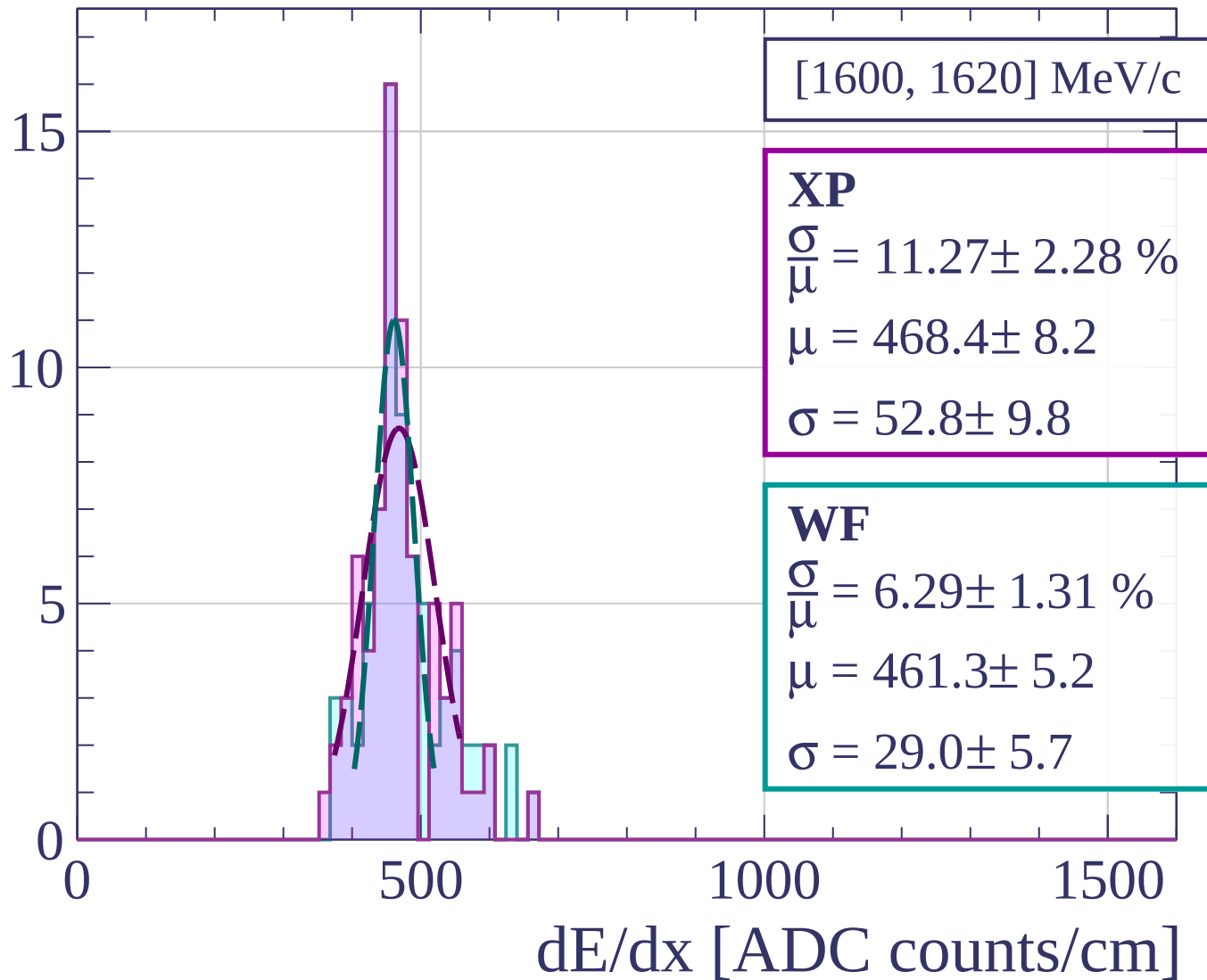
Count



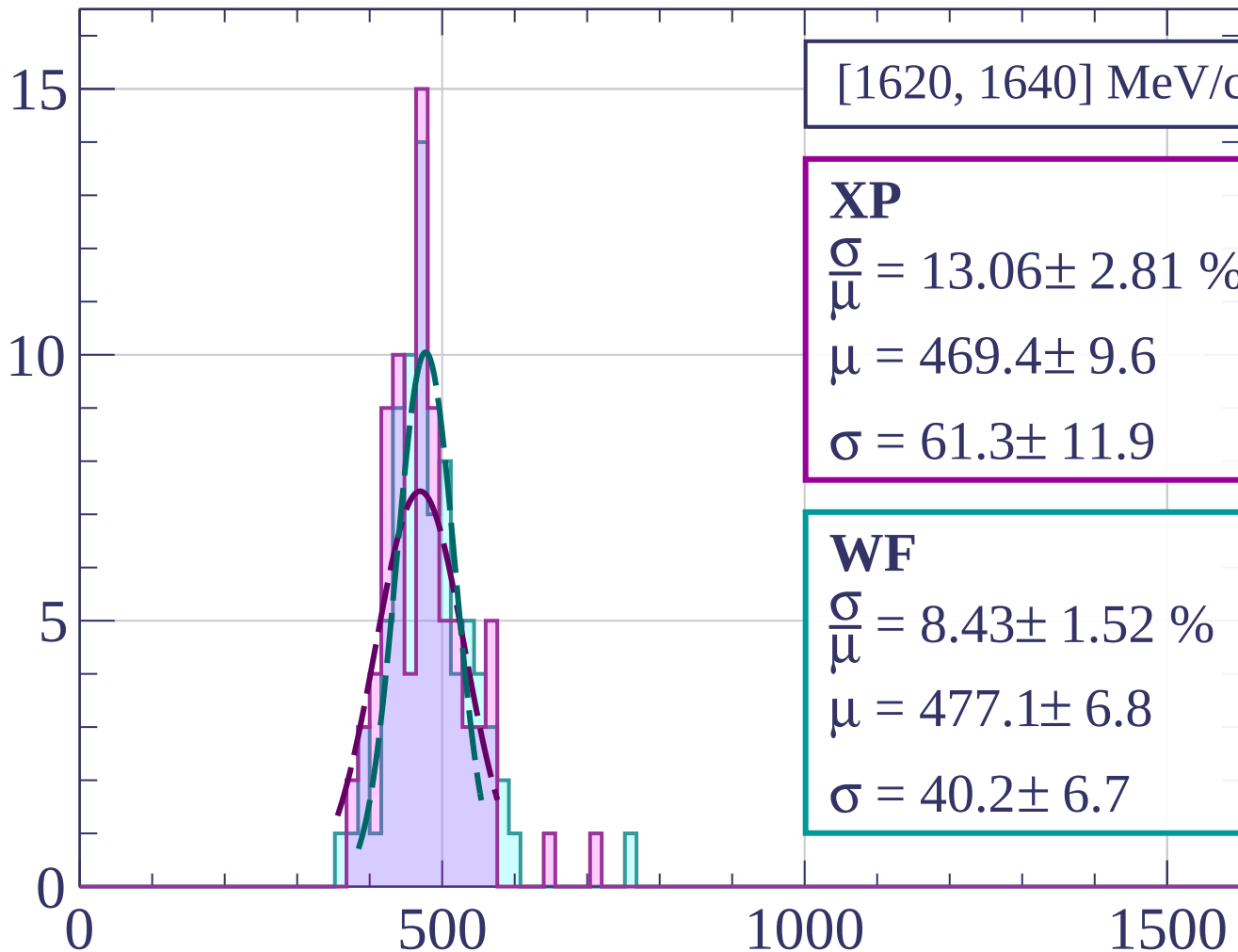
Count



Count



Count



[1620, 1640] MeV/c

XP

$$\frac{\sigma}{\mu} = 13.06 \pm 2.81 \%$$

$$\mu = 469.4 \pm 9.6$$

$$\sigma = 61.3 \pm 11.9$$

WF

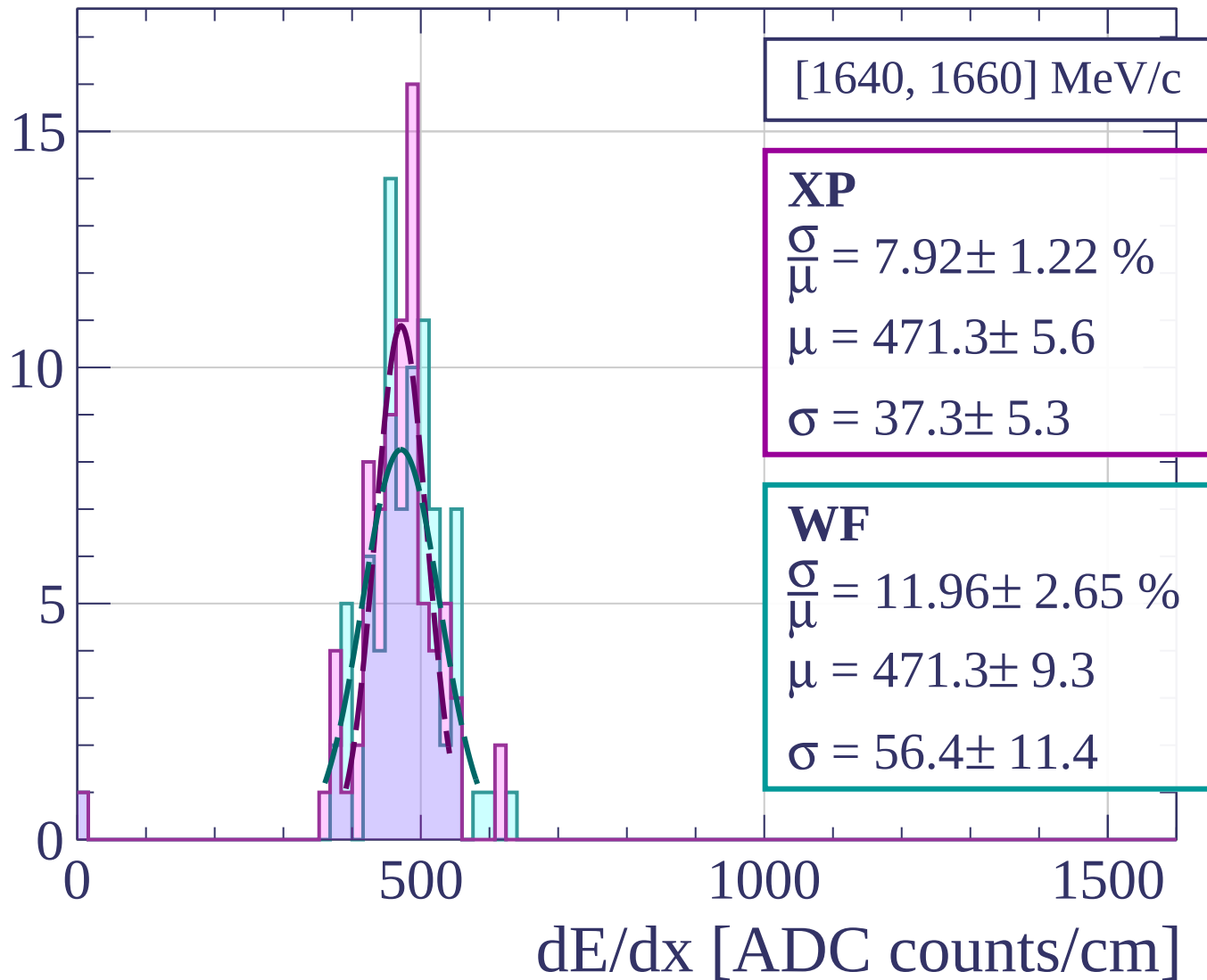
$$\frac{\sigma}{\mu} = 8.43 \pm 1.52 \%$$

$$\mu = 477.1 \pm 6.8$$

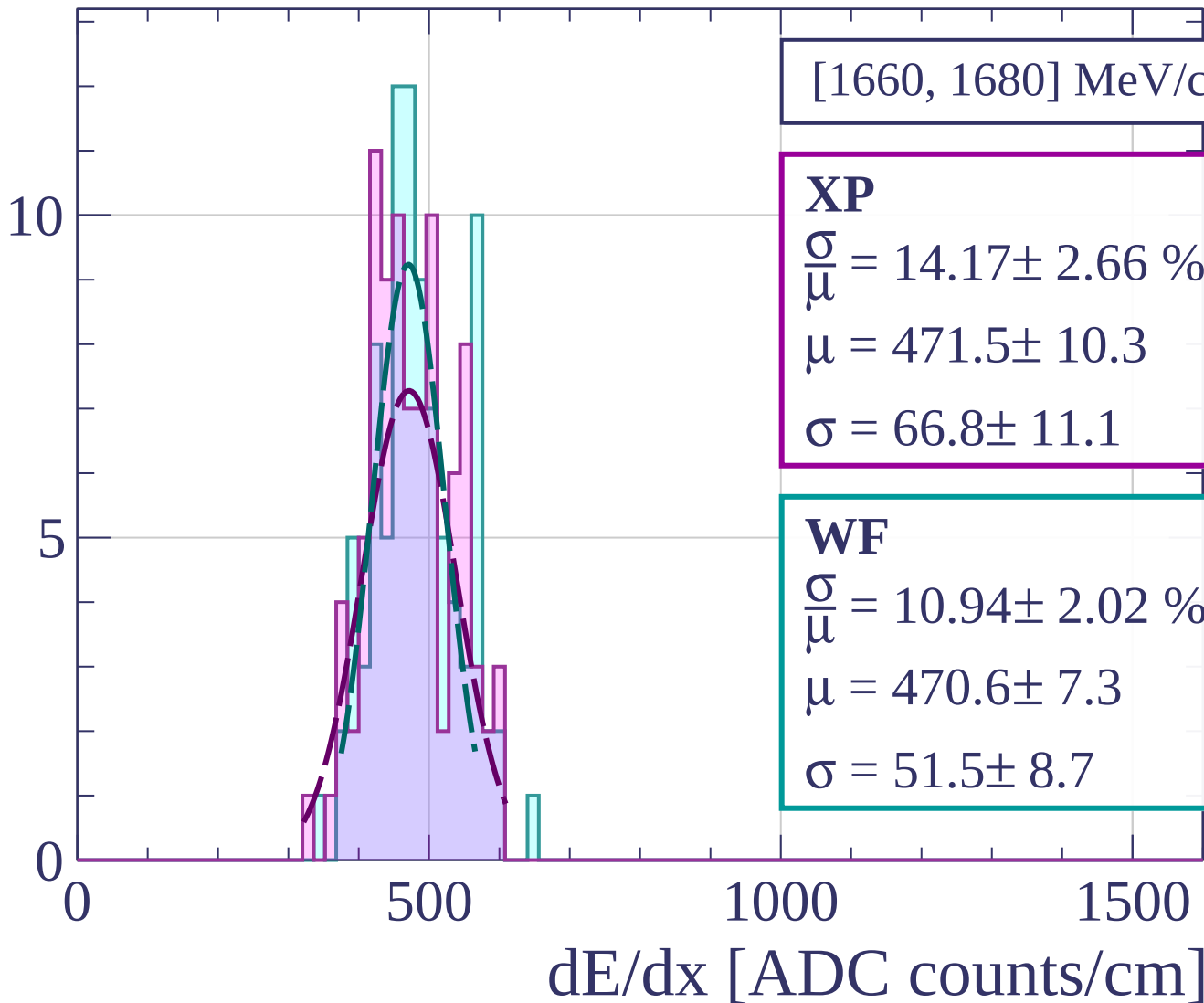
$$\sigma = 40.2 \pm 6.7$$

dE/dx [ADC counts/cm]

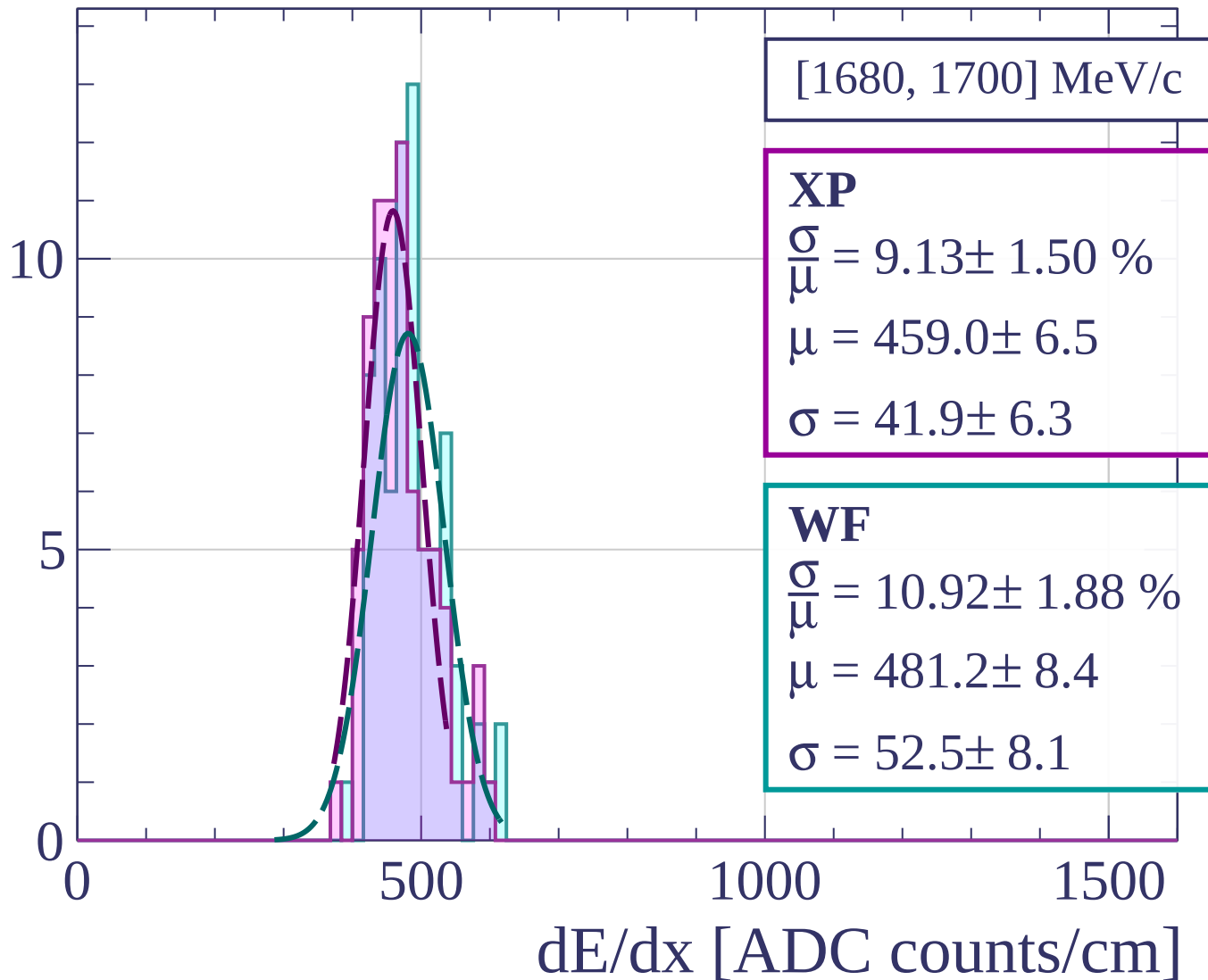
Count



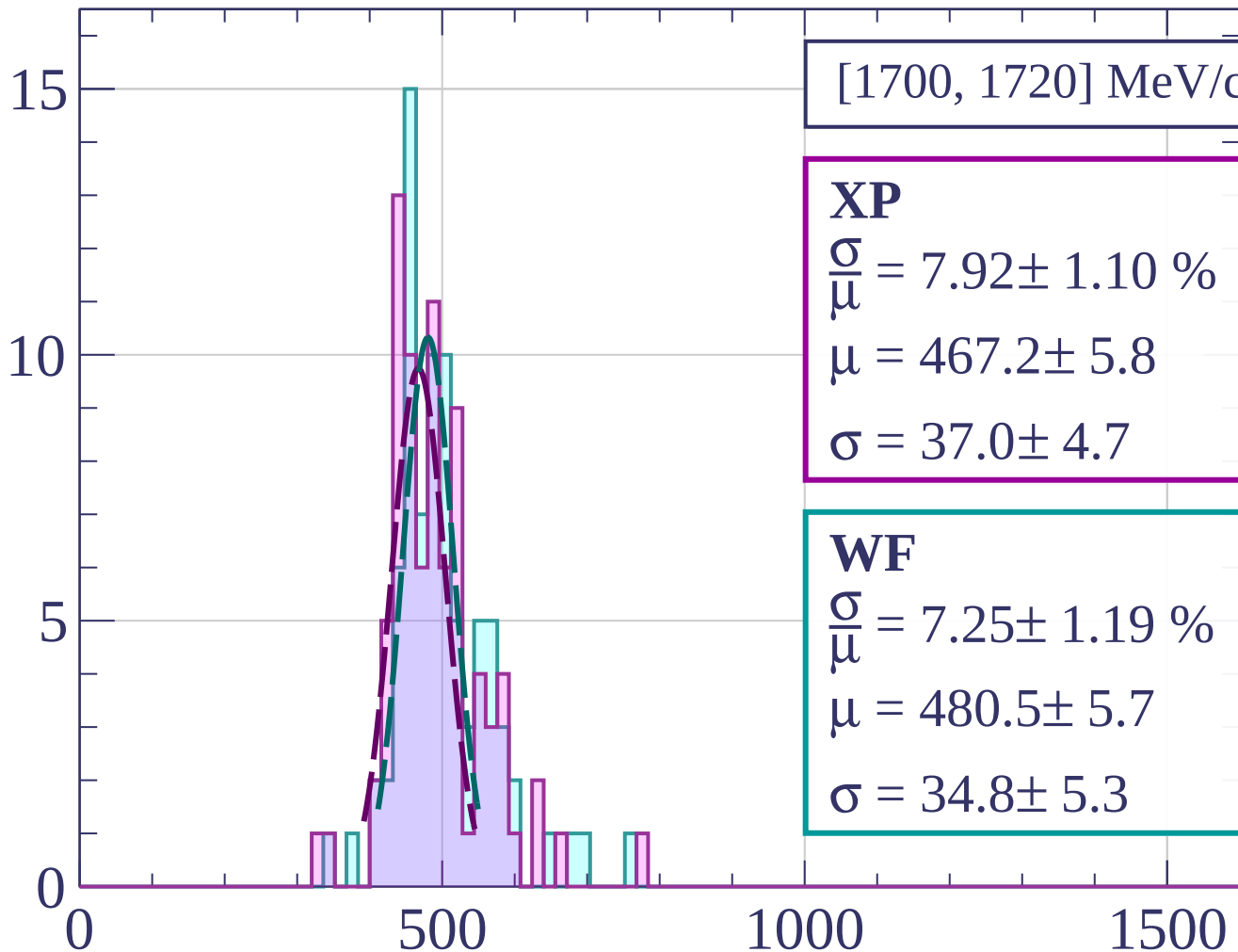
Count



Count



Count



[1700, 1720] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.92 \pm 1.10 \%$$

$$\mu = 467.2 \pm 5.8$$

$$\sigma = 37.0 \pm 4.7$$

WF

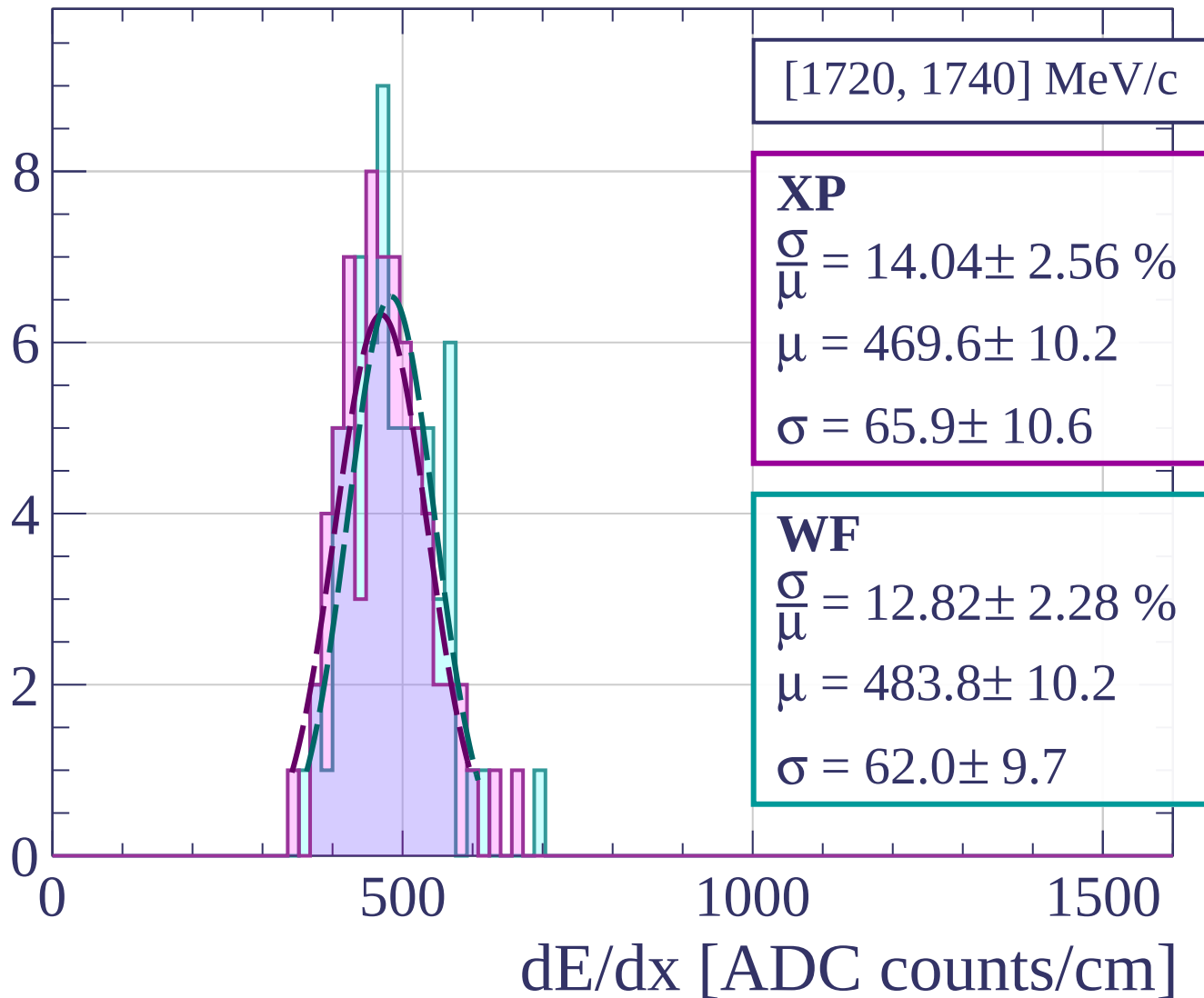
$$\frac{\sigma}{\mu} = 7.25 \pm 1.19 \%$$

$$\mu = 480.5 \pm 5.7$$

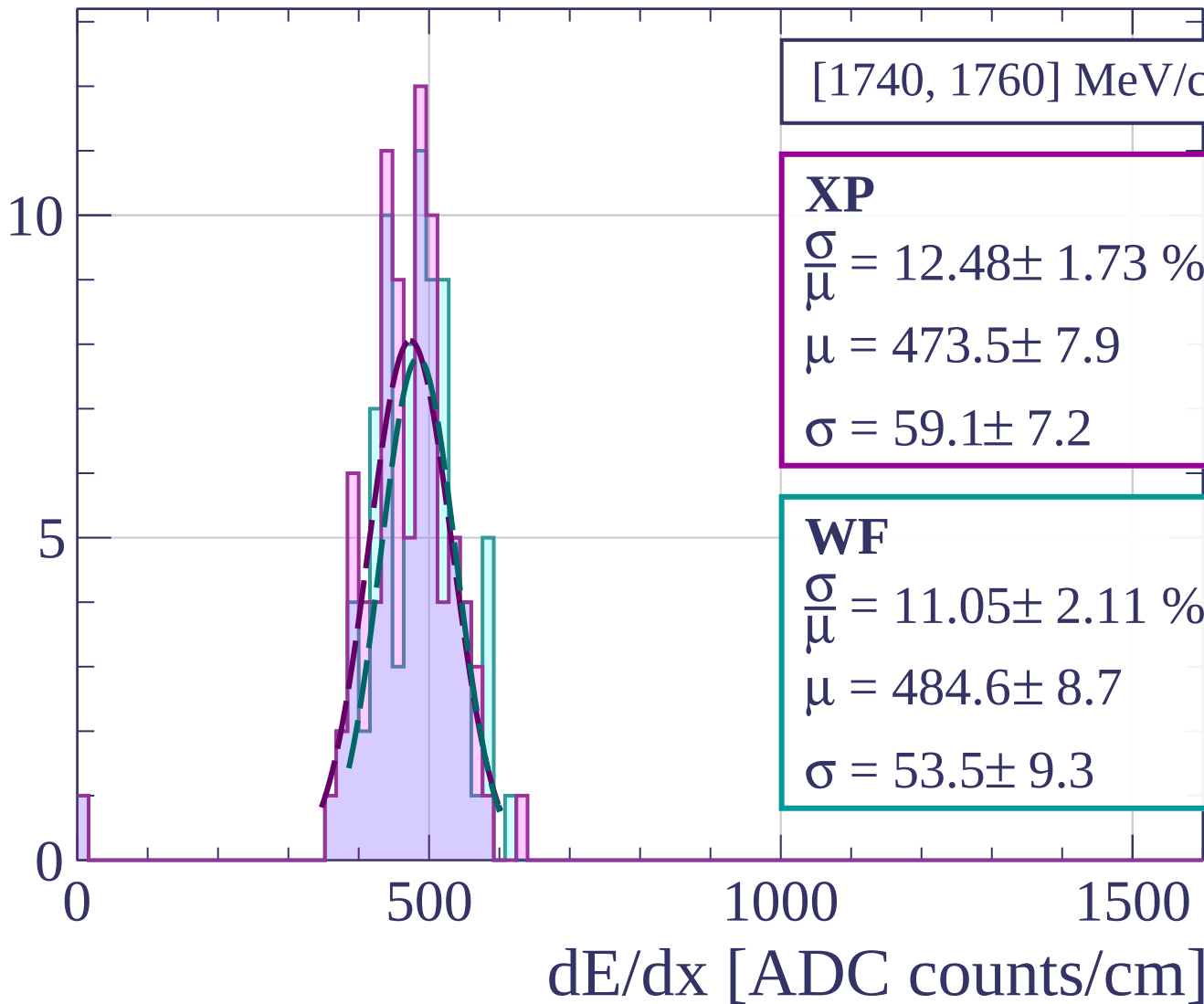
$$\sigma = 34.8 \pm 5.3$$

dE/dx [ADC counts/cm]

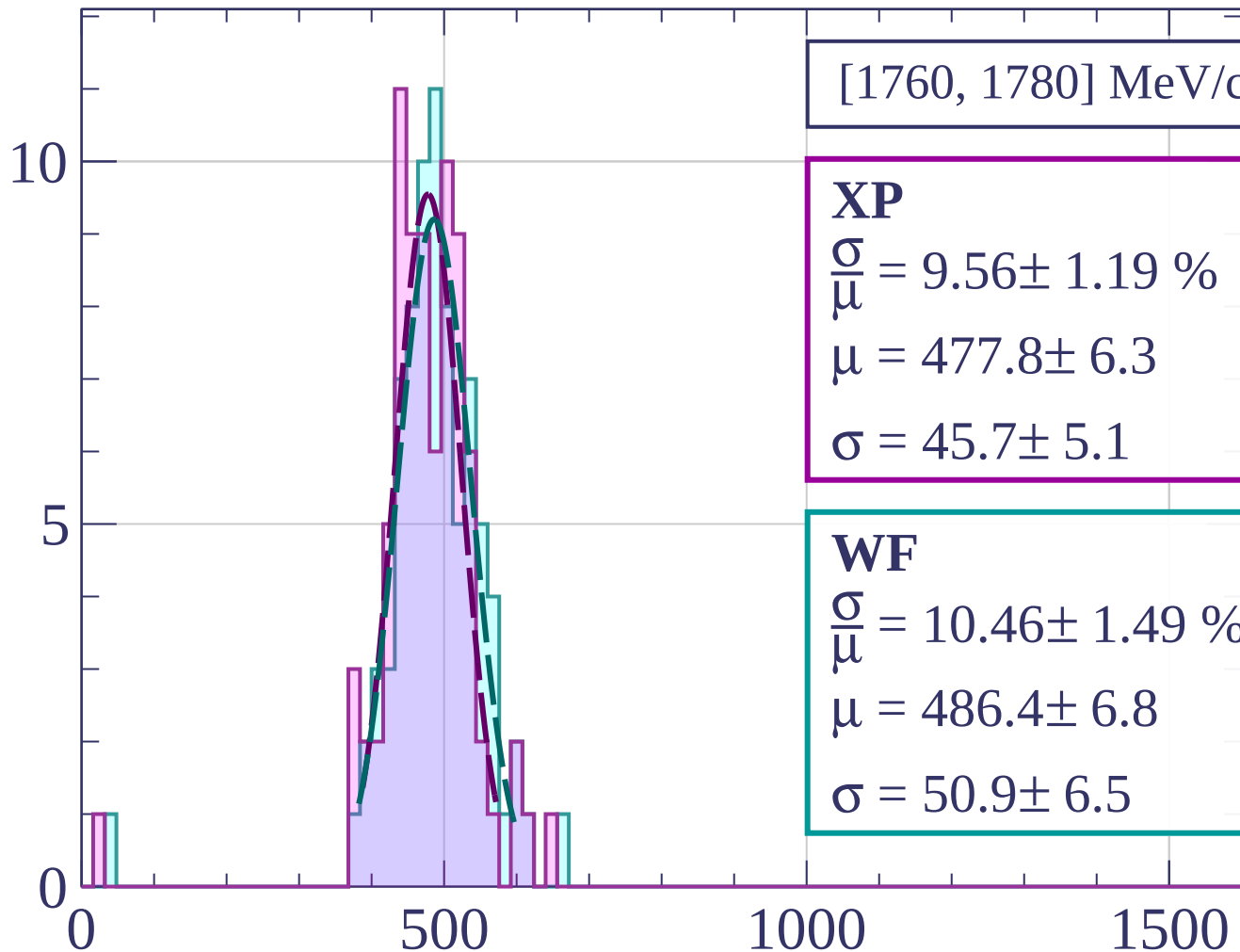
Count



Count

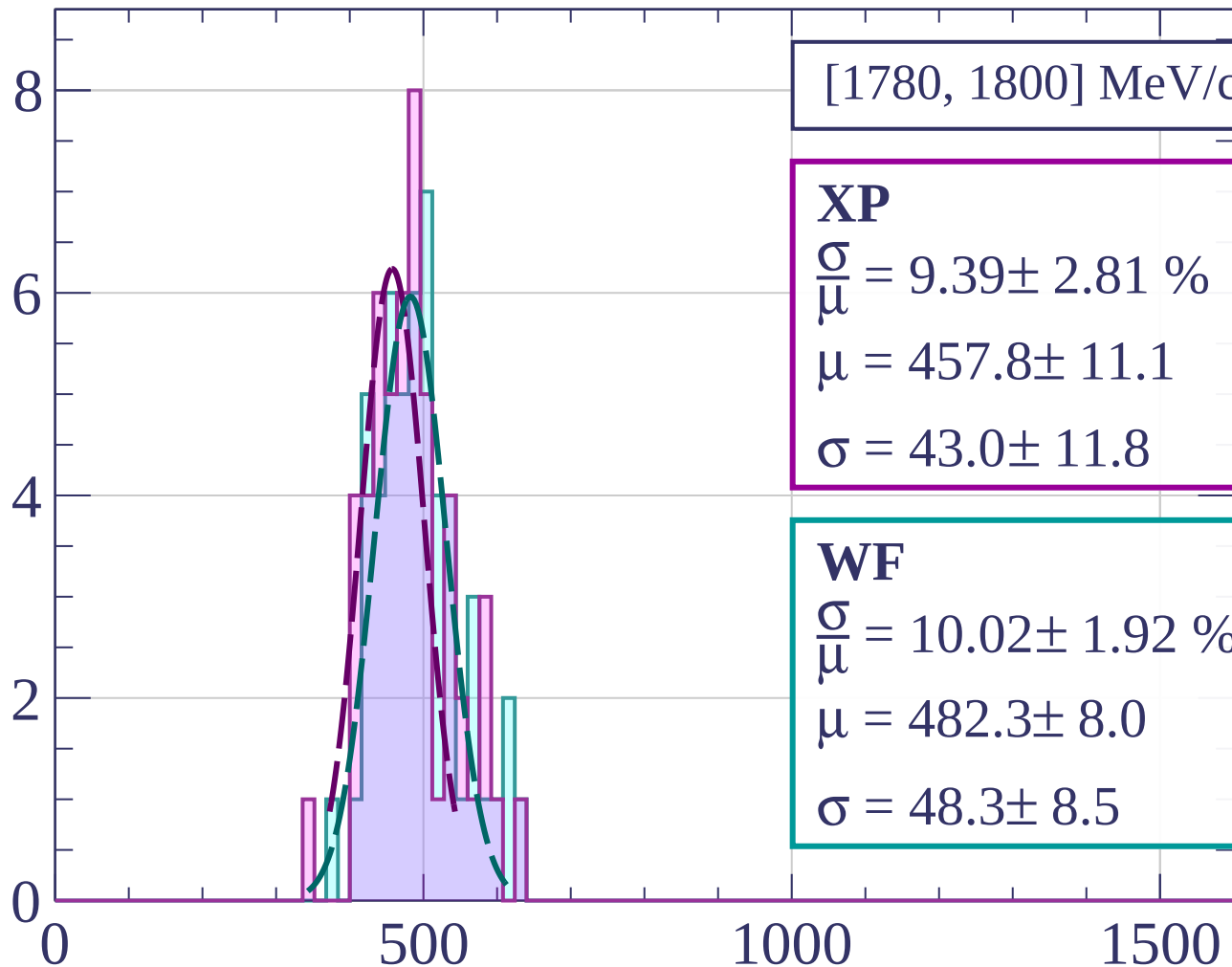


Count



dE/dx [ADC counts/cm]

Count



[1780, 1800] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.39 \pm 2.81 \%$$

$$\mu = 457.8 \pm 11.1$$

$$\sigma = 43.0 \pm 11.8$$

WF

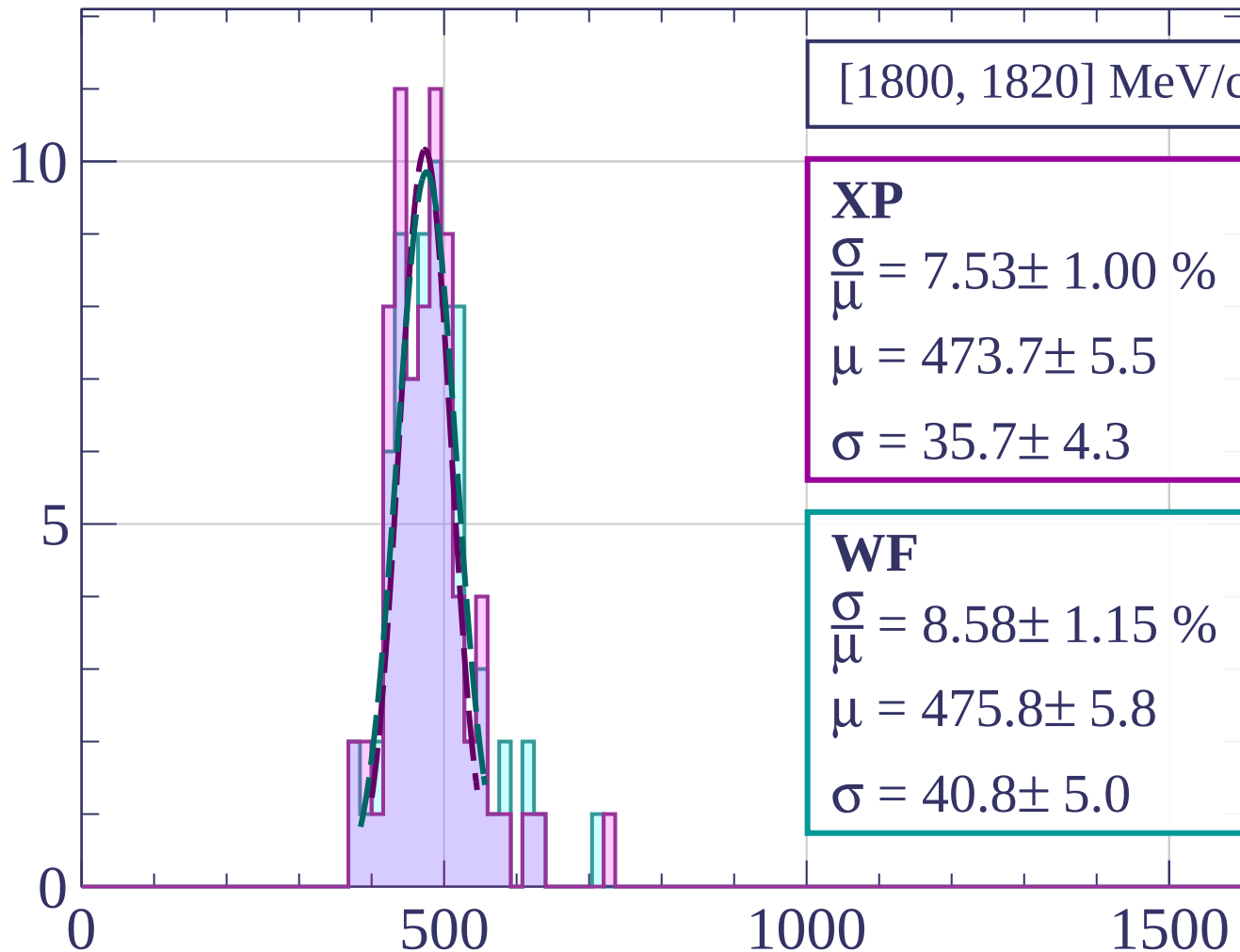
$$\frac{\sigma}{\mu} = 10.02 \pm 1.92 \%$$

$$\mu = 482.3 \pm 8.0$$

$$\sigma = 48.3 \pm 8.5$$

dE/dx [ADC counts/cm]

Count



[1800, 1820] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.53 \pm 1.00 \%$$

$$\mu = 473.7 \pm 5.5$$

$$\sigma = 35.7 \pm 4.3$$

WF

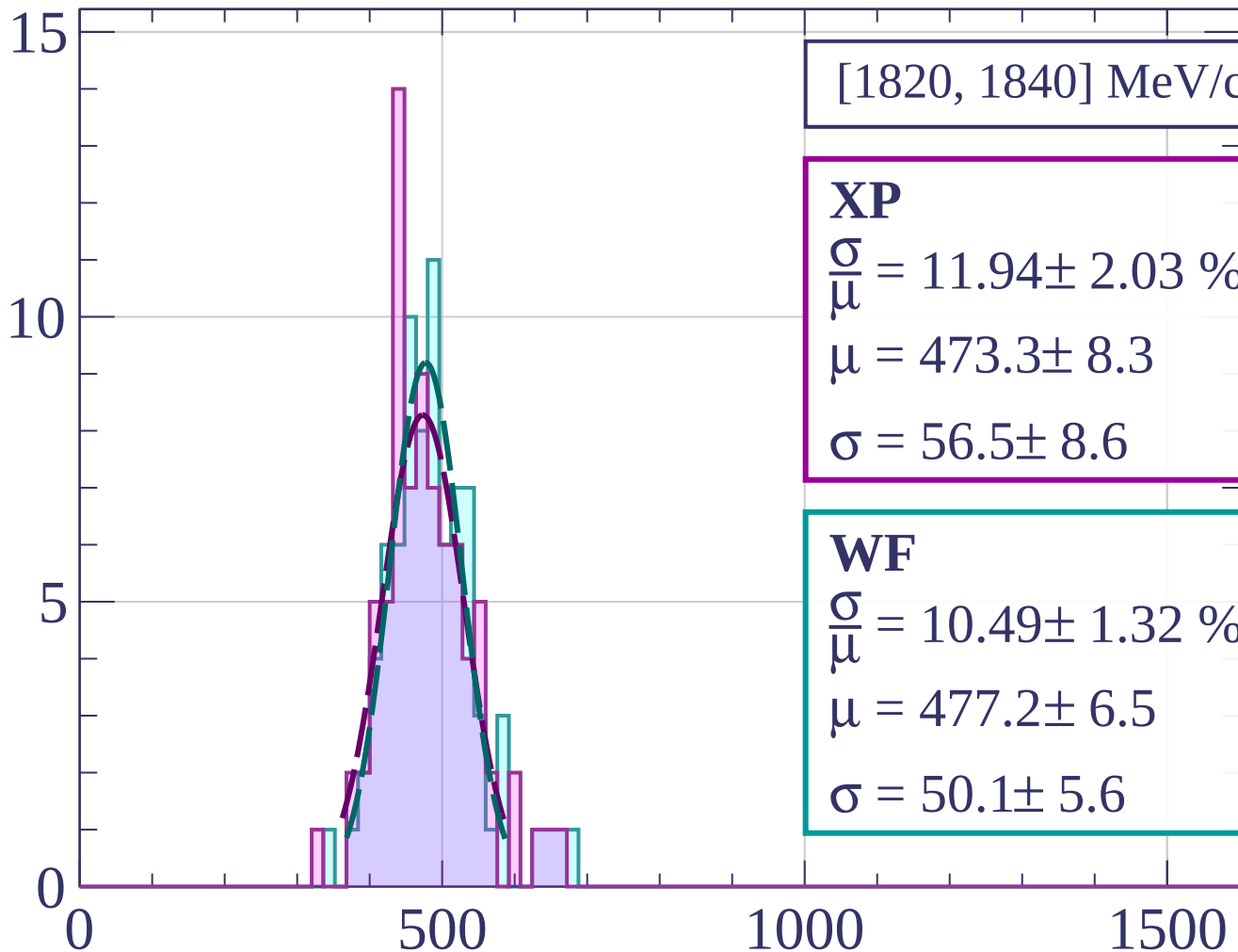
$$\frac{\sigma}{\mu} = 8.58 \pm 1.15 \%$$

$$\mu = 475.8 \pm 5.8$$

$$\sigma = 40.8 \pm 5.0$$

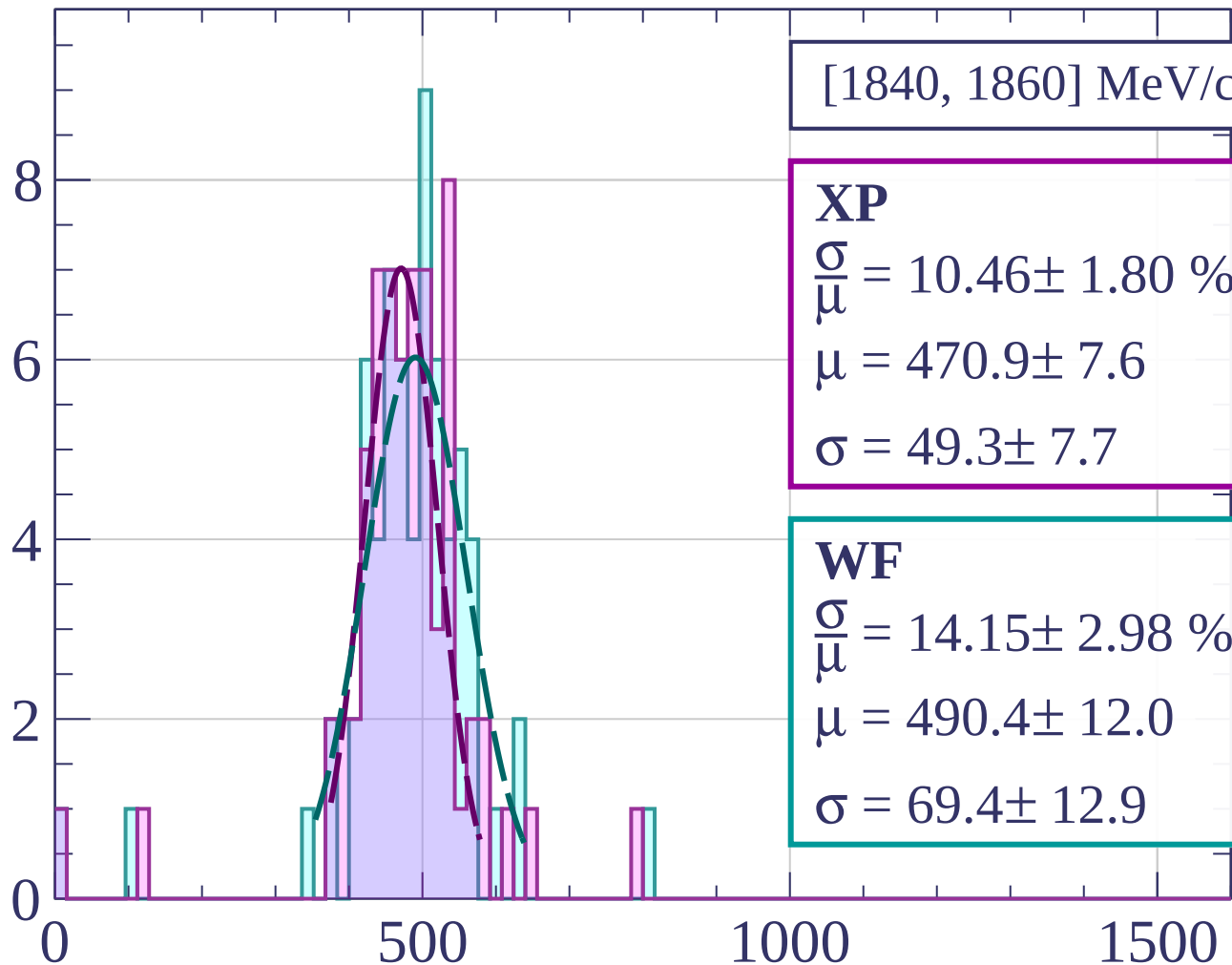
dE/dx [ADC counts/cm]

Count



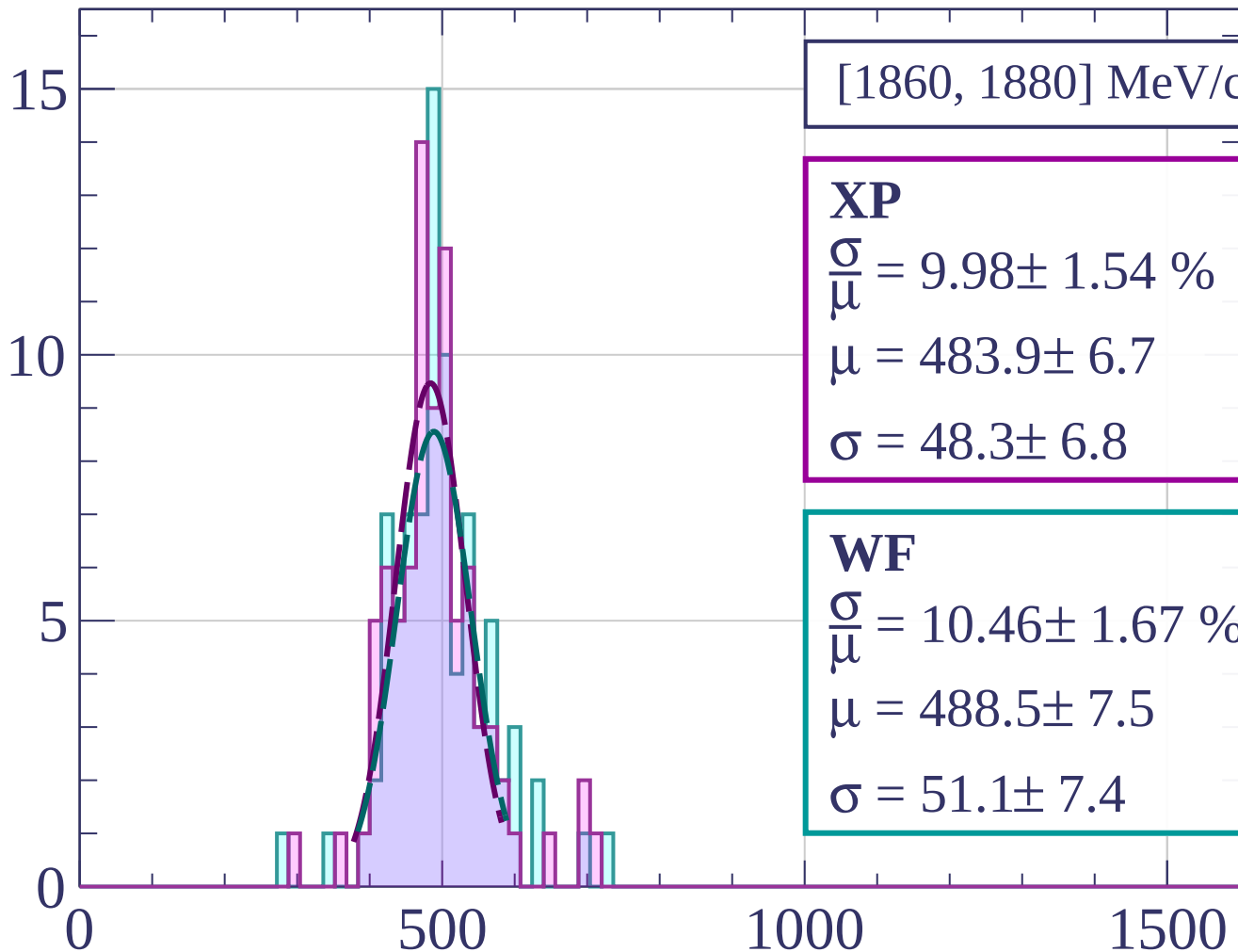
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[1860, 1880] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.98 \pm 1.54 \%$$

$$\mu = 483.9 \pm 6.7$$

$$\sigma = 48.3 \pm 6.8$$

WF

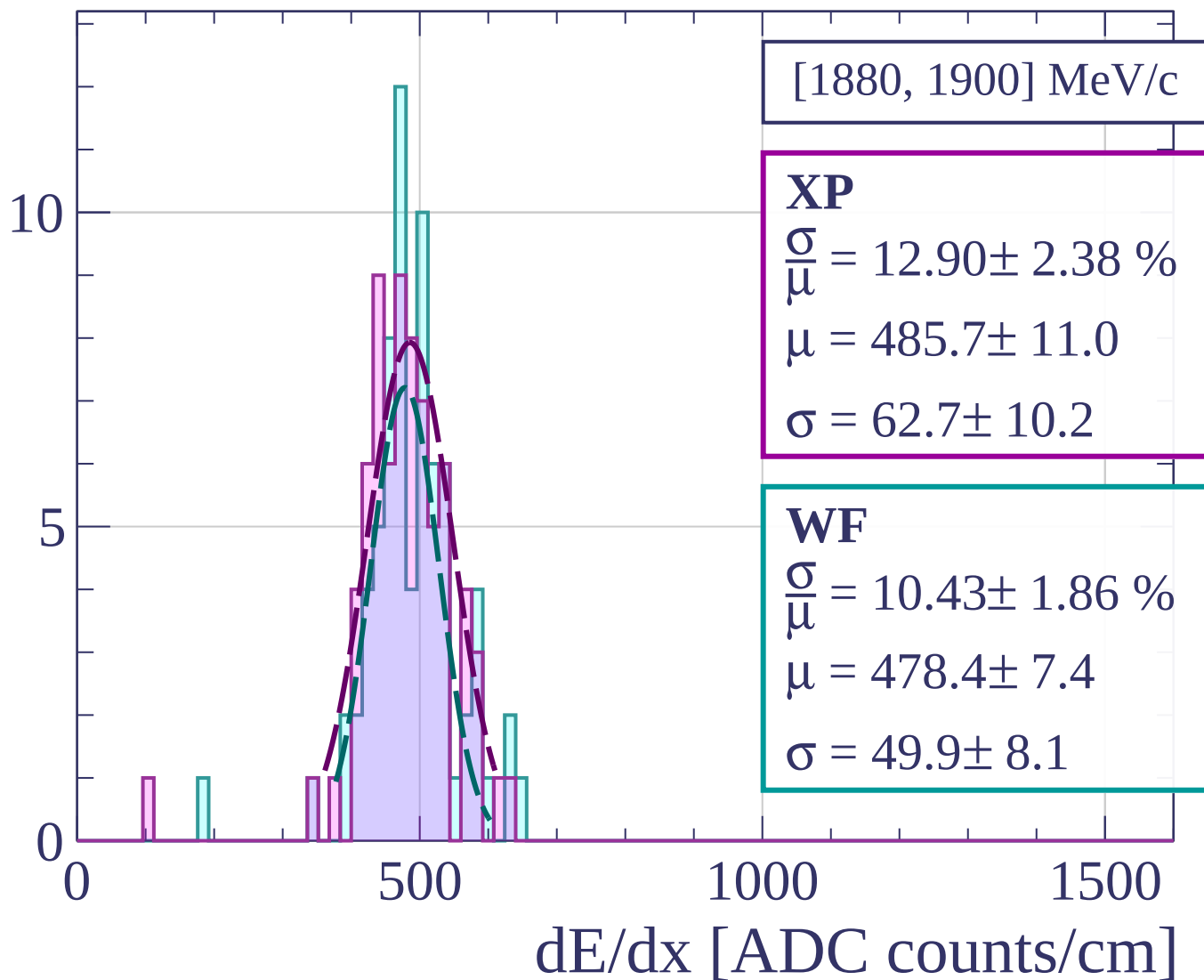
$$\frac{\sigma}{\mu} = 10.46 \pm 1.67 \%$$

$$\mu = 488.5 \pm 7.5$$

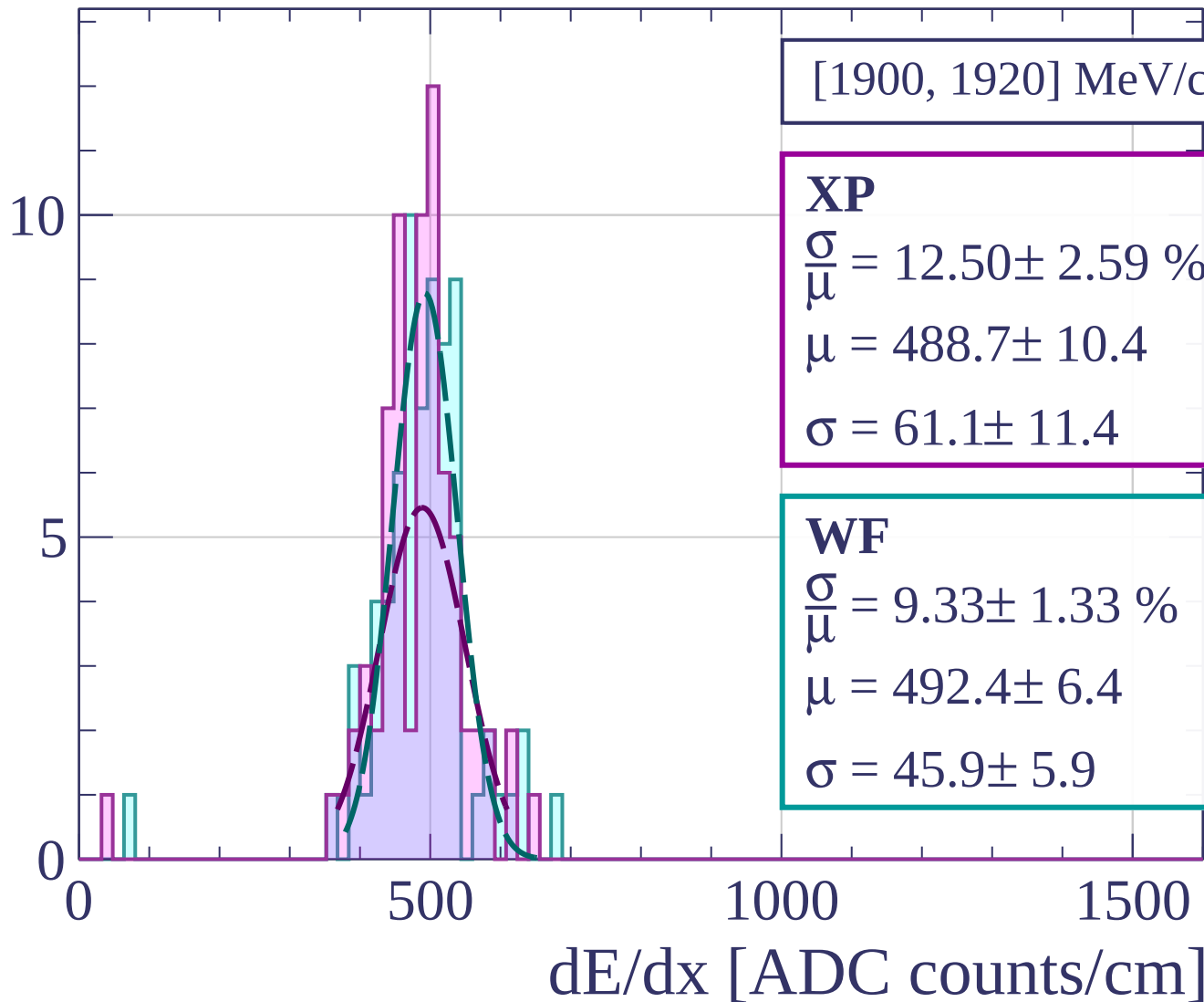
$$\sigma = 51.1 \pm 7.4$$

dE/dx [ADC counts/cm]

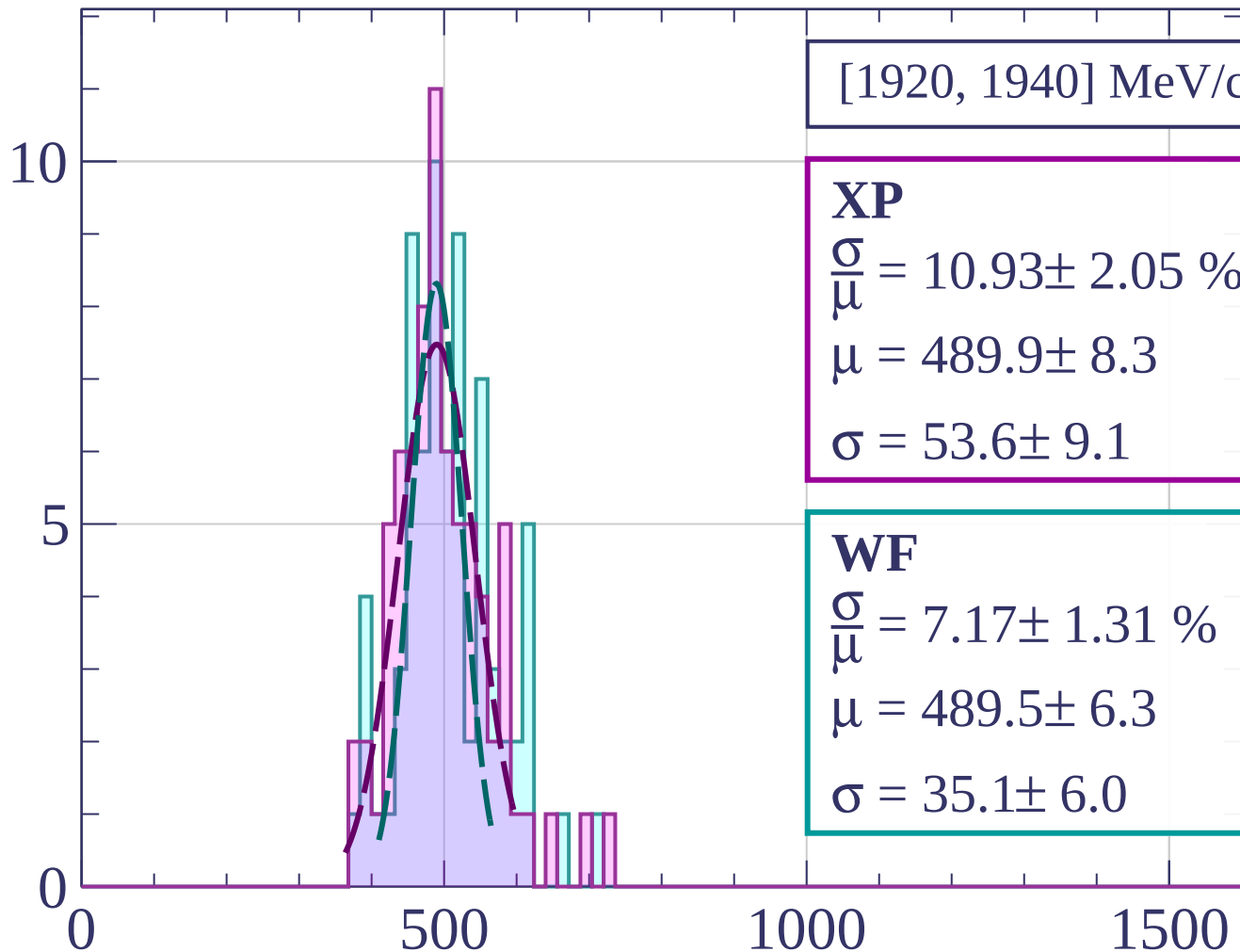
Count



Count



Count



[1920, 1940] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.93 \pm 2.05 \%$$

$$\mu = 489.9 \pm 8.3$$

$$\sigma = 53.6 \pm 9.1$$

WF

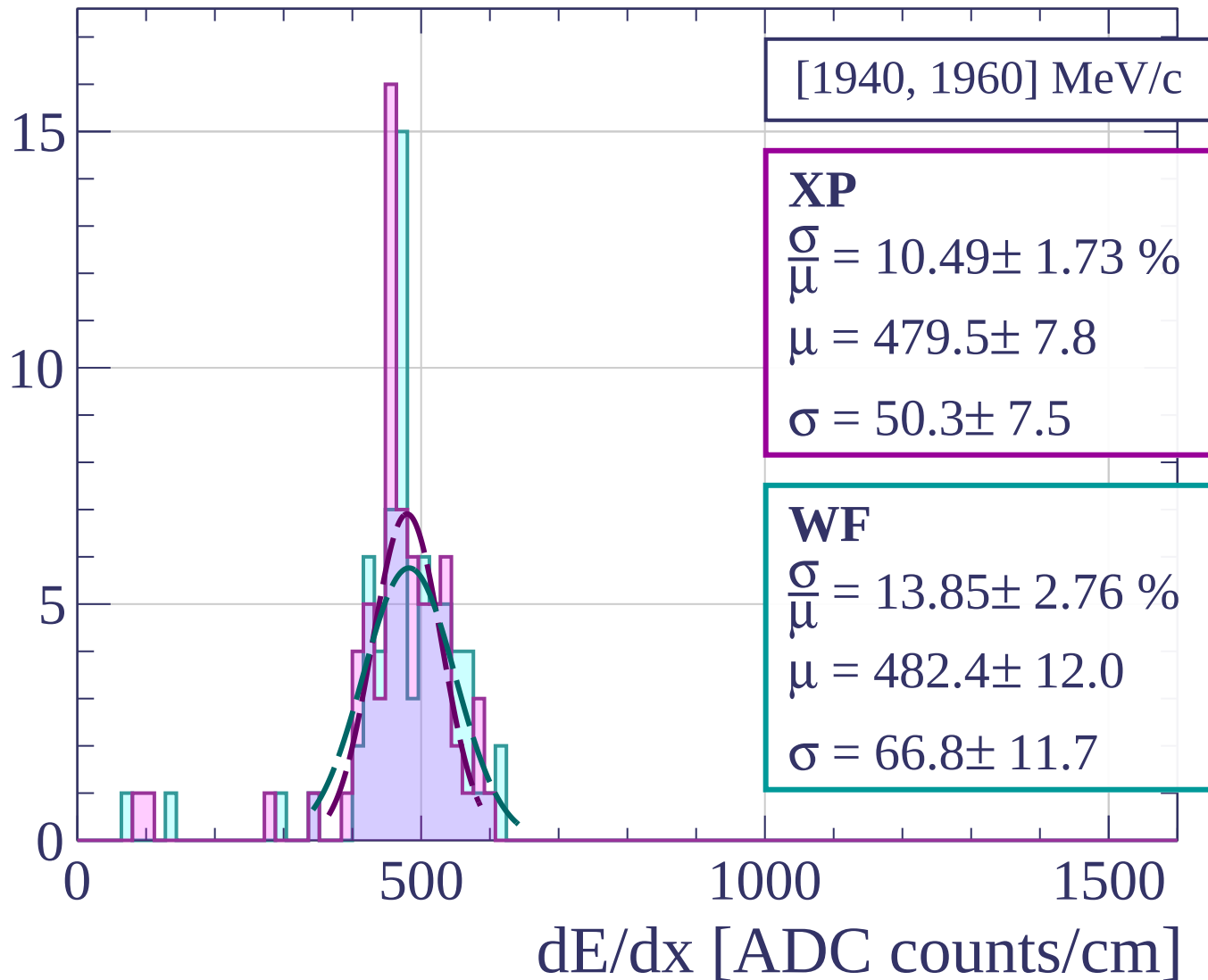
$$\frac{\sigma}{\mu} = 7.17 \pm 1.31 \%$$

$$\mu = 489.5 \pm 6.3$$

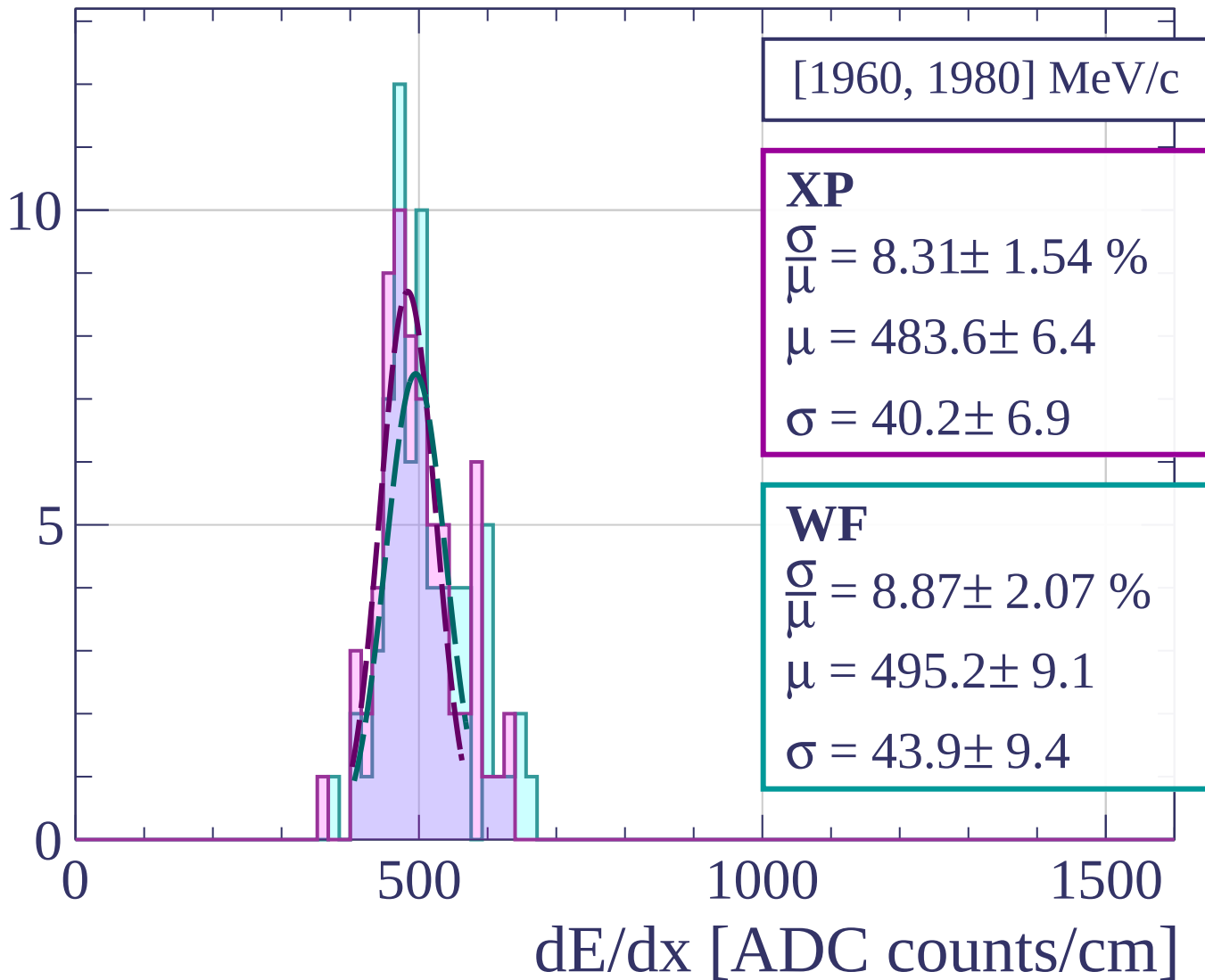
$$\sigma = 35.1 \pm 6.0$$

dE/dx [ADC counts/cm]

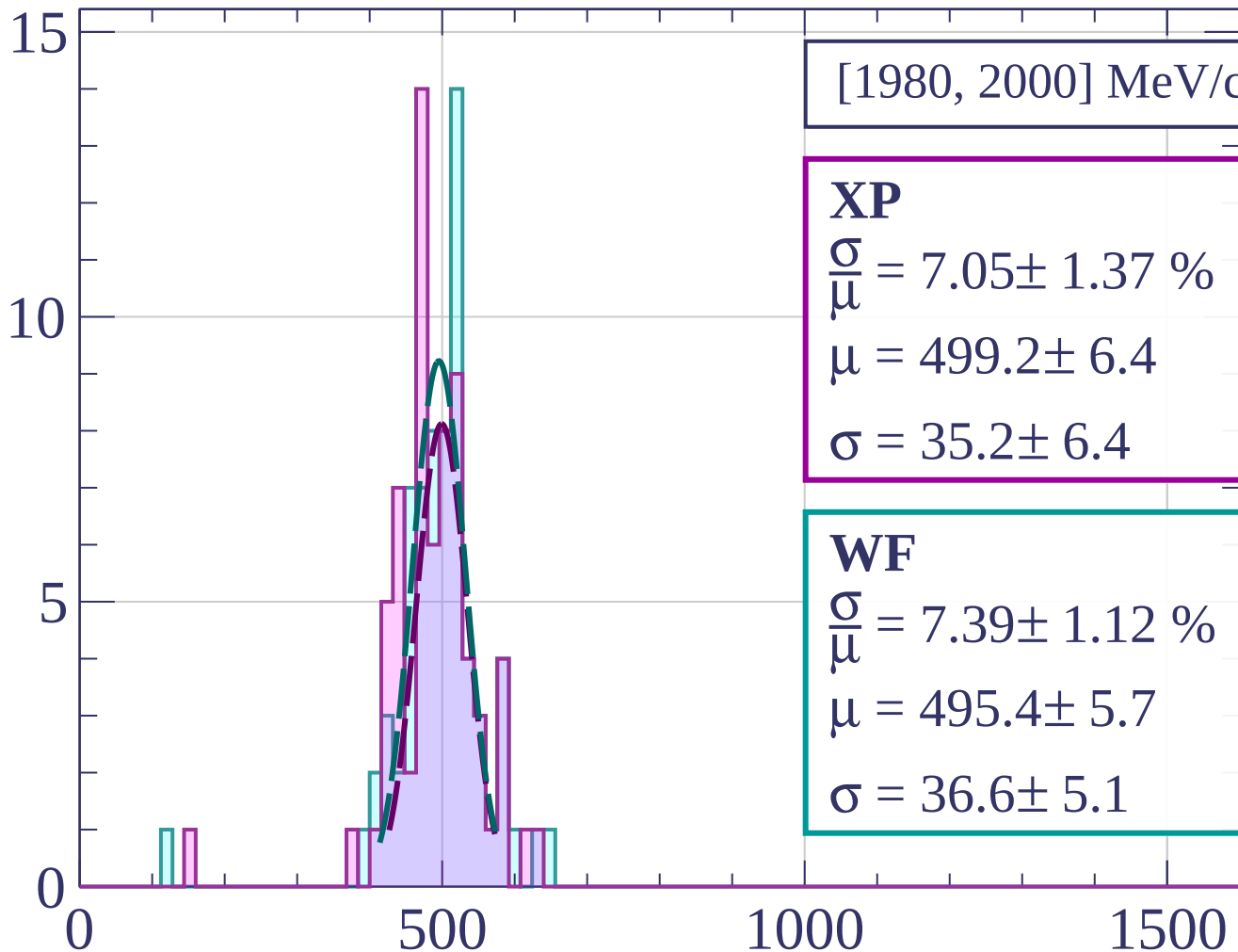
Count



Count

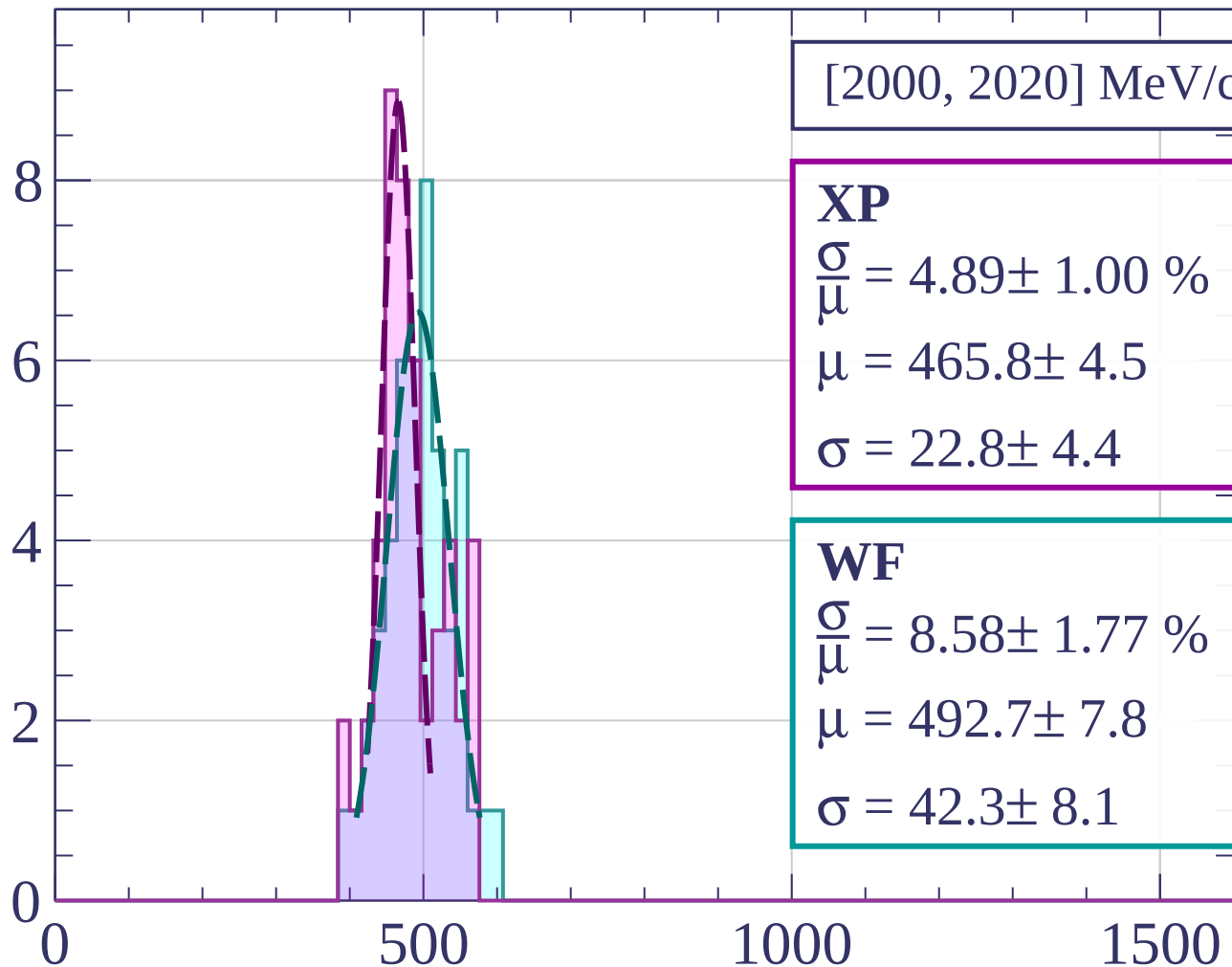


Count



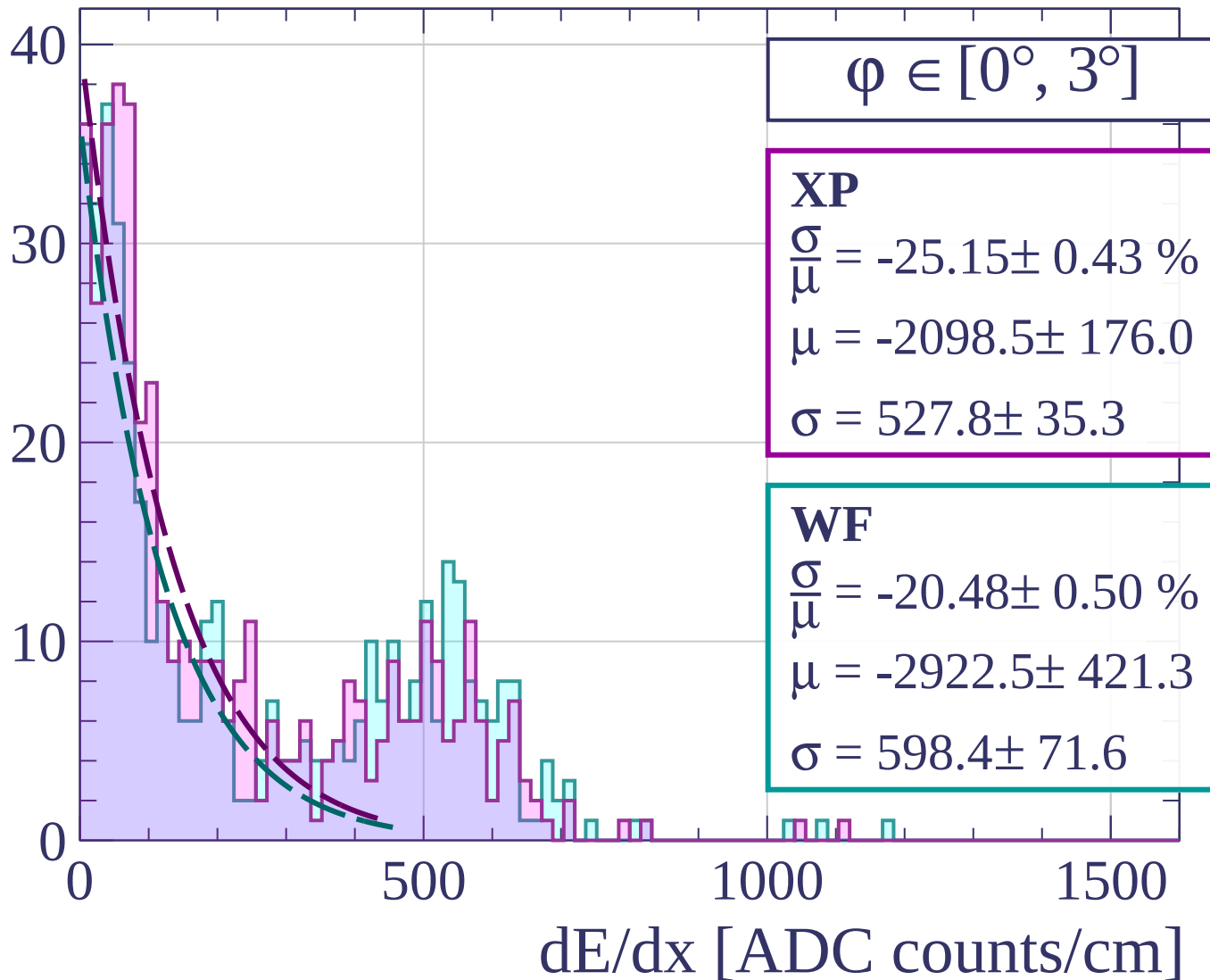
dE/dx [ADC counts/cm]

Count

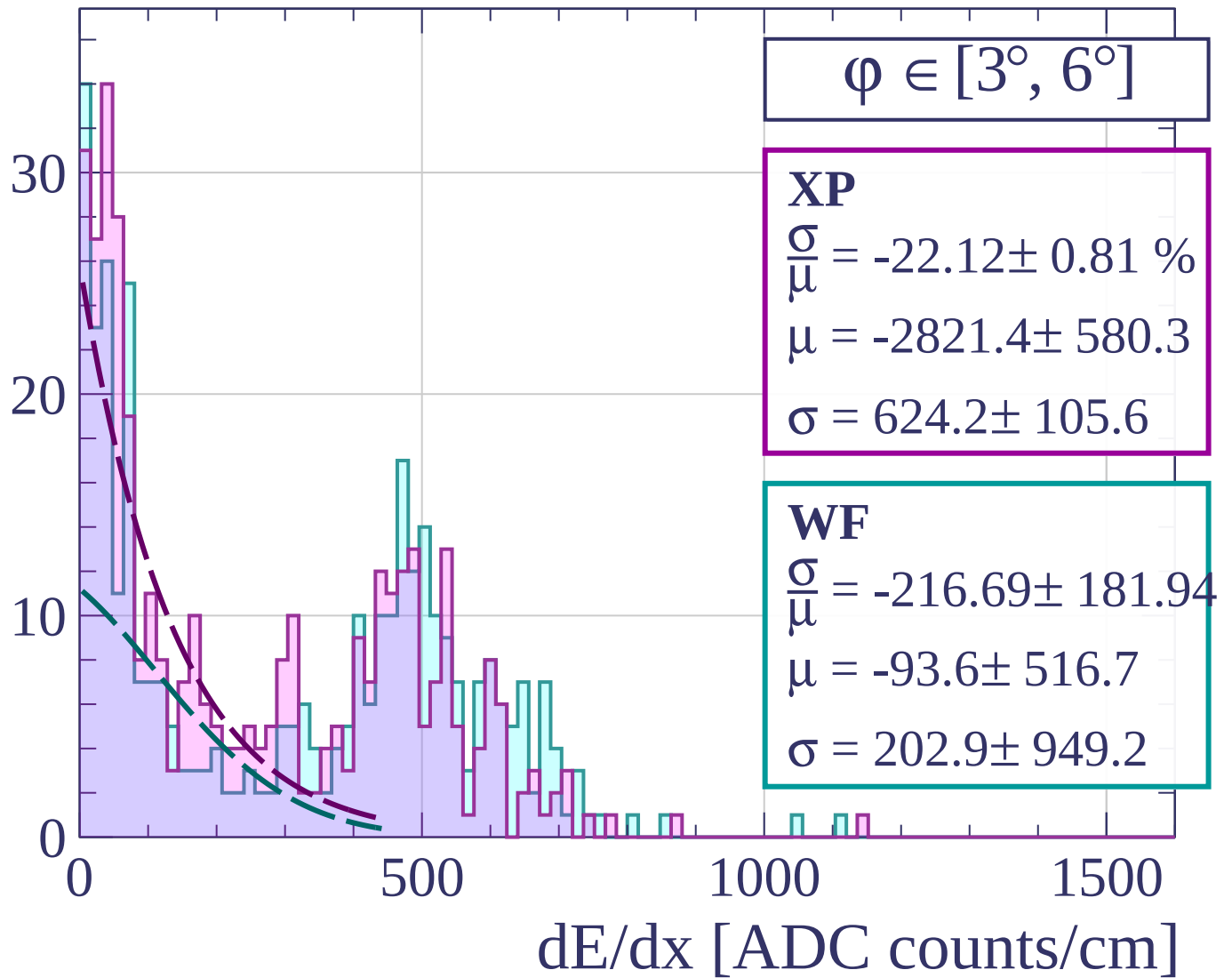


dE/dx [ADC counts/cm]

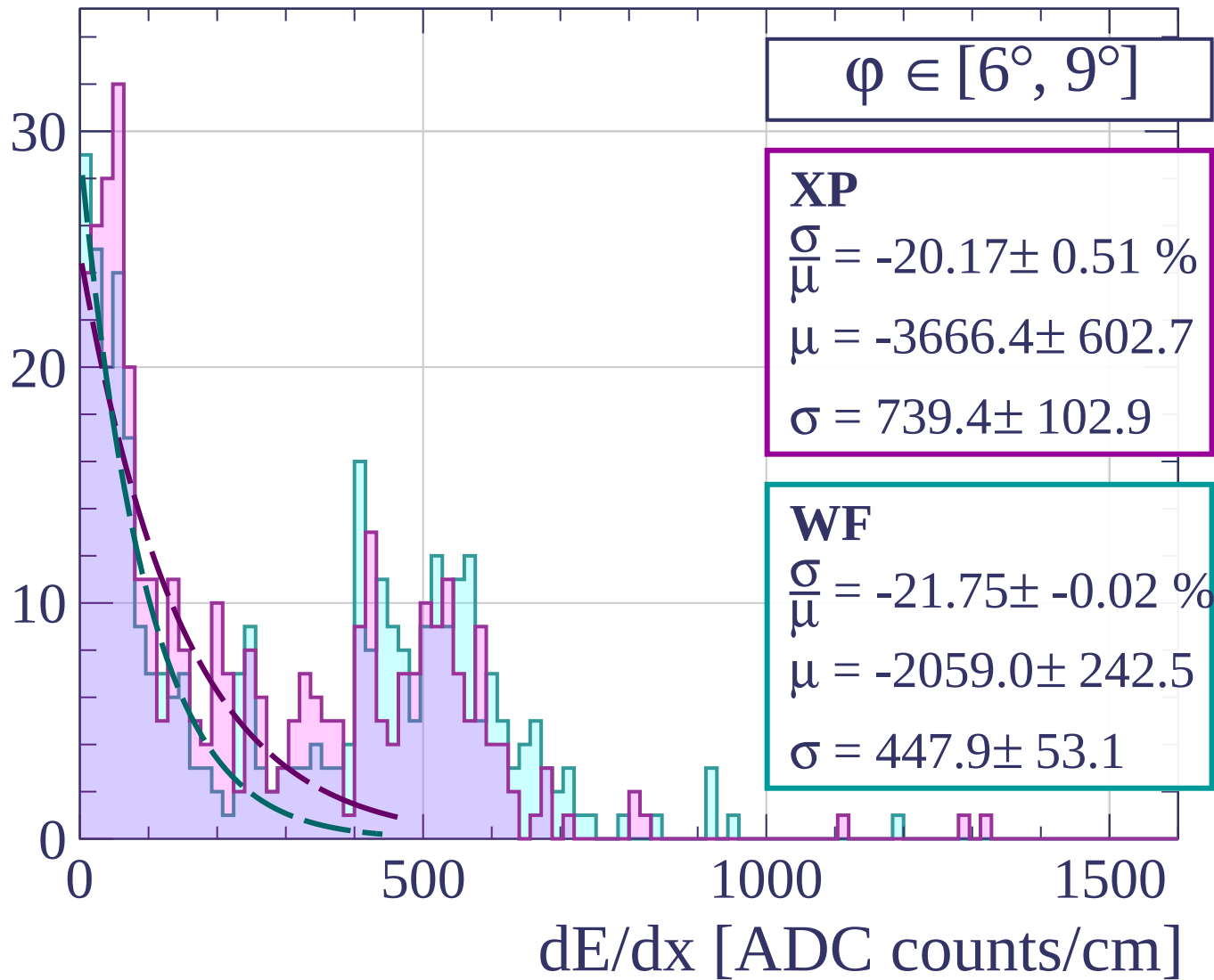
Count



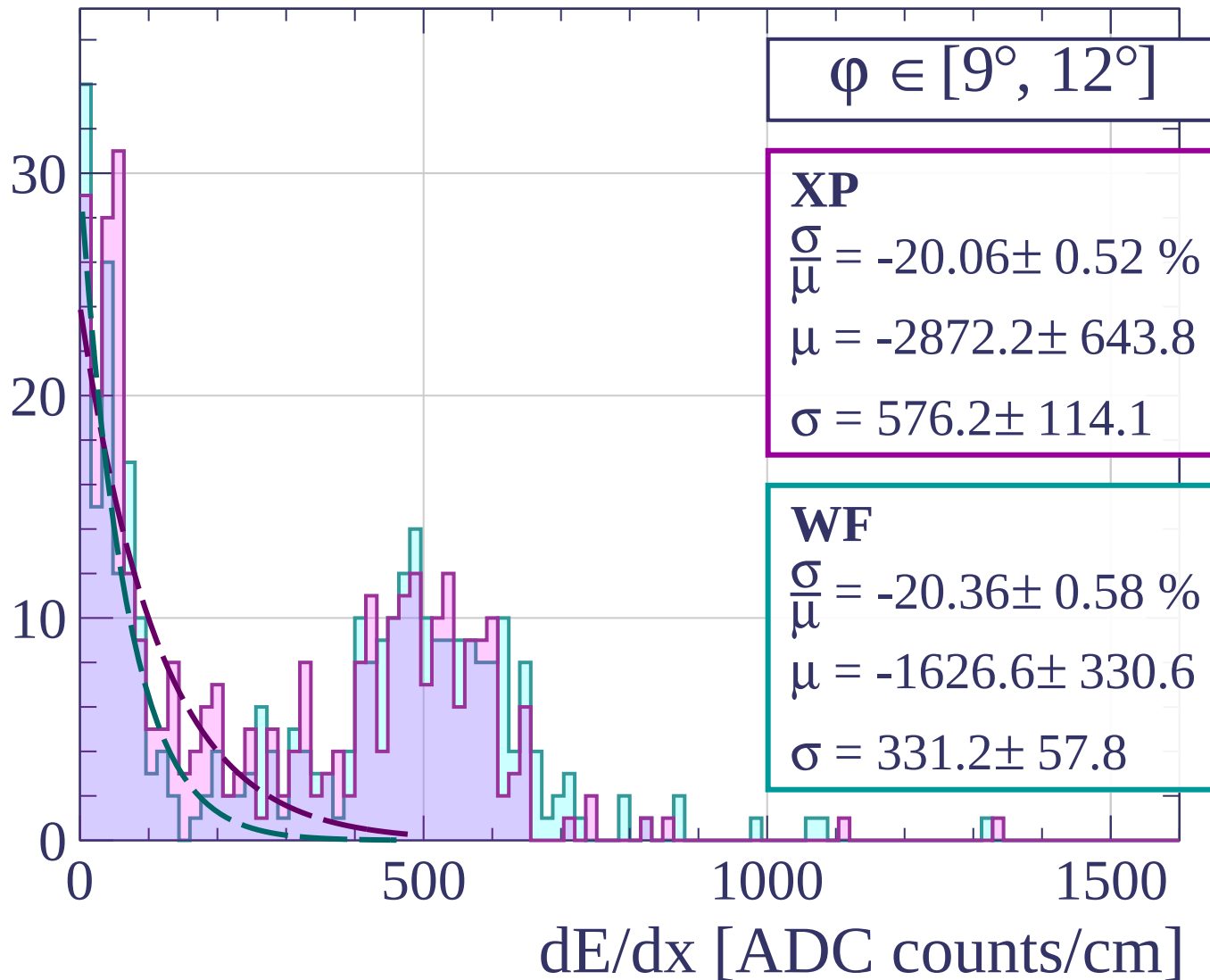
Count



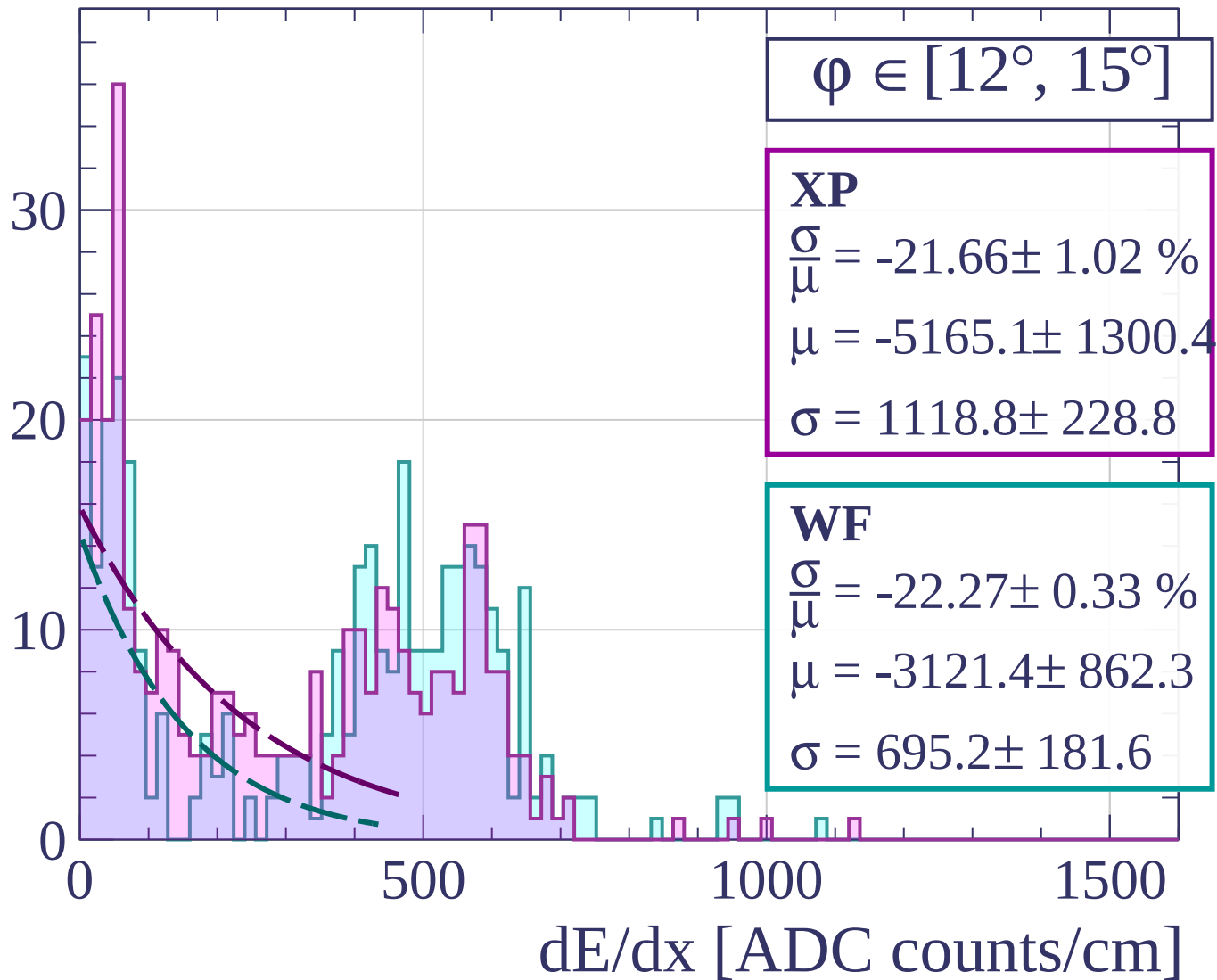
Count



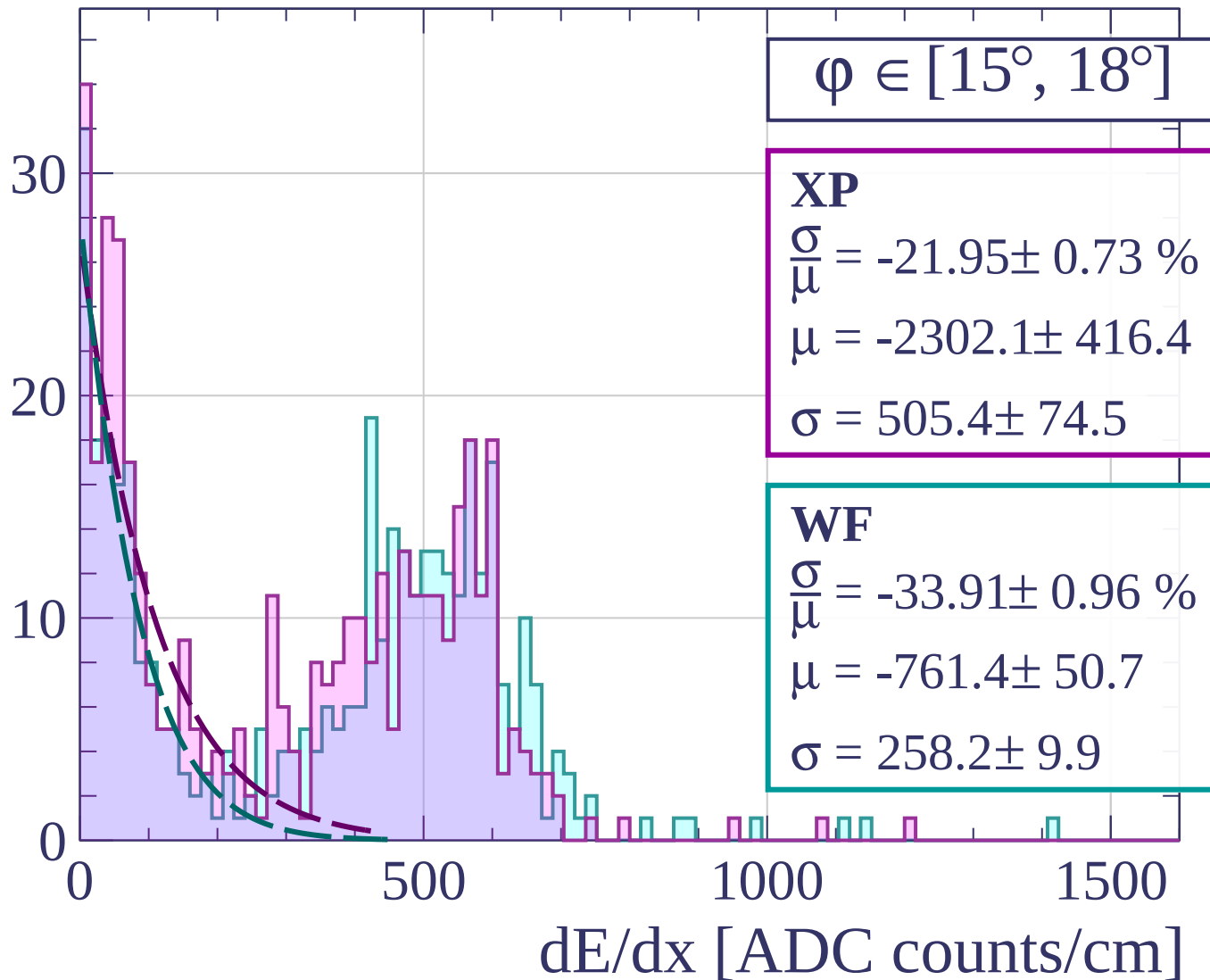
Count



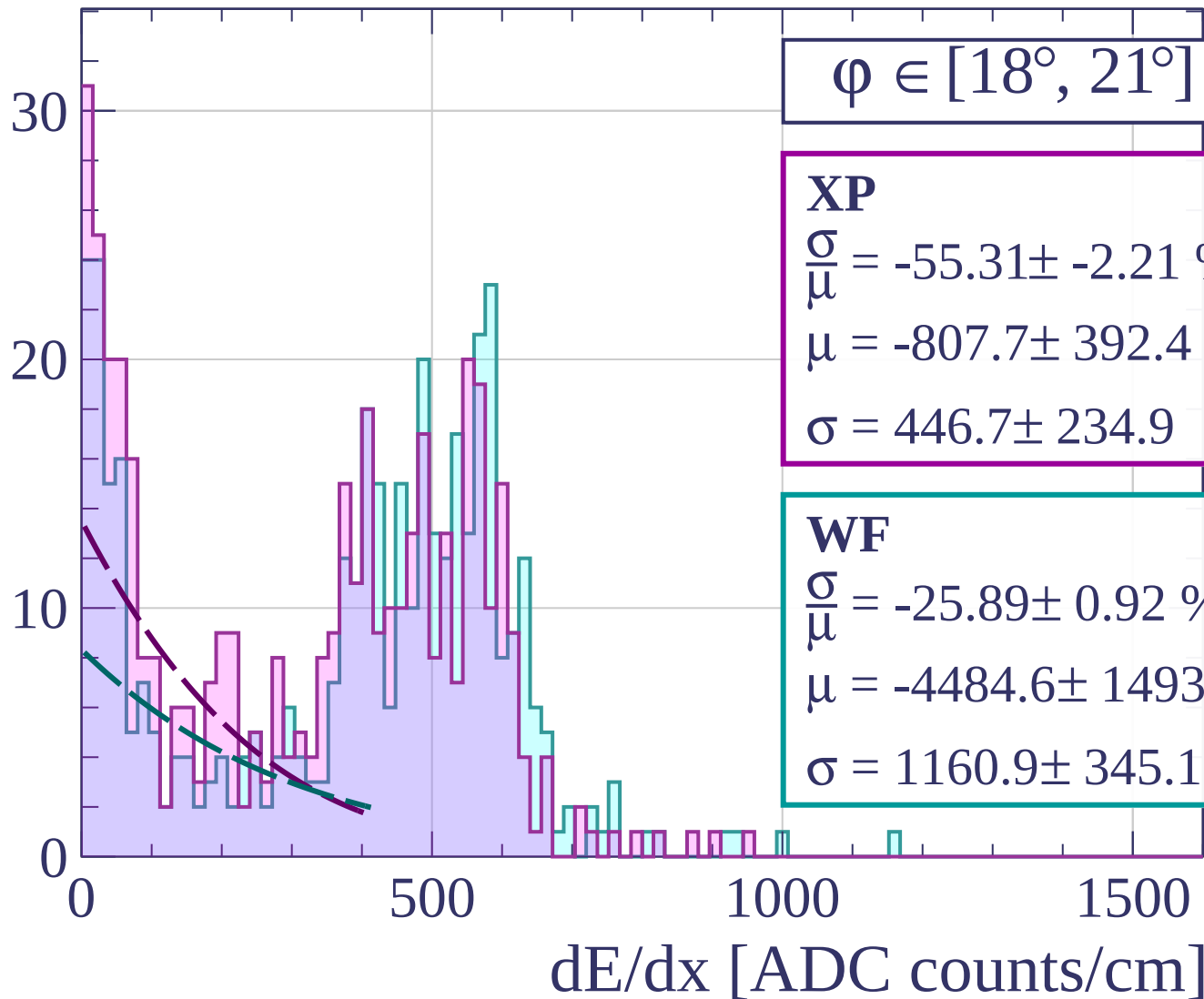
Count



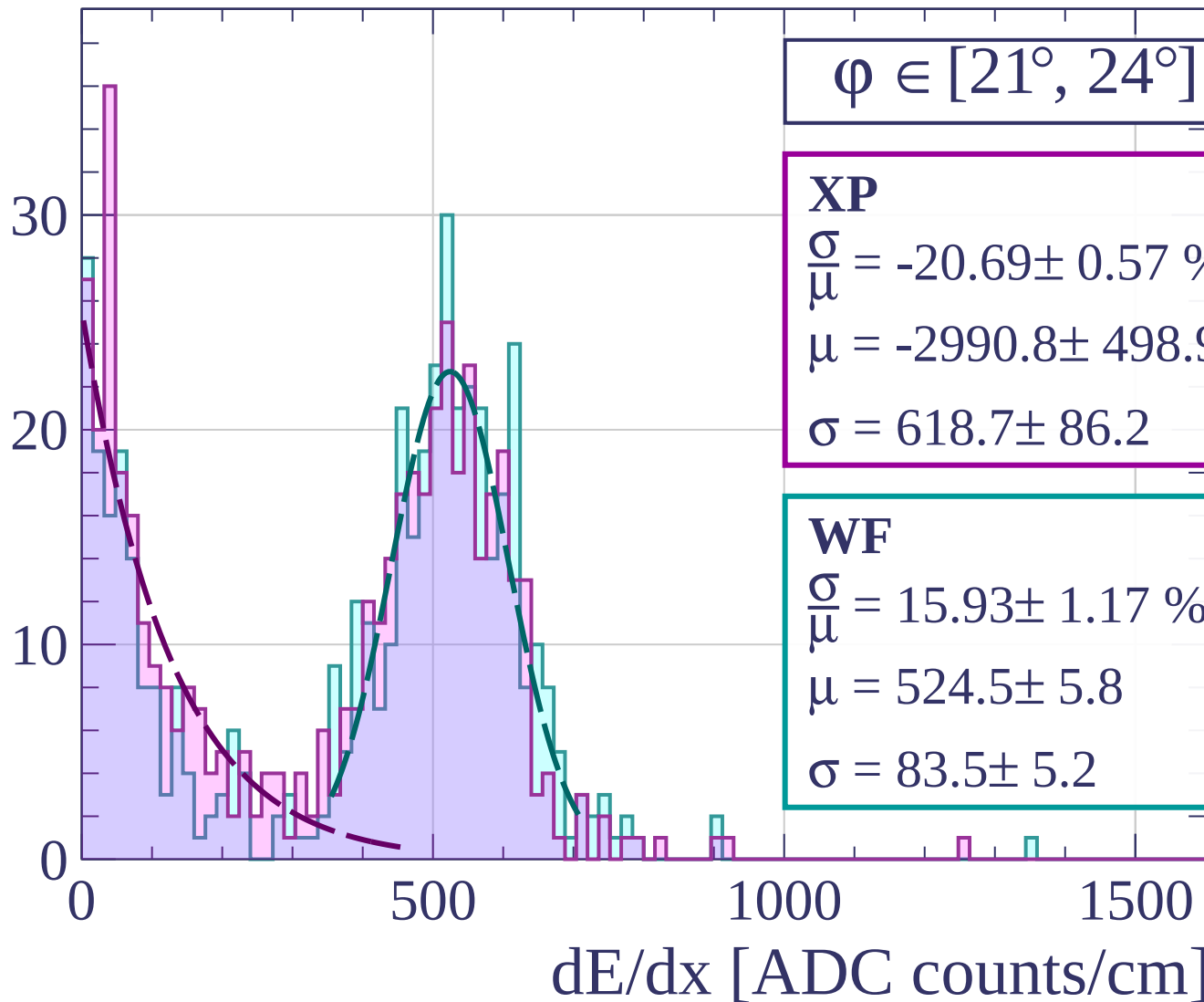
Count



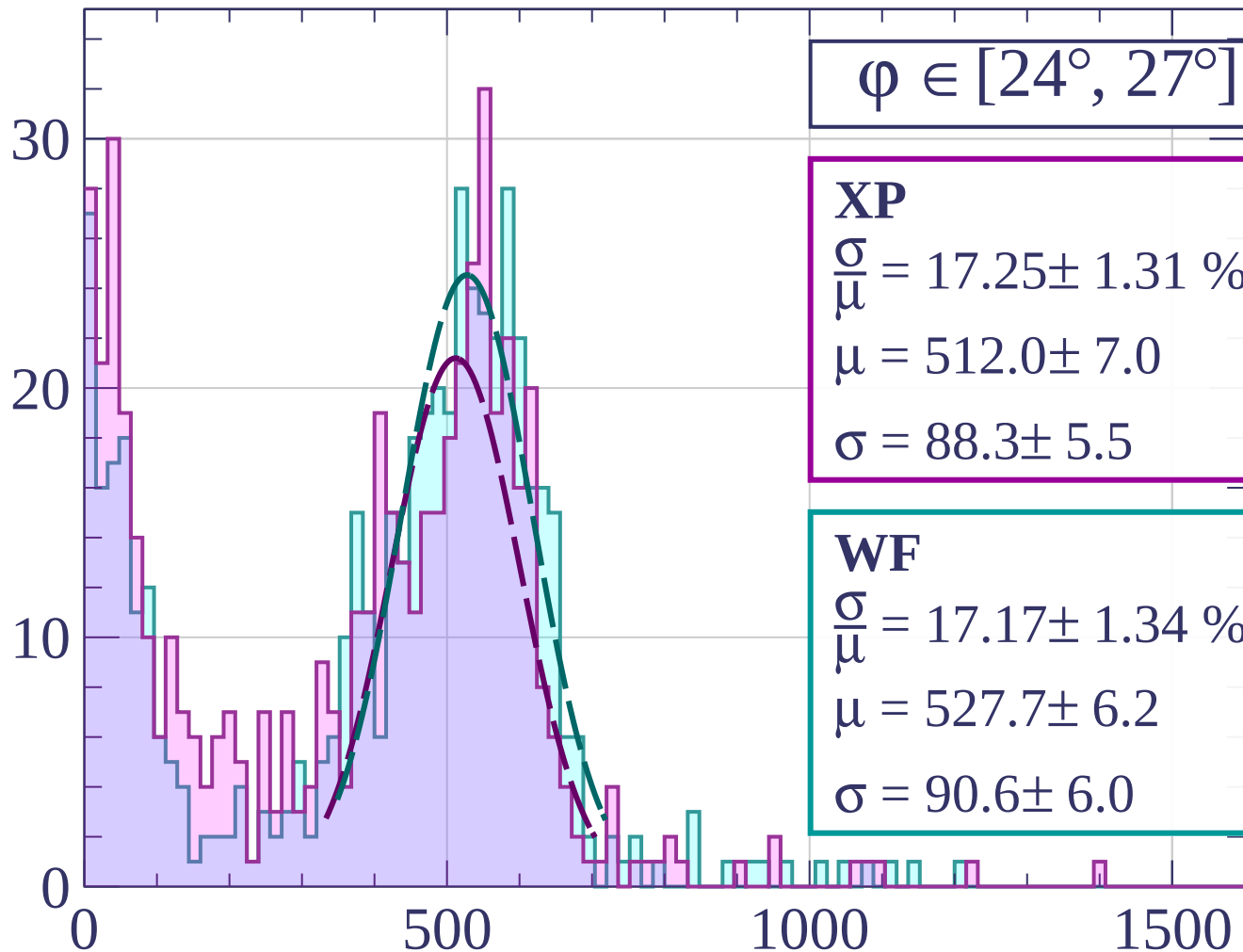
Count



Count



Count



dE/dx [ADC counts/cm]

Count

$\phi \in [27^\circ, 30^\circ]$

XP

$$\frac{\sigma}{\mu} = 17.14 \pm 1.11 \%$$

$$\mu = 501.8 \pm 5.4$$

$$\sigma = 86.0 \pm 4.7$$

WF

$$\frac{\sigma}{\mu} = 17.28 \pm 1.02 \%$$

$$\mu = 509.3 \pm 4.9$$

$$\sigma = 88.0 \pm 4.3$$

30

20

10

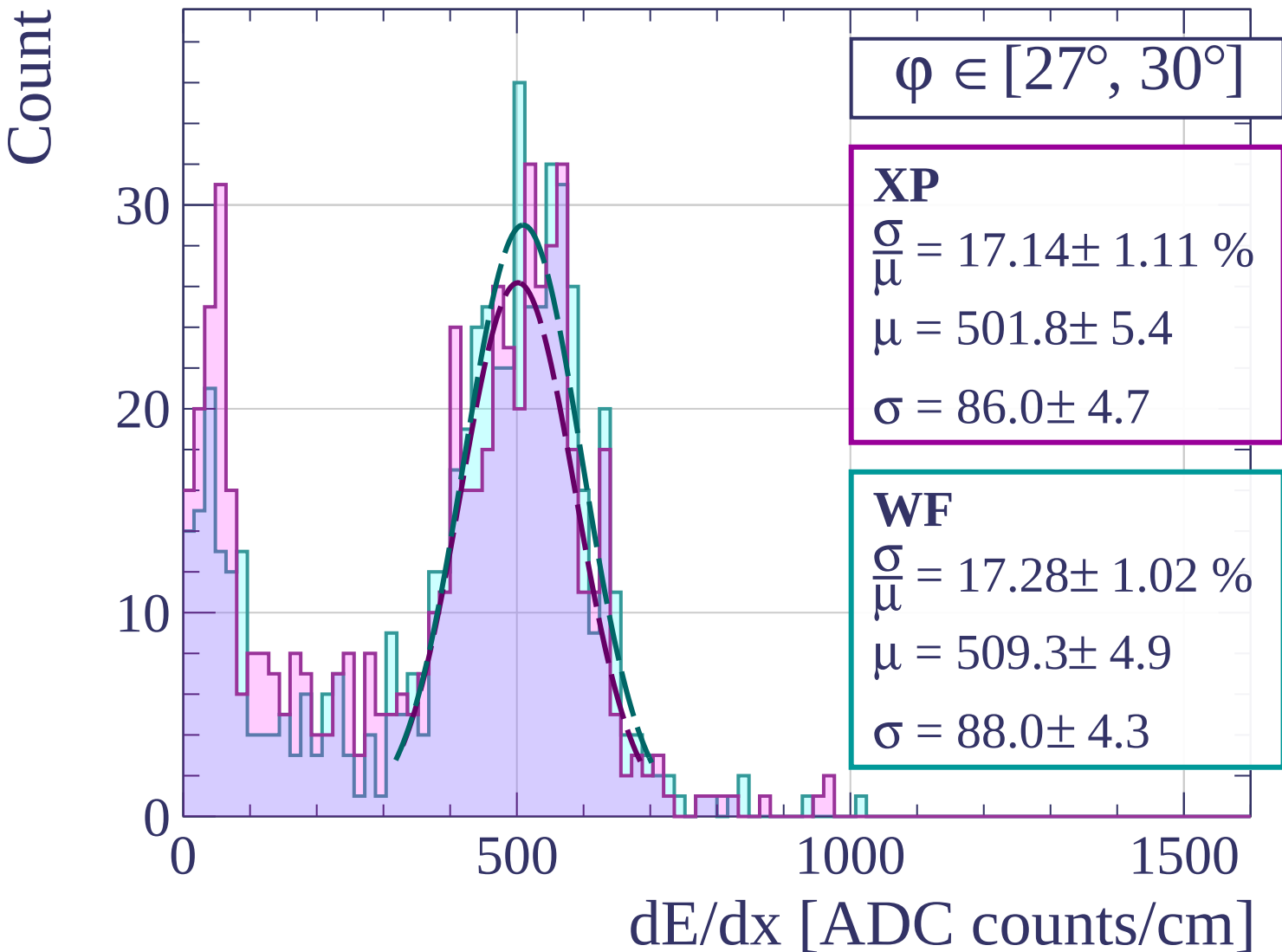
0

500

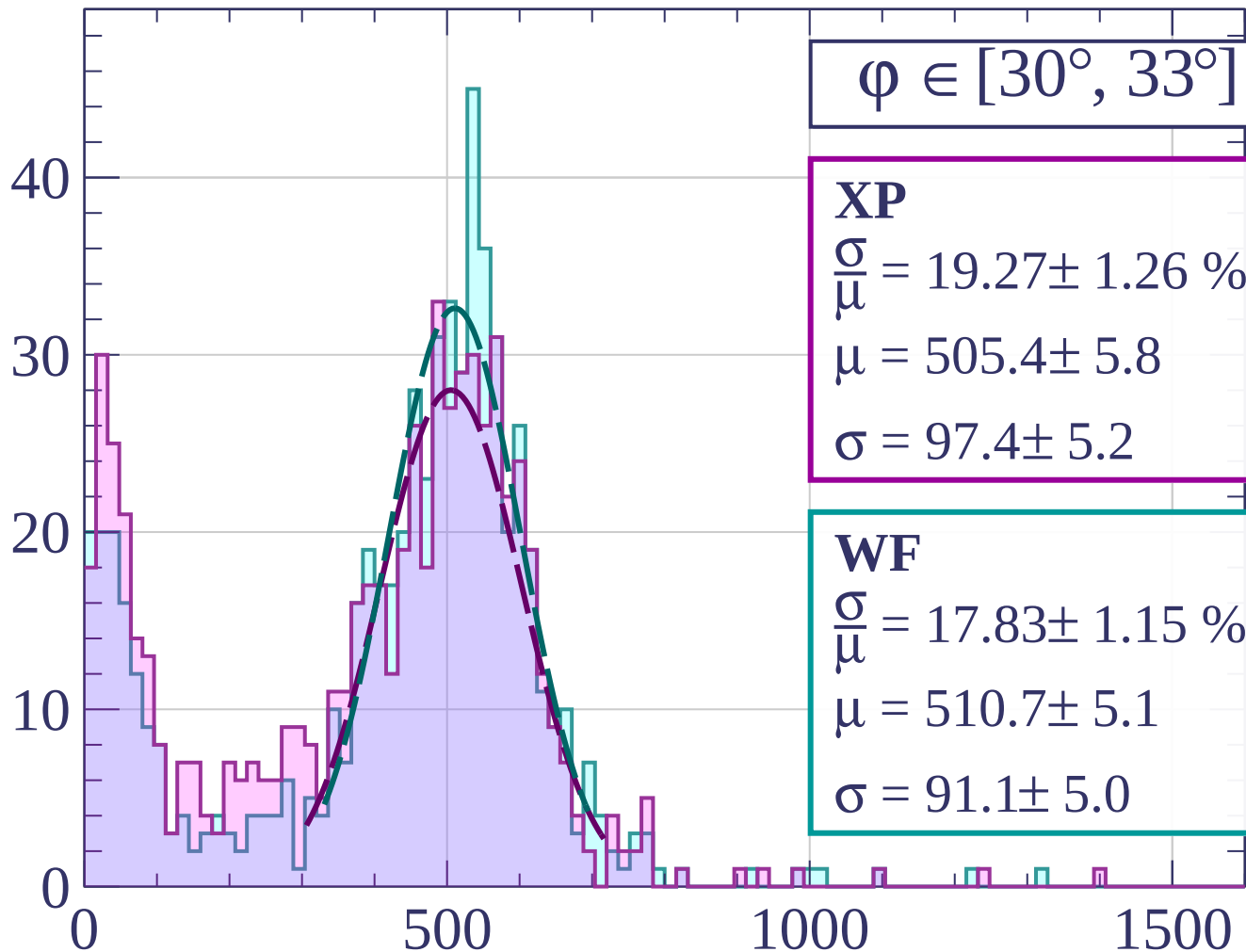
1000

1500

dE/dx [ADC counts/cm]

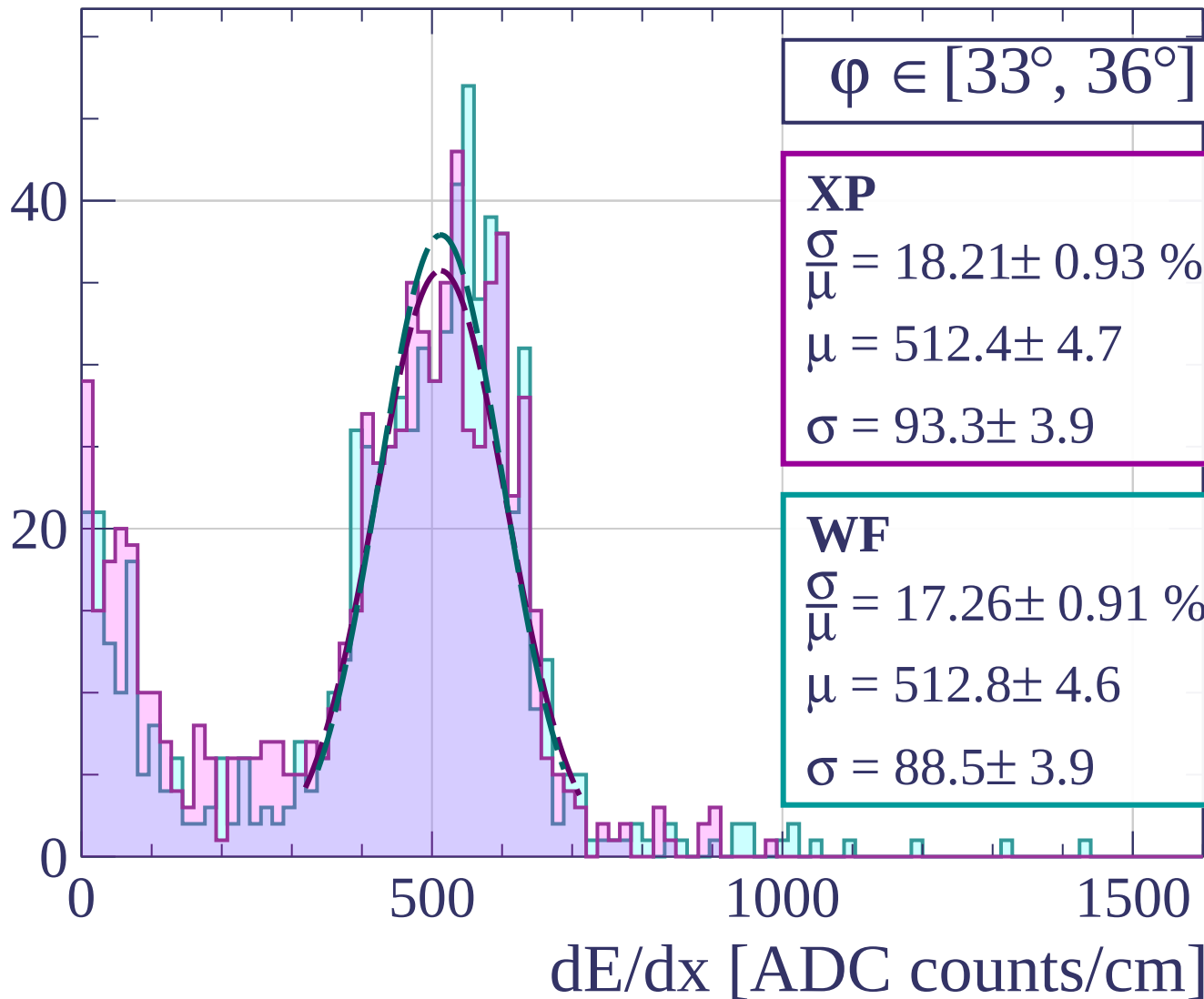


Count

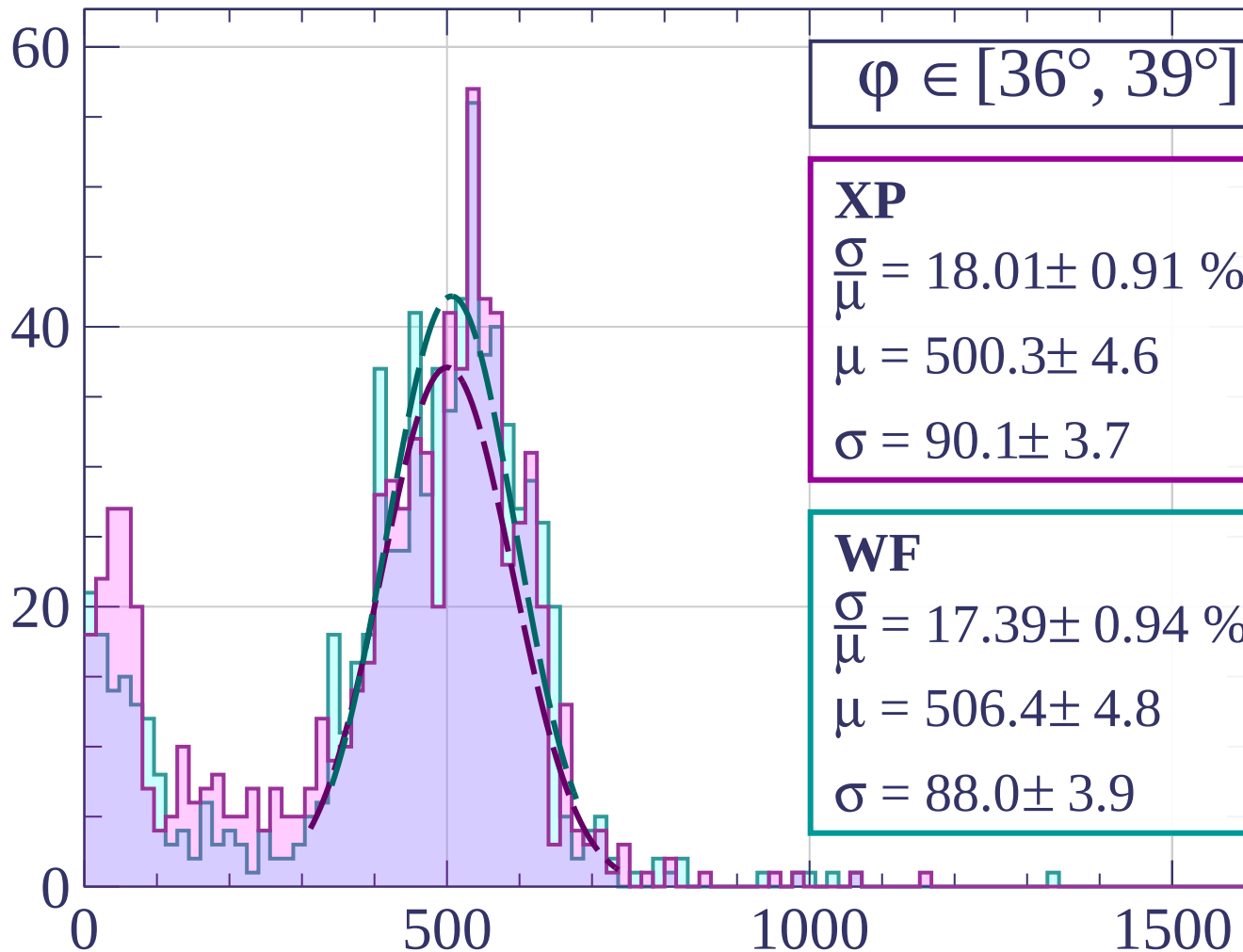


dE/dx [ADC counts/cm]

Count

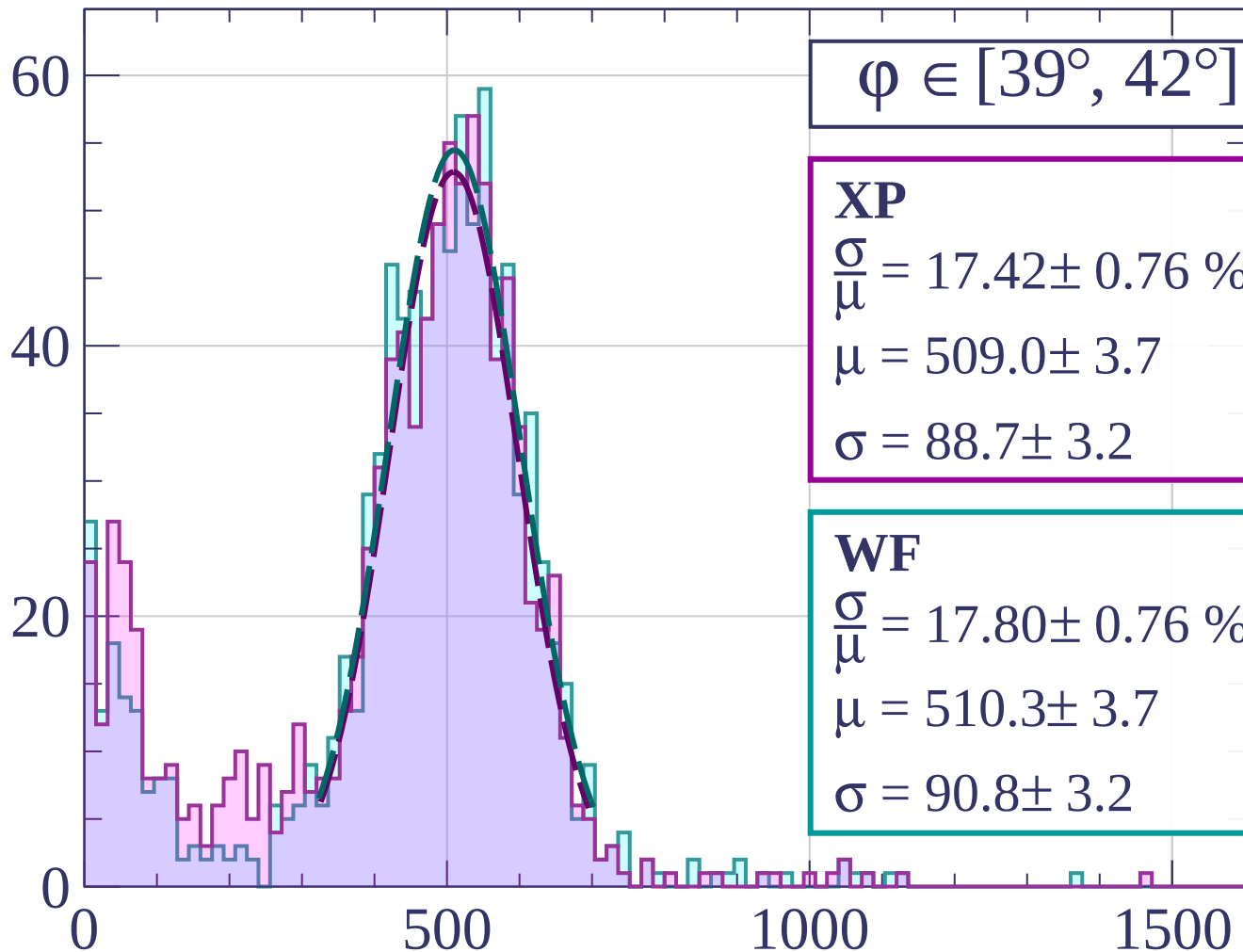


Count



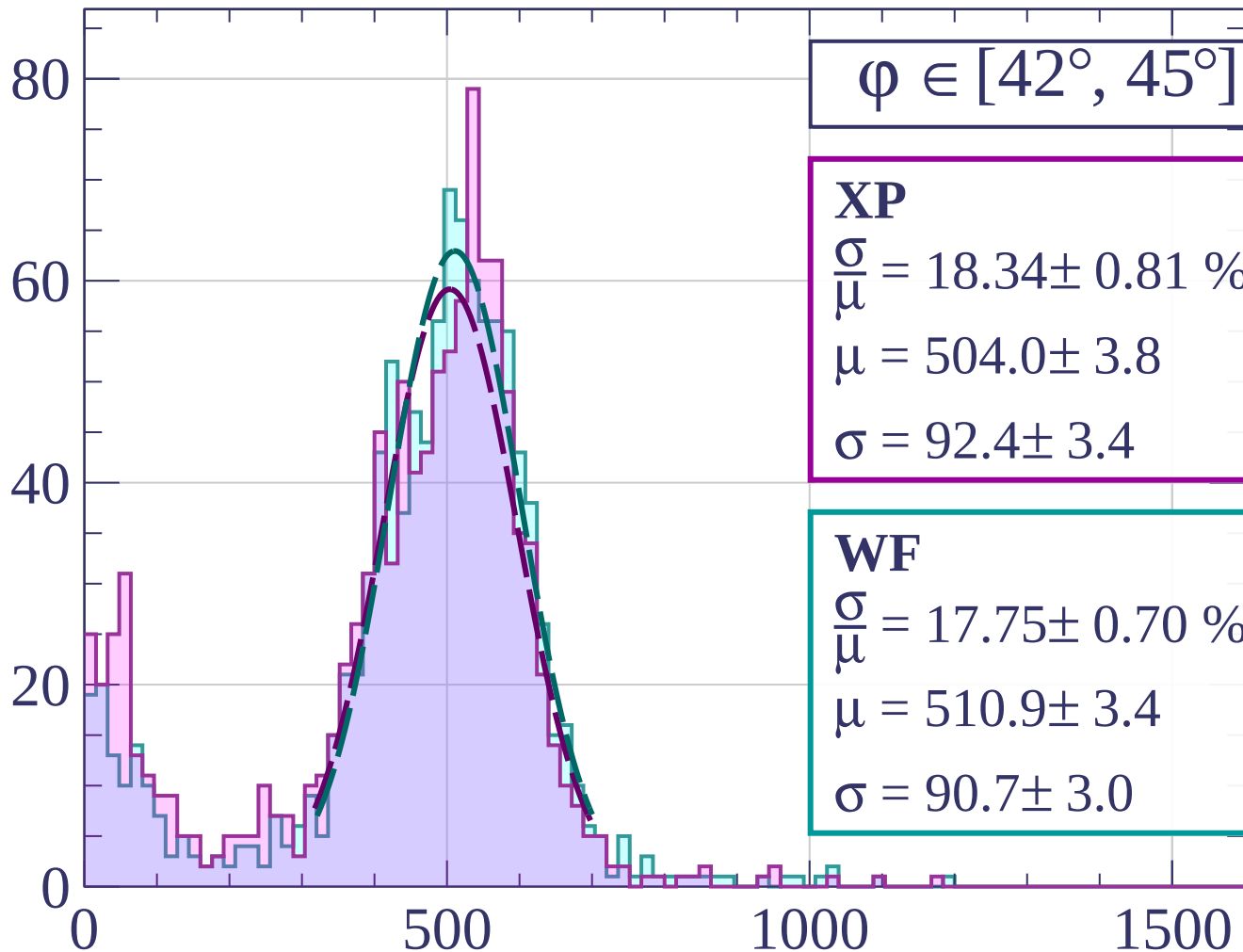
dE/dx [ADC counts/cm]

Count



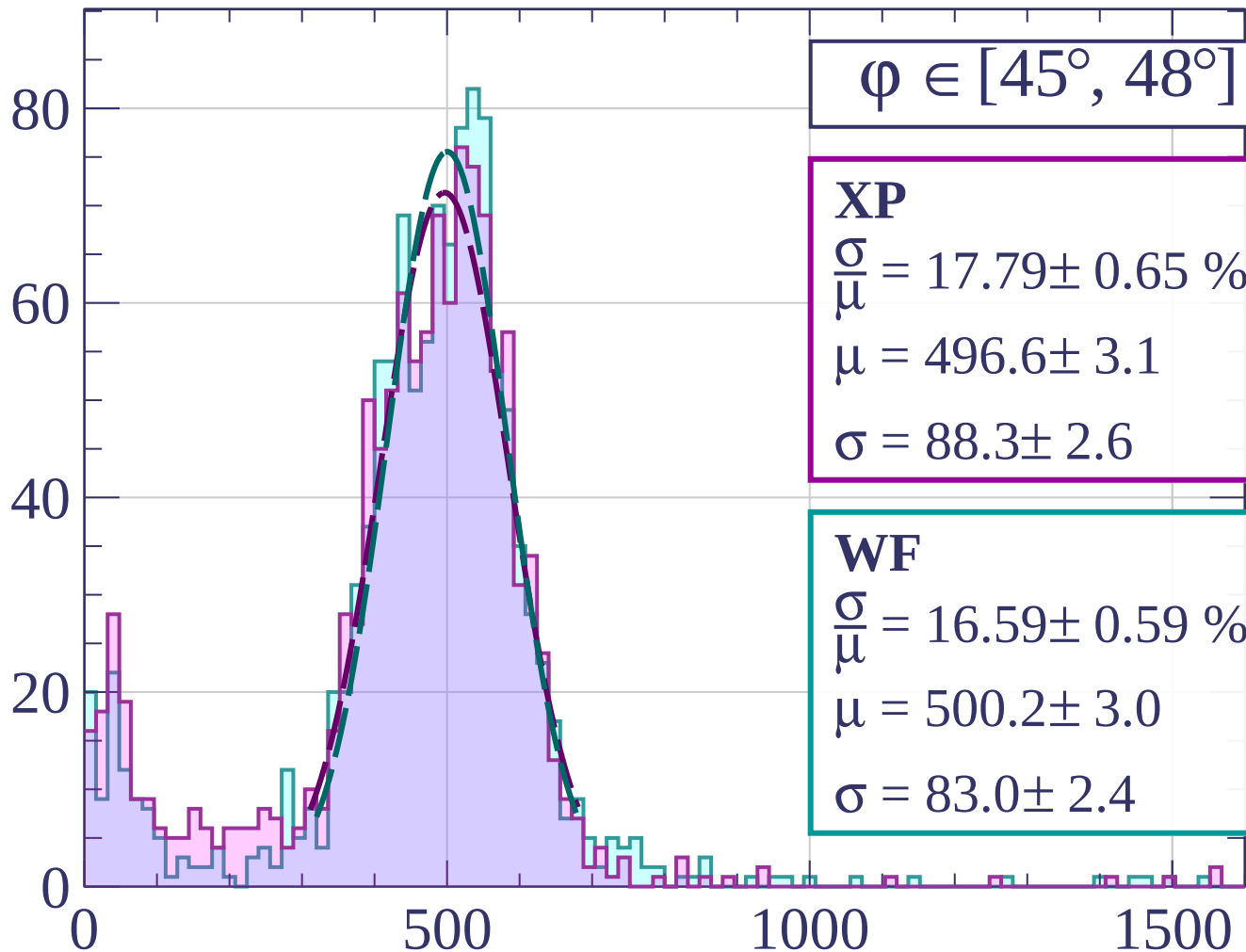
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count

$\varphi \in [48^\circ, 51^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.99 \pm 0.61 \%$$

$$\mu = 499.0 \pm 2.8$$

$$\sigma = 84.8 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 16.70 \pm 0.56 \%$$

$$\mu = 500.3 \pm 2.7$$

$$\sigma = 83.5 \pm 2.3$$

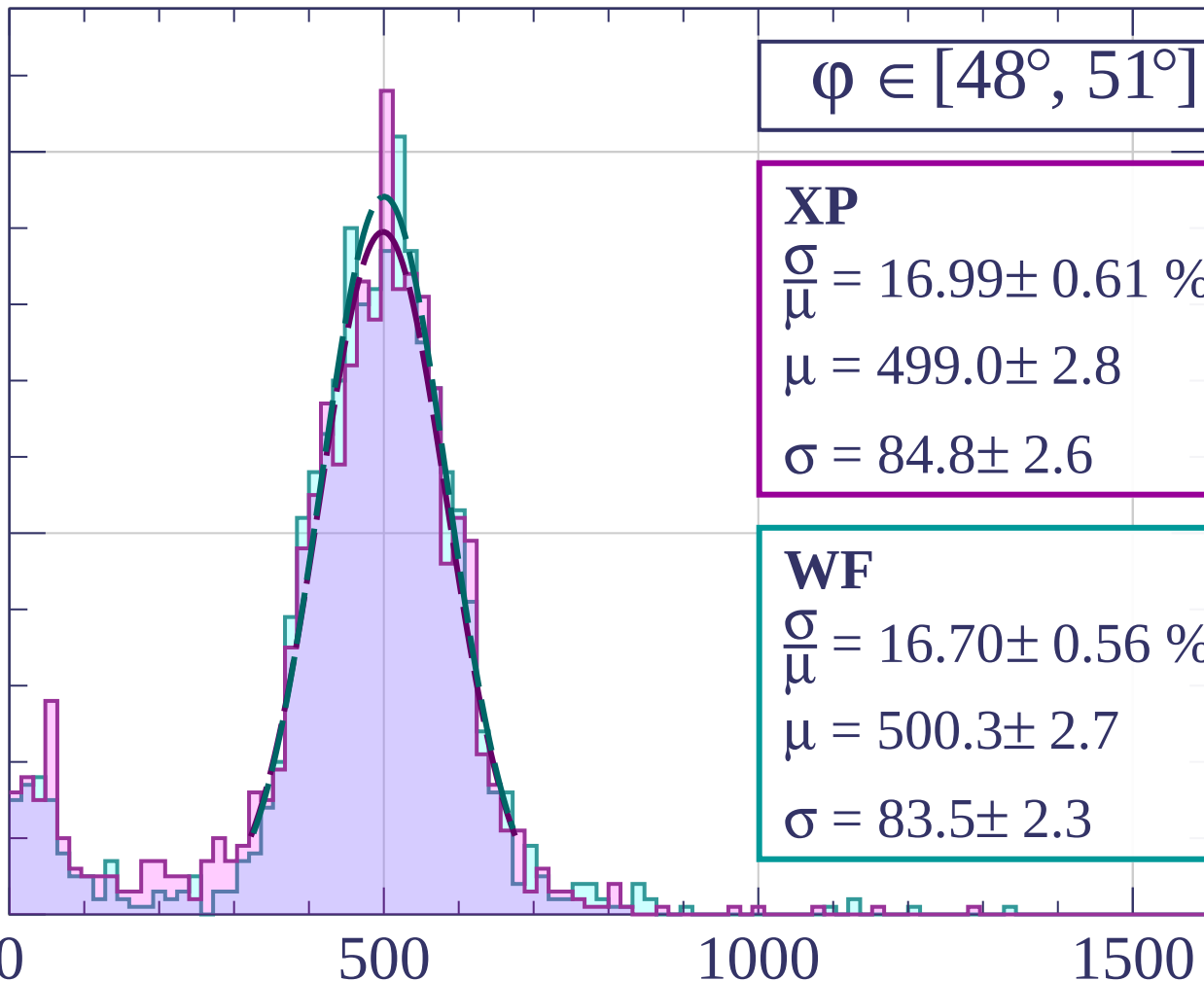
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\varphi \in [51^\circ, 54^\circ]$

XP

$$\frac{\sigma}{\mu} = 17.31 \pm 0.54 \%$$

$$\mu = 493.2 \pm 2.6$$

$$\sigma = 85.4 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 17.60 \pm 0.56 \%$$

$$\mu = 492.0 \pm 2.6$$

$$\sigma = 86.6 \pm 2.3$$

0

500

1000

1500

dE/dx [ADC counts/cm]

100

50

Count

$\phi \in [54^\circ, 57^\circ]$

XP

$$\frac{\sigma}{\mu} = 17.88 \pm 0.53 \%$$

$$\mu = 492.1 \pm 2.5$$

$$\sigma = 88.0 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 17.63 \pm 0.52 \%$$

$$\mu = 489.7 \pm 2.4$$

$$\sigma = 86.3 \pm 2.1$$

100

50

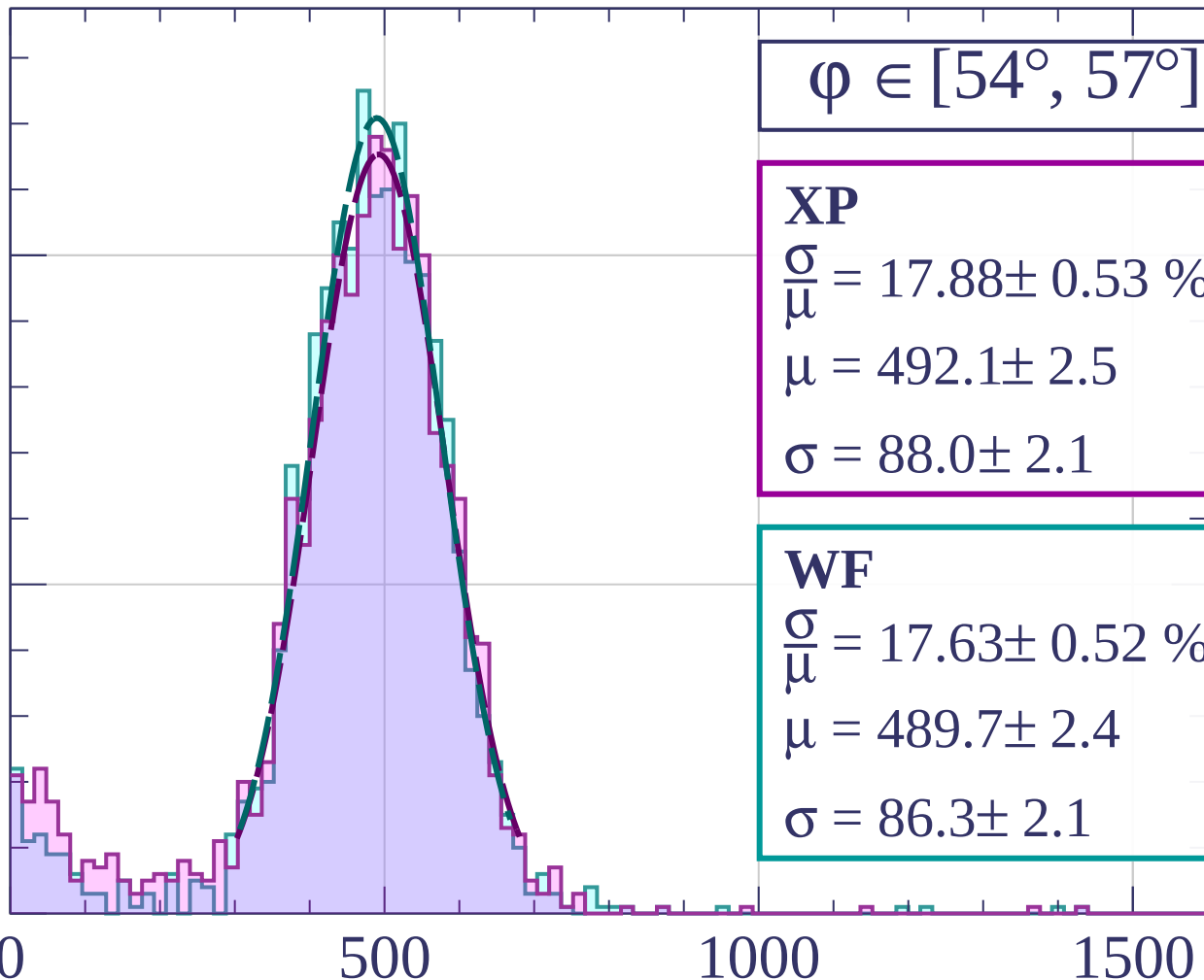
0

500

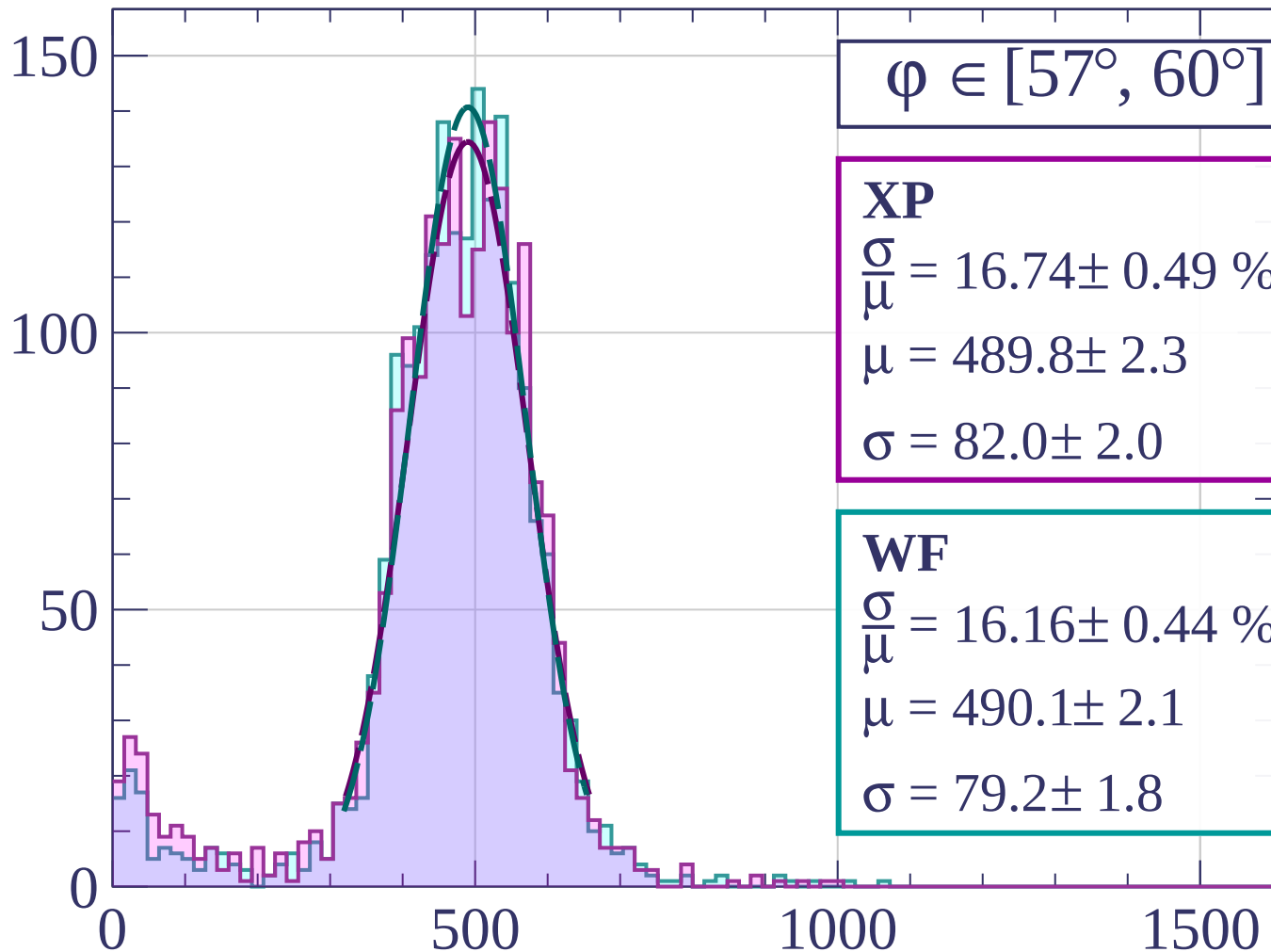
1000

1500

dE/dx [ADC counts/cm]



Count



dE/dx [ADC counts/cm]

Count

$\phi \in [60^\circ, 63^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.21 \pm 0.41 \%$$

$$\mu = 488.7 \pm 2.2$$

$$\sigma = 79.2 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 16.68 \pm 0.43 \%$$

$$\mu = 498.9 \pm 2.1$$

$$\sigma = 83.2 \pm 1.8$$

150

100

50

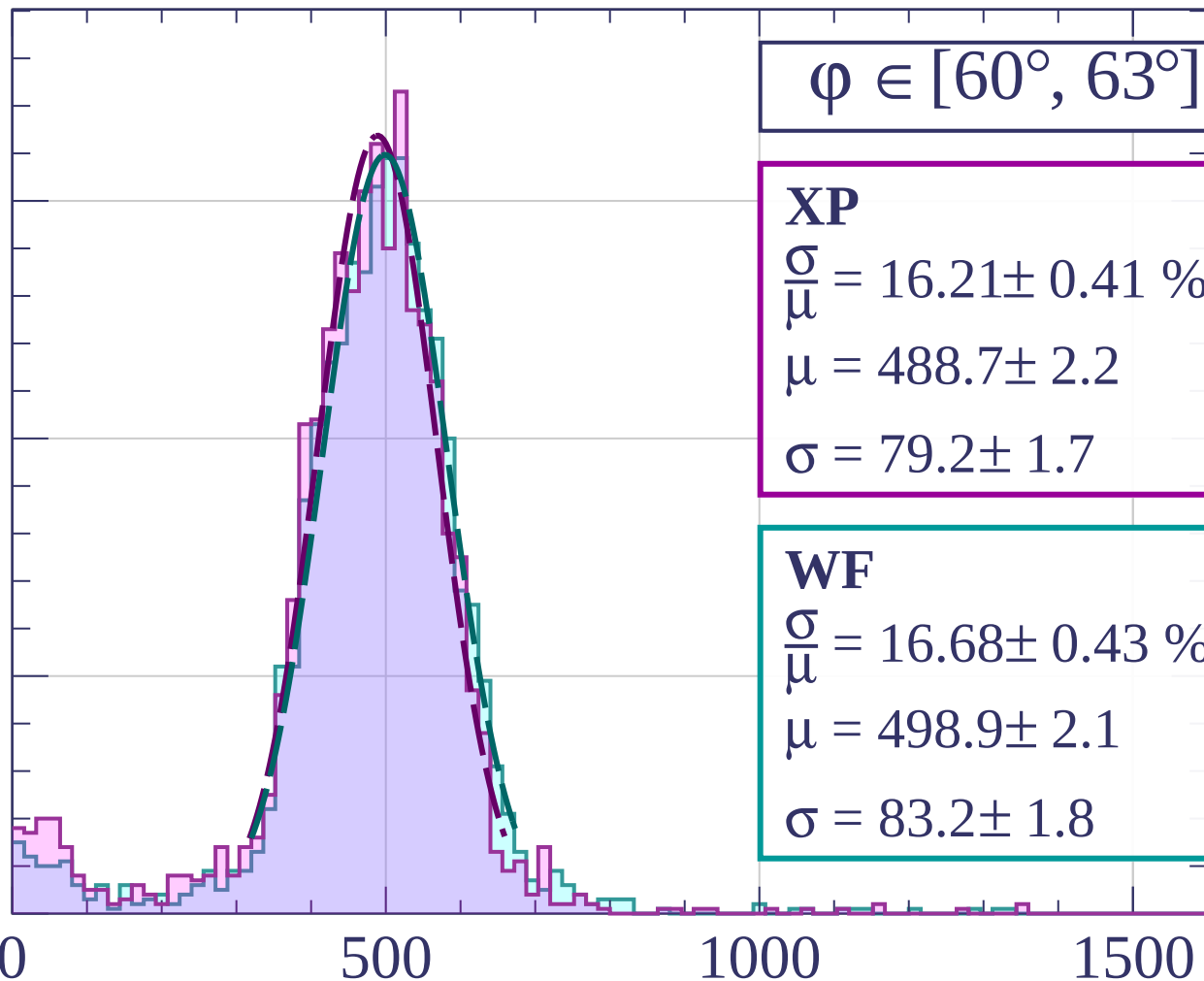
0

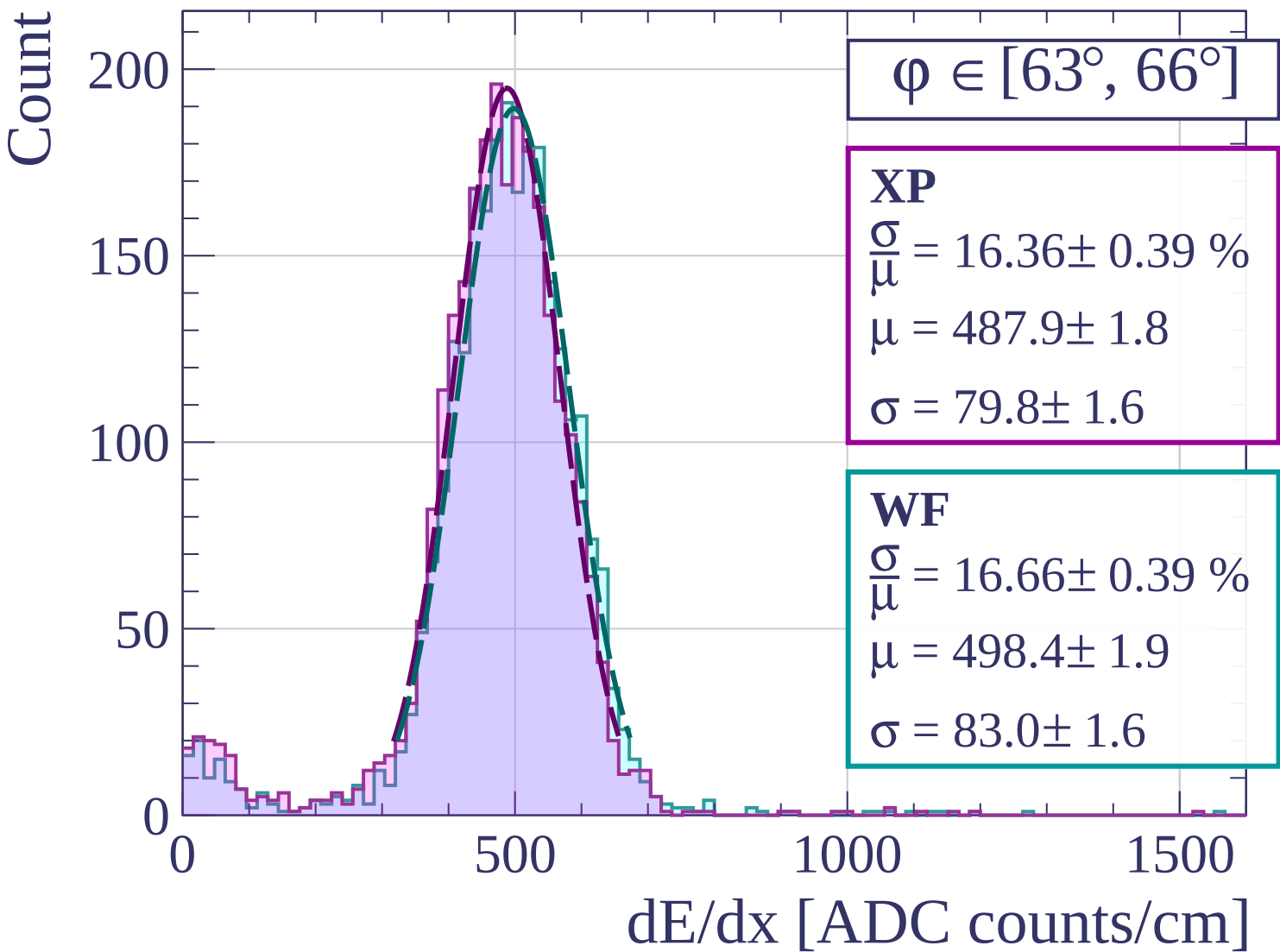
500

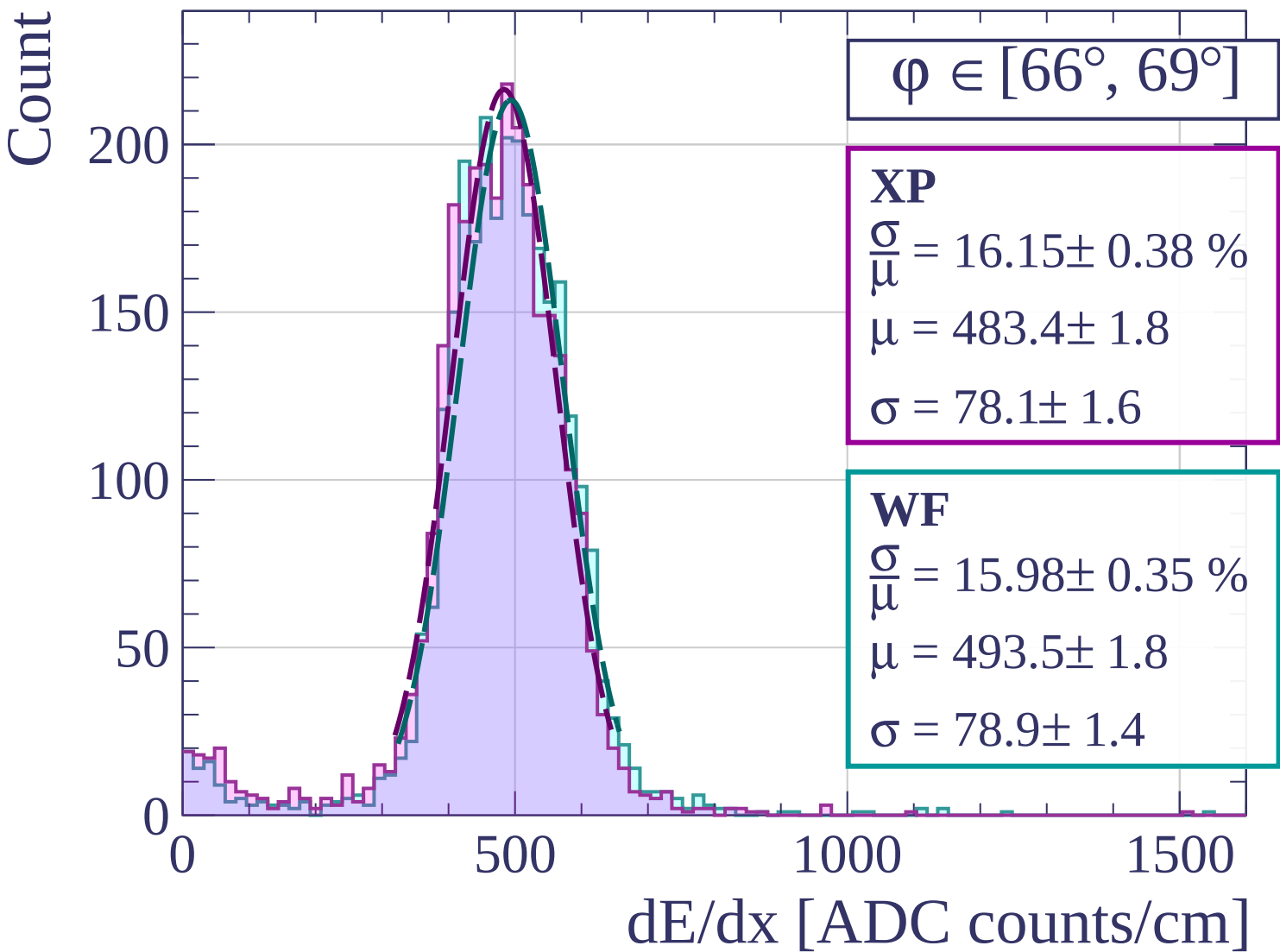
1000

1500

dE/dx [ADC counts/cm]







Count

$\phi \in [69^\circ, 72^\circ]$

XP

$$\frac{\sigma}{\mu} = 15.57 \pm 0.33 \%$$

$$\mu = 484.0 \pm 1.7$$

$$\sigma = 75.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 16.09 \pm 0.34 \%$$

$$\mu = 494.9 \pm 1.7$$

$$\sigma = 79.6 \pm 1.4$$

200

100

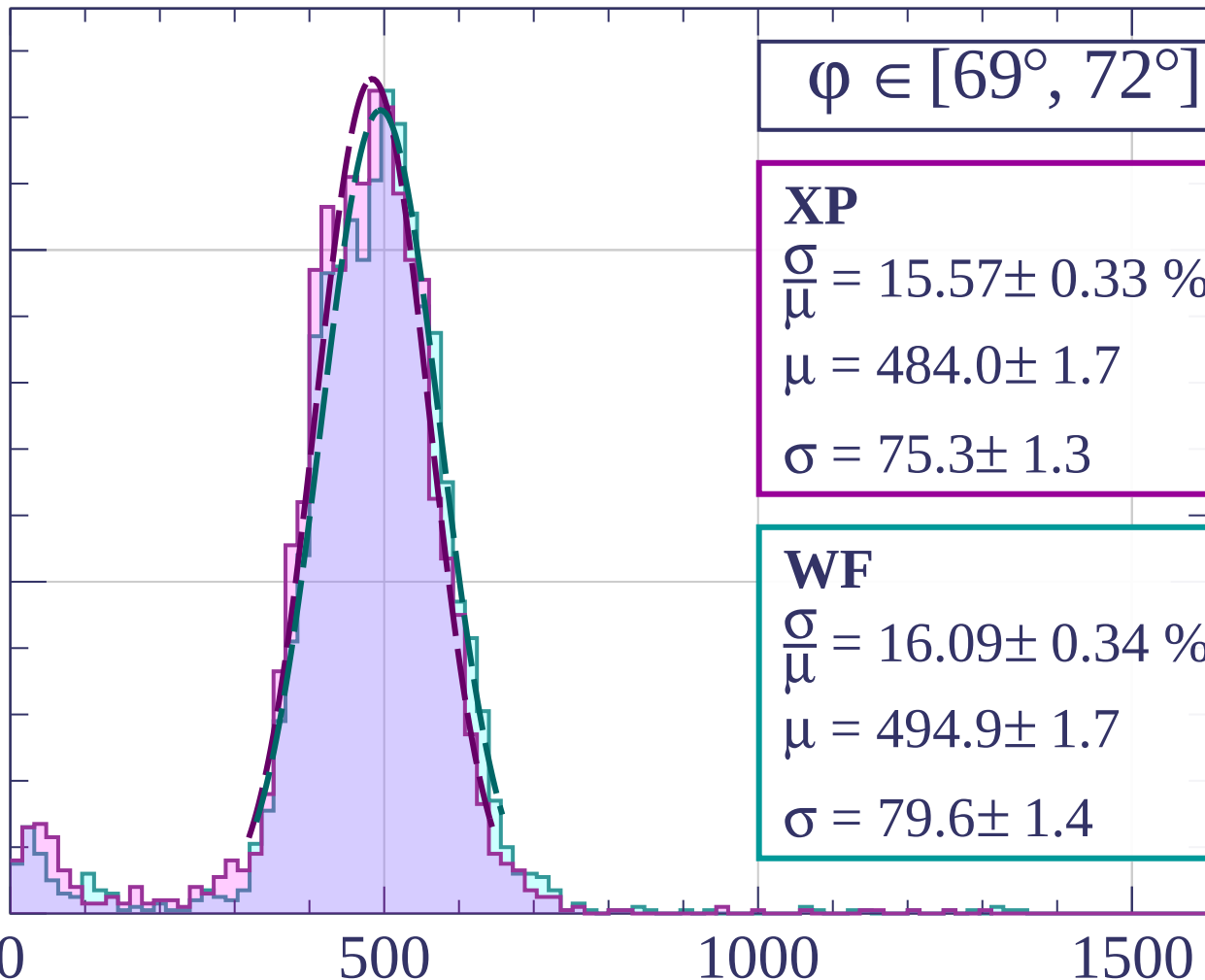
0

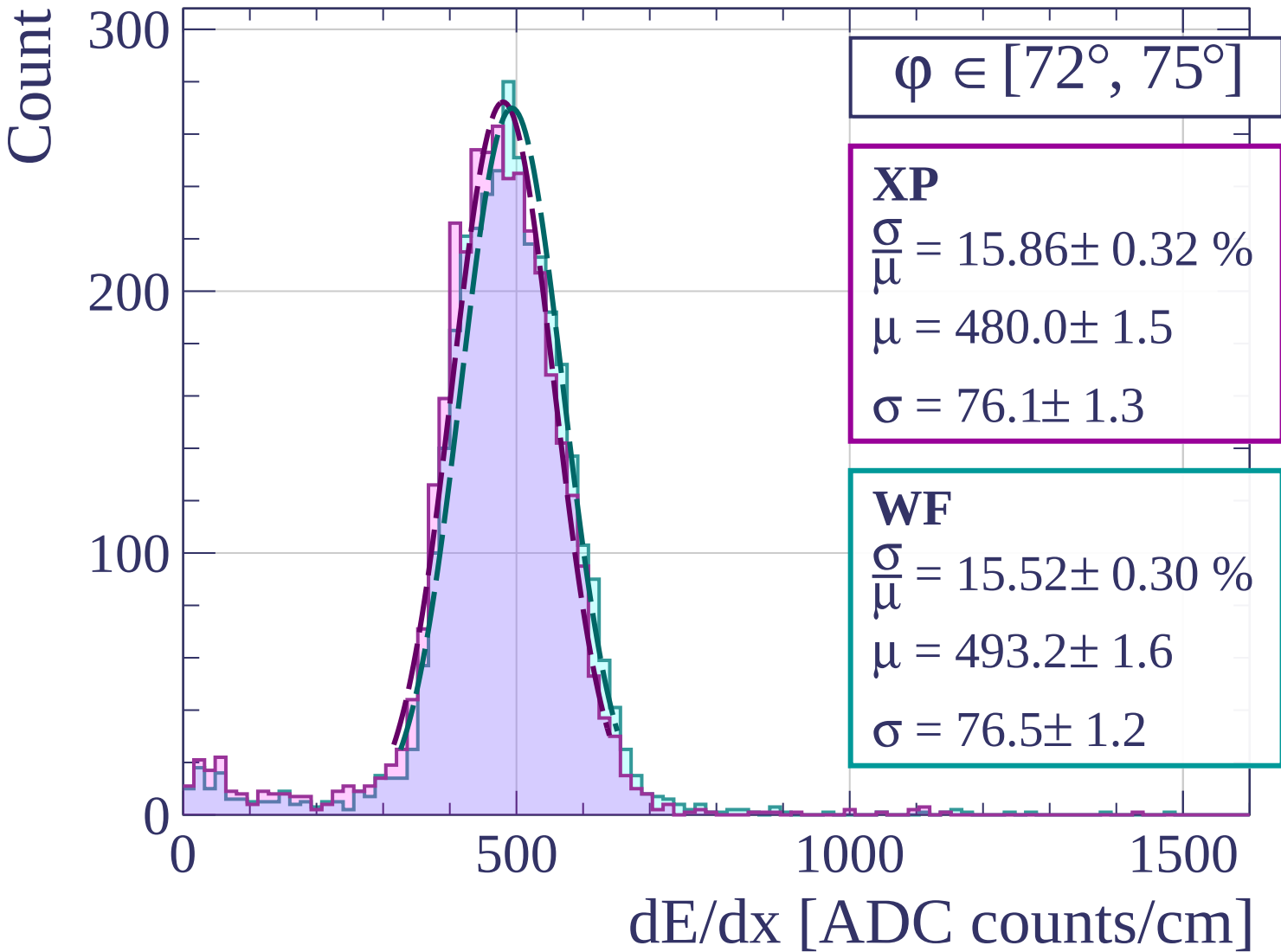
500

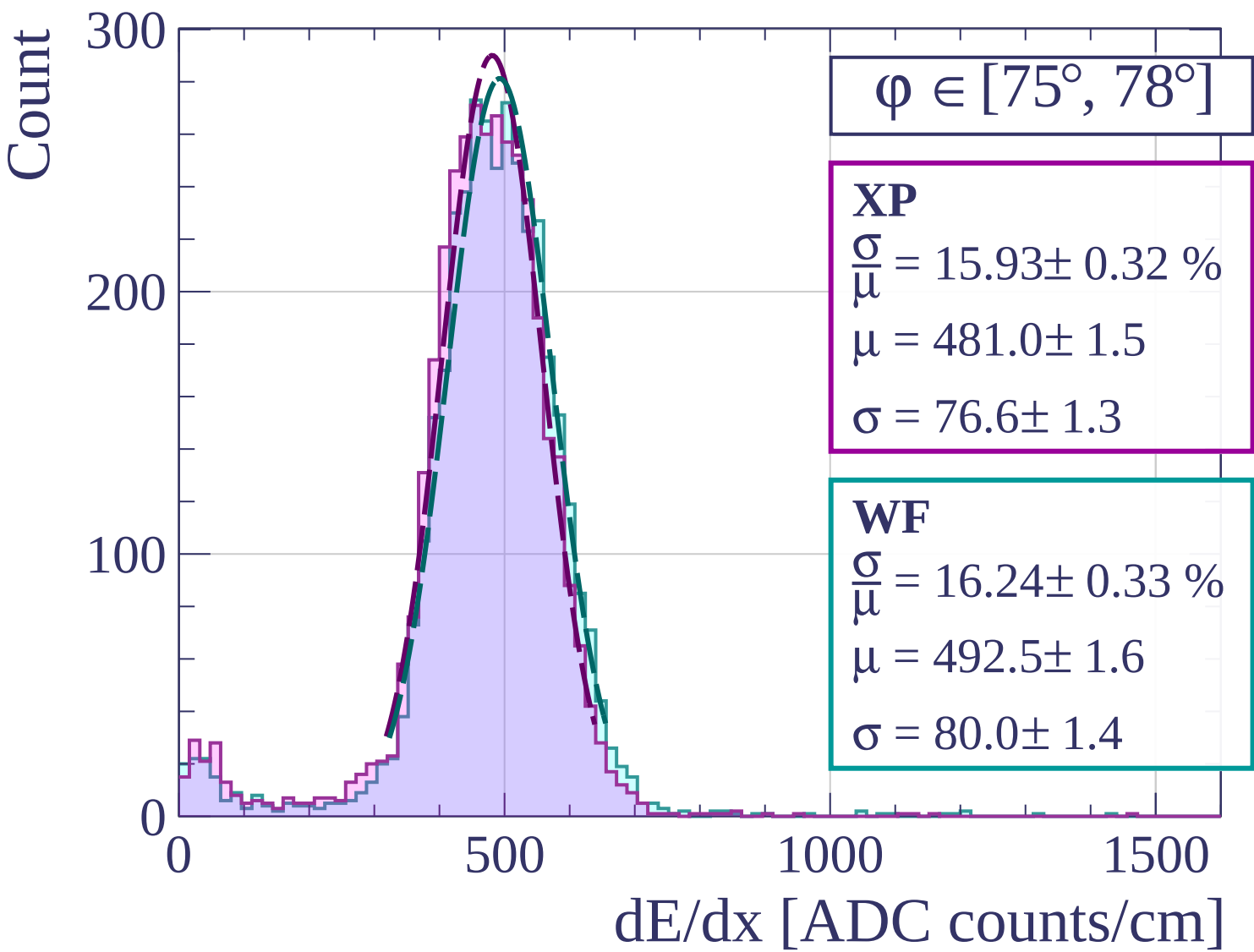
1000

1500

dE/dx [ADC counts/cm]







Count

$\varphi \in [78^\circ, 81^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.04 \pm 0.32 \%$$

$$\mu = 481.3 \pm 1.5$$

$$\sigma = 77.2 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 16.05 \pm 0.31 \%$$

$$\mu = 493.5 \pm 1.5$$

$$\sigma = 79.2 \pm 1.3$$

300

200

100

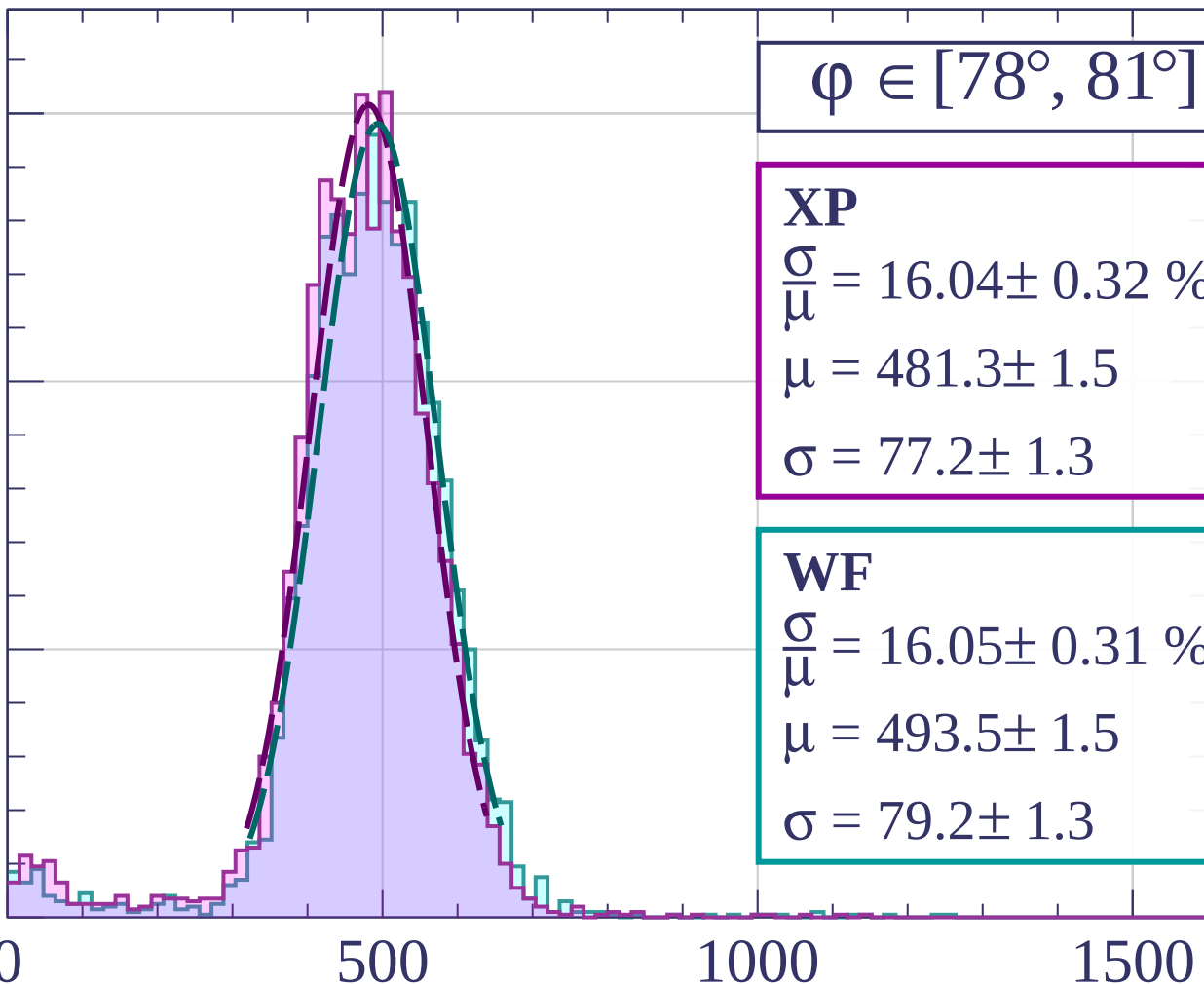
0

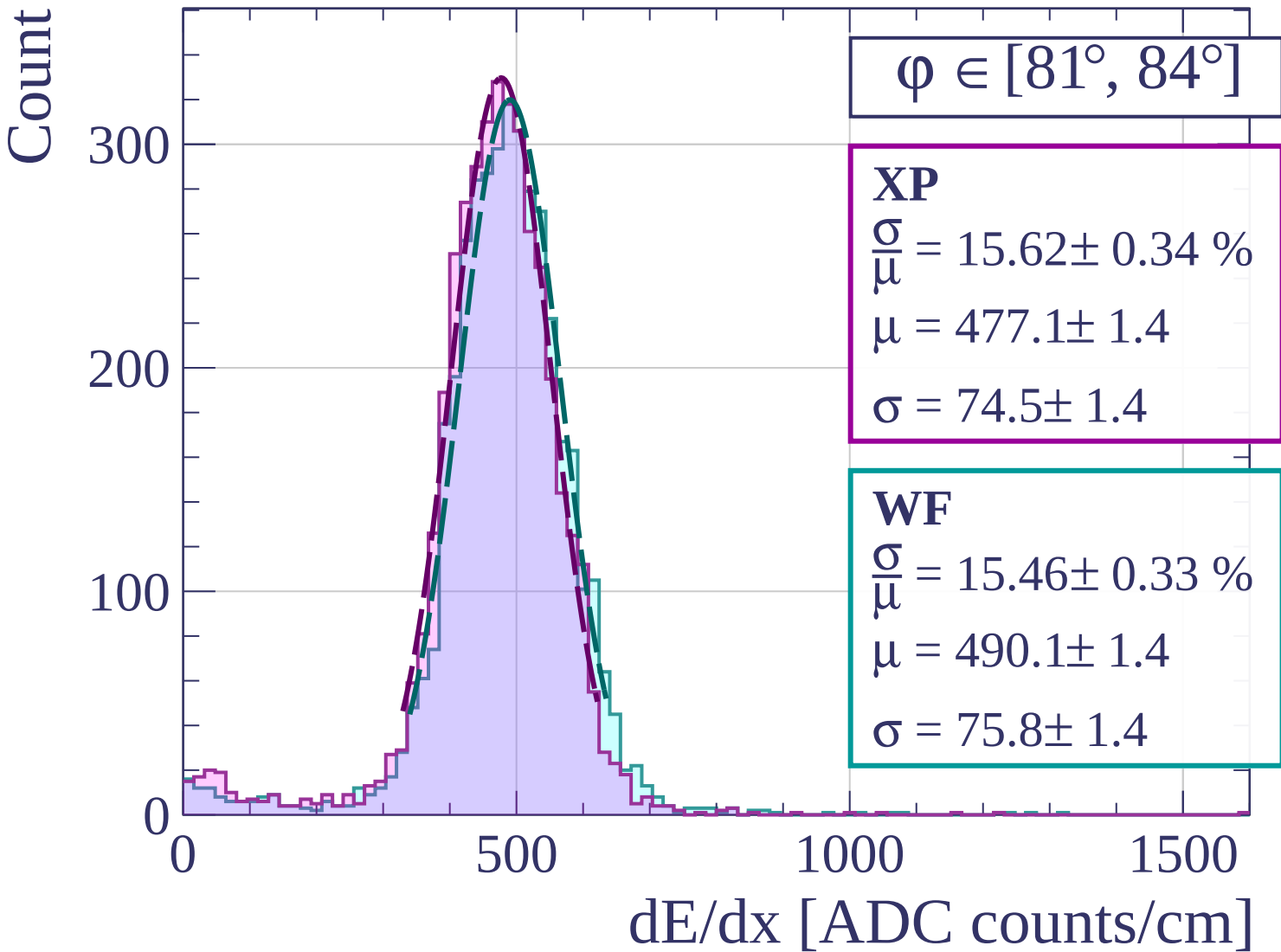
500

1000

1500

dE/dx [ADC counts/cm]





Count

$\phi \in [84^\circ, 87^\circ]$

XP

$$\frac{\sigma}{\mu} = 15.45 \pm 0.31 \%$$

$$\mu = 475.5 \pm 1.4$$

$$\sigma = 73.5 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 15.91 \pm 0.32 \%$$

$$\mu = 488.8 \pm 1.5$$

$$\sigma = 77.8 \pm 1.3$$

300

200

100

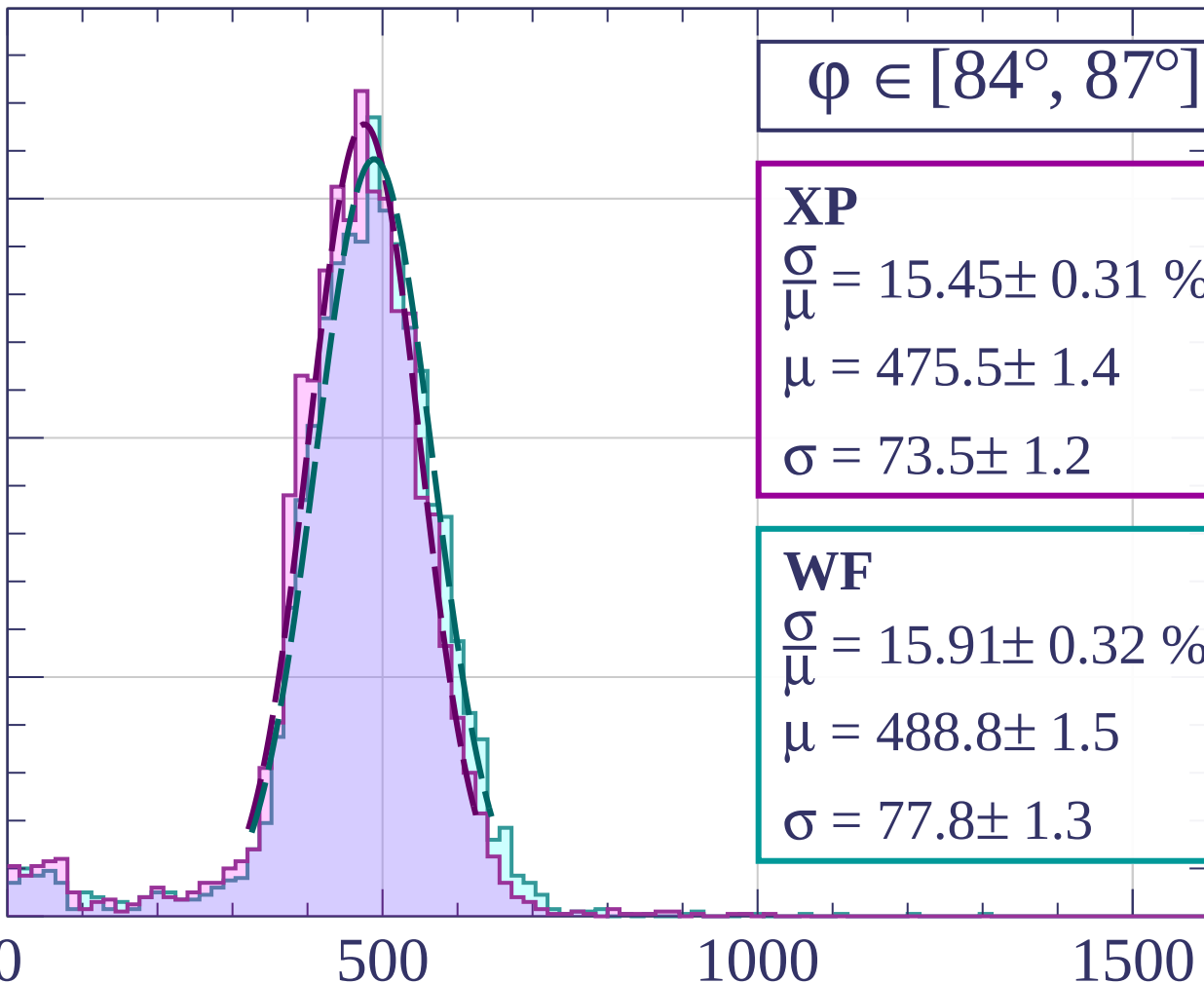
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [87^\circ, 90^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.14 \pm 0.28 \%$$

$$\mu = 475.0 \pm 1.4$$

$$\sigma = 76.7 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 16.73 \pm 0.32 \%$$

$$\mu = 489.8 \pm 1.5$$

$$\sigma = 81.9 \pm 1.3$$

300

200

100

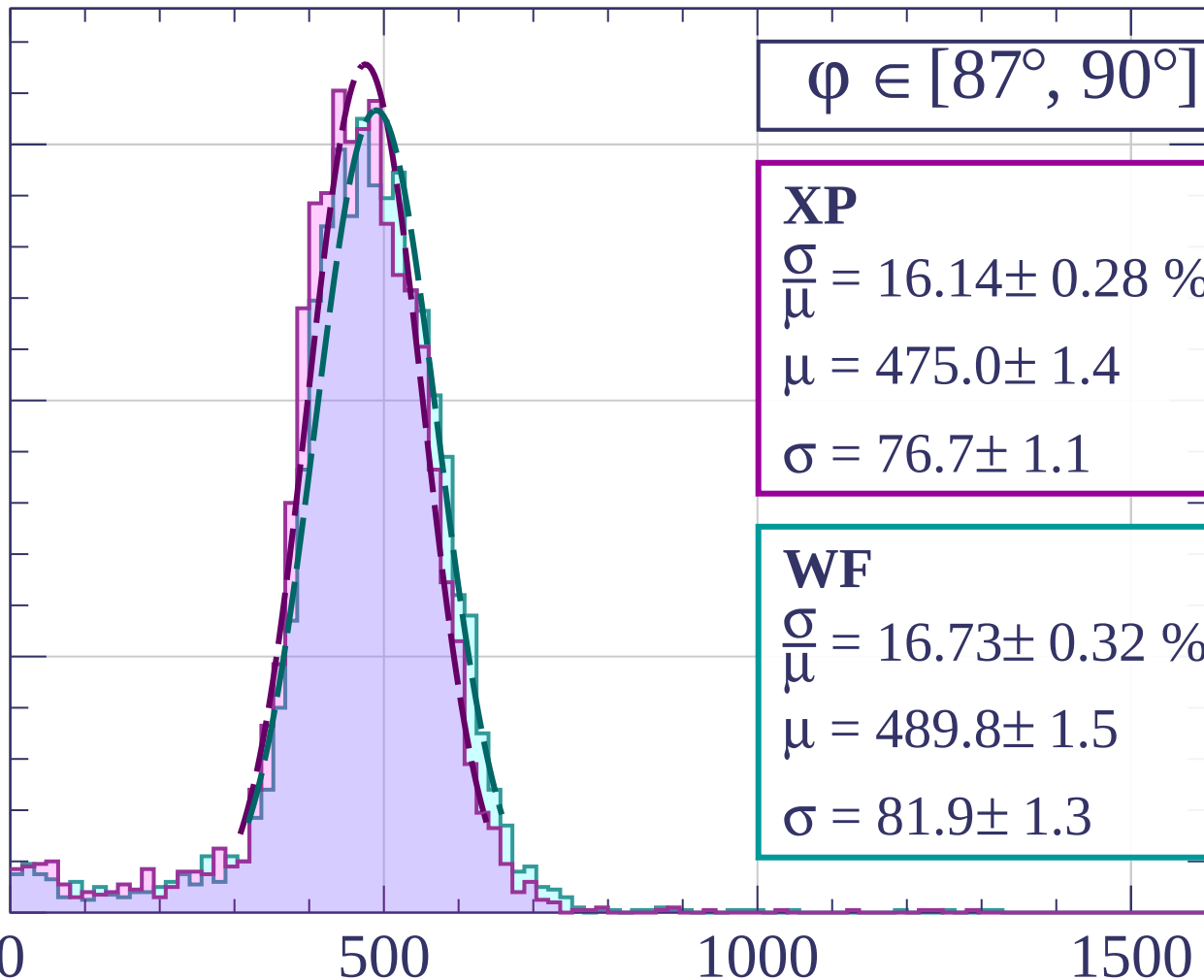
0

500

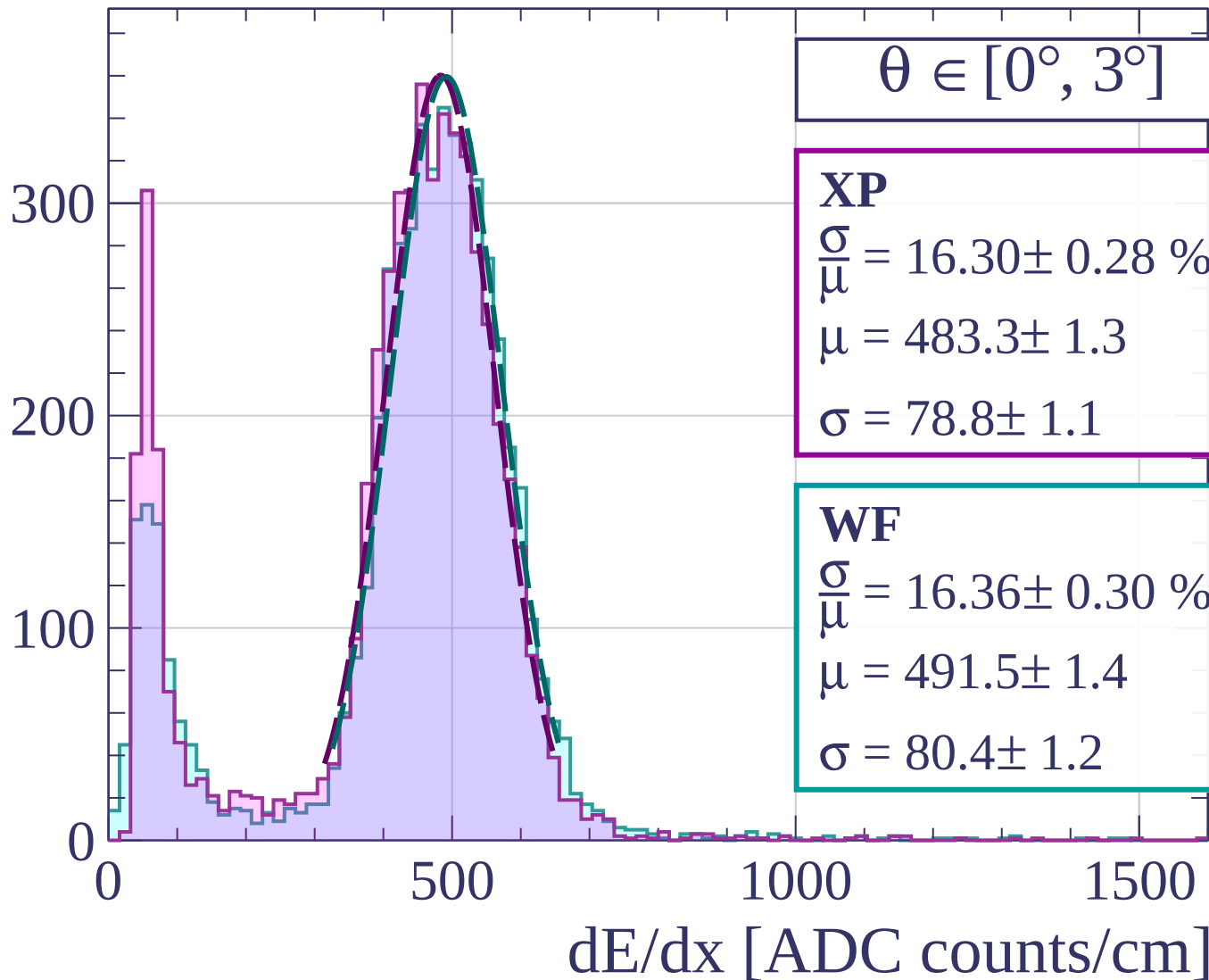
1000

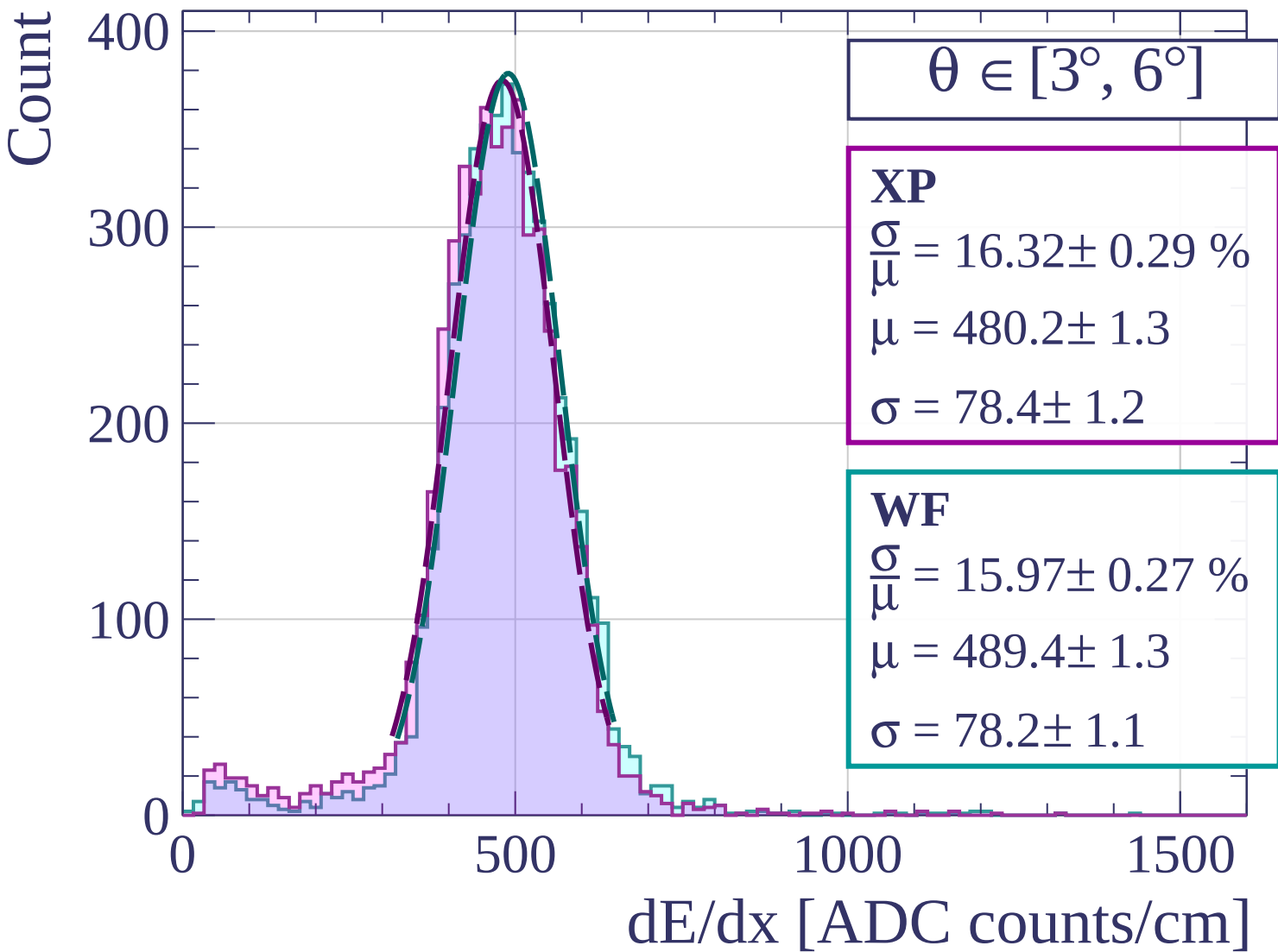
1500

dE/dx [ADC counts/cm]

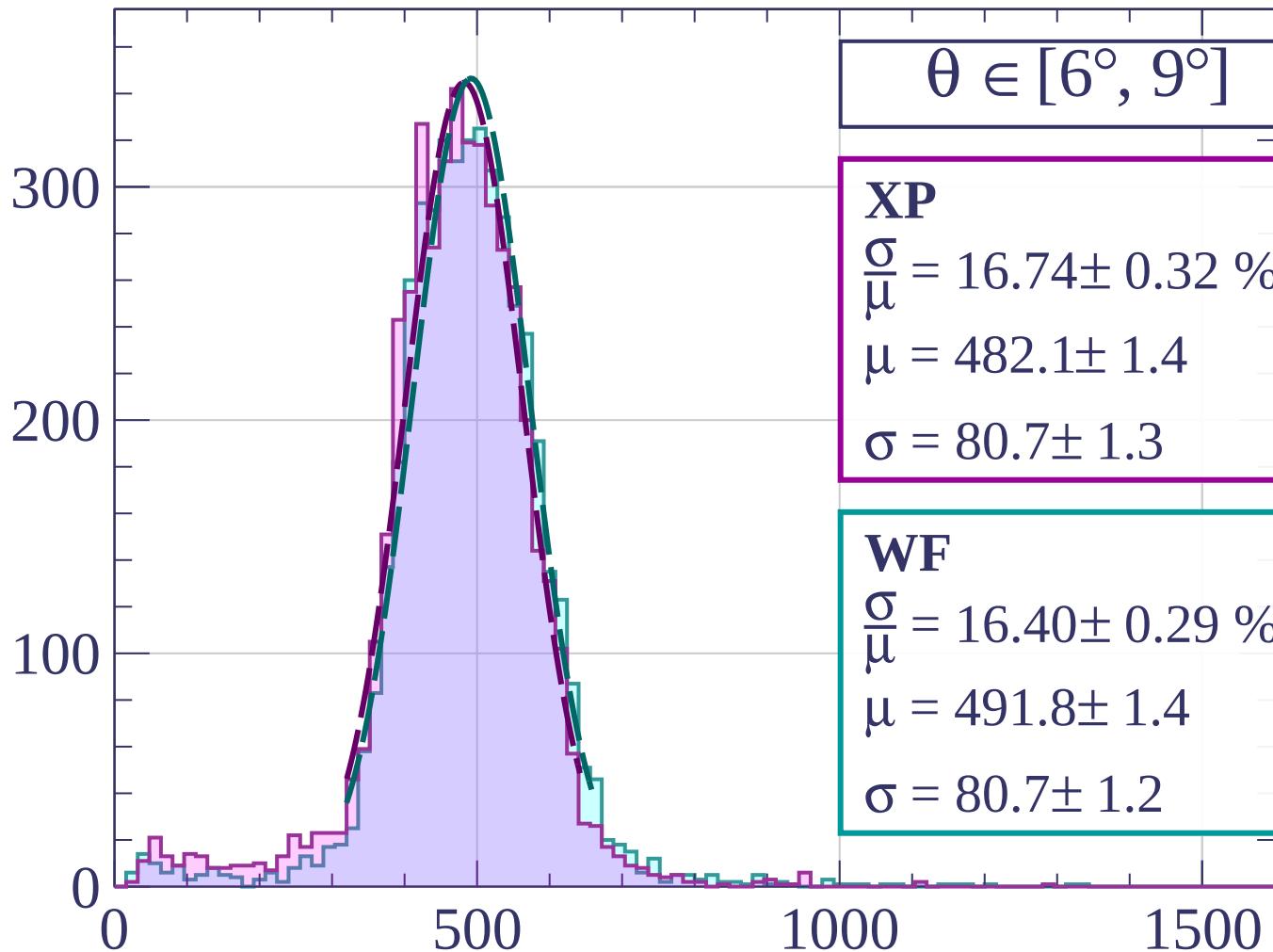


Count





Count



dE/dx [ADC counts/cm]

Count

$$\theta \in [9^\circ, 12^\circ]$$

XP

$$\frac{\sigma}{\mu} = 16.80 \pm 0.32 \%$$

$$\mu = 479.5 \pm 1.4$$

$$\sigma = 80.6 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 16.66 \pm 0.30 \%$$

$$\mu = 491.5 \pm 1.5$$

$$\sigma = 81.9 \pm 1.2$$

300

200

100

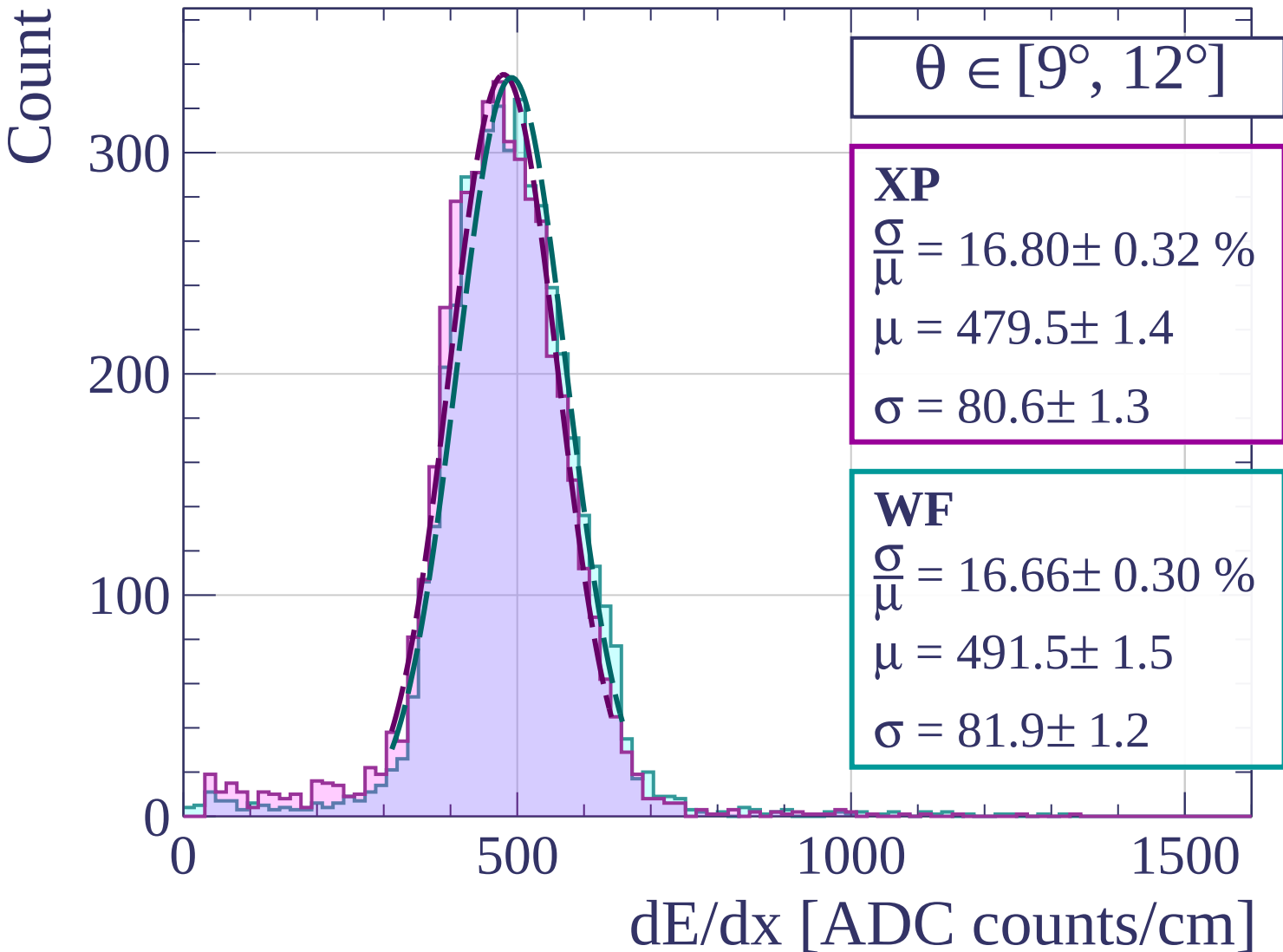
0

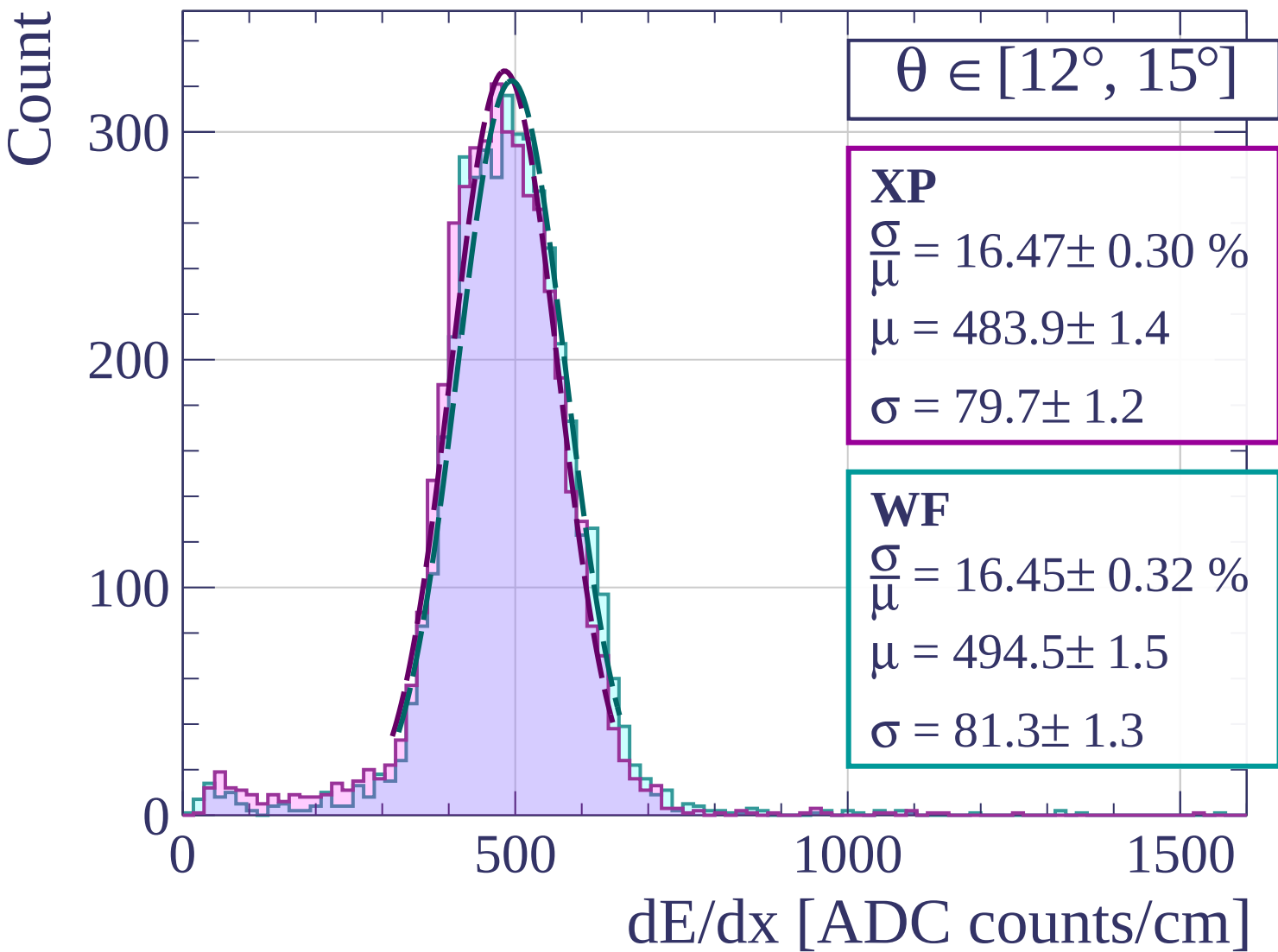
500

1000

1500

dE/dx [ADC counts/cm]





Count

$\theta \in [15^\circ, 18^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.53 \pm 0.30 \%$$

$$\mu = 480.4 \pm 1.5$$

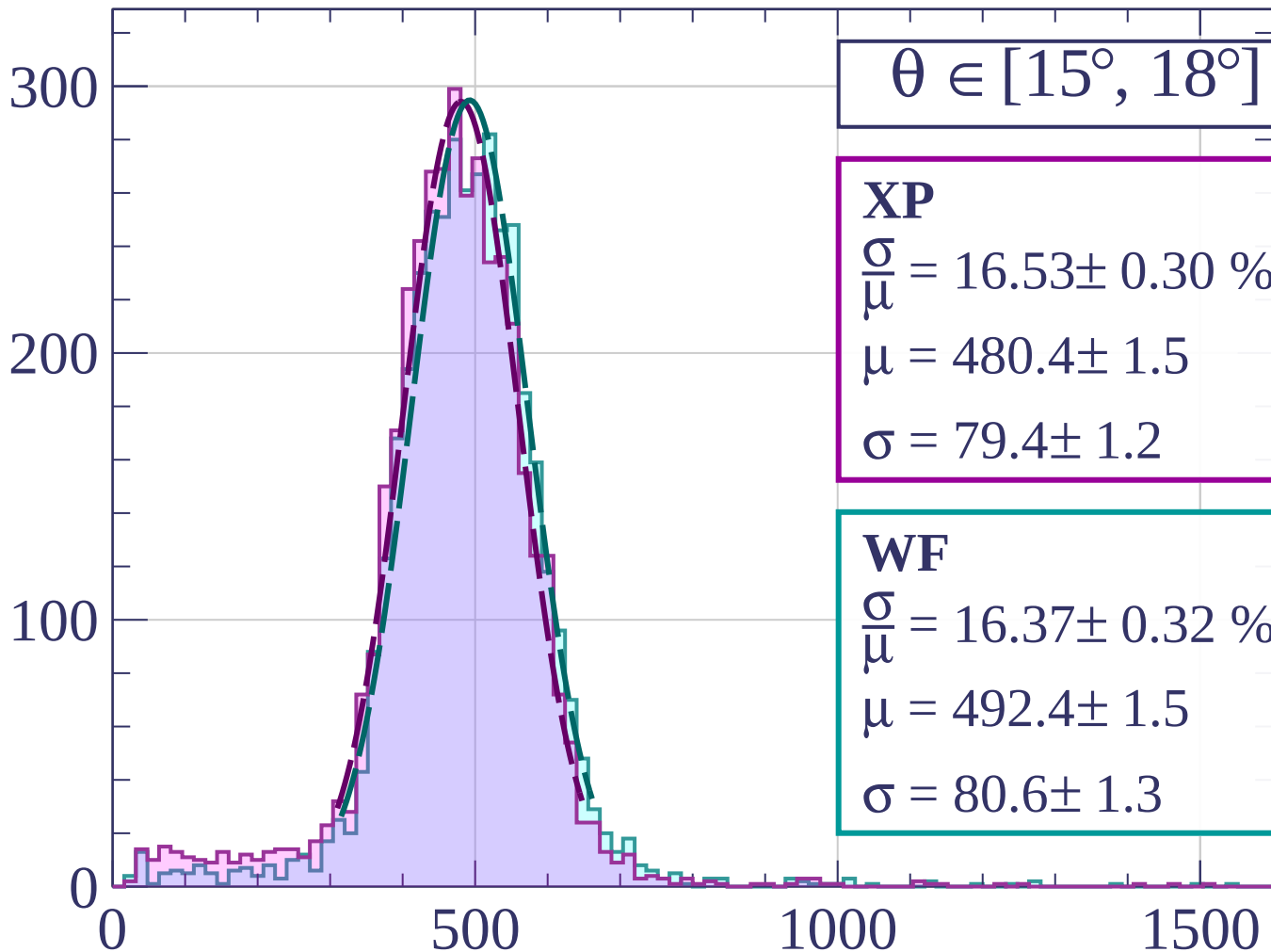
$$\sigma = 79.4 \pm 1.2$$

WF

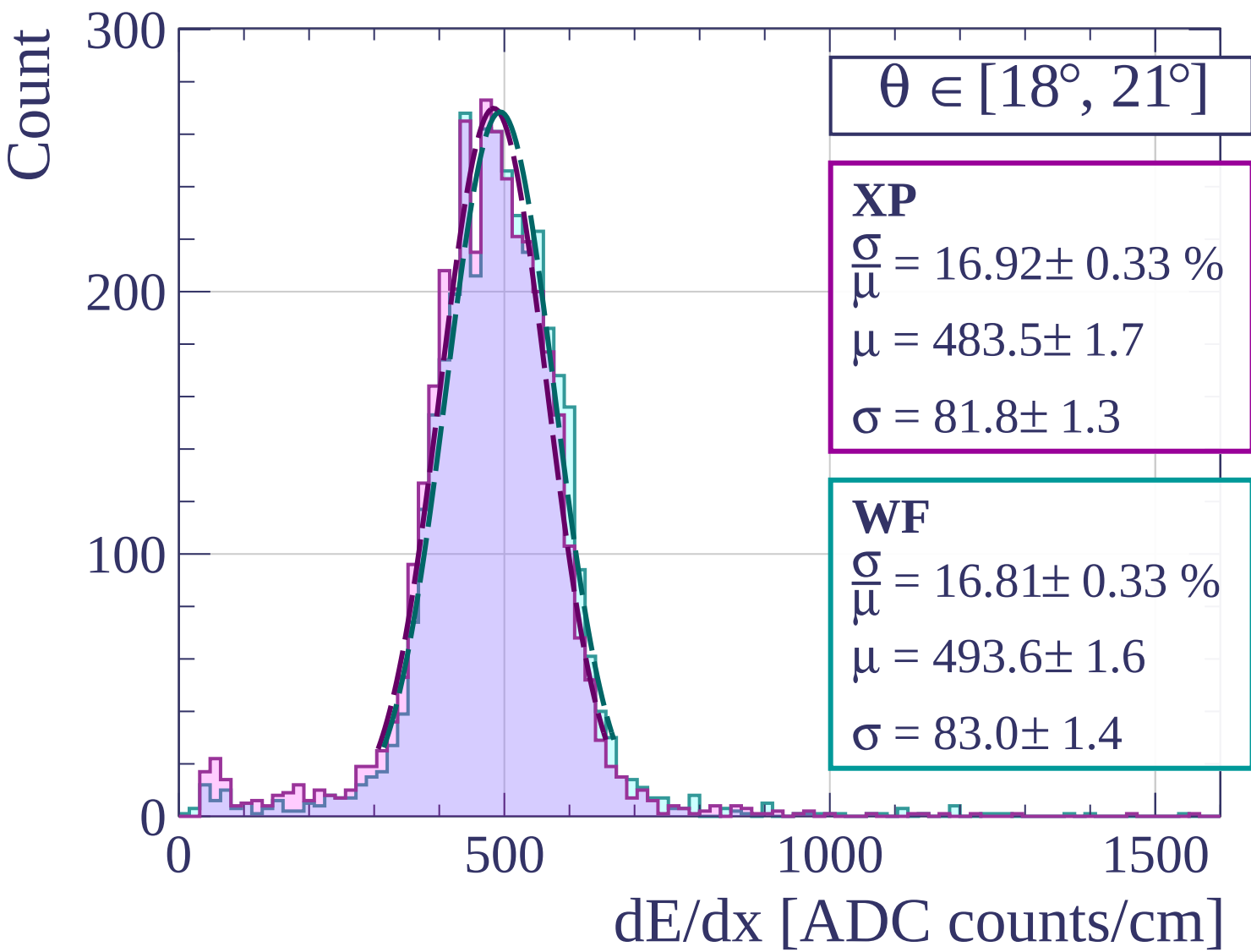
$$\frac{\sigma}{\mu} = 16.37 \pm 0.32 \%$$

$$\mu = 492.4 \pm 1.5$$

$$\sigma = 80.6 \pm 1.3$$



dE/dx [ADC counts/cm]



Count

$$\theta \in [21^\circ, 24^\circ]$$

XP

$$\frac{\sigma}{\mu} = 16.62 \pm 0.36 \%$$

$$\mu = 483.8 \pm 1.7$$

$$\sigma = 80.4 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 16.21 \pm 0.35 \%$$

$$\mu = 494.6 \pm 1.7$$

$$\sigma = 80.2 \pm 1.5$$

200

100

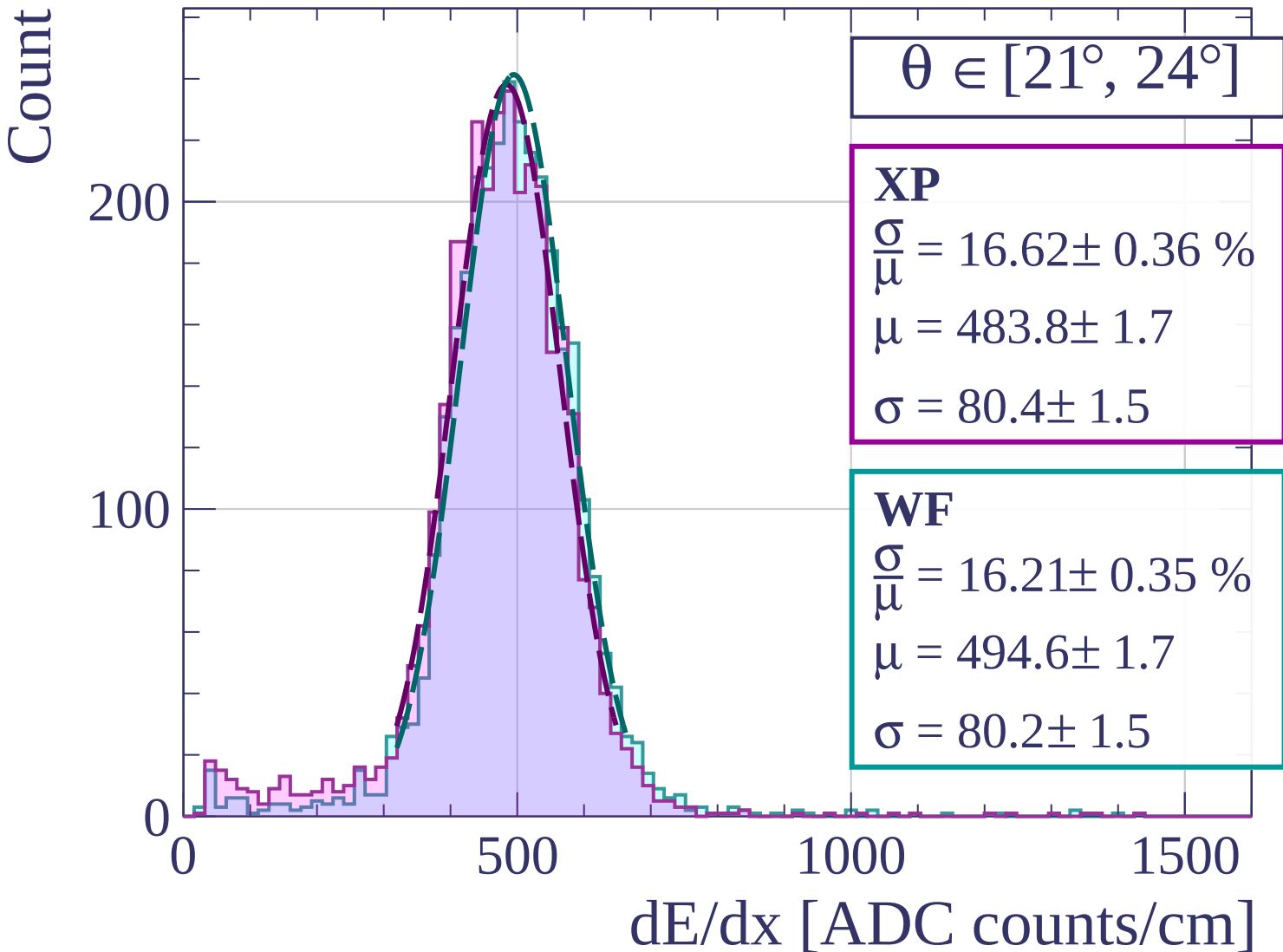
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [24^\circ, 27^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.10 \pm 0.37 \%$$

$$\mu = 486.3 \pm 1.8$$

$$\sigma = 78.3 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 16.00 \pm 0.39 \%$$

$$\mu = 494.2 \pm 1.8$$

$$\sigma = 79.1 \pm 1.6$$

200

150

100

50

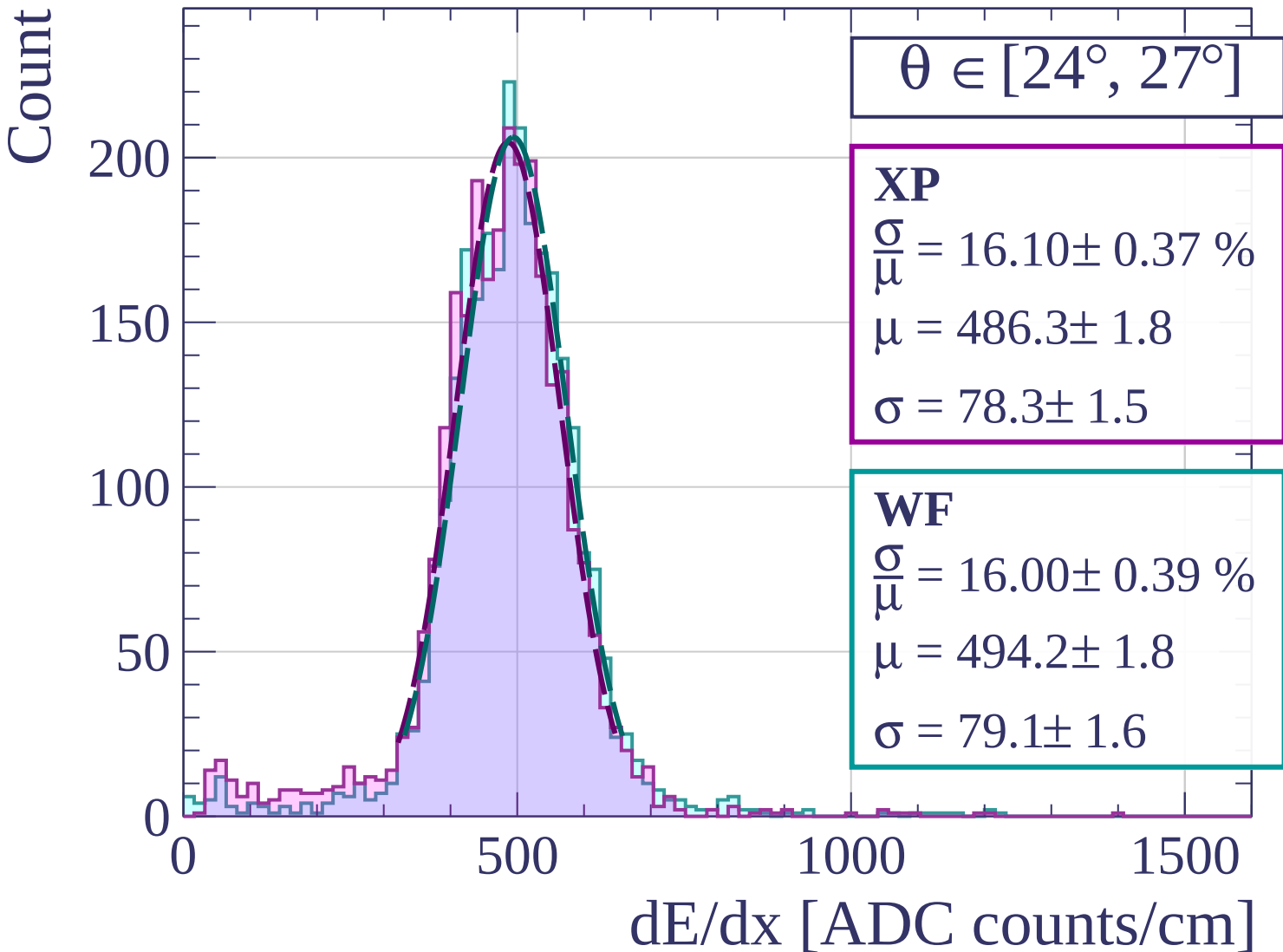
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [27^\circ, 30^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.26 \pm 0.40 \%$$

$$\mu = 489.8 \pm 1.9$$

$$\sigma = 79.6 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 16.23 \pm 0.39 \%$$

$$\mu = 497.8 \pm 1.9$$

$$\sigma = 80.8 \pm 1.6$$

150

100

50

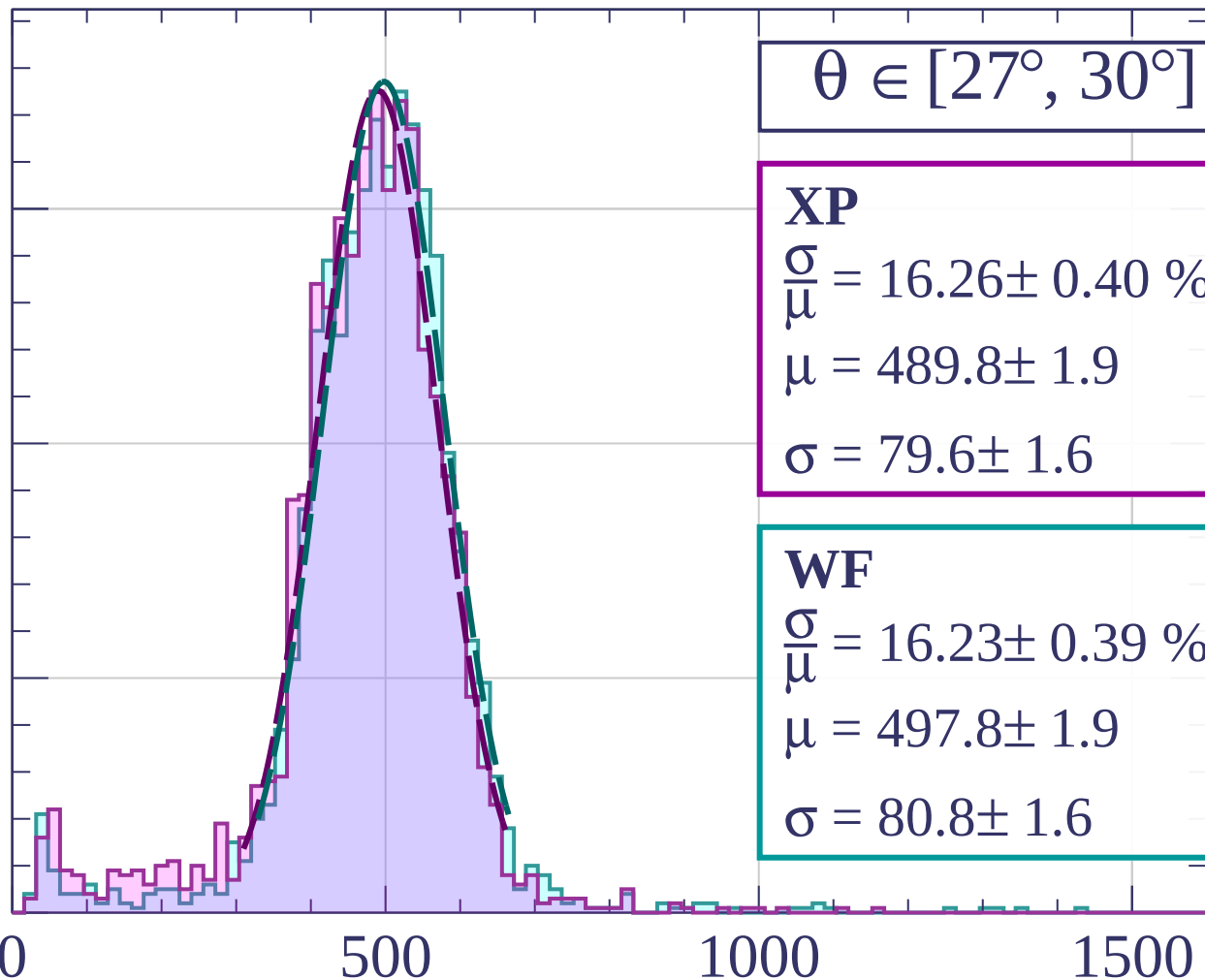
0

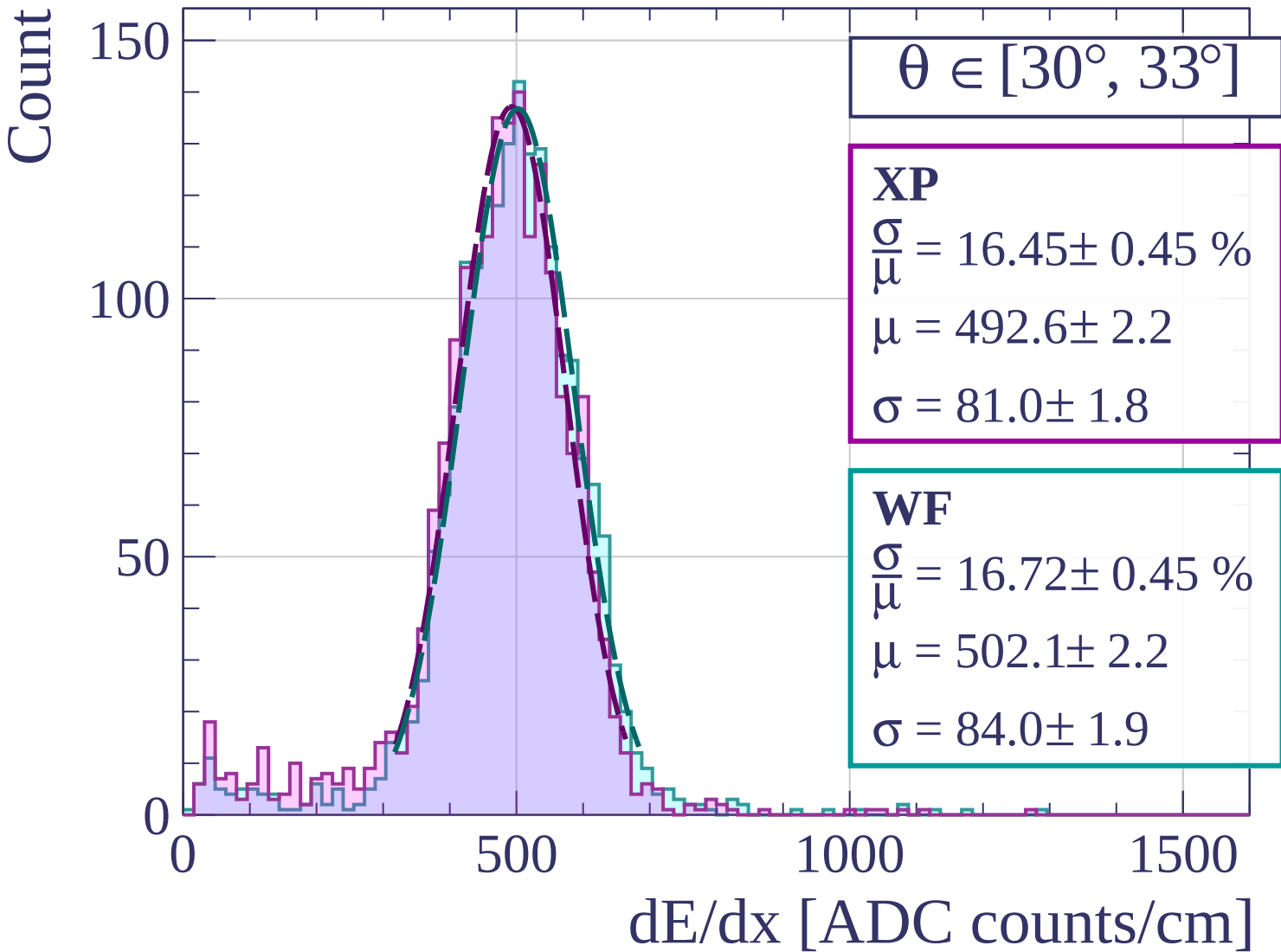
500

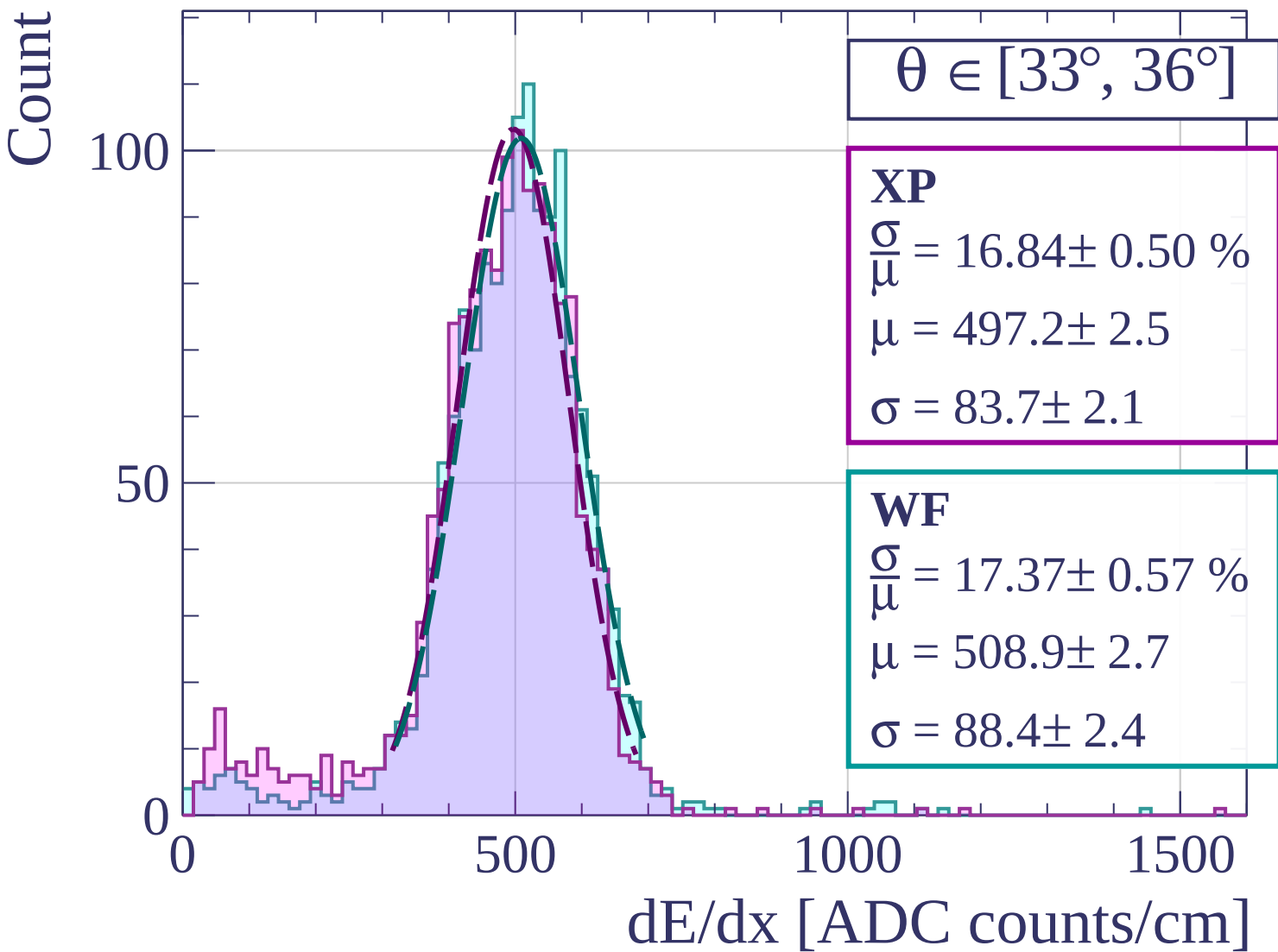
1000

1500

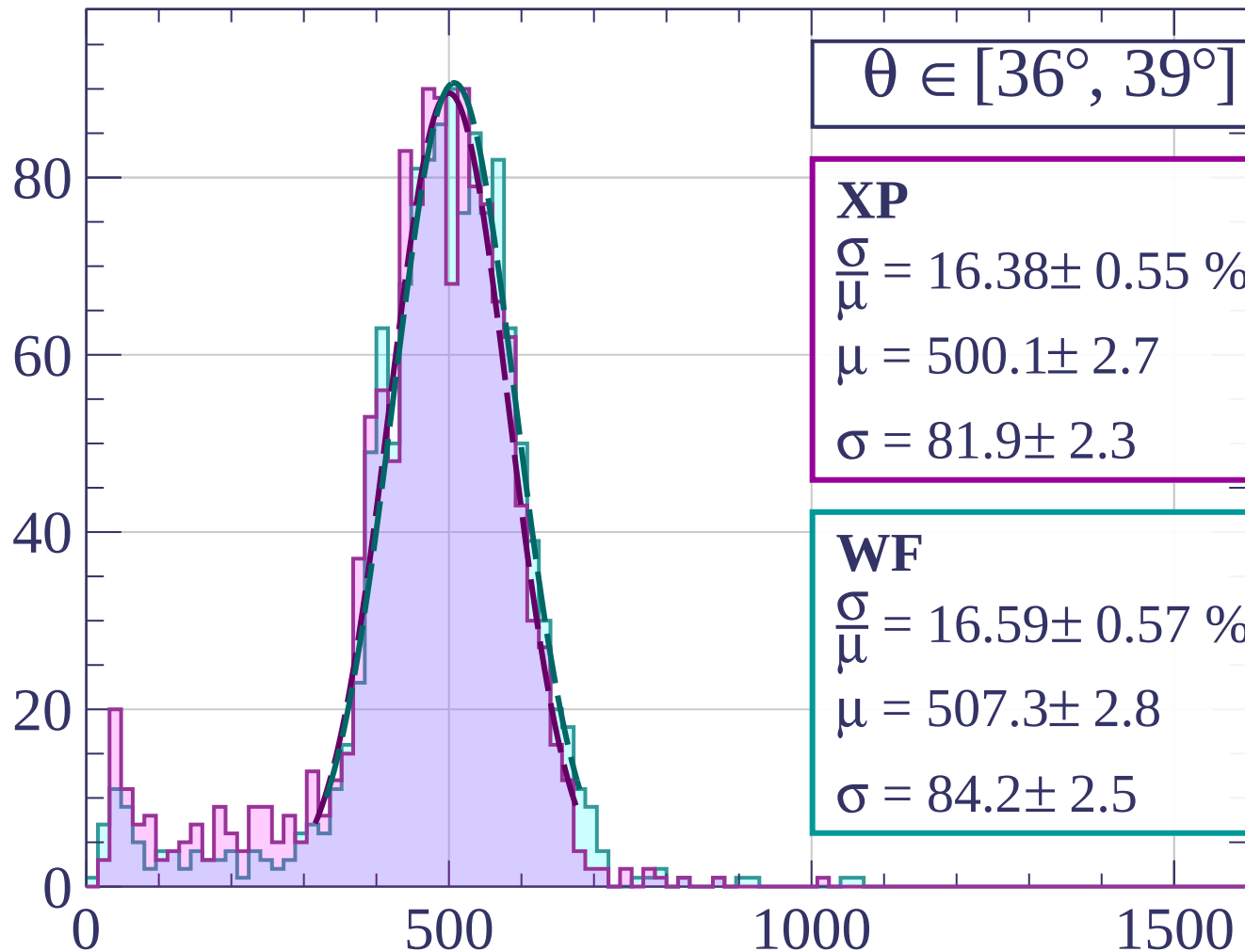
dE/dx [ADC counts/cm]





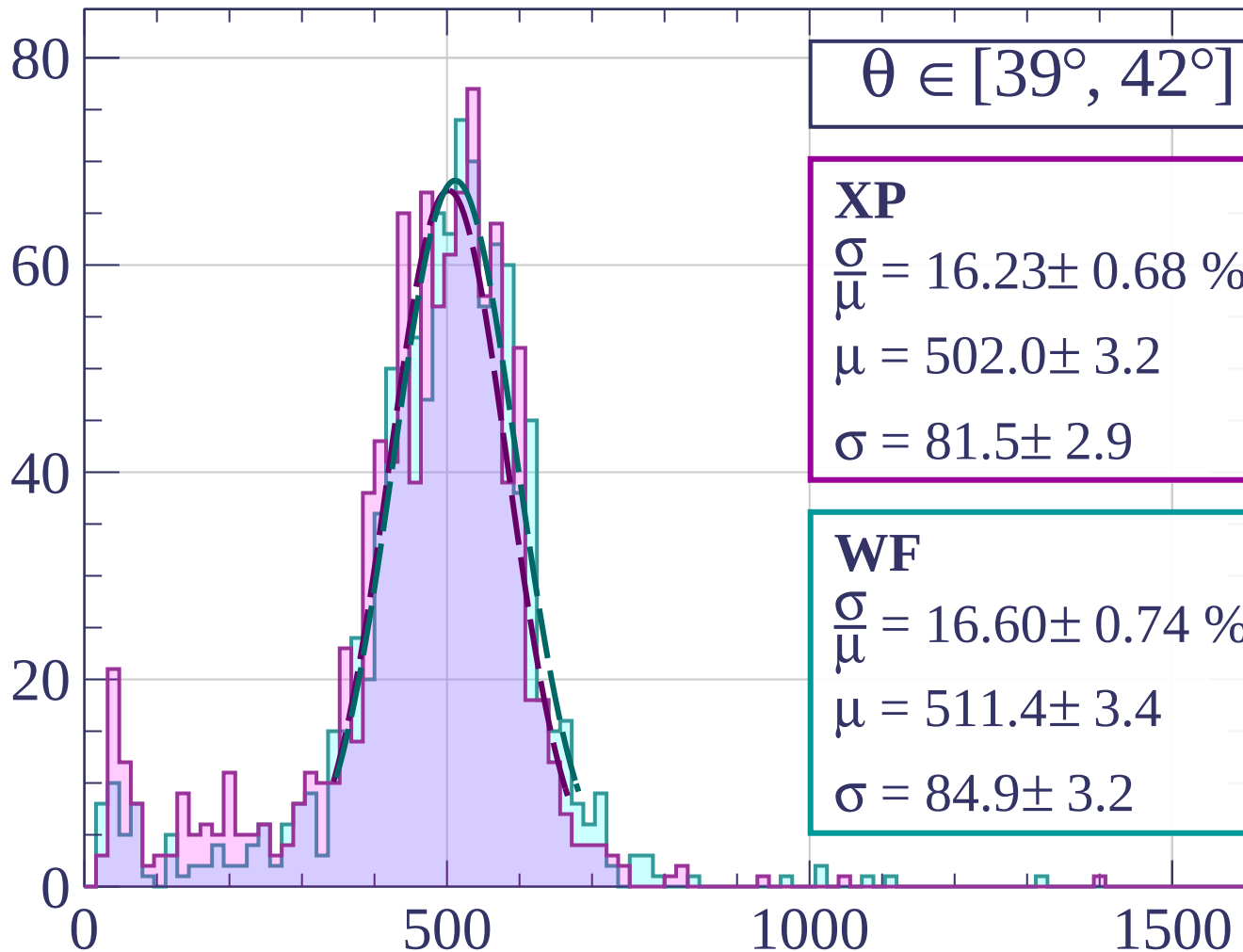


Count



dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count

$\theta \in [42^\circ, 45^\circ]$

XP

$$\frac{\sigma}{\mu} = 17.23 \pm 0.82 \%$$

$$\mu = 494.0 \pm 4.0$$

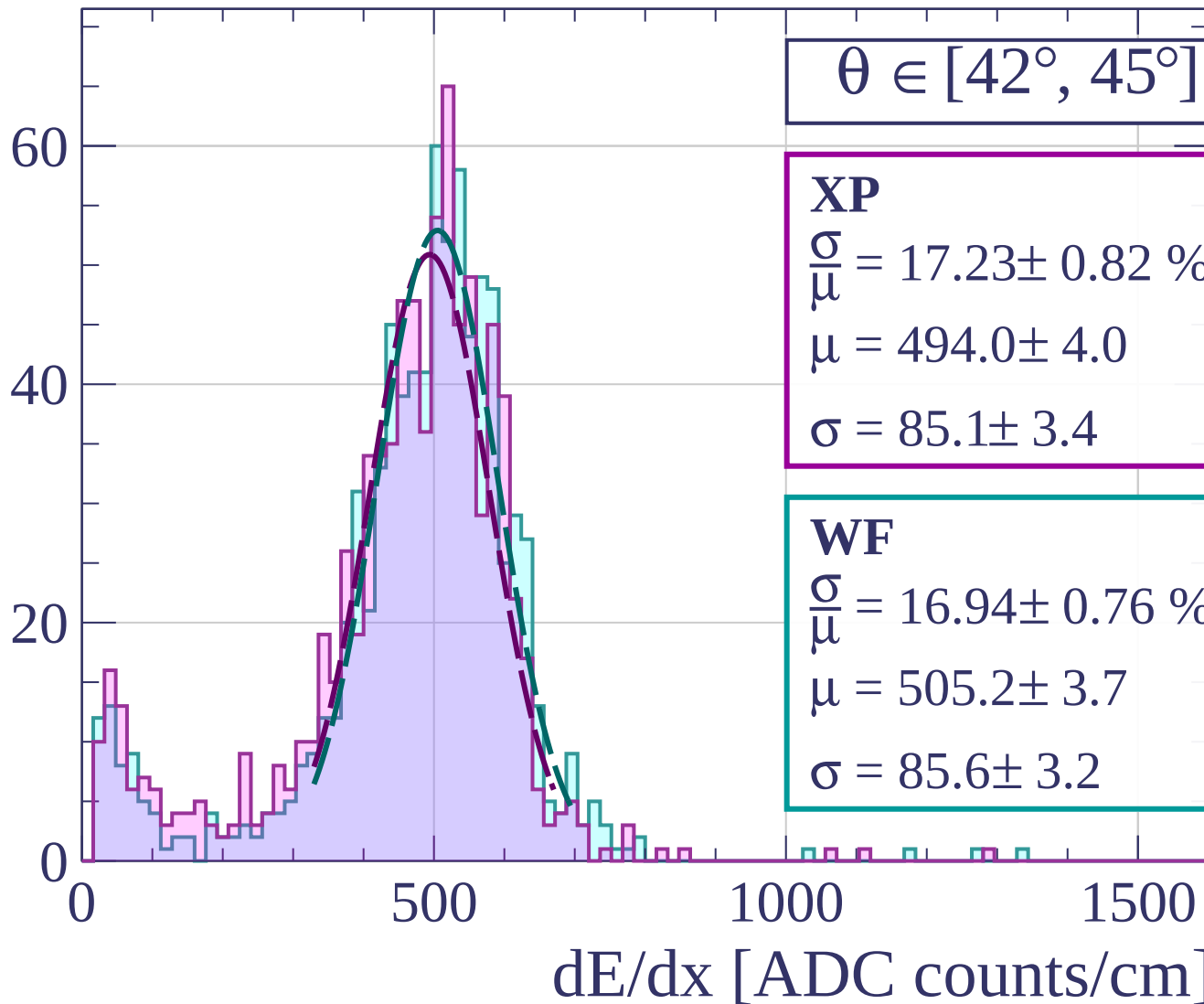
$$\sigma = 85.1 \pm 3.4$$

WF

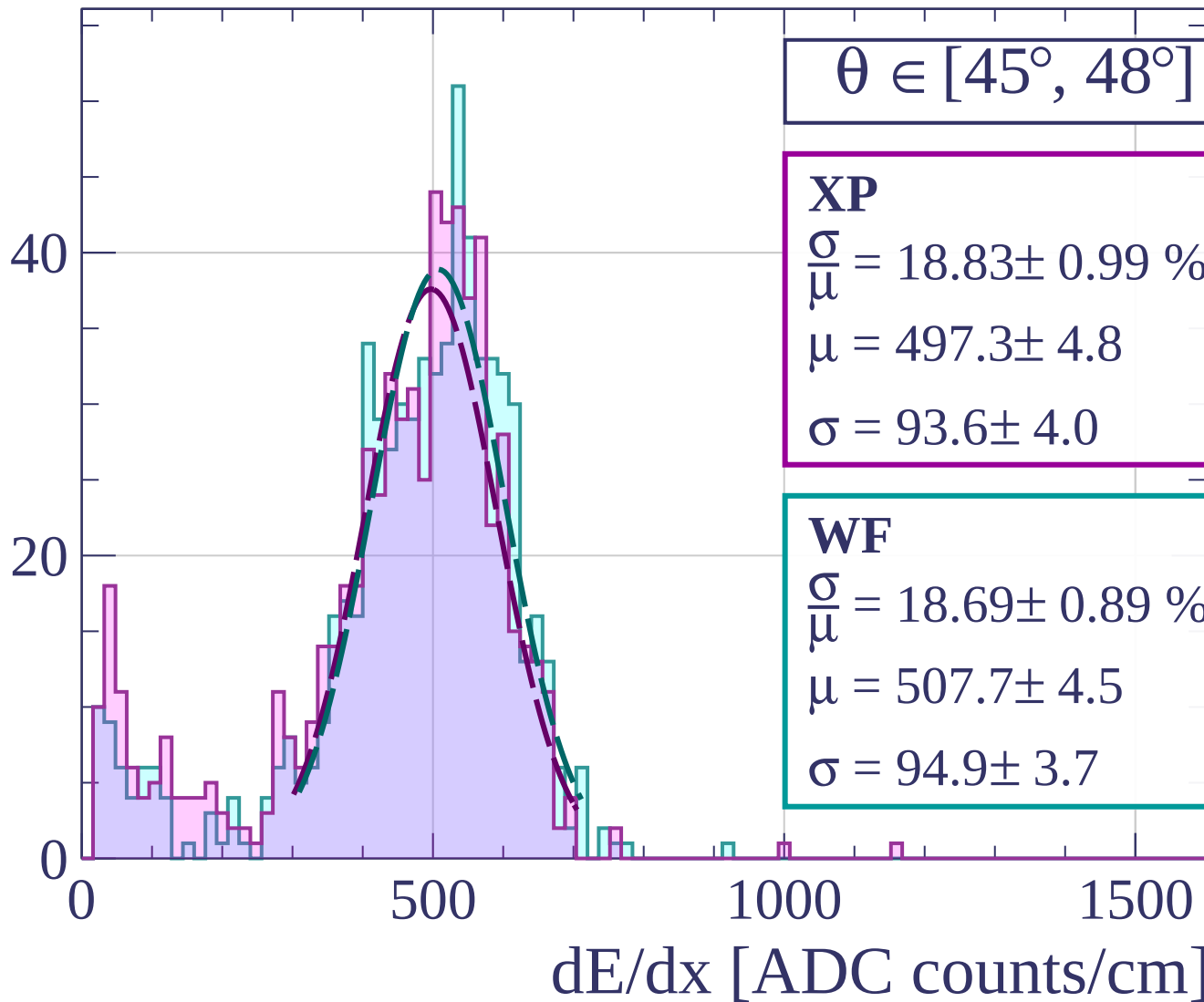
$$\frac{\sigma}{\mu} = 16.94 \pm 0.76 \%$$

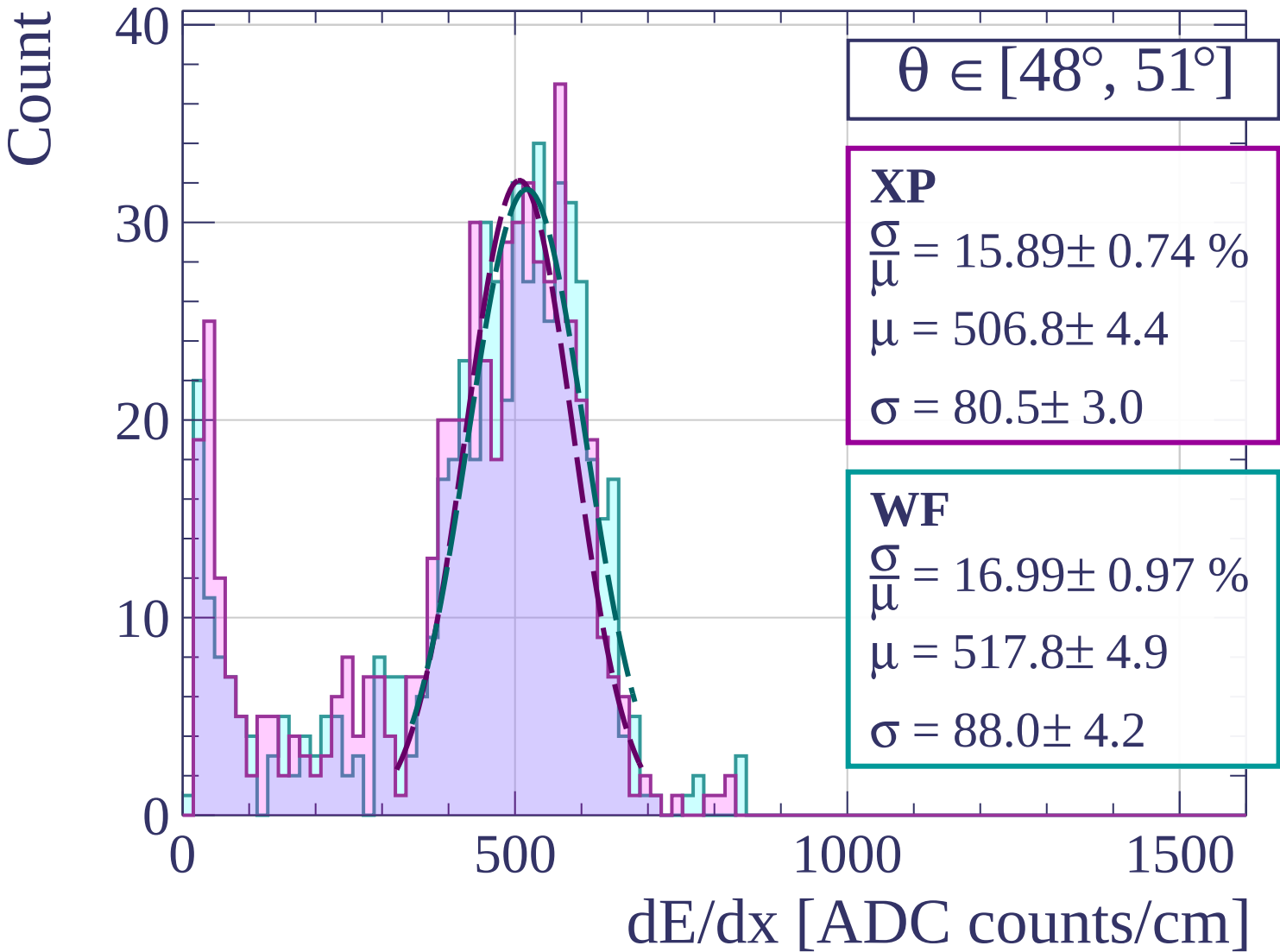
$$\mu = 505.2 \pm 3.7$$

$$\sigma = 85.6 \pm 3.2$$

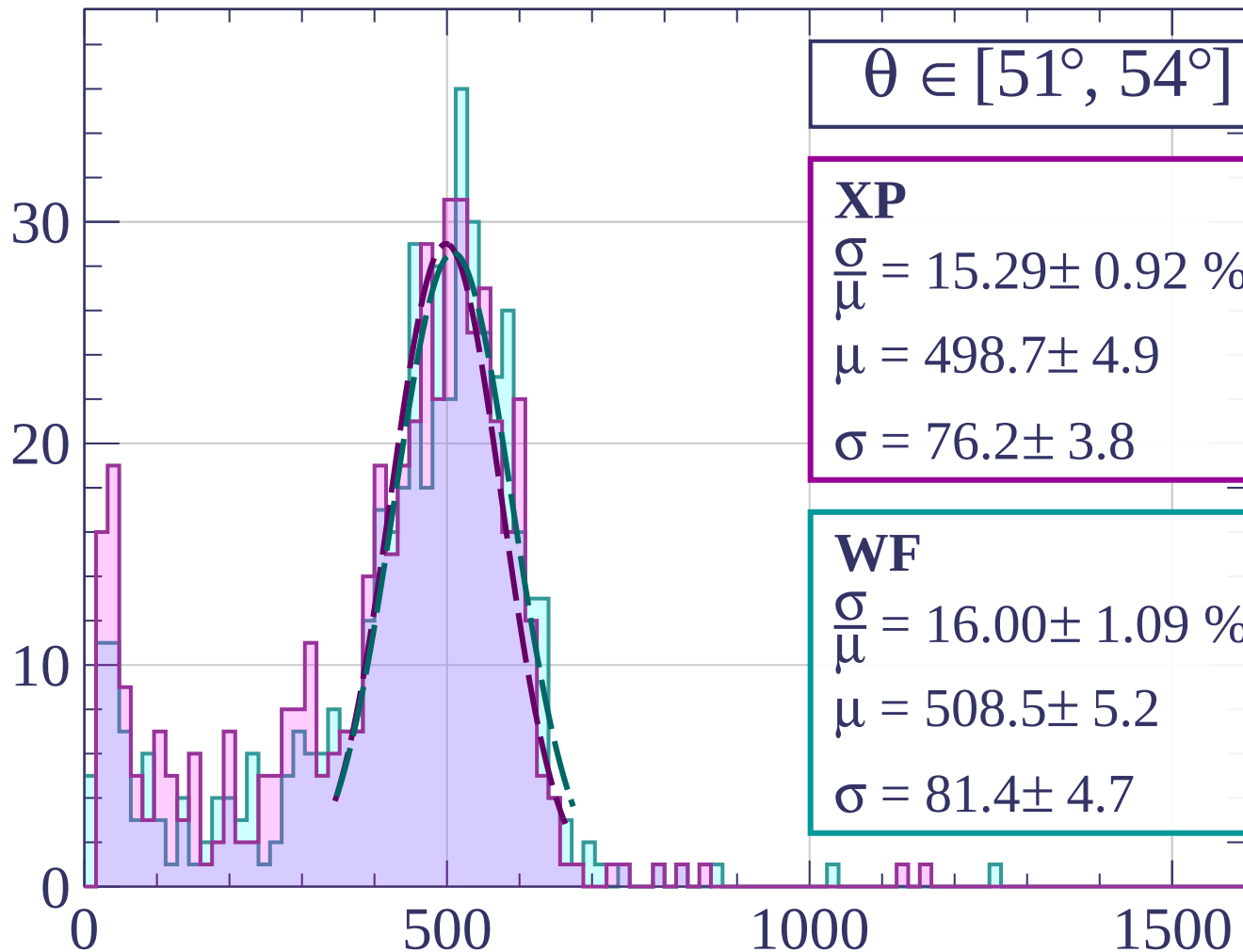


Count





Count



$\theta \in [51^\circ, 54^\circ]$

XP

$$\frac{\sigma}{\mu} = 15.29 \pm 0.92 \%$$

$$\mu = 498.7 \pm 4.9$$

$$\sigma = 76.2 \pm 3.8$$

WF

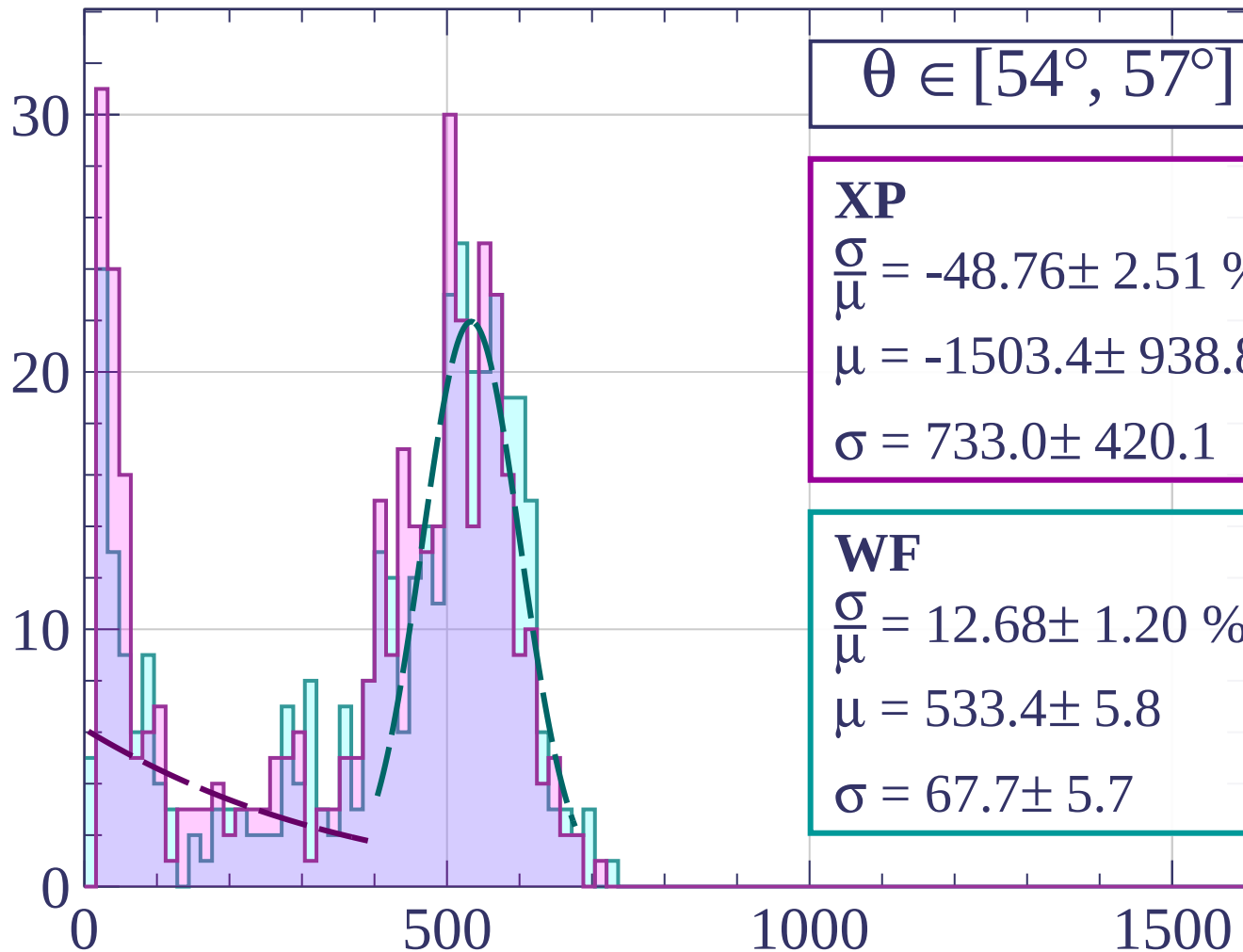
$$\frac{\sigma}{\mu} = 16.00 \pm 1.09 \%$$

$$\mu = 508.5 \pm 5.2$$

$$\sigma = 81.4 \pm 4.7$$

dE/dx [ADC counts/cm]

Count



$$\theta \in [54^\circ, 57^\circ]$$

XP

$$\frac{\sigma}{\mu} = -48.76 \pm 2.51 \%$$

$$\mu = -1503.4 \pm 938.8$$

$$\sigma = 733.0 \pm 420.1$$

WF

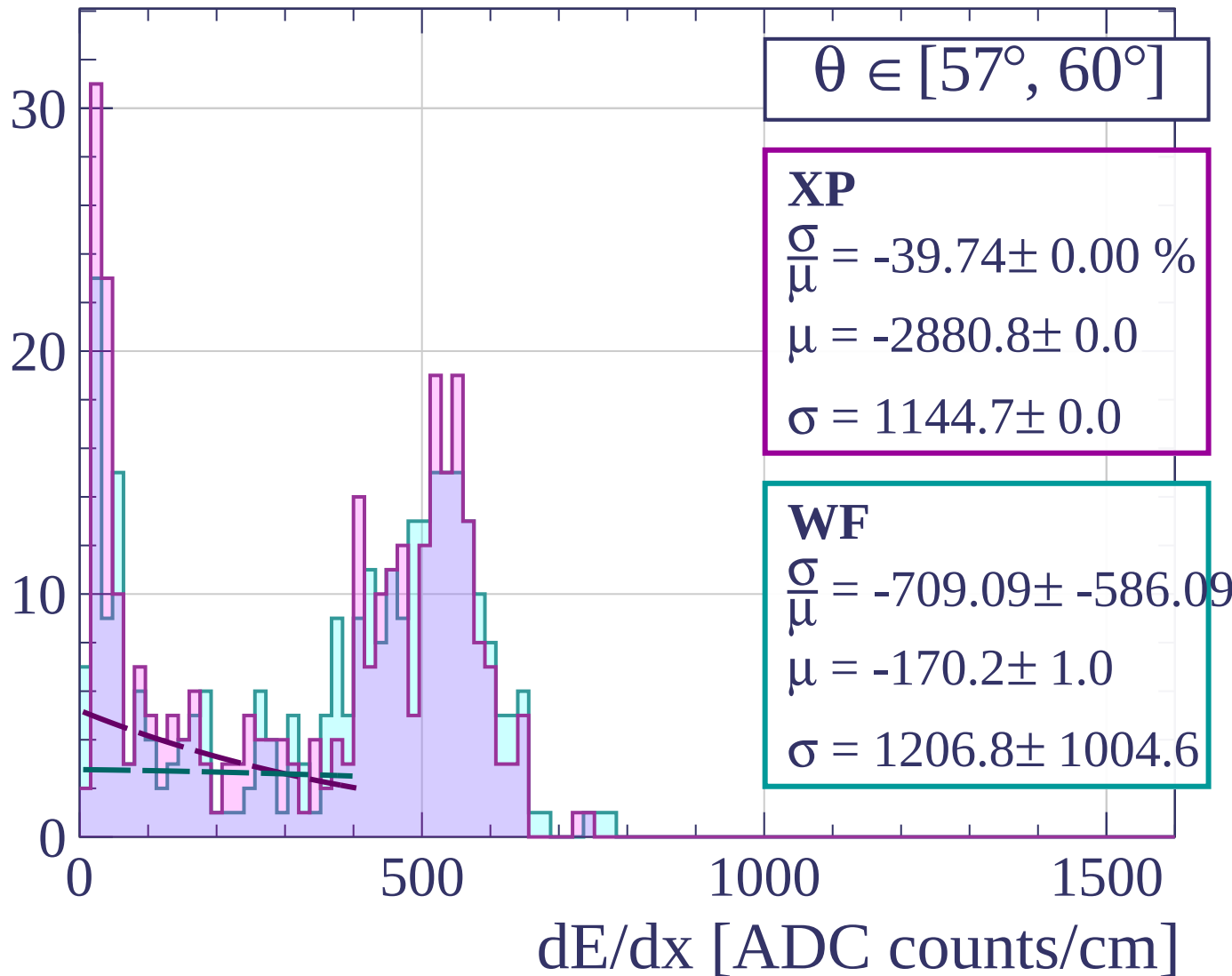
$$\frac{\sigma}{\mu} = 12.68 \pm 1.20 \%$$

$$\mu = 533.4 \pm 5.8$$

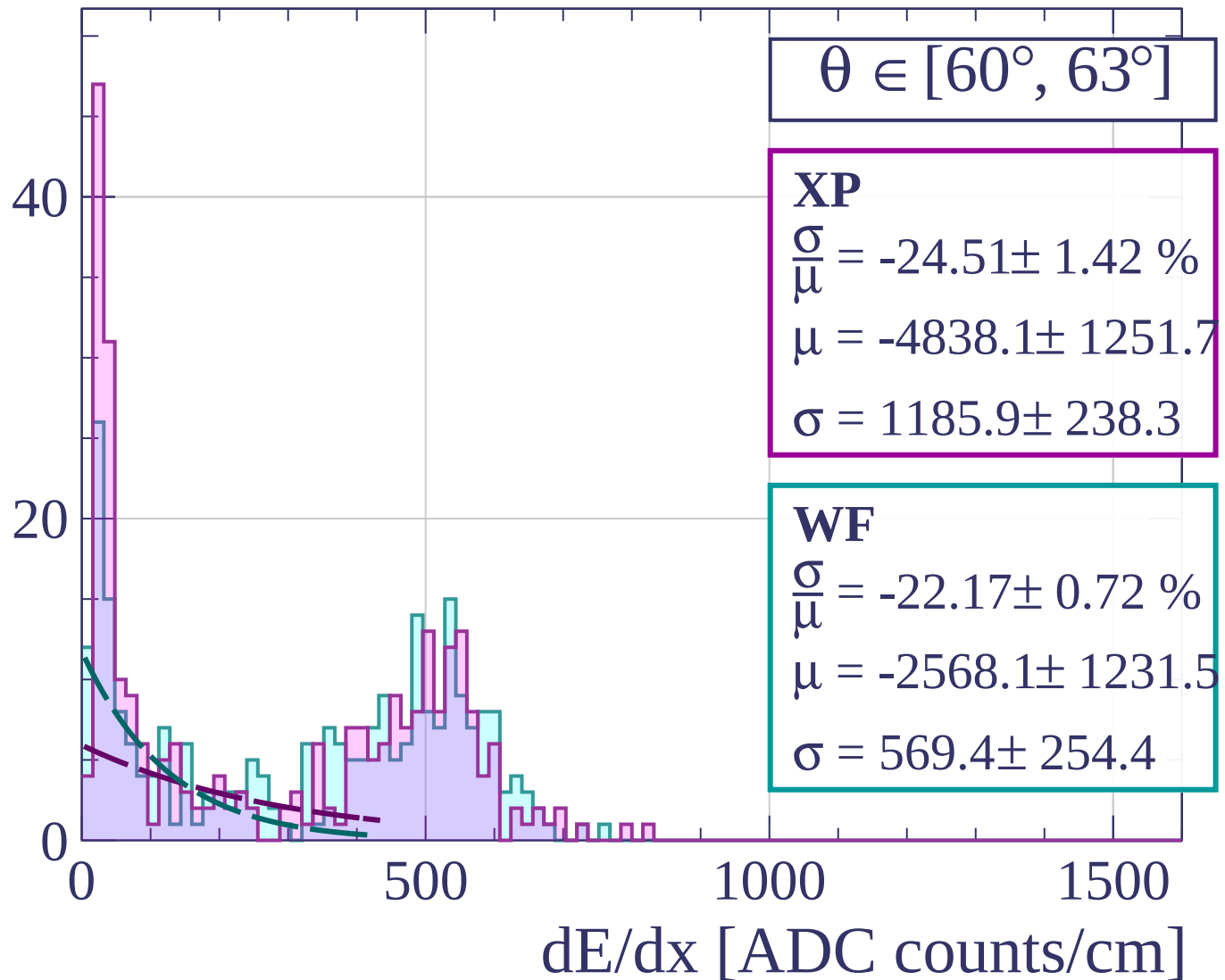
$$\sigma = 67.7 \pm 5.7$$

dE/dx [ADC counts/cm]

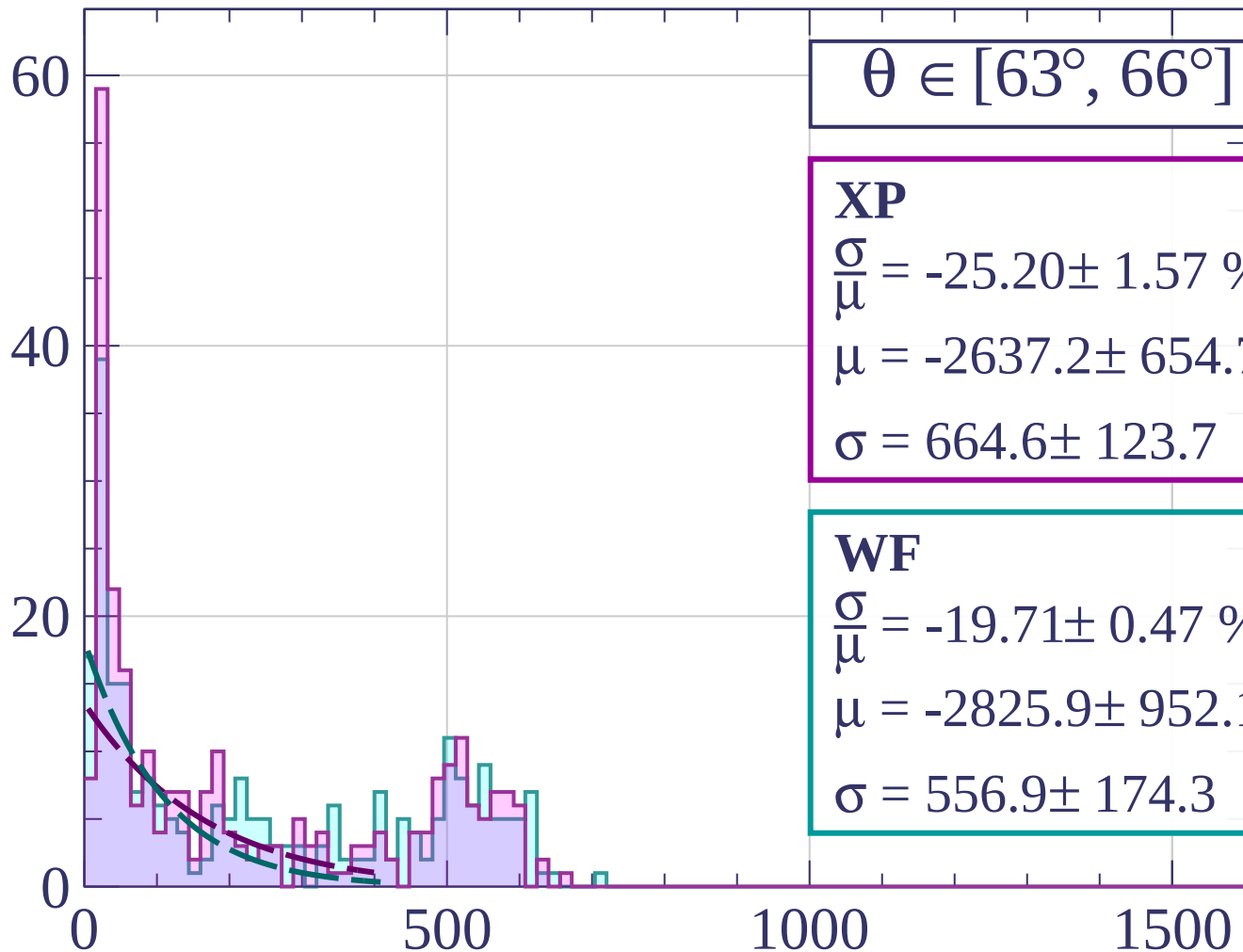
Count



Count

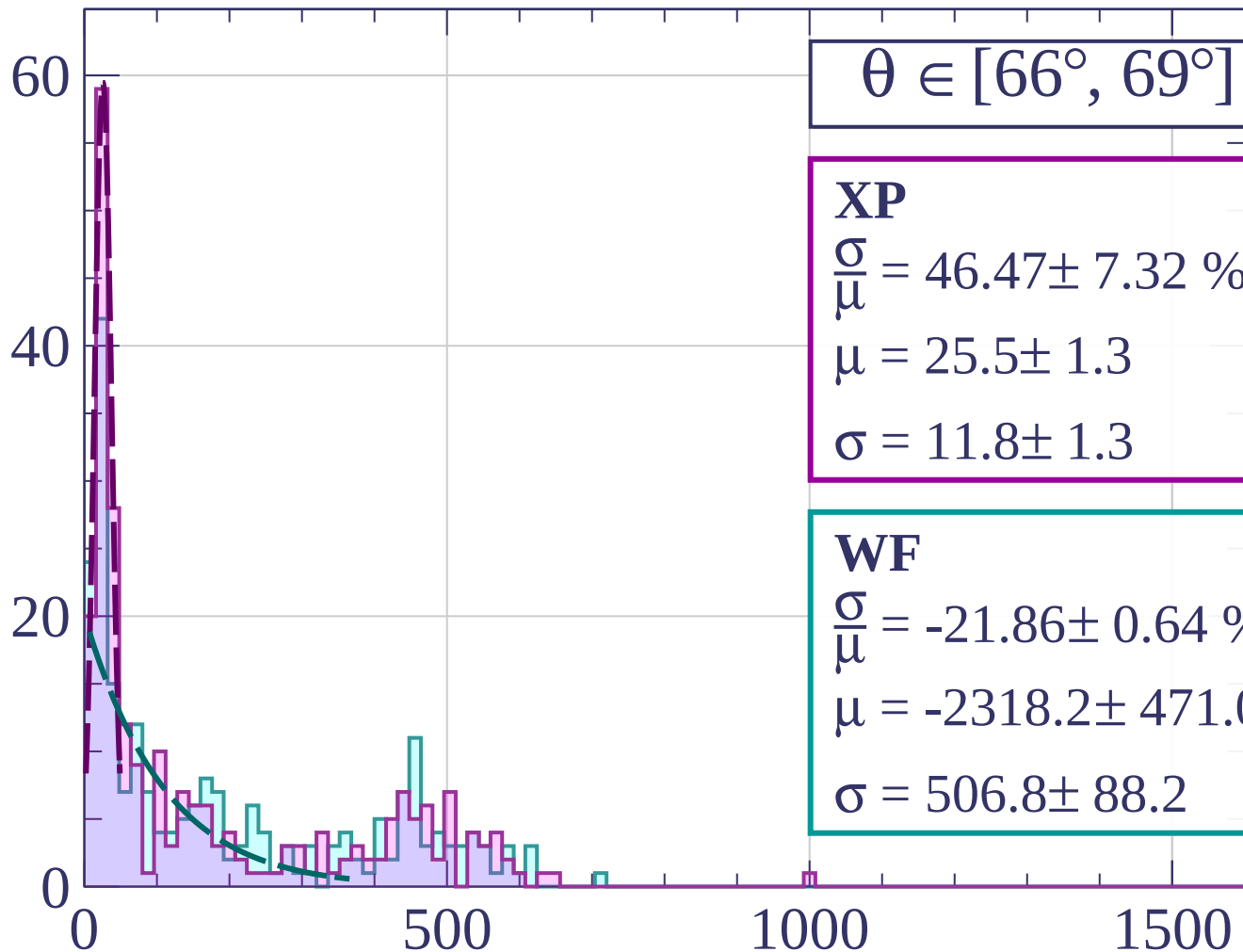


Count



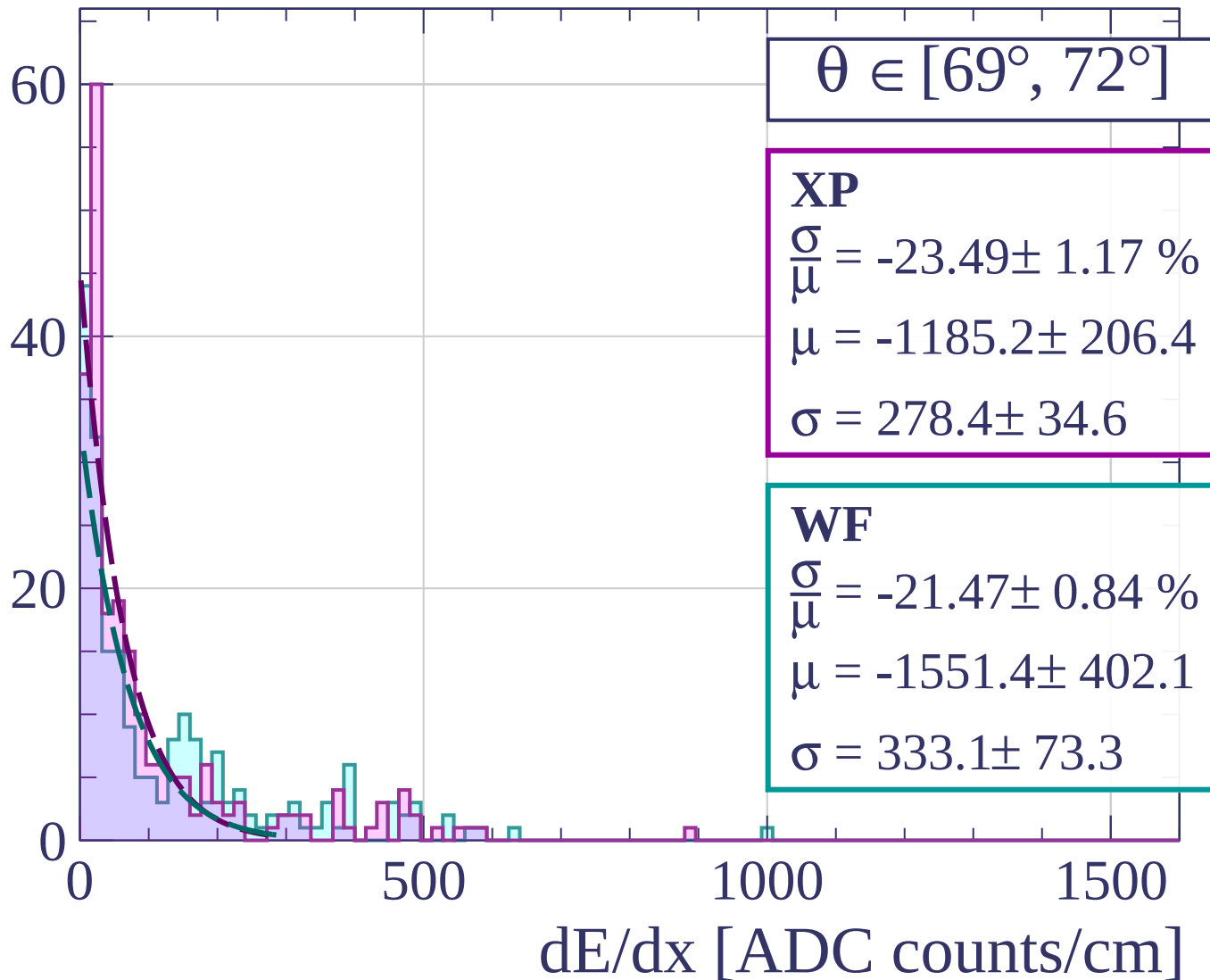
dE/dx [ADC counts/cm]

Count

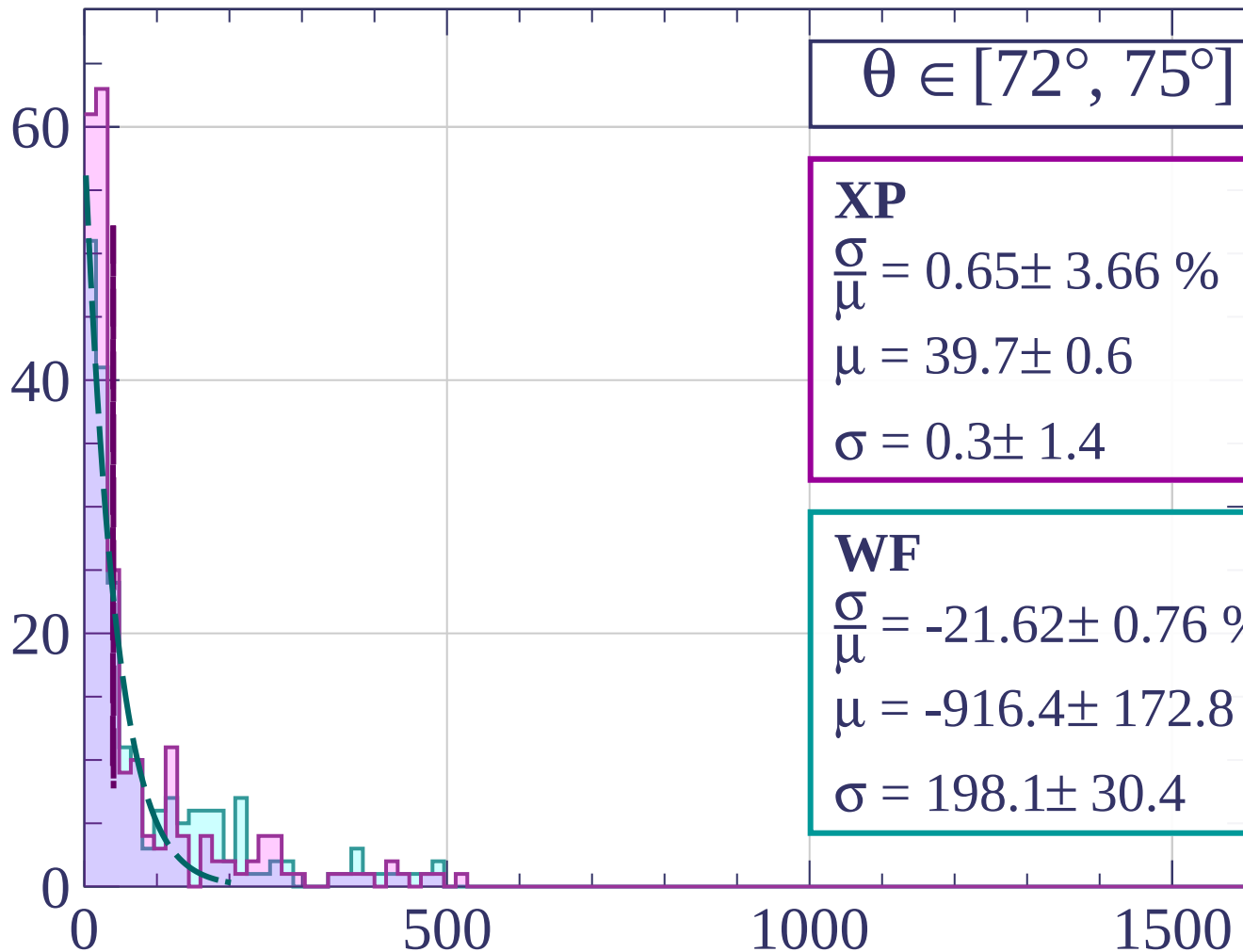


dE/dx [ADC counts/cm]

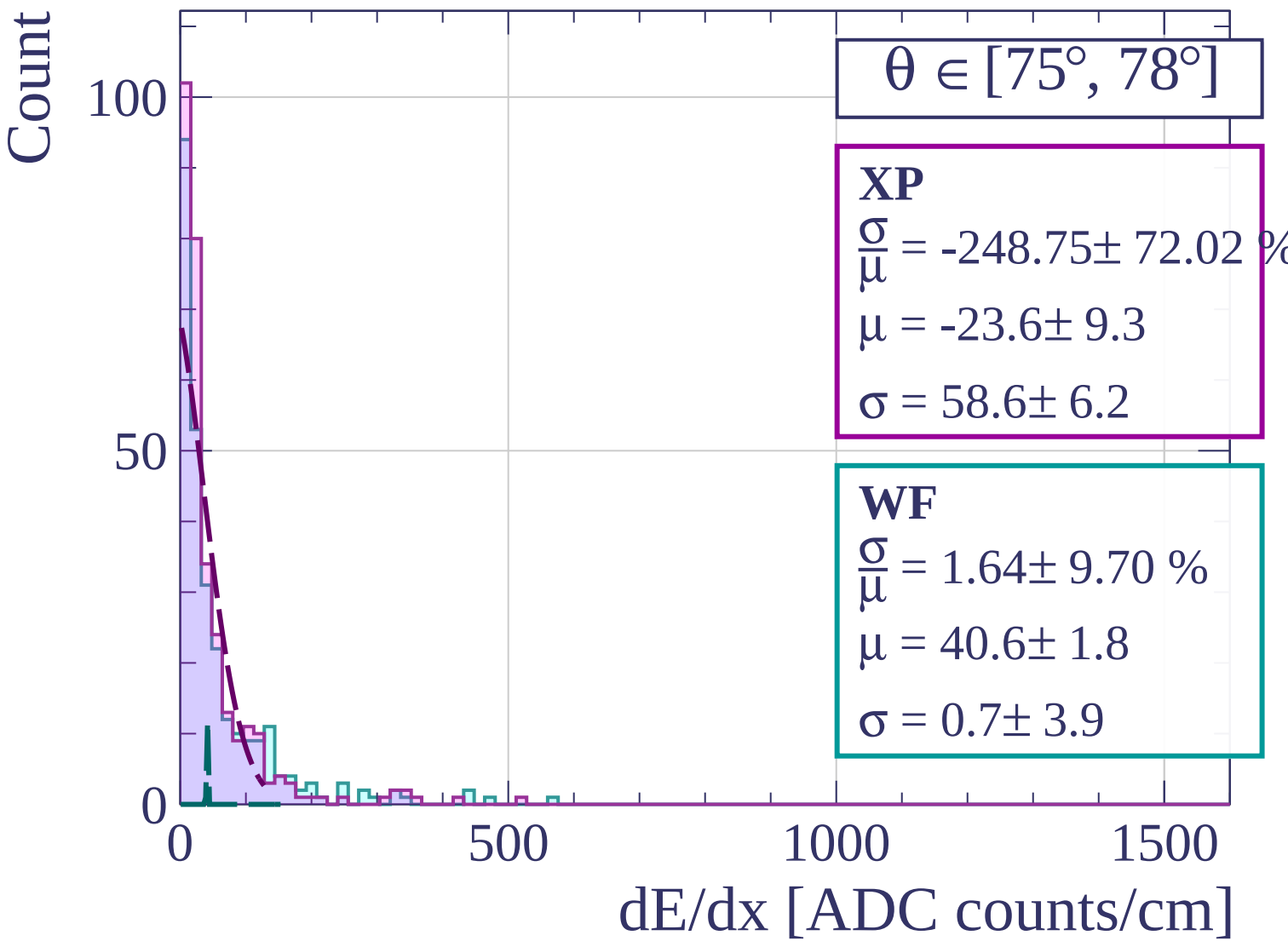
Count

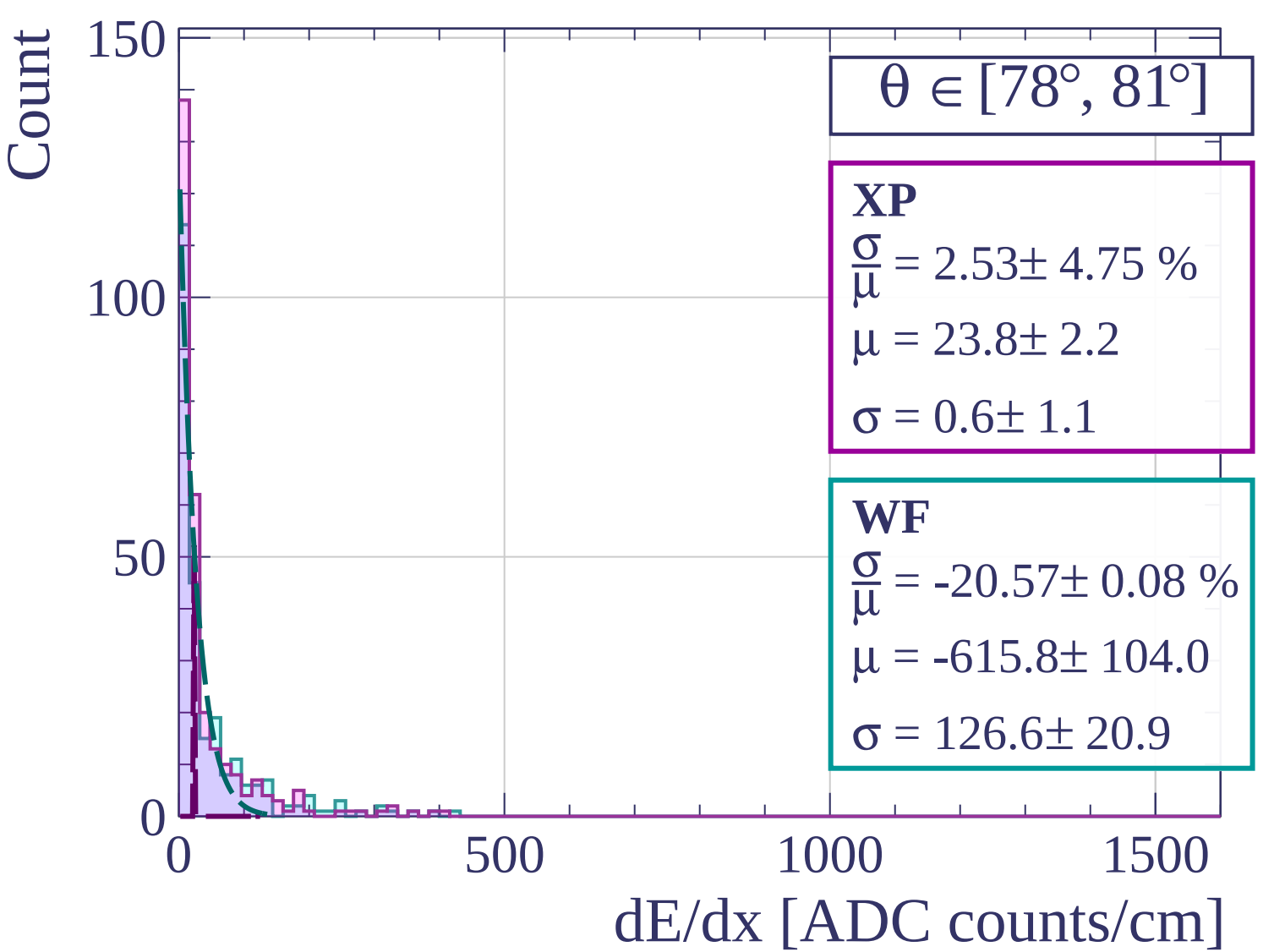


Count

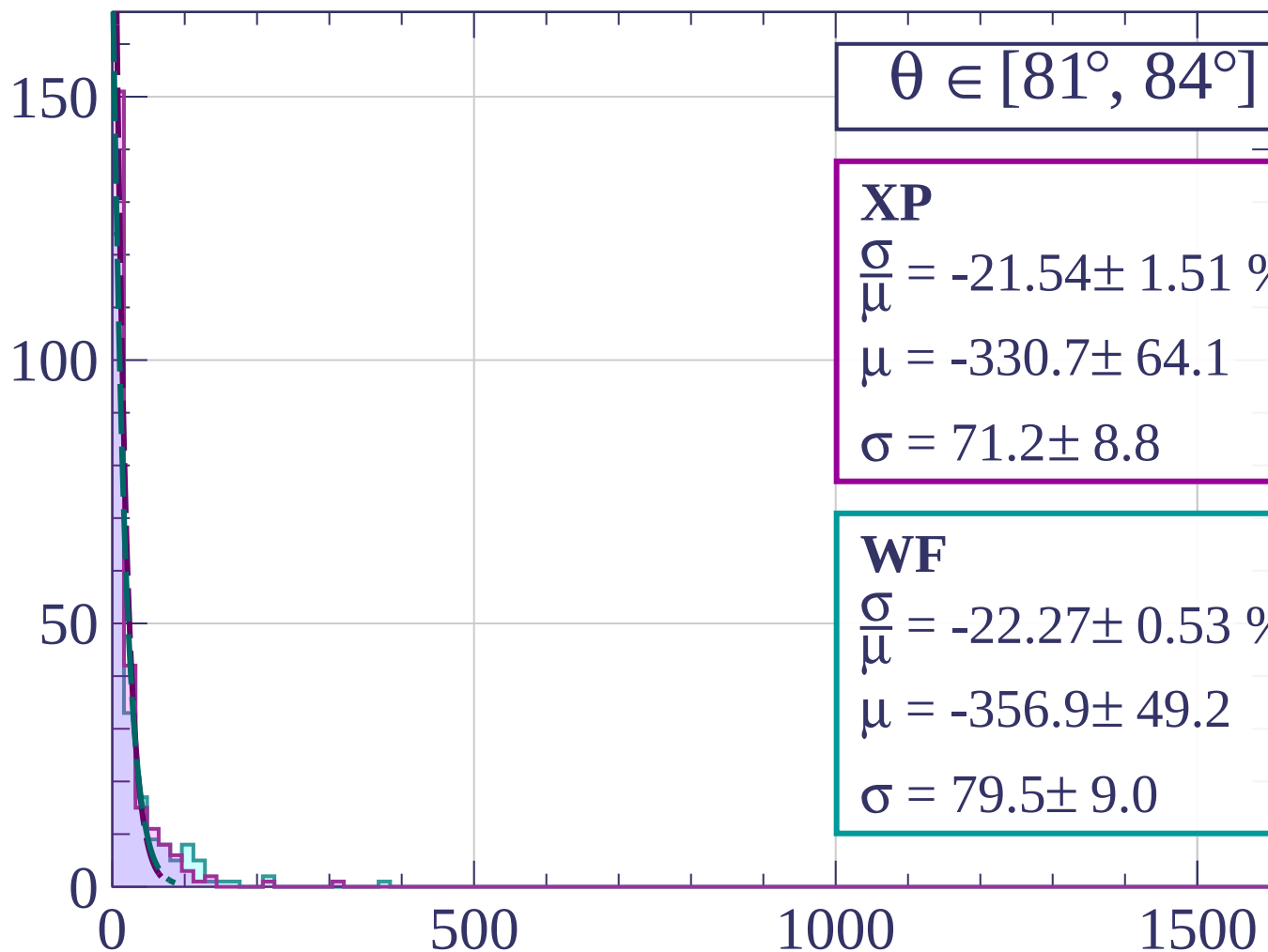


dE/dx [ADC counts/cm]





Count



dE/dx [ADC counts/cm]

Count

$\theta \in [84^\circ, 87^\circ]$

XP

$$\frac{\sigma}{\mu} = 107.19 \pm 319.73 \%$$

$$\mu = 8.0 \pm 16.1$$

$$\sigma = 8.6 \pm 8.4$$

WF

$$\frac{\sigma}{\mu} = -24.14 \pm 1.35 \%$$

$$\mu = -198.0 \pm 43.5$$

$$\sigma = 47.8 \pm 7.8$$

0

50

100

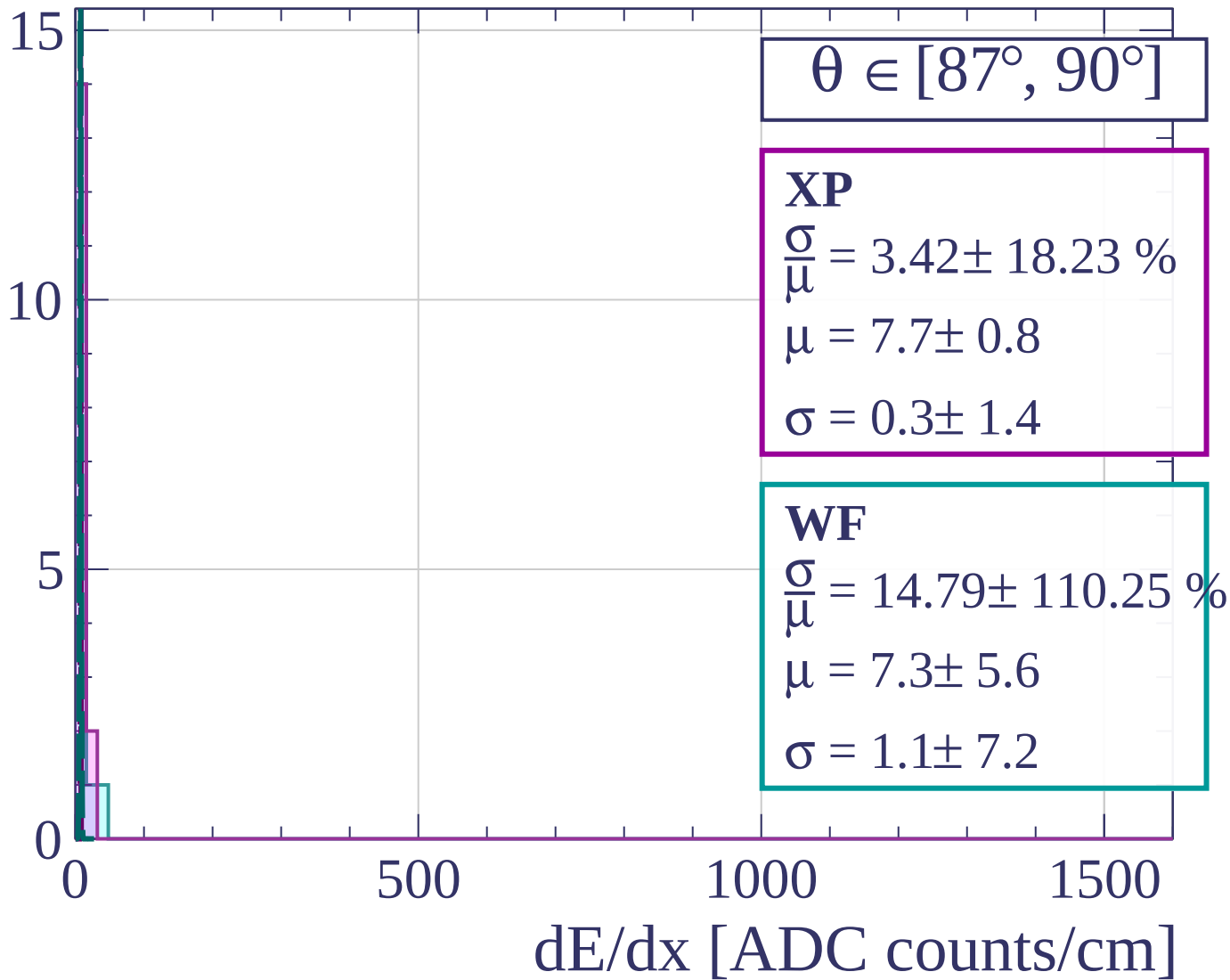
500

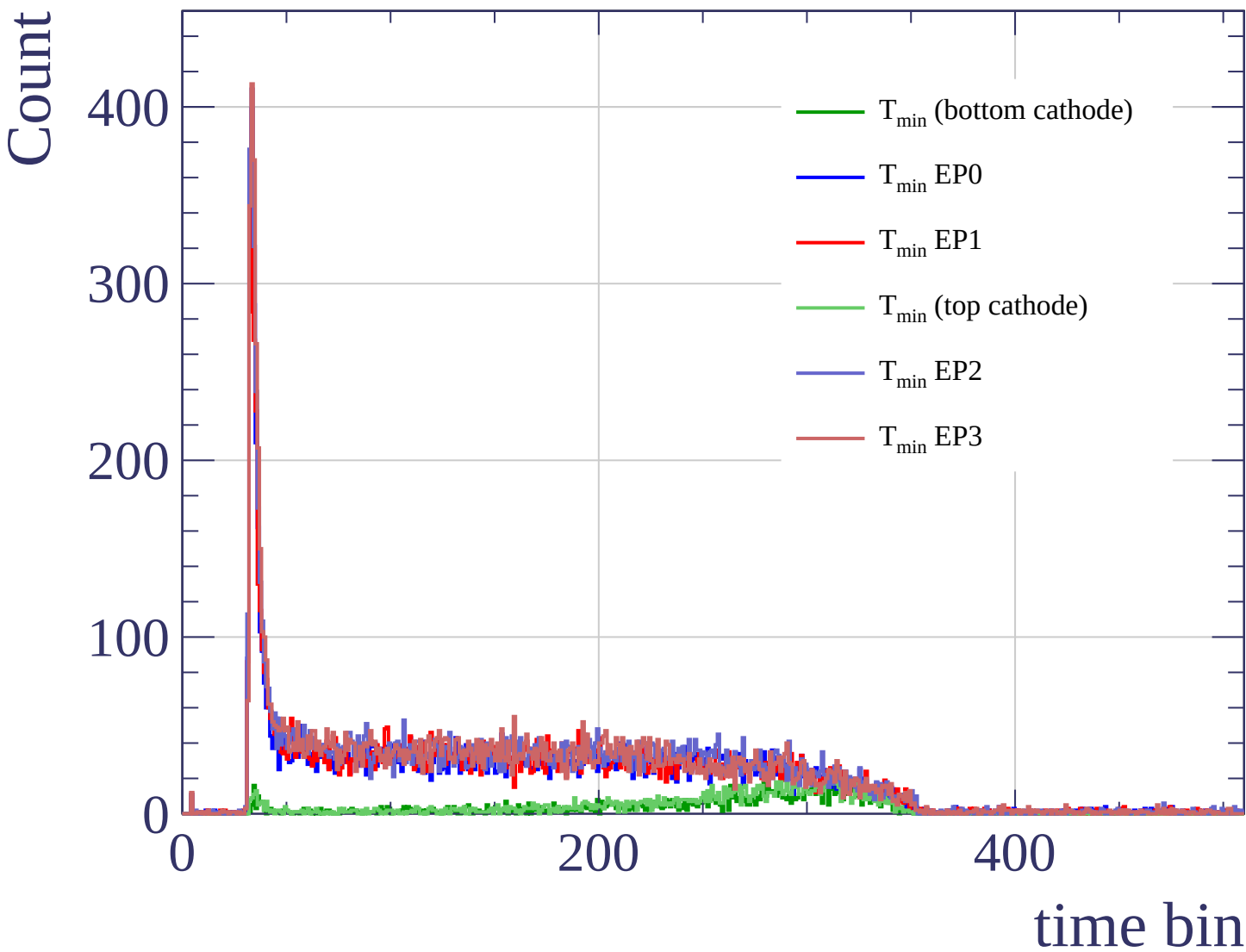
1000

1500

dE/dx [ADC counts/cm]

Count





Count

600

400

200

0

$T_{\max}$ (bottom cathode)

$T_{\max}$ EP0

$T_{\max}$ EP1

$T_{\max}$ (top cathode)

$T_{\max}$ EP2

$T_{\max}$ EP3

200

400

time bin

